User's Manual for EFDC_Explorer8.4: A Pre/Post Processor for the Environmental Fluid Dynamics Code

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1 Introduction

EFDC_Explorer (Figure 1-1) is a Microsoft Windows™ based pre-processor and post-processor for the three-dimensional (3D) hydrodynamic model, Environmental Fluid Dynamics Code (EFDC), initially developed at the Virginia Institute of Marine Science (Hamrick, 1992 & 1996). The US Environmental Protection Agency (EPA) has continued to support its development. The EFDC code is public domain and part of a family of models recommended by EPA for environmental and regulatory applications (https://www.epa.gov/ceam/environmental-fluid-dynamics-code-efdc).

Figure 1-1 EFDC_Explorer Splash Screen.

EFDC is a general-purpose model for simulating three-dimensional (3-D) flow, transport, and biogeochemical processes in surface water systems including: rivers, lakes, estuaries, reservoirs, wetlands, and near-shore to shelf-scale coastal regions. EFDC is capable of simulating cohesive and non-cohesive sediment transport, near-field and far-field plume discharge from multiple sources, and the transport and fate of toxic contaminants in the water and sediment phases. A dissolved oxygen/nutrient process (i.e. eutrophication) sub-model (HEM-3D) was added later (Park, *et al.,* 2000). Special enhancements to the hydrodynamics of the code now include vegetation resistance, drying and wetting, hydraulic structure representation, wave-current boundary layer interaction and wave-induced currents. The EFDC code has been extensively tested and the code is currently used by many university, government, and engineering and environmental consulting organizations.

The EFDC model has been released in the past in various forms and versions. This user's manual focuses on only two versions, the 2010 GVC Version (EFDC_GVC) (EFDC+ Theory Guide, DSI, 2017) and the EFDC_DSI Version 2013_06_07. The EFDC_GVC version has an implementation of the Generalized Vertical Coordinate (GVC) system (Stacey *et al*., 1995; Adcroft and Campin, 2004, EFDC+ Theory Guide, DSI, 2017) that is supported in the current version of EFDC_Explorer.

This latest version of EFDC_Explorer continues to load input and output files from older versions of the EFDC model. However, after loading an older project into EE, users are must save the project into a new folder before running the model. This is because the EFDC input files used by the latest EFDCPlus/EE is not always compatible with older versions of EFDC_DSI/GVC until EE updates them to the new format by saving the model. EFDCPlus has been updated and expanded with a number of enhancements and new features that required the input files to be modified. For this reason, the user should also ensure that EE is correctly linked to the corresponding version of EFDCPlus (see below).

This user's manual provides guidance in the use of EFDC_Explorer. This manual is NOT a user's manual for EFDC. It is assumed that the user is familiar with the types of data and information required by EFDC. *EFDC_Explorer is a tool to assist qualified engineers and scientists in the development, testing, calibration and interpretation/analysis of the model.*

1.1 EFDC Version Compatibility Summary

EFDC has been undergoing continuous updates and bug fixes since its earliest development in the 1990's. During that time, the EFDC.INP file (i.e. the master control file for EFDC) as well as other input files, have undergone a range of changes and additions of card groups and parameters. EFDC_Explorer has been designed to completely load as many of these older models as possible. Table 1 summarizes the EFDC versions that EFDC_Explorer can successfully load.

Again, it should be noted that legacy models need to be loaded into EE8.1 and then saved prior to running to allow the models to be updated to the latest input file format.

Table 1-1 EFDC – EFDC_Explorer Compatibility Table.

Acronym	Description	Dates	Read	Write
EFDC_1D	A sub-model of EFDC that uses cross section data.		No	No
EFDC	The "Full" version of EFDC prior to the implementation of GVC.	2000 to 2007	Yes	No
EFDC_Hydro	Simplified version of EFDC with only hydrodynamics using the 3 time level solution. Intended for linkage to WASP.		Yes	No
EFDC_GVC	The "Full" version of EFDC with the GVC implementation. EPA released this version in 2008.	2007 to 2010	Yes	Yes 2010 Version
EFDC_DSI	The DS-INTL version of EFDC including dynamic memory allocation, particle tracking and internally generated wind waves.	2000 to 2015	Yes	No
EFDC_SGZ	Modified version of EFDC_DSI to handle Sigma Zed layer.	2014 to 2015	Yes	No
EFDC+ (EFDCPlus)	Includes EFDC_OMP and EFDC_SGZ as well as updates to EFDC_DSI	2016 forward	Yes	Yes

Beginning with EE7.1, when the user tries to start a new EFDC run, EE will check the EFDC_DSI executable version number to ensure that the correct version of EFDC_DSI is being used with the current version of EFDC_Explorer. If the user tries to run EFDC_DSI with an incorrect version, EE will report to the user that the versions are incompatible and will not allow the run to start. The EFDC_DSI executable file will have the same version number as that of EFDC_Explorer. For example, EE8.3 will only run EFDCPlus _083_OMP_YYMMDD.exe. Where YYMMDD is the release date such 170925.

Likewise, if a user is still using EE6, then it must be linked to the corresponding release version of EFDC_DSI, but the user must keep track of which EFDC executable works with which version of EE. The EFDC_DSI executable release date allows the user to correctly match the software releases.

1.2 EFDC_Explorer Capabilities

1.2.1. Recent Enhancements to EFDCPlus/EFDC_Explorer Capabilities

- Habitat analysis with Critical Limits Time Series (EE8.4)
- Habitat Suitability Analysis with Instream Flow Incremental Method (IFIM) (EE8.4)
- Time Variable sluice gate operations (EE8.4)
- Rule-based sluice gate operation (EE8.4)
- Time variable for withdrawal/return boundary condition (EE8.4)
- Spatially varying AHO and AHD (EE8.4)
- Improved structure for EFDC+ output files for increased reading speed and logic (EE8.4)
- Momentum option for the flow boundary type (EE8.4)
- Groundwater linked to mass balance (EE8.4)
- Whole channel rating curve option (EE8.4)
- Elevation specified – Free tangential open boundary condition (EE8.4)
- Toxics capability added to the SEDZLJ sediment sub-model (EE8.3)
- Sigma-Zed sub-model enhanced to support one vertical layer (EE8.3)
- Sigma-Zed sub-model linkage files to exclude unused vertical layers (EE8.3)
- Improvements to the Ice Model (EE8.3)
- Marine & Hydrokinetics Sub-model in EE (EE8.2)
- netCDF Output from EFDC+ (EE8.2)
- Full implementation of SEDZLJ Sediment sub-model in EE and EFDC+ including display of initial conditions and model results, and full mass balance in EFDC+ (EE8.1-EE8.3)
- Enhanced hydraulic structures including culverts and bridges (EE8.1)
- East-West connectors (EE8.1)
- Import structures in 3D

1.2.2. Summary of Capabilities

The following lists provide a summary of the major features of EFDC_Explorer. The lists are grouped into three primary categories based on the general use of each feature. The first group contains general purpose features while the other two groups summarize the major pre- and post-processing features.

It is recognized that many more options and features could be added to EFDC_Explorer. It is anticipated that many new features will be added as resources are available.

General

- Graphical interface to most of the commonly used EFDC features.
- Graphical interface for EFDC (Sigma, GVC and SGZ versions) sub-models:
  - Hydrodynamics
  - Density dependent flow state variables: Salinity/Temperature
  - Tracer
  - Sediment Transport including the SEDZLJ sub-model
  - Toxics including linkage to EFDC and SEDZLJ sediment sub-models
  - Water Quality with Sediment Diagenesis
  - Lagrangian Particle Tracking and Oil Spill
  - Rooted Plant and Epiphyte Model
  - Linkage of External Wave Model Results to EFDC
  - Wind Generated Waves
  - Ice formation and melt
- Extensive visualization and point and click inquiries of input and output data.
- Extensive use of popup tips to help the user select the proper inputs.
- Included EE User Manual Help file coupled to forms (F1).
- Extensive error and range checking for user inputs.
- Many functions work with Calendar date and/or Julian dates.
- Binary file access method to allow the access to files > 4.2 GB.
- Any number of snapshots that can be written by EFDC and managed/used by EFDC_Explorer.
- Output Plots and Tables in either Metric or English units.
- Ability to read, write and display ESRI® .SHP files.
- Ability to read, write and display Mapinfo® .tab files.
- Included EE User Manual Help file coupled to forms (F1).
- Ability to write KML files for grid and model 2D fields.
- Ability to user multiple processors (OMP) to significantly reduce run times.
- Double click to run a project (*.EE extension).
- SWAN control and linkage to EE system
- No need for template files for model setup
- Continuing support and development of the utility.

Pre-Processor-General

- Pre-Processor for the EFDC+ & EFDC_GVC 2D-3D Versions.
- Import many previous versions of the main control file (i.e. EFDC.INP).
- Courant # and Courant-Fredrick-Levy calculator and display tools.
- Run logging.
- Status error window

Pre-Processor-Model Generation

- Build Cartesian or simple Curvilinear models.
- Cartesian models can use expanding grid spacing and grid rotations.
- Easily increase or decrease vertical layering.
- Import CVLGrid grids
- Import complex Curvilinear models generated by third party utilities:
  - Delft RGFGrid formatted file (i.e. GRD file) ,
  - Grid95,
  - SEAGrid, and
  - Any generic cell based nodal coordinate file.
- Import grids from different hydrodynamic models:
  - CH3D-WES,
  - CH3D-IMS,
  - ECOMSED, and
  - Prior versions of EFDC.
- Import grids with multiple sub-domains

Pre-Processor-General Grid Tools

- Grid Orthogonality statistics and plots.
- Export any EFDC model grid as an RGFGrid formatted GRD file.
- Export the model domain outline as a XY file (P2D format).
- Export the model cells as a XY file (P2D format).
- Transpose and/or flip the I and J indexes.
- Clean and repair any DX/DY and cell angle problems using a repair utility.
- Triangular Cells on Border toggle option
- Model Domain Rotation
- Connect sub-domains with N-S and/or E-W connectors

Pre-Processor-Initial Conditions

- Easy and fast plan views of the model domain with model option specific viewing options.
- Develop/Edit bathymetry from Digital Terrain Models (DTM's) or irregularly spaced elevation data.
- Build/Edit sediment beds with toxics.
- Set and edit cell properties by "point & click" on the model grid.
- User defined polygon cell selections for editing.
- Use simple operators to edit one or any number of cells.
- View/Set Vegetation mapping (if used).
  - Link Vegetation to Polygons for rapid vegetation assignments.
- View/Set Groundwater mapping (if used).
- Refine grid manually by activating/deactivating cells from the cell map.
- Rapid setting of the initial conditions water surface or depths.
- Use of 3D polylines to assist in setting initial conditions.
- Create/Read a compact binary "archive" file for a model run.
- Assign initial conditions using spatially varying vertical measured/estimated profiles.
- View/Assign/Edit roughness field.
- View Courant #/CFL map.
- View/Assign/Edit "Channel Modifier" information/configuration (if used).
- Set particle seeding and Lagrangian Particle Tracking control options.
- Base Date Updating
- Automated Atmospheric and Wind Series Weighting

Pre-Processor-Boundary Conditions

- Define/Edit/Plot flow, hydraulic structure, open and withdrawal/return type boundary condition groups based on flow series.
- Define hydraulic structures including culverts and bridges governed by equations
- Define tides with harmonic constants
- Set and edit boundary conditions by “point & click” on the model grid.
- Boundary conditions time series intelligent editor and one-button plotting.
- Use familiar names to identify and label boundary cells and input time series.
- Label boundary groups on 2D maps and/or export group labels to a file for more control over labeling of maps in EFDC_Explorer or GIS applications.
- HSPF model boundary condition interface to quickly import HSPF results to EFDC.
- Generate spatially interpolated time series for open boundary conditions.
- Use concentrations instead of mass loading (HEM3D default) for the water quality flow type BC groups.
- Updated EFDC_DSI to internally compute bed shear stress and currents due to wind generated waves.
- Updated EFDC_DSI to include recent EFDC enhancements by EFDC development partners including sediment transport, SEDZLJ and hydro-mechanical devices (HMK).
- Addition of Withdrawal/Return Boundary Condition for reversing flows (i.e. bi-directional flows)
- Updated Hydraulic Structures Boundary Condition (EFDC and EE)
- Jet /Plume Boundary Condition Editor
- N-S and E-W connectors by groups

Pre-Processor-Mass Balance/Boundary Loadings

- Compute mass balance of various model constituents with plotting and tabular output of time series.
- Plot the boundary condition loadings for various model constituents or some derived parameters like Total Phosphorus, Total Nitrogen or Total Carbon.
- Compute average and cumulative loadings using the “Averaging” and “Integration” features of the time series plotting utility.
- Generate mass loading summary tables for each simulated parameter.
- Water Quality Boundary Conditions can be loaded as Concentrations rather than just Mass Loadings

Pre-Processor-Model to Model Comparisons

- Plan view plots of 2D model input fields compared by subtracting a “Comparison” model from the “Base” model.
- Compare water column initial conditions between models.
- Compare sediment bed conditions and bed elevations between models.
- Grids do not need to be identical; however, they do need to overlap horizontally.

Pre-Processor-Utilities

- Tracer configuration utility.
- A bitmap geo-referencing tool.
- Perform QA checks on input data prior to model runs.
- Merge multiple continuation runs into single data sets
- Create new model runs from any saved results from previous runs.
- Unix to Windows CR/LF conversion.
- Gradient fill as a display option for 2D plan view plots.
- Ability to Drag & Drop EFDC.INP files or Project Directories onto the Main EFDC_Explorer form to open a project.

- Ability to Drag & Drop model annotation files (overlays, labels and postings) while viewing the model

Post-Processor-General

- Post-Processor for the 3D Sigma Stretch, Sigma Zed and GVC versions of EFDC.
- Post-Processor for SEDZLJ sediment model including toxics linkage
- Calendar day/Julian date labeling of plots and animations.
- EFDC output file management utility for resampling (i.e. reducing snapshot frequency) or merging continuation runs into a single output file.
- High Frequency Snapshot capability to insert high resolution results snapshots into the standard EFDC snapshot frequency.
- Optionally plot the color ramp in grey scale (for publications).
- Toggle on/off the display of titles on plots (for reports and publications).
- Ability to output sediment bed layer data to the EE_BED.OUT file with a skip frequency to save disk space.
- Automated statistics and plots at end of model run
- Automated folder structure for saving output files, plots etc.
- EFDC Restart Options Improvements (EFDC and EE) for Continuation Runs

Post-Processor-ViewPlan (2D Plan)

- View/label cell maps.
- Quickly animate many of the results to the screen or an AVI file.
- Export results to the commercial graphics package TECPlot® or Google Earth.
- Compute model results statistics for current view or by polygon.
- Compass (north arrow)
- Overlay the model with line drawings and labels.
- Export NetCDF Output
- Display one or more geo-referenced bitmaps as a background to the model grid.
    - Several views allow for transparent grid cells to view the background and model results.
- Capability to output bitmap images at user defined resolutions
- Display multiple “Timing Frames” to temporally reference results to model boundary conditions for the following types:
    - Open boundaries
    - Inflows/outflows
    - Withdrawal/Return
    - Winds
- Display spatial scales in various units.
- Generate a new model from the output of an existing model for any selected time,
- View water surface/depth maps, animations and time series.
    - Transparent Cells for water depths/elevations and other water column results.
    - Flood Inundation Mapping
    - Compute/Display areal extents based on specified a minimum depth and duration.
    - Compute/Display areal extents based on a Depth*Velocity Flood Hazard Factor.
    - Compare up to 3 models on the same plot showing Areal Extents.
    - Energy Mapping
        - Compute/Display total head (WS Elevation + $v^2/2g$).
        - Compute FEMA Overtopping parameter (depth + v^2).
    - Low Pass Filter for Model Analysis
- View Sediment/Bottom results for any time output or animation, (by layer or averaged/totaled over the number of sediment layers):

    - Bottom topography elevations,
    - Bottom scour/deposition,
    - Bottom grain size distributions,
    - Bottom sediment mass distributions (by layer or totaled over the total number of layers (i.e. KB),
    - Bottom sediment mass fractions (by layer or averaged over KB),
    - Bottom sediment porosity (by layer or averaged over KB),
    - Bed surface shear stress.
- View Velocity at multiple layers as magnitude or vector
    - Vector Plots – anchor point for vectors
    - Regular grid extraction for velocity plots
    - Downstream Projection
- View Water Column results for any time output or animations
    - Water Column reporting options
        - By layer or averaged/totaled over the number of water column layers
        - By specified depth or elevation
    - Salinity,
    - Temperature,
    - Dye or a computed Age of Water (EFDC_DSI Only),
    - Toxics,
        - By class or Totals
        - DOC Complexed
        - Dissolved, and
        - POC Adsorbed.
    - Sediments (by class or totaled).
    - Water quality parameters
        - 22 EFDC modeled constituents
        - Derived parameters like Chl-a, algal growth limiters, and TSI's.
    - Anoxic volumes (User specified DO cutoff)
    - Light penetration (Secchi Depth, Extinction Coefficients, % Irradiance).
- View boundary conditions map and view boundary condition time series.
    - View profiles of sediment bed and water column properties with time.
    - Store and quickly display time series calibration comparisons.
    - Compare velocity data to other model runs or field data (e.g. ADCP)
    - Compare model results from two different models to each other.
    - Plot time series plots for boundary conditions
- View Sediment Diagenesis concentrations and nutrient fluxes
    - View by class or total concentrations of PON, POP & POC
    - View concentrations by diagenesis layer or totals
    - View nutrient fluxes
- Model Metrics
    - Grid Orthogonality map & statistics,
    - Cell angle maps,
    - Courant Number,
    - Courant-Friedrichs-Levy (CFL) time step limits,
    - Froude Number,
    - Densimetric Froude Number,
    - Celerity, and
    - Richardson Number.
- View Longitudinal Profile plots
    - Generate longitudinal plots of water column and bottom sediments results.
    - User specified layer averaging (e.g. "1-3" will average the bottom three layers).
    - Water column or bottom sediment data can be overlaid with bottom topography and/or water surface elevation/depth results and/or bed shear.

    - Animations of the Longitudinal Profiles can be generated to the screen or AVI.
    - Display timing frame in Time Series and Profile Plotter
- Overlay plan view plots with the following Calibration Data/Information:
    - Station ID,
    - Date/Time coordinated data values,
    - Date/Time coordinated data residuals, and
    - Data values and residuals can use the same depth averaging or specified layer options specified for the model results.
- View plotting and animation of Lagrangian Particle Tracking and Oil Spill.
    - Ability to export one or more particle tracks to ASCII files for linkage to 3rd party applications.
    - Particles can be assigned to specific groups for separate treatment and display.
    - Particles can have an assigned settling rate or free floating.
    - Lagrangian Particle Tracking (LPT) by time release
    - Group Method for LPT implemented
    - External LPT module integrated into EE
    - Group Method for LPT implemented.
    - Oil evaporation and biodegradation
- View plotting and animation of the Rooted Plant and Epiphyte Model (RPEM).
    - Shoot carbon
    - Root carbon
    - Detritus carbon
    - Epiphyte carbon
- Save plot scales, positions for the model and annotation boxes for reuse later to allow plots to be formatted exactly the same from run to run.
- Specialty Analysis Tools including:
    - Habitat analysis based on critical limits, suitablility criteria, water depth, temperature and other water column parameters
    - Ability to compute the location of a “target concentration” of any water column constituent along a user specified polyline .e.g. river mile location of 2 ppt salinity.
- Output time series at a specified depth.
- Contour lines
- Wind and Velocity Rose Plots
- Photo Viewers in Time Series Forms
- Annotated Plan View of Model Domain in Time Series Form
- Flight Path Animations

Post-Processor- ViewSlice (Horizontal Profiles)

- View Velocity at multiple layers as magnitude or vector
- View Water Column results for any time output or animations
- Contour lines
- View ice thickness

Post-Processor- Vertical Profiles

- View 2D vertical section plots along any I or J index or user defined polyline.
- For any vertical section view/animate:
    - Velocities,
    - Salinity,
    - Temperature,
    - Dye or a computed Age of Water (EFDC+ Only),
    - Toxics,

- Sediments, and
- Water quality parameters.

Post-Processor- View3D (3D Visualizations)

- View 2D vertical section plots along any I or J index or user defined polyline.
- For any vertical section view/animate:
- Velocity Banners
- Blanking and Clipping
- Background Images
- Sediment Bed Viewing
- Mouse Inquire Information
- Import COLLADA structures for display in 3D

Post-Processor-Miscellaneous

- Water flux calculator by layer or totaled for all layers.
- Mass flux of any water column constituent by layer or depth averaged.
- Ability to calculate current time or time series of Mass Fluxes across any user defined polyline(s).
- Sediment mass balance/sediment mass loadings.
- Single cell sediment/water column process plots and animations showing scour, deposition and consolidation.
- Output time series at a specified depth.
- High and Low Pass Filter for Time Series
- Mean Mass Transport - Averaging of 2D/3D fields
- Velocity Profiles

Post-Processor-Calibration Comparisons

- Time series comparisons for water column data of measured to modeled data for any layer, depth averaged and/or Min-Avg-Max model results.
- Vertical profile comparisons.
- Correlation Plots
- Produce report ready graphics.
- Calibration statistics using:
  - RMS Error,
  - Average Error,
  - Absolute Error,
  - Relative Error, and/or
  - Nash-Sutcliffe Efficiency Coefficient.
- Automatic generation of the calibration plots.
- Automatic generation of calibration statistics.
- Model Results Analysis Macro for consistent plots and tables.
- Cruise Plot Comparisons
- Forced evaporation analysis

Post-Processor-Model to Model Comparisons

- 2D plan view plots of model results compared by subtracting a “Comparison” model from the “Base” model.
- Compare water column input and model results between models.
- Ability to compare sediment bed conditions, bed shear stress and scour/deposition between models.
- Compare computed layer or depth averaged velocities between models.

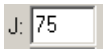
- Grids do not need to be identical; however, they do need to partially overlap horizontally.

1.3 Conventions

1.3.1. Windows Interface

While the general use of EFDC_Explorer is fairly standard with respect to a user interface for the Windows® operating system, some basic conventions will be explained here that should help the user.

Table 1-2 EFDC_Explorer user interface conventions.

	A black box with green text provide information only, it cannot be edited from that location. The information/data may be modified elsewhere.
	A while box with black text is the primary data/text input interface.
	A radial button indicates a range of options. Only one can be selected for each operation requested.
	The “Browse” button is used extensively in the program to allow the user the ability to navigate to the requested/desired file(s) rather than typing in the adjacent text box.

The user will notice several “grayed out” or disabled features in EFDC_Explorer. This indicates that a particular feature is not available for the currently applied version of EFDC or unavailable based on the user selected options. The “grayed out” options are not available to the user.

1.3.2. Message Boxes and the Clipboard

During the use of EFDC_Explorer various informational message boxes will be displayed, presenting the results of some calculation or other message. Most of these messages are also placed into the Windows clipboard for ease of transferring the information to some other applications. The data placed onto the clipboard are generally tab delimited.

1.3.3. Tooltips

When the cursor is passed over a button or field EFDC_Explorer will often show a tooltip that will help to explain the function of that button.

1.3.4. Operators

At several places in EFDC_Explorer the user has the option of entering a value to replace the current value of some input parameter (e.g., bottom elevation) or to use an "operator". The latter is a simple mathematical function that will be applied to the current value of the parameter. A field that allows *operators* recognizes the inputs described in Table 1-3. Operators must be followed by a space then the value, unless it is a simple replacement value.

Table 1-3 Operator descriptions.

Input	Description	Example		
		Current Value	Input "Operator"	Result
A Number	Replacement value	300	310	310
+ Number	Additive Operation	300	+ 1	301
- Number	Subtraction Operation	300	- 20	280
* Number	Multiplication Operation	300	* 1.1	330
/ Number	Division Operation	300	/ 1.1	272.73

1.3.5. Units

EFDC uses the metric system to define the space and concentration variables. Meters are used for all of the length related parameters and g/m^3 are used for concentrations, with the exception of salinity (kg/m^3) and toxics. For toxics, the units must be consistent between concentrations and the water column and sediment bed sorption parameters. Generally, these are in mg/m^3 or g/m^3.

The internal EFDC time units are in seconds. For EFDC, most of the input files can use any units along with a conversion factor to change the input units to seconds. EFDC_Explorer has fixed the user input units to days. All of the timing input should be in days and EFDC_Explorer generates the necessary conversion factors to seconds.

1.4 Terms & Abbreviations

The following is a description of common terms and abbreviations used in this report:

LMC "Left Mouse Click". On a standard mouse this refers to the left side button. Some Windows configurations can reverse the function of the left and right mouse buttons.

RMC "Right Mouse Click". On a standard mouse this refers to the right side button.

Ctrl The "Control" key on the keyboard.

Alt The "ALT" key on the Keyboard.

1.5 General Program Operation

Upon starting EFDC_Explorer the user will be presented with the form shown in Figure 1-2. This form is basically divided into three parts. At the top of the form, the main Toolbar is shown

providing quick access to many of the primary EFDC_Explorer features and actions. Beneath this is some general information about the location of the model. The main section of the form presents the pre- and post-processor inputs and options.

The number and function of the tabs shown along the left column of the form are automatically updated based on the user selected computational options. The operation and further instructions for each of these groupings will be described in the following sections.

It is strongly recommended that the user keep a run log. EE provides a simple text interface to record changes to a run from a previous run and/or run notes.

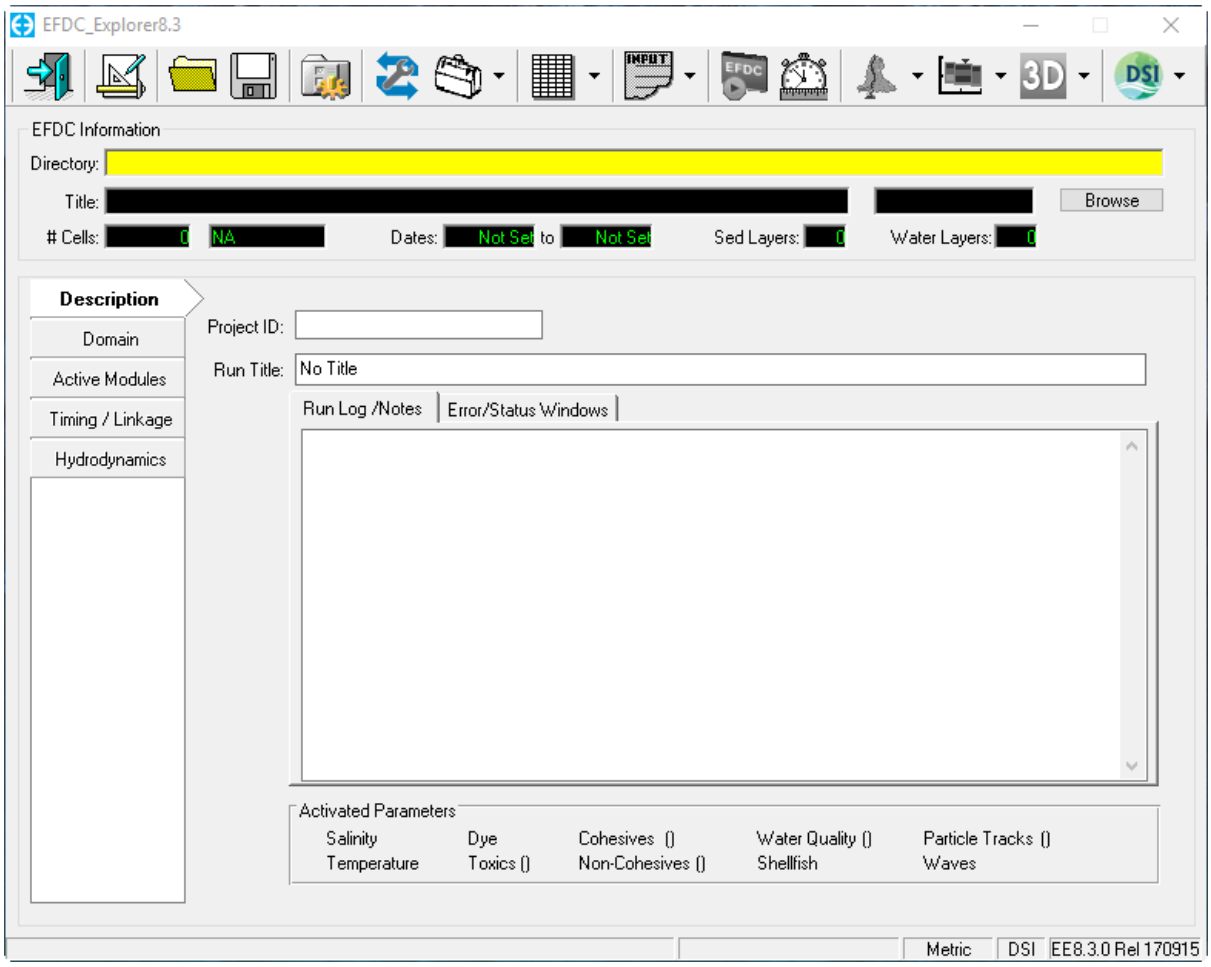


Figure 1-2 Main EFDC_Explorer form at startup.

1.6 EFDC_Explorer Files

1.6.1. EE Project Files

The main control file for every EFDC application is the EFDC.INP file. EFDC.INP is an ASCII file structured into card groups that generally have the same basic objective, e.g. card group 8 (C8) contains the settings for the run time but it also contains other miscellaneous parameters. This file contains almost all of the computational options and data settings.

The EFDC model uses fixed file names (e.g. DXDY.INP) based on the type of information each file contains. The files that are required for a model application vary based on the computational and grid options selected. For example, if the ISVEG flag (C5) is >0 then the VEGE.INP file, which contains vegetation information to compute vegetation based flow resistance, must be supplied. EFDC_Explorer reads and writes these same files and reduces the need for the user to remember exactly which file and/or which card group has what flag or setting.

EFDC_Explorer requires additional information and data to perform its pre-processing and visualizations. When saving an existing or new project, EFDC_Explorer automatically generates these files based on the project's settings. The following provides a list of EFDC_Explorer specific files and their functions:

EFDC.EE	This is the main EFDC_Explorer project file. This file contains much of the labeling, formatting, boundary condition, and model-data linkage information. This file is REQUIRED for EFDC_Explorer to correctly manage boundary groups
CORNERS.INP	This file contains the corner coordinates for each cell. EFDC_Explorer computes these using the cell center coordinates, DX, DY and cell rotation. It then matches the cell corners and builds the nodal list. By default, EFDC_Explorer displays the 2D plan view cells using these corner coordinates. However, the user can choose to view the rectangular cells if desired.
EFDC_LOG.EE	This file contains the run log that is displayed in the main EFDC_Explorer form. This file is ASCII and can be viewed with any ASCII editor.
CalForm_TS.EE	This file contains formatting and labeling information for the time series calibration plots.
CalForm_VP.EE	This file contains formatting and labeling information for the vertical profile calibration plots.

EFDC_Explorer now use the "*.EE" file extension rather than the old project file extension of "*.DS". EFDC_Explorer will still load .ds files but only .EE files can be double clicked to open the project. EFDC_Explorer will only update the files if the user saves the project, otherwise the old files will be kept.

1.6.2. EFDC – EFDC_Explorer Linkage

In order for EFDC_Explorer to **post-process** the data the following files must be generated by EFDC:

EE_WS.OUT	Required. This file contains the water depths.
EE_VEL.OUT	Recommended. This file contains the 3D velocity field.
EE_WC.OUT	This file contains the water column results as well as the information for the top layer of the sediment bed.
EE_BED.OUT	This file contains the sediment bed data for each layer, including the toxics associated with each layer.
EE_ARRAYS.OUT	This is an optional file that the EEXPOUT subroutine in EFDC optionally generates. This file contains snapshots of almost any internal EFDC array desired. EFDC_Explorer automatically loads this file and provides visualization, if it exists. (See Appendix A for more details).
EE_WQ.OUT	This file contains the water column results for water quality constituents simulated.
EE_SD.OUT	This file contains the sediment diagenesis results if the full sediment diagenesis option is turned on.
EE_DRIFTER.OUT	This file contains the time history of the location of the particles from the Lagrangian particle tracking feature, if turned on.
EE_RPEM.OUT	This file contains the Rooted Plant and Epiphyte Model results.
EE_BC.OUT	This file contains flows for boundary conditions.

These files are created by EFDC+ and the version of EFDC_GVC that DS-INTL distributes on our web site. EFDC_Explorer allows the user to set the appropriate options to view EFDC output. EFDC_Explorer generates the EFDC input files needed when the project is saved. EFDC then reads the input files and writes the EFDC_Explorer linkage files.

The following **subroutines** are used by EFDC to write the EFDC_Explorer linkage files:

SURFPLT Writes the depths for each cell for each snapshot into the file EE_WS.OUT. Water surface elevations are computed by adding the bottom elevations of the cells at the same model snapshot time.

VELPLTH Writes the U & V velocities in IJ space to the file EE_VEL.OUT. This allows EE the use of velocities on U and V cell faces for flux calculations. For visualizations, EE transforms the velocities by computing a cell centroid U & V then rotating them to the XY space using the cell rotations.

EEXPOUT This is the main subroutine to write the linkage files for EE. Depending on the options selected, this routine will write the files, EE_WC.OUT, EE_BED.OUT, EE_ARRAYS.OUT, EE_WQ.OUT, EE_SD.OUT and EE_RPEM.OUT.

DRIFTER This is the Lagrangian Particle Tracking module. It writes the particle positions to the EE_DRIFTER.OUT at an independent snapshot frequency to the other output data.

1.7 EFDCPlus Project Folder Structure

From release of EFDC_Explorer7 there has been a pre-set folder structure for model outputs and EE generated files. All input files needed by EFDCPlus and EE are stored in the main project folder. This makes it easier to identify which files are needed by EFDC/EE and which ones are not. To send a colleague/regulator the required files the user only needs to copy all the files (not subdirectories) in the project folder. All other EFDCPlus and EE generated files are placed in a defined sub-folder. EFDC model output is contained in the #output folder with other folders being used to organize and manage EE user created files. These include:

- #output for all EFDC output files.
- #analysis for general EE plot files and EE exported data.
- #animations for all EE generated .AVI animations
- #calib_plots for all calibration plots.
- #calib_stats for the calibration statistic files.
- #calib_export for the EE exported files from the calibration plotting utility.

When the user requests some plot or data extraction EE will still bring up a file browsing dialog allowing the user to write the files to any location, however, the default locations will be those identified above.

2 Installation & License Management

2.1 Operating System and Hardware Compatibility

It should be noted that EFDC_Explorer is designed for Microsoft Windows based operating systems (OS). However, EFDC and EFDCPlus can be run on either Windows or Linux OS. This page describes the various operating systems we have tested and regularly use to run the EFDCPlus/EFDC_Explorer Modeling System. Recommendations on the hardware specifications are also provided.

EFDC_Explorer

- The application is 32 bit. It is compatible on XP, Vista, Windows 7, Windows 8 and 8.1, Windows10, and Windows Server (2012) for either 32 or 64 bit.
- EFDC_Explorer can run on Apple Mac computers running the Windows OS
- EFDC_Explorer (v.8 and up) can be run on Virtual Machines running the Windows OS
- Modern Intel i5 and i7's as well as Xeon systems all work with EFDC_Explorer. The faster the CPU, the better the speed.

Note that Versions of EFDC_Explorer using a legacy licensing system (prior to EE8 release) are not compatible with Virtual Machines & issues associated with attempting to use the legacy systems on a VM may not be supported.

EFDCPlus

- The application is 32 bit. It is compatible on XP, Vista, Windows 7, Windows 8 and 8.1, Windows 10, and Windows Server (2012) for either 32 or 64 bit. It may also be compiled to run on Linux OS and DSI regularly use CentOS and Redhat to run EFDCPlus models to support real-time systems.
- EFDCPlus is recommended to be run on an Intel system in order to take full advantage of the efficiency gains for the OMP (multi-threading) capabilities. For versions of EFDC prior to EE8, the single threaded version of EFDC_DSI works on an AMD system, but EFDC_DSI_OMP should be run on an Intel system for efficiency gains.
- EFDCPlus has been run with various cloud-computing services such as Amazon AWS.
- Users of EFDCPlus should run the software on a CPU with as many cores as possible. DSI uses systems with 24 threads (12 cores) using multiple Xeon chips. The more threads/cores the greater the efficiency in model run times.
- Memory is not particularly critical as the memory for the EFDCPlus model is dynamically allocated and most models take < 250 MB of RAM. Four or eight gigabytes should be plenty of memory for most modeling systems.

2.2 Installation

1) If you received EEMS8.3 in a zip file (EEMS8.3.ZIP), unzip the zip file into a *temporary directory* on your hard drive. To simplify the cleanup of files later it is recommended that the *temporary directory* be empty before unzipping.

2) You should use the EEMS8.3 Installation program to install EEMS8.3 (EEMS8.3_SetupRel170927.exe). After opening the setup program the EEMS Install Wizard will run as shown in Figure 2-1 below.

Figure 2-1 Installation Wizard for EEMS.

3) If you used a *temporary directory* you should delete the files that were unzipped. Keep a copy of the zip file in case you need to install the program again.

4) As part of the conditions of use of EEMS, the user must accept the End-User Agreement before installation.

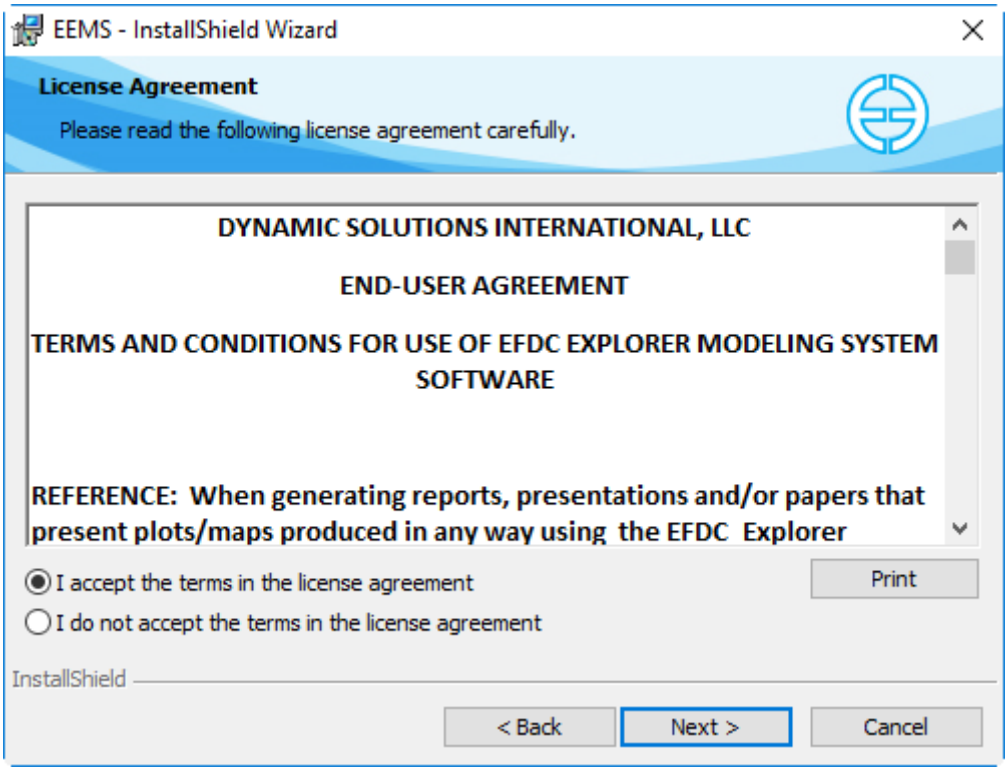


Figure 2-2 EEMS License Agreement prompt.

5) The user is then prompted for customer information. Please enter your User Name and Organization in the fields provided then click *Next* button.

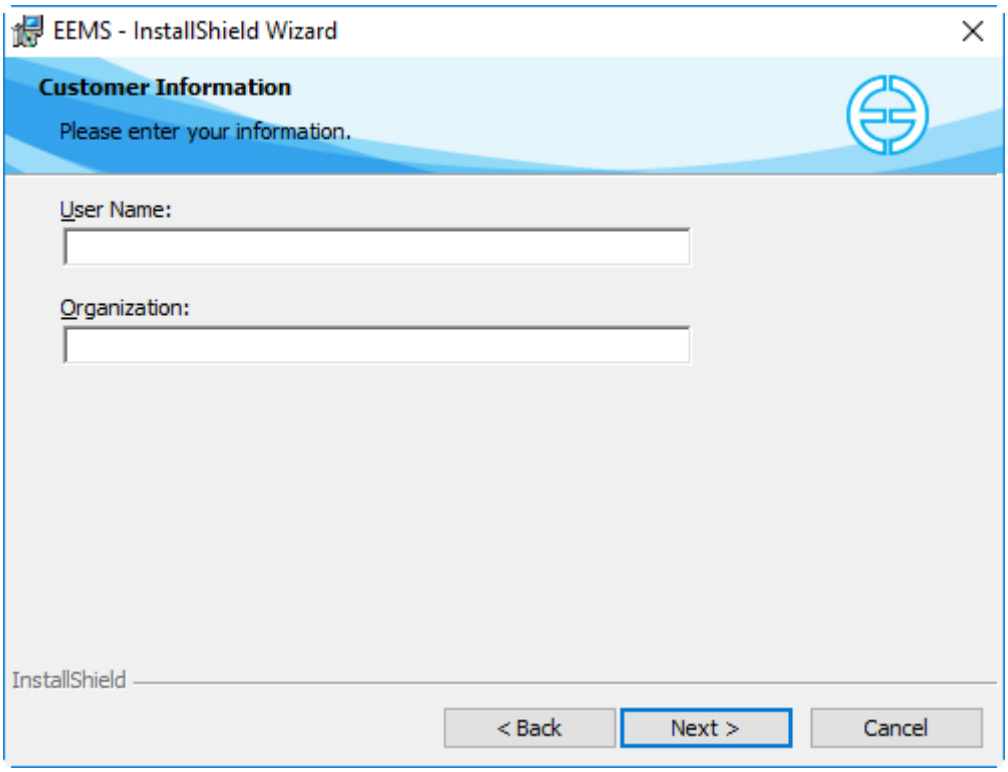


Figure 2-3 User information form.

6) After clicking *Next* button, a form for setting the destination folder for installation of EEMS and the example models appears. The user either choose default destination or change the destination as needed by clicking the *Change* button.

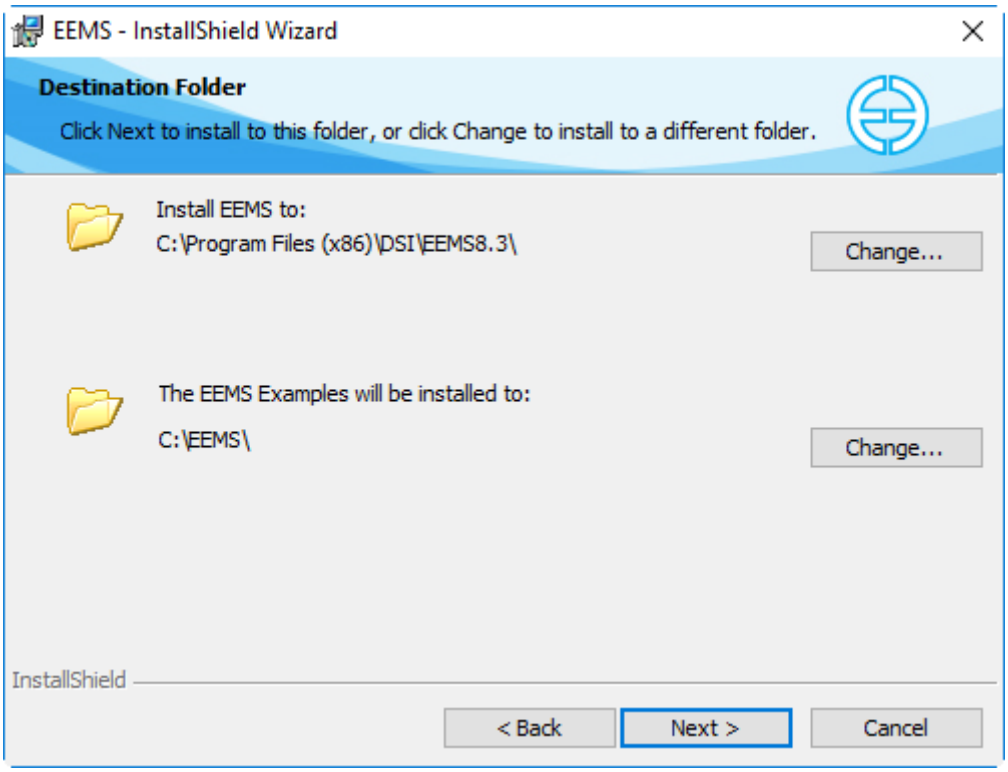


Figure 2-4 Destination folder selection.

7) After the destination folder has been selected, the user should click the *Next* button where they will be prompted to select a setup option. A choice of “Complete” or “Custom” setup options are available. “Complete” means that all setup features will be installed, requiring more disc space. The “Custom” setup option allows the user to install just one or a combination of files, including EEMS program file, data files, documents and extra files.

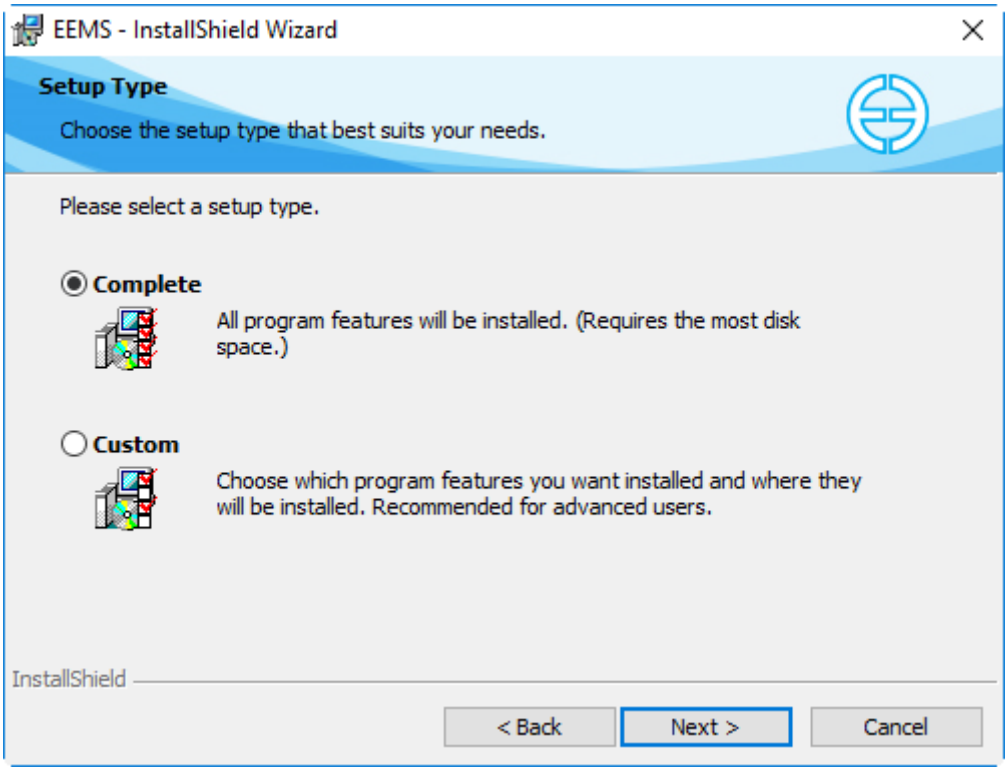


Figure 2-5 EEMS Installation Wizard – Setup Type.

8) If the "Custom" option is used a number of features may be selected to be installed. Here the user may also decide on the destination folder for the setup by selecting the *Change* button. The default folder is: C:\Program Files (x86)\DSI\EEMS8.3\. An example test case will also be saved to C:\EEMS.

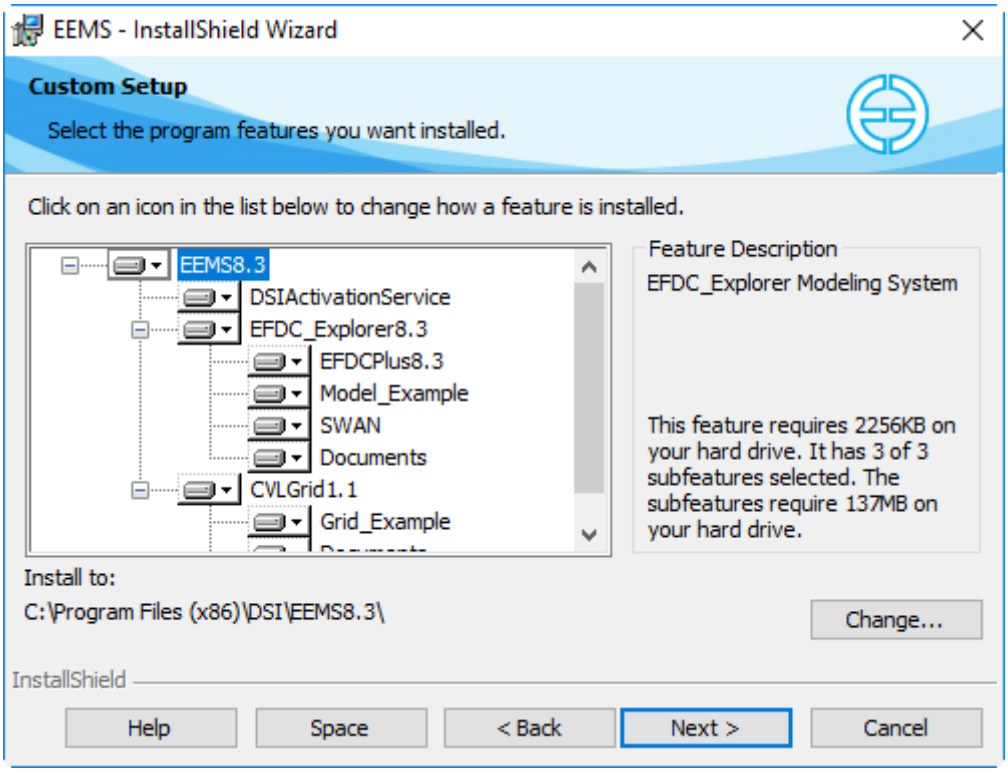


Figure 2-6 EEMS Installation Wizard – Custom Setup.

9) Upon completion of the Installation Wizard, the user is prompted with the message “Install Wizard Complete”. On the form of Install Wizard Complete, there are two checkboxes for launching EE8.3 and CVLGrid1.1. If those checkboxes are unchecked, the software will not be opened after clicking *Finish* button. The user is returned to the desktop.

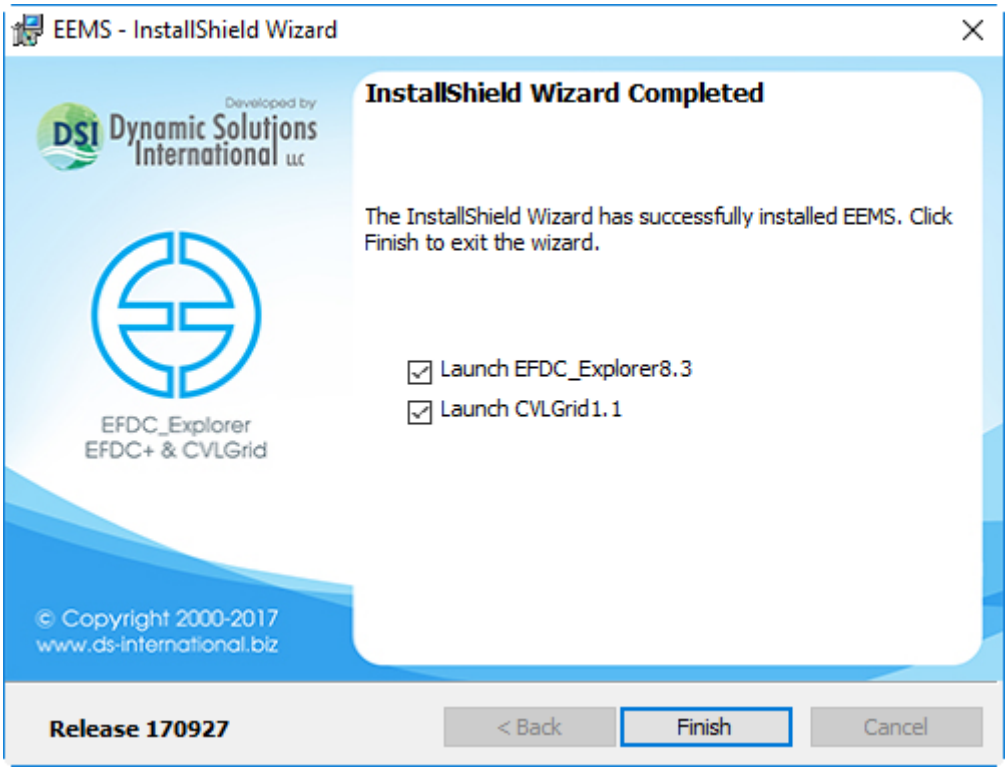


Figure 2-7 EEMS Installation Wizard completion screen.

2.3 License Models

EEMS utilizes a new licensing system. This system provides increased stability and ease of use for the users. Rental licenses are allowed and offline licenses may also be provided for users who don’t connect their machines to the internet. The following license models are now supported by EEMS:

1) Subscription License

A subscription license provides the full capability of EE and EFDC+ for a certain fixed period of time. Once this time has been exceeded the license will expire. During the subscription period the user may transfer the license to another machine by deactivating the license and then reactivating on the new machine. The EEMS License Manager form is shown in Figure 2-8.

Figure 2-8 EEMS License Manager: Subscription License.

2) Perpetual License

*Perpetual li*censes have no time limit in their use and provide the full capability of the EE modeling system without any kind of limitation. However, users will only be able to update their software to the latest version of EEMS within a certain time-frame agreed to in the sales arrangement. After that period of time, EEMS will still function normally but any bug fixes or new features which are added to EEMS after this date will not be accessible to the user. In order to access these updates the user will be required to have a valid maintenance contract with DSI. The *EEMS License Manager* will inform the user when this maintenance period is finished as shown in Figure 2-9. The user may then contact DSI to have their maintenance period extended if required.

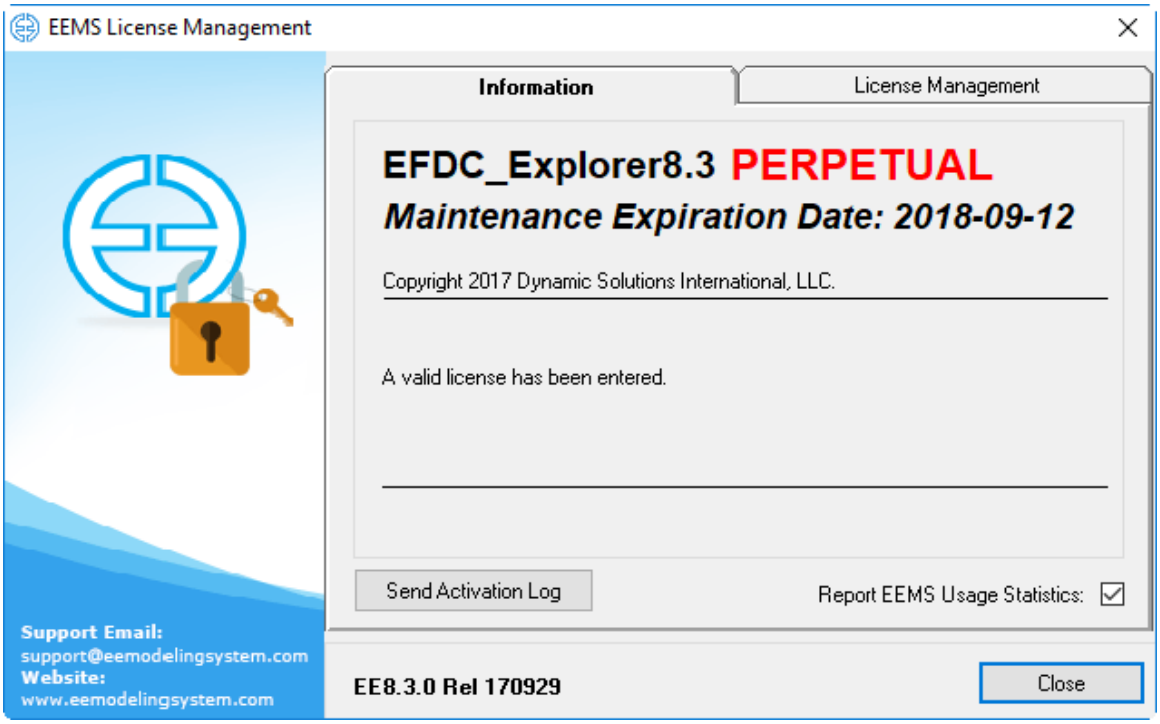


Figure 2-9 EEMS License Manager: Perpetual License.

2.4 Online Activation Process

After installation, the user must activate the software before first use. When you purchase an EEMS subscription license, you will have access to the entire EEMS modeling suite for the subscription period you have purchased. You will receive your license code. Please note that to activate EEMS, the user **must be connected to the internet unless using offline/dongle activation.** If the user is not connected to the internet, or if there is a firewall on the network or installation computer, an error message may be given. In this case, please adjust your internet virus checker or firewall settings and try activating again. If this fails please contact support.

1) To activate EEMS the user must start EE from the EE desktop icon or using the Start menu: "Start\DSI\EE8.3\" (as example for Windows 10)
2) The user should then review the "Terms and Conditions" associated with use of EEMS software and then select "I Accept" option, if they do.
3) If this is the first time you have run EEMS on your computer you will be see the EEMS be prompted for your license code, as shown in Figure 2-10. The user enter the license code then click *Activate* button. EEMS will automatically activate the license version based on your license code.

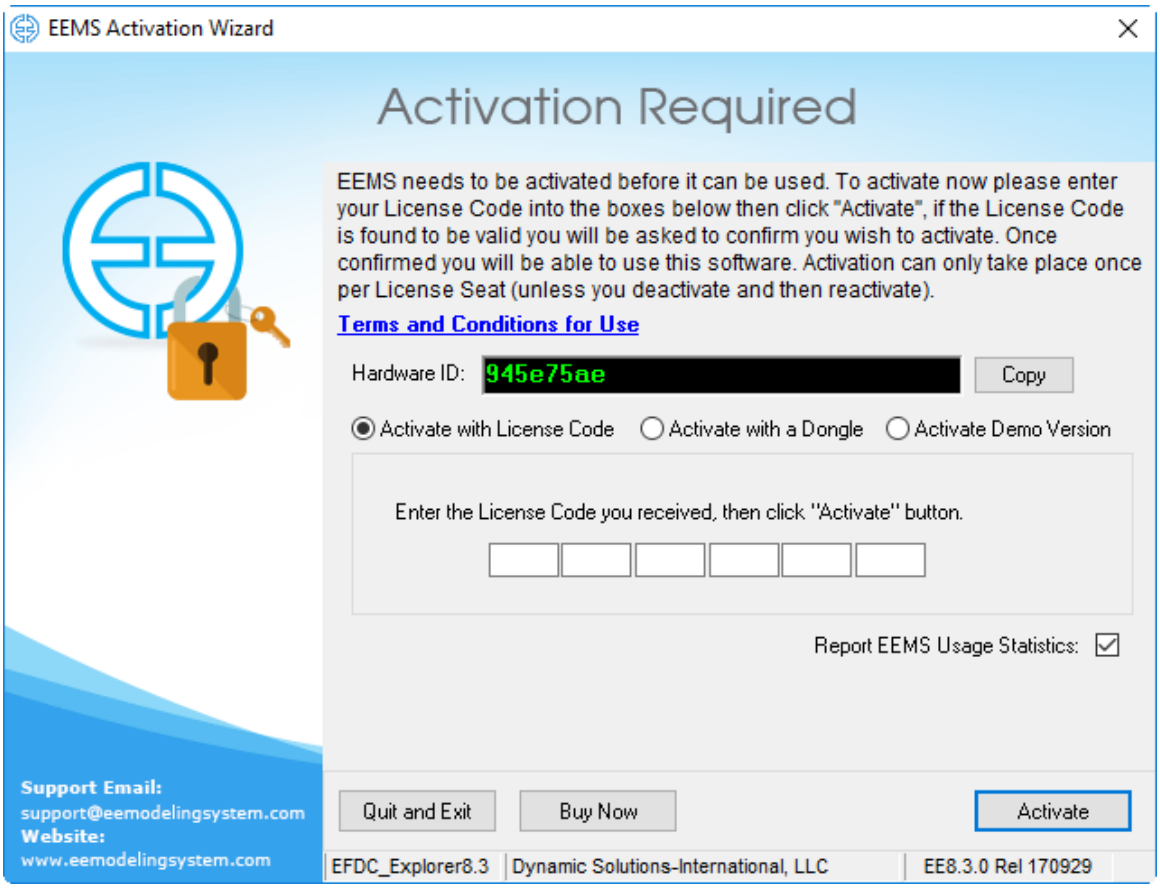


Figure 2-10 Activation with license code.

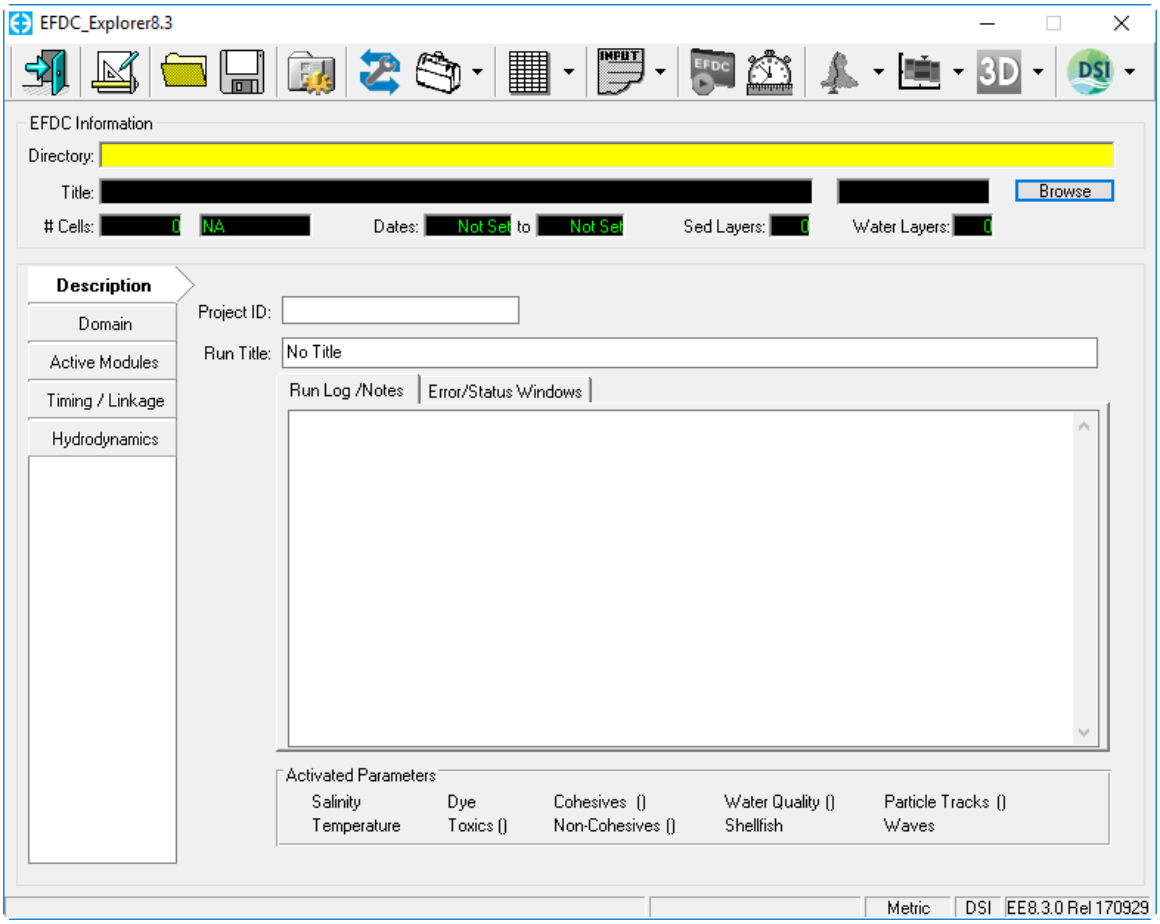


Figure 2-11 EE main form for license version.

4) The user can use EEMS Demo version by selecting *Activate Demo Version* on the form (Figure 2-10). If the user is a new user of EEMS, click *New User* button which takes you to the EEMS website to register. After the registration is successfully completed, the user must check their email then activate the account.

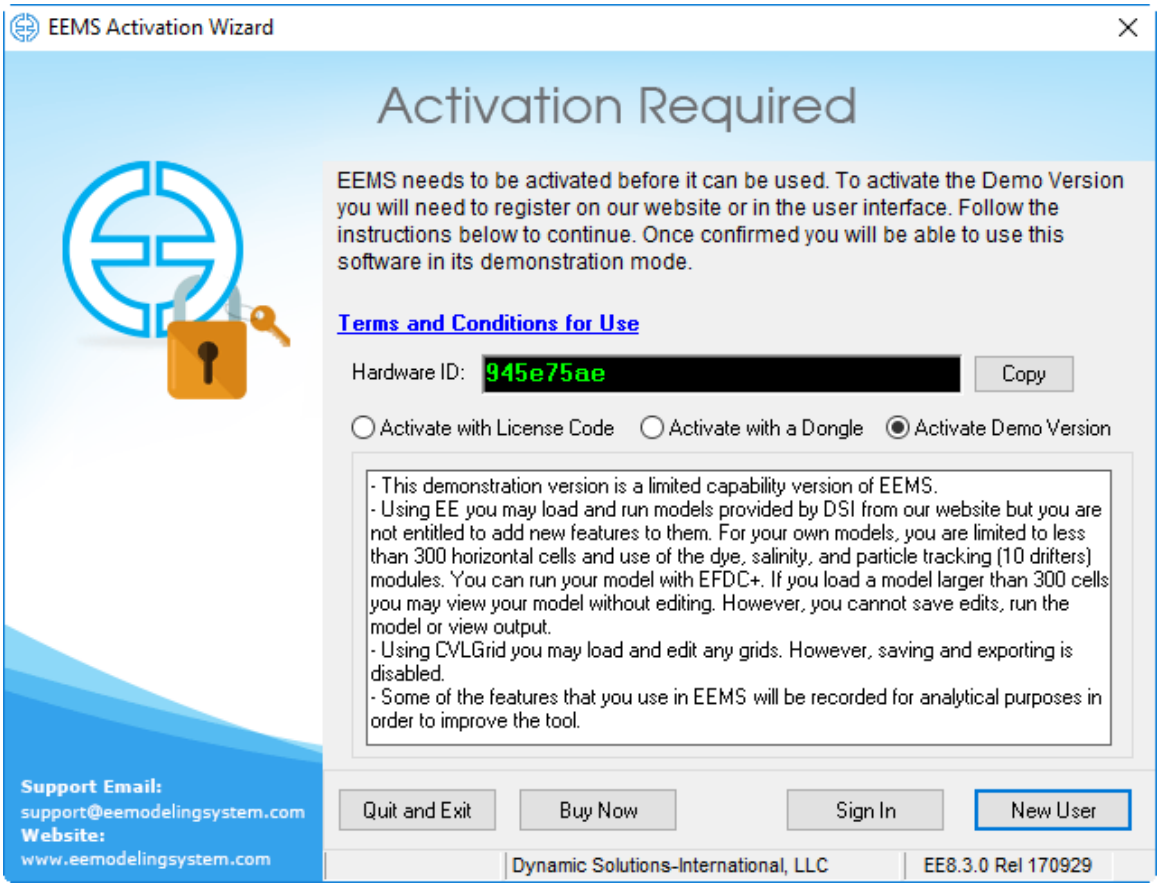


Figure 2-12 EEMS Demo Version activation options.

5) When the user has an account on the EEMS website they may activate EEMS. Click *Sign In* button in the Figure 2-12, the form *Sign in for EEMS Demo Version* is displayed as shown in Figure 2-13. Enter email and password and click the *Sign In* button.

6) A survey form for EEMS Demo Version will appear as shown in Figure 2-14. The user should fill in the survey form in order to use EEMS Demo version.

7) On completing the survey the main EE form will be displayed as shown in Figure 2-15 .

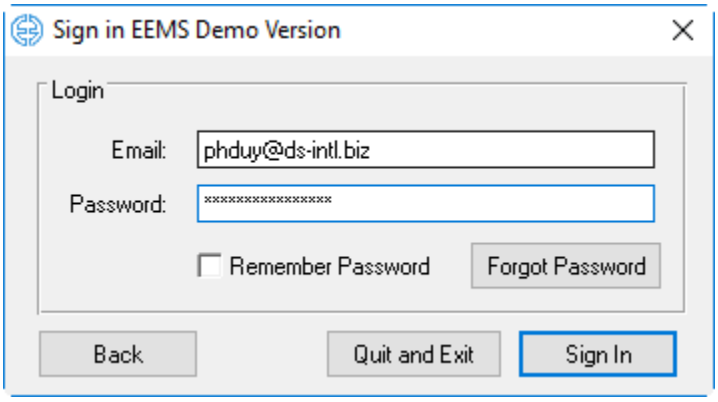


Figure 2-13 EEMS Demo Version sign in form.

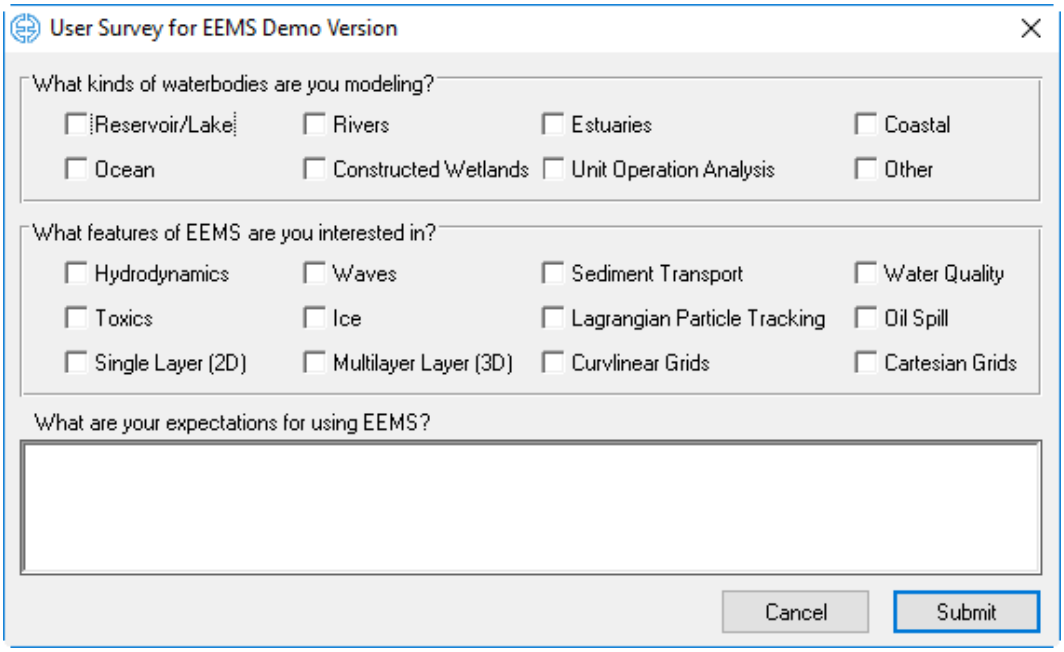


Figure 2-14 User survey for EEMS Demo Version.

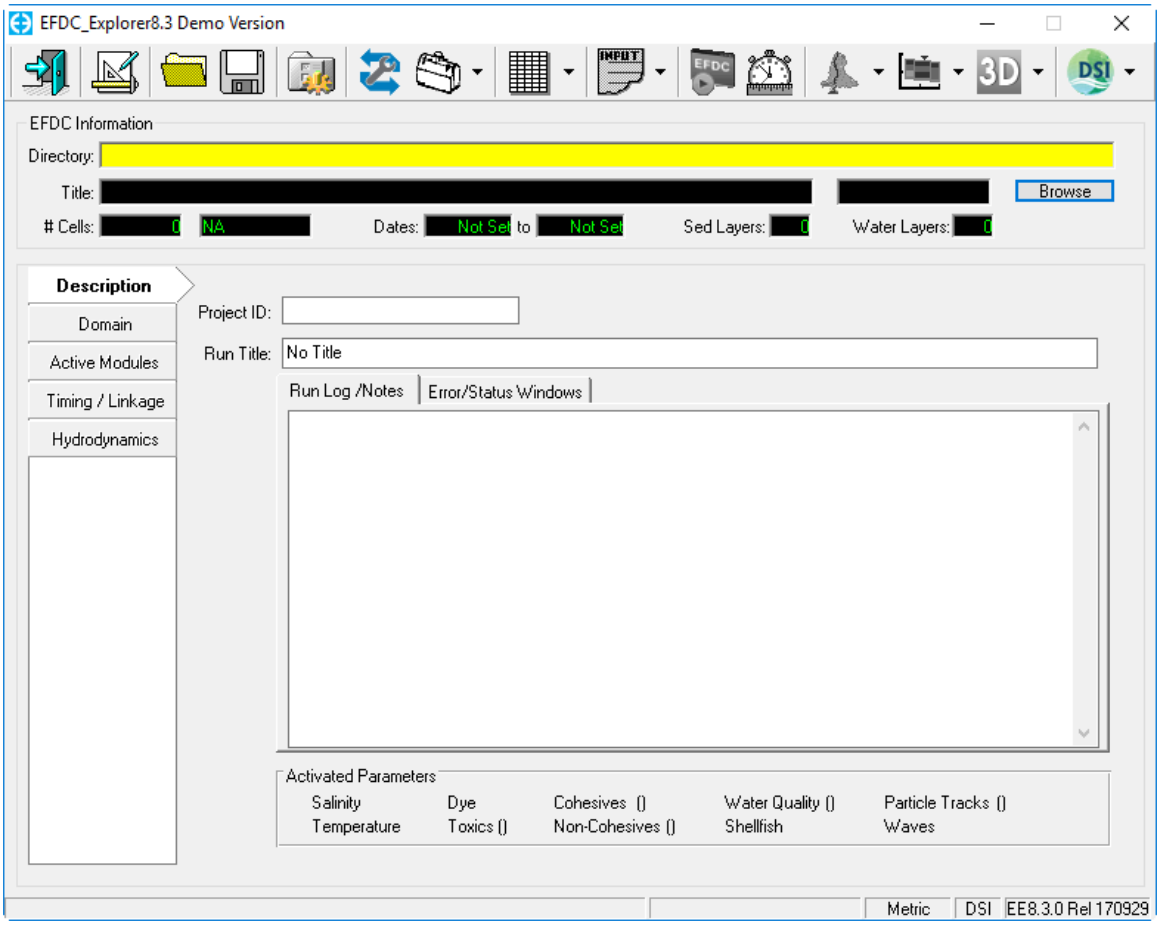


Figure 2-15 EE Demo Version main form.

2.5 Online Deactivation and License Management

Deactivation of EEMS is required when the user wishes to move their seat to a different computer, or plan to make some major changes to the hardware or operating system of their computer. Deactivation allows the seat to be retained and then reactivated on the same computer or a different computer later.

As for the activation process, deactivation of EEMS requires **connection to the internet unless using the offline/dongle license.** If the user is not connected to the internet, or if there is a firewall on the network or installation computer, an error message may be given. In this case, please adjust your internet virus checker or firewall settings and try activating again. If this fails please contact support.

1) To deactivate EEMS the user must click on the DSI button on the top right of the user form to the drop-down menu and select *License Manager* as shown in Figure 2-16 (you may need to enlarge your EE window to reveal the DSI button or select function key, F3). The *License Manager* provides the user with basic license information, including the user of the total number of seats available, the number of allowed activations and deactivations remaining, license code as shown in Figure 2-17.

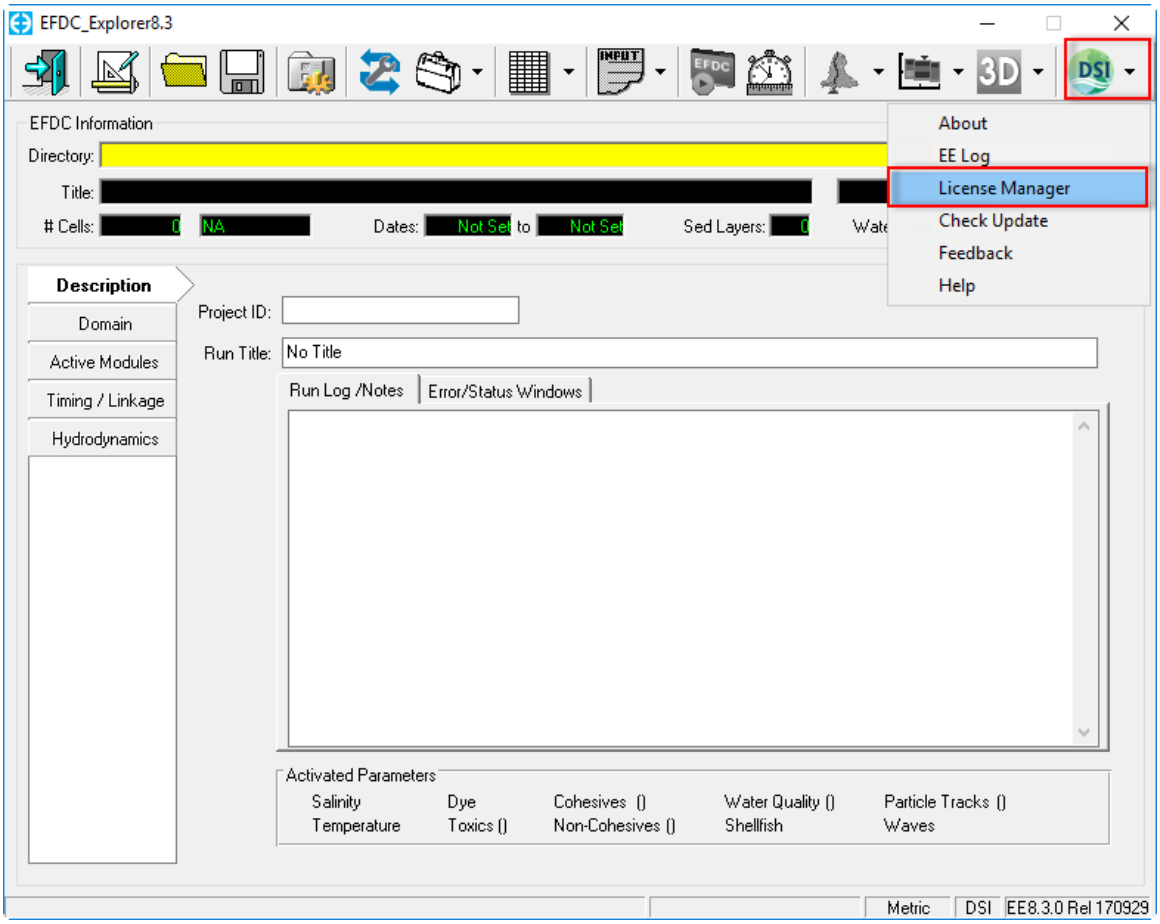


Figure 2-16 License Manager option selection.

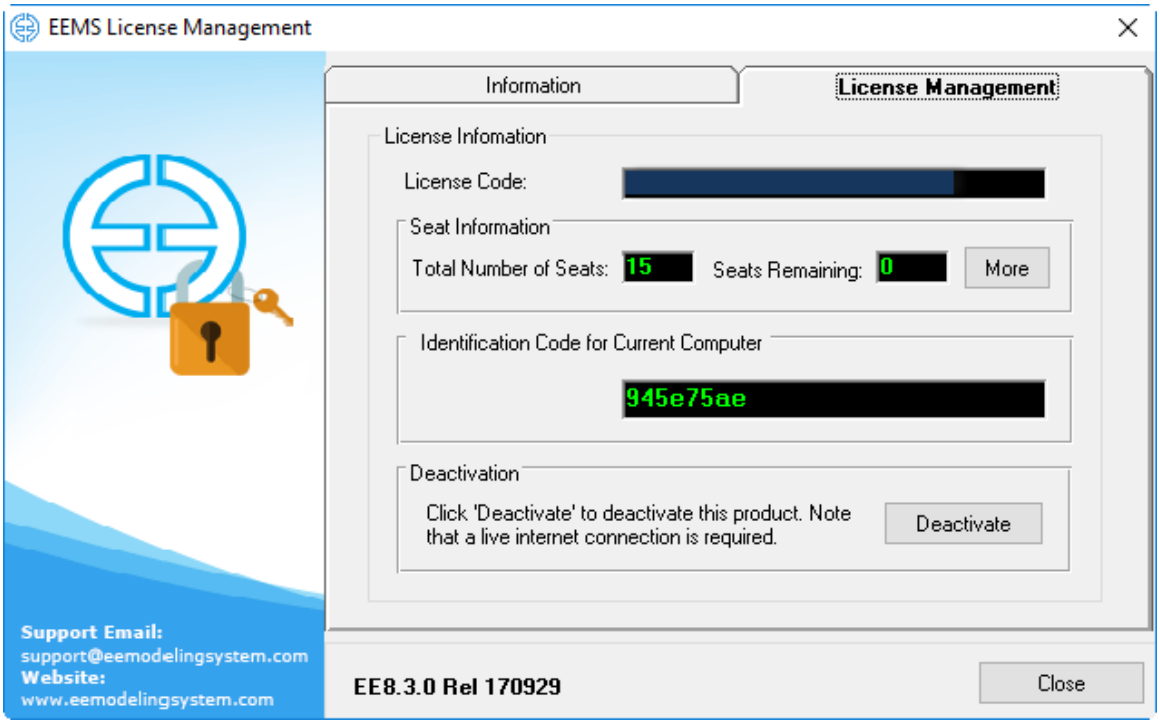


Figure 2-17 License Management tab option.

2) At the bottom of the *License Management* form is the *Deactivation* button. Clicking this button will bring up the *Deactivation* form as shown in Figure 2-18. Selecting *Next*, and selecting *OK* at the prompt will deactivate your current license seat, and the number of license seats available for activation will increase by one.

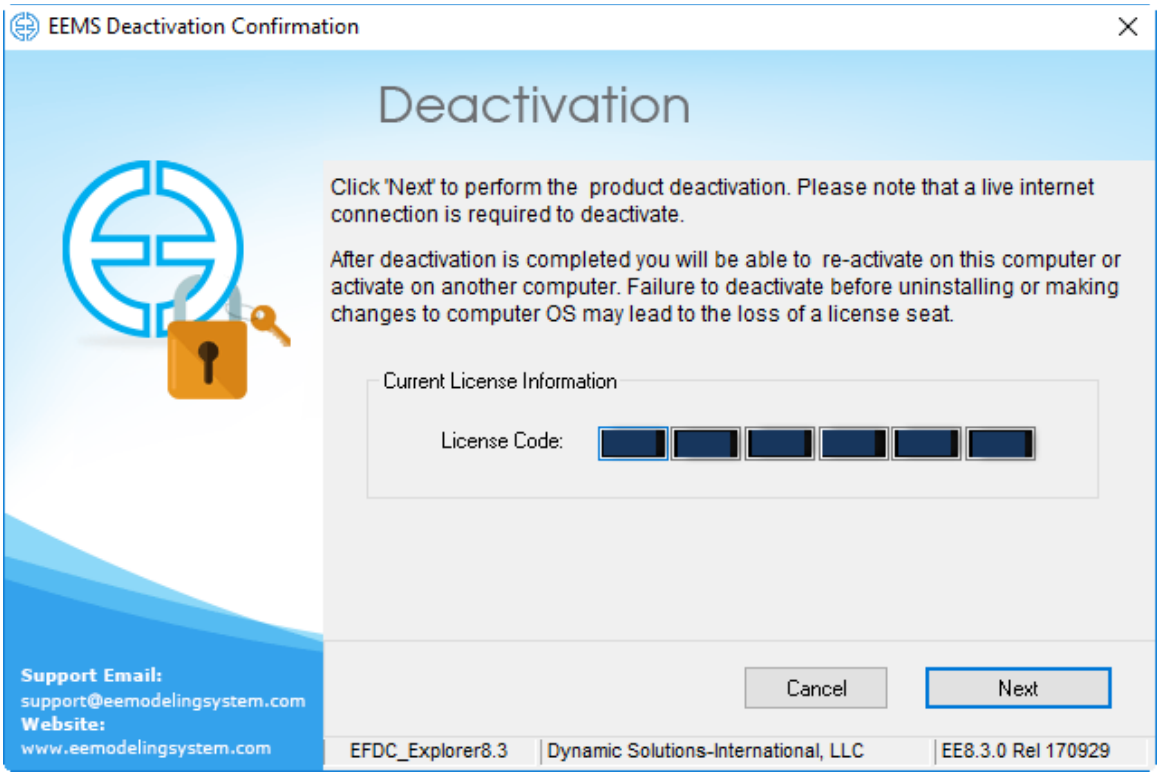


Figure 2-18 EEMS Deactivation Confirmation form.

3) After clicking the *Next* button, a message box will appear to ask the user to confirm the deactivation. Click *Yes* to proceed or *No* to cancel the process of deactivation as shown in Figure 2-19.

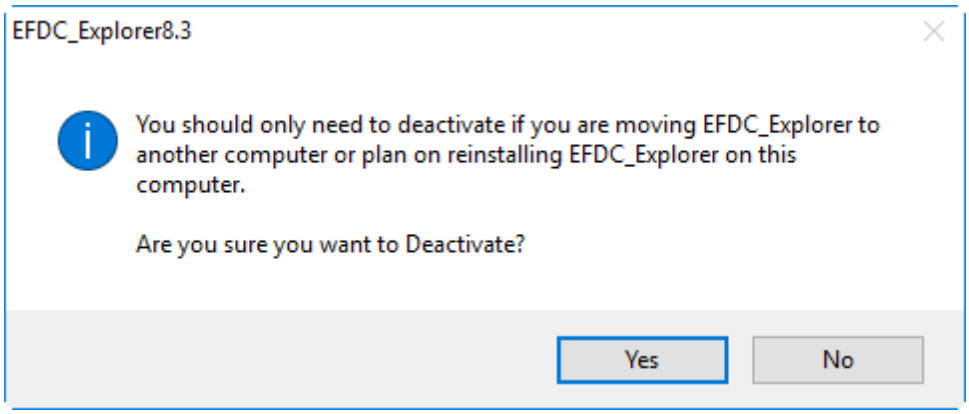


Figure 2-19 Deactivation confirmation message.

4) DSI usually provides the license code via email. The best way to enter the license code is to copy and paste these from the DSI email (sometimes you may need to copy and paste into a text editor first). When pasting the license code, click on the first code field from the left and then paste the entire code. EEMS will parse the code into the appropriate blocks.

2.6 Offline Activation

Users who wish to have a license of EEMS without requiring a connection to the internet can do so with an offline license. DSI will provide the user with a USB dongle which can be plugged into their computer to provide the license file required for operation. Offline licenses can be provided for both subscription licenses as well as perpetual licenses. The steps for activation using a dongle are described below.

1) In order to use dongle, a Flexnet USB dongle driver may need to be installed. This may be downloaded from the EEMS website.

2) Plug the dongle into the USB port.

3) Open EE8.3 from shortcut on desktop or with the Start button and the *Activation Wizard* form will appear as shown in Figure 2-20. The user should select the *Activation using a Dongle* option.

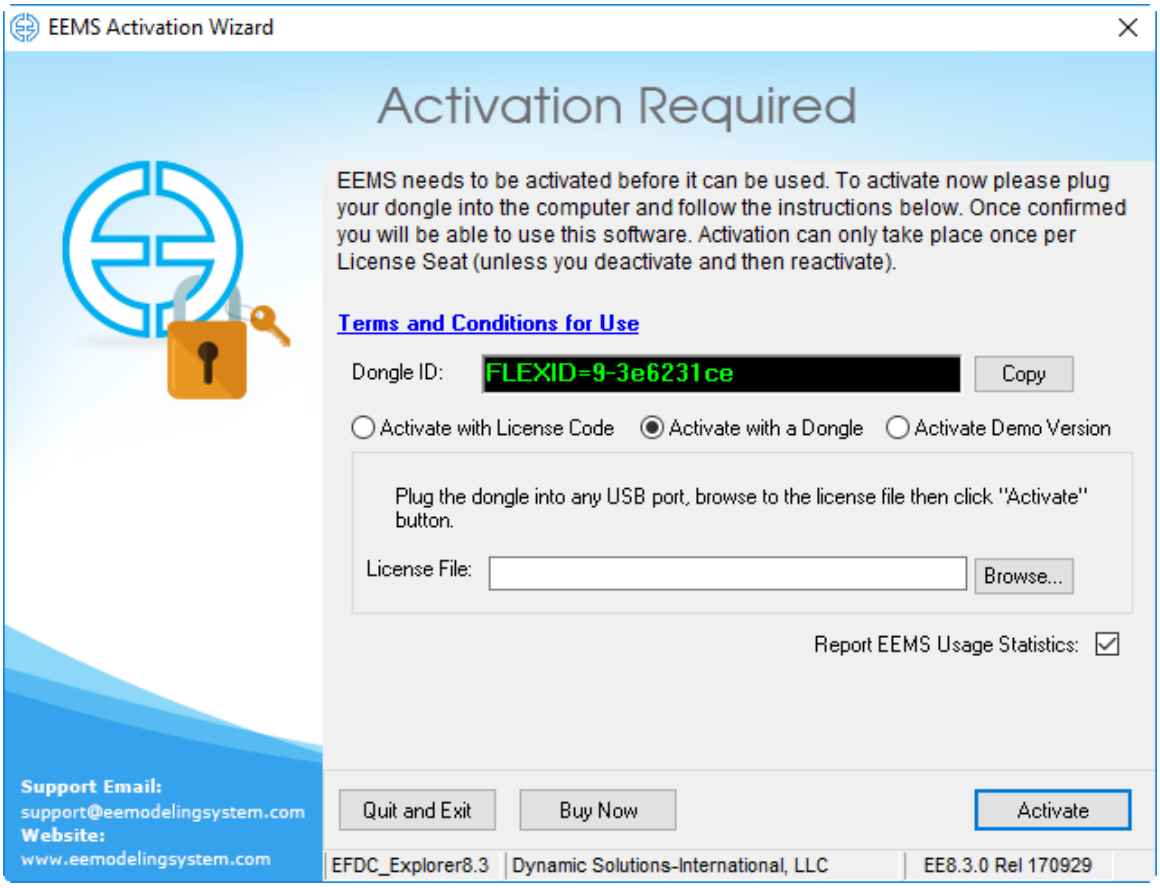


Figure 2-20 EEMS Activation Wizard form for dongle.

4) Select the *Browse* button to browse to license file (*.lic) in on the USB dongle drive as shown in Figure 2-21 and select *Open*.

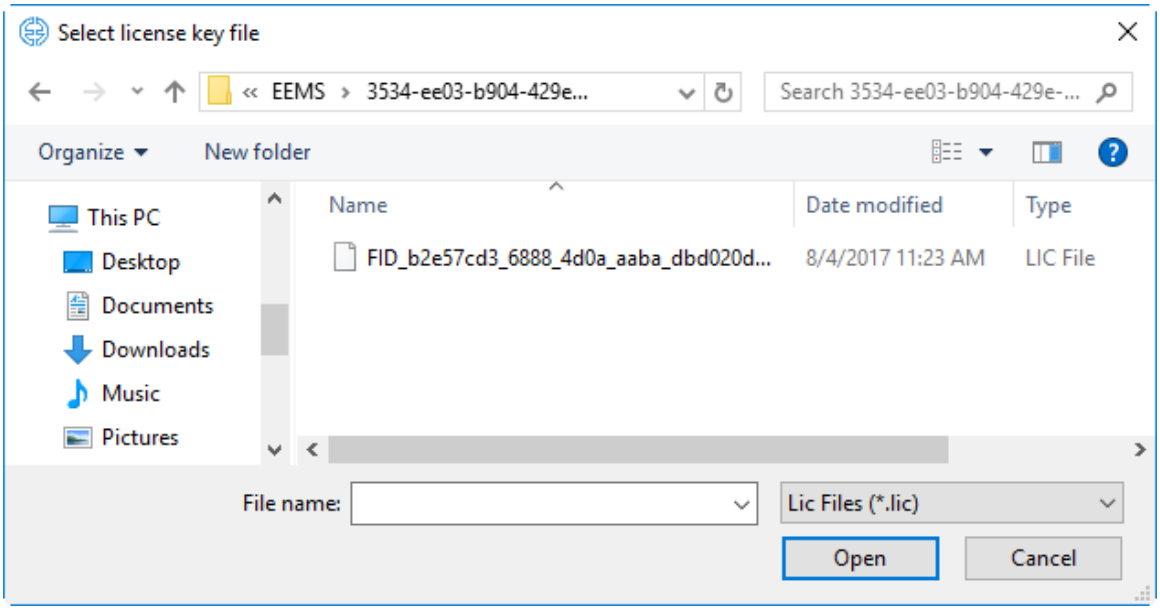


Figure 2-21 License key selection.

5) Select the *Activate EE* button. The EFDC_Explorer Main form will then be opened.

6) If the user unplugs the USB dongle during use of EFDC_Explorer, a warning form will appear as shown in Figure 2-22 and Figure 2-23. The user has two options: to save the current work and close EE or keep using EE by reinserting the dongle into the USB port. If the user select the upper option, then click OK button, EE will count the time up to 5 minutes then close automatically.

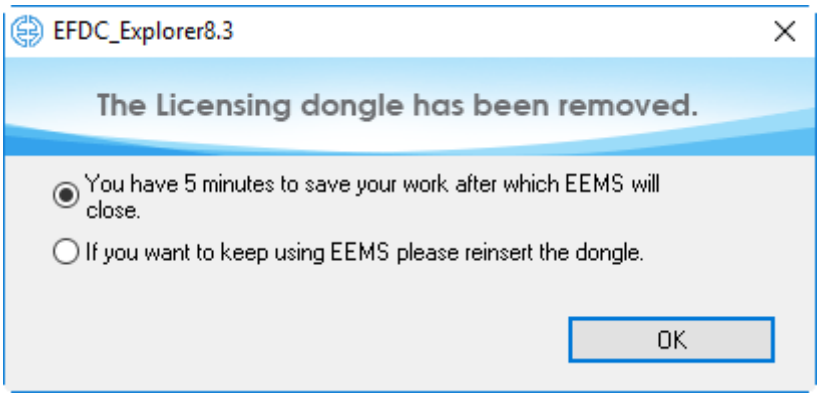


Figure 2-22 USB dongle License removed warning (1).

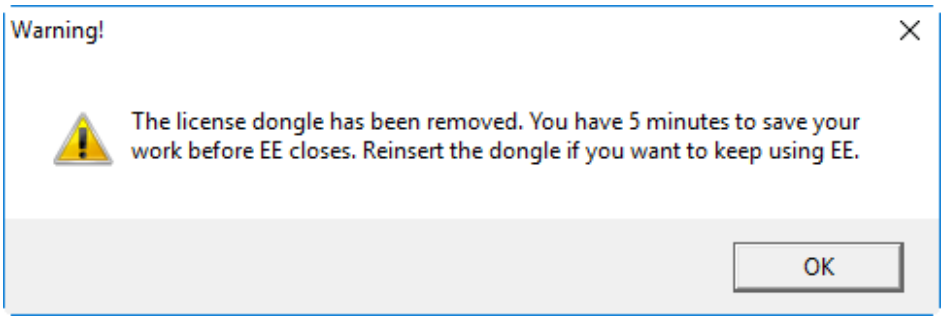


Figure 2-23 USB dongle License removed warning (2).

2.7 Offline Deactivation

1) To remove the Dongle key, the user must click on the DSI button on the top right of the user form to the drop-down menu and select *License Manager* as shown in Figure 2-24 (you may need to enlarge your EE window to reveal the DSI button or select function key, F3). The form for *EEMS License Management* will appear as shown in Figure 2-25.

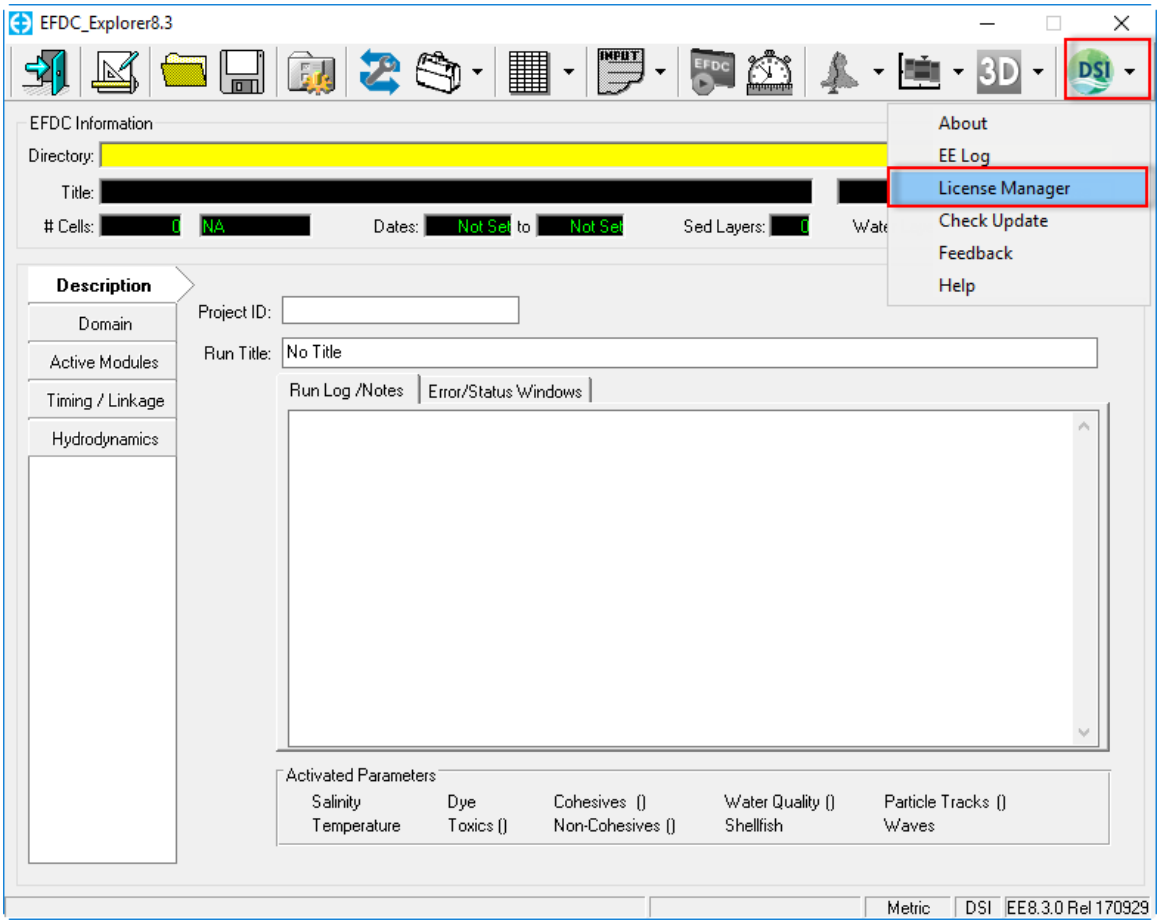


Figure 2-24 License Manager option selection.

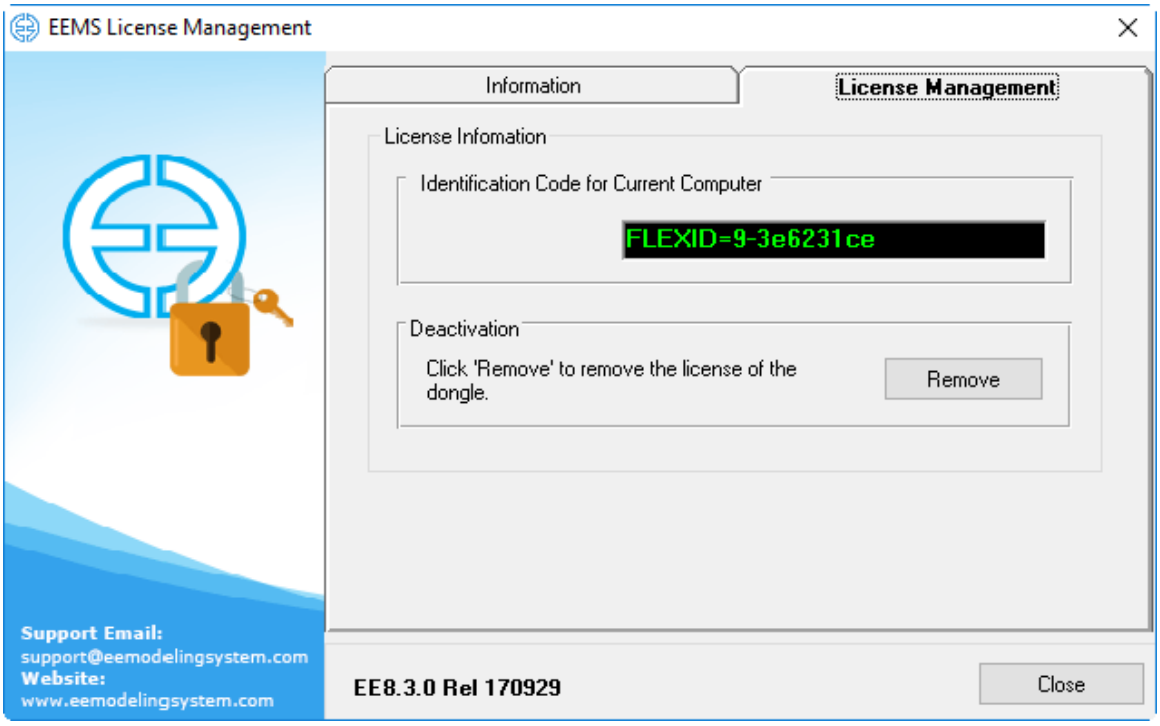


Figure 2-25 License Management tab option.

2) At the bottom of the *License Management* form is the *Remove* button. Selecting this button will bring up the confirmation form shown in Figure 2-26. Select *Yes* to proceed. If the license has been removed successfully the EE Main Form will close immediately and user will need to reactivate.

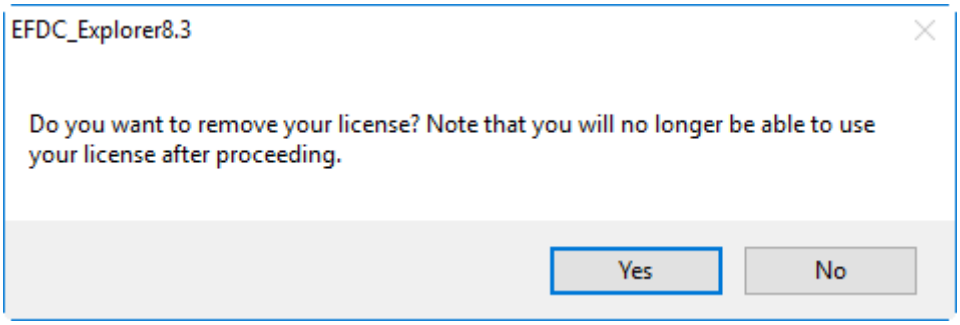


Figure 2-26 Remove license confirmation message.

3 EFDC_Explorer Primary Toolbar

Upon starting EFDC_Explorer the user will be presented with the form shown in Figure 1-2. This form is basically divided into three parts. The middle section represents the pre-processor functions and the lower section provides access to some of the post-processor functions. Another major section of EFDC_Explorer is the toolbar located at the top of the form (Figure 3-1). This provides access to the program configuration options and the main model viewing functions. This section contains brief descriptions of several miscellaneous functions that are available from the main toolbar of EFDC_Explorer. Help files from the User Manual may be accessed at any time by pressing F1.

Figure 3-1 EFDC_Explorer main form toolbar.

3.1 EFDC Project Management

EFDC uses fixed file names for its input files; therefore, each run/project should be stored in separate directories. EFDC_Explorer operates in the same manner. EFDC_Explorer reads and writes the files with the standard fixed file names to/from the specified subdirectory (called a "project" by EFDC_Explorer). Figure 3-2 shows the main file management toolbar and *Browse* buttons to access the opening and/or saving of a project. Table 3-1 provides a quick summary of each of the toolbar functions.

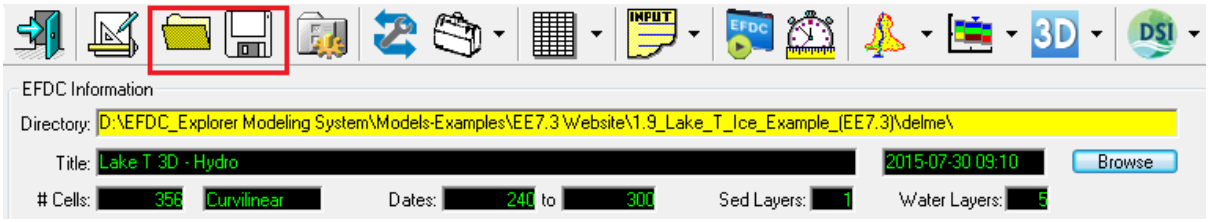


Figure 3-2 Project Open (Main Form).

Usage Tip

EFDC projects can be opened by **dragging and dropping** either the EFDC.INP file or the folder which contains the project on to either the ***Directory*** field or the ***Title*** field.

A project may also be opened by **double clicking** on the EFDC.EE file or any of the other ***.EE** extension files.

Table 3-1 Main toolbar summary of functions.

	Exit EFDC_Explorer. Does not save project, only Pre-/Post-processor settings.
	Generate an EFDC model using a template.
	Open/Read an EFDC model.
	Save current EFDC model into the same or new directory.
	Setup current printer.
	EFDC_Explorer configuration options, including where the EFDC executable files are located, one for the EPA version and one for the EFDC+ version.
	Convert between Julian days and Gregorian calendar dates
	Toolbox of miscellaneous features and utilities.
	View/Edit main EFDC.INP file for the current project.
	Various tools and utilities for analyzing and adjusting the grid.
	Run EFDC using the current project. Does not save the project first.
	Get runtime and other timing information for a completed model run.
	ViewPlan. Display the model in plan view. This is used for some pre-processing tasks (e.g. setting boundary conditions and modifying cell properties) and post-processing results. The dropdown menu allows the user to control whether the model results should be loaded and displayed or only initial conditions.
	ViewSlice. Display the model profile view along an I or J or a user defined section. This is used for post-processing results. The dropdown menu allows the user to control whether the model results should be loaded and displayed or only initial conditions.
	View3D: Display the model with 3D view. Viewing options include cut-aways (blanking), animations and light effects.
	Access Help files (User Manual); License Management

3.1.1. Open Operation

To open an existing project click on either on the toolbar folder button or the browse button highlighted. They accomplish identical tasks. The *Select Directory: Open Operation* form is then displayed. An example is shown in Figure 3-3. The directory displayed will be the last project directory opened. The last 20 projects are available in the dropdown list located near the bottom of the form. The panel on the right shows the files contained in the selected directory. For Open operations, the EFDC.INP file must exist in the directory.

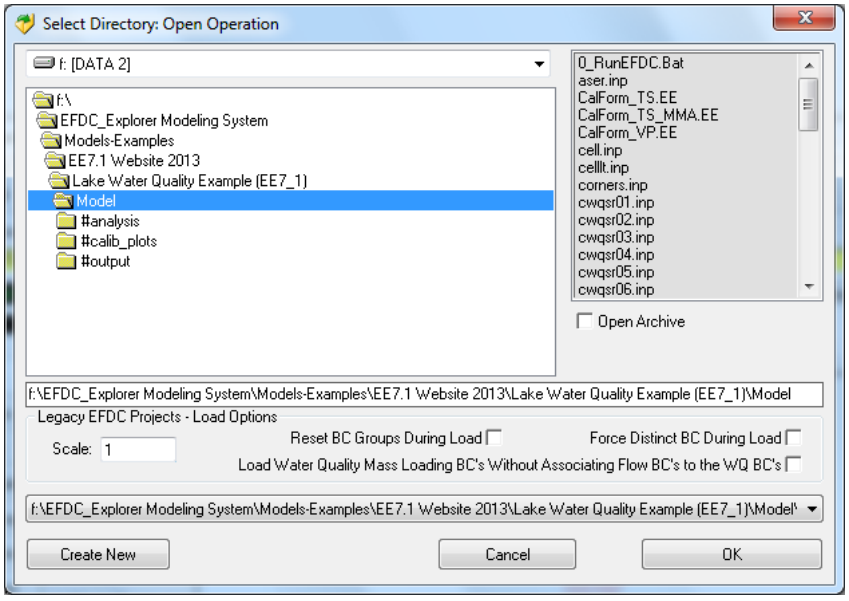


Figure 3-3 Select Directory: Open Operation.

An option to open a previously EFDC_Explorer saved archive file is given. These files all have an extension of “EFDC”, for example “CedarRiver.efdc”. When you select the *Open Archive* check box, the right panel only shows the available archive files in the selected directory.

3.1.1.1 *Drag & Drop Method*

If EE is already loaded, the user can load a new EFDC project using Windows Explorer. The user can browse to the project folder (or parent folder) and the click on one of the following and then drag the item to the yellow *Directory* box or the *Title* box. EE accepts:

- EFDC project folder
- EFDC.EE File
- EFDC.INP file

3.1.1.2 *Double Click Method*

During EE installation, the file extension “EE” was associated with the EFDC_Explorer application. Therefore, the user can start a new instance of EFDC_Explorer by double clicking on any file with the EE extension. EFDC_Explorer will start and then load the EFDC project contained in the same folder as the EE file double clicked on.

3.1.1.3 *Legacy EFDC Loading Options*

The options provided in the *Legacy EFDC Project – Load Options* frame can be used when loading old/legacy EFDC projects that were developed without EFDC_Explorer. Generally, these only need to be used the first time EE loads the project. When EE writes a project it saves the project out in the updated formats so these options do not need to be used but once for a legacy model application.

Historically, different versions of EFDC used different CELL.INP formats. EFDC_Explorer automatically handles the file format to correctly load the CELL.INP file. Similarly, the main EFDC.INP file and other input files have changed over the recent development history of EFDC. EFDC_Explorer attempts to correctly read most of the historical input files, while ensuring the latest version works and is the standard.

The “Scale” input box allows the user to apply a conversion factor to the centroid units used in the legacy LXLY file. The EFDC_Explorer default XY unit is in meters. Many applications use kilometers or miles as the units for the cell centroids provided in the LXLY file. For EE to correctly display the model these cell centroid coordinates must be converted to meters. EE can perform this function by entering the conversion factor in the “Scale” box when loading the model for the first time. Note: When a model is loaded and then viewed but looks like a bunch of large cells stacked on top of one another, it is likely to be a LXLY unit conversion issue. Try reloading the model with an appropriate Scale factor.

The check boxes concerning resetting boundary condition groups apply to existing projects that have been managed by EFDC_Explorer. During the initial loading of a project, or if the *Reset* check box is selected, EFDC_Explorer tries to logically group boundary condition cells into groups by type and location. EFDC_Explorer then manages the boundary conditions using this group approach. If the user has modified the boundary conditions somehow and wants a different logical grouping, they should select one of these options. The *Load WQ Mass Loadings BC’s Without Associating Flow BC’s to the WQ BC’s* option is applicable when the user has no flow boundary condition assigned for a specific WQ mass loading boundary. Note that this is not recommended to be done in practice but has been seen in some models.

3.1.2. Write Operation

To save a currently opened project (i.e. Write Operation), click on the disk button highlighted on the toolbar shown in Figure 3-2.

The *Select Directory: Write Operation* form will be displayed (Figure 3-4). The user has the option to select which files are written by selecting the appropriate *Save Option* button. For a complete save of all the input files select the *Full Write* option. If you have only made changes to the formatting options in EFDC_Explorer and want those saved, select the *Save Profile* option. The profile is always saved for the other save options also.

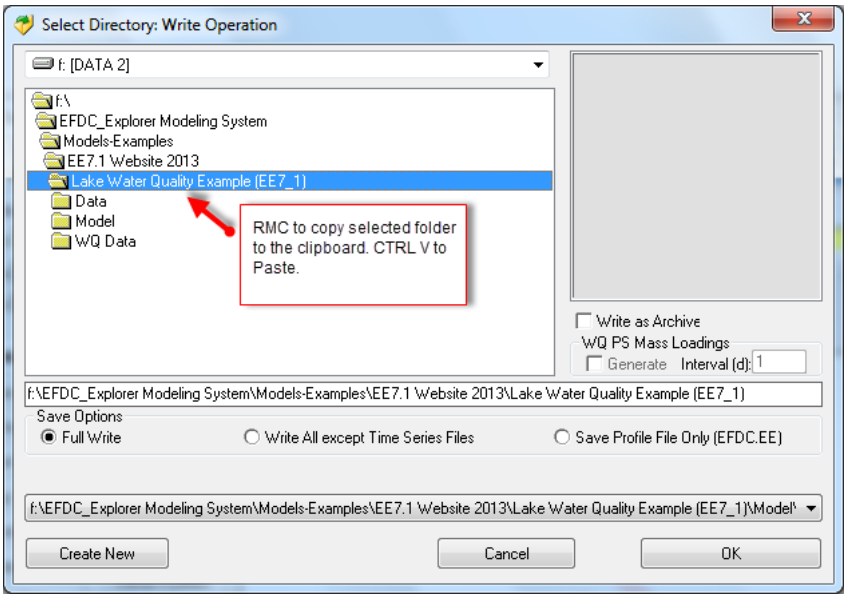


Figure 3-4 Select Directory: Write Operation.

If the user only has the "*.EFDC" archive file and wants to create a set of files that EFDC needs to run that project, the user must select the *Full Write* option to create all the input files required.

To create a new project using the existing project, use the *Create New* button to create a new subdirectory under the currently displayed directory. All the .INP files will be copied to the new directory after the user selects OK on the *Write Operation* form.

If the user wishes to save in the EPA GVC Model format rather than EFDC+ model format this should be selected in the *Active Modules* tab *Model Selection* frame on the main EFDC_Explorer form. This method allows quick reformatting of the EFDC.INP file for the different models. Care must be exercised to ensure that all the parameters have the desired values when switching models (see Section 6.4).

The project specific settings for information/data that the EFDC model does not use (i.e. labels, plot formatting, etc.) are saved in the EFDC.EE file, which is located in the project directory. If the user wants to manually copy a complete EFDC project data set (together with all the ASCII .INP files or the binary archive file) the user should also copy all of the *.EE files. However, the most important EE project file is the EFDC.EE file. The EFDC.EE file is an ASCII file that can be edited with any ASCII editor, though care should be taken not to corrupt the file.

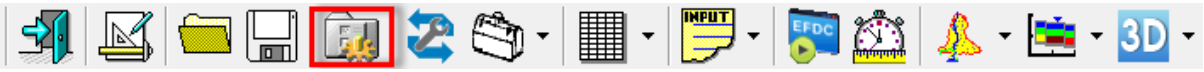
Cross Platform Note

Many users may want to use EFDC on both a PC and a UNIX based computer. When transferring the input files from the UNIX machine to the PC, the carriage control MUST be reset to the Windows/DOS carriage control.

*EFDC_Explorer has the ability to convert non-Windows/DOS carriage control to Windows/DOS, via the **Toolbox** or when loading the UNIX project for the first time.*

Also, the user may use one of several ASCII editors that have this capability as well.

3.2 EFDC_Explorer Settings



This toolbar button allows the user to specify some installation specific parameters, like the location of the EFDC executable to use, as well as project specific settings like default precisions and some other miscellaneous settings. Figure 3-5 shows the settings form.

The settings for EFDC_Explorer are saved in the EFDC_EXPLORER.INI file that is located in the same directory as the EFDC_Explorer executable. This is an editable ASCII file, though care should be exercised to not corrupt the file. The file structure follows the Windows standard INI file using groups and tags.

The user must first specify the path to the *EFDC executables* before EE can be used to launch an EFDC run using the Run button. The *Browse* button is used to browse to the corresponding EFDC executable. If the EFDC executable selected does not exist, EE will change the associated path background to light red.

The *Default Precision for Write Operation* settings are for setting the output/display precisions for the indicated data types. The default settings shown are appropriate for most applications. However, for special cases (e.g. flume studies or other types of research applications) the user will likely have to make adjustments to the defaults. The settings are stored in the project specific EFDC.EE file.

Other miscellaneous settings include: *Secchi Conversion Factor* (to convert light extinction coefficients to a Secchi Depth in meters); the *Minimum Inorganic Sediment Thickness* allowable; *Carbon/Dry Weight Ratio* (ratio of mg of carbon to mg of dry weight to convert POC and Algae to a solids weight); and the *Maximum Number of Cells to Auto Refresh*.

The company/organization's name and address as well as the default fonts can be set by selected the *Edit* button in the *Company Data & Defaults* frame.

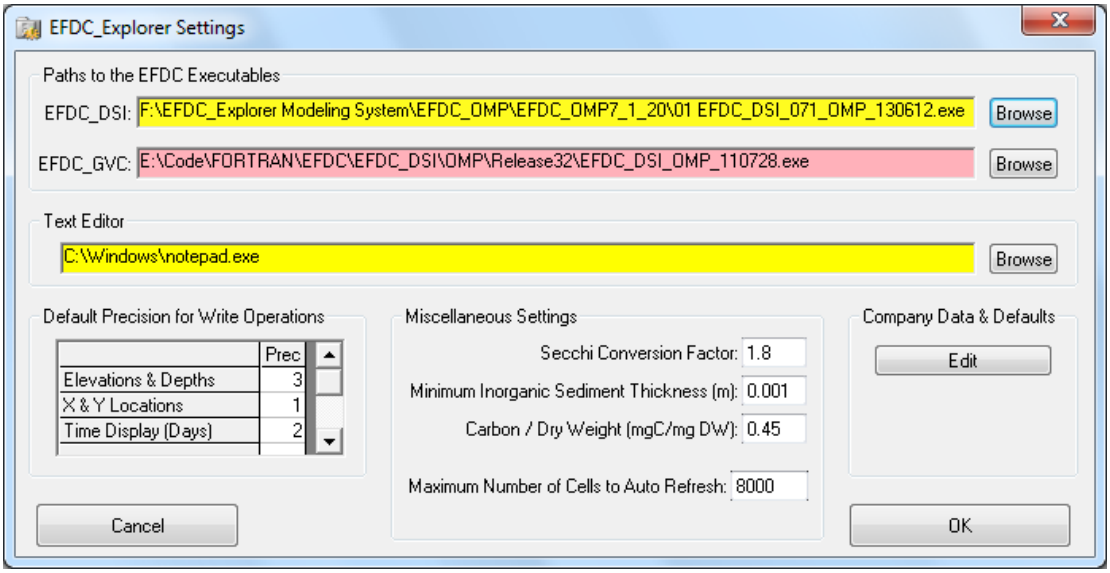


Figure 3-5 EFDC_Explorer settings form.

3.3 Date and Coordinate Conversion Tool

This toolbar button brings up a calendar and coordinate conversion tool. There are three tabs on this tool, *Date Conversion*, *IJ/XY Conversion*, and *UTM Conversion*. The first, shown in Figure 3-6 allows the number of days to be calculated from the time of a Base Date to the time of a specified (Gregorian) calendar date. If the calendar date entered is before the Base Date a negative number of Julian days in given. This tool can also be used to determine a Calendar Date, given any number of days before (<0) or after (>0) a given Base Date. The default Base Date is the Project's Base Date. However the user can change the Base Date in this utility without impacting the Project's Base Date.

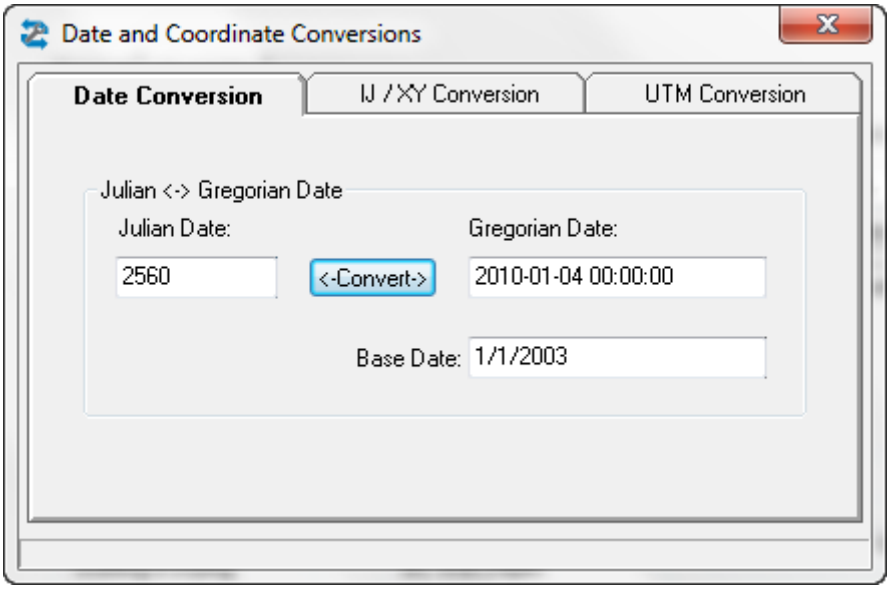


Figure 3-6 Julian to Calendar Date conversion.

The *UTM Conversion* tool is in the second tab is shown in Figure 3-7. An L index or IJ pair can be entered into the text box and EE will automatically display the corresponding L, IJ or model grid coordinates.

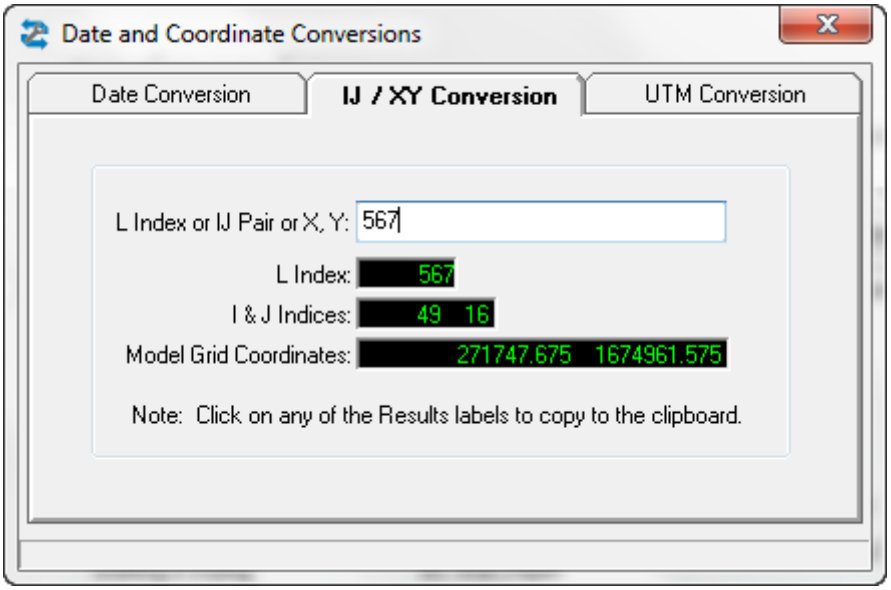


Figure 3-7 Date and Coordinate Conversions.

The user may also select the *UTM Conversion* tab to convert from longitude and latitude coordinates to UTM or vice versa. The user should enter the coordinates in the text box and ENTER keystroke for this conversion to take place as shown in Figure 3-8. The user may also copy this information to the clipboard with the *Copy to Clipboard* button. A file containing coordinates may also be converted using the *Convert File* button and browsing to the file to be converted.

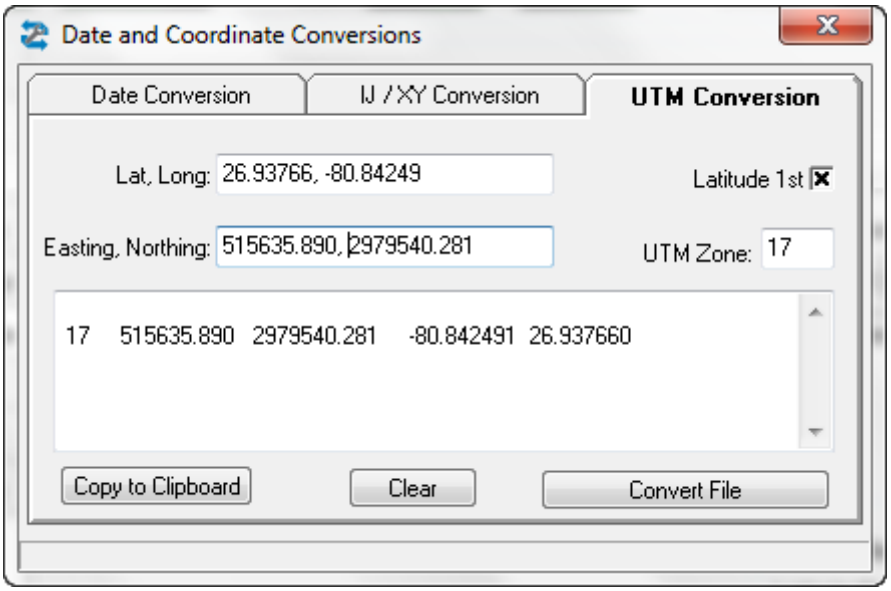
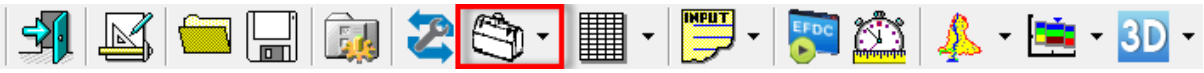


Figure 3-8 Date and Coordinate Conversions.

3.4 Toolbag: General Utilities



The *Toolbag* function provides access to a range of general utilities that support the modeling process but may not be directly related to the EFDC model. Figure 3-9 shows a screen capture of the current functions available under the *Toolbag*. The following list provides a summary of these functions.

Figure 3-9 Toolbar functions.

Bitmap Georeferencing: This utility can be used to create or edit the configuration file that EFDC_Explorer uses to provide bitmap images and maps as a background for the display of models in *ViewPlan*. To use bitmaps you must first create a “geo” file. This is an ASCII file with the extension “geo” that contains the pixel and project coordinate information. It is very similar to MapInfo’s TAB format. One “geo” file can reference multiple bitmaps to build mosaics, if needed. When this function is selected a form is displayed with the bitmap filenames and coordinate information. Any changes must be saved into the same or new “geo” file for later use in *ViewPlan*.

Unix -> Windows CRLF Conversion: This utility scans a specified directory and determines if any of the files contained in the directory do not use the Windows CRLF standard. The user then has the option of automatically converting all non-Windows CRLF files to the Windows standard. EFDC_Explorer uses the Windows CRLF convention, so files created or edited on a Unix platform need to be converted to Windows before use by EFDC_Explorer.

Delete EFDC Generated Files: This utility allows the user to specify certain groups of output files from EFDC to be deleted. The main purpose of this function is to clean up all the project directories and save disk space by deleting all the files in a project directory that are not needed by EFDC or EFDC_Explorer. This utility works on the specified directory and ALL subdirectories under the top level directory specified. The utility scans the directory structure and then lists all the files that may be deleted if the user presses the *Delete Matched Files* button. If a few files are listed that the user doesn’t want deleted the user can delete their file names from the list. This will keep them from being deleted as the utility uses the files in the list to delete.

Merge Continuation Runs: This utility merges EFDC_Explorer specific output files from two EFDC runs into a single output file. Multiple runs can be merged by starting with the earliest runs and sequentially appending each subsequent run. At the end of each merge process, the EE_WS, EE_Vel, EE_WC, EE_WQ and EE_Bed from the base

run will be saved in the base run project directory, but with an ".org" extension. The newly merged files will have the ".out" extension and can be used by EFDC_Explorer. The models/projects must be of the same model domain and discretization.

Re-sample Output: Allows the user to reduce the number of saved output snapshots in the EFDC_Explorer output files (i.e. the EE_WS.OUT, EE_VEL.OUT, EE_WC.OUT, EE_WQ.OUT and EE_BED.OUT). At the end of the resampling process, the original output files will have been saved in the project directory but with an ".org" extension. The newly resampled files will have the ".out" extension and can be used by EFDC_Explorer. To resample the output the user runs this utility and enters a snapshot skip interval. Once the user verifies that the resampling produced the desired results, the user may want to delete the *.ORG files to save disk space. This function can also be used to delete all the model results after a specified time (i.e. truncate).

Modify ModChan File: This option provides access to a channel modifier global editor that recalculates channel lengths, change orientation from U to V and vice-a-versa. It also has a Q/A function to ensure all the upstream and downstream I & J's point to valid cells. The *ViewPlan* function allows the user a graphical point and click approach to create and edit these "pipes".

Categorize Bottom Shears: This utility scans the EE_WC.OUT file for the entire simulation period and builds a list of categorized shears into predetermined bins. The results are displayed in a message box and placed onto the clipboard for pasting into Excel® or some other display/plotting package.

Compute HSPF FTables: This utility computes the data necessary for creating an FTable for the HSPF model. It assumes that the current project has been designed to be able to generate the necessary information for the FTable. This type of project is simply the base model, but without the actual inflows into the domain. Instead, all the flow boundaries have a step flow to allow the system to achieve steady state at that flow, and then all the flows must be stepped up again to the appropriate level. Generally, this can be done with just one upstream flow BC, but each case needs to be evaluated. After the run is complete the EFDC results should be a series of steady state flow regimes throughout the model domain. This utility then uses these results along with the reach polygons to build the FTable.

3.5 Grid Tools and Utilities

The *Grid Tools* function on the toolbar contains a range of different functions and utilities that have been needed. Figure 3-10 shows a screen capture of the current functions available under the *Grid Tools* button.

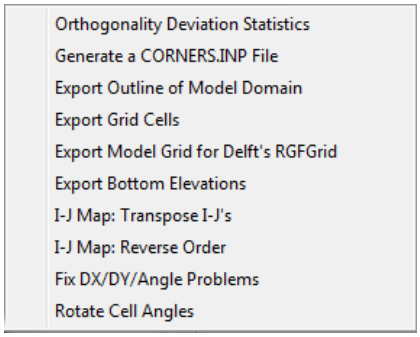


Figure 3-10 Grid Tools functions.

Orthogonality Deviation Statistics: This utility provides summary statistics on the orthogonality of the grid. A perfectly orthogonal grid will have a zero deviation angle at every grid intersection. The goal during the gridding process is to produce a representative and computationally efficient grid of the model domain with as low as practical deviation angles (e.g. < 3 degrees). The user can view a map of the deviation angles using the ViewPlan function.

Generate a CORNERS.INP File: This utility allows the user to generate a CORNERS.INP file. This is which is in the DRIFTER module, and determines the centroid of a cell from the coordinates of the four cell corners. This file is generated automatically by EE when a new model is built, but this tool allows the user to force EE to recompute the cell corners, if needed

Export Outline of Model Domain: Use this function to output a bounding polygon in the coordinate system used by the model. The output file format is P2D. This file can be used as an overlay in ViewPlan or imported into GIS systems.

Export Grid Cells: Use this function to export the cells by cell polygons in the coordinate system used by the model. The output file format is P2D. This file can be used as an overlay in ViewPlan or imported into GIS systems.

Export Model Grid for Delft's RGFGrid: Use this function to export the currently loaded EFDC model grid out to a Delft RGFGrid formatted GRD file. This file can then be directly loaded into RGFGrid for editing. RGFGrid edited grids can be imported into EFDC_Explorer by selecting "*Generate New Model* | *Import Grid* | *Delft RGFGrid".*

Export Bottom Elevations: Use this function to export an XYZ data file at the cell center coordinates of the bottom elevations.

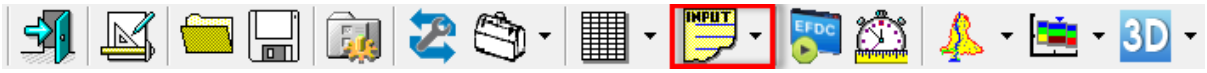
I-J Map: Transpose I-J's: These *I-J Map* functions are used during the grid generation-importing process to correct any model I-J orientation issues. This function switches I to J and J to I. If boundary conditions are already assigned, the IJ mapping is remapped to the new IJ space.

I-J Map: Reverse Order: These *I-J Map* functions are used during the grid generation-importing process to correct any model I-J orientation issues. This function flips the numbering direction for either the I or the J index. If boundary conditions are already assigned, the IJ mapping is remapped to the new IJ space.

Fix DX/DY/Angle Problems: This routine attempts to correct DX, DY and Cell Angle problems before computing cell metrics. EE picks a South cell face for cell L=2 then assigns DX/DY/Angles. The user may enter a face shift of n, where n=0 (the default) means no shift but correct the angles (if needed), and n can vary from -2 to +2. Entering a -1 EE reorients the IJ map and corresponding DX/DY -90 degrees.

Rotate Cell Angles: This function applies a rotation to the cell rotation angles. It does not rotate the actual cell. If using the CORNERS.INP file it just applies a user specified rotation to the cell rotation matrix. In EFDC the cell's rotation matrix is used for velocity plots and interaction with the wind field.

3.6 Text Editor

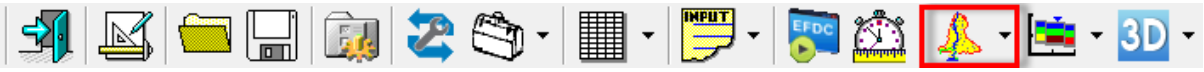


This button on the toolbar starts the ASCII editor (configured in *EFDC_Explorer Settings*, see Section 3.3) and directly loads certain input files for the current project. From the drop down window the user can directly access the input files: EFDC.INP, DXDY.INP, LXLY.INP, CELL.INP, WQ3DWC.INP, WQPSL.INP or WQ3DSD.INP. In addition, the user can select "Other INP" to access any of the input files in the current project.

3.7 Run EFDC Model

This is the function that actually runs an EFDC project/model. See section 5 for details of this feature.

3.8 ViewPlan Viewer



The *ViewPlan* button of the main toolbar provides access to the utility for viewing two dimensional (2D) plan views of the model. This function is used for pre-processing visualization and editing of the grid, initial conditions and boundary conditions. Once the model has been run and output files generated (see Section 8.1). *ViewPlan* can be used to view 2D plots, extract model cell time series and vertical profiles and other post-processing visualizations and analyses. Section 8 contains a more complete description of the features and use of *ViewPlan*.

ViewPlan has many options and features for displaying background maps (geo-referenced bitmaps), line features (e.g. shorelines), ESRI© .SHP files, labels, measured data, model-data residuals and many more. An example plot is shown in Figure 3-11 of Perdido Bay's bathymetry with the 250K scale USGS topo map in the background, a timing frame showing the Gulf of Mexico tide level at the date/time indicated in legend and the location/ID of the defined calibration stations.

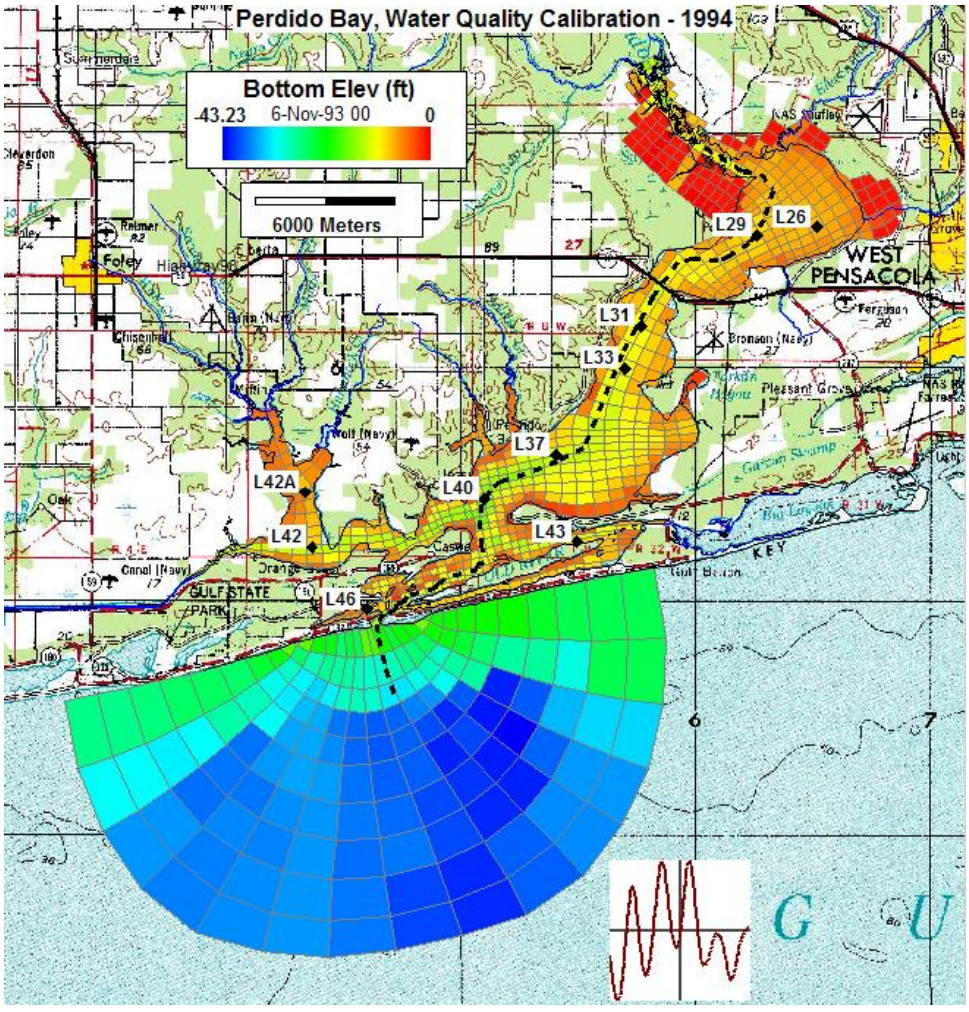


Figure 3-11 Example of ViewPlan output for the Perdido Bay water quality model.

ViewPlan provides the user with a dropdown menu comprising the following options:

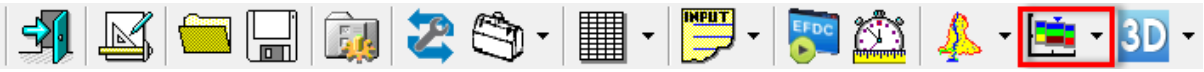
View Initial Conditions: This option may be used when the user is setting up the model or wants to view only the IC's, even if model output has already been loaded.

View Model Results: This option is used after the model has been run and the user wishes to analyze the model results. This is the default action if the user clicks on the *ViewPlan* button and not the dropdown arrow.

View Model Results (Re-scan Output): This option should be used if the user has paused an EFDC model run, viewed the results then resumed the model run. EE will only have the data from the originally paused run loaded. To later view the complete model run results, the user would then use this option to reload the model results.

Drag & drop files such as ArcView polylines or ASCII data posting files to provide quick annotations on the model.

3.9 ViewSlice Viewer



The *ViewProfile* button of the main toolbar provides access to the visualization of the 2D vertical section post-processing utility of EFDC_Explorer. An example of the type of plot available is shown in Figure 3-12. This plot shows a 2D profile of dissolved oxygen along the user defined transect shown previously in Figure 3-11 (see the black dotted line) for Perdido Bay. Section 9 contains a more complete description of the features and use of *ViewSlice*.

The *ViewSlice* button also provides the user with a dropdown menu with functions identical to those described above for the *ViewPlan* button.

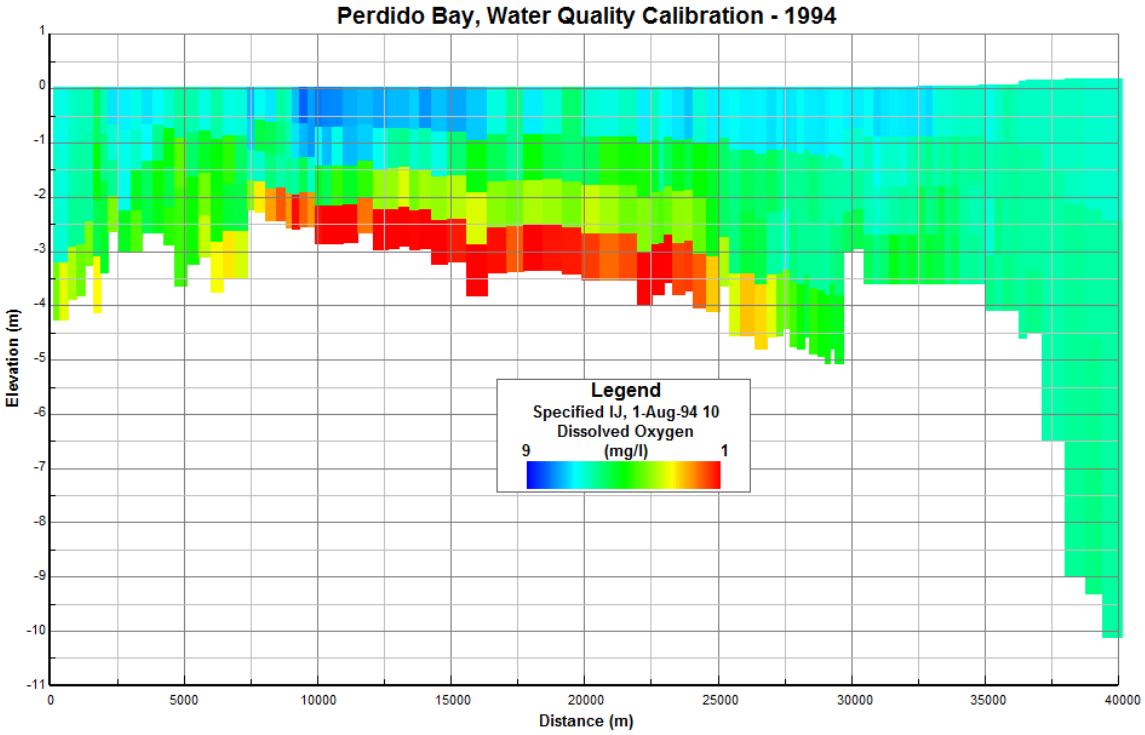
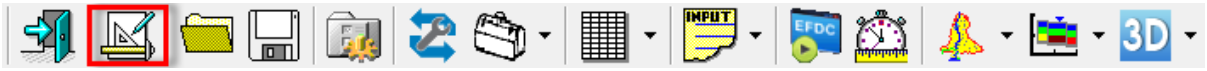


Figure 3-12 Example of ViewSlice for dissolved oxygen for WQ model.

4 Generate New Model



EFDC_Explorer can generate a new grid and write to disk the essential EFDC model files to begin building an application. Figure 4-1 shows an example model generation options form.

There are many settings that need to be assigned when building a new model. The approach taken by EFDC_Explorer is to use a master input file for EFDC which is stored in a non-editable database. EFDC_Explorer automatically updates the EFDC default initial conditions from this MS Access file (EFDC.MDB) for the following files:

- EFDC.INP
- WQ3DWC.INP
- WQ3DSD.INP
- WQRPEM.INP

From EE7.3 onward this method supersedes the use of the EFDC.INP file which had been used to update the basic features the user required such as computation options, hydrodynamic settings, and a range of miscellaneous settings. Formerly a valid EFDC.INP had to be selected prior to being able to generate a new model.

EFDC_Explorer can quickly generate simple to complex Cartesian grids. These grids can be uniform or expanding, rotated and trimmed to match the physical domain. For more complex grids, EFDC_Explorer can import grids generated with many tools or other models as described further below. In addition to these grid generation/import options, EFDC_Explorer can import a generic set of nodal points.

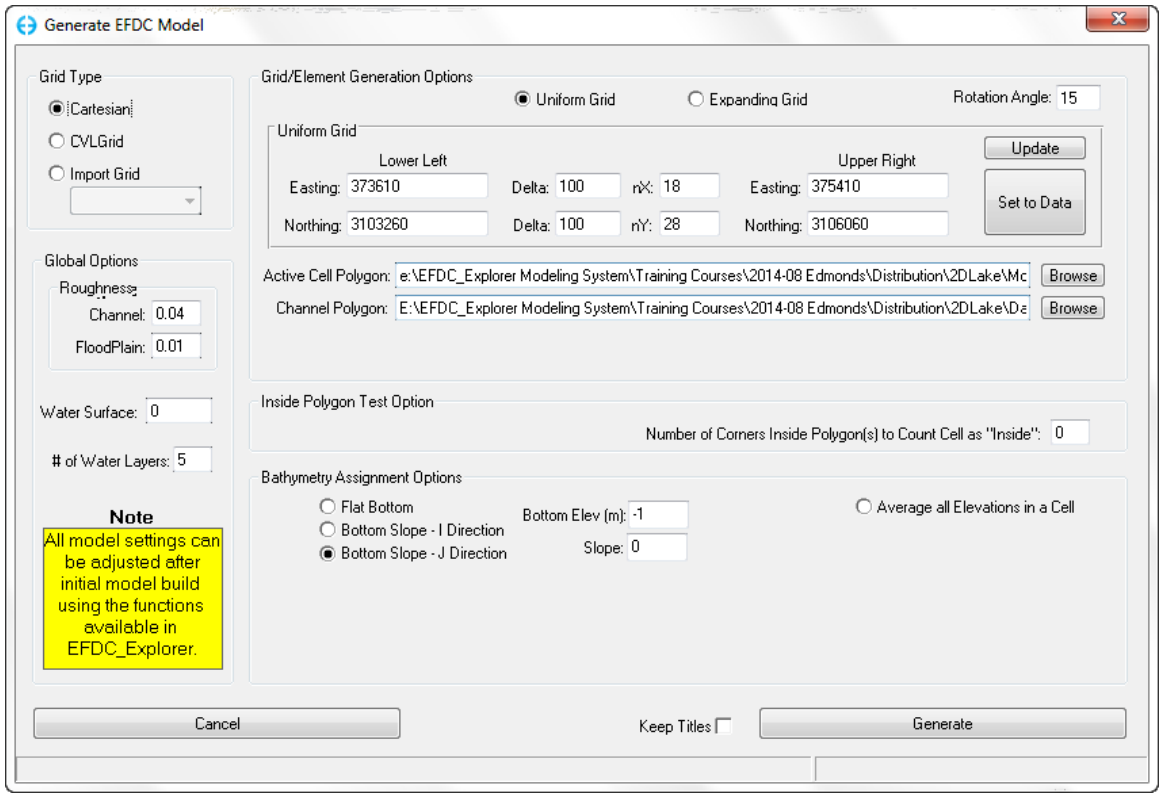


Figure 4-1 Generate new model options form, with Cartesian uniform grid option.

4.1 Creating and Importing Grids

The *Grid Type* frame contains the primary selection of the various grid generation functions: *Cartesian,* for creating a Cartestian grid in EE, *CVLGrid*, for loading a grid built by DSI's CVLGrid 2D grid generator, and *Import Grid*, for loading third party grids. Each for these are described in detail in the following section.

The contents of the center frame, *Grid/Element Generation Options*, will vary depending on the option selected. Shown in Figure 4-1 is the frame contents Cartesian grid type which is further described below.

4.1.1. Creating a Cartesian Grid in EE

There are two options for generation of Cartesian grids, uniform grid spacing and variable grid spacing. The user selects the desired option and then fills in the appropriate settings as outlined below.

The coordinate system used in the topographic file, if used, boundary polygons and specified in the various required coordinate parameters must be the same.

Typically the user is not building a rectangular model domain. Therefore the user has the option of entering a polygon file that specifies the outline of the model domain. Any cells that are inside the polygon will be set as active for the model. The *Cell Test* frame allows the user to set the inside polygon test options to determine if the cell is inside the model domain or not. An option value of "0" will cause the inside cell test to use the cell centroid only. If the # *Corners* is between 1 and 4, the inside cell test will require the specified number of cell corners to be inside the polygon to be considered "inside" and therefore an active cell.

If the user is modeling a riverine system with overbank/floodplains, the user may provide a *Channel Polygon* file. This file assigns all cells outside this polygon but inside the model domain as floodplain cells (Cell type = 7). Those cells inside the polygon receive a standard computational cell type (Cell type = 5). This allows some of the tools in EFDC_Explorer to be quickly applied, e.g. Roughness Options.

The *Rotation Angle* can be entered to rotate a Cartesian grid about its centroid. If a grid is to be rotated, the coordinates entered in the following sections are based on the unrotated grid. However, the application of the *Active Cell Polygon* and *Channel Polygon* are applied after the cell rotation.

4.1.1.1 *Uniform Grid*

For the Cartesian grid option, the user must specify the model corner coordinates and spacing. Several utilities have been provided to speed up this process. The *Set to Data* button simply copies the data from the topographic data corners to the grid corners. The user can edit them afterwards as desired. Secondly, the *Update* button computes empty fields. For example, if the user enters the lower left (LL) corner, the DX, DY and NX (IC-2) and NY (JC-2) then presses the *Update* button, the upper right (UR) coordinates are set. Likewise, if the LL and UR coordinates are set and the NX and NY values are set the DX and DY are computed.

If the user has specified 3 of the 4 entries per dimension, pressing the Ctrl-U key in the field needed will cause EFDC_Explorer to compute the required entry.

4.1.1.2 *Expanding Grid*

An expanding grid is simply a variable spaced Cartesian grid that expands in all directions from a focal point. The form shown in Figure 4-2 shows the parameters needed for developing an expanding grid. The user must specify the coordinates of the focal point, the initial (smallest) delta X and delta Y's, the rate at which the user wants them to expand at and the maximum cell sizes. Specify an *Active Cell Polygon* to trim the cells to the desired model domain. The spatial limits of the grid are set either by the topographic data or the active cell polygon.

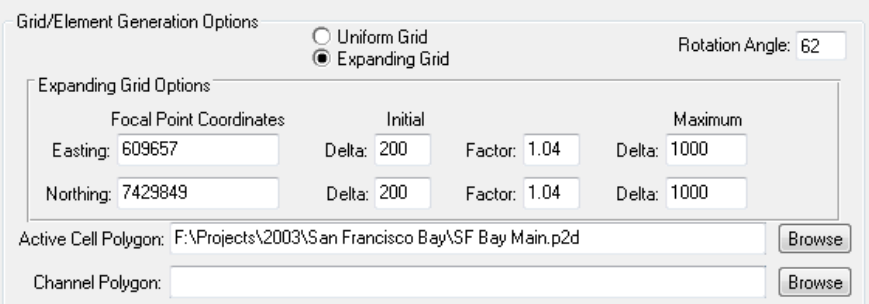


Figure 4-2 Cartesian gridding: Expanding & rotated grid.

Figure 4-3 shows an example of the expanding grid developed for the San Francisco Bay. The focal point is just offshore from Hunters Point.

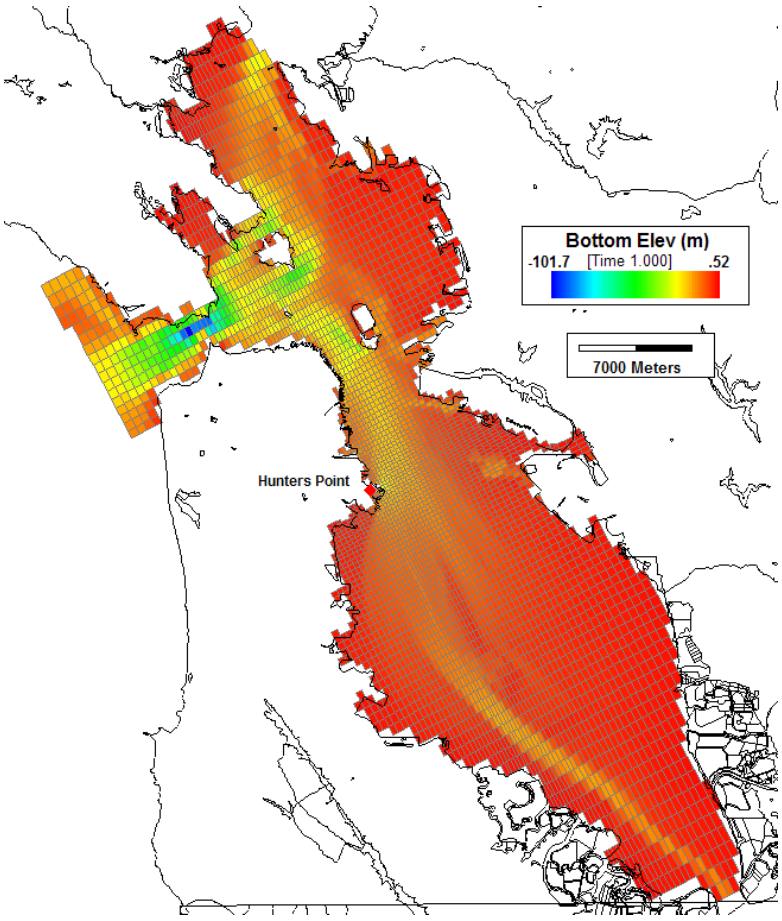


Figure 4-3 Expanding Cartesian grid example of San Francisco Bay.

4.1.2. Importing from CVLGrid

CVLGrid is 2D grid generation software developed by DSI for use with curvilinear orthogonal models, and in particular for use with EFDCPlus, EFDC_SGZ, EFDC_EPA, and EFDC_Hydro models. As all EFDC models require 2D curvilinear orthogonal grids this tool has been developed to meet this need for DSI and our clients and customers. This tool has been further optimized for use with the EFDCPlus/EFDC_Explorer Modeling System. CVLGrid can be used to refine and modify existing grids and if the I, J map is not changed then model can be loaded back into EE without needing to update the boundary conditions settings.

In EE main form, the user should click *Generate New Model* button ( ), and the *Generate EFDC Model* form will appear as shown in Figure 1. In the *Grid Type* frame, the user selects the *CVLGrid* option and then clicks *Browse* button on the right as shown in Figure 1 to browse to the CVLGrid grid file.

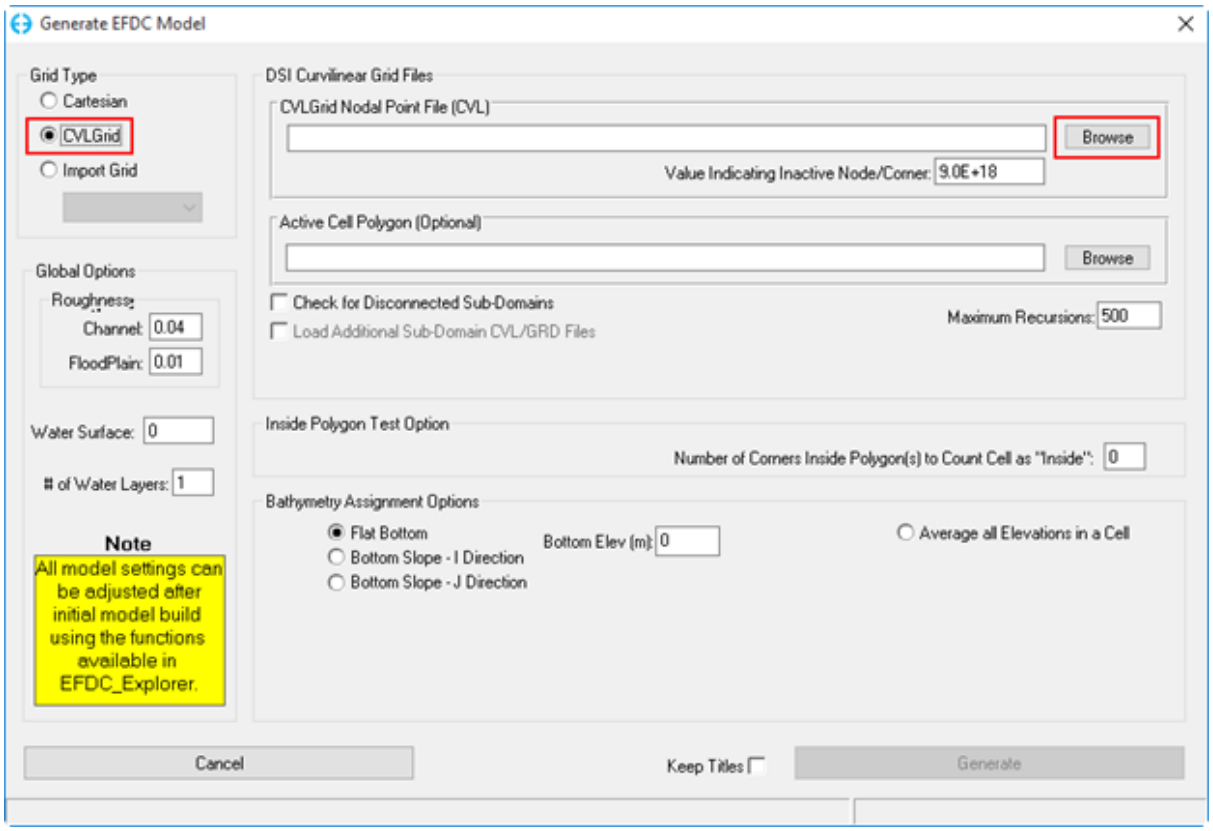


Figure 4-4 Generate new model options form, with CVLGrid option.

After clicking the *Browse* button, the *Import a CVLGrid Node File* form will appear to allow the user to select CVLGrid grid file. A CVLGrid grid file has extension .CVL. CVLGrid project files, which CVLGrid uses to save the splines, overlays and background maps, has the extension .CVP. EE only opens .CVL extension files. Click the *Open* button as shown in Figure 2 and the *Generate* button on the *Generate EFDC Model* form will be enabled. If this is not enabled then some information is lacking so that EE cannot generate the grid. The user should check that grid has been properly loaded if this is the case.

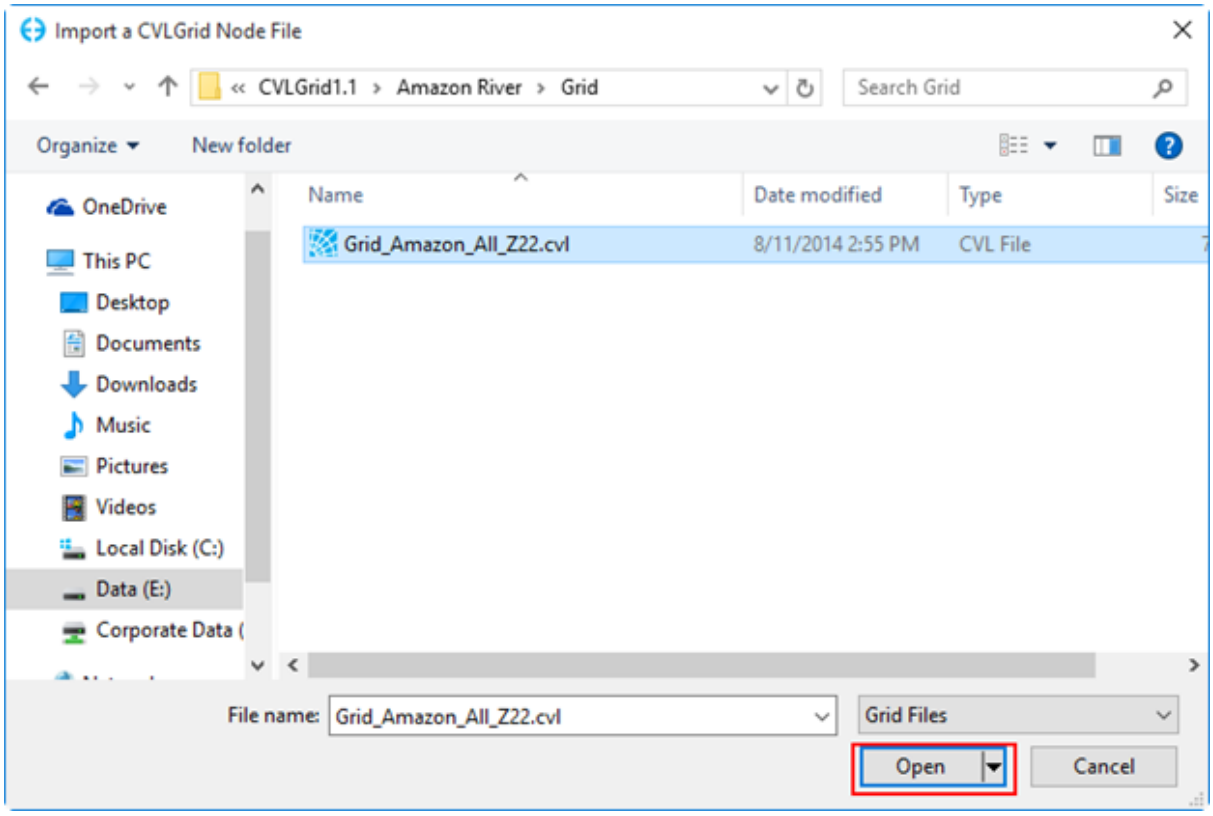


Figure 4-5 Import CVLGrid file form.

The user should then click the *Generate* button to generate an EFDC model as shown inFigure 4-6. A message will appear to inform the user that an EFDC model has been created with the number of grid cells as shown in Figure 4-7. Click *OK* button and the EFDC model with grid map will be generated as show in Figure 4-8. The user is now ready set the initial and boundary conditions for the model as outlined in the Section 4.3.

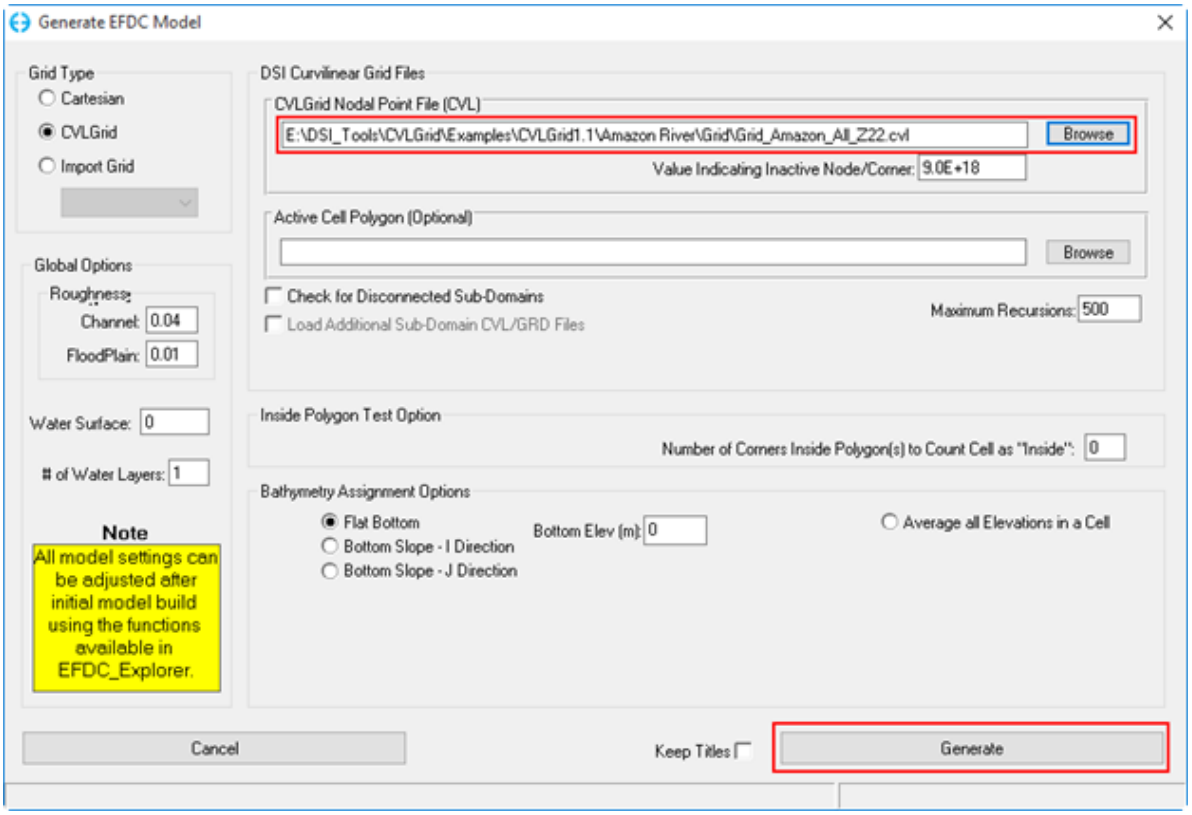


Figure 4-6 Selected CVLGrid file form.

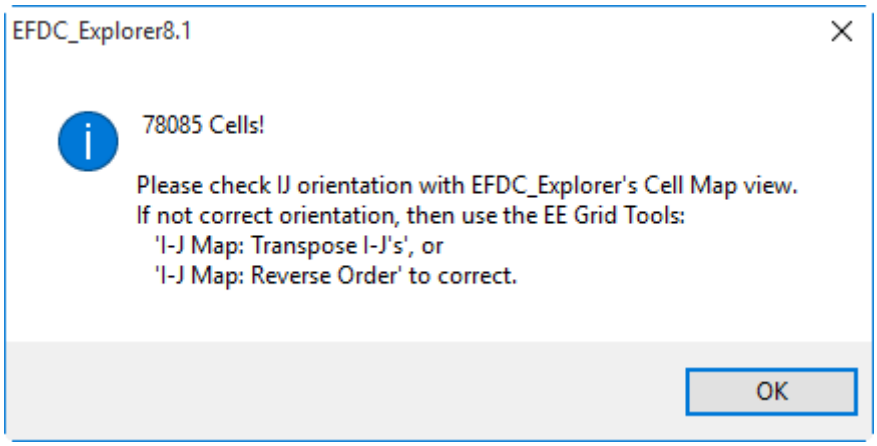


Figure 4-7 EE Informing message form.

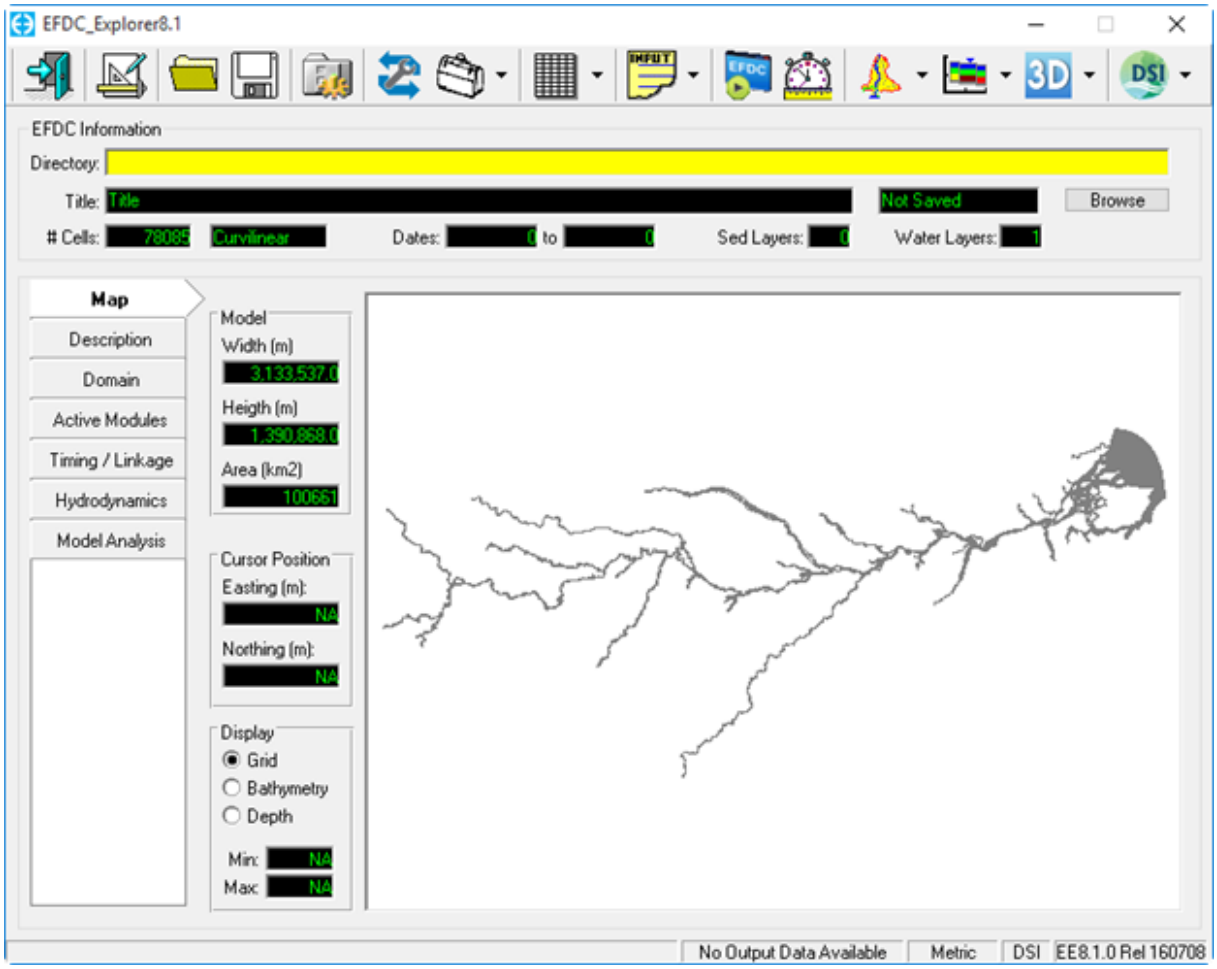


Figure 4-8 EFDC Model generated on EE main form.

4.1.3. Importing Third Party Grids

This feature allows the user to import an existing model grid from another hydrodynamic model or a grid generated using a supported third-party grid generator. The types of third party grids that can be imported are:

- Delft RGFGrid (Delft 2006),
- Grid95, or
- SEAGRID (Signell, 2007).

EFDC_Explorer can also import the grids from the following models:

- GEFDC (Hamrick, 2007)
- CH3D (WES version and University of Florida version), and
- ECOMSED.

In the *Grid Type* frame the user should select the *Import Grid* option. The dropdown list below the option is then enabled. The user then needs to select the appropriate import type. Import type dependent options will then be displayed. The user should specify the files and enter any options required. The import file(s) which are optional are labeled as such. Figure 4-9 shows an example of the import form with the Delft RGFGrid option. Generally, it is

recommended to use CVLGrid to develop grids best suited to use with EFDC and edit at a later time if required.

For an existing model grid import, the topographic data does not need to be separately processed as the cell bottom data should already exist in the model grid files.

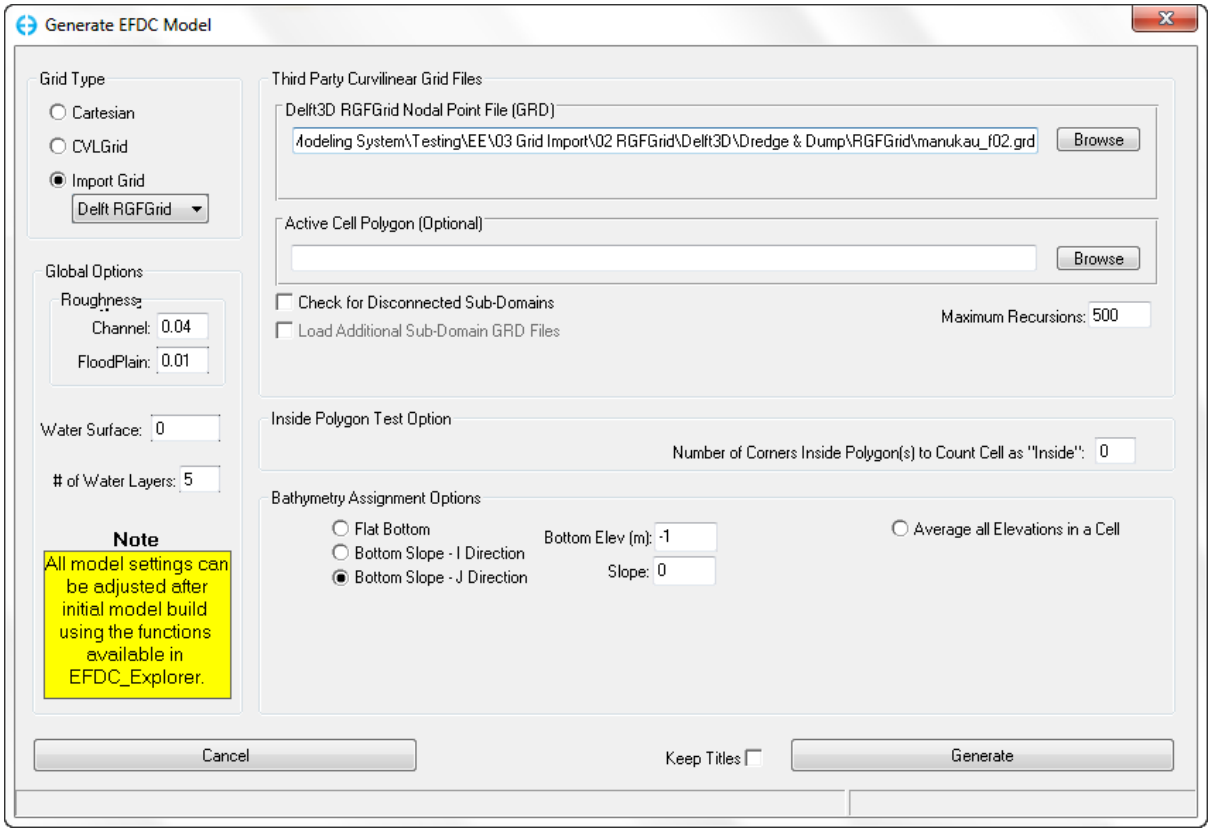


Figure 4-9 Grid generation: Import Delft's RGFGrid.

4.1.4. Importing Disconnected Sub-domains

EE has the ability to import grids with multiple sub-domains. When the user checks the box labeled *Check for Disconnected Sub-Domains* the MAPGNS file will store the various sub-domains. The user is then able to post-process the grid to allow non-adjoining cells to connect. This is useful when importing complex grid systems.

To import multiple subdomains the user should first browse to load one subdomain. After this they may select the *Check for Disconnected Sub-Domains* checkbox as well as the *Load Additional Sub-Domains CVL/GRD File* as shown in Figure 4-10. The user may then select the *Generate* button. Only after selecting the *Generate* button will EE allow the user to navigate to the other subdomains. The remaining subdomains should all be selected at the same time by combination of *Ctrl* key and *LMC* then select the *Open* button. In this way several unconnected sub-domains can be loaded into EE.

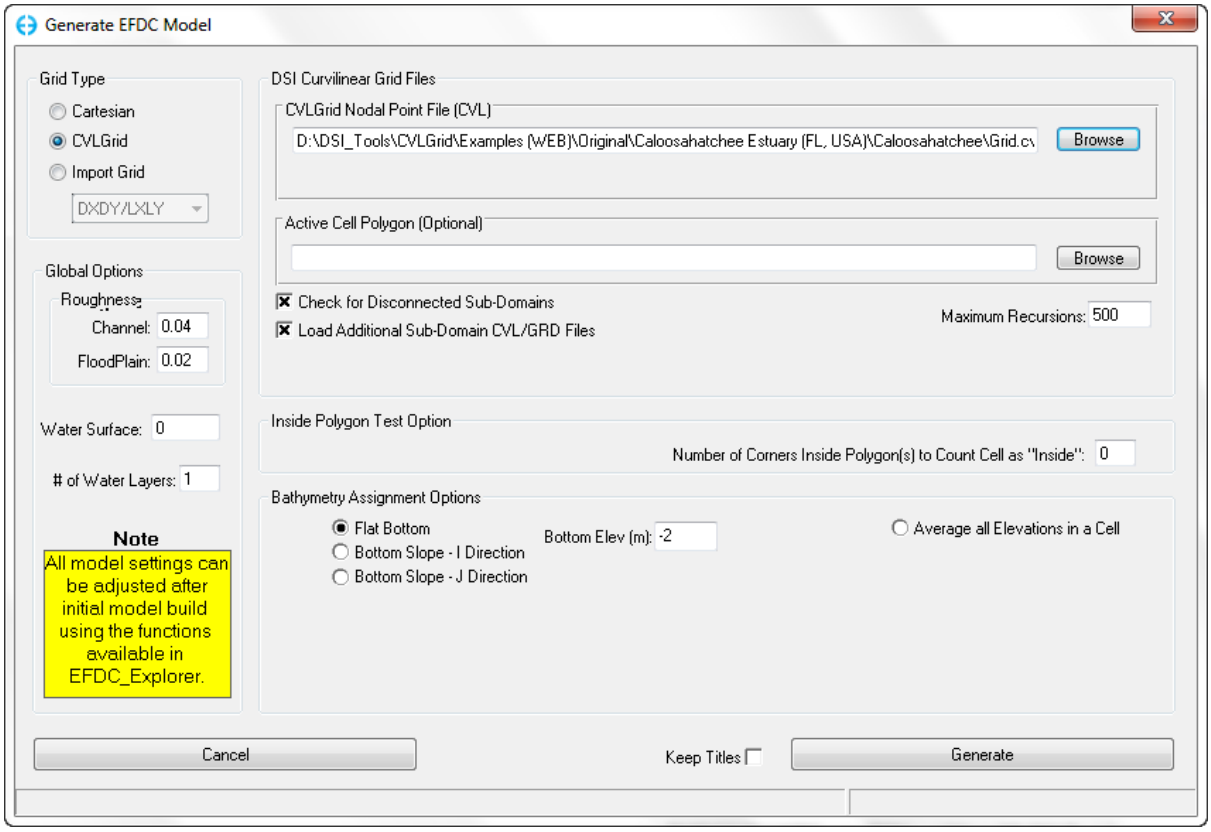


Figure 4-10 Generate Model: Loading multiple sub-domains

See Section 8.6.6.1 on how to connect the sub-domains in *ViewPlan*. An example of the MAPPGEW.INP file that EFDC uses to map the connectors is provided in the appendix for the user's reference.

4.1.5. Triangular Cells on Border

EFDC has long had the capability to simulate triangular cells on the boundary of the domain. However, previously the editing and display of these cells was not supported by EE. From release of EE7.2 triangular cells are allowed on the border.

In the CELL.INP file, active cells are designated as 5, and border cells as 9 and inactive cells as 0. Triangular cells are 1 or 2 on the western face and 3 or 4 on the eastern face of the grid. The user may set all border cells as triangular by select ALT-T from the *ViewPlan* | *Viewing Options* | *Bottom Elevations*. The user will be prompted to *Convert the staggered border cells to triangular cells.* Selecting "Yes" will automatically place triangular cells on corner cells. To change individual cells the user should then *Enable Edit* from the *Viewing Options* frame and RMC on a cell. Selecting *Edit* allows the user to alter the cell as shown in Figure 4-11 with options *Quad* or *Triangular: NE*.

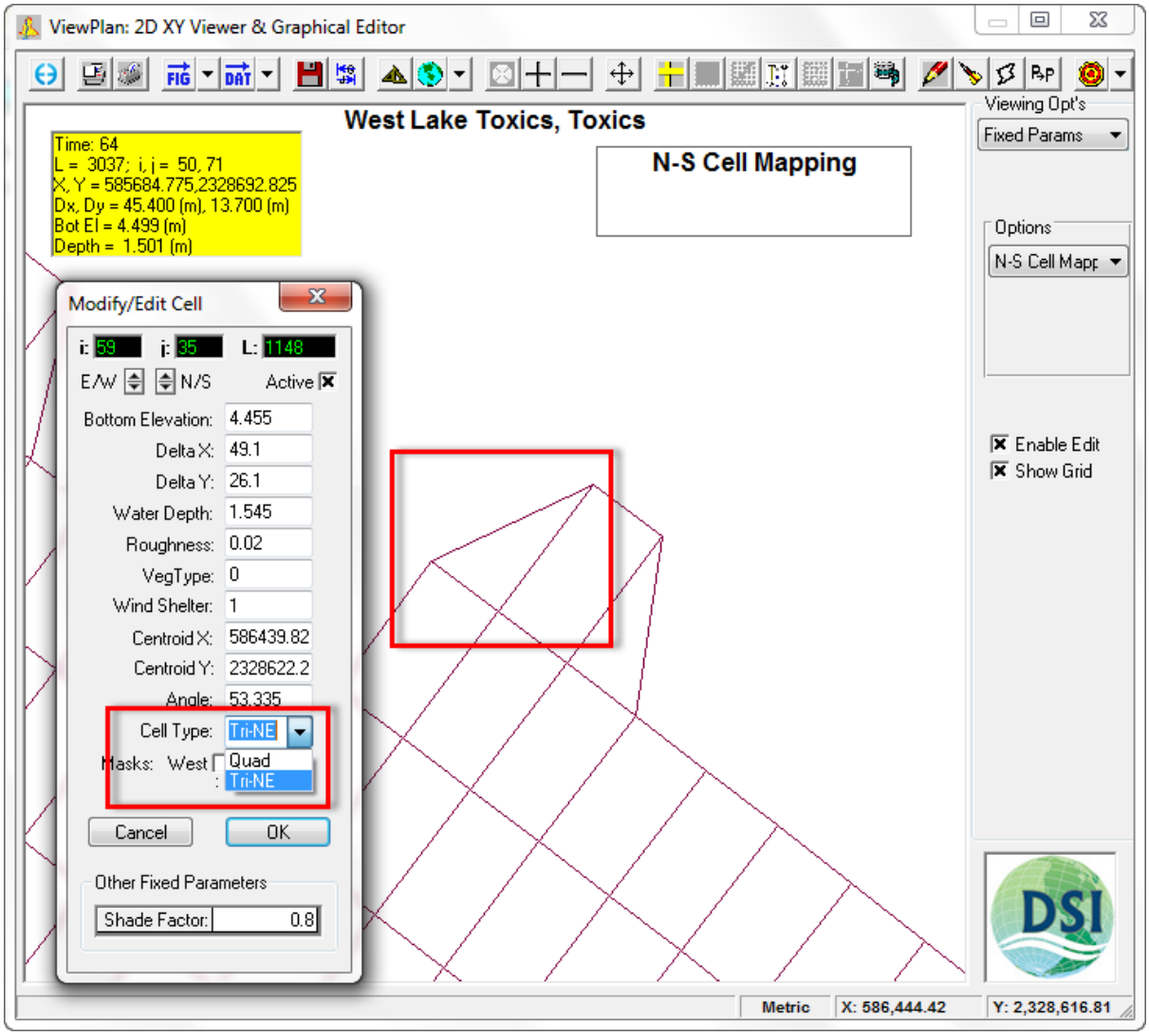


Figure 4-11 ViewPlan: Triangular Boundary Cells.

The triangular edge is also displayed with exporting /graphic images such as shapefiles and Google KML files as shown in Figure 4-12.

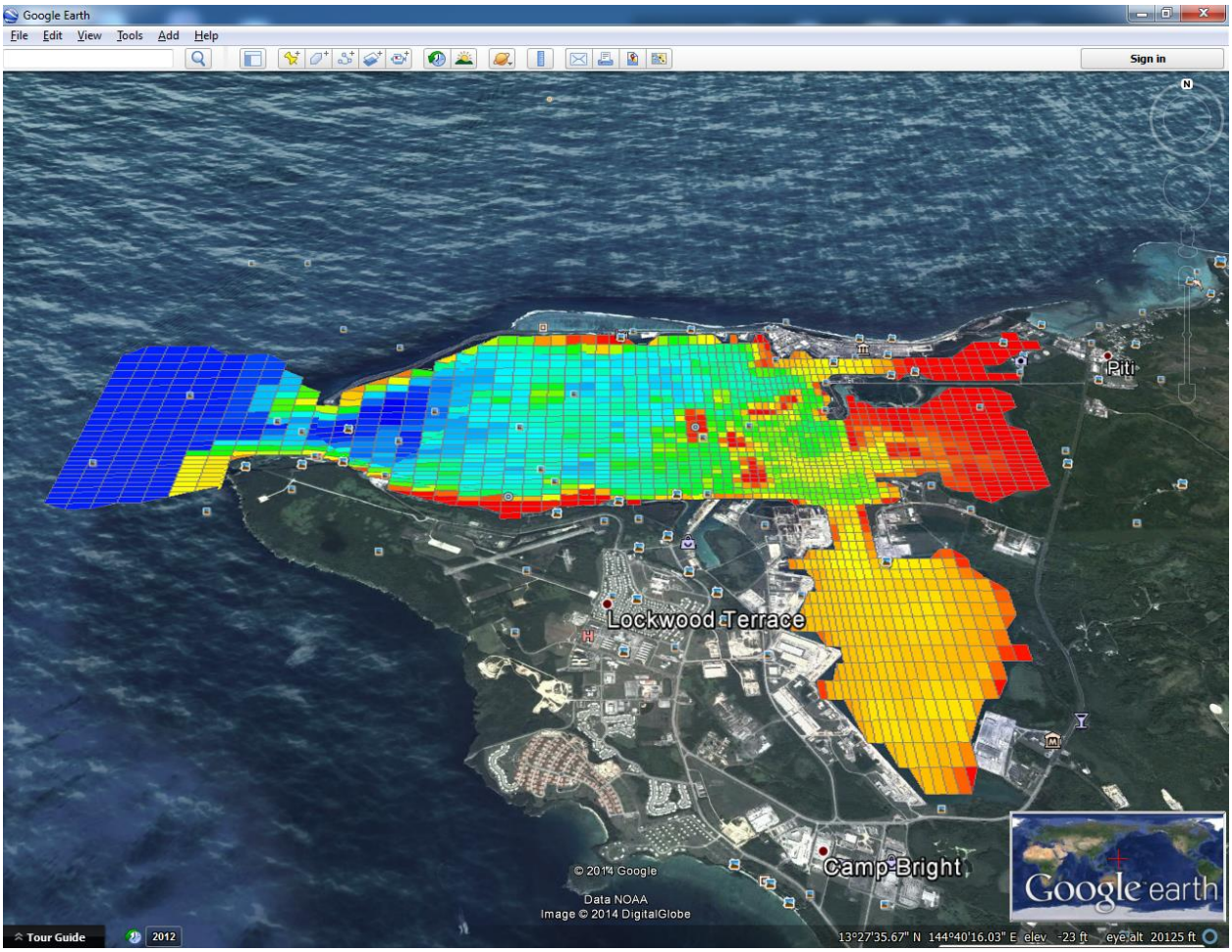


Figure 4-12 Triangular Edge Cells in KML Plot.

4.2 Elevation Options

EFDC_Explorer provides the user with several methods of generating the cell bottom elevations from the input topographic data. The following discussion addresses each option.

The following three options do not require a topographic data file. They can be used for quicker model generation or building test or evaluation models.

Flat Bottom: Applies the specified constant bottom elevation to all cells. Due to the iterative nature of grid generation, this is the recommended option. Actual Model bathymetry can be applied once grid generation is complete.

Bottom Slope – I Direction: Applies a constant bed slope to the cells in the I direction, starting with a specified initial bottom elevation. If the entered slope is positive the bottom elevations will decrease with higher I's. The inverse is true if the entered slope is negative.

Bottom Slope - J Direction: Applies a constant bed slope to the cells in the J direction, starting with a specified initial bottom elevation. If the entered slope is positive the bottom elevations will decrease with higher J's. The inverse is true if the entered slope is negative.

The following three options do require a topographic data file to be specified.

Average all Z's in Cell Use this option if the topographic data is very dense, relative to the grid.

Use Minimum Z This option scans the data and applies the lowest elevation for any point located inside the model cell.

Use Z at Cell Centroid: Use this option if the topographic data is relatively sparse as compared to the grid. This approach uses the inverse distance interpolation scheme to compute the elevation at the X and Y of the cell centroids.

4.2.1. Topographic Information File

Fundamental to any model is an understanding of the topography/bathymetry of the region. If bathymetry is to be applied at this phase, a file must be selected that contains topographic data over the region to be modeled. The formats of this type of file are provided in Appendix B and listed in Section 4.2.2. Once selected, EFDC_Explorer loads the file and displays the corner coordinates in the *Upper Right* and *Lower Left* frames. The units displayed are assumed to be in meters with direct correspondence to the coordinates to be used for the LXLY file. An example digital topographic data file using colors to represent the elevation range is displayed in Figure 4-13.

The user should note that this file is not necessary to generate a new model. EFDC_Explorer provides the user the ability to reapply revised or new topographic information to an existing model using the *Assign* button on the *Domain* tab, *Initial Conditions and Bottom Roughness* Tab in the *Bathymetry* frame.

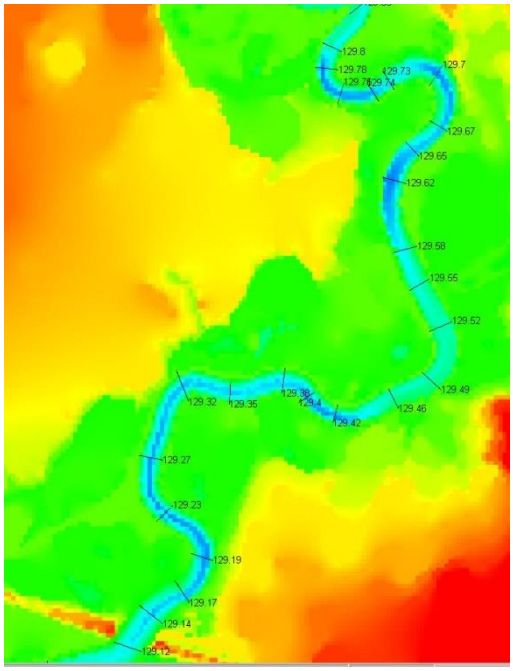


Figure 4-13 Example digital topographic data.

4.2.2. File Formats

Some of the data file formats that EFDC_Explorer currently reads include:

DAT/TXT	ASCII XYZ files with space, comma or tab delimited (Appendix B)
TB2/TP2	ASCII regular grid file (Appendix B)
P2D	ASCII XYZ files grouped into polylines. Space, comma or tab delimited (see Appendix B)
2DM/3DM	Surface Water Modeling System (SMS) mesh geometry file
PLT	TecPlot formatted file
E00	ArcInfo Coverage Export file
BIL/DEM	USGS BIL format for Digital Elevation Model (DEM)
GRD/ASC	ArcInfo ASCII grid file

Furthermore, some of EFDC_Explorer polygon/polyline formats supported include:

P2D ASCII XY or XYZ files grouped into polylines. Space, comma or tab delimited (see Appendix B)

DAT, TXT, XYZ, OUT, ASC ASCII XYZ files with space, comma or tab delimited (see Appendix B)

2DM/3DM Surface Water Modeling System (SMS) mesh geometry file

PLT TecPlot formatted file

E00 ArcInfo Coverage Export file

GRD/ASC ArcInfo ASCII grid file

KML Google Earth formatted file. Model units must be in UTM WGS84

SDF HEC-RAS GeoRas File

G0 HEC-RAS Geometry File

DXF AutoCad drawing exchange format

SHP ESRI ArcView Shapefile (units must be in model units or Lat/Long)

TAB Mapinfo (units must be in model units or Lat/Long)

LDB, SPL Delft land boundary or spline format

DLG, OPT USGS Digital Line Graphs

4.3 Model Generation Process

The model generation process is:

1. Select the *Generate New Model* button from the main form of EFDC_Explorer. EFDC_Explorer will automatically set many of the standard coefficients and parameters. The user can change any of these later in the model generation process. (In versions of EE prior to EE7.3 an EFDC.INP template file was required. A simple EFDC.INP was supplied in the original EFDC_Explorer setup package (C:\EE_EFDC). In EE7.3 and later versions it is no longer required to use a template EFDC.INP).

2. Bathymetry is required for the model generation process, but not required for the gridding process. Because the grid generation process is often iterative, it is recommended to skip specifying the *Topographic Information File* and, instead, use the flat bottom option in the *Elevation Options* frame.

3. Select the file containing polygons to trim the model cells (optional). The first polygon in the file should be the main model domain outline. Subsequent polygons in the file are interpreted as cutouts/islands.

4. Set the number of layers.

5. Set the initial water surface elevation. Should be above the flat bottom elevation

6. Set the default bottom roughness height.

7. Select the gridding approach and import or generate a grid.

8. At this point the *Generate* button should be enabled. If it is still grayed out then there is still an essential piece of information that has not been set. The required inputs vary depending on the gridding option selected. Press the *Generate* button to create a new EFDC model using the options specified. Even though a project directory is requested, this process does not write the files, only creates the model in memory.

9. When the model generation process is complete, EFDC_Explorer pops up a message informing the user as to how many active cells were created and what the maximum I and J were.

10. The last pop up displayed informs the user to review the model and make sure the I and J orientations are reasonable. For some convoluted grids, the L=2 lower left cell may not be correctly assigned. After reviewing the grid in *ViewPlan* and determining that the IJ mapping needs to be adjusted, the user can use the IJ mapping tools from the main EFDC_Explorer form to flip either I or J and/or transpose the I and J mapping.

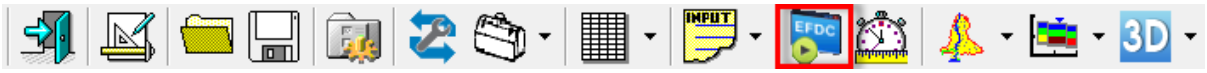
11. Review the generated grid in *ViewPlan*. Load background images and/or polyline overlays to visually check the grid. View and check grid orthogonality.

12. If the grid is acceptable, then apply the bathymetry to this new grid using the *Bottom Elevations* button on the *Initial* conditions tab on the main EFDC_Explorer form.

13. Review the model grid with bathymetry added to make sure important bathymetric features are properly represented. Check the reasonableness of the CFL time step.

14. Repeat these steps as necessary to obtain the proper balance of grid resolution, computational speed and other project specific factors.

5 Run Model



This is the function that actually runs an EFDC project/model. It does not first save to disk the currently loaded EFDC project prior to running. Therefore, if the user has made changes that they desire the run to reflect, the user must first save the project.

Figure 5-1 shows the options box for the *Run Option*. The *EFDC Run* form provides quick access to the most pertinent run time settings for the EFDC simulation. Some of these data have already been set in the main form but EE allows the user to review and update these data as needed just prior to running the model. EE will save the EFDC input files automatically, if changes are made.

5.1 General

If the user selects the *Overwrite?* check box then existing EFDC+ model results files will be overwritten. If model results exist in the project folder, EE will not start a new run unless this checkbox has been checked. If the user does not want to overwrite the results, then save the project out to a new folder then run the model.

The user should ensure that the correct EFDC executable has been selected in the EFDC_Explorer Settings in the main form (see Section 3.2). The run option builds a batch file, saves it in the EFDC project directory, and then launches the batch file. The file name is 0_RunEFDC.Bat. This DOS batch file launches the EFDC executable EFDCPlus or EFDC_GVC as a DOS window with the current project directory in the Title bar.

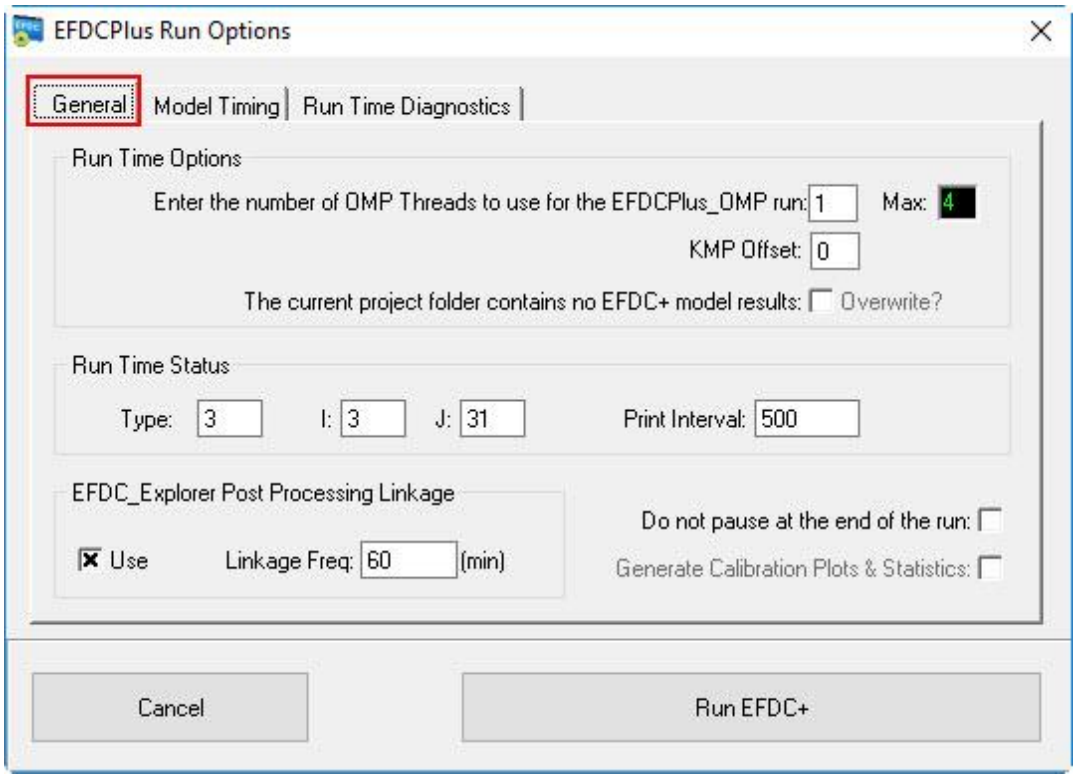


Figure 5-1 EFDCPlus Model Run Option Form: General.

Run Time Status frame contains the settings for feedback to EFDC's runtime screen during the model run. The user can simply type in the desired I and J or the user can set them using the mouse. To set using the mouse, select *ViewPlan* | *Cell Map* or *ViewPlan* | *Bottom Elev* views, then right mouse click on the desired cell and select *Set as Show I J*. The data displayed on the screen depends on the *Type*, whose options are: 0-TSS, 1-Salinity, 2-Temp, 3-Dye, 5XX-Toxic, 6XX-Seds, 7XX-Snds, 8XX-WQ Var, where XX refers to a specific constituent in the sub-model group.

For more information on EFDCPlus_OMP, setting the KMP offset, and selecting the number of threads for multithreading, please see below.

When EFDCPlus finishes execution, the model waits for the user to press a key to continue/exit. If the user does not want the pause function, they make check the box *Do not pause at the end of the run*.

EE now has the option of automatically generating calibration plots and statistics. In order to do this the file "0run" is created as a flag which is tells EE that the model is still running or has crashed. When this is not the case EE will automatically generate the plots and statistics. If the user opens a new EFDC model with the same instance of EE while the EFDCPlus model is running, the automatic plot/statistics generation function is aborted. A new EE window should be opened if the user needs to open another EFDC model.

5.1.1. Open Multi-Processing Functionality

In EFDC+ the EFDC code has been upgraded to support multi-threading capabilities and provide vastly improved model run times. Open Multi-Processing (OMP) is an API designed to facilitate the use of multi-threaded programming. It does this by creating and synchronizing threads automatically at the users command. OMP is an unofficial standard for FORTRAN, and is supported by IBM and Intel among others. Both Microsoft and Intel's most recent compilers provide excellent support. With OMP the compiler is informed that a section of code may be parallelized through use of certain directives. The compiler is then able to generate an application that consists of a single thread, which forks into several threads for a parallel region; these threads are then synchronized and all but one terminates at the end of the parallel region as the program execution reverts to a single master thread. Depending on the machine topology, application, and operating system, thread affinity can have a dramatic effect on the application speed. Both the Two Time Level (2TL) and Three Time Level (3TL) solutions subroutines in EFDC+ have been updated to allow for Open Multi-processing, thereby providing significant speed up in model run time for users running with of these solution.

The OMP capability of the EFDCPlus/EFDC_Explorer was formerly an optional extra for users. From EE8 and EFDC+ it is the default option. In order to use it effectively it is important to note the following features. The OMP option is only required to be set when the user runs the model. In the *Run Options* frame shown in Figure 5-2 below, the user is able to set the number of OMP threads for the model run. This will rewrite the EFDC.INP file.

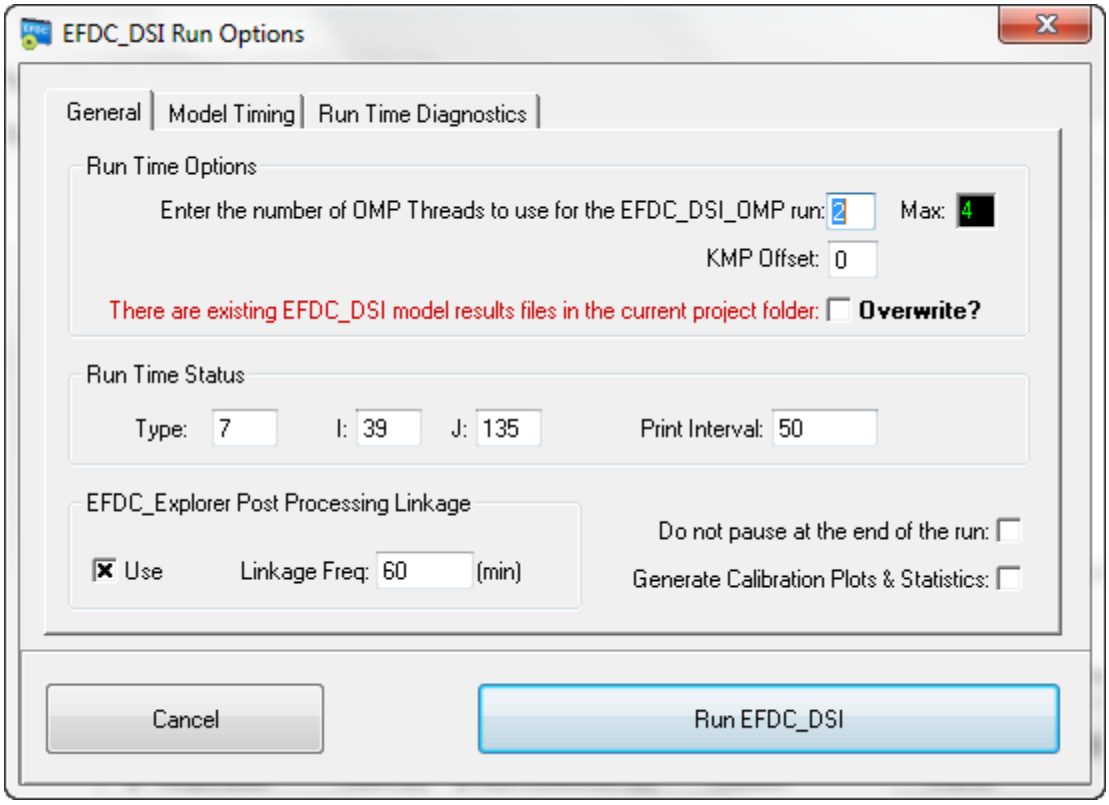


Figure 5-2 Run Options for OpenMP model runs.

5.1.2. KMP Offset

The user may also set the *KMP Offset,* which is the core offset for the current run. The *KMP Offset* relates to the number of cores already being used by the system, and is therefore usually set to zero. The *KMP Offset* should only be adjusted if the user wishes to run more than one model on the same machine. In order to run second model the *KMP Offset* should be set to the number of cores being used in the already running model. For example, assume a system comprising a Xeon Processer with 6 cores (12 threads), is already running a model on 2 cores (where KMP Offset had been set to 0). If the user wishes to run a second model, they then should set the KMP Offset to 2 and the OMP threads to 4.

A typical Intel I7 computer had from has 4 to 6 cores. The way these cores are handled in the EFDC code is with the KMP_AFFINITY variable, defined as:

KMP_AFFINITY=[<*modifier*>,...]<*type*>[,<*permute*>][,<*offset*>]

For example, to list a machine topology map, specify “KMP_AFFINITY=verbose,none” to use a modifier of verbose and a type of none.

“KMP_AFFINITY = granular = fine, compact, 1,0”, will assign individual threads to different cores. When it assigns the threads it also spreads them out. If the modifier is set to verbose, then a map of threads will also be produced.

Note that a “0_RunEFDC” file is created in the model folder each time the EFDCPlus model is run. This file tells EE whether the model ran to completion or not as it is deleted on successful model completion. If the user opens that file then they can see these parameters.

Figure 5-3 illustrates the binding of OpenMP thread to hardware thread contexts when specifying “KMP_AFFINITY=granularity=fine,compact”.

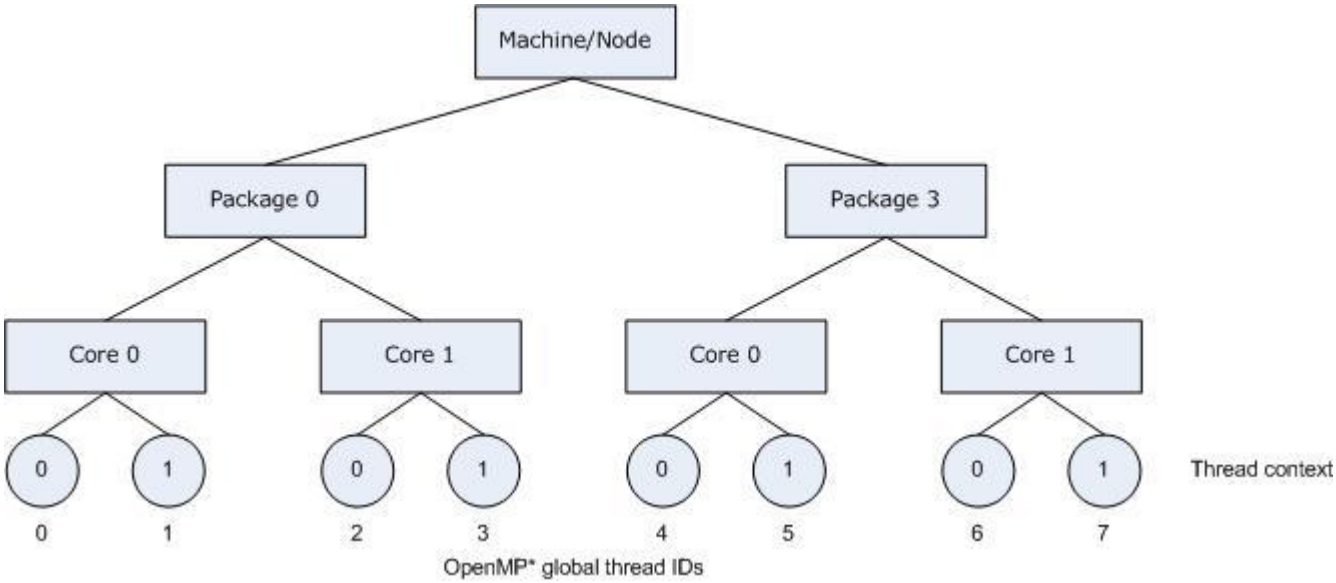


Figure 5-3 Open MP thread IDs (graphic courtesy of Intel).

To learn more about the KMP_AFFINITY environment and the specific syntax which must be used please visit the following website:

http://software.intel.com/sites/products/documentation/hpc/composerxe/en-us/cpp/mac/optaps/common/optaps_openmp_thread_affinity.htm

5.1.3. Speed Gains with OMP

The Intel® runtime library binds OMP threads to physical processing units. Depending on the machine topology, application, and operating system, thread affinity can have a substantial impact on the application speed. Users typically produces **run times up to 4 times faster** on a six core processor than the conventional single-threaded EFDC model. The figure below shows the speed gains from one processor (OMP1) to six processors (OMP6) for various computational groups in an EFDC model setup for the Sacramento-San Joaquin Delta. As can be seen in the first column, "Elapsed Total", overall run times are now up to 4 times faster with 6 threads than what they were running a single thread.

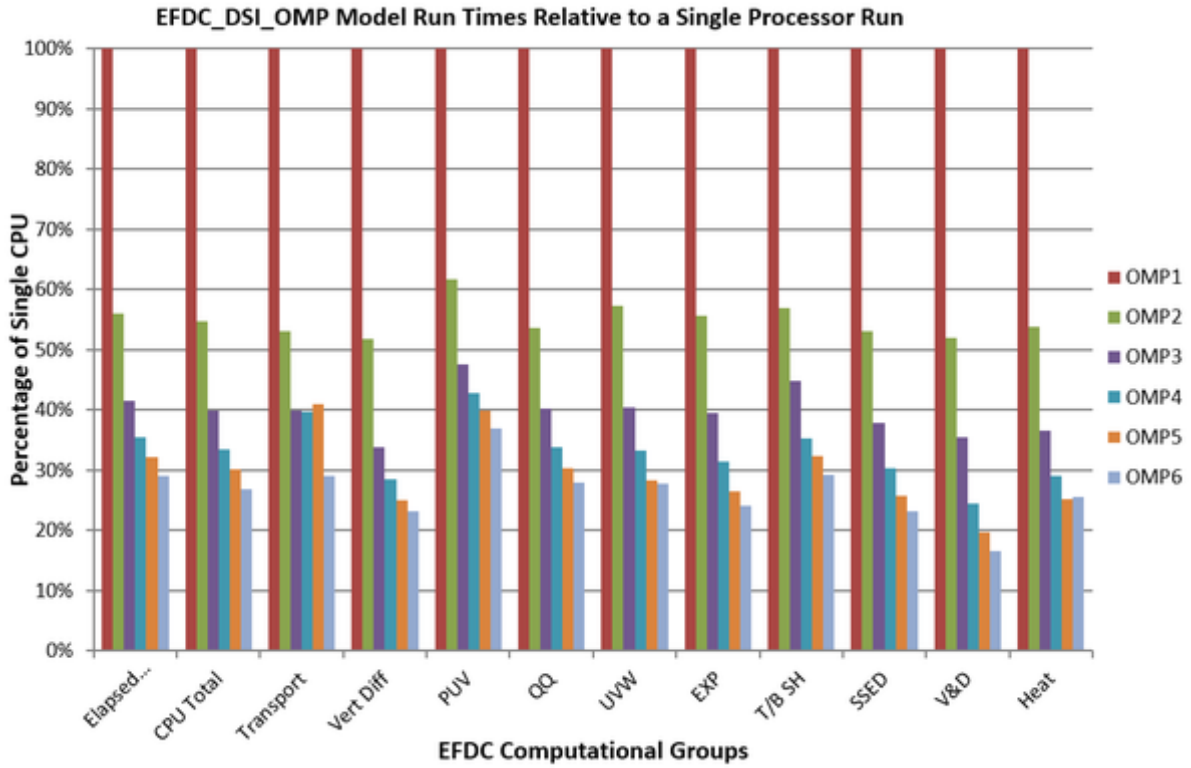


Figure 5-4 EFDC_DSI_OMP Model Run Times relative to a single processor run.

Note that the **computational groups**, or sub-models, within EFDC have been individually timed in order to track the internal process that uses the majority of the run times. The sub models shown here are:

- **Elapsed Total** = total elapsed time;
- **CPU Total** = total CPU time;
- **Transport** = advective transport calculations;
- **Vert Diff** = vertical diffusivity calculations;
- **PUV** = pressure fields for 2D solution of water surface elevation;
- **QQ** = turbulence intensity;
- **UVW** = internal mode momentum calculations;
- **EXP** = explicit terms in momentum calculations;
- **T/B SH** = top and bottom shear calculations;
- **SSED** = sediment transport calculations;
- **V&D** = horizontal and vertical diffusivity calculations;
- **Heat** = heat transfer calculations.

5.1.3.1 *Results Independent of Number of Threads*

OMP in EFDC+ has been implemented in such a way as to ensure that the models produce exactly the same results no matter how many threads are used. To check this a model can be run with different sub-models activated. This model is run with a range of threads from 1 (referred to as "OMP1") to 16 threads (OMP16). Using the *Model Comparison* feature the model results can be compared. Using the *Model Comparison* to subtract the 2D results from one model from a base model for the same time snapshot produces a 2D view of the differences between models. For the OMP comparisons, the differences should be equal to 0.0.

Plots from an example that has full 3D hydrodynamics, sediments and water quality activated are shown in Figure 5-4. This model was run five days for OMP1 and OMP2. The two models were then compared and plots below generated. A plot is shown of the model results (left) and the comparison for salinity (right). As can be seen, the model differences are equal to zero, showing that the number of threads does not impact model results.

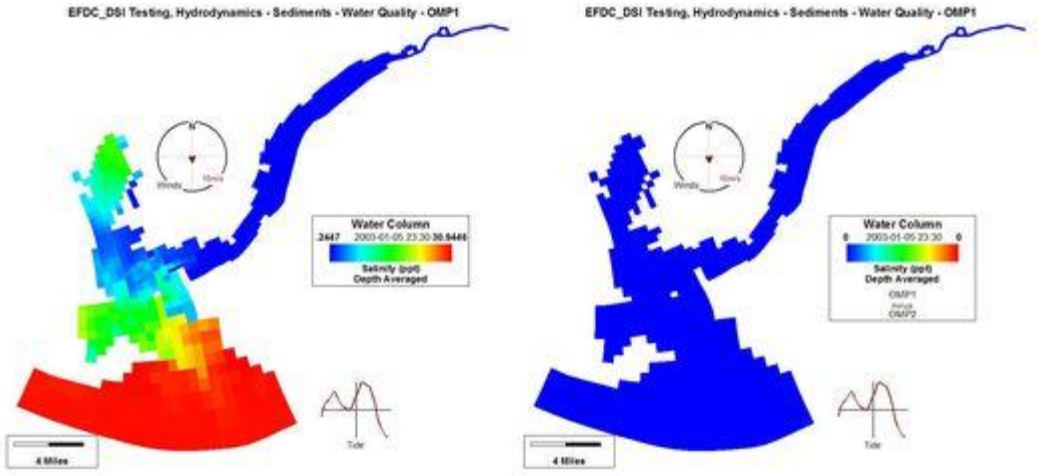


Figure 5-5 Salinity results (left) and comparison of OMP1 minus OMP2 (right).

5.1.4. Model Timing

Model Timing options may also be set by the user with the *Run Options* form as shown in Figure 5-6. These options are also available within the *Timing* tab further described in Section 6.5.

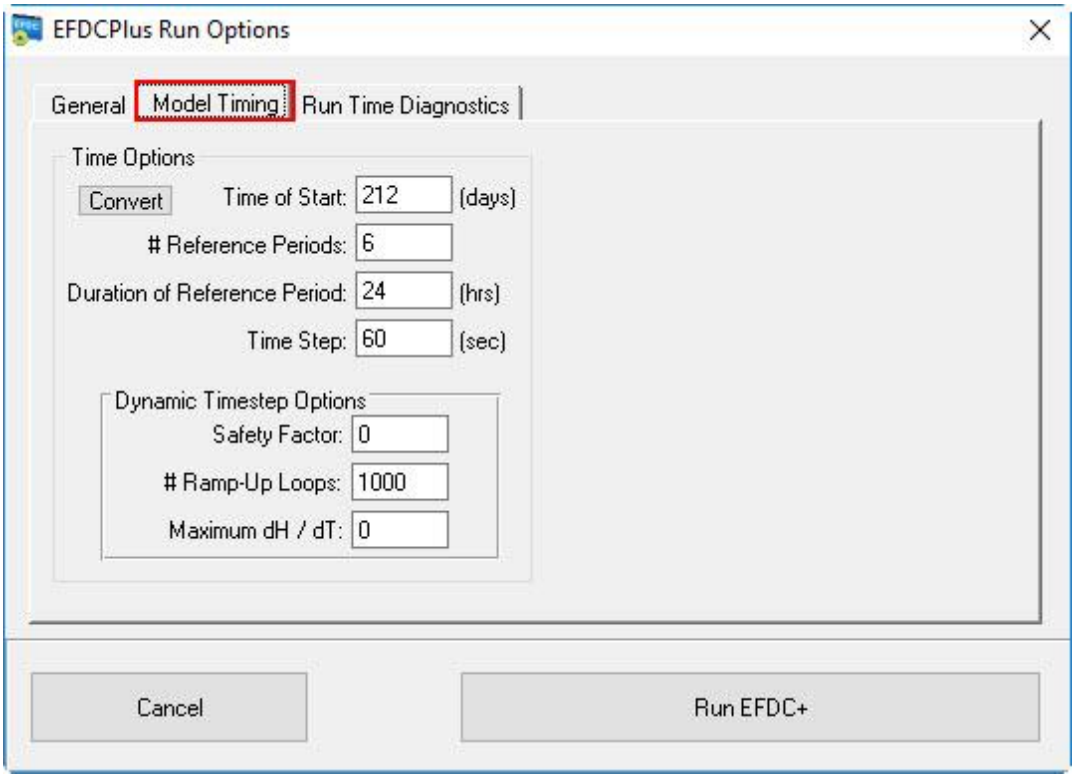


Figure 5-6 EFDCPlus Model Run Option Form: Model Timing.

5.1.5. Run Time Diagnostics and Mass Balance Reporting Options

Figure 5-7 shows the *Run Logging Options* frame with various check box options to aid error diagnosis. Checking the *Enable Diagnostics* causes EFDC to produce many extra output files but will add to runtime. The diagnostic files depend on the EFDC options, and the source code needs to be consulted to determine the content of the diagnostic files. The diagnostic files can usually be turned off after initial model testing. Other options include the ability to write Negative Depth and Courant diagnostics; and to write the optional EE_ARRAYS.OUT file, which contains data from the arrays in the EEXPOUT.FOR subroutine.

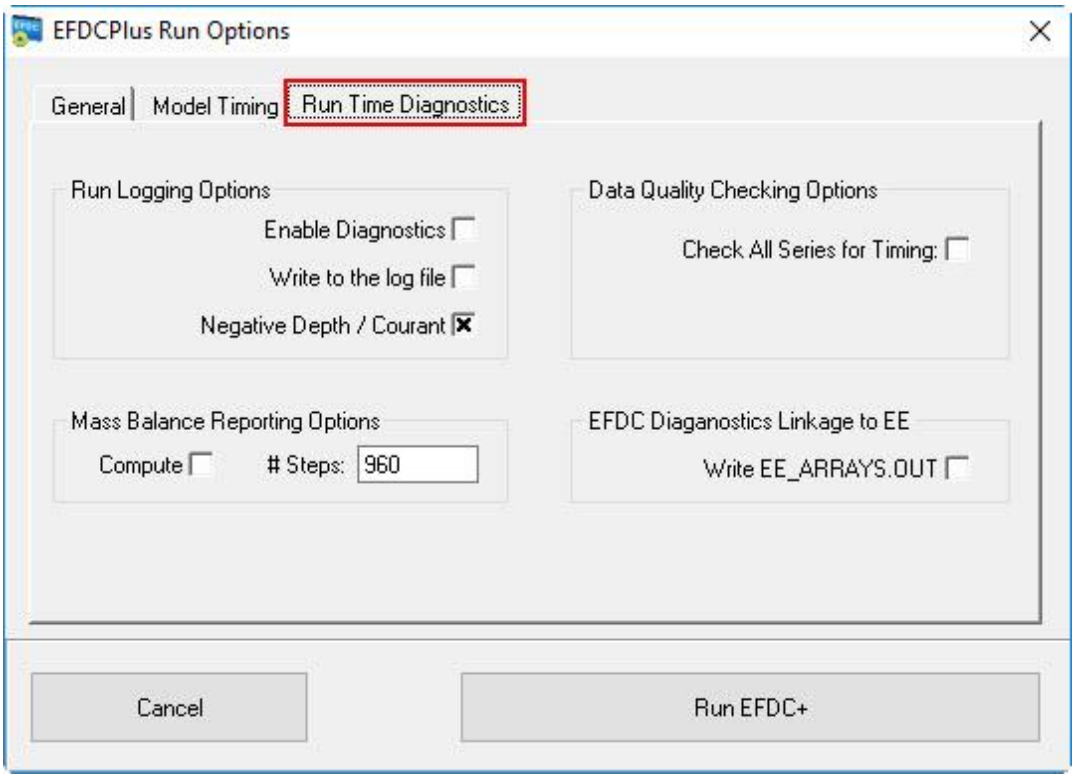


Figure 5-7 EFDCPlus Model Run Option Form: Run Time Diagnostics.

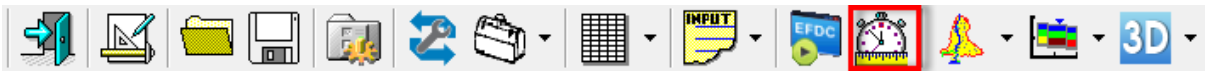
The *Mass Balance Reporting Options* frame enables mass balance checking and reporting for EFDC. The number of steps relates to the number of time steps accumulated between reporting. Consistent with the fixed naming convention, the mass balance file written by EFDC is BAL.OUT for the 3 time level solution and BAL2T.OUT for the 2 time level solution. The user must refer to the FORTRAN code to interpret these outputs.

In the *Data Quality Checking Options* frame, the *Check All Series for Timing* frame will run a check on the EFDC input files to ensure that all of the boundary condition time series files start and end times fit within the model run time, even if they are not used by a boundary condition. EFDC_Explorer ALWAYS checks for boundary condition timing for time series used by the model. However, if the user is running the model in DEBUG model (e.g. VS2010 debug mode), the model will report an out of array bounds error for any time series that starts late or ends too soon. This option helps the user either fix the error or delete the series, since it is not being used.

5.1.6. EFDCPlus Runtime Pause/Resume

The EFDCPlus model allows the user to *pause* and *resume* the EFDC model during a run by pressing any key when the model's DOS window has focus. This is often used to check the model performance during a run. The user can pause the model, load the results, perform any level of review including calibration statistics and plots, and then resume the model. This can be repeated as many times as desired during an EFDCPlus run. Note that pressing the same key that was used to pause the run will exit the run. Pressing any other key will resume the model run.

5.2 Run Times



The *Show Runtime Output Summary* (Clock) button provides a summary of the project's run times in hours. As EFDC runs it keeps track of its run time, and when the run finishes it writes the information to the file TIME.LOG. When this button is pressed, EFDC_Explorer reads the TIME.LOG file and provides a summary of the information. Figure 5-8 shows an example timing output. This function is operational only for the EFDCPlus version. The summary may be copied for pasting into another application with the *Clipboard* button.

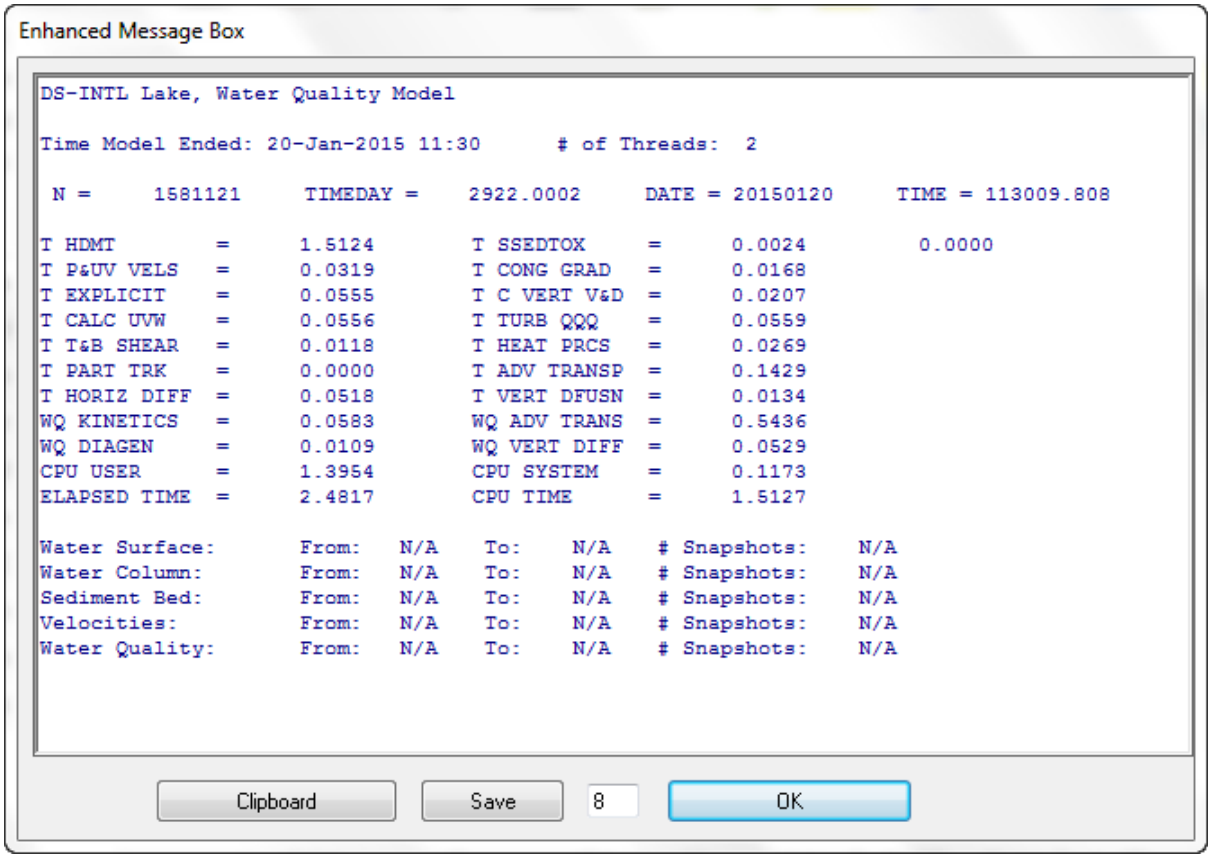


Figure 5-8 Example of the model run times.

6 Pre-Processor Form Operations

In the pre-processor section of the main form, EFDC_Explorer provides a simple user interface to most of the commonly used options in EFDC that are stored in EFDC.INP. Some parameters and settings are adjustable directly from this form while some others, usually some specific subgroup, are available from buttons located on the main form.

Many of the input boxes have tips/values that pop up as the user holds the mouse pointer over the input box. In addition, many of the input boxes have internal range checks, but as they are so broad as to cover a large range of applications they should not be relied upon in any way to validate user inputs. For further user help, F1 will display the EE User Manual linked to the currently displayed form. F2 will generally display the short-cut keys associated with the particular function being viewed at that time.

The *ViewPlan* function also provides some pre-processing features and functions that will be described later in Section 8.

6.1 Map Tab

The *Map* tab is shown in Figure 6-1 and provides the user with basic model grid information such as width and height of the grid as well as a simple display. The user may select from grid, bathymetry and depth display. For display and editing of the detailed grid and most pre-processing related to the model the user should use the *ViewPlan* mode (available by RMC on the Map view).

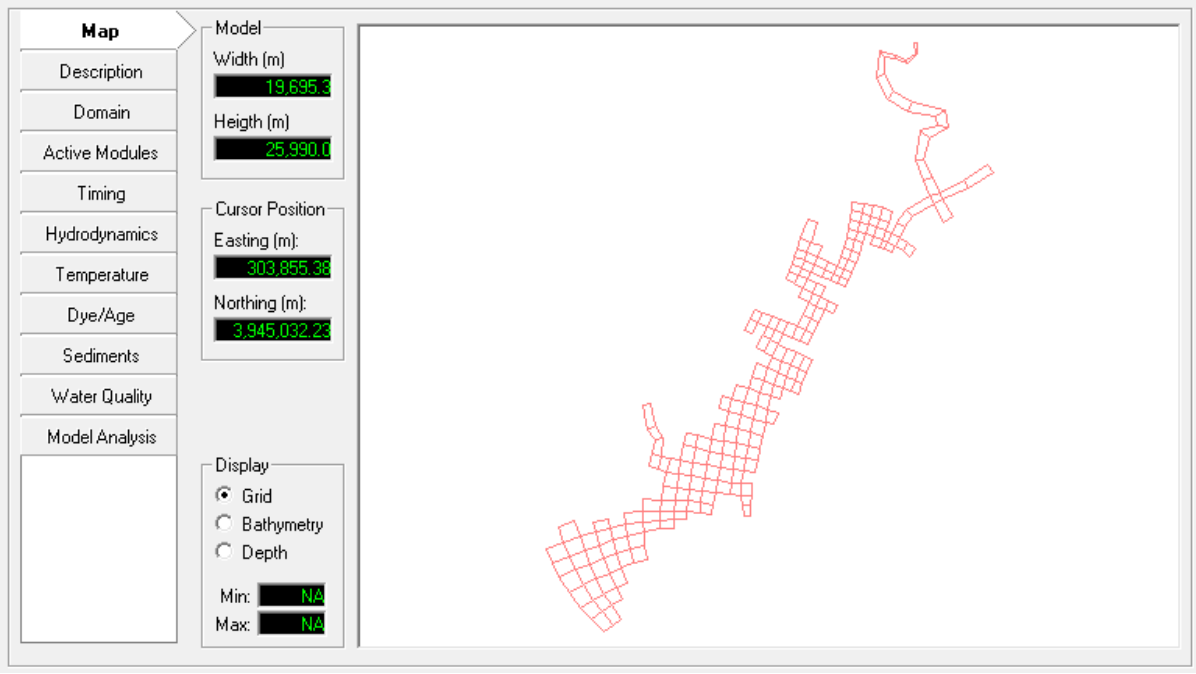


Figure 6-1 Map Tab

The user can drag and drop an EFDC project to the map the same way they drag and drop a folder to the yellow *Directory* bar on the main form.

6.2 Description Tab

The *Description* tab is shown in Figure 6-2. Here the *Project ID* and the *Run Title* are optional labels that will be displayed on plots and tables. Many of the EFDC input files will be stamped with the Project ID and Run Title and the date/time of the file creation.

The *Run Log/Notes* text box provides a readily accessible free form notepad to record changes and notes concerning each run. If used, this can provide a complete run history during a calibration process. For each run, it is recommended to note which prior run you are using for the current run. For example "Run012 Based on Run011" provides a clear run path. After that entry the user should then note any changes made to the current run. These data are stored in an ASCII text file in each run directory called "EFDC_LOG.EE".

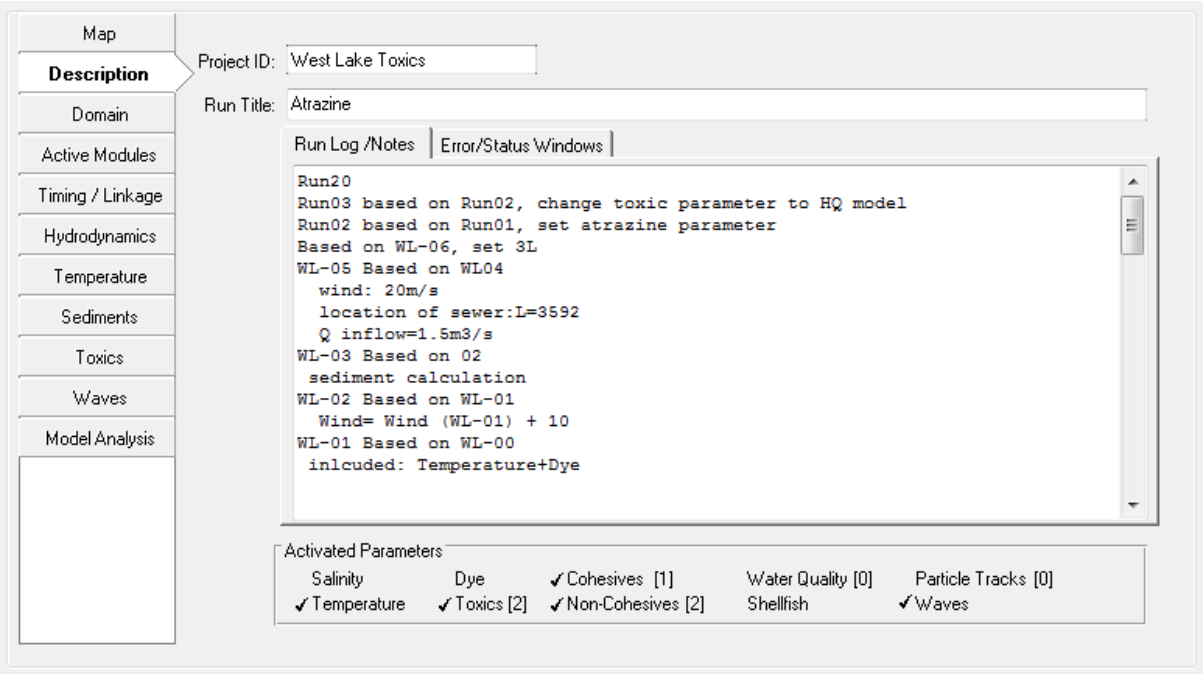


Figure 6-2 Description Tab: Run/Log Notes

The *Activated Parameters* form indicates with a check mark which parameters have been activated in the current model. This may be changed in the *Active Modules* tab (Section 6.4).

The *Error/Status Windows* features is new with EE7.1. This function has been included for the purpose of recording any error or warning messages found by EE when loading a model and it is shown on the Main EE Form in Figure 6-3. Any error or warning messages are displayed in red.

The *Show Dialog* box is off by default. If *Show Dialog* is on it serves the function of recording each error or warning message as a comprehensive list which is then displayed in its entirety each time a new warning is given. If the *Show Dialog* is off the typical error message is shown instead of the full list of errors.

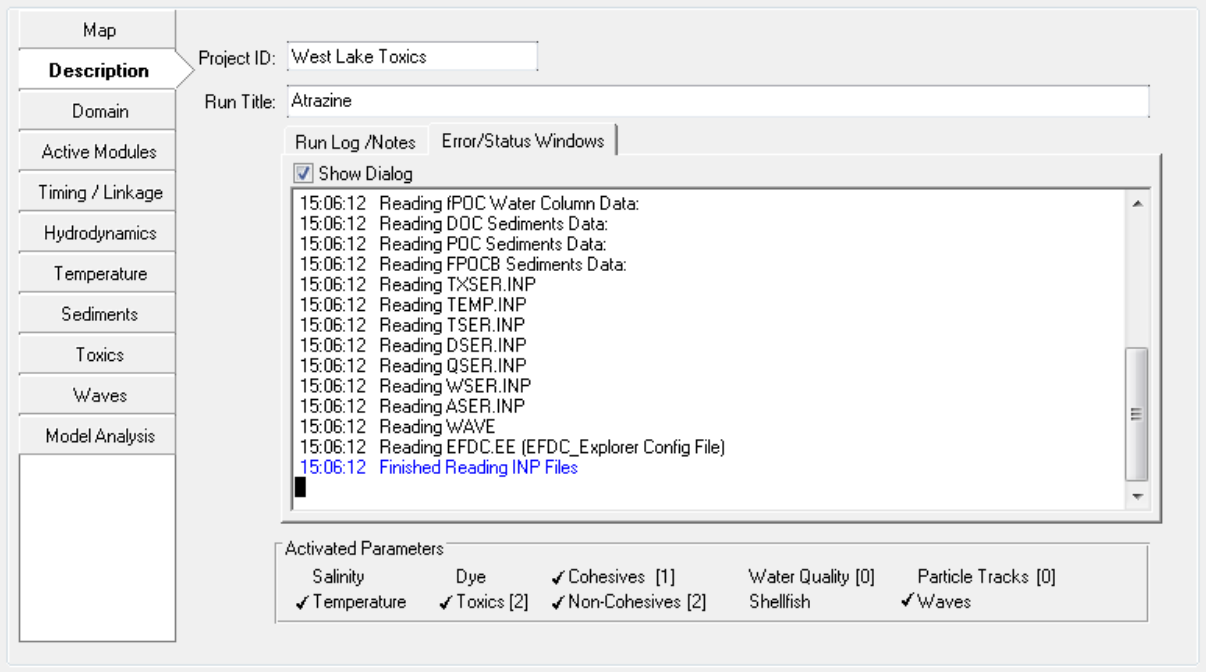


Figure 6-3 Description tab: Error/Status Window.

6.3 Domain Tab

The *Domain* tab shown in Figure 6-4 contains three sub-tabs: *Grid*; *Initial Conditions and Bottom Roughness*; *Boundary Conditions* and *MHK Devices* (Marine and Hydrokinetic devices), each of which is described below.

6.3.1. Grid Tab

The *Grid* tab provides the options by which the user may set the vertical layering system as well as modify the horizontal grid as shown in Figure 6-4. The Sigma Zed (SGZ) vertical layering option is implemented with EFDC+ alongside the previous option of Standard Sigma grid (SIG). A dropdown option for the *Water Layers* is provided and each option described below.

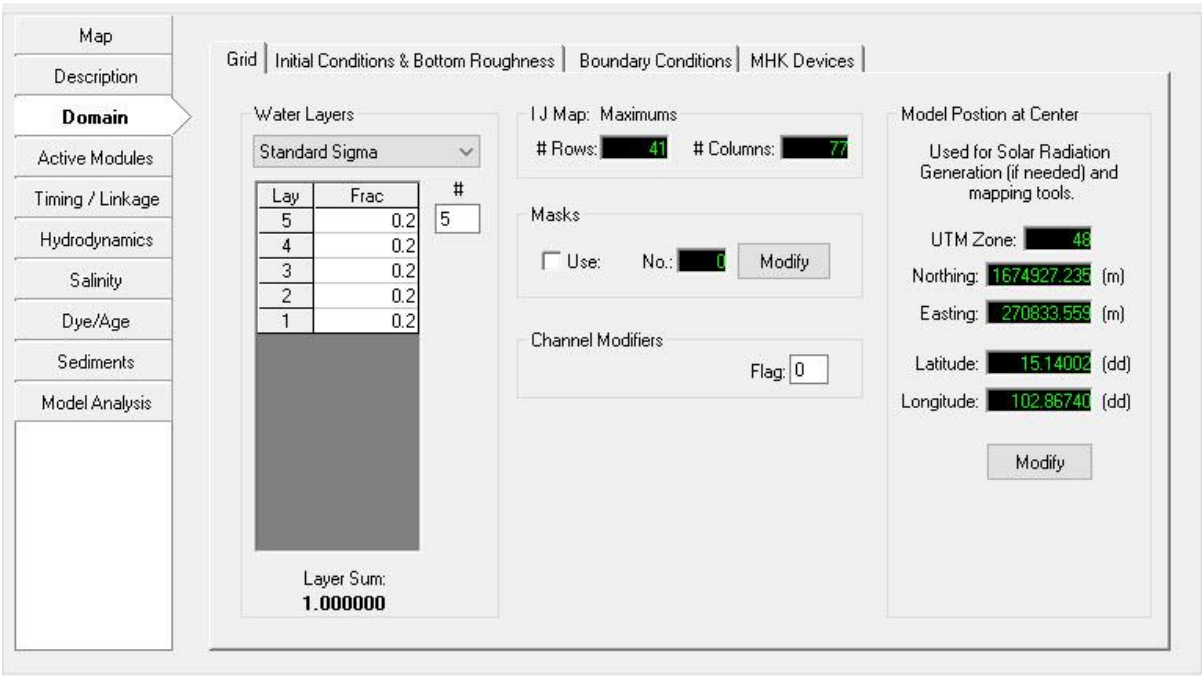


Figure 6-4 Domain Tab: Grid

6.3.1.1 *Standard Sigma Stretch Grid*

The *Standard Sigma* option for *Water Layers* is the previous option provided in all versions of EFDC+. When this is selected the SGZ options are not displayed.

The table in the *Water Layers* frame shows the primary water layer settings (KC = number of water layers) and the distribution of the relative layer thicknesses.

The relative thicknesses must add to 1 (or very close). EFDC_Explorer checks this for the user. EFDC_Explorer will automatically divide the water layers into equal fractions based on the number of layers specified by the user. Note that layer 1 is the bottom most layer and that the highest layer number is the surface layer (i.e. KC).

A feature here that is very useful during preliminary testing of a model application is the rapid adjustment of layering simply by changing the number of layers. EFDC_Explorer simply reallocates the layering based on the new KC and then adjusts all the boundary conditions to reflect this change. Use this feature with care, but it has been used on many applications and found to work well.

The *I J Map*: *Maximums* frame display the maximum number of rows and columns in the I J map.

6.3.1.2 *Sigma Zed Vertical Layering*

EFDC_Explorer8.1 and later releases implement DSI's improved version of the EFDC code, the EFDCPlus_SGZ. DSI has developed this version to deal with pressure gradient errors that occur in models that have steep changes in bed elevation. This new version of the code is called the Sigma Zed code, or EFDC_SGZ. This version contrasts with the conventional sigma stretch version (SIG) of the EFDC code in terms of vertical layering in the model. SIG uses a sigma coordinate transformation in the vertical direction and uses the same number of layers for all cells in the domain. To accommodate the varying depths over the model domain in the SIG approach, the thickness of the layers vary from cell to cell but the number of layers and fraction of depth for each layer are constant. SIG approach introduces a well-known error in the density gradient terms, otherwise known as the pressure gradient error (Mellor, et. al. 1994). These errors are most pronounced in regions with steeply varying bathymetry.

The new vertical layering approach (SGZ approach) developed and applied to the EFDC model is computationally efficient and significantly reduces pressure gradient errors. In the EFDC_SGZ model, the vertical layering scheme has been modified to allow for the number of layers to vary over the model domain. Each cell can use a different number of layers, though the number of layers for each cell is constant in time. As with the Sigma Stretch approach, the thickness of each layer varies in time to accommodate the time varying depths. The z coordinate system varies for each cell face, matching the number of active layers to the adjacent cells. The new version is computationally more efficient than a similarly configured Sigma Stretch grid, thus making models with 20 to 50 layers or more practical for typical projects.

A comparison of SIG and SGZ grids is shown in Figure 6-5.

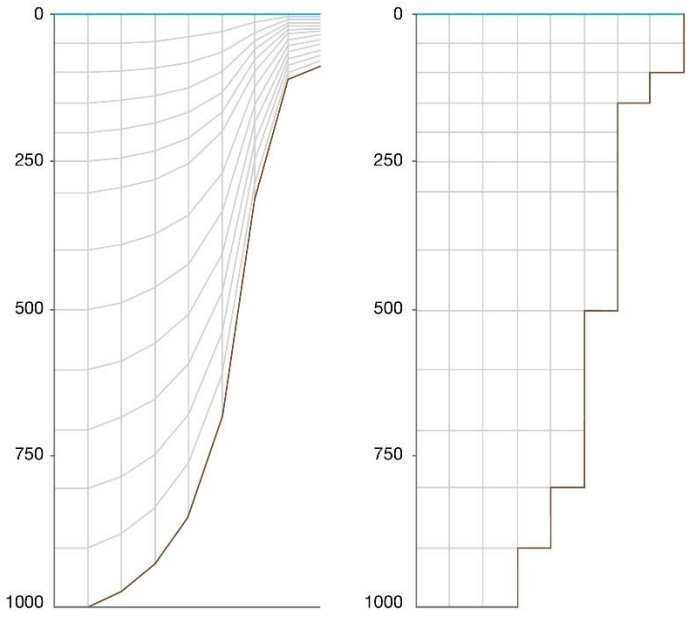


Figure 6-5 Vertical grids for Sigma stretch (left) and Sigma Z (right).

The model grid specification as SIG or SGZ can be made in the main form of EE from the *Domain tab* | *Grid sub-tab* as shown in Figure 6-6 The first option *Standard Sigma* is the conventional gridding scheme of EFDC model.

The two options for implementing SGZ option in EE8 are a) Specified Bottom Layer and b) Uniform Layer Thickness. The second two options i.e., *SGZ Specified Bottom Layer* and *SGZ Uniform Layer Thickness* are both SGZ model options with the following attributes.

(1) Specified Bottom Layer:

In *Specified bottom layer*, the layer thickness of the adjacent cells in the vertical do not align in the same plane. Maximum depth in the model is first calculated and the maximum layer thicknesses are determined based on maximum depth. Zonation can be applied using this option. This is the robust option and can be applied on any type of systems such as stratified systems, systems with rapidly changing bed elevation.

(2) Uniform Layer Thickness:

Layer thickness of the cell is mostly of the same size as the neighboring cell. The bottom layer thickness can be from 20-120% of the overlying layer. The maximum thickness is first calculated throughout the domain. Thickness at each layer for the cell with maximum depth is then calculated by multiplying DZCK with maximum thickness. Zonation cannot be applied in *Uniform Layer Thickness* option. *Uniform Layer Thickness* is suitable is quiescent systems where there are very little external flow influence.

When computing buoyancy integrals in the model, there are differences between layer thickness and difference between elevation. Using the uniform layer thickness, the differences between layer thicknesses is very small except for the regions with minimum sigma layers.

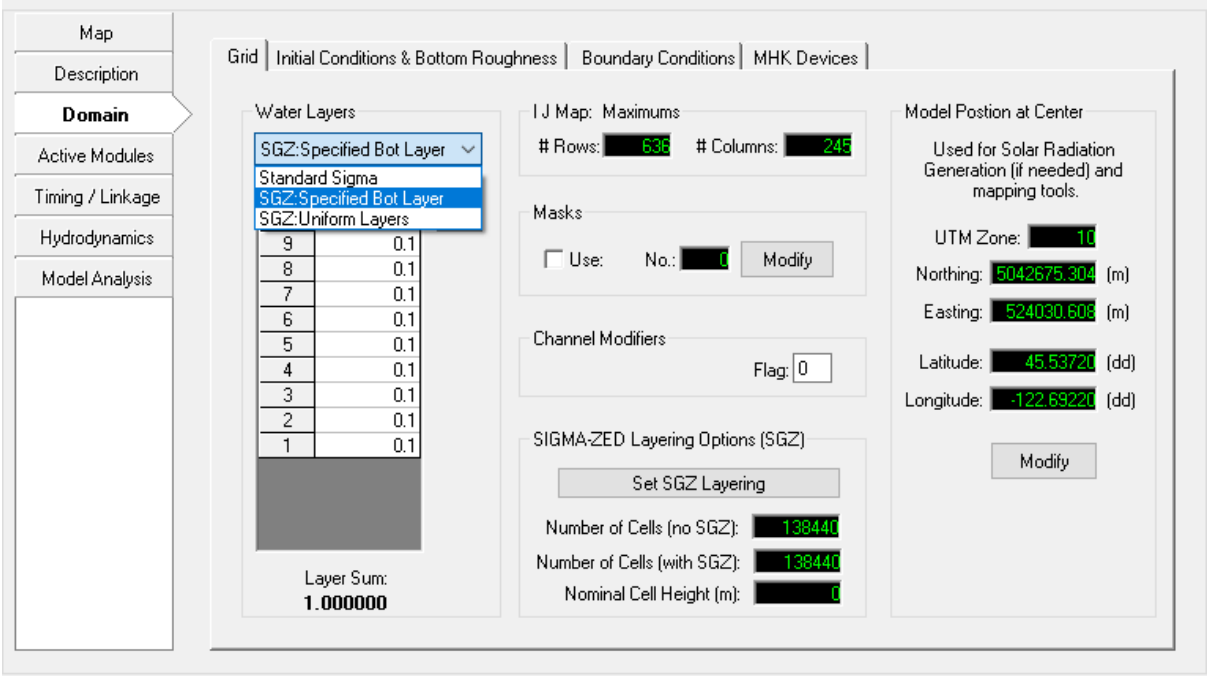


Figure 6-6 Domain Tab: Layering Options (SIG & SGZ)

The *SIGMA-ZED Layering Options* form displays user information on the number of cells with and without Sigma-Zed and the nominal cell height. When the user selects the Set SGZ Layering button the form shown Figure 6-7 provides the user with the option of specifying maximum and minimum layers to be used in the model. Sigma-Zed requires a minimum of one layers for all model simulations. Being able have one vertical layer is useful in situations

when there are 1-D rivers feeding into a many layered reservoir or lake. After selecting *OK,* the layers are all automatically updated to the Sigma-Zed layering.

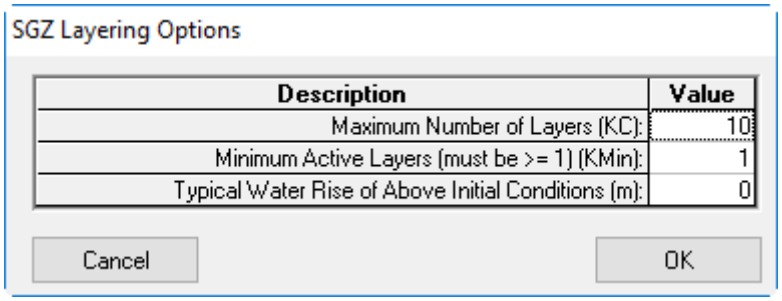


Figure 6-7 Sigma-Zed Layer Options setting form.

6.3.1.3 *Masks*

In the *Masks* frame the *Use* check box enables the user to enable masks (zero thickness flow barriers) in EFDC. The MASK.INP file will then be required by EFDC. EFDC_Explorer can generate and modify masks using the *Modify* button as shown in Figure 6-8. The *Create Masks* utility does not automatically turn on the mask computation, it just generates the masks and writes the MASK.INP file. The user must manually click the mouse cursor inside the *Use Masks* check box to use the masks generated.

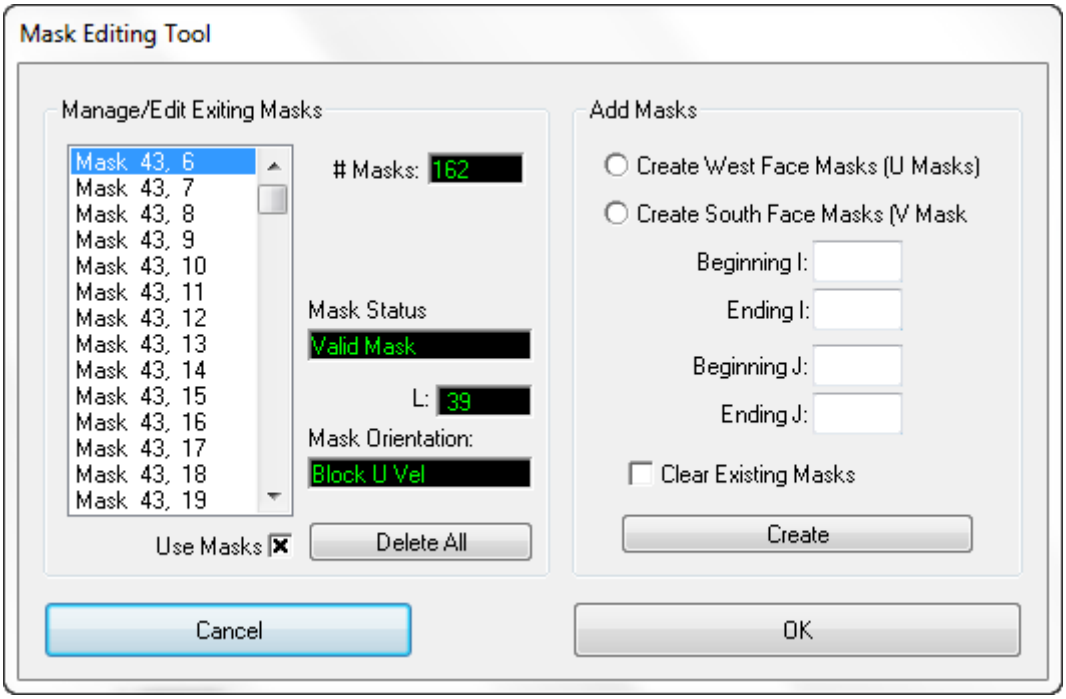


Figure 6-8 Mask Editing Tool.

Masks can be viewed, added and deleted in the *ViewPlan* function. When viewing the *Bottom Elevations,* RMC on a cell (*Enable Editing* must be checked first) and the cell properties are displayed, along with the mask options.

The *Channel Modifiers* frame allows the user to select the use of EFDC channel modifiers. These are like *pipes* that connect two cells. The *Channel Modifier Flag* option box is used to

enable the use of the channel modifiers capability of EFDC. The normal "on" option is when this flag is set to 2. If set, EFDC requires the MODCHAN.INP file to be input.

The most flexible approach to create and edit channel modifiers is to:

1. turn on this option (i.e. ISCHAN=2),
2. enter the *ViewPlan*,
3. select *Modchannel*,
4. enable edit, and
5. use the right mouse click to add, delete or modify channel modifiers (i.e. "pipes").

Care should be exercised in using channel modifiers as model instability is sometimes increased and mass balance errors can occur.

6.3.1.4 *Model Position at Center*

Latitude and longitude settings the center of model are accessed in *Domain* | *Grid* | *Model Position at Centre* as shown in Figure 6-4 above. These settings are required for EFDC+ to calculate solar radiation as well as mapping features in EE. The user should select the *Modify* button to update these values. If the user has not entered a UTM zone for the model the default will be UTM zone 10 and the Northing and Easting will be set to the model centroid. EE reads the UTM zone from C91 (netCDF settings) in the EFDC.INP and updates to the latitude and longitude are stored in C46C (atmospheric location and wind function coefficients). Northing and Easting values are calculated based on the UTM and lat/long values.

6.3.2. Initial Conditions and Bottom Roughness Tab

Figure 6-9 shows the *Initial Conditions* frame tab that contains access to the functions that assign initial conditions (IC's) that may be applied for bottom elevations, water depths, and bottom roughness. This is done through the use of the *Assign* button in the pertinent frame.

Other water column parameters, sediment transport bed, salinity and bed thermal sub-model use a common approach to interpolate data onto the model grid. This approach is discussed in the following section.

Note that each input parameter also has *Legacy* setting frames. These are ways of making global adjustments to the data sets, through addition or multiplication of all the data. These are no longer required as EFDC_Explorer provides more capability through the *Assign* button. However, they are included in the EE interface so the user can make sure how they are being used in legacy models.

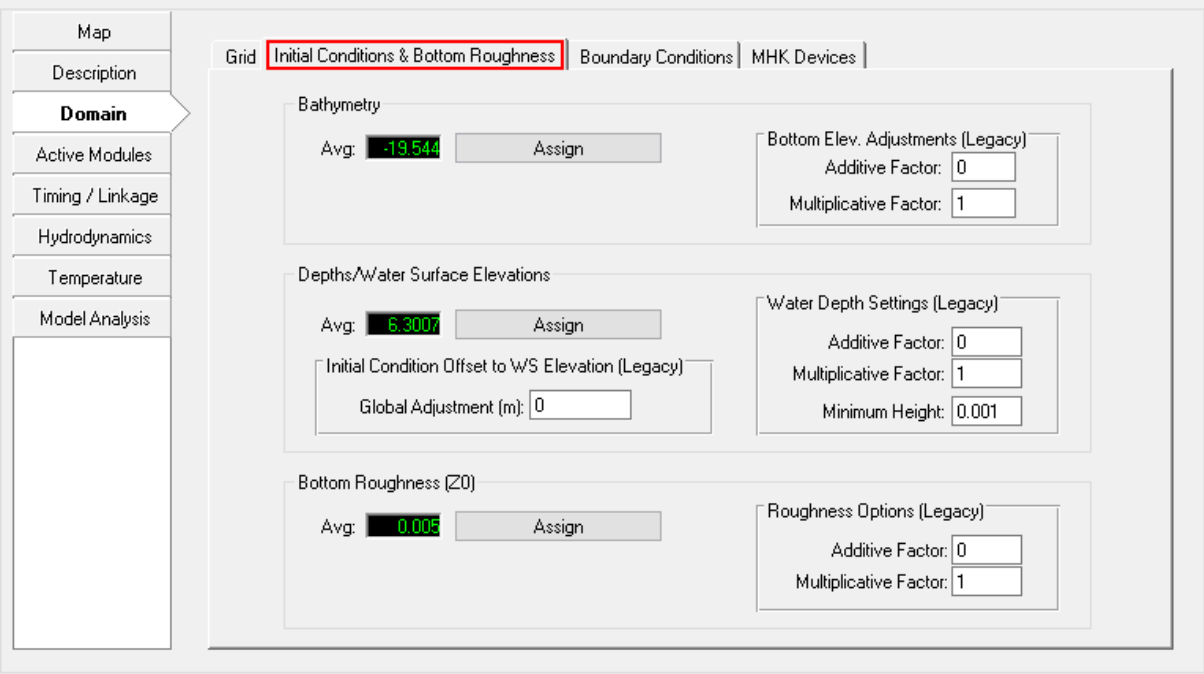


Figure 6-9 Domain Tab: Initial Conditions and Bottom Roughness

6.3.2.1 *Apply Cell Properties via Polygons*

EFDC_Explorer often needs to assign constant or spatially varying values across the entire model domain or within subsets for a variety of water column, sediment and other spatially variable parameters and concentrations. The *Initial Conditions* frames in the *Domain* tab: (Bathymetry; Depths/WS Elevation; Bottom Roughness) use the same process as the *Salinity*, *Temperature*, *Dye/Age*; *Toxics*, *Sediment* and *Water Quality* tabs, each containing an *Assign* button for: Surface Elevation, Salinity, Temperature, Dye, Bottom Elevations, Thermal Depth and Bed Temperature. Selecting any of these buttons will bring up the form *Apply Cell Properties via Polygons*. As shown in Figure 6-10. This form provides the general purpose horizontal spatial assignment function needed. However, this utility is context/parameter sensitive and displays different options and features, based on what the user requested and/or specified for data. Figure 6-10 shows an example of this utility for bottom elevation.

.

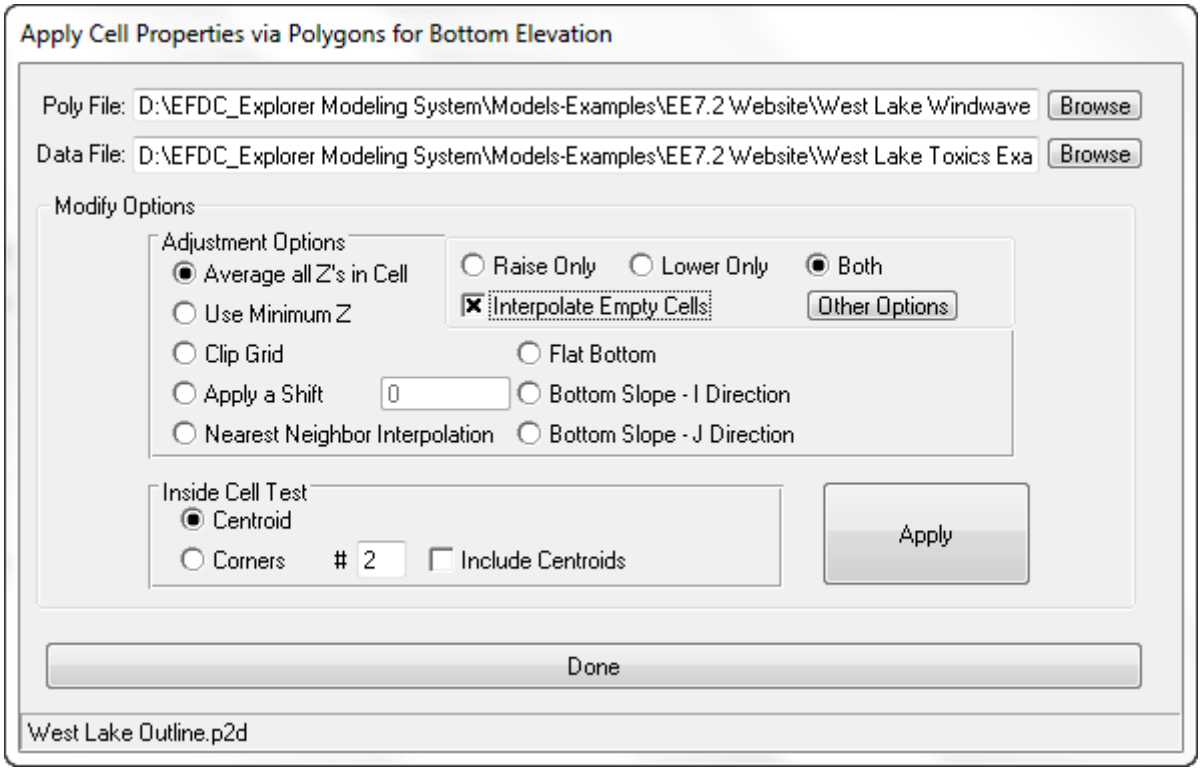


Figure 6-10 Apply Cell Properties via Polygons.

The *Poly File* field is an optional field used to specify a polygon file (in P2D or DAT formats) that contains one or more polygons. Cells that are inside the polygon(s), using the *Inside Cell Test* options, will be adjusted according to the options specified in the *Modify Options* frame. If the *Poly File* is missing, EFDC_Explorer will use all of the cells in the model.

The *XYZ File* is an optional file that identifies the measured or otherwise determined data that the user wants to assign to the EFDC cells. Generally, this file is required unless the user is simply applying a constant to the cells or is applying an operator.

Under the *Modify Options* frame various functions and features will be displayed, based on what parameter or process the user is assigning or modifying. In this case (Figure 6-10), bathymetry is being assigned. The user can select either to apply an operator to the currently defined field or interpolate the data in the *XYZ File* to the selected cells (via the Poly File). Nearest neighbor approach is used to interpolate the horizontal spatial data onto the grid. If it is for the water column, the user must specify which layer this assignment is for, one layer or all the layers.

If data is spare the user is recommended to select the *Interpolate Empty Cells* option so that bathymetry for the whole domain is generated from interpolated data. The *Other Options* or *Options* button brings up a form allowing the user to change the interpolation options. Depending on whether bathymetry or wave parameters are being set the *Option* button also allows the user to define raw data zonation options. The zonation option simply automatically divides the raw data into a specified number of X and Y zones (with overlap to handle zone edges) before applying the requested interpolation process. This significantly speeds up many applications with no loss in accuracy. Zonation is recommended if the raw data has a large number of points. EFDC_Explorer informs the user of the range in number of data points in each zone before conducting the actual assignment.

For bathymetry and wave data the user has two choices for cell assignment, spatial interpolation via nearest neighbor or averaging data into each cell. If the XYZ data is very

dense, relative to the model cell size, then the best approach will be to use cell averaging. However, if the XYZ data is not dense enough to completely assign at least one value for every applicable model cell, then the nearest neighbor interpolation option is better.

Pressing the *Apply* button causes EFDC_Explorer to perform the requested operations. You can perform a series of operations, one after the other by changing the options and pressing *Apply*. For example, you can set the water surface elevation from a data file with units in centimeters then change the *Constant* options and change the *Operator* to "/ 100" to convert the elevations to meters. Once assigning/editing the current parameter is complete, pressing the *Done* button returns to the main EFDC_Explorer form. The *Done* button does not do anything except return the user to the main EFDC_Explorer form.

(1) Example Problem

The user has generated a poly line, and now wants to set a constant bottom elevation to all the cells inside the poly line.

1. Go to the *Domain/Initial and Boundary Conditions* tab.
2. Click on the *Assign* button for bathymetry. From the bathymetry assignment form, type in or browse to the polyline file.
3. Select the *Flat Bottom* option and enter the elevation in the text box
4. Click *Apply*. If the polyline is not closed, EE will ask you if you want to treat it as open or closed. Open polylines select cells along the line. Closed polylines select all the cells whose corners/centroids are inside the polygon, based on the *Inside Cell Test* options.
5. EE will build a list of the cells and apply the specified flat bottom to those cells and report how many cells were modified.

6.3.2.2 *Set Initial Conditions – Water Column*

This section contains methods to assist the user to assign the initial conditions for the water column parameters. The check boxes next to each parameter set the flags for EFDC to read and use these initial condition fields. When an *Assign* button is pressed for a parameter (e.g. *Bathymetry*), the *Apply Cell Properties via Polygons* form is displayed with appropriate options for that parameter (e.g. Figure 6-10). Depending on the options selected on the spatial assignment form, the IC's can be initialized or modified, in one or more operations. Each water layer must be specified or the user can select the *Set for All Layers* to assign vertically uniform ICs.

The *Assign Value Using Vertical Profiles* option on the *Apply Cell Properties via Polygons* form allows the user to specify a data file with vertical profiles (data format in Appendix B) of the water quality parameter being assigned. In this case, the *XYZ File* will be the file that contains the vertical profile data. When the *Apply* button is pressed, EFDC_Explorer interpolates this data onto the water column initial conditions horizontally and vertically (all active layers).

6.3.2.3 *Initial Condition to Offsets*

The *Initial Condition Offset to WS Elevation* frame contains an input box that allows the user to apply a global offset to the Water Surface elevation that will be applied during the initialization.

6.3.2.4 *Roughness Options*

Under *Bottom Roughness*, other than the standard additive and multiplicative factors that EFDC will apply to whatever z_0's are input in the DXDY.INP file, EFDC_Explorer has several methods to set and/or modify the z_0's that will be written in the DXDY file.

Floodplain Z_0's and Channel z_0's: If the model was constructed with channel and floodplain cell ID's (i.e. 5 for channels and 7 for the floodplain in the cell map file, i.e., CELL.INP) then the user can set each one quickly using these buttons.

Polygon Set: The user can employ a user defined polygon to modify/set the z_0's. This tool may be applied as many times as needed anytime during the model construction and calibration process.

6.3.2.5 *Legacy Options*

EFDC_Explorer allows the user to make changes to older (legacy) models whose functions have been superseded by changes in the EFDC_Explorer/EFDCPlus modeling system. It is not recommended users edit these fields unless the user wants to maintain consistency with some legacy model.

6.3.3. Boundary Conditions Tab

Figure 6-11 shows an example of the *Boundary Conditions* tab that contains boundary condition information and editing features. The *Number of Boundary Groups* frame provides the user with a summary of the current assigned boundary groups. The user can also view, create and edit boundary conditions from the *ViewPlan* | *Boundary Conditions* feature (Section 8.6.5).

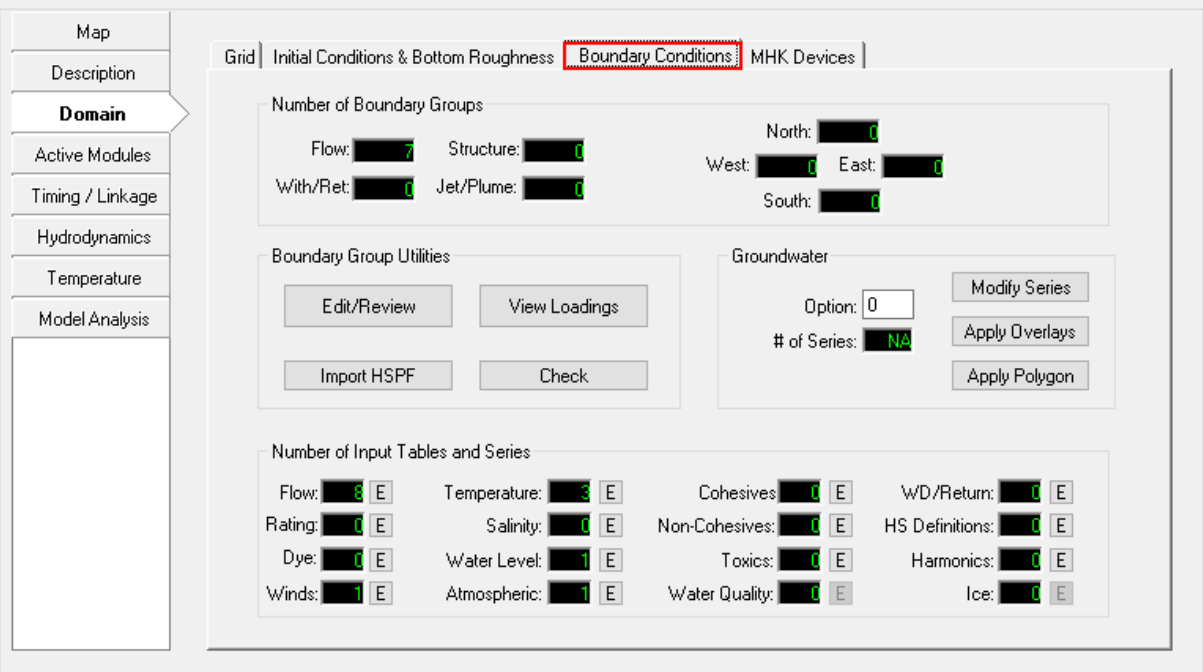


Figure 6-11 Domain Tab: Boundary Conditions.

EFDC applies the boundary conditions in a cell by cell manner. However, EFDC_Explorer takes a more physically based approach to handle boundary conditions. It groups boundary cells into logical groupings, e.g., a river inflow, a tributary or an open boundary along one face. Ideally, the user will create the group, name it something meaningful to the project, assign all

the cells included in that group to the group (this can be done manually or via polylines/polygons), and then set the boundary condition. The boundary group information is stored in the EFDC.EE project file located in the project directory. However, if there are no current groupings when EFDC_Explorer loads a project (which is the case for existing EFDC models not managed with EFDC_Explorer), it groups the existing boundary cells into groups based on the type of boundary condition and their locations. Within a group, the flow, head and pressure settings can vary from cell to cell but the water quality parameters must be the same. This limits the complete specification of each cell that EFDC provides, but gives a much more logical way to manage the boundary conditions. If the user needs to specify the water column parameters on a cell by cell basis the EFDC_Explorer user needs to manage each cell as a separate group.

The boundary condition time series available in EFDC are displayed in the *Number of Input Tables and Series* frame (see Figure 6-11). The number of currently defined tables and series are displayed alongside the button labeled “E”. A time series boundary condition editor is displayed if the user clicks the “E” button. Jet type boundary condition uses a flow series. Once the user has created a Jet type boundary conditions from *ViewPlan* | *Boundary Conditions*, the GUI is accessed from the *Edit/Review* button in *Boundary Group Utilities* .

6.3.3.1 *Boundary Group Utilities*

The *Boundary Group Utilities* contains *Edit/Review, View Loadings, Import HSPF, and Check* shown in Figure 1.

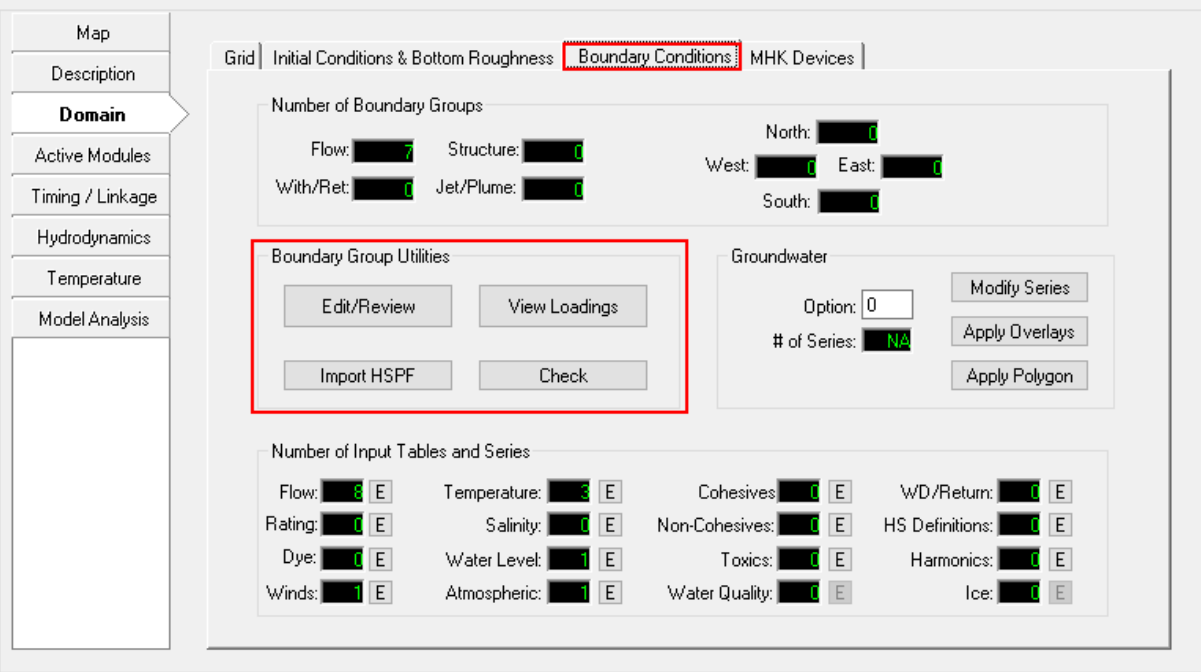


Figure 6-12 Boundary Group Utilities.

(1) Edit/Review Boundary Conditions

The *Edit/Review* button provides access to the general *Boundary Conditions Definitions/Groups* form shown in Figure 6-13. This form provides a listing of all the defined boundary groups by group ID. The list of boundary groups can be sorted by ID or listed in the order they are defined in the EFDC.INP file. In the top frame, *Number of Boundary Groups*, the number of the currently defined boundary groups, by type, are displayed.

A method to quickly define one or more boundary groups (where boundary condition details are to be added later) is available using the *Batch Define* button. This approach uses a P2D file with one or more polylines/polygons in a single file. If a polyline is open (i.e. the 1st and last point are not the same) EFDC_Explorer will select all of the cells the line intersects as one group. If the polyline is closed (i.e. the 1st and last point *are* the same) EFDC_Explorer will select all of the cells whose centroids are inside the polygon as one group.

The *Momentum Corrector for BC Cells Located on the Model Edge* is used to set the appropriate model response for the EFDCPlus or EFDC_GVC models. It should be 0 for the former and 1 for the latter, but may set to some number between 0 and 1.

As the user selects a boundary group from the list, summary information about the boundary group and its linkages to various boundary forcings is displayed. When using the time series plotting function from this tool, the model simulation start and end times define the minimum and maximum date range. If the time series is longer, the entire time series can be viewed from the Time Series editor.

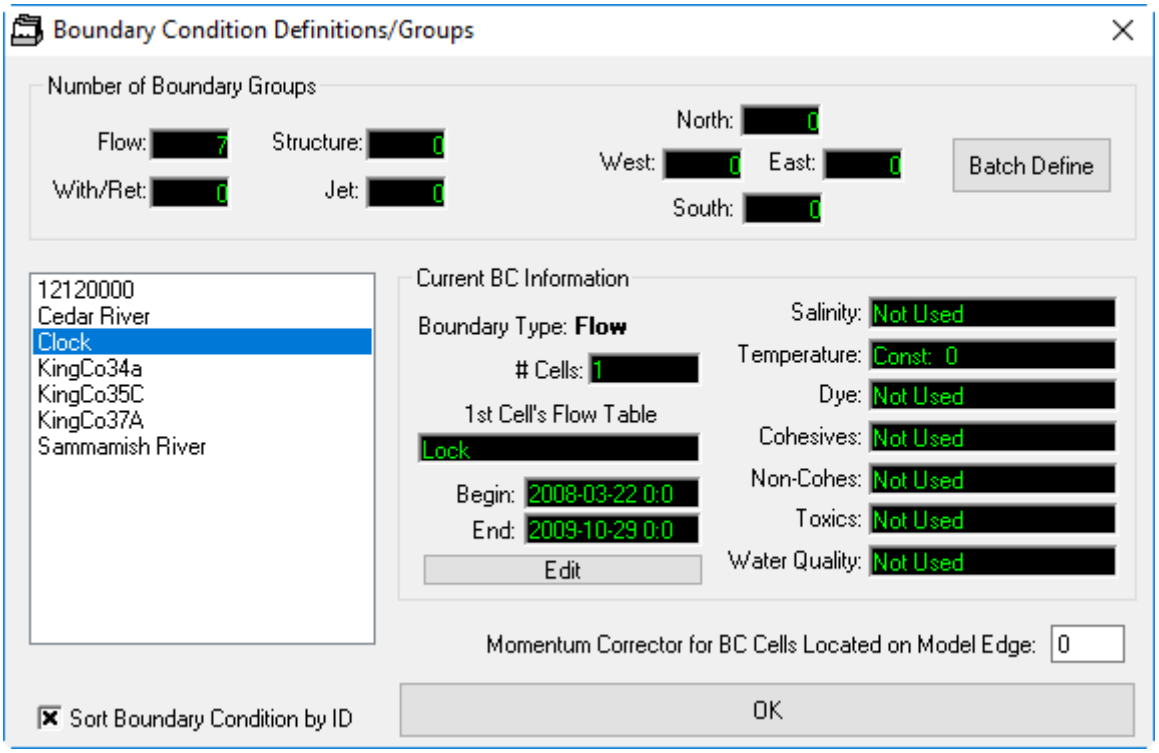


Figure 6-13 Boundary Condition Definitions/Groups.

To edit a boundary group the user can double click on the group ID or press the *Edit* button. When the user RMCs on a group ID a menu is displayed as shown in Figure 6-14.

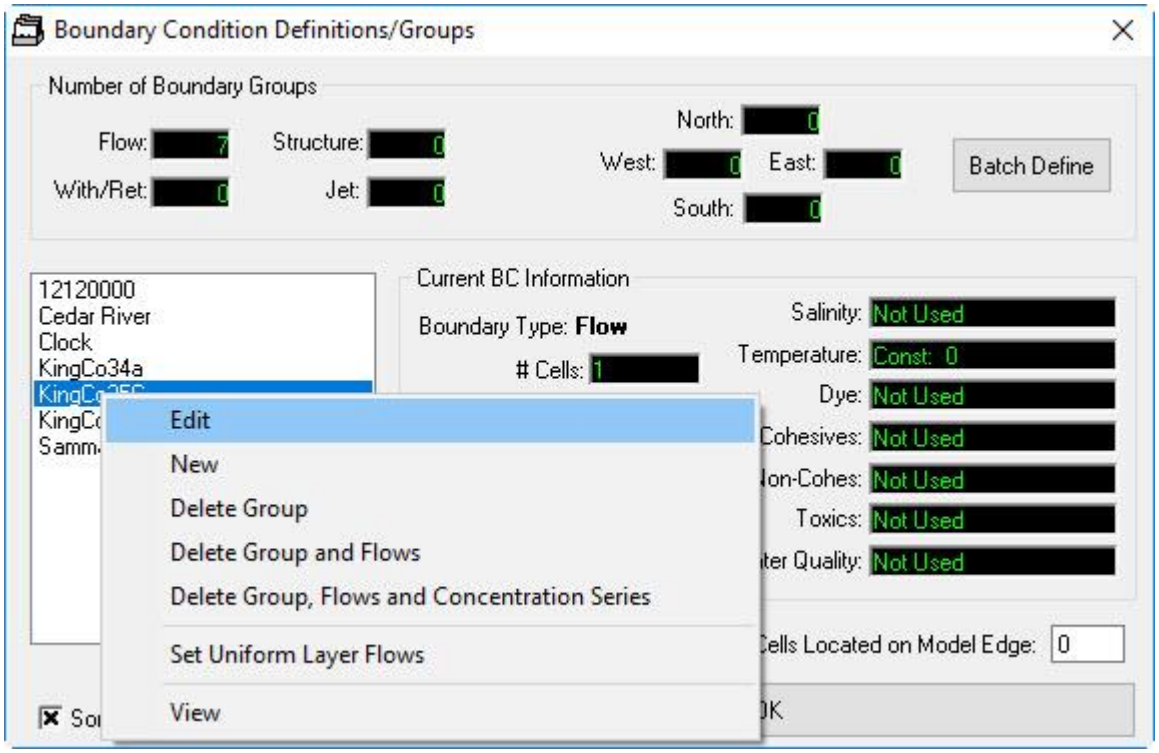


Figure 6-14 Options when RMC on a boundary group ID.

The options displayed in Figure 6-14. Include the following:

Edit	Modify the select group BC
New	Add a new boundary group (see below for details).
Delete Group	Delete the currently selected group only
Delete Group and Flows	Delete the currently selected group and its flow series
Delete Group, Flows and Concentration Series	Delete the currently selected group with its all associated series
Set Uniform Layer Flows	Uniform flows by layer for the selected Boundary group (see below for details)
View	Plot the time series of the primary forcing

When the user selects *New* from the a drop-down menu, a form will appear as shown in Figure 6-15. Here the user should enter a boundary condition ID and click *OK*. A form to select the of boundary type is then displayed. There are eight boundary types as shown in Figure 6-16. The user should enter the number corresponding to boundary type required e.g. if the boundary type is flow, just enter 1, then click *OK*.

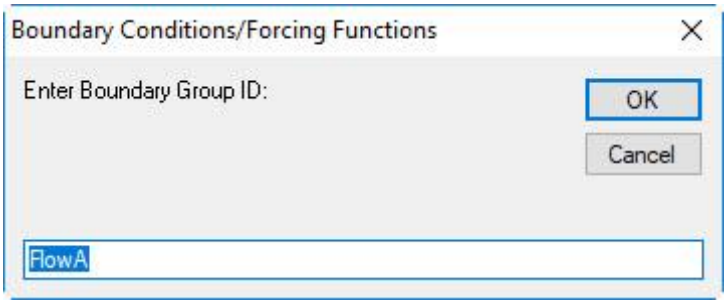


Figure 6-15 Add a new Group BC.

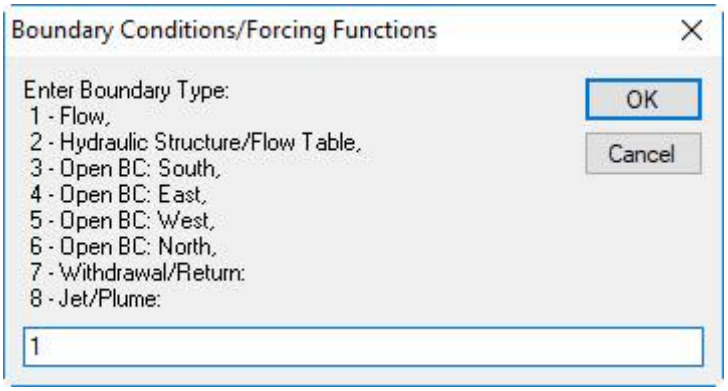


Figure 6-16 Assign Boundary Type to a Boundary Group.

The *Set Uniform Layers* option is required to reduce files size when editing Sigma-Zed models. When using EFDC+ model with Sigma-Zed the user should be aware that changing the number of cells will impact the QSER.INP file. When the number of cells is changed EE will recalculate all the layers for flow and constituents. If this is not done carefully it can lead to large file sizes as outlined below.

EFDC+ supports two kinds of QSER.INP files as set in the header file. If the flag for the input type (InType.EQ.1) is 1 then EE uses layer splits. This means it will read the depth weighting for each layer and then a single value for the flow series. However, if the flag is 0 then the time stamp and flow series will be read for each individual layer. This second option can lead to large files, sometimes many gigabytes.

In many case the user will have built their project and then need make a change to the bathymetry. For example, a model initially has 4 cells vertically, a change to the bathymetry by the user removes one cell from the bottom and so model requires an update to the number of cells vertically. EE will automatically reset the layering for the Sigma-Zed sub-model. EE will move all the layers so that the one bottom cell is no longer activated and the others are moved up into the top 3 cells. However, EE will see this as a user defined layer split and rebuild the QSER.INP file. This will increase the file size each time performed and the file size can get very large.

Selecting the *Set Uniform Layer Flows* will total and then recalculate the layer split so that it is uniform by layer. This will reduce space and allow the model to load more quickly for Sigma-Zed models.

(2) Import HSPF Data

The *Import HSPF* button provides access to the import function for the Hydrologic Simulation Program in Fortran (HSPF) (Bicknell et al., 2001). This is a hydrologic watershed modeling tool that is commonly used to predict flows and some water quality parameters. If this tool is used to predict the flows in a basin/watershed in which the EFDC model is being applied, then these results can be imported as boundary conditions for EFDC.

In addition to HSPF results, the *Import HSPF* tool can import any time series data whose flow and or water quality parameters are contained in columns (i.e. Excel™ type csv or tab delimited data). The dates can be Julian or Gregorian calendar dates. If Julian dates, the Base Date used for the imported time series should be the same as that used in the EFDC project.

The import tool, shown in Figure 6-17, works by linking the time series/HSPF files to specific boundary groups. The only groups that can have HSPF data linked to them are flow boundaries. Each group can have its own HSPF file and import options. Flow, temperature, solids and water quality are optionally selected in the Import Parameters Frame.

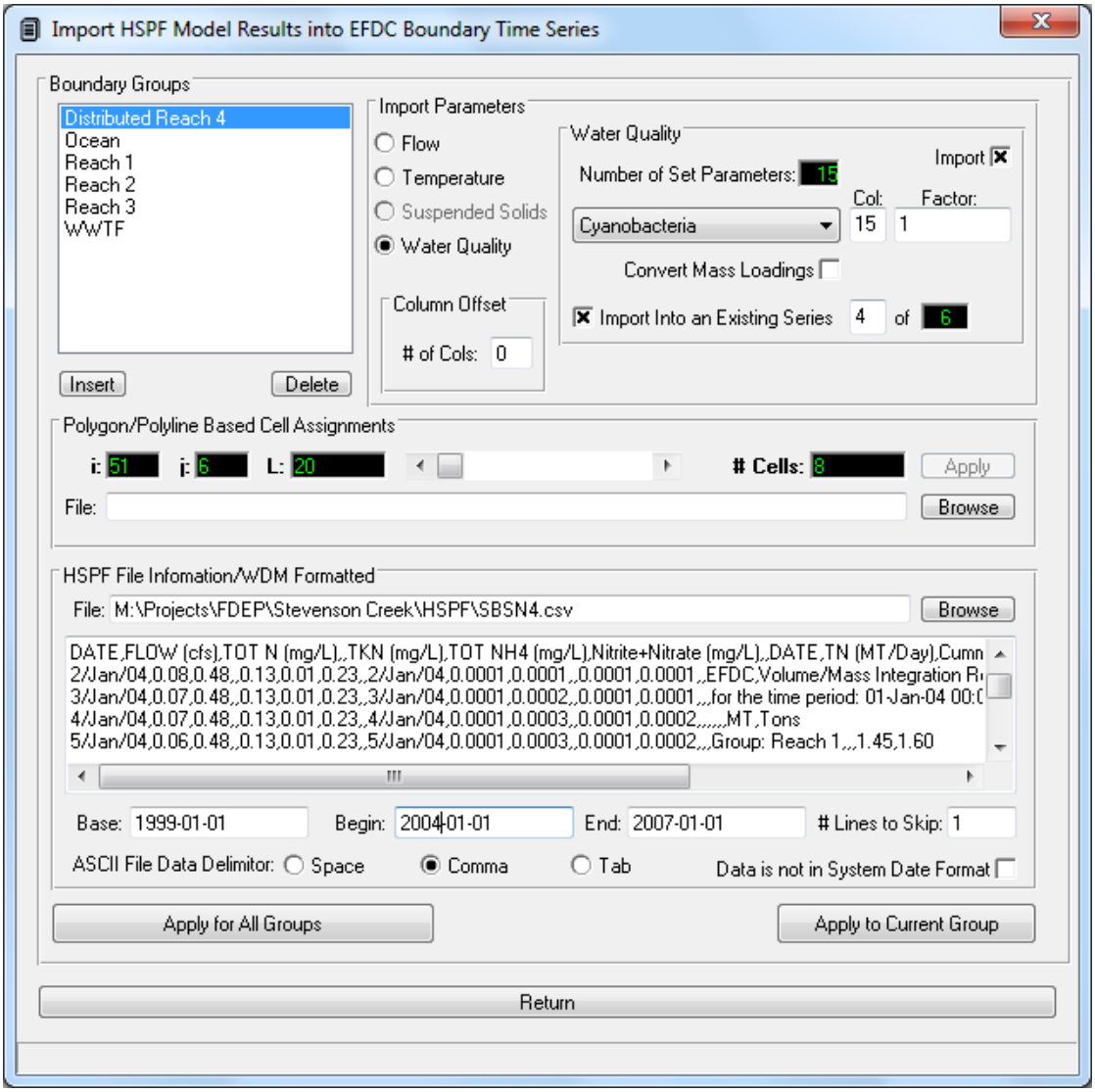


Figure 6-17 HSPF model results import utility.

To view each parameter's options, select the corresponding radial option in the *Import Parameters* frame. When the parameter is selected, the options frame is shown to allow the user to:

- Select whether to import that parameter for the current group using the *Import* check box. To activate all the groups for the current parameter, press Ctrl-A.
- Set the column which contains the data. Column 1 is the 1st column after the column offset.
- Define a conversion factor to convert whatever units are in the file to the appropriate units for EFDC (i.e. m^3/s for flow, °C for temperature and mg/l for solids, mg/l for water quality and MPN/100 ml for fecal Coliform). In addition, the conversion factor can be used to split a single number into its sub-components. For example, total organic nitrogen from and HSPF run can be split into the dissolved and particular (refractory and labile) nitrogen components.
- Select which time series number to import the series into. The first time the user imports an HSPF file, the *Import into Existing* checkbox should not be checked. EFDC_Explorer will import the series and add a new series ID and assign it to the appropriate boundary group. However, if the HSPF results are being re-imported or updated based on new HSPF results then the user will want to import the data into an

existing EFDC series. EFDC_Explorer uses the current boundary information to assign the series number to import to, but the user can change it if needed. When the *Import to Existing* checkbox has focus, pressing Ctrl-A will cause the all of the boundary groups to use the same option as the current group. Importing into an existing series does not overwrite the entire series, rather it inserts into the specified series between the *Begin* and *End* dates.

- The # *of Cols* in the *Column Offset* frame provides a global column offset. This may be needed if there is an "E" in column 1 (some HSPF export series use this) or some other information in columns before the date column. EFDC_Explorer uses the date/time columns as the 1st columns from the left, after accounting for the column offset.

The solids are a special case. Along the rows the solids classes for HSPF are displayed. In the 1st grid column ("Col") the user specifies the data column number that contains the clay, silt and sand. Then the user assigns weights or factors to each grain size class modeled in EFDC to the data columns. In this manner the user may combine HSPF sizes into whatever EFDC class they desire. If the class weights totaled for each row add up to 1, then the same mass that is predicted by HSPF will be input to EFDC. It is possible to assign more of any HSPF size class to EFDC than HSPF computed by using weights that add up to more than one. This same approach is true in reverse as well.

The period of extraction from the HSPF files is controlled by the *Begin* and *End* dates specified in the input boxes located at the bottom of the HSPF preview pane.

The HSPF File preview pane shows the some of the data from the HSPF File for the user's reference. It can be seen in this example that the flow is contained in column 2 (0.01 cfs) and dissolved oxygen is contained in column 7 (5.58 mg/l).

EFDC_Explorer interprets a calendar date using the Windows date settings. This is important if files are processed on different computers with different date settings (i.e. files from Asia being processed on computers in America). The issue here is if the dates are ambiguously formatted as MM-DD-YYYY or DD-MM-YYYY. A date of "01-02-2008" can be interpreted as 01-Feb-2008 or 02-Jan-2008, depending on the windows date format. To correct a date interpretation problem between windows configurations, check the *Data is not in System Date Format* checkbox. This simply reverses the order of the first two date fields.

The delimiter must be specified as space, tab or comma. The correct delimiter must be selected before processing the linkage.

The time series/HSPF File that will be linked to the current boundary group is shown in the *HSPF File Information/WDM Formatted* frame. The user should *Browse* to the file. If the same file will be used for all boundary groups the user may press Ctrl-A when the HSPF File field has focus to assign the same file for all the groups.

To actually link the HSPF data to the EFDC boundary condition group(s) the user must press one of the following:

- *Apply to Current Group*: This button processes all the parameters designated for import for the currently selected boundary group only. Only those parameters selected for import are processed.
- *Apply for All Groups*: This button processes all the parameters designated for import for every flow type boundary group. All the options defined for each boundary group are used.

Groups are added and deleted by setting the focus to the group list and then pressing “INS” or “DEL” as is appropriate. If deleting a group, make sure the correct group is selected.

(3) Check Boundary Conditions

The *Check* button runs a series of checks all the currently defined boundary conditions. Some of the validation checks include:

- I & J cell indexes point to active cells
- Table series are valid
- Matching head/concentration cells for open boundaries,
- Etc.

If errors are found they are reported and can be copied to the clipboard for pasting into other reports.

(4) View Loadings

The *View Loadings* button brings up the loadings option form shown in Figure 6-18. This form shows the valid model timing (based on the model start and end times) and allows the user to select a date range to process within the model dates. The parameter to view is also shown. This list varies, depending on the computational options.

When the *View* button is selected, EFDC_Explorer computes the mass loadings time series for all of the flow type boundary groups selected and displays them in a plot. The units displayed will vary, depending on whether the user is configured to display metric (metric tons (MT) per day) or English (tons (2000 lbs.) per day). Figure 6-19 shows an example of mass loading (by group) for Total Phosphorus. To then compute the *total mass loading* for the time period displayed on the plot, press Ctrl-I to integrate the series (see Section 10.2 for more details). Since the time units are days and the mass loading is MT/day, entering a 1 for the conversion factor results in the mass loading being reported in metric tons. Note that the user is now able to use concentration loadings rather than mass loadings as discussed in Section 6.14.5.

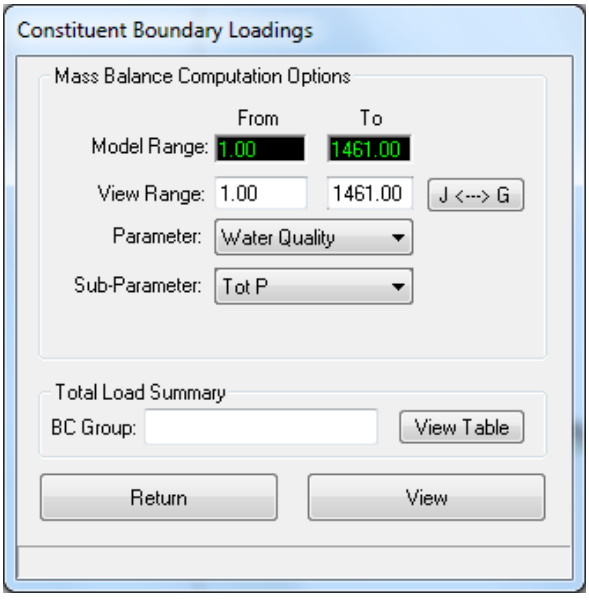


Figure 6-18 View loadings options form.

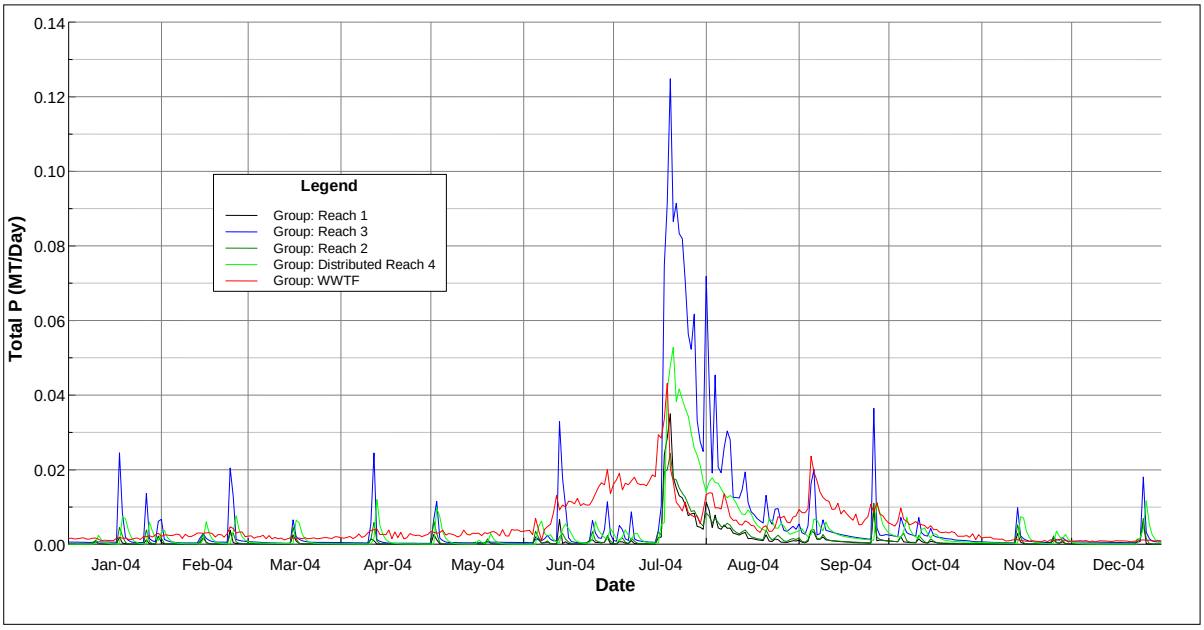


Figure 6-19 Example of mass loading plot for Total Phosphorus (TP).

6.3.3.2 *Groundwater Boundary Condition*

In the *Groundwater* frame (Figure 6-20) the user can select a range of groundwater interaction options. The user should review the EFDC code to ensure the option specification and EFDC function for the groundwater parameters. The current options are:

Option 0 Disable groundwater interaction;
Option 1 Allow soil wetting and drying;
Option 2 Allow groundwater interaction (using GWMAP and GWSER), and
Option 3 Apply a constant groundwater flux out of the model by zones (using GWSEEP and GWMAP, and available for EFDC+I only).

The *Modify Series* button gives access to the groundwater class data (Option 3). The number of series is input in the appropriate box and then the grid allows the user to specify the groundwater class data. As with the vegetation class data, the ID and the Description are used only by EFDC_Explorer.

The ID, as with the vegetation classes, is again used to match the groundwater classes to the EFDC cells using the *Apply Overlays* button. The same process as discussed for the vegetation classes is applied here.

The *Apply Polygon* button is similar in function to the *Apply Overlays* button but does not require a matching ID in the polygon file. Open the polygon file and then specify the groundwater class, and then click the *Apply* button to assign all the cells that are *inside* the polygon to that groundwater class.

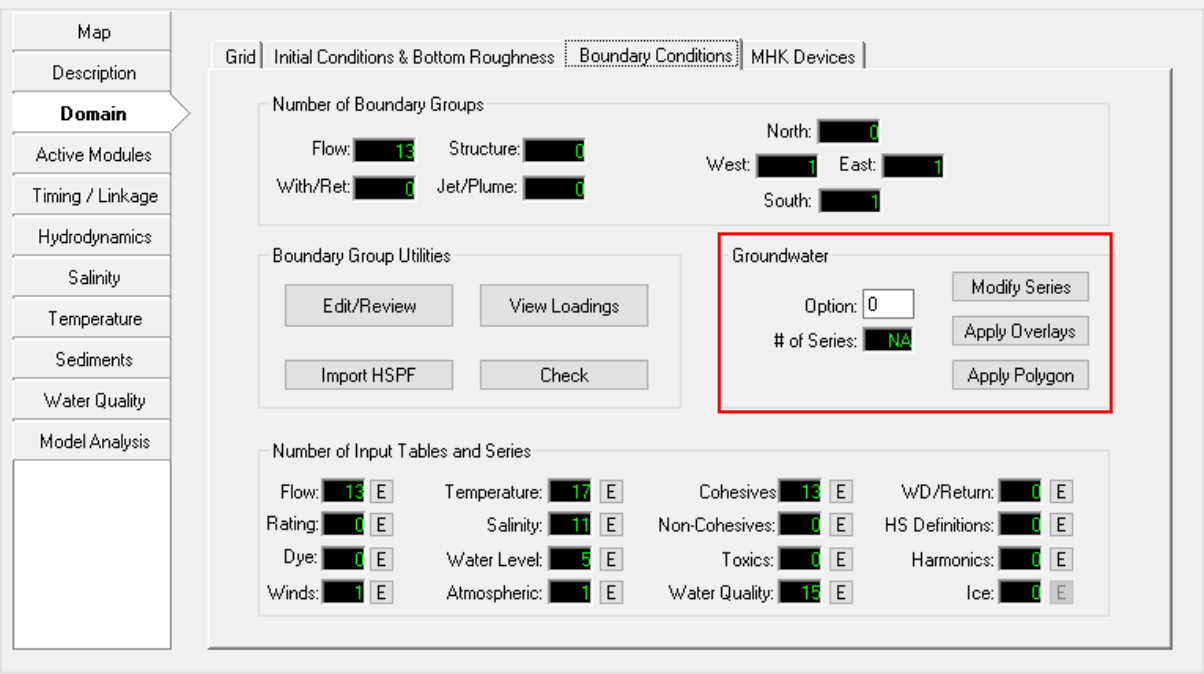


Figure 6-20 Domain Tab: Boundary Conditions for Groundwater.

6.3.4. Boundary Condition Group Editing Form

The editing of boundary condition groups were formerly all handled by one *Modify/Edit BC Properties* form. From release of EE7.2 onward each BC has its own specific form which is accessed by the *Edit* button shown in Figure 6-13 . The individual boundary condition types are explored in the sections following. However, two frames are generic across all the BC types, the *Current Boundary Group Information* and *Setting for Current Boundary Cell* as shown in Figure 6-21.

The top frame, *Current Boundary Group Information*, contains general information on the total number of BC groups, the number of groups for this BC, the current group number and number of series for this BC. The next frame, *Setting for Current Boundary Cell*, contains the cell specific information for the current group. The user may add a cell or remove a cell from a boundary group using the scroll bar to select the *Current Cell* or set the cell location. The user may also use *Add by Polygon* option to select a polygon or polyline to assign the cells that are to be included in the group.

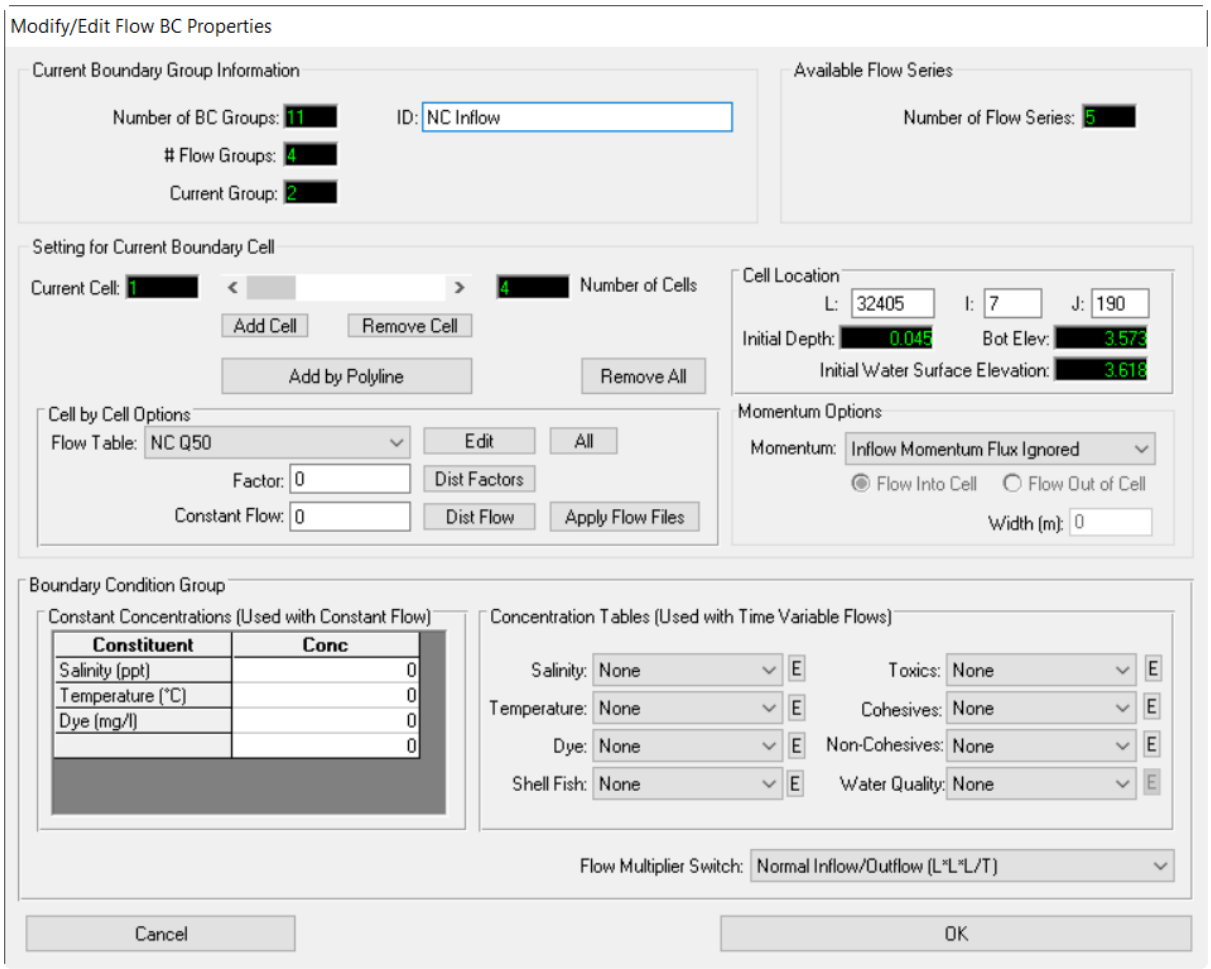


Figure 6-21 Boundary Condition Settings – Flow Boundary.

6.3.5. Flow Boundary Condition Editor

For flow BC's the *Cell by Cell Options* frame has several useful functions. The *Edit* button allows the user to edit the associated time series. The *All* buttons next to the fields in this subframe allow the user to quickly assign all the boundary cells in the current group to the parameter specified in the adjacent field. For example, pressing the *All* next to the *Flow Table* field shown in Figure 6-21 would set the flow series number for all cells in the group to the current "NC Q50" table settings. If the user has an input flow series for a river that requires distributing the river flow across several cells, the single time series of flow can be specified in the *Flow Table* and the *Factor* can be set to split the flow appropriately among the cells. The *Dist Factors* button sets the *Factor* based on cell volumes. A constant inflow can also be distributed in a similar manner by using the *Dist Flow* button next to the *Constant Flow* field.

The *Cell by Cell Options* frame also provides the *Apply Flow Files* button, allowing the user to import flow with the flow files to build a cell by cell variable inflow series. This feature is predominantly used in wetting and drying models.

The user interface for boundary conditions has a uniquely designed editor with drop down menus for concentration table assignment to flow tables. When the user selects *Flow Table* as shown in Figure 6-21, they can use drop down menu items to link flow tables with concentration tables. As with earlier versions of EFDC_Explorer, the concentrations specified in the *Constant Concentrations* frame are only used with constant flows. Specifying a constant

salinity concentration of 5 ppt in the *Constant Concentrations* frame but with zero constant flow, even if the boundary has flows assigned via a time series, the salinity input to the model will be zero. If using a time series for flows, the user must use a time series to assign salinities, even if they are constant. Note that the water column concentration settings, for both time series and constant, are fixed for all cells for the same group.

The *Momentum Options* frame for the flow boundaries provides the user with more control over how EFDC handles momentum flux on the cell faces. The options now include:

- Inflow Momentum Flux Ignored (default option)
- Momentum Flux on West U Face
- Momentum Flux on South V Face
- Momentum Flux on East U Face
- Momentum Flux on North V Face

The *Flow Multiplier Switch* subframe provides a drop-down box with a number of options, from normal inflow to various lateral inflow/outflow options. Constant concentrations and time variable Concentration Tables may also be edited from this frame.

6.3.6. Hydraulic Structures Boundary Condition Editor

To create and configure hydraulic structures the user should open the *Hydraulic Control Structure BC* form shown in Figure 6-22. This may be accessed from the *Domain* | *Boundary Conditions* | *Edit/Review* button and then selecting a hydraulic structure boundary (type 2) or RMC on the BC from *ViewPlan* when in editing mode.

In the *Flow Derived From* dropdown menu there a range of options based on rating curves or equations. In the case of the sluice gate, combinations of these to support rule-based and time variable control of the gate. Only when the sluice gate option is selected is the *Time Control* dropdown menu enabled. Hydraulic structure flow options currently comprise:

- Upstream Elevation Whole Channel Rating
- Upstream Depth
- Elevation Difference
- Elevation Difference with Flow Accelerations
- US and DS Elevations
- Upstream Depth with Low Chord
- Elevation Difference with Low Chord
- Equation: Culvert
- Equation: Sluice Gate
- Equation: Weir
- Equation: Orifice

The details of how to configure and use the equation-based structures are described in the next section. Configuration of the flow table based hydraulic structures are described in this section.

In the *Cell Options* frame, the *Rating Table/Flow Lookup Table* shown in Figure 6-22 allows the user to set the head for each table. After the user has selected the table number, the *All* button applies the current flow table setting for cells in the current group. The user may also enter an *Offset* which will apply an offset to the result as described in the *Head Lookup Table Equations* frame.

The user sets boundary flow multiplier options in the *Boundary Condition Group* frame. The dropdown menu allows selection from the following equations for the boundary flow multiplier, where q(h) are unit discharges and need to be multiplied by a dimension:

Equation	**Units for q[1]**
$Q = q(h)$	L^3/T
$Q = q(h)DY$	L^2/T
$Q = q(h)DX$	L^2/T
$Q = q(h)(DX + DY)$	L^2/T

[1] Where L is length and T is time

The user may also input the cell downstream from hydraulic structure in the *Downstream I and J* boxes. The length “L”, bottom elevation and initial depth are all displayed for the downstream cell.

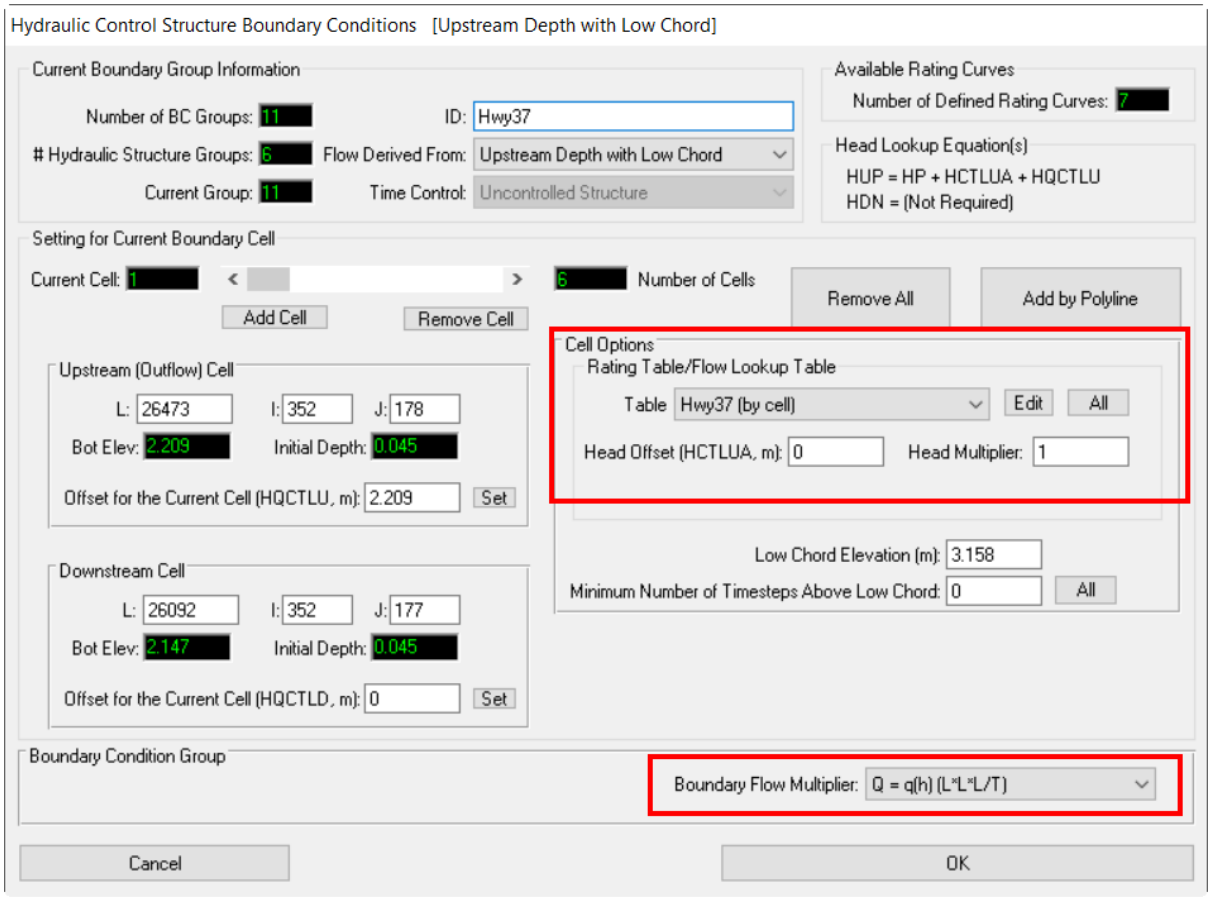


Figure 6-22 Hydraulic Structures BC: Rating curve type.

6.3.6.1 *Whole Channel Rating Curve*

EEMS provides the option of setting a boundary that is defined by a single rating curve to represent the river. This feature is useful if the user wants to create stepped flows for a IFIM habitat analysis.

The *Hydraulic Control Structure BC* form is shown in Figure 6-23. In the *Flow Derived From* dropdown menu the user can select *Upstream Elevation with Whole Channel Rating Curve*. This option acts much like a flow table, or flow time series, which may be split with a flow multiplier across different cells, however in this case it uses the same rating curve for all the cells assigned. Using the scroll bar in the *Setting for Current Boundary Cell* frame allows the user to see the how the multiplier changes.

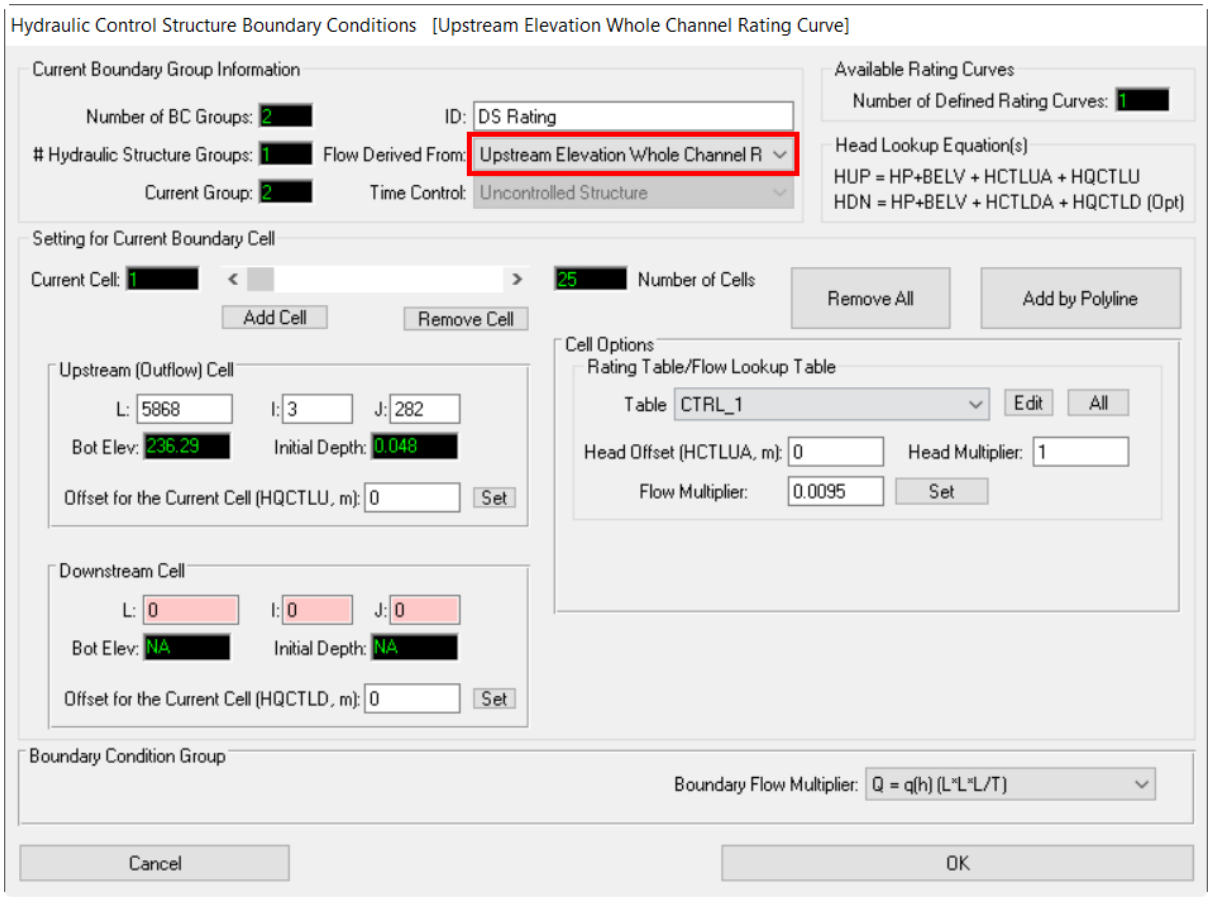


Figure 6-23 Hydraulic structure: Whole channel rating curve option.

The user also has the option of setting a downstream cell that the flow will return into in the *Downstream Cell* frame. If this is left blank, then EE ignores the downstream end.

When the user selects the *Set* button next to the *Flow Multiplier*, several options are provided for the split approach as shown in Figure 6-23. The first option is to set the water surface elevation to use when the flow splits are applied. By default this is the maximum value in the rating curve.

The second option is to select one of three approaches for calculating the flow splits. The user may split flow based on DX, on DY, or on DX*DY. In the case shown in Figure 6-24 the cells are split based on DX.

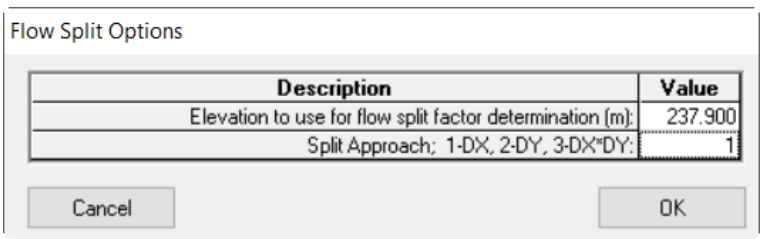


Figure 6-24 Flow split options.

6.3.6.2 *Upstream Depth and Elevation Difference Options*

The hydraulic structure boundary condition option that uses flow derived from upstream depth, or elevation difference, uses a head lookup table to describe the relationship between head

and flow for that cell. This is selected with the *Flow Control Derived From* menu as shown in Figure 6-25. Several options are available for the flow control type including:

If the *Elevation Difference with Flow Acceleration* option is selected the flow acceleration parameter flows are now squared and multiplied by an acceleration factor when passing through an inlet.

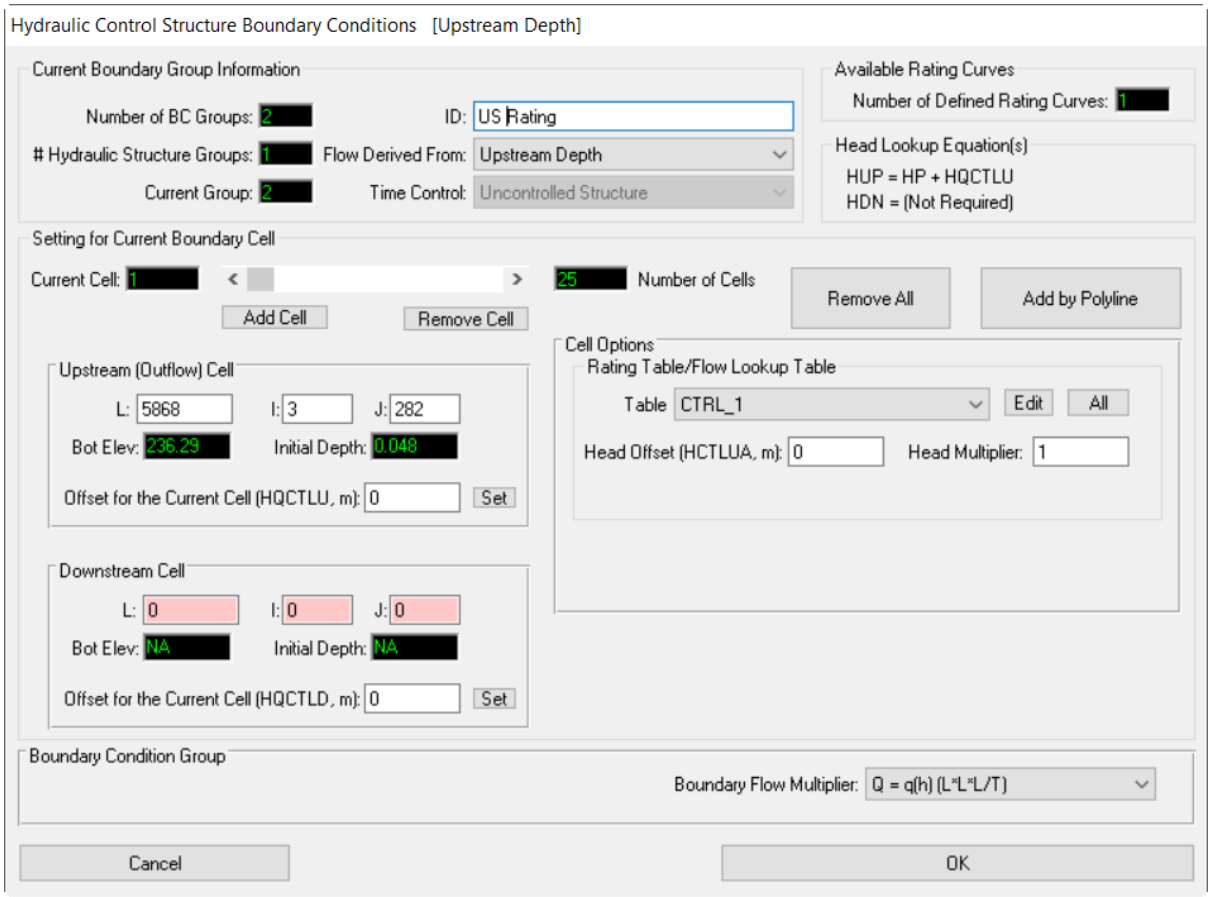


Figure 6-25 Hydraulic Structure BC: Flow derived from upstream depth.

6.3.6.3 *Flow Derived from Upstream and Downstream Elevations*

Another option is to use flow derived from upstream and downstream elevations. In the example shown in Figure 6-26, Pacheco Outlet uses such a control type. In this case it is necessary to have a matrix to describe the relationship between head and flow as shown in Figure 6-27. Most data tables, such as those by the US Corp of Engineers, use the same downstream and upstream head. Therefore, if the user wants to change the table, they should select *Edit Heads*, and add a new table. When *Accept Heads* is selected a new row and column will be created.

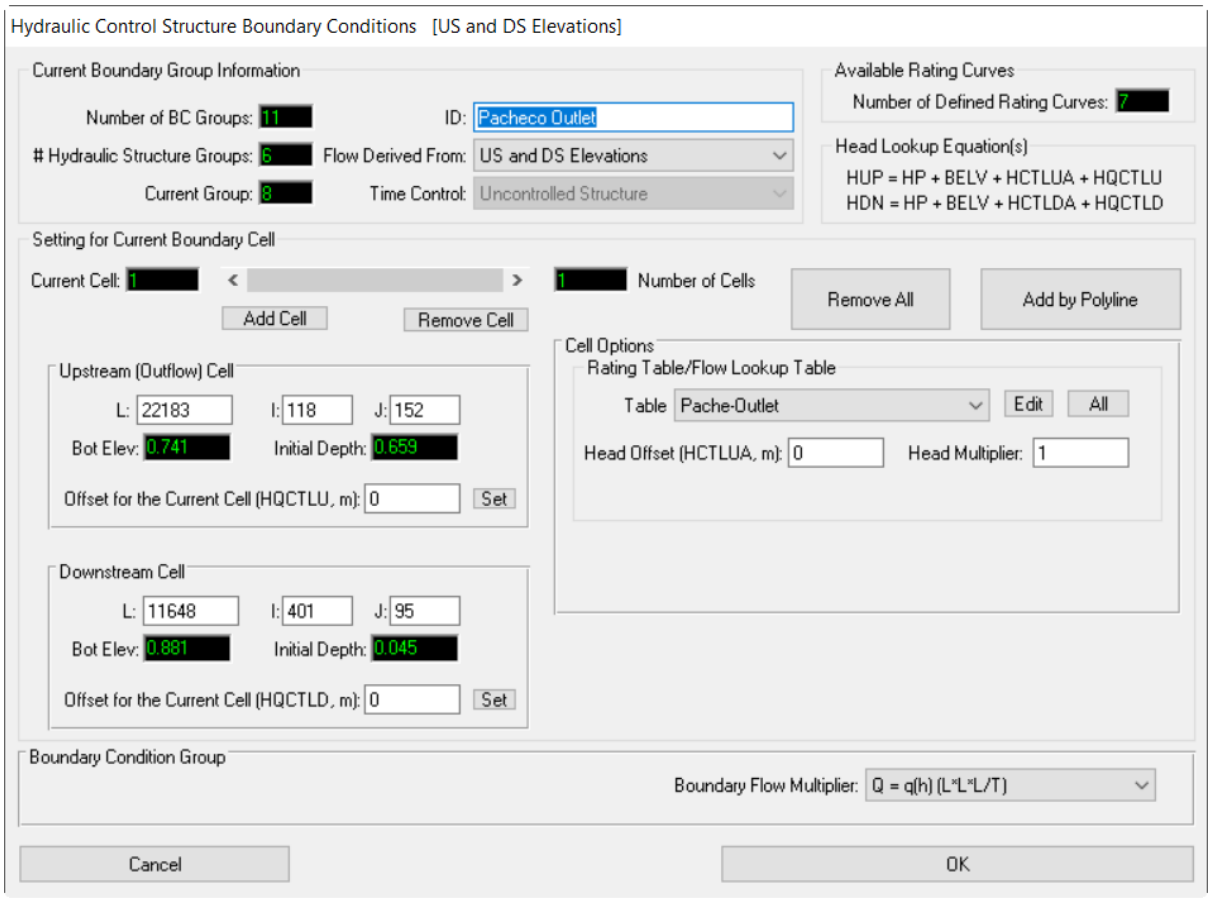


Figure 6-26 Structure Boundary Conditions: Flow from US and DS Elevations.

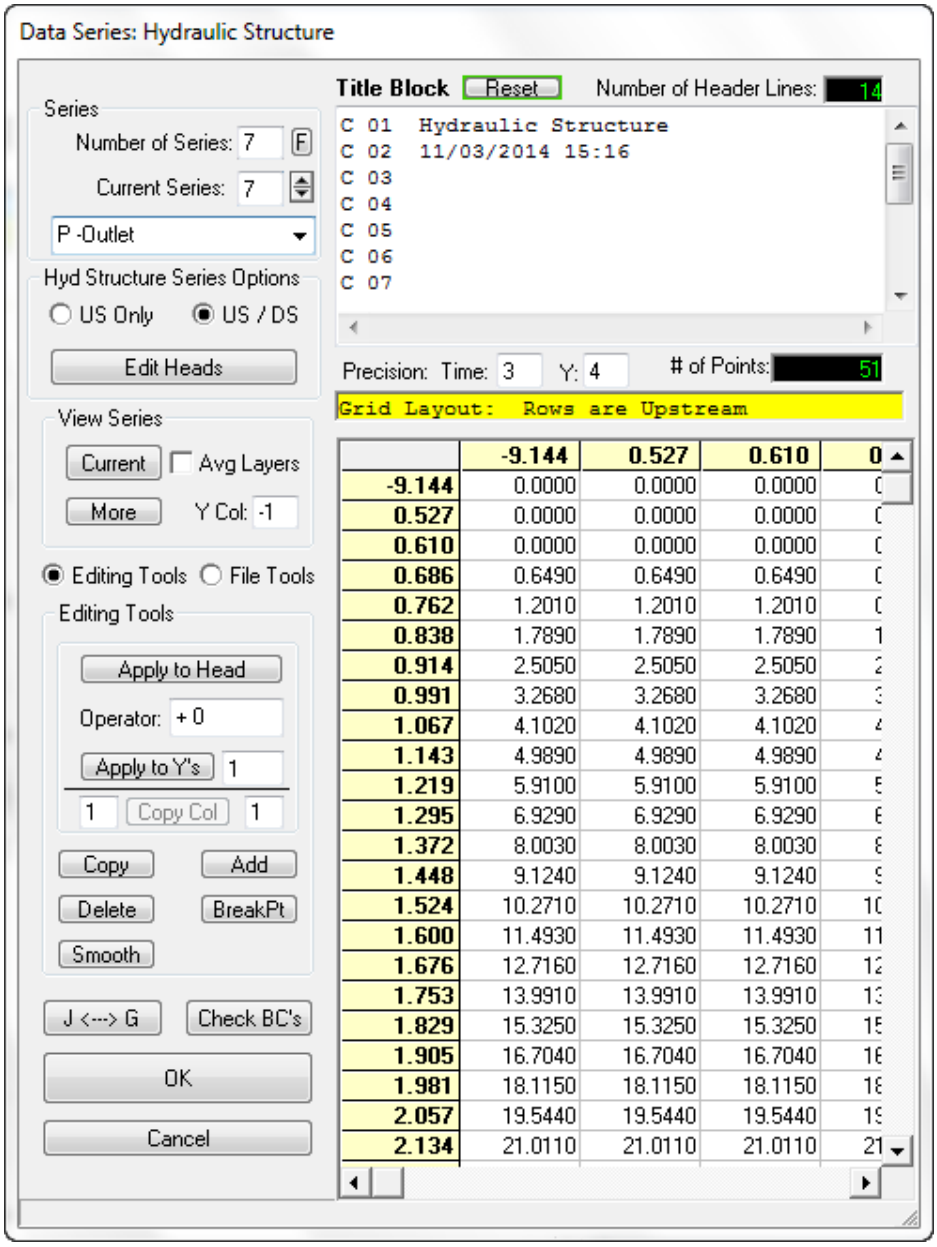


Figure 6-27 Hydraulic Structure BC Editing: Flow from US and DS Elevations.

This matrix may be plotted by selecting the *View Series* | *Current* option. This displays the graph as shown in Figure 6-28. As there are often too many series to plot, the user is prompted for a skip interval so that some lines are not displayed. The user may also select a specific upstream head to plot.

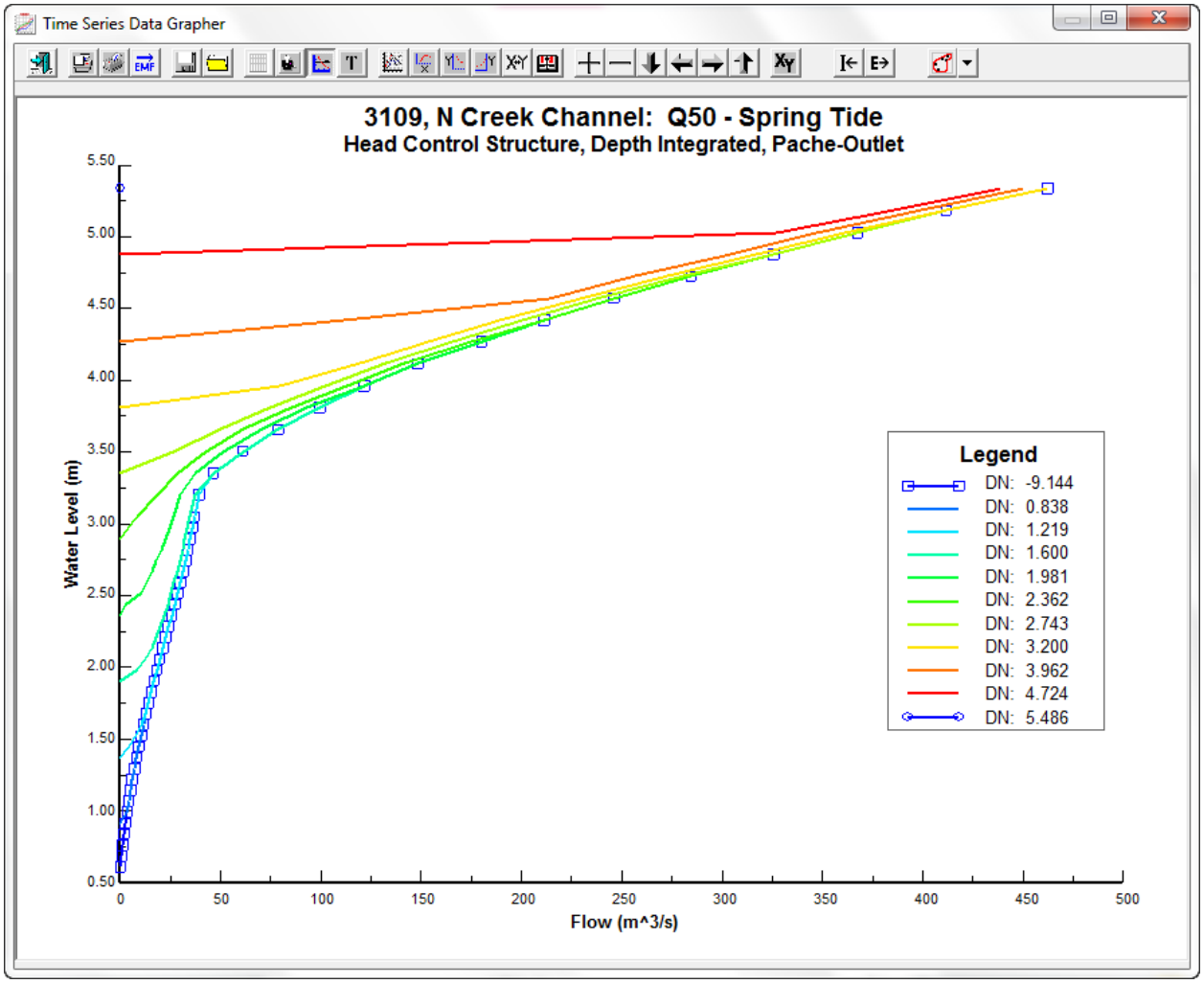


Figure 6-28 Flow from US and DS Elevations.

6.3.6.4 *Flow derived from Elevation Difference with Low Chord*

One use of hydraulic structures is to simulate a low chord, i.e. the bottom of a bridge. In this case, when flows are below the bridge deck they may be bi-directional, i.e. flows can be going upstream or downstream. However, once the bridge is overtopped flow is only upstream to downstream.

EFDC uses the total flow rate for the flow calculations, therefore it requires the flow to be at the actual time when the cell reaches the low chord elevation. The elevation is then subtracted from the value obtained from the rating curve as this curve defines the relationship for the total flow around the bridge for the whole range of depths. It is necessary to subtract the actual flows from the curve to prevent a large jump in flow. To prevent instability at the transition the minimum number of time steps above the low chord may be provided by the user. An example of how this is setup is shown in Figure 6-29.

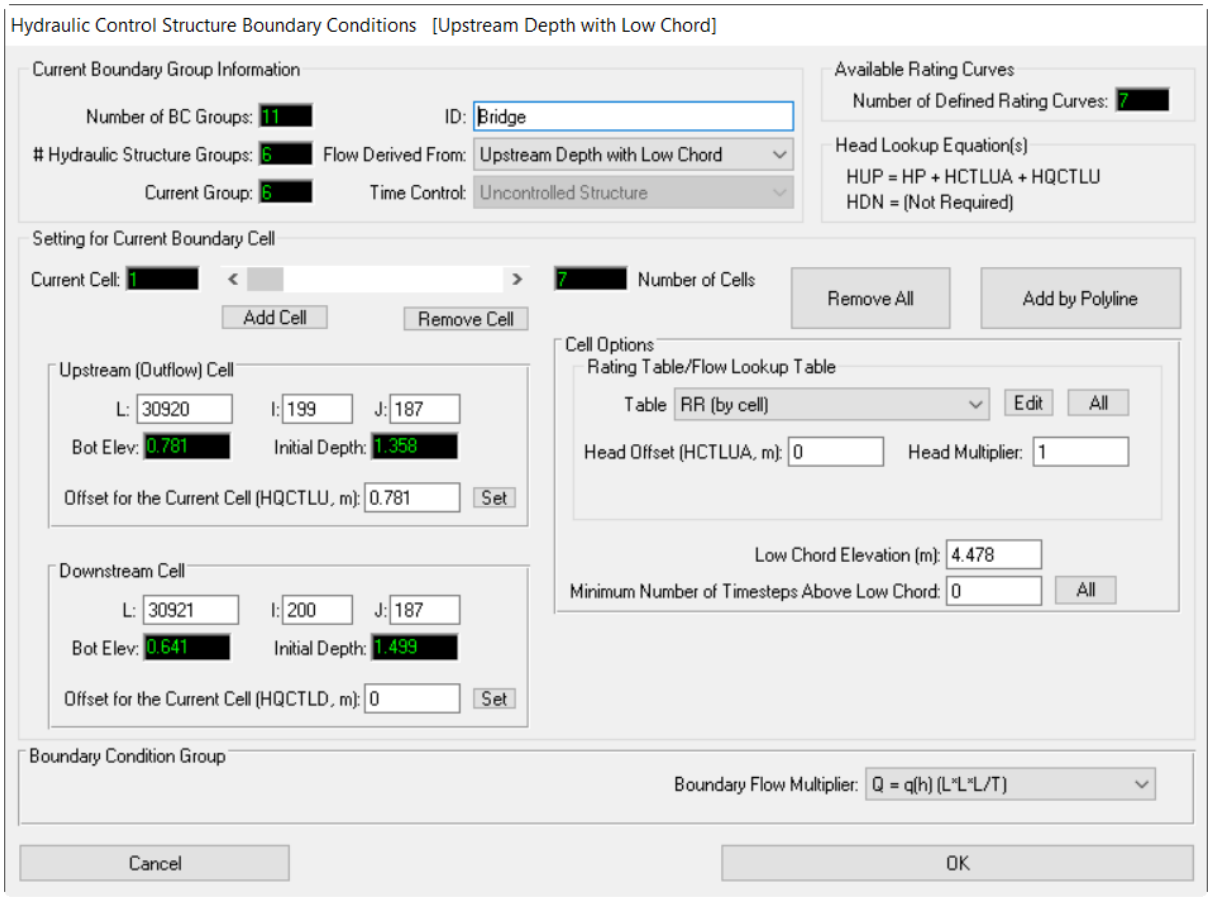


Figure 6-29 Hydraulic Structure BC: Low Chord Option.

In the equations for the head look up tables (Figure 6-29), HQCTLU is set on a cell by cell basis. HCTLUA comes from cell options offset.

The table for CTRL_1 is defined by the user as shown in Figure 6-30.

This boundary condition needs to be turned on for each cell. However, to enter one value for all the cells in a group the user may select all (CTRL A) and then apply to all the cells in that group. The 3D view of the backwater effect from the bridge is shown in Figure 6-31. Note that the bridge must be of a size larger than the grid size for this option to work effectively.

It is recommended that users set a common low chord elevation for adjoining cells and pay attention to the bathymetry for those cells. While it is possible to set a different low chord elevation for adjoining cells, if they have different values then flow will pass from cell to cell across the bridge and cause model instability. In the same way, sudden change of bathymetry between the cells may also create oscillations. To prevent this it is suggested to use masks.

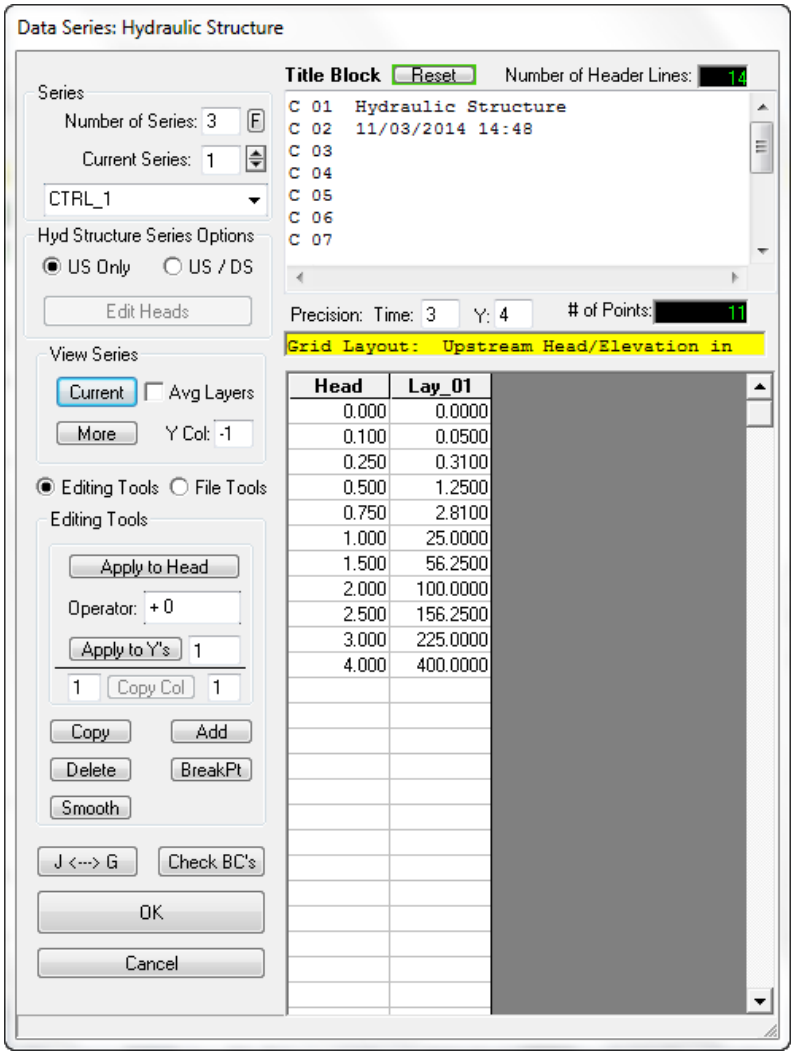


Figure 6-30 Hydraulic Structure Editing Form: Low Chord.

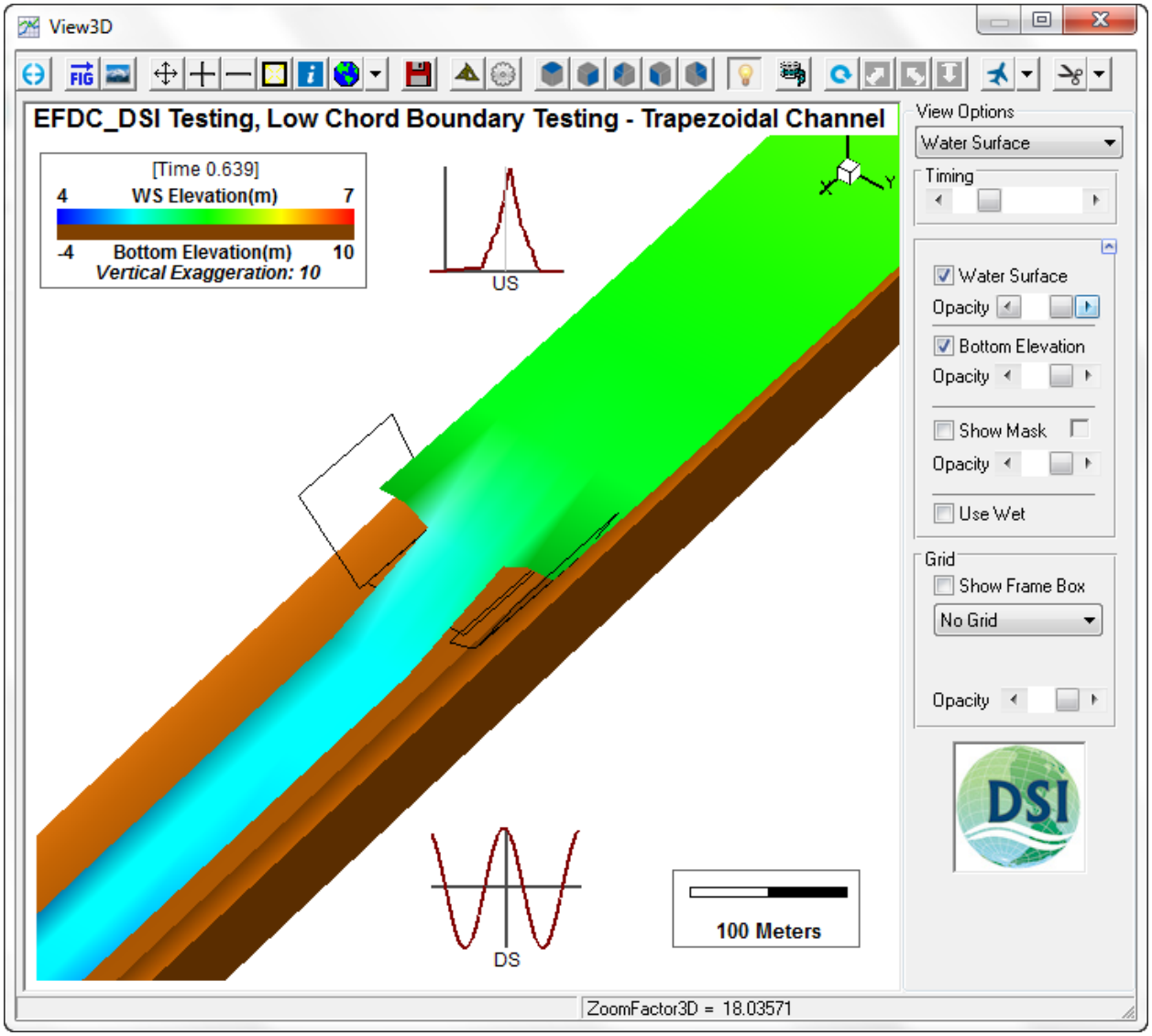


Figure 6-31 Bridge (Low Chord) Hydraulic Structure in View3D.

6.3.7. Hydraulic Structures Boundary Condition Editor - Equation Based

EE8 and EFDC+ can simply and quickly setup up hydraulic structures (HS) such as culverts, weirs, sluice gates, and orifices with the physical dimensions of the structures and use the corresponding HS equations (Dill, 2012), as well as the previous option of defining a flow series for the structure. EE allows the user to view summary of the type of the hydraulic structure and the equation definition (i.e. parameter definition) as shown in Figure 6-32.

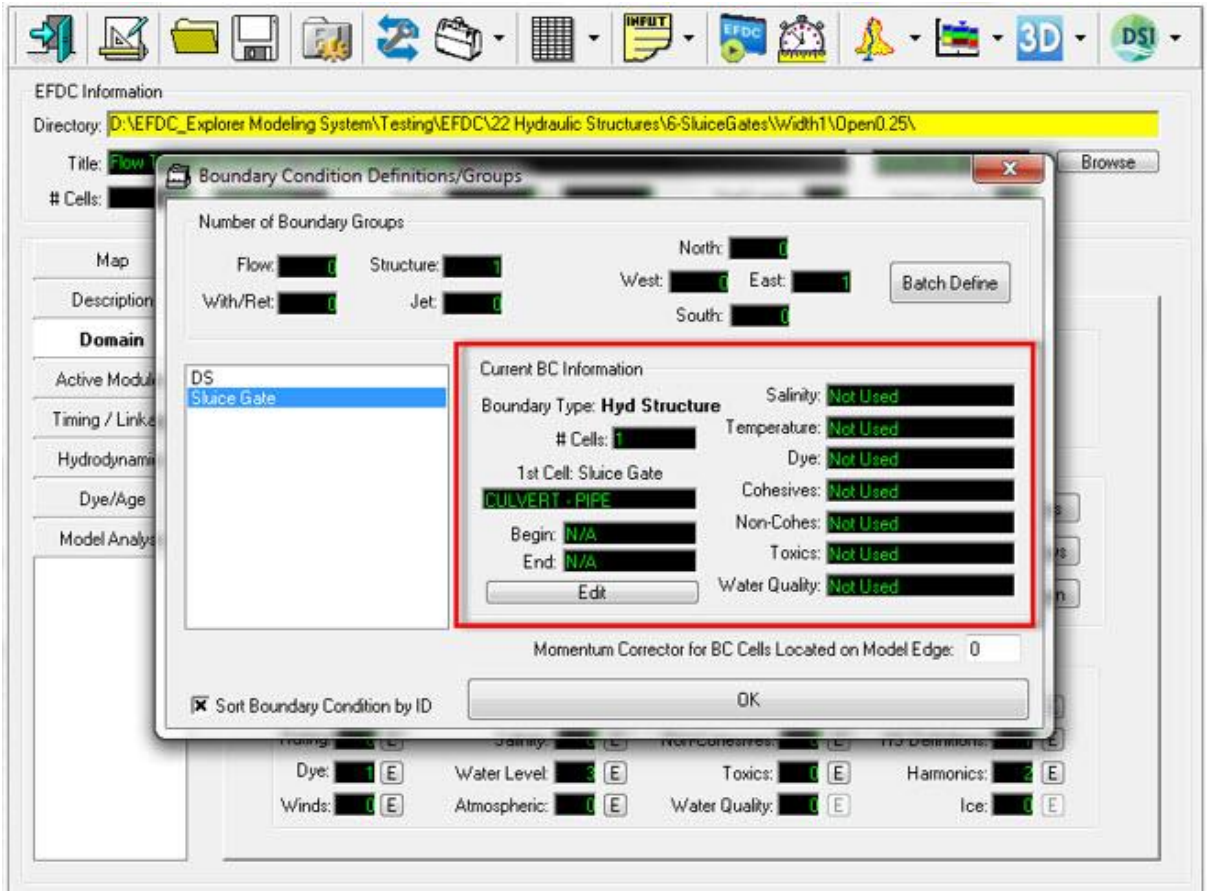


Figure 6-32 Boundary condition groups for Hydraulic Structures.

Using the *Edit* button shown in Figure 6-32 the user may load the *Hydraulic Control Structure Boundary Conditions* form as shown in Figure 6-33. On this form (Figure 4-2), there is a dropdown selection of all available *Flow Control Type* (NQCTYP). When the user has defined a hydraulic structure the EFDC.INP file creates the card C32A to handle these variables. Some of the parameters in C32A are not used, depending on which equation has been selected.

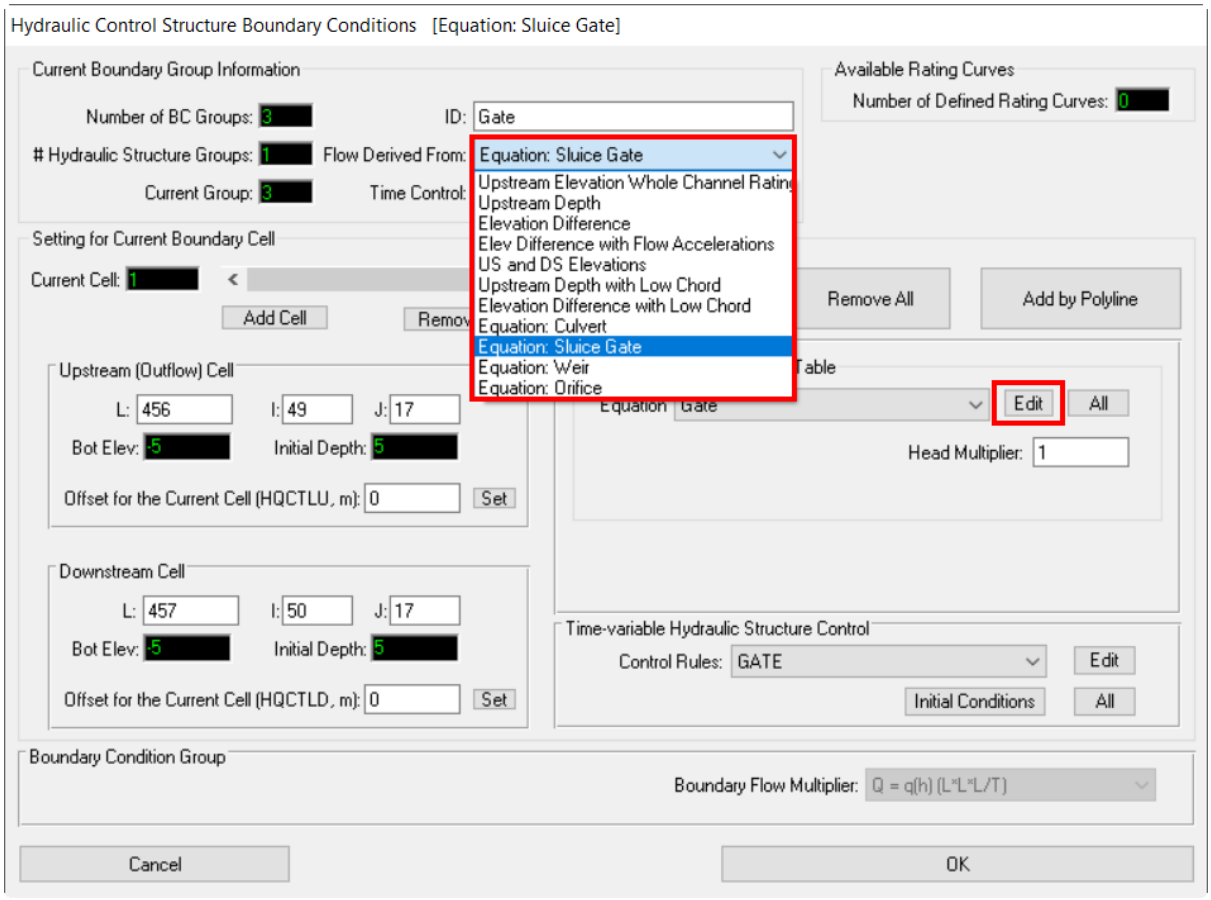


Figure 6-33 Hydraulic Structure BC: Flow control types.

The dropdown list defines the flow derived control structure (NQCTYP < 5) and the new hydraulic control structures (NQCTYP > 4). Based on what the user selects the "*Head Lookup Equations*" and the "*Cell Options*" frames are updated based on the selection.

Selecting the *Edit* button shown in Figure 6-33 opens the equation editor for the flow control type selected. Each hydraulic structure type is described further below. Note that for every equation, there is a check box to indicate whether or not to allow reverse flows.

6.3.7.1 *Culverts*

To assist the user define the input parameters for the culvert, a *Definition Editor* is provided as shown in Figure 6-34. This form can be accessed by clicking on *Edit* button in Figure 6-33. Here the user may define the *Equation ID,* allowing various culverts dimensions and type to be defined.

The user should select the *Hydraulic Structure Type* from the dropdown menu. In this case, the option chosen is "culvert". The user should then select the *Cross-section type*. Depending on the options chosen the image of the culvert will change to match the user's selection. The user can then specify the culvert dimensions in meters, including upstream and downstream elevations of the pipe, length of the pipe, Manning's roughness coefficient and diameter of the pipe as shown in Figure 6-34. The user may also import 3D structures to better visualize the hydraulic structures in EE8 as shown in Figure 6-35.

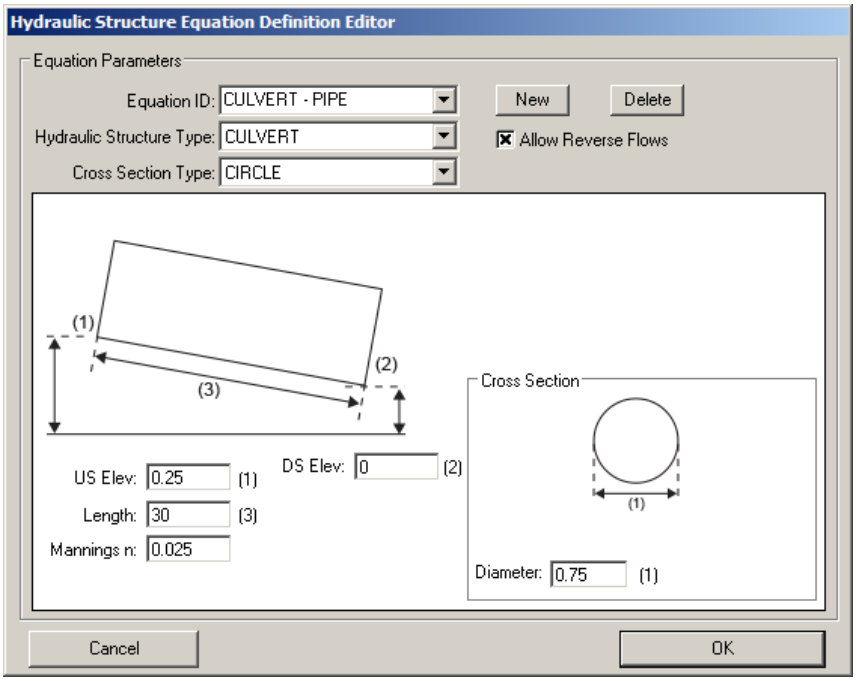


Figure 6-34 Hydraulic Structure Equation Editor: Culverts.

The methods for determining flow through culverts are based on the culvert flow type classification and analysis from Chow (1959). This methodology describes six different types of culvert flow based on the location of the control section within the culvert and the relative elevations of the headwater, tailwater, and culvert invert and crown elevations in meters. The discharge is primarily computed using Manning's equation which can be expressed as:

$Q = KS^{1/2}$

Where conveyance, $K = (A/n)R^{(2/3)}$ and where A = cross-sectional flow area (m); n = Manning's roughness coefficient, and R = hydraulic radius (m).

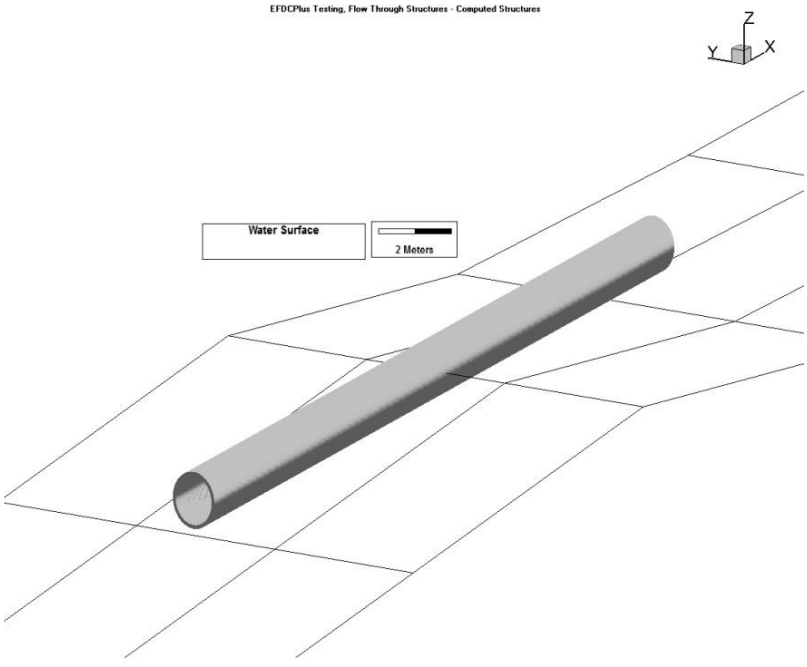
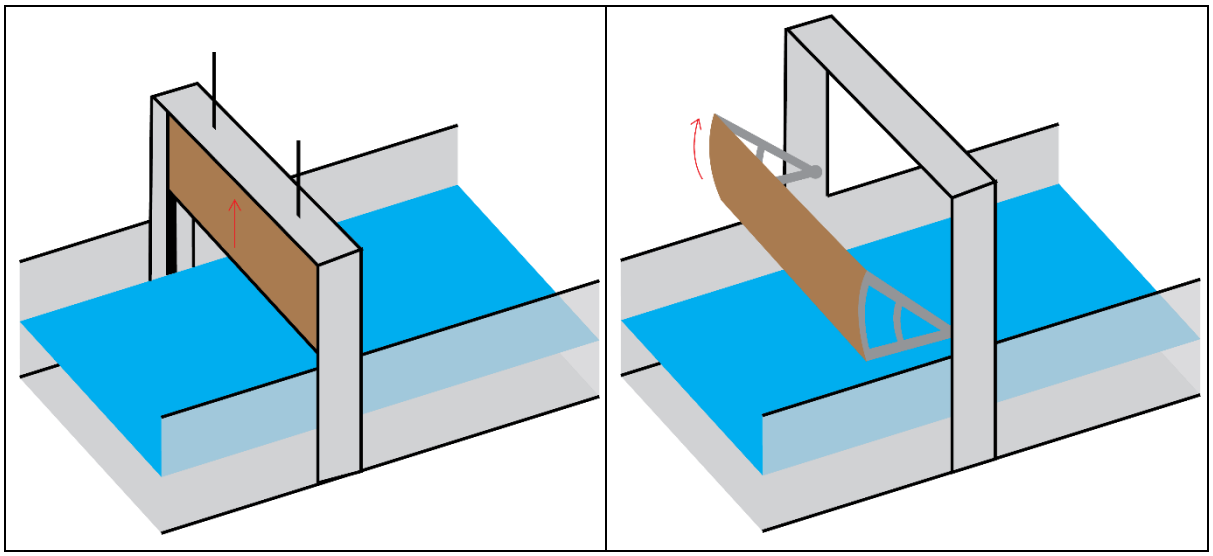


Figure 6-35 Image of culvert in View3D.

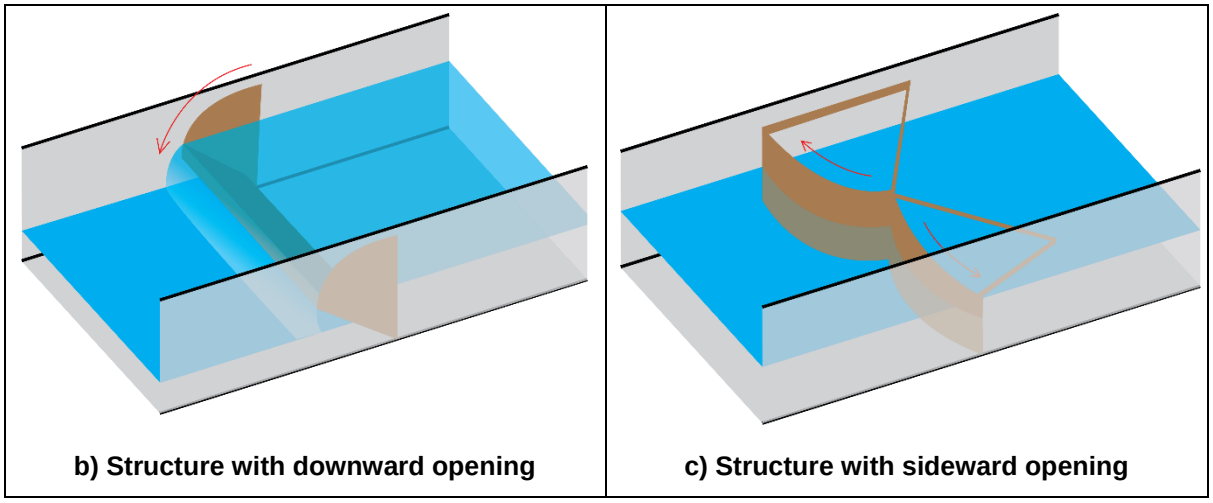
6.3.7.2 *Sluice Gates*

The sluice gate hydraulic structure boundary has been enhanced to simulate the opening and closing of gates. Currently two types of sluice gate operation are supported: operational time series, and operational rules.

With the operational time series option, the gate is opened or closed during the simulation based on a defined time series of the operational state and settings such as rate of gate opening, and maximum opening height. Examples of the types of gates may be shown in Figure 6-36.



a) Structures with upward opening



b) Structure with downward opening

c) Structure with sideward opening

Figure 6-36 Operation for different types of gates.

When using the equation based hydraulic structures, if the user selects *Equation: Sluice Gate*, the *Time Control* dropdown menu will be displayed as shown in Figure 6-37.

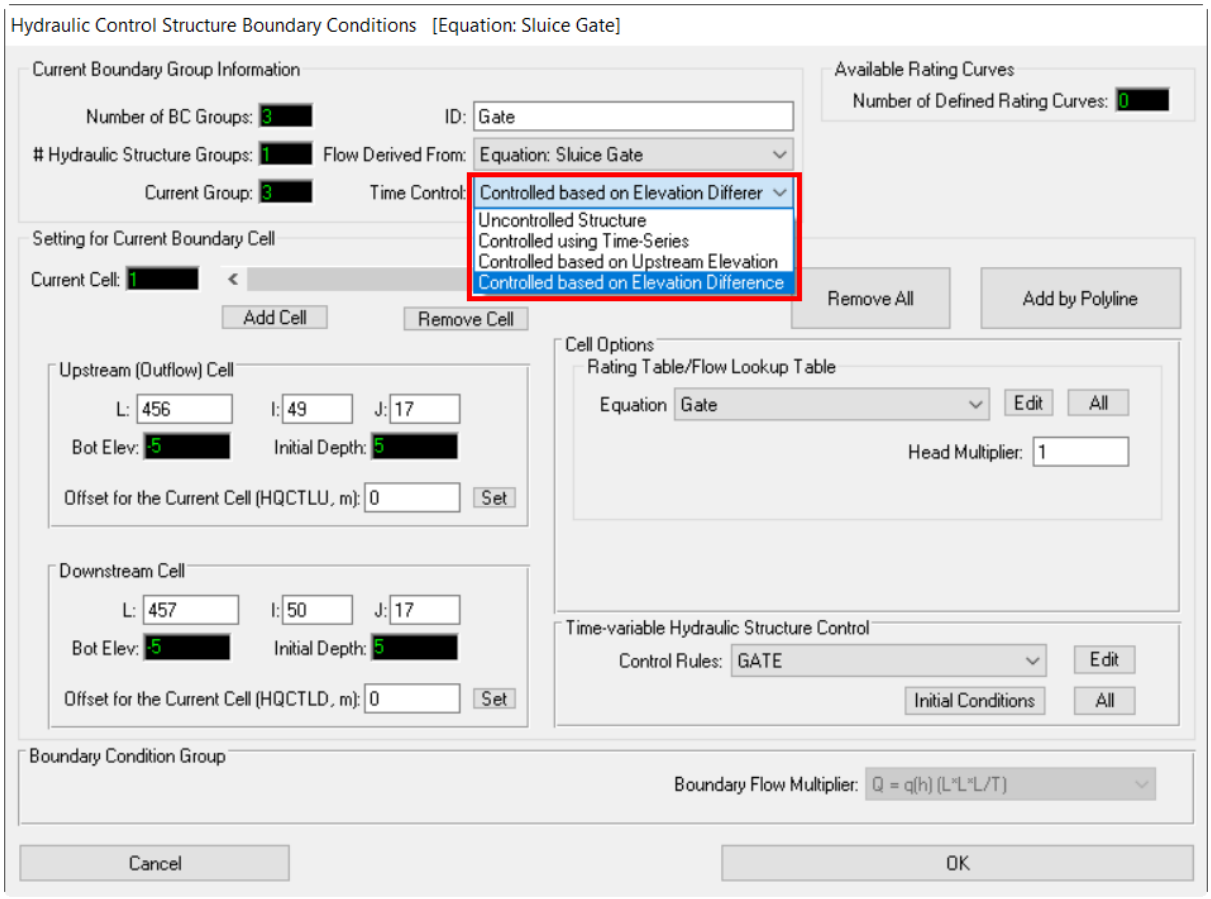


Figure 6-37 Hydraulic Structures: Sluice gate controls.

The options available in this menu are as summarized in the table below and the final three options described in detail in the following sections.

Name	Meaning
Uncontrolled Structure	The hydraulic structure is uncontrollable type. The previous method of flow computational using lookup table or equation will be used
Controlled using Time-Series	The operation of the hydraulic structure is controlled using a time-series which defines the changes with time of gate openings or rating curves
Controlled based on Upstream Elevation	The operation of the hydraulic structure is controlled using control rules defined based on water surface elevation at an upstream location. This location can be different from the upstream cell of the structure.
Controlled based on Head Difference	The operation of the hydraulic structure is controlled using control rules defined based on the difference of water surface elevations at two locations upstream and downstream of the structure. These locations can be different from the upstream and downstream cells of the structure.

(1) Time-Series Control of Sluice Gates

When the flow control type of *Controlled using Time-Series* is selected for sluice gate, the user can select which time-series will be used for the structure by selecting the corresponding item in the *Table* dropdown list in *Time Variable Hydraulic Structure Control* group (Figure 6-38). To create new time series, or edit an existing one, the user can click on *Edit* button beside the Table dropdown list to edit the control time-series which is shown in Figure 6-39.

Selecting the *Show Params* button provides the user with more control over which parameters from the time series will be used to control the gate. Parameters that can be selected for the sluice gate include the initial opening height, opening width, sill level change, number of units, flow discharge and rating curve. A setting of "1" means that this parameter is used in control calculations. A setting of "0" means this parameter is ignored. In this example only gate height is set to 1. Other parameters are left to 0, meaning only the opening height of the gate will change with time.

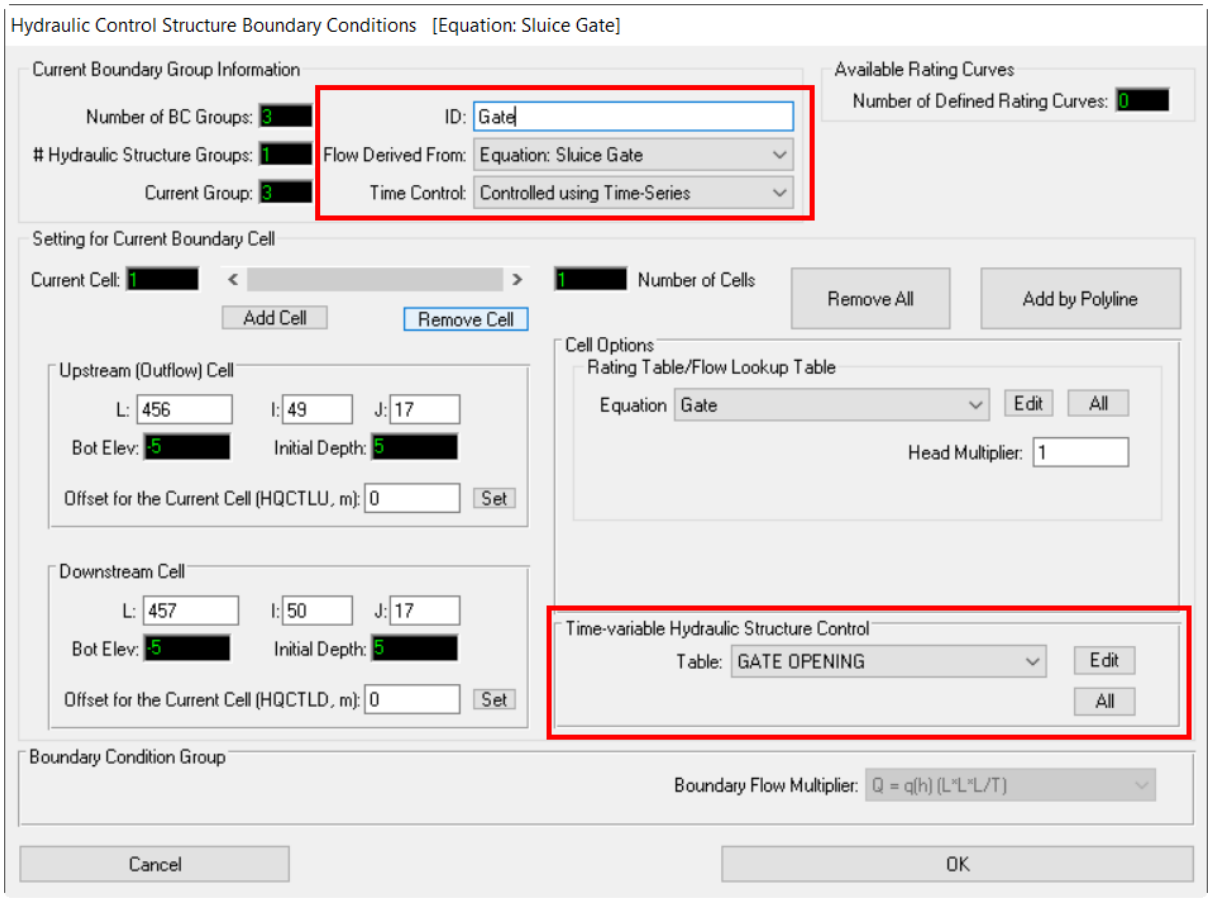


Figure 6-38 Sluice gate control using a time-series.

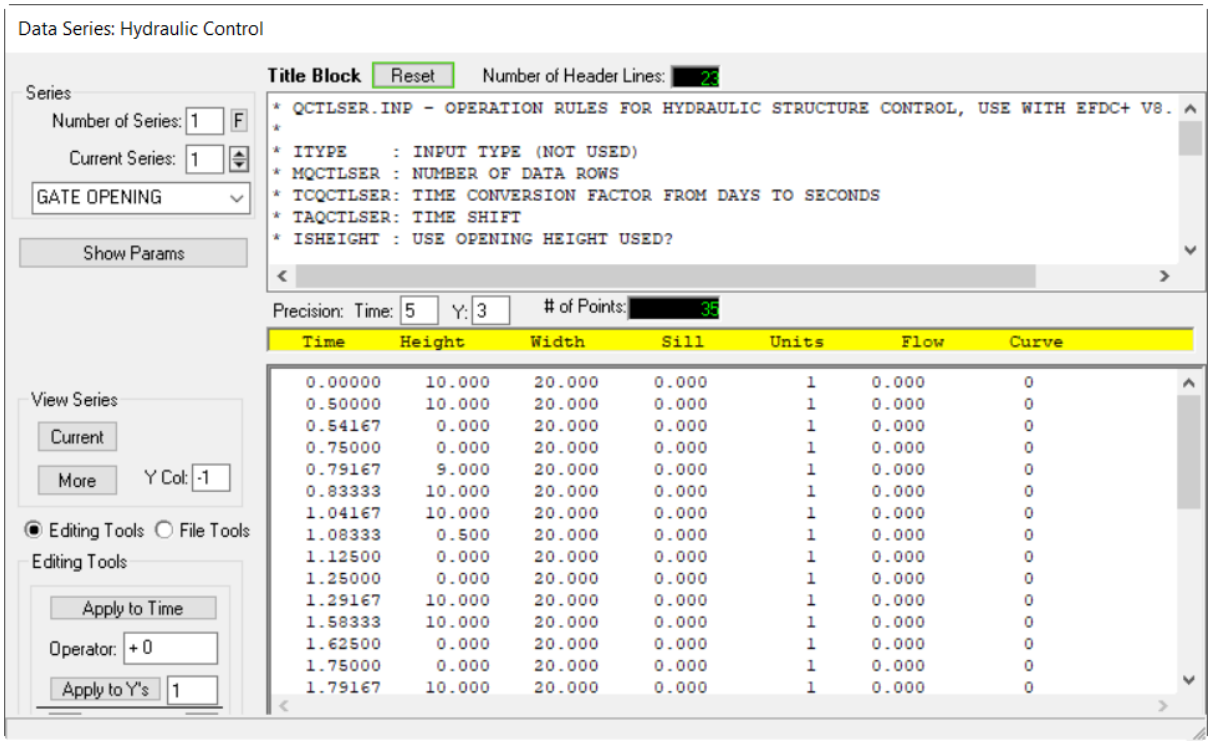


Figure 6-39 Data Series: Time series for sluice gate control.

(2) Control Rules from Upstream Elevation or Head Difference

When the flow control type of "*Controlled based on Upstream Level*" or "*Controlled based on Head Difference*" is selected, the user can select which control rules will be used for the structure by selecting the corresponding item in the *Control Rules* dropdown list in *Time Variable Hydraulic Structure Control* group shown in Figure 6-40.

To create a new set of control rules, or edit an existing set, the user can click on *Edit* button beside the *Control Rules* dropdown.

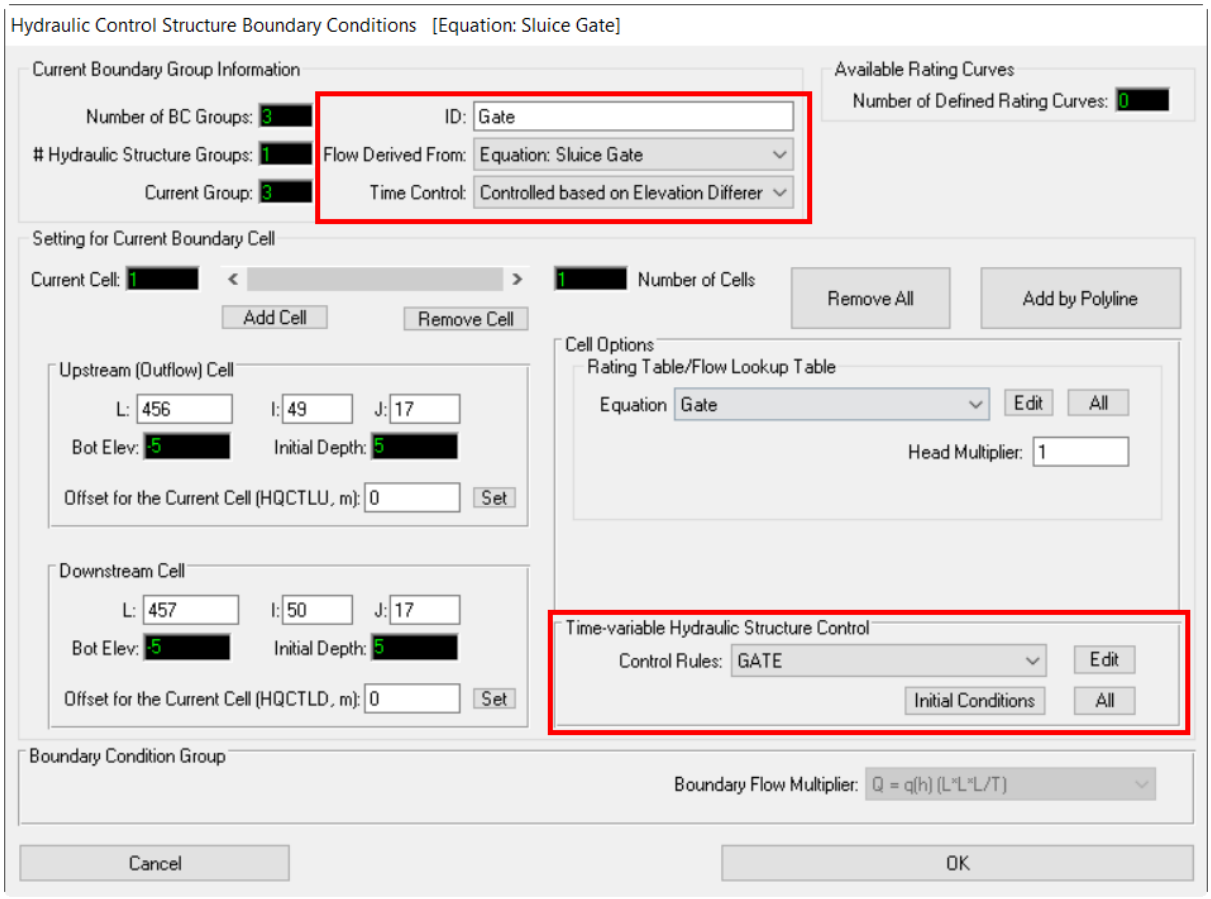


Figure 6-40 Hydraulic structure: Sluice gate control using operational rules.

The hydraulic structure control rules are shown in Figure 6-41. Here the user can configure the number of sets of rules with *Number of Rules*, and then scroll between the sets with *Current Rule* menu. For each set of control rules, the user must configure which parameters the rules are to be set for: opening height, opening width, and sill level (height above bottom of structure) change. The actual rules require the following settings:

- Control value: the elevation or (head difference) which will trigger the gate to start to open or close
- State: the direction of the opening or closing. Opening is configured with a "1" and closing is configured with "0"
- Height: opening height to which the structure can rise or fall
- Width: the opening width (m) of the gate at (for sideward opening).
- Sill Level Change: the sill level change (m) of the gate. Default is zero.
- Rate: the rate of increase or decrease of the gate opening.

EE uses the "state" of the structure to allow the user the high level of control required to simulate realistic scenarios. The logic of the process is that if the state of the structure is "1" then it is currently opening, and the rate of opening determined by the *Rate.* This will occur from that trigger level until it reaches another trigger level, or it reaches the *Height* assigned (presumably the maximum opening). Once it reaches the maximum opening the rate no longer applies and the final flow rate is maintained. When a trigger is reached with state of

"0" the gate will close, and flow decrease at the set *Rate.* This continues until it reaches the *Height* assigned (presumably fully closed).

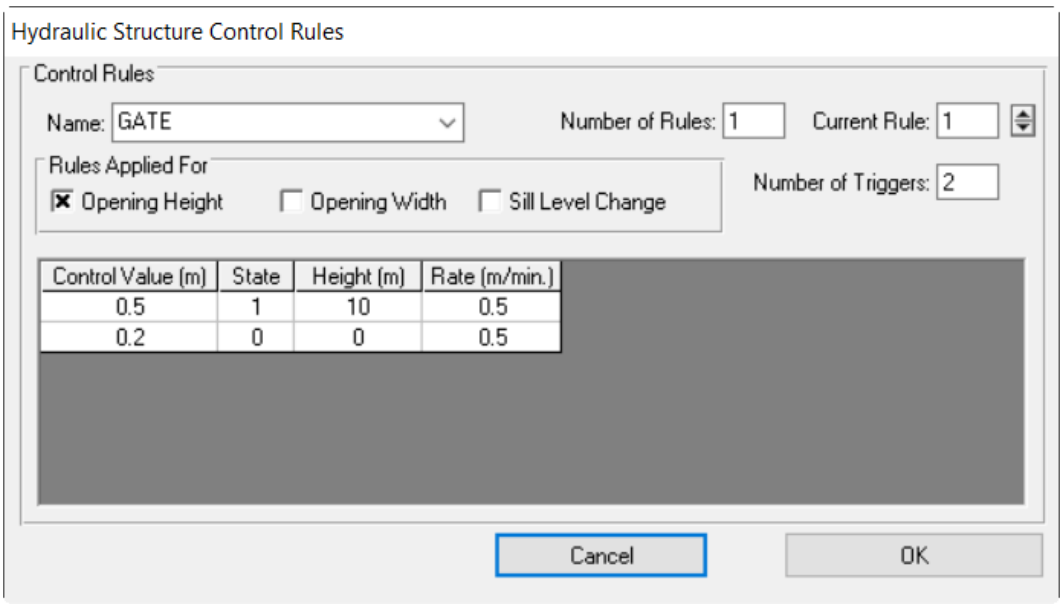


Figure 6-41 Operational rules configuration for sluice gate.

Figure 6-42 shows the form for setting the initial condition of the hydraulic structure. As described above, the location of the hydraulic structure trigger does not have to be in the same cell as the actual structure. Here the user can configure the upstream and downstream locations of the control triggers in the *Time Control Locations* frame. The user should also configure the initial conditions: whether the gate is opening or closing, and the initial height width and sill level change, if any.

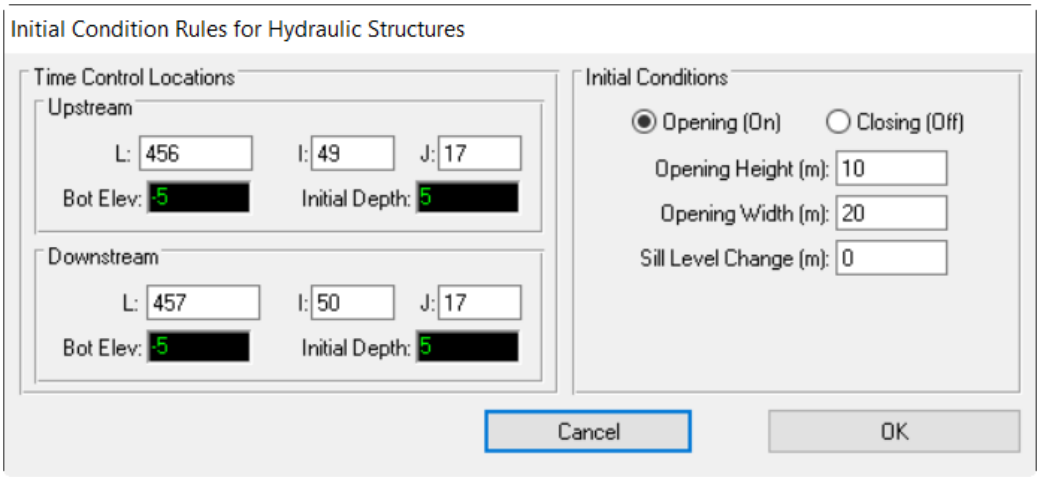


Figure 6-42 Initial Conditions for gate operation.

(3) Sluice Gate Equation Editor

To assist the user define the equation parameters for the sluice gate, a *Definition Editor* is provided as shown in Figure 6-43. The user should define the *Equation ID,* for various sluice gates to be defined.

The user should select the *Hydraulic Structure Type* from the dropdown menu, in this case it is “sluice gate”. The user should then enter the sluice gate dimensions, including sill elevation, height, width, super-critical weir flow, sub-critical weir flow, free sluice flow and submerged orifice flow parameters.

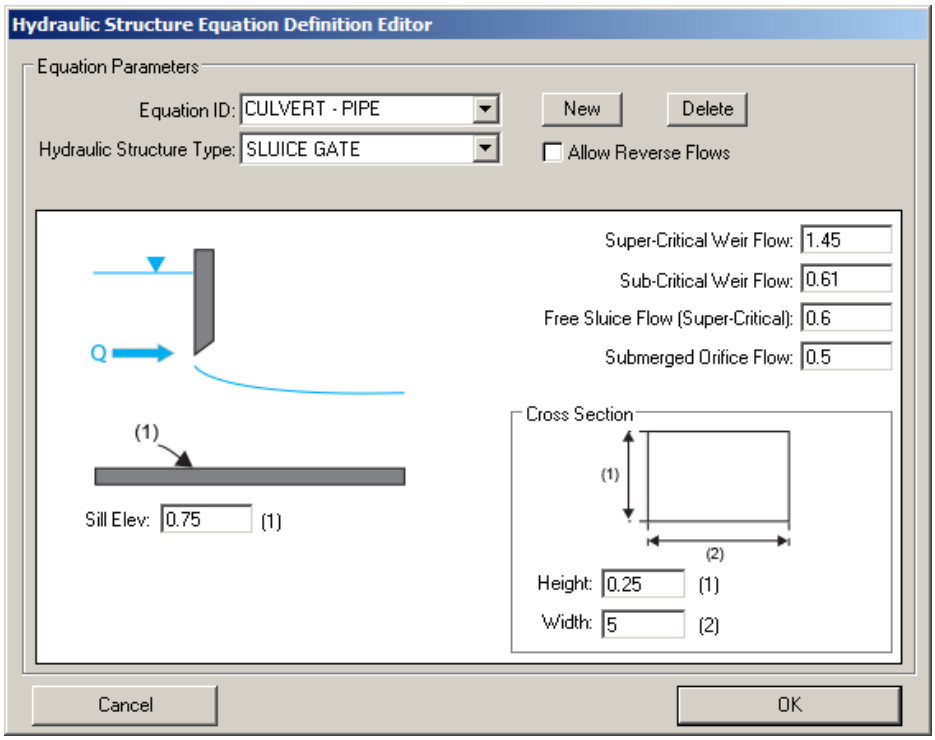


Figure 6-43 Hydraulic Structures Equation Definition: Sluice gate.

Flow through a sluice gate can be characterized by two basic parameters: the tranquility of the flow (i.e., subcritical or supercritical flow) and the water depth (i.e., gate submerged or not).

The equation for free sluice gate flow is

$$Q = C_3 W B \sqrt{2gH_u}$$

And for submerged orifice flow

$$Q = C_4 W B \sqrt{2g(H_u - H_d)}$$

Where C_3 and C_4 are discharge coefficients for free flow and submerged orifice flow respectively, *W* is the width of gate, $H_u$and H_d are headwater and tailwater, respectively.

6.3.7.3 *Weirs*

To assist the user define the equation parameters for a weir, a *Definition Editor* is provided as shown in Figure 6-44. The user should define the *Equation ID,* for various weirs to be defined.

The user should select the *Hydraulic Structure Type* from the dropdown menu, in this case it is "weir". Four cross-section types are available to the user: rectangle, v notch, trapezoid and broad crested. This example shows a broad crested weir definition form which requires the crest elevation, width of the weir, and the co-efficient of discharge.

An imported 3D structure image (COLLADA file), used to better visualize the hydraulic structure in *View3D* is shown in Figure 6-45.

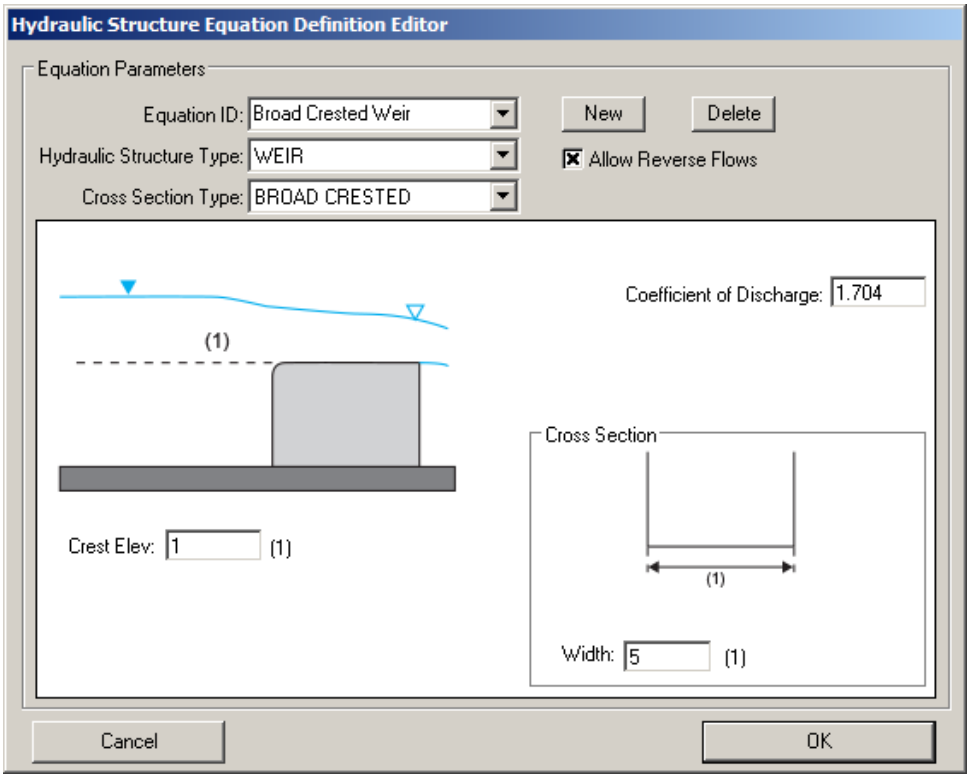


Figure 6-44 Hydraulic Structures: Weir definition form.

The equation to define the flow for weirs are as follows:

$$Q = \alpha C W \sqrt{2g} H_u^{\beta}$$

Where α is the ratio, C discharge coefficient and β is a exponent coefficient depending on the shape of flow cross-section.

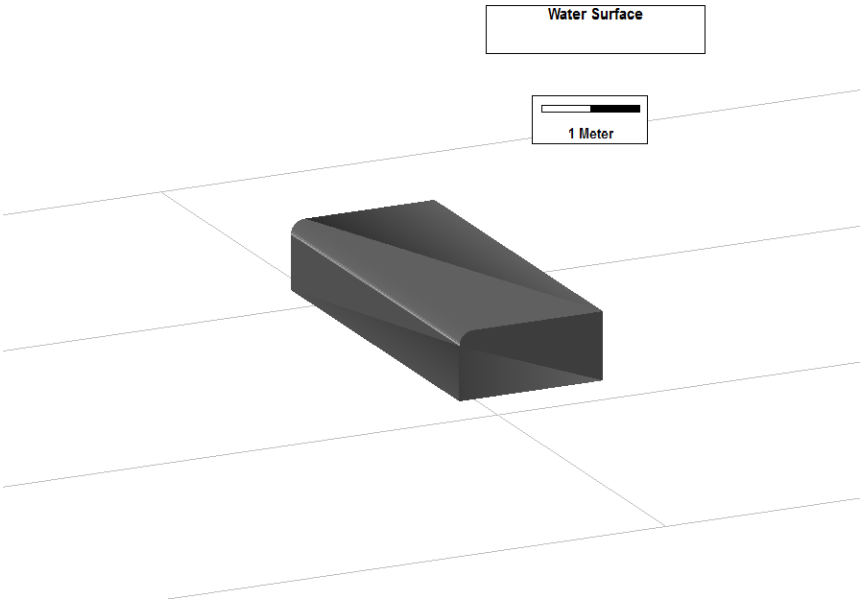


Figure 6-45 Image of broad crested weir in View3D.

6.3.7.4 *Orifices*

To assist the user define the equation for an orifice, a *Definition Editor* is provided as shown in Figure 6-46. The user should define the *Equation ID,* for various orifices to be defined.

The user should select the *Hydraulic Structure Type* from the dropdown menu, in this case it is “orifice”. Six cross-section types are available to the user: circle, half-circle, ellipse, half-ellipse, rectangle, v notch, and trapezoid. This example shows an orifice with a half circle cross section. The user should send the upstream elevation, diameter of the cross-section in meters, coefficient of discharge for a weir, and coefficient of discharge for an orifice.

An imported 3D structure image (COLLADA file), used to better visualize the hydraulic structure in View3D, is shown in Figure 6-47.

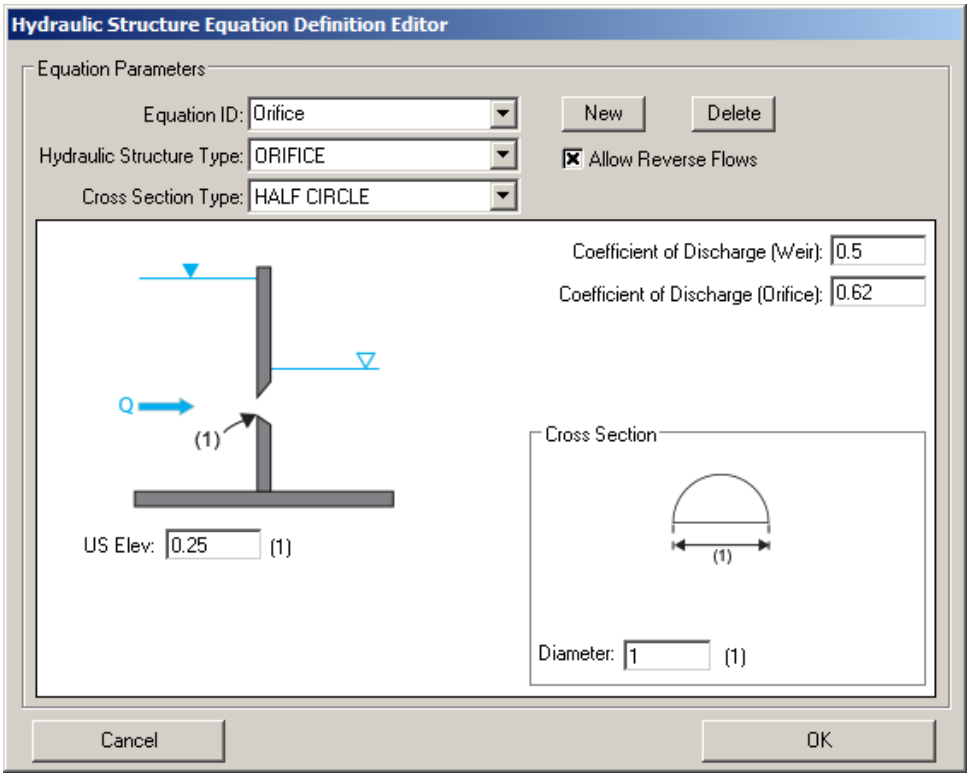


Figure 6-46 Hydraulic Structures: Orifice definition form.

The flow equations for the submerged orifice as defined as follows.

$$Q = CA\sqrt{2g(H_u - H_d)}$$

And for free orifice

$$Q = CA\sqrt{2gH_u}$$

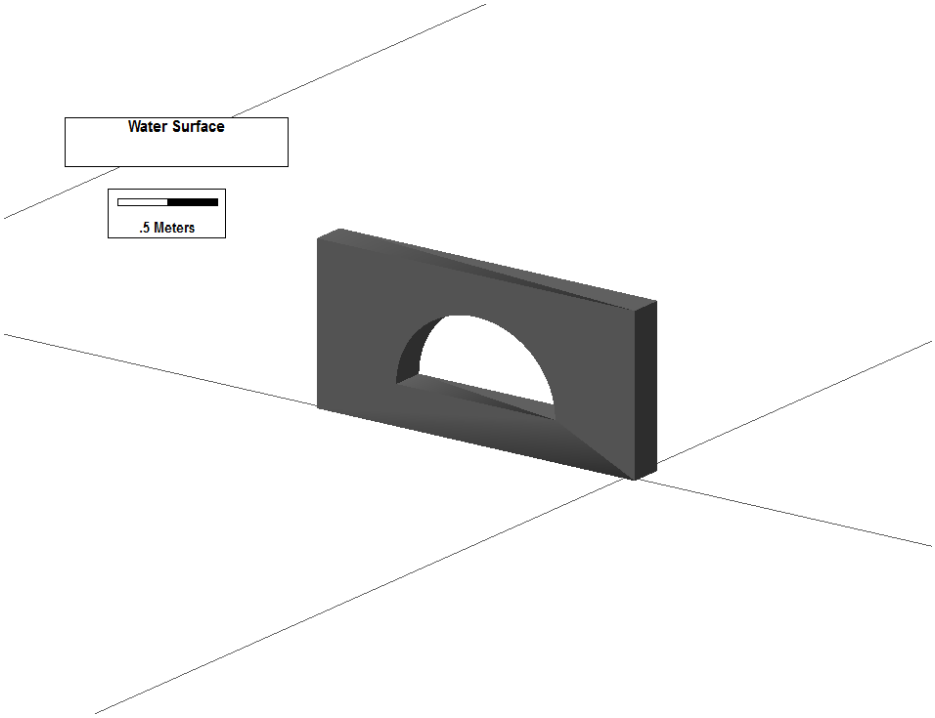


Figure 6-47 Image of half circle orifice in View3D.

6.3.8. Open Boundary Condition Editor

The form for the open boundary condition is shown in Figure 6-48. In the *Current Boundary Group Information* frame the user may assign the *Open Face Direction* and *Type of Open BC* with drop down menus.

There are four *Types of Open BC* that the user may select from:

- Elevation Specified
- Radiation Separation – Zero Tangential
- Radiation Separation – Free Tangential
- Elevation Specified – Free Tangential

Elevation Specified is the standard approach for water surface elevations, using a combination of harmonic constituents and time series. With the *Radiation Separation* boundary condition the incoming wave at an open boundary is separated from the outgoing wave and the incident wave is assumed to be twice the surface elevation required (Zhou, 2014). By default, the outgoing characteristic is left undefined, allowing waves generated interior to the model domain to pass outward across the boundary with no reflection. When using the *Elevation Specified* option the user should deactivate one cell in the corner of the grid where two open boundary directions meet. When using a *Radiation Separation* boundary then the user must have 2 cells deactivated in the corner as shown in Figure 6-49. This is because an open boundary always sets a mask in E-W direction for a S boundary, likewise an open boundary sets a mask in N-S direction for a W boundary.

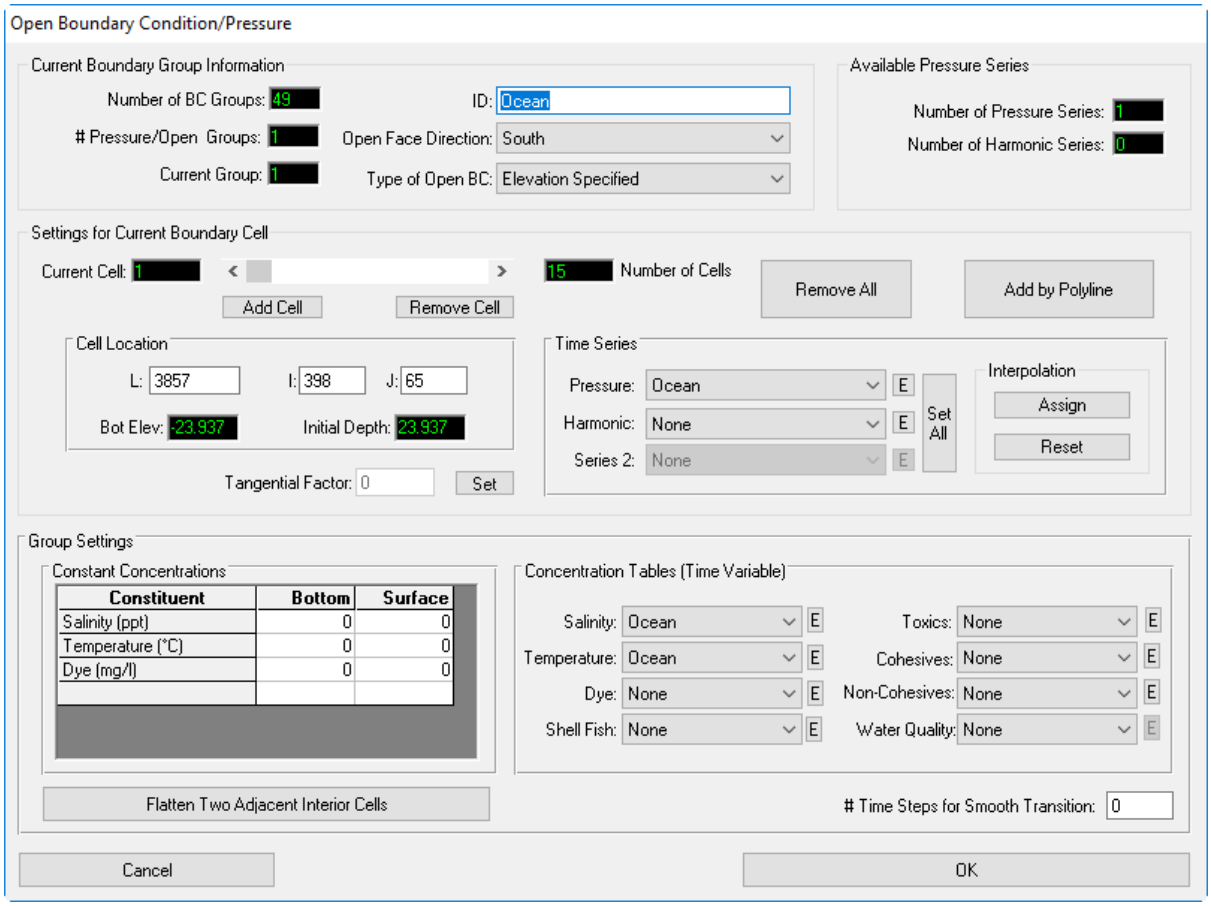


Figure 6-48 Boundary Condition Settings – Flow Boundary.

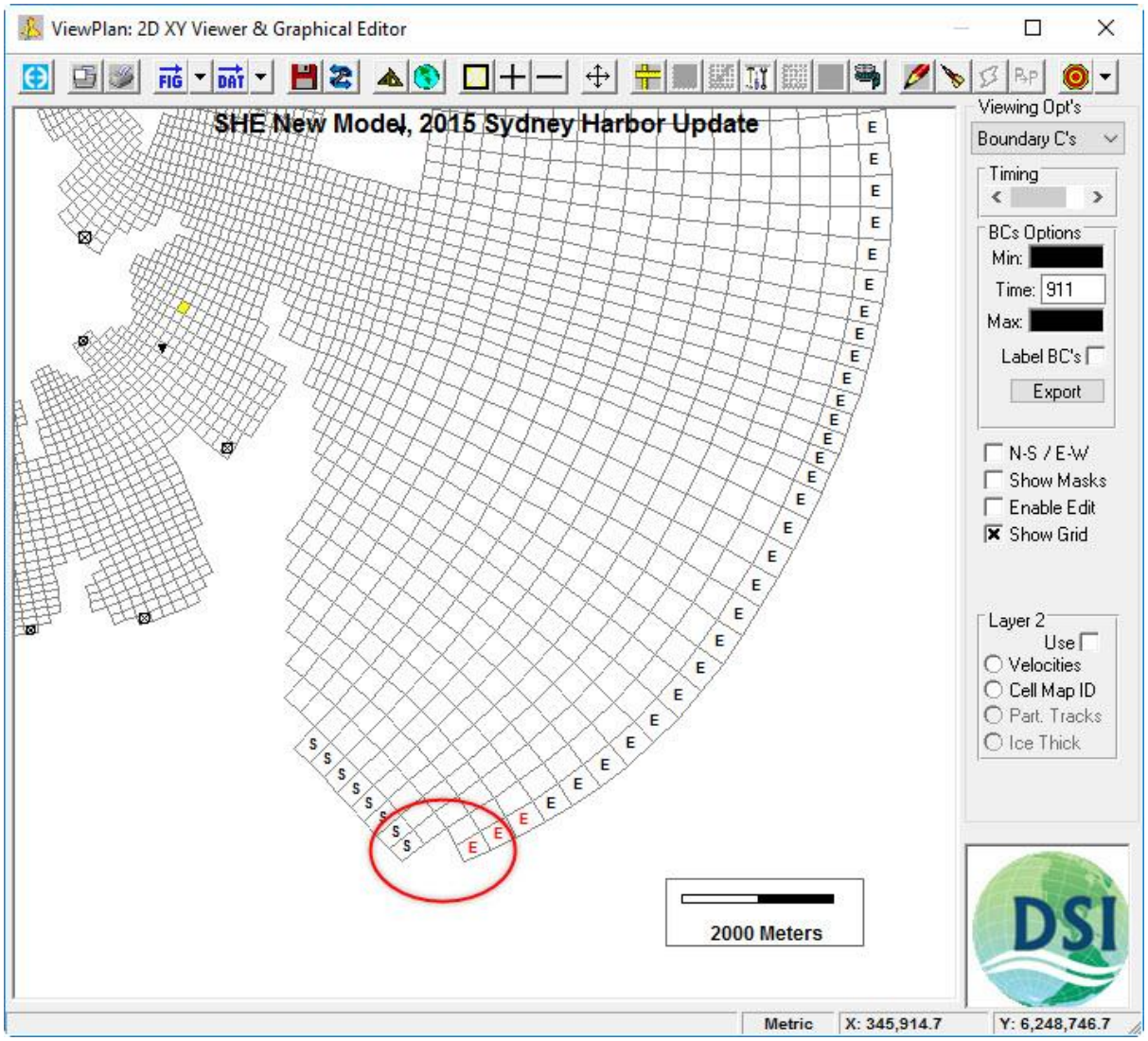


Figure 6-49 Open Boundary Condition Settings – delete corner cells for Radiation Separation option.

Users should also be aware that if they have rapidly changing bathymetry at the open boundary then it is possible to get mass balance errors. Rapidly changing bottom elevation at the open boundary can lead to EFDC becoming unstable. To easily resolve this issue it is recommended that the first two cells going out in the direction of the open boundary i.e. in southward direction for the southern open boundary, be at the same elevation. To facilitate this a new tool has been added to the *Open Boundary Condition* form as shown in Figure 6-48. Selecting the *Flatten Two Adjacent Interior Cells* button will set all the first two cells the same bottom. This feature works on a group basis and uses the type ie the direction N, S, E, W, to know which cells need to be smoothed.

In EE8.4 a BC option has been which allows the use of an elevation specified flow but allows for a “free tangential”. Normally an open BC requires two cells at the border. In some models where there are tight boundaries this can be restrictive. The *Elevation Specified-Free Tangential* option allows EFDC to only use one cell at the border.

Group Settings Frame

For open BC's, unlike flow BC's, the concentrations specified in the *Constant Concentrations* frame are always used for boundary assignments. The constant concentrations will be added to any time series concentrations defined. For example, if the user has a tidal time series of salinity that varies from 5 to 10 ppt and specifies a concentration for salinity of 10 ppt in the *Constant Concentrations* frame, EFDC will apply time variable salinity concentrations ranging from 15 to 20 ppt.

Interpolation Sub-frame

The *Interpolation* frame provides access to two functions:

- *Assign* generates new pressure series for the unassigned cells interpolated from the cells along the boundary that were assigned prior to pressing *Assign*. Interpolated series always have an ID that begins with "I:" to identify the series as an interpolated series. Each unassigned cell has its own interpolated series generated, then that series is assigned to that cell. Two or more prior assigned cells to existing series is required. The cells at either end of the open boundary must be defined for this process to work properly.

- *Reset* scans the current boundary group and finds any cells whose pressure series ID begins with "I:". If it finds cells with an interpolated series, it deletes the series and sets the pressure series index equal to zero (i.e. resets the assignment).

6.3.9. Harmonic Tidal Boundary Series

EFDC_Explorer provides two approaches for representing tidal forcing boundaries, the simple harmonic forcing boundary and the astronomical tidal forcing.

6.3.9.1 *Simple Harmonic Forcing*

With the simple harmonic forcing boundary, harmonic constants are specified and EFDC computes the forcing time series using a simple harmonic formulation. This time series then can be combined with a normal pressure time series for different correction purposes, such as storm surge simulation or datum correction.

This forcing boundary type can be selected in the *Domain | Boundary Conditions* tab and clicking "E" (edit) button for *Harmonics* which displays the form shown in Figure 6-50. The user can specify the forcing of an arbitrary number of combinations for harmonic components from 234 predefined tidal constituents. This is done by LMC to select the *Global Tidal Constituents* and then pressing the arrow button on the form to populate the Harmonic Constituents to the right.

These constituents may be sorted either by angular speed (frequency) (degrees/hours) or period (days) with the radial button in the *Sorted by* frame.

In the *Harmonic Coefficient Forcing Type* frame the user has three options; constant, linear variation and quadratic variation. It is generally recommended to use a constant forcing. With the other two options the user must specify additional one or two amplitude and phase for each tidal constituent, respectively. EFDC will then vary the final (i.e. applied) phase and amplitude for each tidal constituent using the specified method and values.

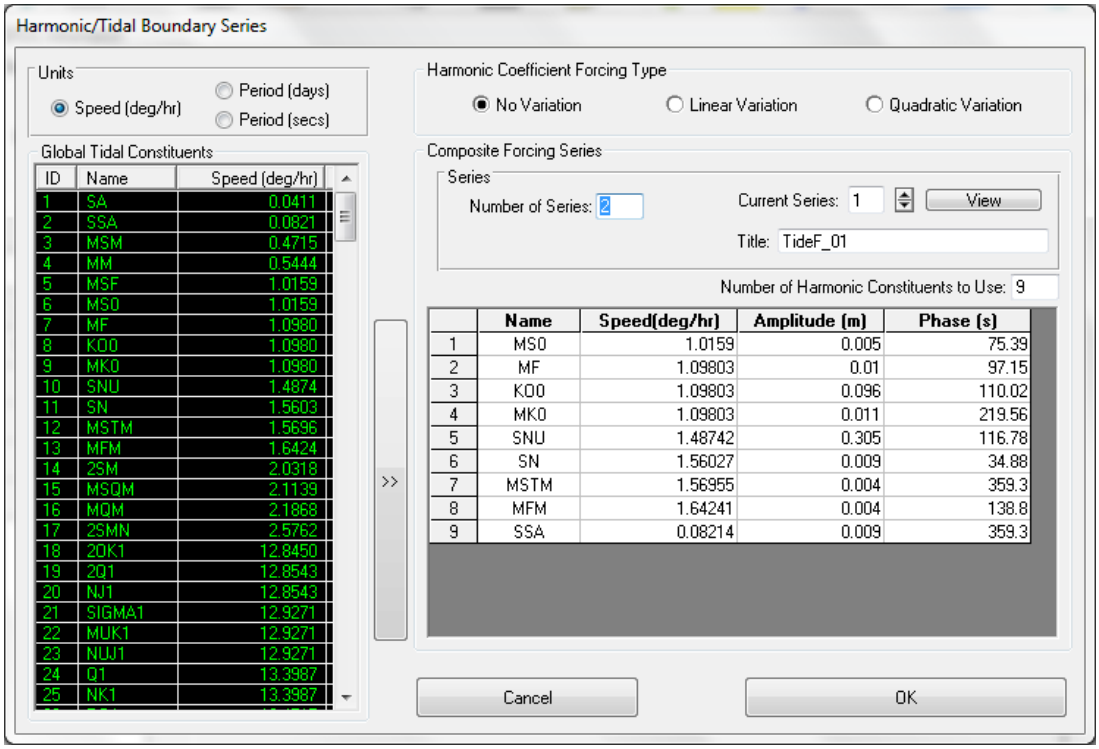


Figure 6-50 Domain Tab: Simple Harmonic Forcing.

Note that when specifying an open boundary, it is possible to specify harmonic tide as an independent boundary condition or both a harmonic series and a pressure series. In the latter case the harmonic series is added to the pressure series for the final pressure to be used by EFDC in the solution. That is how the user could adjust for a datum shift (i.e. the pressure series would just be a constant offset).

6.3.9.2 *Astronomical Tidal Forcing*

EFDC_Explorer provides a *Tide Generator Tool* for generating a pressure time series computed as astronomical tidal forcing.

The basic formula for the oscillation of water level due to astronomical tides is based on the combination of the tidal constituents as follows:

$$\eta = z_0 + \sum_{i=0}^{N} \mathrm{f_i H_i cos}\{\omega_i t + (V_0 + u)_i - g_i\}$$

In which

η = tide level (m)
z_0 = mean sea level above the datum (m)
$\mathrm{f_i}$ = nodal coefficient to adjust amplitude for the i^{th} component
$\mathrm{H_i}$= amplitude of the i^{th} component (m)
ω_i= speed or angular velocity of the i^{th} component (radians per second)
t = time (s)
$(V_0 + u)_i$= astronomical arguments of the i^{th} component
g_i= phase lag of the i^{th} component from Greenwich longitude

It should be noted that the units used in EE can be changed so that they are convenient for the users.

The *Tide Generator Tool (EFDC Input Tides)* provides the ability to generate pressure time series from astronomical tidal forcing. This is accessed via the "E" button for the *Pressure* series in the *Domain tab | Boundary Conditions.* In the *Data Series: Water Surface* form the user should click the *Generate from Harmonics* button. This will display the form shown in Figure 6-51. The user should complete the form in a manner similar to that described for simple harmonic forcing above. The user must specify the number of constituents, and import the data file for the constituent parameters (name, amplitude, phase) if available.

Each series can have its own name, and the user may apply a datum shift (z_o) in meters or time shift in hours to the series. A row may be deleted by selecting Shift and RMC. An example of the data structure is provided in Appendix B.

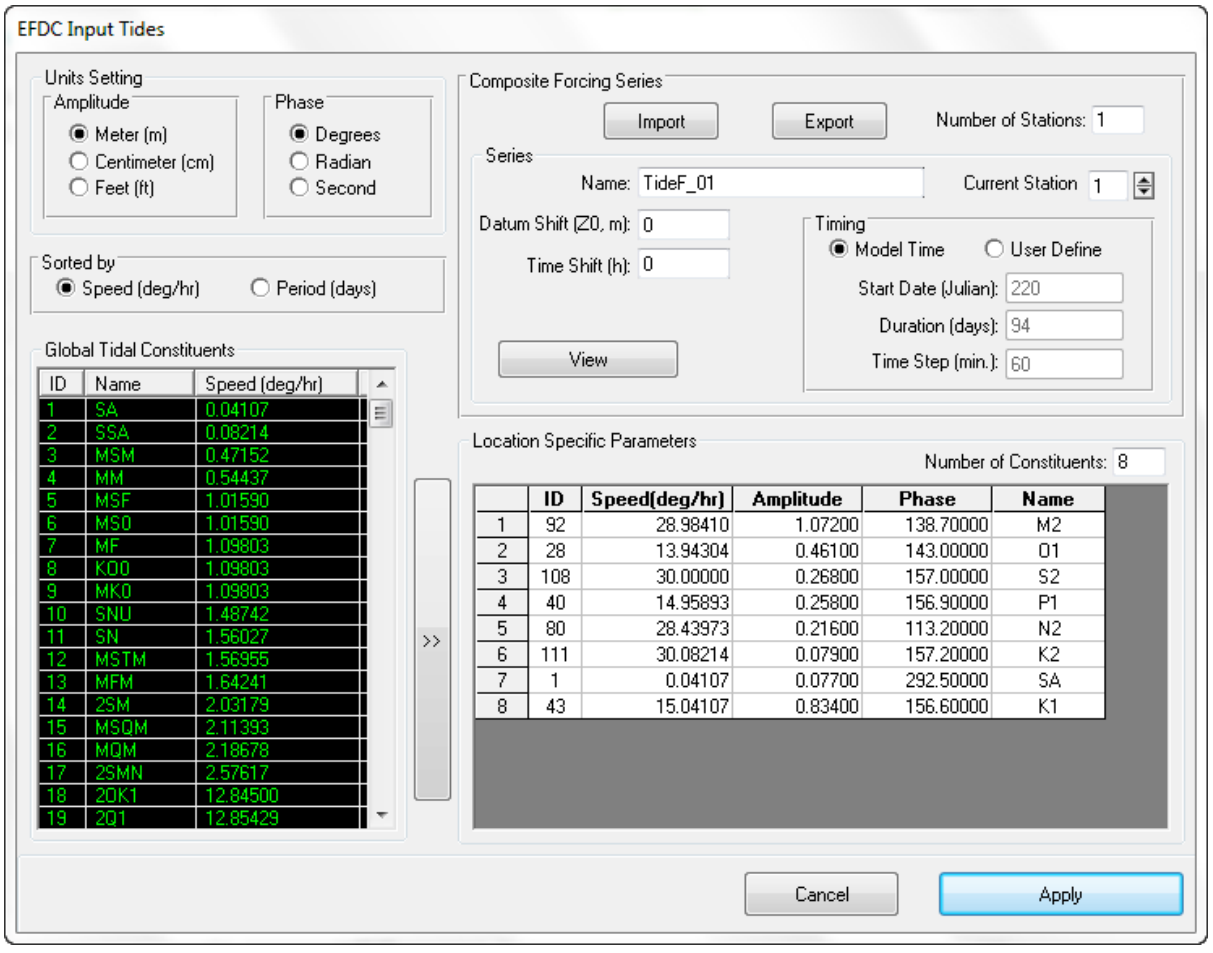


Figure 6-51 Domain Tab: Astronomical Tidal Forcing

6.3.10. Withdrawal/Return BC Editor

EFDC_Explorer fully supports the withdrawal/return flow boundary condition of EFDC. The user interface for the withdrawal/return boundary condition editor is shown in Figure 6-52. As shown in the *Time Control* drop-down menu, EE allows the operation of withdrawal/return structures using time-series of flow (and constituent rises), as well as operation of the structures based on operational rules.

The user may specify the location of the withdrawal and return cells directly for each cell in the group in the *Withdrawal Cell* and *Return Cell* frames.

In the past EFDC did not allow the specification of the rise/fall for the water quality constituents. A "0" rise was always assumed. EFDCPlus was modified to allow the rise/fall for each water quality constituent to be specified, if using the time series option (i.e. QWRS.INP file). The constant rise/fall feature is only available for the original set of EFDC parameters of salinity, temperature, dye, toxics and sediments. These may be set in the *W/R Flow and Concentration Rise/Fall Settings* frame.

The user may also specify the momentum options for the withdrawal and return cells in the bottom frame. Use the drop down menu to select momentum flux to be ignored, or positive or negative in the U and V faces. The user may also set the momentum width for both withdrawal and return flows. In the case of the return flow the user may also set *Return Flow Horizontal Angle.*

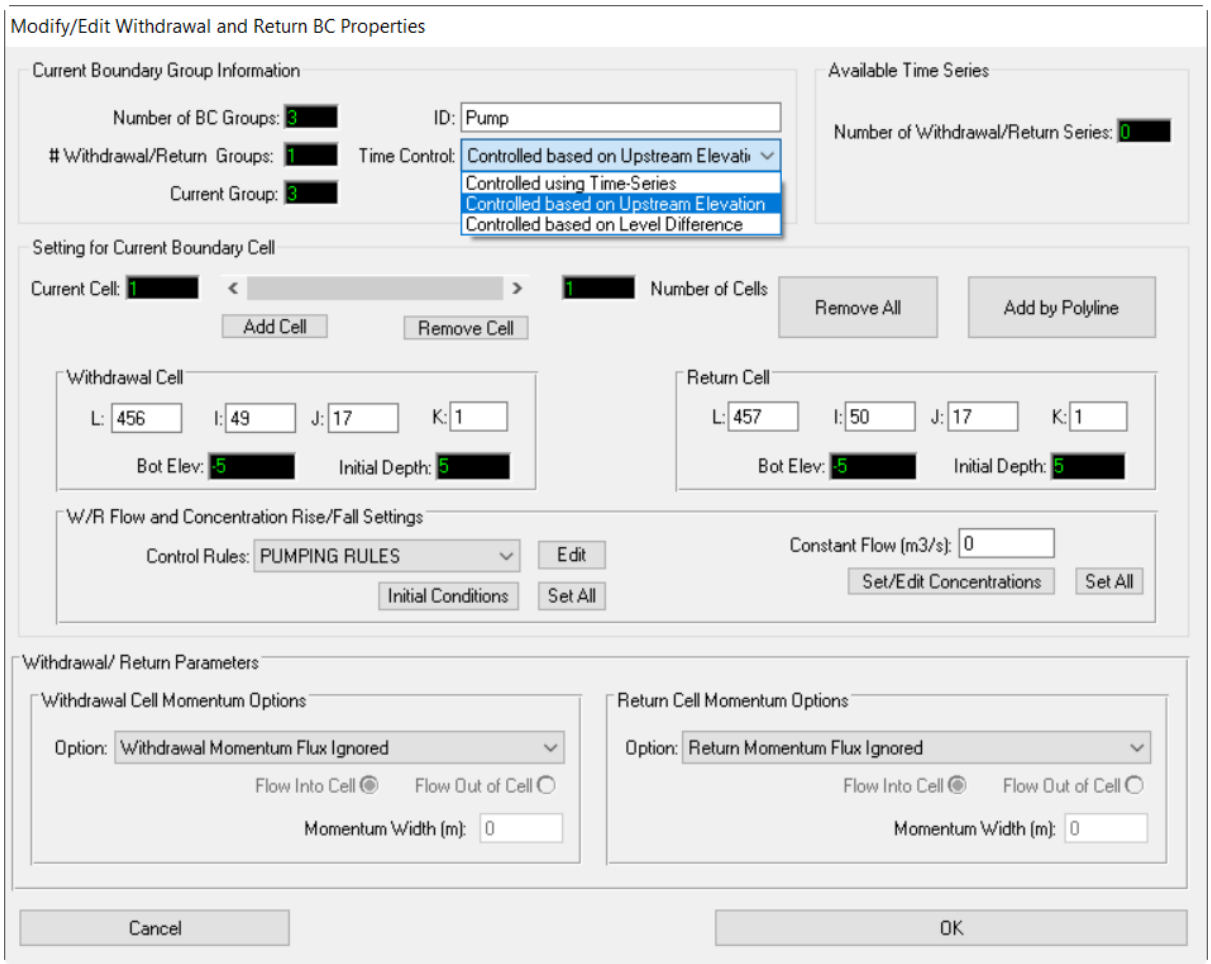


Figure 6-52 Boundary Condition Settings – Withdrawal Return.

With release of EE8.4 the W/R boundary condition has been modified to allows the operation of the structures based on operational rules. This has been implemented in an approach similar to that of the rule-based operations of sluice gate for hydraulic structures. One purpose of this enhancement is to allow the user to simulate rule-based pumps. It should be noted that at this stage this feature only acts on the flow series, and maintains the concentrations as constant i.e. there is no rise or fall for the WQ constituents.

The interface for the withdrawal/return boundary condition setup for a based on control rules rather than a time series is shown in Figure 6-52. The two control rule approaches require the user to configure rules for the structure. These rules are set in the *W/R Flow and Concentration Rise/Fall Settings* | *Control Rules*. Here the user may edit existing rule or create new ones as well as set the initial conditions

Figure 6-53 shows the settings for the rule-based withdrawal/return boundary. In this case a pumping station has been configured using the withdrawal/return boundary conditions. The approach is similar to that described for rule-based sluice gate operations: the user configures the number of sets of rules, and uses the up and down arrows to select the *Current Rule* set. Each set can have a number of different triggers, in this case two. The triggers are determined by the control value. This is the upstream elevation, or the difference in upstream and downstream head, depending on which option the user has selected.

EE uses the "state" of the structure to allow the user the high level of control required to simulate realistic scenarios. The logic of the process is that if the state of the structure is "1" then it is on, discharging the *Flow Discharge* set. This will continue from that trigger level until it reaches another trigger level. When a trigger is reached with state of "0" the pump will stop, and flow changes to the new *Flow Discharge.* The prevent instability in the system with sudden turning on and off of large flow rates within one time step, EE allows the user to set a rate of increase and decrease of the flow discharge with the *Rate* variable, in cms/minute.

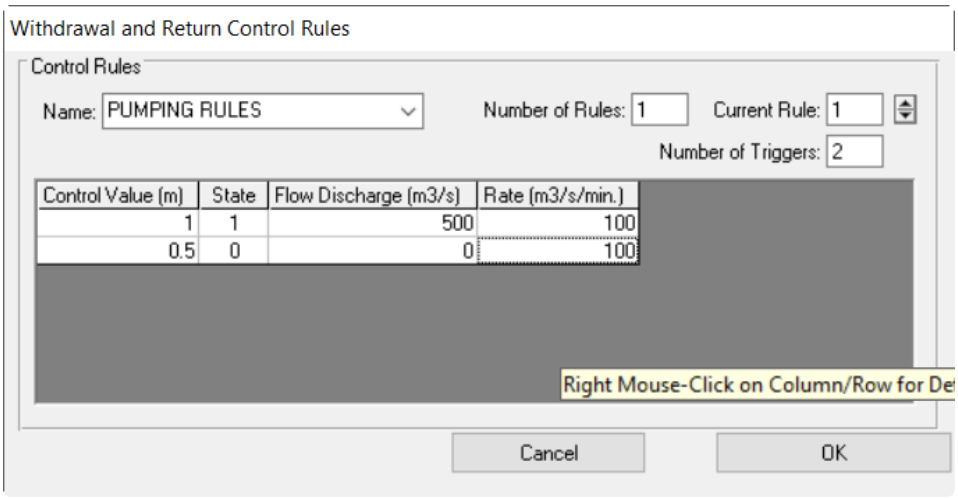


Figure 6-53 Withdrawal/Return BC: Control rules settings.

Figure 6-54 demonstrates the effect of the operation of the pumping station. In many cases the user will configure a simple system as described here. In this case it would seem the "state" setting is redundant as the pump is simply on or off. However, in some more complex systems there may be two or more pumps at a particular location. These may be configured as lead and lag pumps, with the lead pump turning on at a low water level and lag pump only turning on at a higher level. In these cases the user may want both pumps to stay on even when water level drops below the trigger level of the lead pump. For this reason the "state" setting is required to fully simulate the complexity of the system.

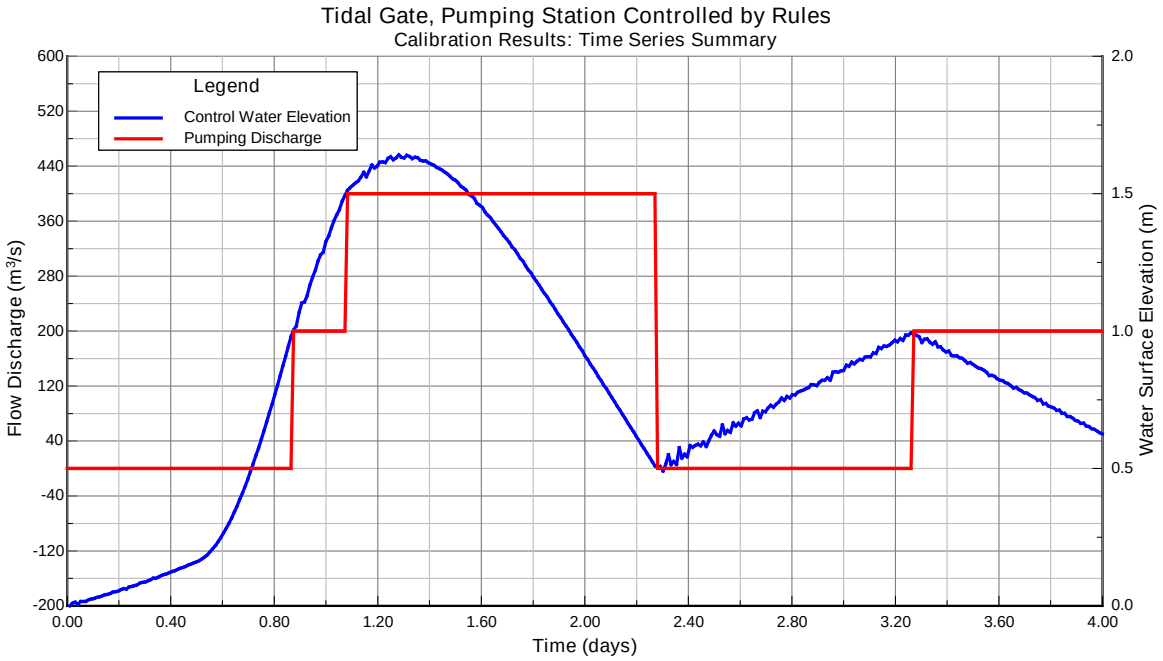


Figure 6-54 W/R boundary condition pump example.

6.3.11. Jet/Plume Boundary Condition

6.3.11.1 *Jet/Plume BC Editor*

One of the options available when a user creates a new boundary condition is type 8, *Jet/plume*. The form for editing the Jet/plume BC is shown in Figure 6-55. EFDC+ has been enhanced to work smoothly with the EE defined jet/plume settings.

Although similar to a flow boundary condition or withdrawal/return user interfaces, the settings for jet/plume boundary conditions are unique to this boundary condition and associated sub-model (described below). In the *Jet/Plume Type* drop-down menu the user specifies the flow data of a jet/plume boundary to link to with the time series options. These options include by-passing (ICAL=0), using a flow time series (QSER, ICAL=1), using a withdrawal return time series (W/R, ICAL=2). The user may also specify the frequency, in number of time steps, to update the jet/plume calculations.

When using the *J/P with W/R Series* option the user should set the location of the withdrawal and discharge locations as they would for a W/R boundary. However, it should be noted that the vertical layer setting for the discharge is not used by EFDC, though withdrawal layer setting is used. This is because J/P boundary automatically distributes the discharge flow from the plume over the vertical layers based on the flow characteristics and ambient temperature.

In the *Diffuser Port Settings for Current Cell* frame the user may specify the settings for each discharge cell including the number of ports, the elevation and the diameters of the port outlets, as well as the horizontal (azimuth) and vertical (altitude) angles. It should be noted that the flow in the QSER file is multiplied times the number of ports for the total flow for that BC cell. For example, if a user has 10 ports with each port having a time series of 0.15 cms assigned for the jet, then the total flow is 10 * 0.15 cms = 1.5 cms.

The setting for the jet/plume discharge cell is not dependent on the K value. It is only dependent on the "Elevation" set for the port. For this reason he K box for the cell is greyed out.

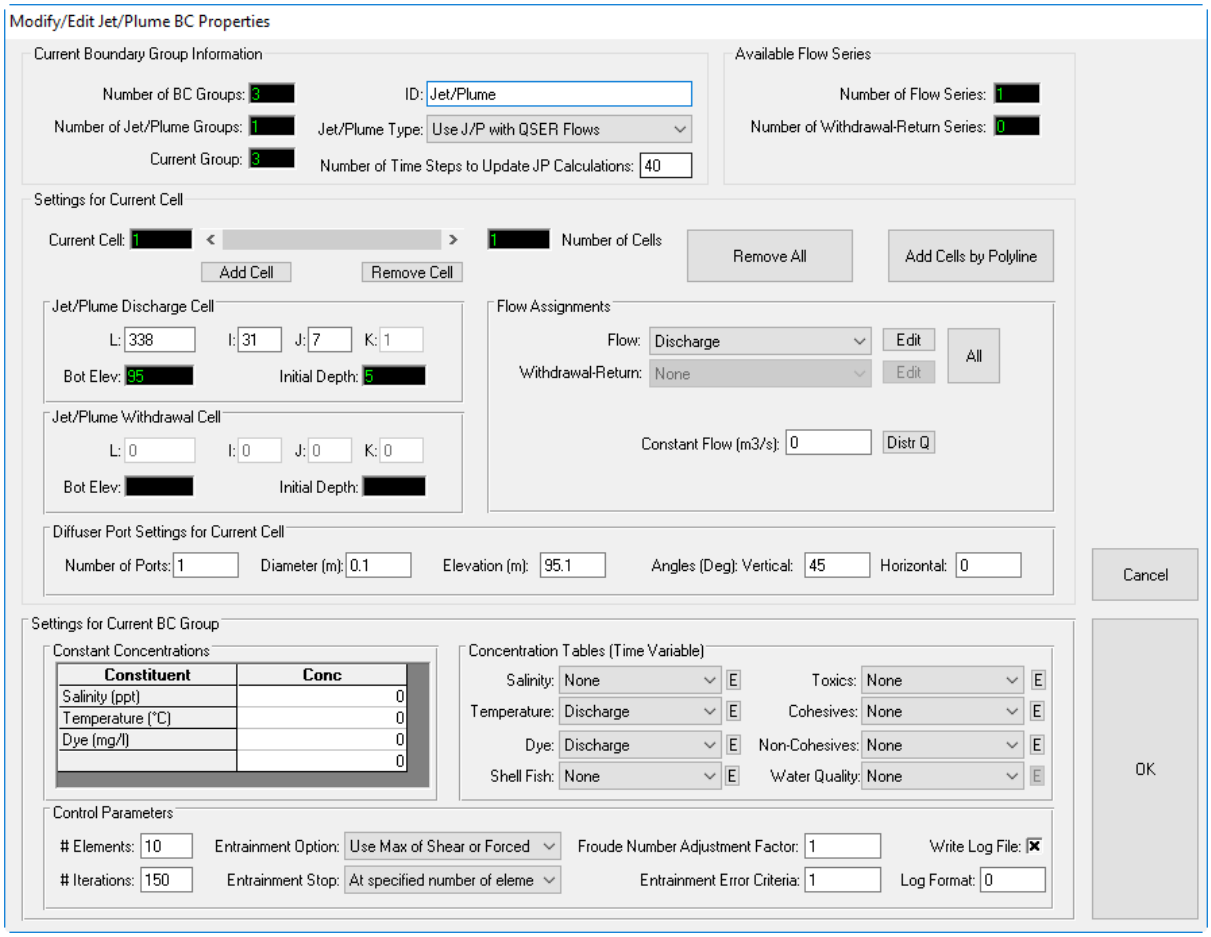


Figure 6-55 Modify/Edit Jet Plume BC Properties.

The *Control Parameters* frame allows the user to set various parameters relating to the plume entrainment. The user can set the *#Elements*, which is the maximum number of vertical computational elements along the length of the jet or plume. The entrainment calculation is an iterative process, the maximum number of iterations to be made if convergence is not obtained is set in the *#Iterations* form.

In the *Entrainment Option* menu the user sets whether to use the maximum of shear and forced entrainment, or to use the sum of shear and forced entrainment.

The *Entrainment Stop* menu provides three options for stopping the entrainment: stop at a specified number of elements (ISTJP=0), stop when the center-line penetrates the bottom or the surface (ISTJP=1), or stop when the boundary penetrates the bottom or the surface (ISTJP=2).

Other options include setting the *Entrainment Error Criteria,* which is a factor used in the iterative solution. An adjustment factor for the *Froude Number* may also be set. EFDC can write out a log file of the Jet/plume in various formats. Specify 1 for full ASCII, 2 for compact ASCII output at each update, 3 for full and compact ASCII output, and 4 for binary output.

Figure 6-56 illustrates an application of the jet/plume BC for a temperature distribution in the case of two diffusers in a weak flow stream.

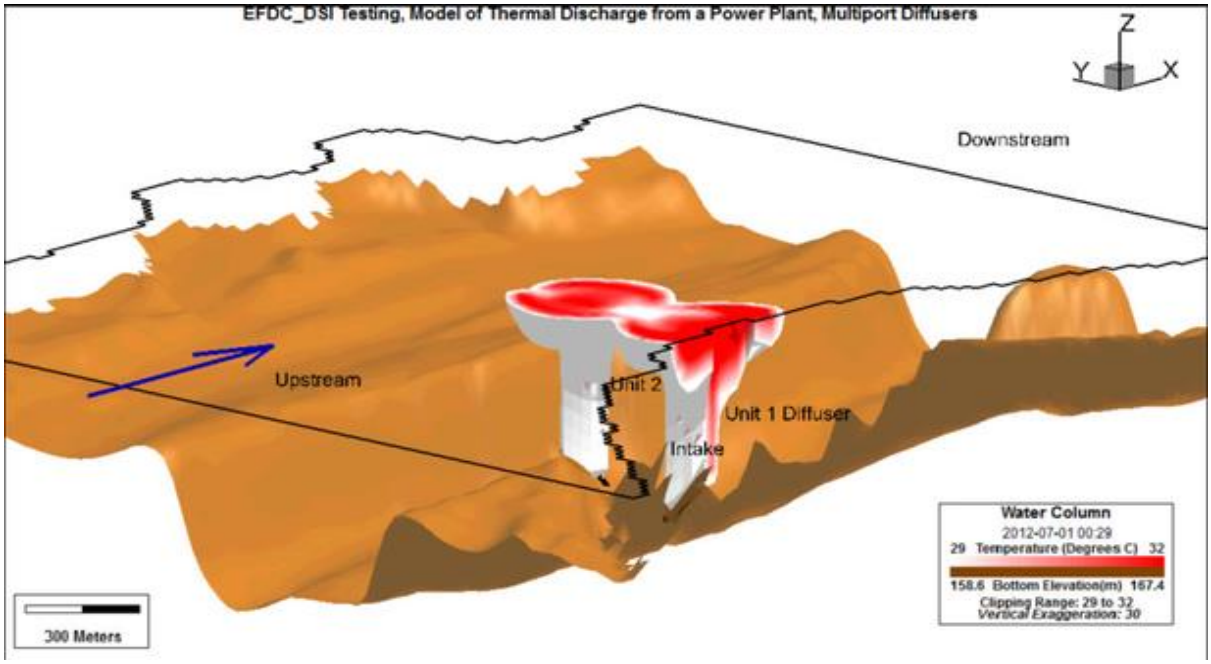


Figure 6-56 Thermal Discharge from Multiport Diffusers using Jet/Plume BC.

6.3.11.2 *EFDC Jet/Plume Sub-model*

The Jet/Plume sub-model in EFDC is partly described in EPA's "Review of potential modeling tools and approaches to support the BEACH program." Here an early version of the sub-model is described as providing a buoyant jet near-field dilution and mixing zone sub-model that is incorporated directly into the EFDC. The model simulates single-port and merging multi-port discharges using a three-dimensional extension of the Lagrangian formulations used in the UM model (Baumgartner et al.. 1994) and the JETLAG model (Lee and Cheung. 1990). The jet/plume model is unique in its use of unsteady fully three-dimensional ambient velocity density and concentration fields and realistic bathymetry for trajectory, entrainment, and dilution calculations. For multi-port discharges the merging of individual port plumes into multiple coalesced plumes is simulated. In addition to simulating the near-field and far-field concentration of dissolved contaminants, the jet/plume sub-model simulates sediment transport and the transport and fate of sorptive contaminants including the settling and bed exchange of the suspended and sorbed material (note that water quality parameters are not simulated at this stage).

The Jet/Plume sub-model automatically updates multiple outfalls as ambient conditions and outfall discharges evolve in time. The near-field solution is automatically coupled to the EFDC model's far-field transport and fate simulation, allowing the ambient concentration field to represent the historical influences of all unsteady discharges. The Jet/Plume sub-model is particularly well suited to near-shore coastal simulations. The hydrodynamic component of the EFDC model is capable of simulating unsteady tide-driven, wind-driven, and wave-driven currents including long-shore and across-shore currents associated with incident wave transformation and surf zone wave breaking.

6.3.12. Data Series Editing Form

A common function needed for all the boundary forcing data series is the need to edit, view and compare these series. These processes and other time series utilities are available using

the *Data Series* time series data management form (Figure 6-57). This form is broken down into several sections.

The *Title Block* displays and allows edits of the title block EFDC requires for the specific boundary file(s). The number of lines varies between boundary forcing files. EFDC_Explorer ensures the correct number of header lines is used for each type of file. Pressing *Reset* will overwrite the current header with a "standard" header for that type of file, using the *Project Title* as the project specific label. The user may also manually edit the header and EFDC_Explorer will maintain those edits. However, only one header is available for all the different series.

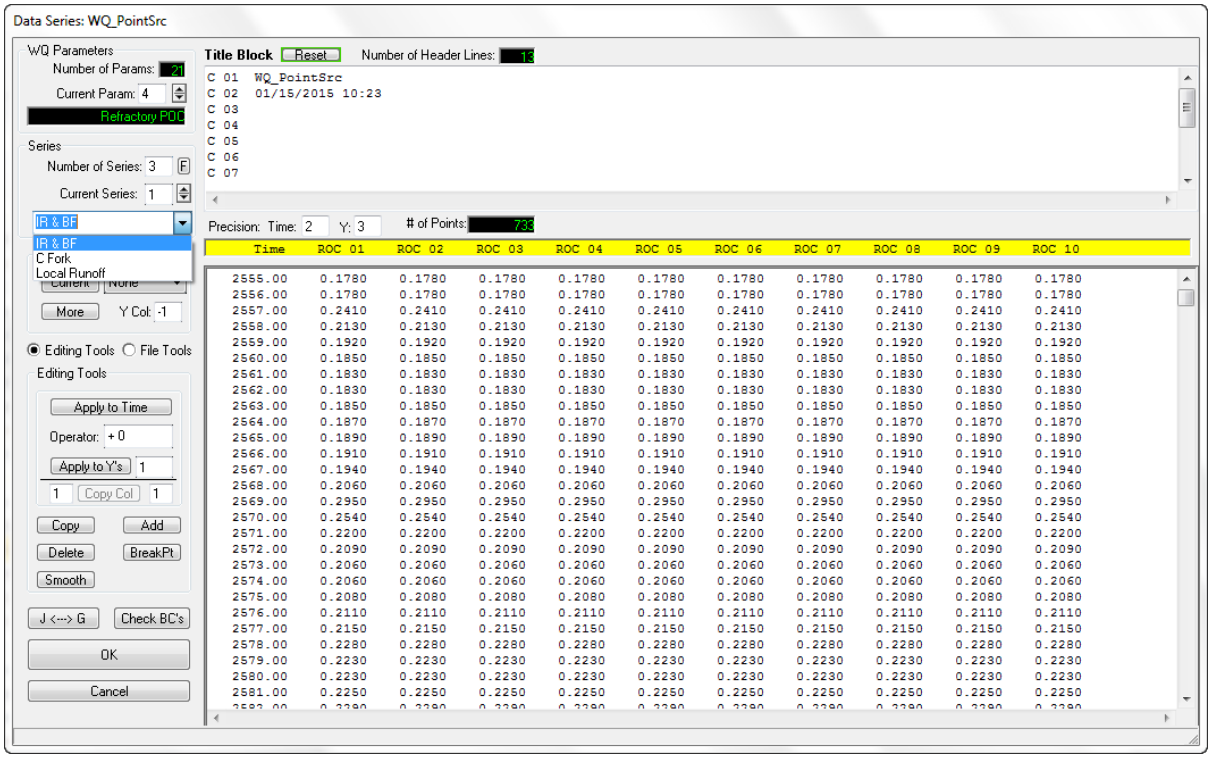


Figure 6-57 Data Series Editor: WQ boundary conditions.

Sometimes, the type of data series has fixed groups for a given series. This is the case for sediments, toxics and water quality. To view and/or edit a series the user must specify the series (e.g. "1" in Figure 6-57) and a group (e.g. "4" for Refractory Organic Carbon). The user can then edit, plot, compare, copy, etc. the currently selected series/group. If the type of series does not use groups (e.g. flows) then the group frame will not be displayed.

The *View Series* frame provides the user with the ability to plot the time series. The *More* button provides access to additional plotting functions. The *Show All Series* button compares already defined series to each other. The user can turn on or off the series by selecting 1 or 0 for each series as shown in Figure 6-58 The *Filters* frame includes *Daily Minimum* and *Daily Maximum.* This allows the user to show any series the user selects on the same plot for the selected parameter in a given column (*Y Col*), also shown in Figure 6-58.

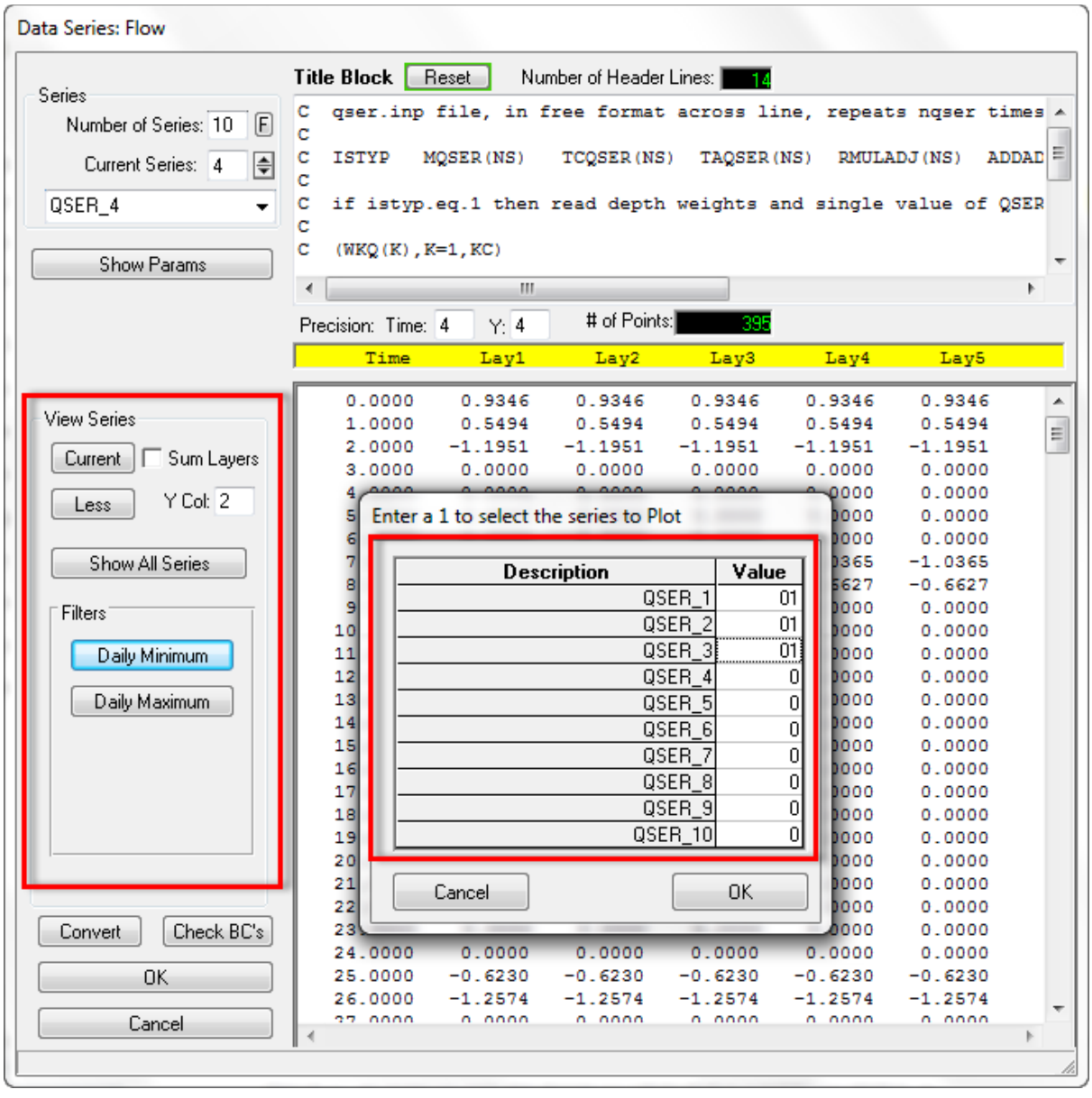


Figure 6-58 Data Series Editor: View Series options.

The option buttons for *Editing Tools* and *File Tools* change the function of the frame below these buttons. The *Editing Tools* is the default frame. This option allows the user to apply *Operator's* to the date column or any/all of the data columns, or a range of columns. The user may enter the column number, or if they want the operator to apply to layers from 1 to 3, then they should enter "1-3" in the *Apply to Y's* input interface. This will apply the operator to these layers for the whole time series. Entering "-1" will apply the operator to all columns and a "P" in front of this will apply to parameters instead of layers.

The following is a description of the various "Editing Tools" functions:

- *Copy* To copy the current series to a new series at the end of the current list of series.

- *Add* Adds a specified series to the current series. This process overwrites the current series with the results. The timing does not need to match. EFDC_Explorer uses the current series' timing and computes the value of the second series at the times of the current series.

- *Delete* Deletes the current series. All the series number above the current series are renumbered to the next lower number and boundary conditions that use these series are updated.

- *BreakPt* This function "breakpoints" the current series. The breakpointing process removes consecutive points that have the same value (within a user specified tolerance), leaving the 1st and last points with the same value. An example of a series breakpointed is shown below.

Raw Series

X	Y
1.0	0.2
2.0	0.3
3.0	0.3
4.0	0.3
5.0	0.3
6.0	0.4

Breakpointed Series

X	Y
1.0	0.2
2.0	0.3
5.0	0.3
6.0	0.4

- *Smooth* Applies a smoothing process to the current series. The weights are 1/6, 2/3 and 1/6. A single pass is applied. The user can apply the smoothing repeatedly to attain the desired smoothing. Since the current series is overwritten with the results of the smoothing process, it is useful to export (via the Time Series plotting function) the unsmoothed time series first. As smoothing is applied, the user can import (via the Time Series plotting function) the original series for comparison.

In the *Series* form there is now also a dropdown menu with a list of boundary conditions that may be selected from as shown.

Note that the time series form has several additional keystroke functions not available on the main form. Figure 6-59 shows a screen capture of the information message box (F2).

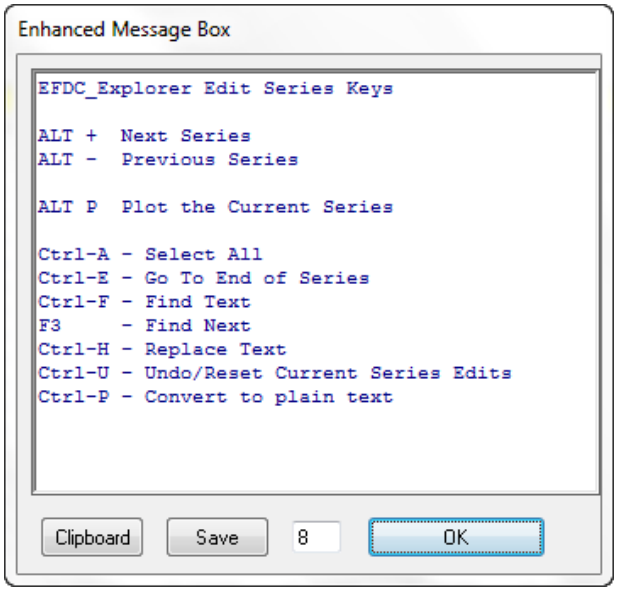
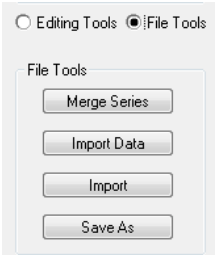


Figure 6-59 Time series keystroke function information message



The *File Tools* option changes the frame to that shown here. The purpose of this option is to provide ways to populate the time series from existing data files. In addition to importing data from existing files, the user can paste data into the time series from another application (e.g. Excel) as long as the formats are compatible.

- *Merge Series* Imports an existing ASCII data file and then interpolates the data series onto the timing of the current series. This is especially useful for time series that have multiple groups for the same series (i.e. sediments, toxics, and water quality). Figure 6-60 shows an example of the ASCII data time series import form.

- *Import Data* Imports an existing ASCII data file and replaces the current series data with the new data. The same import form (Figure 6-60) used for the "Merge Series" is used for this function.

- *Import* Imports an existing EFDC formatted time series for the current time series type (i.e. if editing flows, will import a QSER.INP file). All the currently defined series for the current boundary type are overwritten with the contents of the import file.

- *Save As* Allows the user to write all of the series of the current boundary type to an EFDC formatted file. EFDC_Explorer writes these files during the project save process. However, this function is useful for backup during the editing process or to provide output for other user needs.

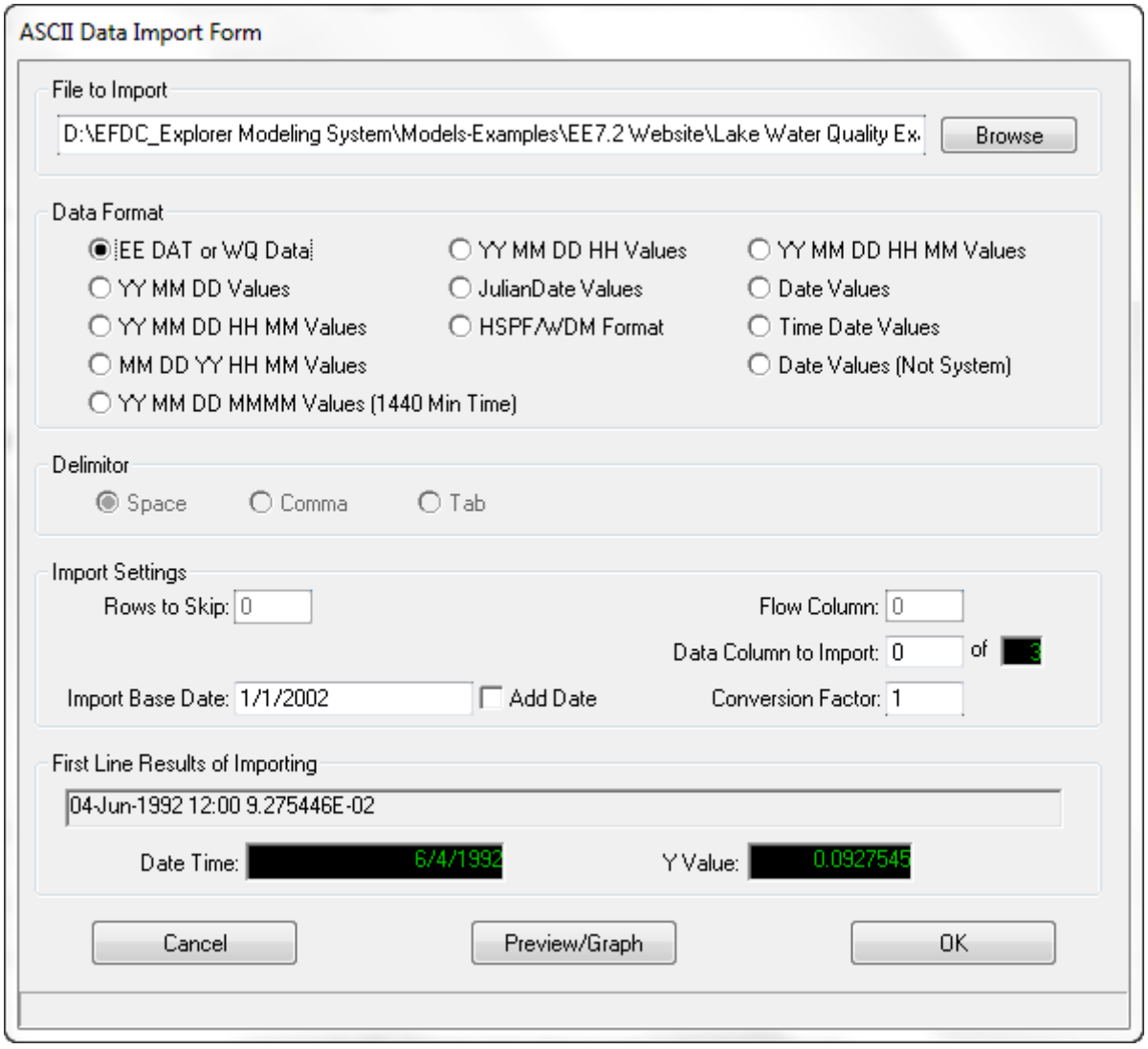


Figure 6-60 ASCII data time series import form.

The following provides a quick guide to the use of the ASCII data import form:

- Select the data file to import using the *Browse* button.
- Select the delimiter.
- Depending on the *Data Format* selected, other options will also need to be selected in the *Delimiter* and *Import Settings* frames. .
- The *Data Column to Import* should point to the column number that contains the Y data.
- As changes are made the information in the *First Line Results of Importing* frame provides the results of the current settings.
- Preview/Graph the results of the current settings to see if the import results are what was expected. If not, make further adjustments, as needed, until the data import plot is correct.
- Press *OK* to finalize the actual importing process.

6.3.13. Marine Hydrokinetic Devices

EFDC+ has now fully incorporated the Marine Hydrokinetic (MHK) module to simulate placement and potential effects of installation and operation of turbines and wave energy converters in rivers, tidal channels, ocean currents, and other waterbodies. The code is implemented based on Sandia National Laboratories modified Environmental Fluid Dynamics Code (SNL-EFDC) (James, 2010).

As shown in Figure 6-61 the *MHK Devices* sub tab may be accessed from the *Domain* tab on the main form. Here the user is informed how many devices have been defined and how many cells contain devices. Select the *Define/Modify* button to setup the MHK devices.

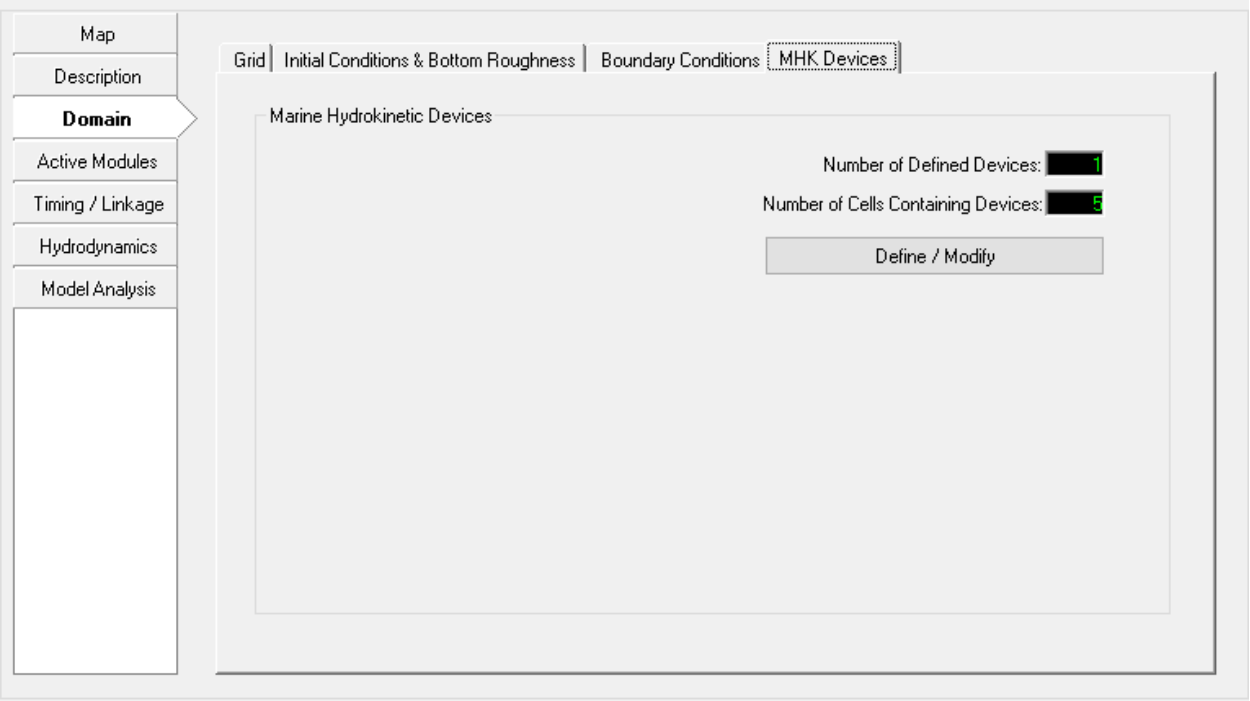


Figure 6-61 Domain Tab – MHK Devices settings.

6.3.13.1 *MHK Devices – General Tab*

In *Marine Hydrokinetics Device Options* form, the *General* tab is where the user specifies whether MHK calculations will be used in EE or not by checking the *Use MHK Devices in the EFDC Simulation* checkbox as shown in Figure 6-62.

For MHK devices, it is important to specify the power extracted from the flow. There are different methods for computing power extracted from flow, though currently only one method is enabled. The method is to use the specified thrust coefficient that is multiplied by velocity cubed. The user may specify the *Velocity Source for Power Calculations required for computation of power*. The available options are: local cell velocity and the upstream velocity.

The *MHK Output Option* frame provides options for text based reporting of the output. These options include:

- Energy fluxes access a transvers section upstream and downstream of the device
- Average velocity and the Z-profile for the tidal reference model at throat
- Average velocity and surface velocity for the reference model
- Outputs for straight-channel calibration model of the wake structure.

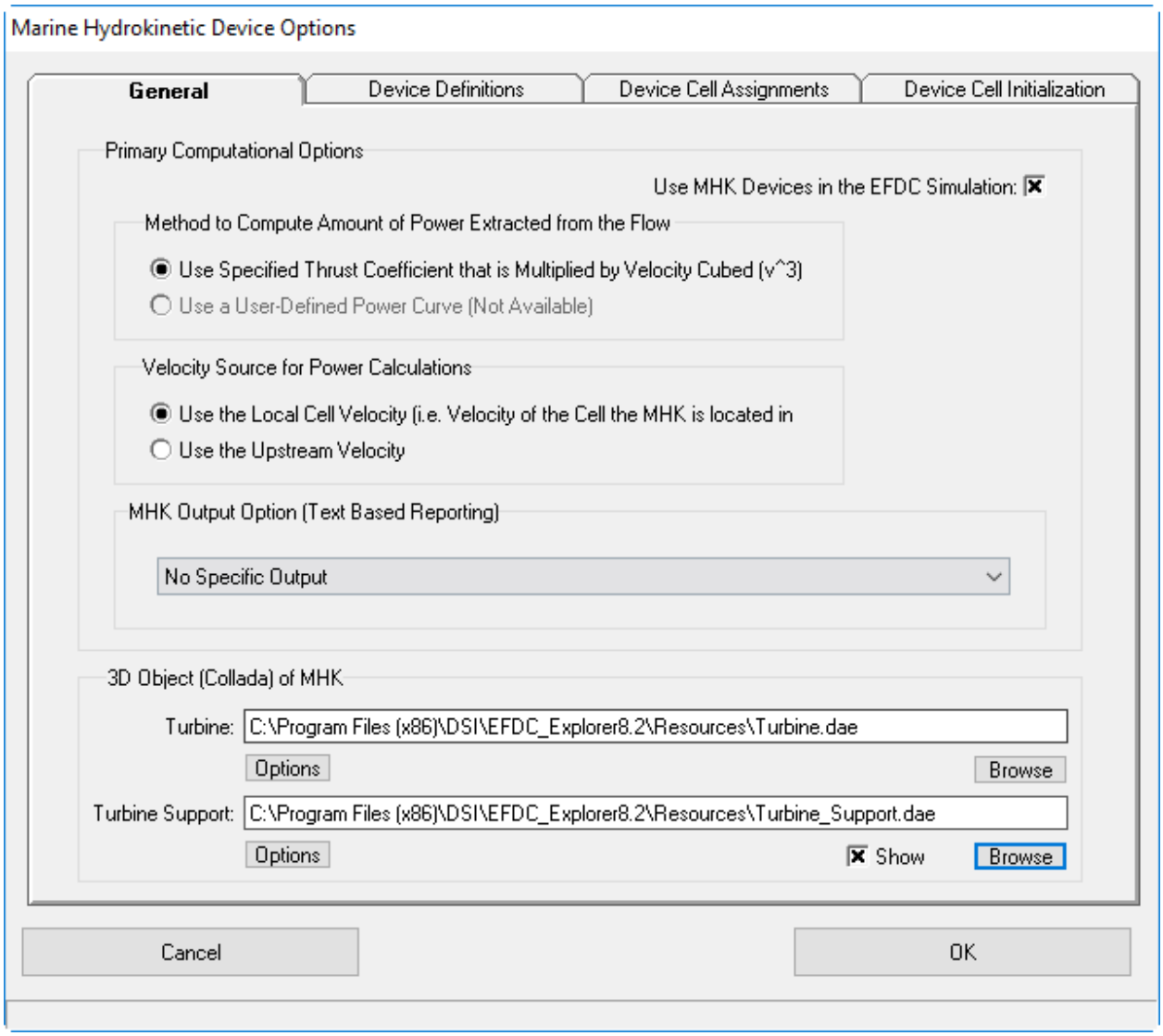


Figure 6-62 MHK Devices Options – General tab.

The *3D Object (COLLADA) of MHK* from provides the option to browse COLLADA files that can be viewed from *View3D*. By default these files are saved here:

C:\Program Files (x86)\DSI\ EEMS8.3\Resources

An example of the default turbine is shown in Figure 6-63. The user can adjust the settings for the MHK device display by selecting Option in the *General* tab or with CTRL K in *View3D* with the options shown in Figure 6-64.

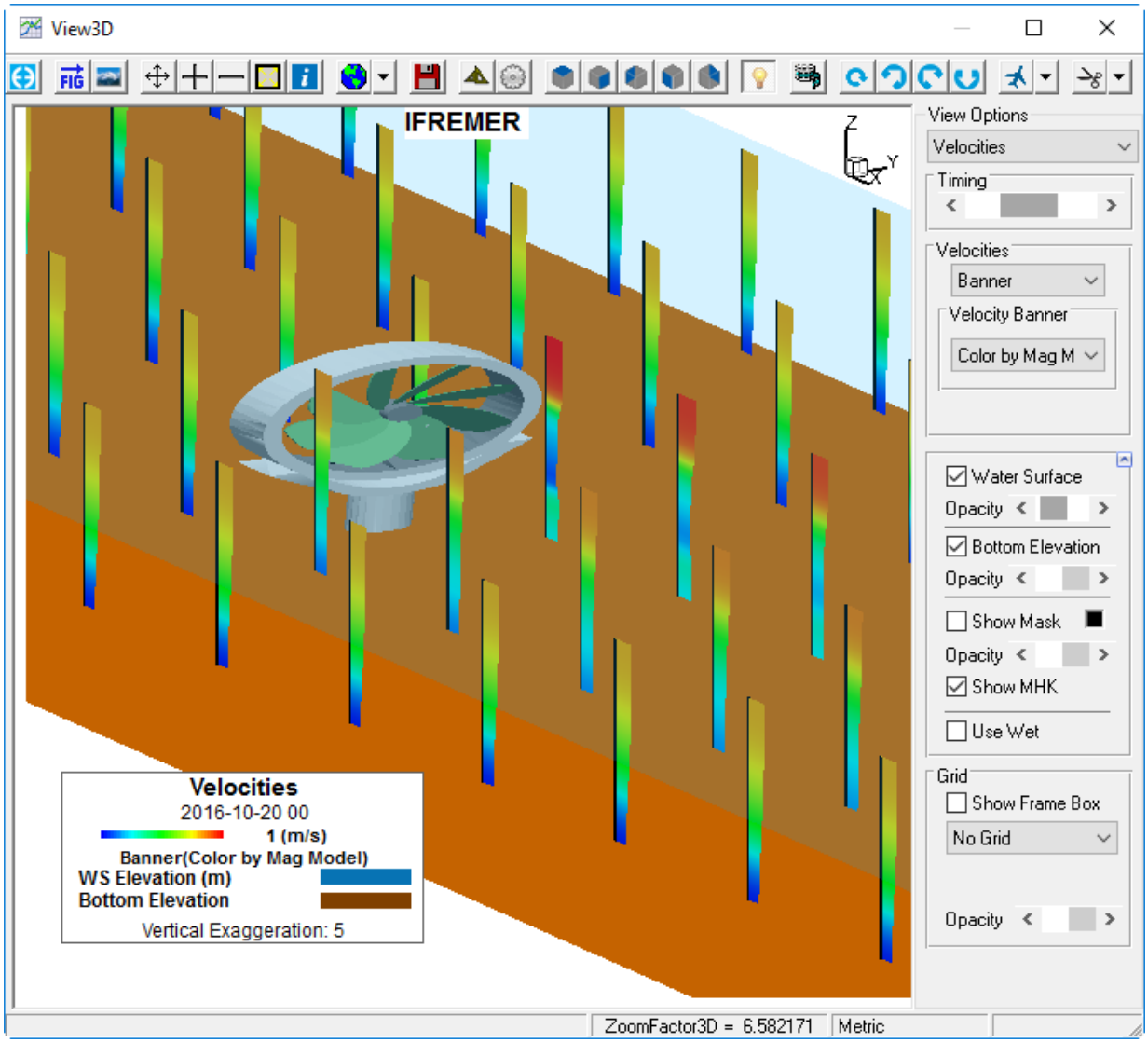


Figure 6-63 MHK Devices in View3D.

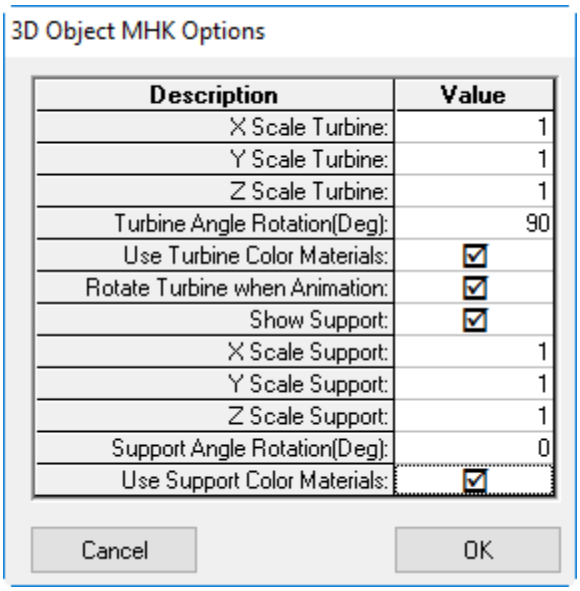


Figure 6-64 MHK Devices viewing options in View3D.

6.3.13.2 *Device Definitions Tab*

The *Device Definitions* tab is shown in Figure 6-65. Here the user specifies the number of MHK devices in the model and the number will be reflected in the *MHK Device Definitions* dropdown box. The device names can be specified by the user, and for each device the user must specify the device dimensions and a range of coefficients (Figure 6-62). Guidance on how to determine these definitions can be found in the SNL user guide (James, 2010).

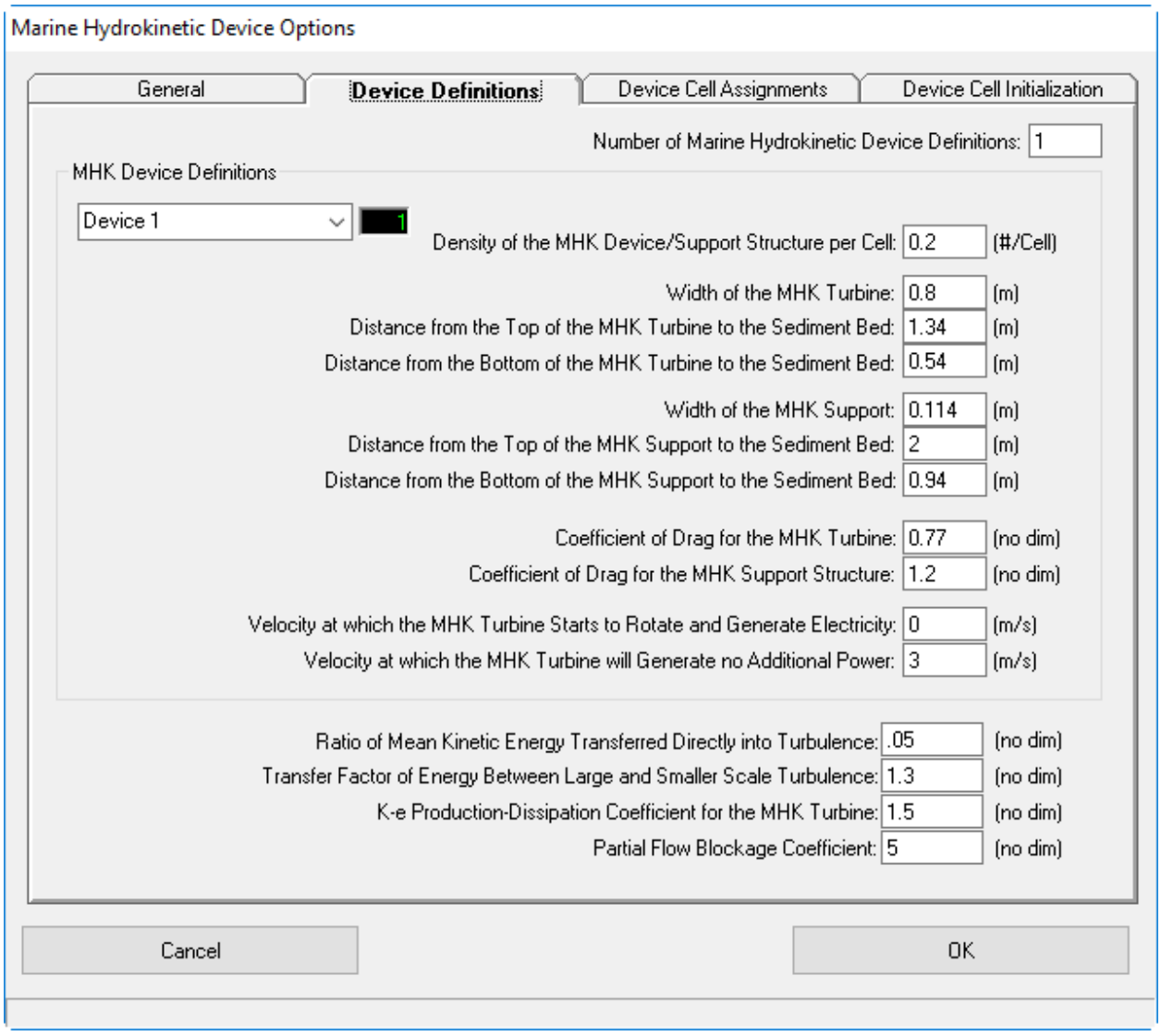


Figure 6-65 MHK Devices Options – Devices Definitions tab.

6.3.13.3 *Device Cell Assignments*

The *Device Cell Assignments* tab is where the user should specify the location for the devices. The user can enter either Easting and Northing or I, J index for the cell to specify locations. The user can type or copy and paste the device number which correspond to that cell as shown in Figure 6-66.

The *Export List* button will export the Easting, Northing and MHK Device number as a .DAT file for use by the user externally to EE.

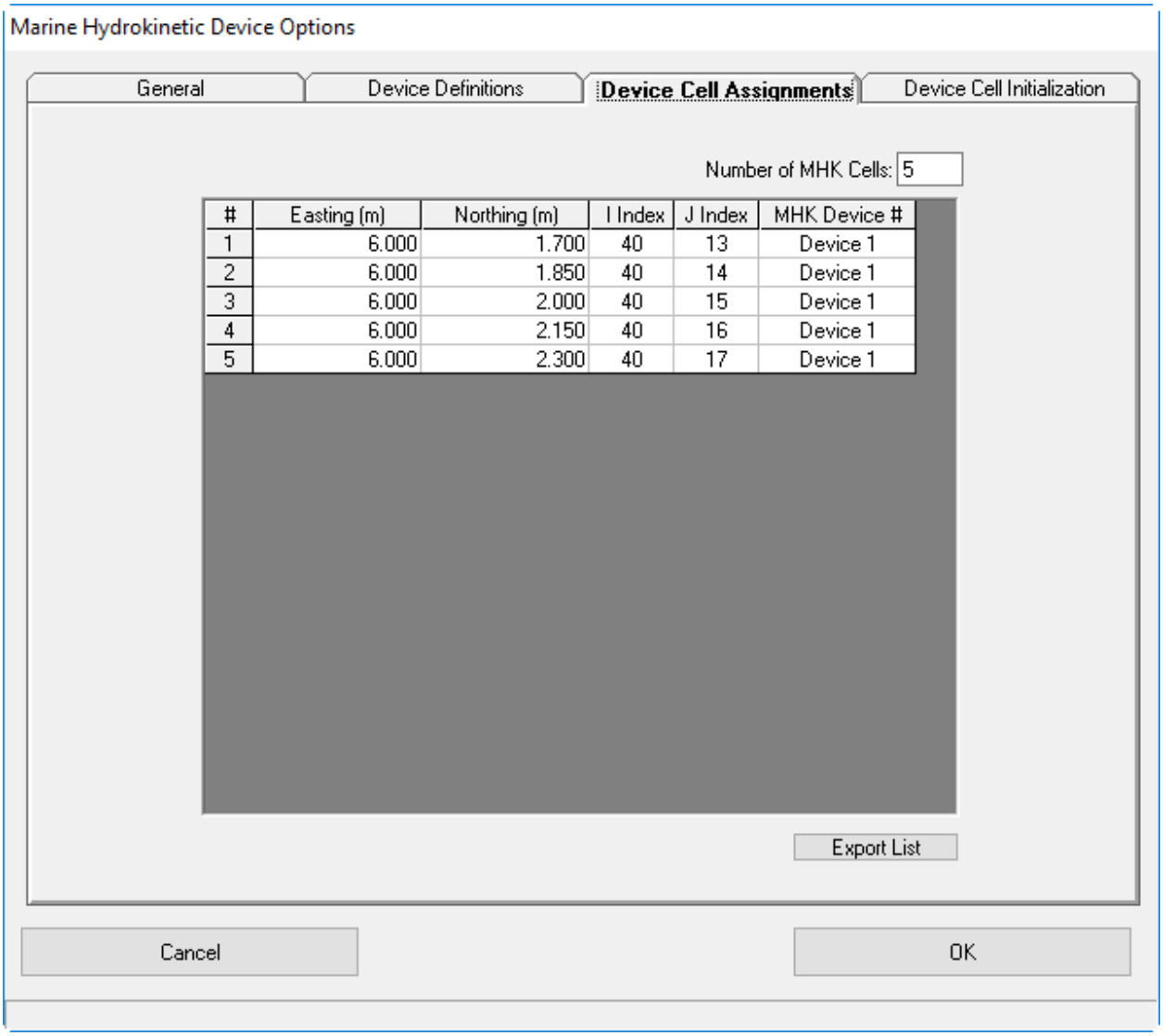


Figure 6-66 MHK Devices Options – Devices Cell Assignment tab.

6.3.13.4 *Device Cell Initialization Tab*

The *Device Cell Initialization* tab allows the user to set the MHK devices from an external file which is based on XY data as shown in Figure 6-67. Two options are provided, *Read XY Pairs from a File for Discrete Cell Assignments*, and *Assign MHK Devices using Polygons with MHK IDs*. This latter file is in the P2D format (native EE file).

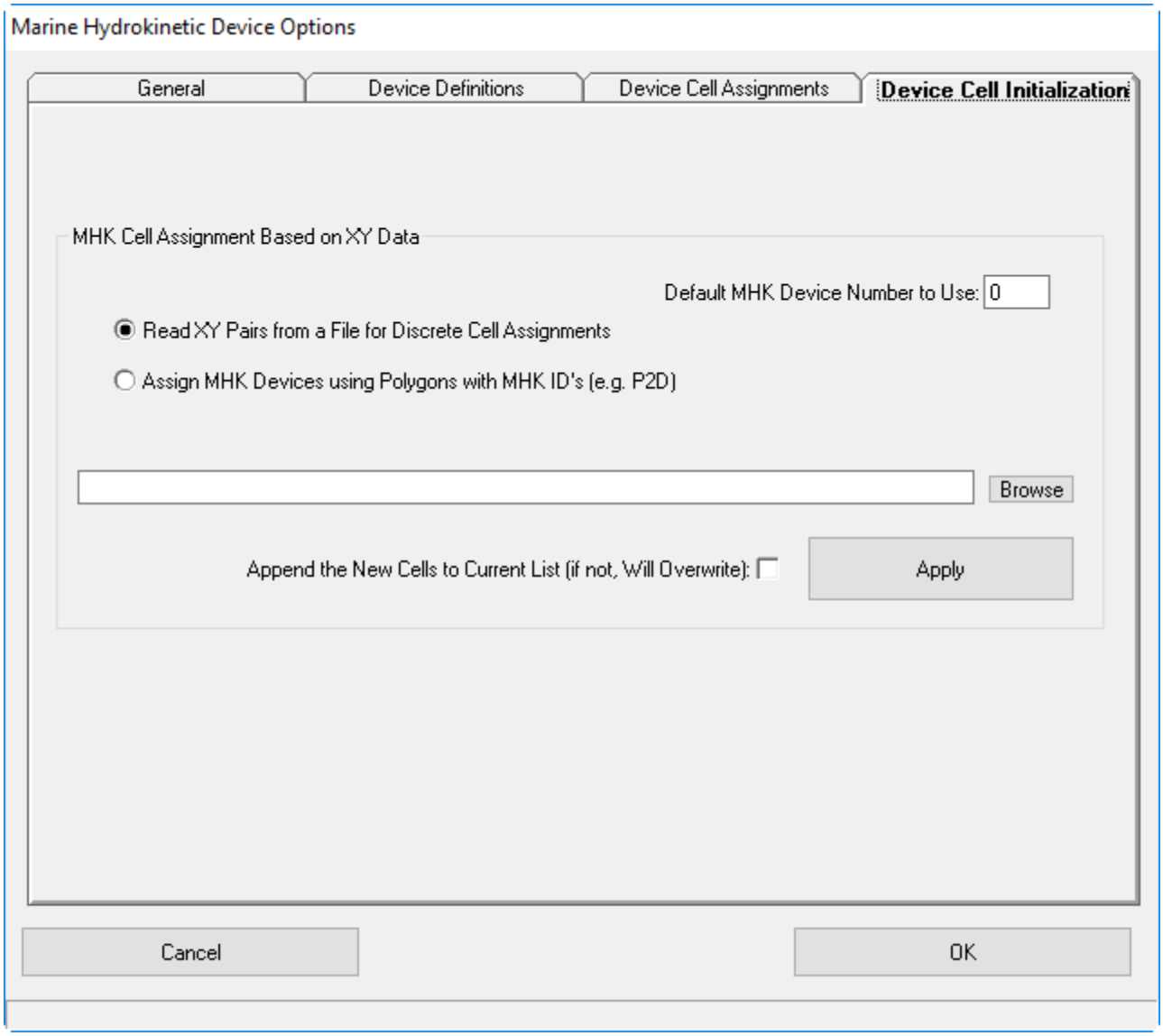


Figure 6-67 MHK Devices Options – Devices Cell Initalization tab.

6.4 Active Modules Tab

The *Active Modules* tab provides access to the computational and transport switches and options that are contained in C6 in EFDC.INP. These switches are the primary activation switches to specify which state variable(s) will be simulated by EFDC. Figure 6-68 displays the maximum number of modules currently allowed in the EFDC_Explorer/EFDCPlus modeling system. Selecting a check box will display a new module in the left hand column, and deselecting a box will hide it.

If the *Activate Cohesive Sediments* or the *Activate Non-Cohesive Sediment* boxes have been selected the user may make changes to the sediments and sediment bed settings.

The user should first define the sediment model intended by selecting the appropriate option in the check box. Two sediment models are available by choosing either *Use EFDC Sediment Model* or *Use SEDZLJ Sediment Model*. The SEDZLJ Sediment sub-model is partially implemented in EFDC_Explorer. EFDC_Explorer can display SEDZLJ input and results including sediment fluxes and active layer thicknesses. However, at this time, the user cannot

create or edit the input data. Further settings for sediments must be made in the Sediments Tab (Section 6.12).

SEDZLJ has been developed by Sandia National Laboratories, and the SEDZLJ EFDC User Manual is downloadable from the EE website. In his paper on this sub-model, the primary developer, Scott C. James, provides a detailed description of how recently-developed sediment dynamics formulations are incorporated into the United States Environmental Protection Agency's EFDC. The new approach is an extension of previous models and accounts for multiple sediment size classes, has a unified treatment of suspended load and bedload, and appropriately replicates bed armouring. The resulting flow, transport, and sediment dynamics model is an improvement to previous models because it may directly incorporate site-specific data, while maintaining a physically consistent, unified treatment of bedload and suspended load.

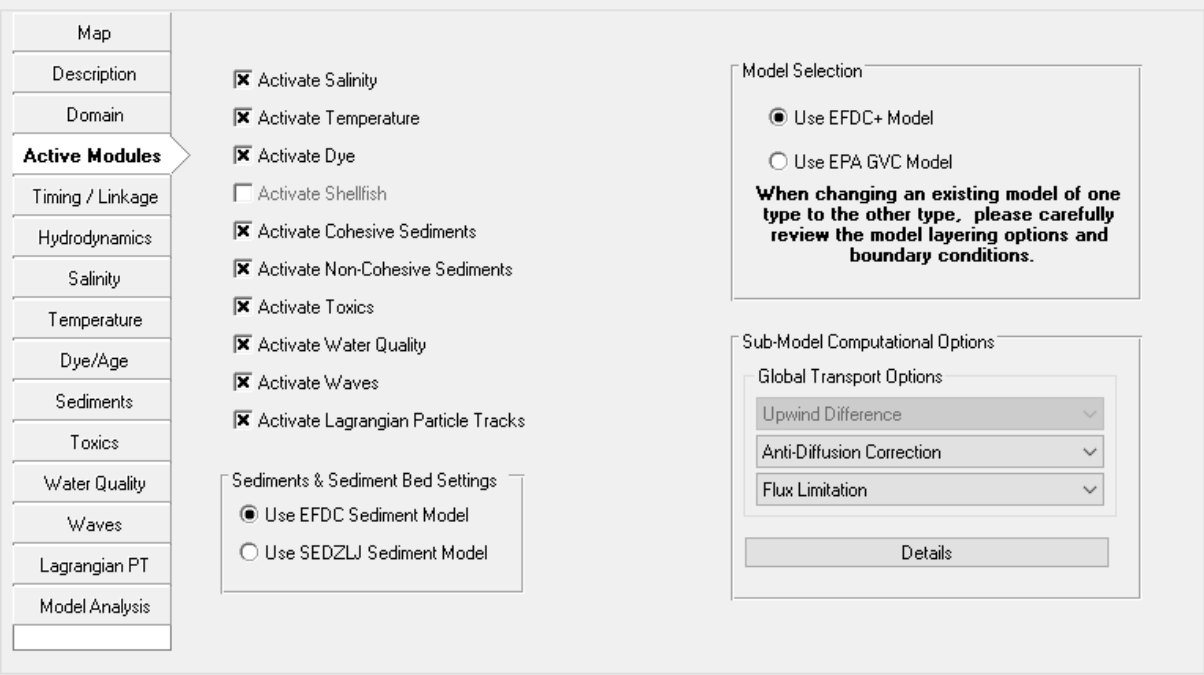


Figure 6-68 Active Modules Tab

The *Model Selection* frame shows the GVC and model selection options. EFDC_Explorer supports two EFDC models, the EFDC_GVC version (with GVC) and the EFDCPlus version.

If the EFDCPlus Model is selected the GVC Model options are not displayed. When the user selected the *Use EPA GVC Model* the Lagrangian Particle tracking and Wave options will be hidden.

When loading an existing model, EFDC_Explorer identifies the type of model and automatically sets the options on this tab, based on the data in the EFDC.INP file. In addition, the user can switch from one type of model to the other using the model selection option in the *Model Selection* frame. When changing from one model type to the other, the user should check the model layering and boundary conditions to ensure the inputs are correct. See Section 6.8 for the settings on the GVC.

The *Global Transport Options* sub-frame allows the user to select from Upwind Difference, Central Difference and Experimental Upwind, as well as to use Anti-Diffusion or Anti-Diffusion Correction methods. This option sets the same mass transport computational approach for all constituents. More granular control is available using the *Details*.

6.5 Timing/Linkage Tab

The *Timing/Linkage* tab provides the user with access to four sub-tabs: *Model Run Timing; EFDC_Explorer Linkage; Legacy Output;* and *WASP Linkage*, each of which are described in detail below.

6.5.1. Model Run Timing

The model timing and output options available to EFDC_Explorer are shown in Figure 6-69. EFDC has many more output options but the majority of them have become obsolete due to the capabilities of EFDC_Explorer.

In the main frame the user must specify the time to start and end the run, along with the time stepping options. *Time of Start* is the Julian day, relative to the *Beginning Date/Time*, to begin the simulation. The *Duration of Reference Period* is used to define a project specific meaningful period. This is often set to 24 hours to select a 1 day long reference period. The duration of the simulation is then set by specifying the number of reference periods. The ending time is computed from the starting time and duration.

The *Dynamic Timestep Options* subframe allows the user to engage auto stepping by setting the *Safety Factor* to a positive number >0 and <1. Generally, the safety factor ought to be less than 0.8 but some runs work with the safety factor>1 and some require a value <0.3. If set to 0 it will result in fixed time steps. Note that if auto time stepping is used then it is important to use a small initial time step, as the auto timestep must a multiple of this number. Therefore, a number like 0.5 or 1 second is advised, not 5 seconds for example.

The number of *Ramp-Up Loops* can also be set by the user; that is the number of initial iterations to for which to hold the time step to constant during ramp-up. The *Maximum dH/dT* option is ignored if set to zero, but if >0 then CALSTEP will use this additional criteria to set the dynamic timestep.

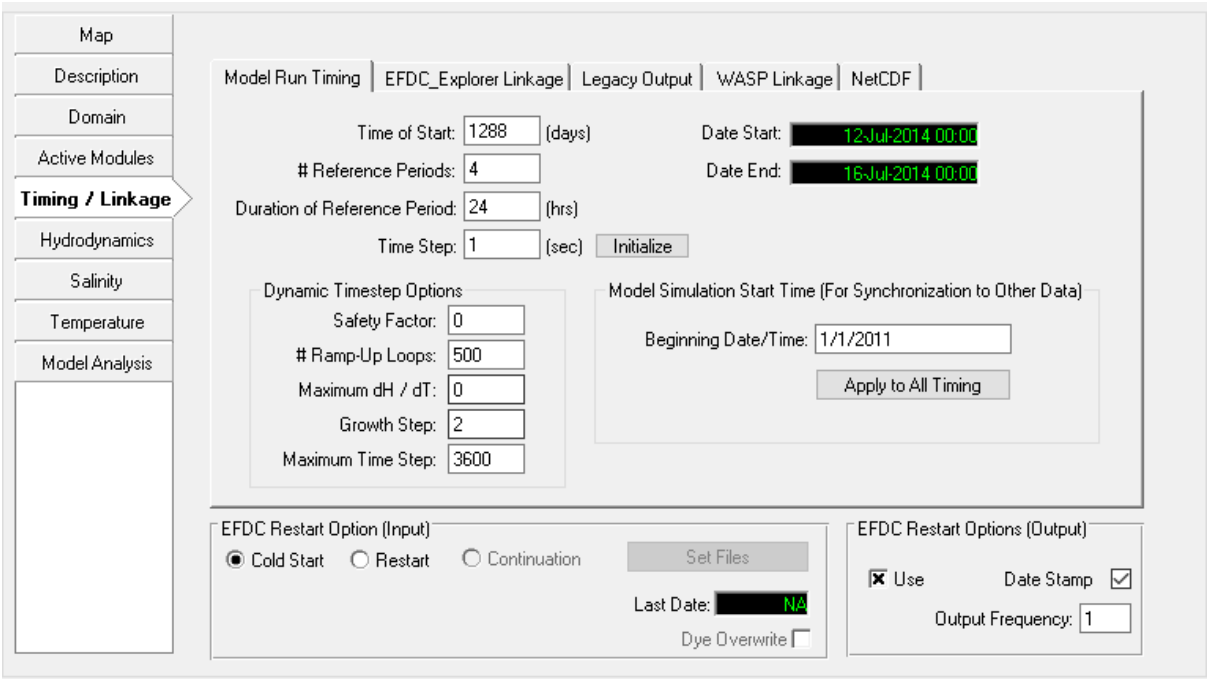


Figure 6-69 Timing/Linkage Tab: Model Run Timing.

A further option is the *Growth Factor*. This feature set the time-steps so that the delta t won't be increased for a specified number of time-steps. For example, if the current delta T is 1 second, then the new time step could be 3 seconds. With the previous options is would increase by the minimum time step. The new option will hold the time step for it jump to the maximum delta T. This makes the model more stable. Previously users could get a very high delta T, but it might become so high that the model is unstable. This option allows the model to stay at the maximum stable time step and not go higher. Generally, it is recommended to set to about 25 for steady growth.

6.5.2. Calendar/Julian Date Linkage

If the user wants to relate the model timing to actual calendar dates, it is required to enter a base date from which the model Julian dates will be added to compute the corresponding Gregorian date. The use of the base date and model Julian dates allow EFDC_Explorer to compare model versus data for a range of plots and statistics.

The base date is entered in the *Model Simulation Start Time* frame. Typically this date is the 1st of some calendar year but this is not required, any date/time is acceptable. The user can use the *J<->G* button to convert Julian to Gregorian or visa-versa.

If using Microsoft Excel dates for the Julian day counter, the EFDC_Explorer base date should be set to "01-Jan-1900". This allows number formatted dates in Excel to be directly used by EFDC_Explorer without adjustment. However, a projected oriented base date is recommended to prevent large numbered Julian dates, e.g. 39234.375 (01-Jun-2007 09:00). If 01-Jan-2007 was used as the base date the corresponding Julian date is 151.375.

From release of EE7.2, when the user changes the base date, it is possible to automatically update all of the Julian timing in the model including time series, starting times, etc.

When the user resets the *Beginning Date/Time* as shown in Figure 6-70 they now have the option to see if they want to update all the timing. When the user presses *Apply to All Timing* a popup will ask the user to confirm whether to make changes to other timing or keep the time series time the same.

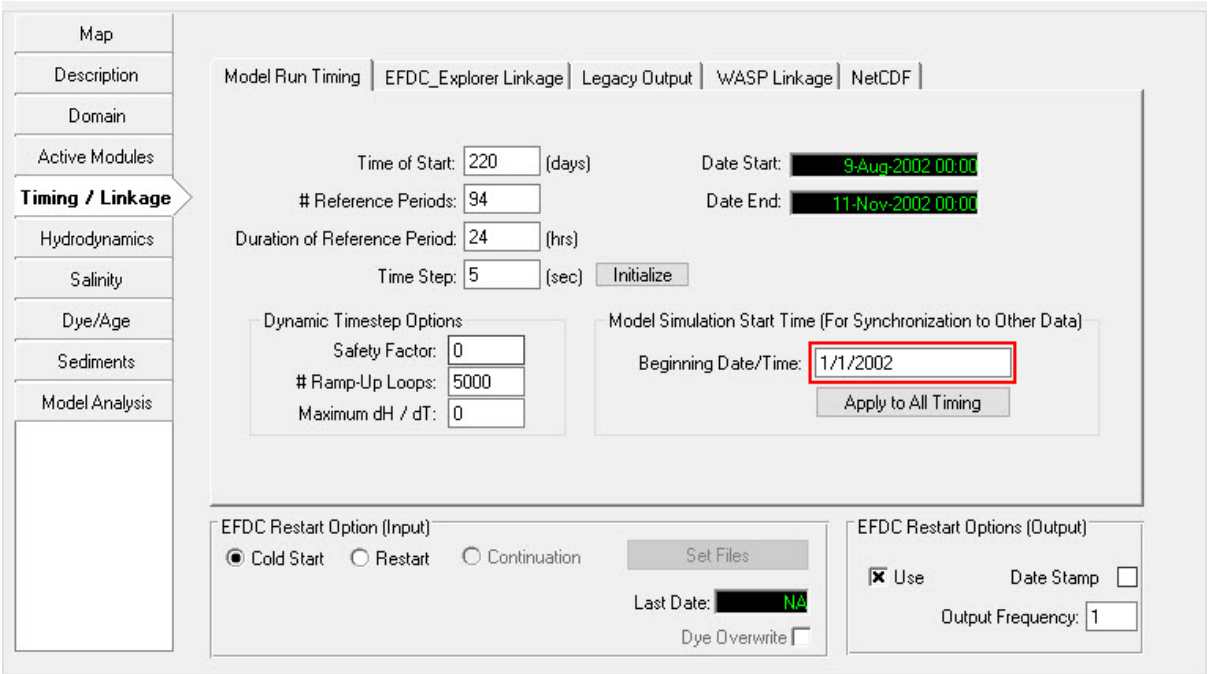


Figure 6-70 Base date update option.

Note that users must manually copy all the *.EE files as this stores the information regarding base date. This is different to EE70 where if the user edited the *.INP file then the *.EE file would overwrite it. Now in EE71 and later versions you can directly edit the input file and the base date is only stored in *.EE file.

It is important to be aware that selecting *Apply to All Timing* will not update all the calibration series files. It is the user's responsibility to update those files. However, if user puts the base date at top of series in the format (1990-01-01 0:00) then EE will allow for the change.

6.5.3. EFDC_Explorer Linkage

In order for EFDC_Explorer to post-process model results, the EFDC_Explorer Output option must be turned on. This is done in the *EFDC_Explorer Linkage* frame by checking the *Link EFDC Results to EFDC_Explorer* check box. The Linkage Output Frequency is used to determine how often EFDC will write the output. This is then multiplied by the Output Interval to determine how often various model results will be written. For example, if the *Linkage Output Frequency* is 60 minutes, then an Output interval of 6 will lead to model results being written every 6 hours for that parameter.

Which data to output is selected in the *Primary Model Results Linkage Options* frame, with the sub-item check boxes, such as *Velocities*. Water Surface *must* be turned on for EFDC_Explorer to post process any of the model results.

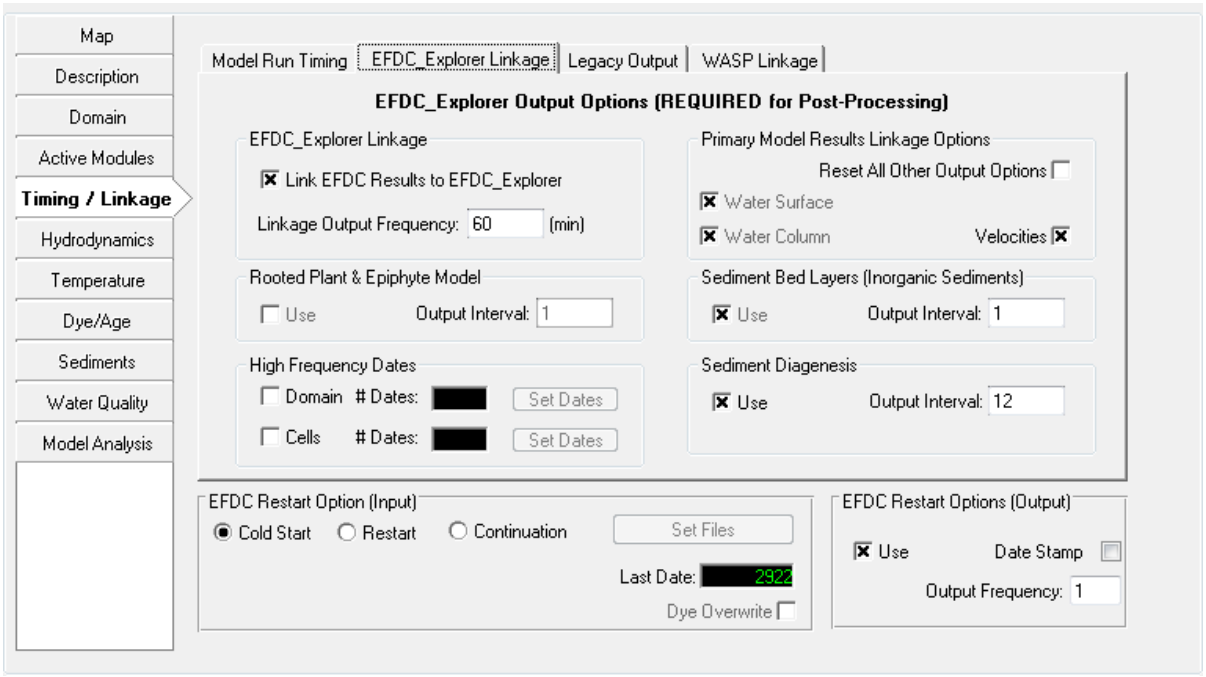


Figure 6-71 Timing/Linkage Tab: EE Linkage.

If simulating water quality with the full sediment diagenesis option turned on, EFDC_Explorer can display the spatial and temporal sediment fluxes and concentrations if the user enables the *Sediment Diagenesis* check box. Because the sediment processes are slow, as compared to water column processes, the user has the option to output the diagenesis data at a slower frequency as described above.

If simulating sediment transport with the maximum number of bed layers > 1, EFDCPlus can write the sediment bed properties by layer. Given that the sediment bed dynamics are generally slow relative to the water column dynamics, the user may want to set the sediment bed layer data output interval to a number greater than 1. For example, if the user is writing the water column data every hour but the user only wants sediment bed date every day, the user would specify an output interval of 24 to tell EFDCPlus to output the sediment data only every 24th water column snapshot.

The user may also manually adjust the output interval for the Rooted Plant & Epiphyte Model when applicable (see Section 6.14.6.1. for details on RPEM).

6.5.3.1 *High Frequency Dates*

The *High Frequency Dates* option (Figure 6-71) is an EFDCPlus option that inserts higher frequency output snapshots into the base snapshot frequency defined in the *Linkage Output Frequency*. This function allows the user to use a larger base snapshot frequency, resulting in smaller output files and faster post processing times, and still capture specific times in detail. This approach is very useful when matching model results to specific sampling events.

Selecting the *Domain* check box and then setting the calendar date (or Julian date), frequency and duration of the snapshots. Creates high frequency snapshots covering the whole model domain for the time specified. This approach will clearly lead to slower run times than the approach described above.

Checking the *Cells* checkbox (as in Figure 6-71) and selecting the *Dates* button provides the user with the option of creating high frequency snapshots of individual cells as shown in Figure 6-72. The user may create any number of subsets of the model domain by inputting a number in the *Number of Subsets* text box. Further subsets may be added one at a time with the *Add Subset* button.

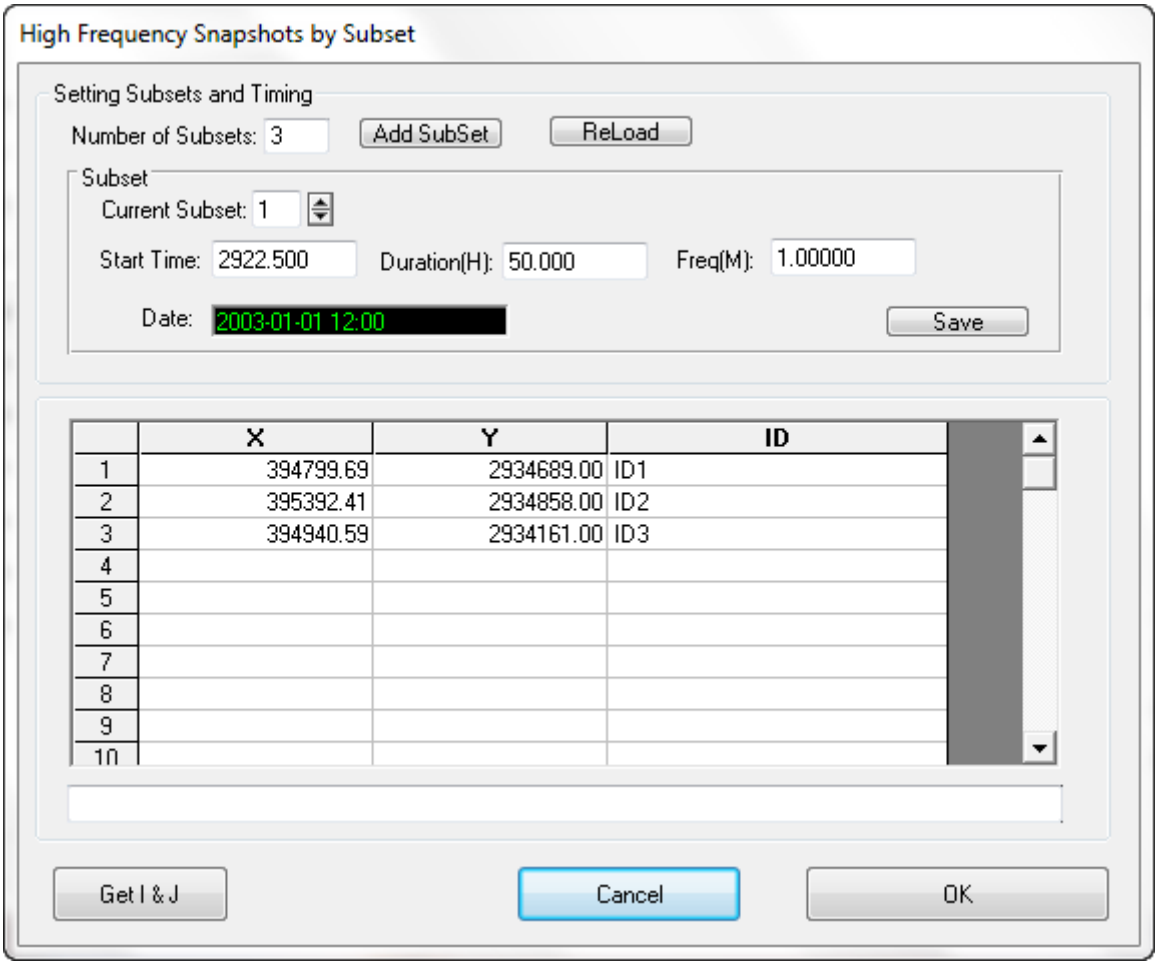


Figure 6-72 Timing/Linkage: High frequency snapshots by subset

The *Subset* frame shows which subset is in view at any given time and allows the user to cycle through the subsets with the *Current Subset* arrows. The Julian start time, duration (in hours) and frequency (in minutes) are set by the user for each subset of cells. Cells are defined by their x, y coordinates and may be designated with an ID for easy reference. The full set of subsets are saved with the *Save* button.

6.5.4. Legacy Output

The *EFDC Generated Time Series* frame is shown in Figure 6-73. It is not recommended to be used as EFDC_Explorer generates its own time series, but the option is available to users.

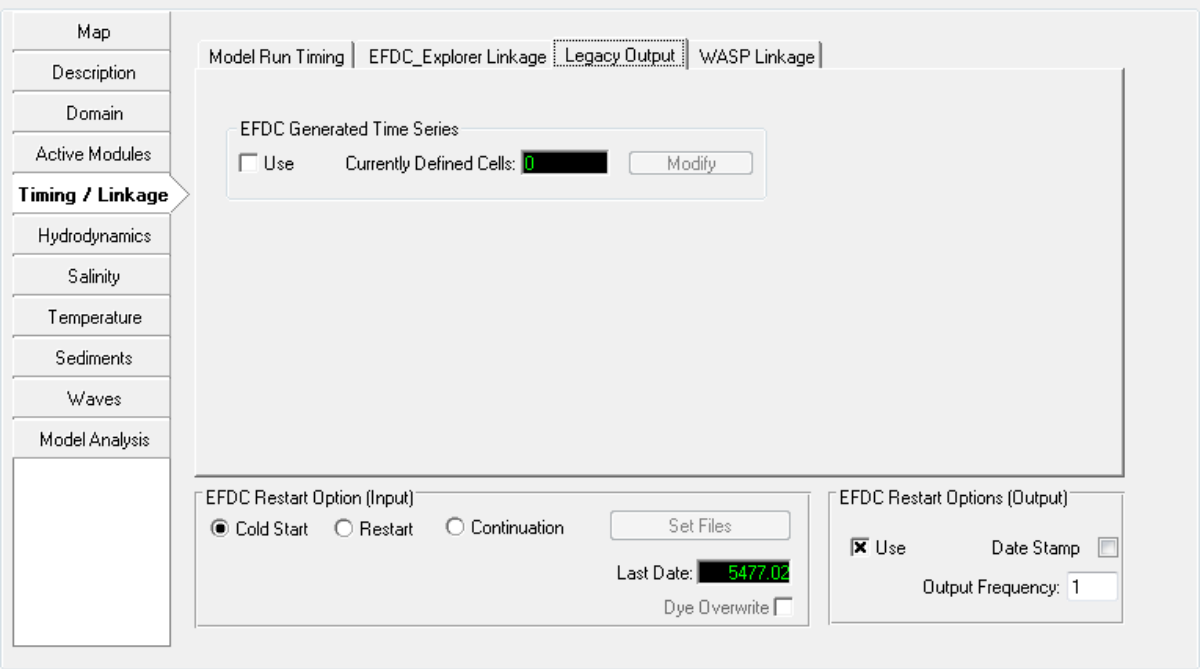


Figure 6-73 Timing Tab: Legacy Output.

6.5.5. WASP Linkage

Linkage to Water Quality Models frame (Figure 6-74) allows the user to bring in data from water quality models such as: WASP, CE_QUAL_ICM and RCA. . The WASP models allows the user to select from EPA DOS Ver 4, EPA DOS Ver 5.1, TT DOS Ver 5 (Old), TT DOS Ver 5 and EPA Ver 6.1 (Windows) and WASP7. The user is referred to the EEMS Knowledge Base for a detailed explanation of the linkage procedures and example models.

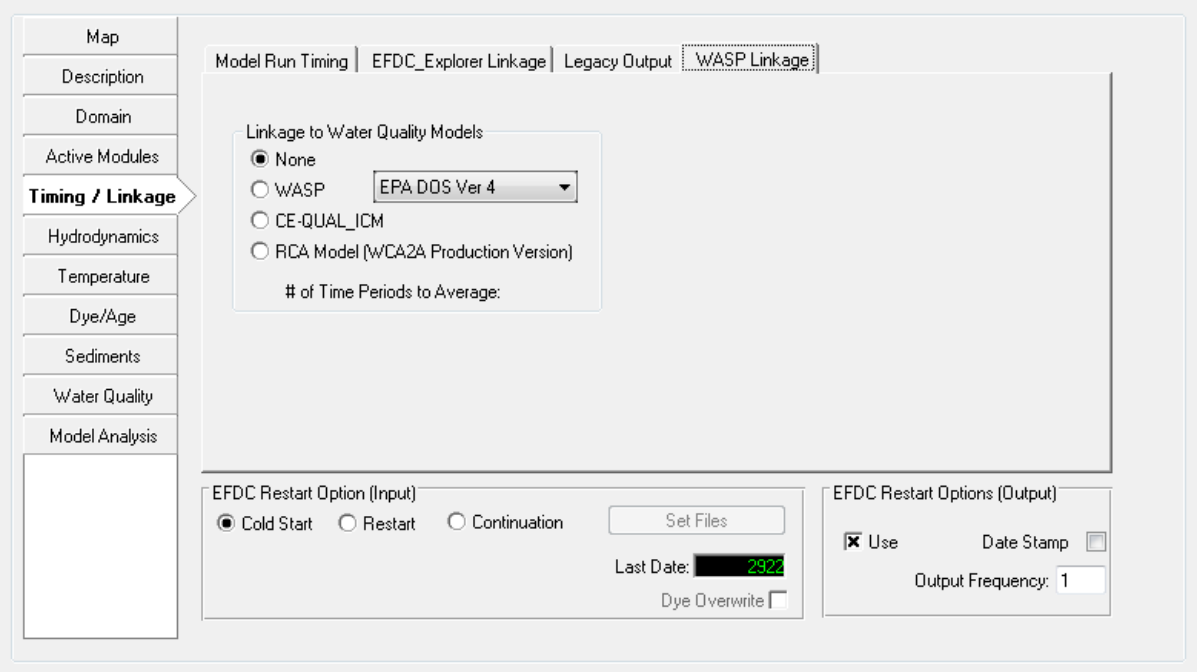


Figure 6-74 Timing Tab: Wasp Linkage.

6.5.6. EFDC Restart Options

The bottom frame of Figure 6-75 shows the *EFDC Restart Option*. An explanation of some of terms is provided below, as well how to use each option.

6.5.6.1 *Cold Start*

Cold start is the normal start of a model run and EFDC will generate the new output files or overwrite the existing output files.

6.5.6.2 *Restart*

The *Restart* is a previously available feature of EFDC and is applicable when a model has stopped prematurely or finished. The user can use it to restart from the point at which it terminated or some other point by selecting the appropriate restart file. In this case there is a hard restart from the time selected. EFDC will create a whole new set of output files which the user will then need to merge later.

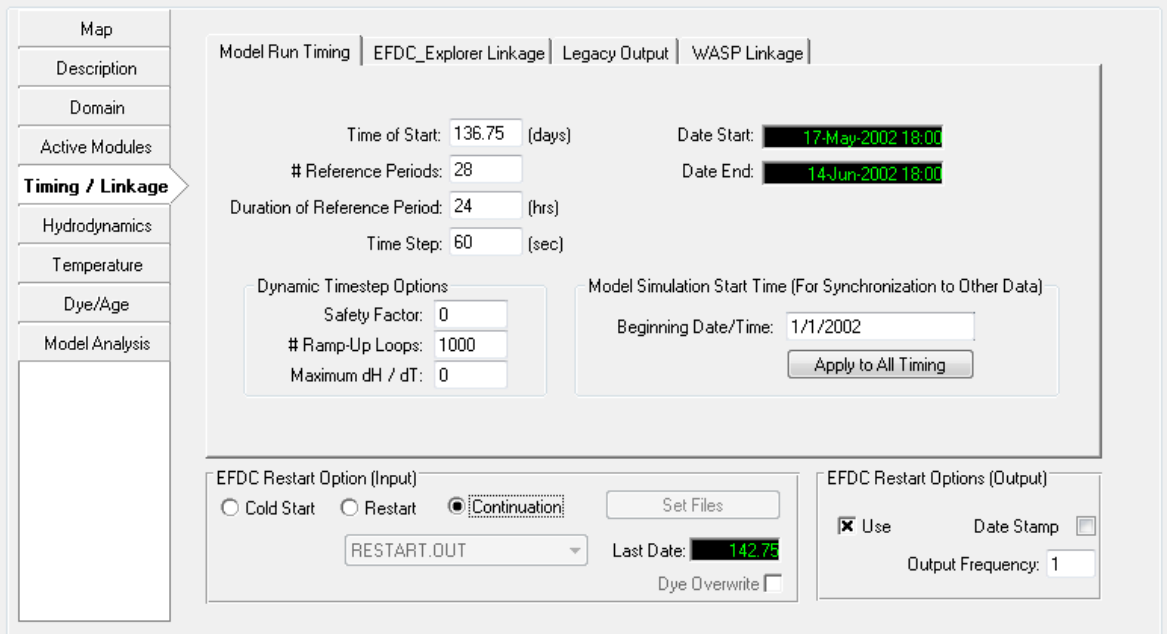


Figure 6-75 Timing/Linkage: EFDC Restart Options.

(1) EFDC Restart Option (Output)

EFDC Restart Options (Output) check box provides options for creation of restart files. If the *Use* option has been selected then EFDC will create a RESTART.OUT file. If *Date Stamp* is not checked then one file called RESTART.OUT will be created at the user defined *Output Frequency,* overwriting the previous file each time. If *Date Stamp* is checked then a file called RESTART_YYYYMMDD_HH.MM.OUT is created at the O*utput Frequency* for each EFDC sub-module being utilized. Note that if *Use* is not checked it is not possible to create restart file should the model stop or crash for some reason.

DSI standard practice is to always select *Use* and set *Output Frequency* to 1 to be the same as the duration of the reference periods. The user must have this option turned on in order to use restart as this creates the files required for the restart. Date stamp can be left unchecked so that the file is written over each time it outputs. A

(2) EFDC Restart Option (Input)

As noted that if *Use* is not checked for *EFDC Restart Option (Output)* then EFDC will not create restart files and restart function is not available. If it has been checked then the user should select the *Restart* radial button in the *EFDC Restart Option (Input)* frame. The output will be separate from the previous output. In this case EFDC has generated the restart file and EE will copy it for the user. Use the *Set files* button to select the restart files to hot start. *Set files* is used to navigate to the folder where the output files are located. EE will automatically copy and rename the EFDC restart files to the current project directory. The user has to select only the RESTART.OUT file and click open then the other restart files (such as, RSTWD.OUT, etc.) will be automatically selected by EE based on the requirement of EFDC model (prior to EE7.2 the user had to select each restart file applicable to that model run in turn). These files will normally be in the *#output* folder, with file name to be selected shown in the header and bottom left of the open file interface. EE will rename the files automatically and save them to the project directory after each one has been selected.

Some of the restart files that are created include the TEMP.RST and RSTWD.INP files if temperature and wetting & drying are being simulated. The *Dye Overwrite* is a legacy option. If the checkbox is selected the dye concentrations in the restart file will be overwritten by DYE.INP file. This is useful when restarting a tracer study with a new initial condition.

6.5.6.3 *Continuation Run*

From release of EE7.2 a number of updates have been provided for restart options. A new option is *Continuation*, whereby when restarting a model, EFDCPlus can now move to the correct location in the EE linkage files and begin writing from the time the model stopped or from the time selected for restart in case of using date stamp. The settings for this restart option are shown in Figure 6-75.

A *Continuation Run* is the same with Restart option except that the output will append to an existing run and so is only applicable when there is already model output in the current model directory. If there is no output and a restart file then this option is not enabled. If no date stamp is used EE will display only one restart file without date label which is the last restart file before EFDC terminated. To create the restart file the *Restart Output* should be set to continuous (i.e. N=1). EE displays the the date range available for the continuation from the most recent binary file output in the *Last Dates* box.

The process for using continuation runs is as follows:

1. Before running the model ensure *EFDC Restart Options (Output)* is set/checked to *Use*. If *Date Stamp* is selected the user may continue from any snap shot as all snapshots saved, not just the final snapshot. However, this is more memory intensive.
2. Reload the model to update *Last Date* box. In this case (Figure 6-75) the model stopped at T = 142.75.
3. Select the *Continuation* radial button (this is not enabled if the model is not reloaded).
4. Do not change the time of start. Save the model in the same folder. If the user wants to save in another folder then they will need to manually copy the output to the new folder.
5. Run the model and select overwrite.

The model will run from the continuation time and automatically merge with the prior output. This removes the necessity of the user having to merge runs as was required in earlier versions of EE. This option is different to the *Append* feature as it is now possible to restart a model at various user defined times.

Note: Whenever a new cold start is run the user should save the model in a new folder. This will prevent restart files from prior runs being available, in which case the *Last Date* may correspond to that prior run.

Other discussed above include: *Warm Start.* This function that allows the user to create the initial conditions for the model at a specified time. This is useful for diagnostic and sensitivity analyses, allowing the user to understand key processes in the system. A *Warm Start* may be created using either the *ViewPlan* or *View3D*. The user goes to the desired time and saves the current model conditions by selecting the highlighted button, shown in Figure 6-76 below. In this case, the new *Warm Start* model would begin with the water column and sediment bed conditions at 271.042. A *Warm Start* model is different from a *Hot Start* model because the momentum is not included. Therefore model would begin from a quiescent system as all velocity values are zero.

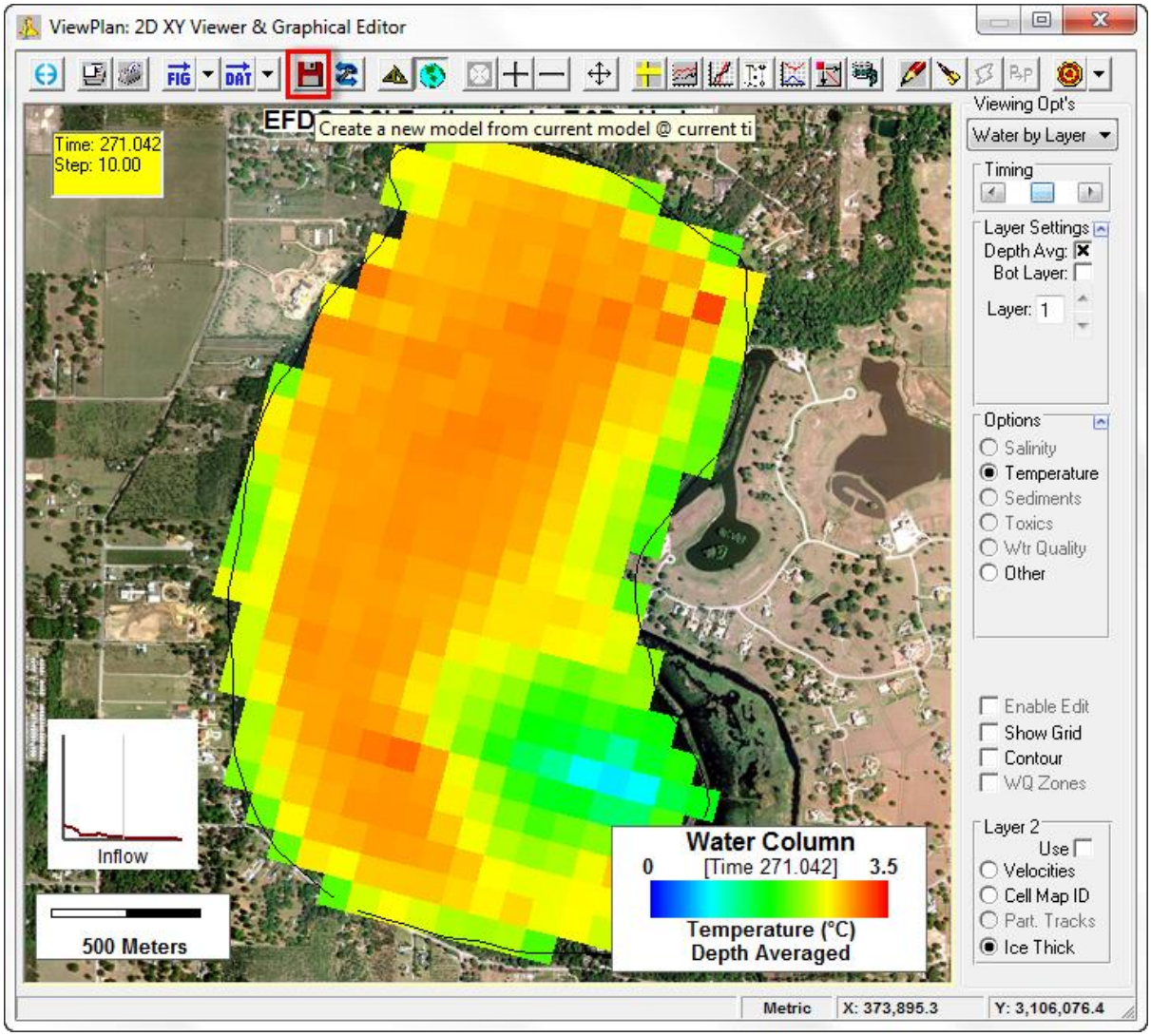


Figure 6-76 EFDC Restart Options: Warm Start.

Hot Start means using the existing output files to restart the model.

6.5.7. NetCDF Output

The EFDC+ code has been upgraded to allow export of netCDF-CF (netCDF- Climate and Forecast) formatted data. This development has been conducted with the goal of providing EFDC+ output that can be read and displayed on web servers.

According to the UNIDATA User Guide the Network Common Data Form (NetCDF) software was originally intended to create, access, and share array-oriented data in a form that is self-describing and portable. These deal with a variety of data types that encompass single-point observations, time series, regularly-spaced grids, and satellite or radar images.

The netCDF software functions as an I/O library, that can be accessed from different programming languages such as C, FORTRAN, Fortran 90, C++, Java, Perl, Python, and other languages for which a netCDF library is available. The library stores and retrieves data in self-describing, machine-independent datasets. Each netCDF dataset can contain multidimensional, named variables (with differing types that include integers, reals, characters, bytes, etc.), and each variable may be accompanied by ancillary data, such as units of measure or descriptive text. The interface includes a method for appending data to existing netCDF datasets in prescribed ways, functionality that is not unlike a (fixed length) record structure. However, the netCDF library also allows direct-access storage and retrieval of data by variable name and index and therefore is useful only for disk-resident (or memory-resident) datasets.

NetCDF is designed to:

- Facilitate the use of common datasets by distinct applications.
- Permit datasets to be transported between or shared by dissimilar computers transparently, that is, without translation.
- Reduce the programming effort usually spent interpreting formats.
- Reduce errors arising from misinterpreting data and ancillary data.
- Facilitate using output from one application as input to another.
- Establish an interface standard that simplifies the design of new software for accessing geo-science data.

EFDC+ now exports the native output in netCDF-CF format for Windows and Linux. EE, the pre- and post-processor for EFDCPlus can also convert EFDC *.OUT files to netCDF .nc files, including the derived WQ parameters. EFDCPlus currently exports 44 parameters to NetCDF format as listed at the end of this document.

EFDC+ for EE8.2, exports the netCDF files at the same interval that it produces the EE linkages files, which is set in the tab for *Timing and Linkage* | *EFDC_Explorer Linkage* | *Linkage Output Frequency*.

The main configuration for netCDF output is in the *Timing and Linkage* | *NetCDF* tabs as shown in Figure 6-77. Here the user can turn on netCDF output by checking the *Activate NetCDF Output* checkbox. The user may select the level at which the netCDF data is compressed with the *Compression* setting. This can range from 1, no compression, to 9 for maximum compression, but is recommended by UNIDATA to be set to 2 for the best balance between file size and speed.

The user may also select to *Use Velocity Rotation.* If this option is checked, the velocity field is rotated to true east and true north and if not checked then there is no rotation to true east and true north.

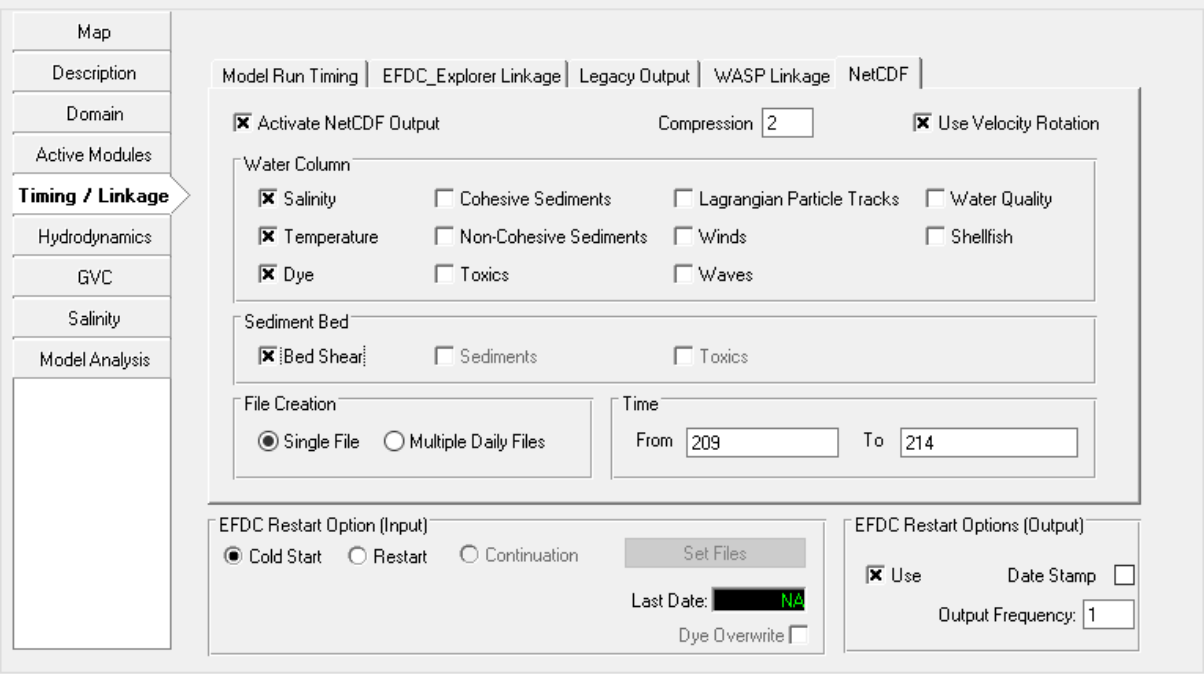


Figure 6-77 NetCDF Settings.

The user should then select which parameter they want to output from the water column or the sediment bed. Some options for the sediment bed are not yet enabled and will be added to EE in future releases.

In the *File Creation* frame, the user is given the option of EFDC+ generating a *Single File*, or *Multiple Daily Files*, The *Multiple Daily Files* option provides one netCDF file for each one day (24 hours) of model output. NetCDF files (.nc files) are placed in the #output folder along with the EFDC output. The snapshots for the netCDF files determined by the setting for the interval for the EE linkage file, and the user can change the snapshot interval by changing the EE linkage options, Note that linkage frequency should be 10 minutes or more for proper display in ArcGIS tools. After setting these options the user should save their model and run EFDC+, which will then export the netCDF .nc files to the #output folder.

Note that calendar date is the default date in the netCDF output

6.5.7.1 *Displaying NetCDF Output*

There are a number of options available to users for displaying netCDF output such as using ArcGIS, or using web based tools such as ncWMS. Some of these are outlined here to help users become familiar with these tools.

(1) Convert to Text

To convert the .nc file to a text file, the user can download the ncdump.exe tool from UNIDATA.

(2) Using ArcMap

A detailed guidance video for how to animate netCDF output in ArcGIS has been prepared by DSI and made available here: https://www.youtube.com/watch?v=vgT7dN4UK44

The basic steps involve selecting the ArcToolbox > Multidimensional Tools > Make netCDF Feature Layer

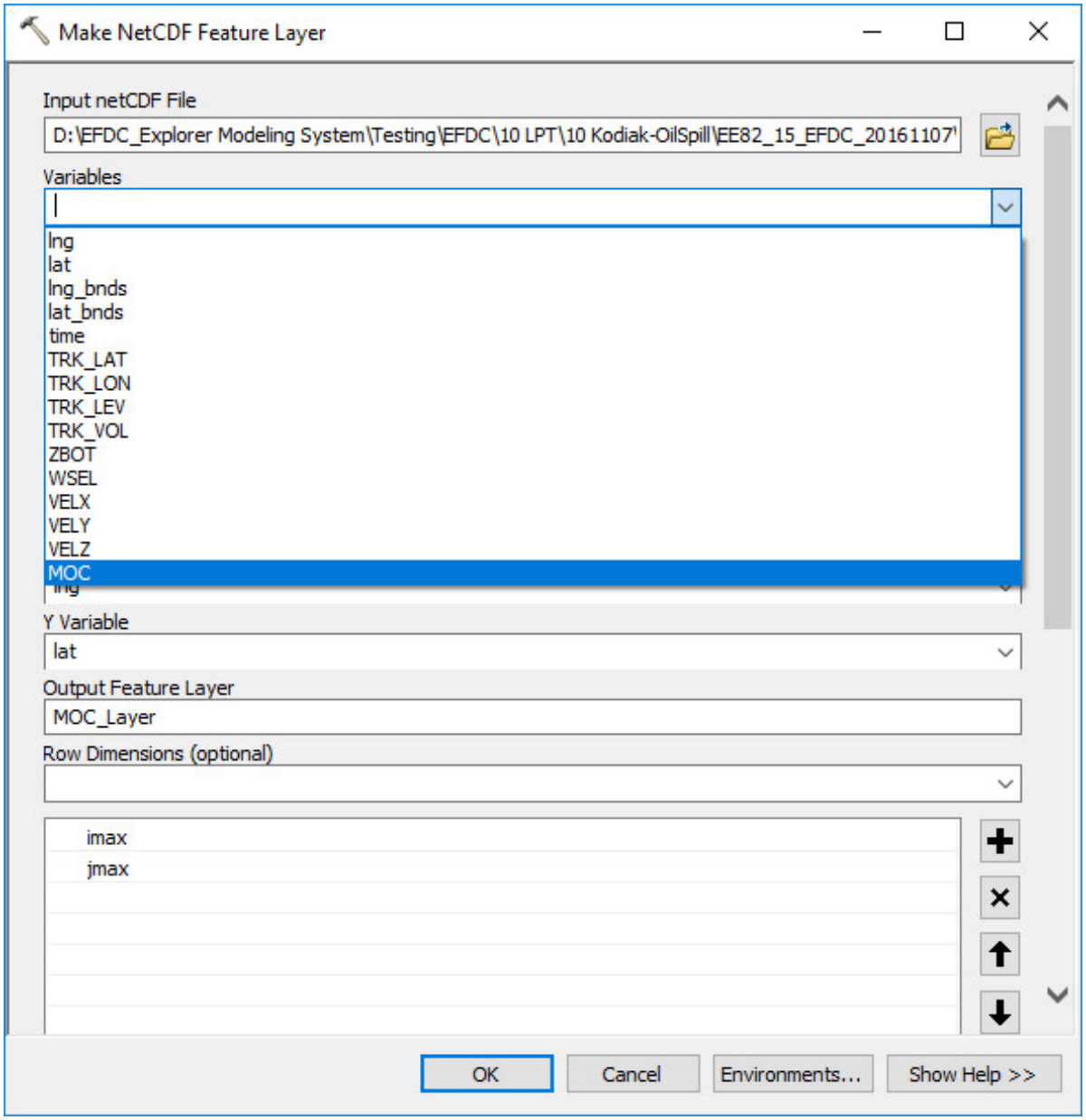


Figure 6-78 ArcGIS reading the NetCDF file.

(3) Using NCWMS

ncWMS is a Web Map Service for geospatial data that are stored in CF-compliant NetCDF files. The intention is to create a WMS that requires minimal configuration: the source data files should already contain most of the necessary metadata. ncWMS is developed and maintained by the Reading e-Science Centre at the University of Reading, UK. The latest version is 1.2, released in August 2014.

ncWMS is implemented in Java as a web application. To install ncWMS you simply place the WAR (Web ARchive) file in your application server of choice (e.g. Tomcat).

ncWMS relies heavily on the Java NetCDF interface from Unidata. This library does a lot of the work of metadata and data extraction. In particular the GridDatatype class is frequently used to provide a high-level interface to gridded geospatial NetCDF files. The library will also

read from NetCDF files on HTTP servers and from OPeNDAP servers. From http://www.resc.rdg.ac.uk/trac/ncWMS/

ncWMS loads native netCDF files and displays them on a web browser. To make the output the same as EE8 would require further adjustment to the tool. For the purposes of this demonstration the basic display of the output is shown in the next section.

(4) Using Ocean Data View

Ocean Data View (ODV) is a software package for the interactive exploration, analysis and visualization of oceanographic and other geo-referenced profile, time-series, trajectory or sequence data.

ODV supports the netCDF format and lets you explore and visualize CF, COARDS, GDT and CDC compliant netCDF datasets. This works with netCDF files on your local machine as well as with remote netCDF files served by an OPeNDAP server.

6.5.7.2 *Examples of Model Output*

(1) Water Quality Model

To demonstrate the netCDF output is consistent with the data viewed in EE, a number of comparisons have been made for different models. Examples of an output from the West Lake WQ model in EE, ArcGIS and ncWMS are shown in Figure 6-79, Figure 6-80, and Figure 6-81 respectively. Note that the West Lake model may be downloaded from the EE website.

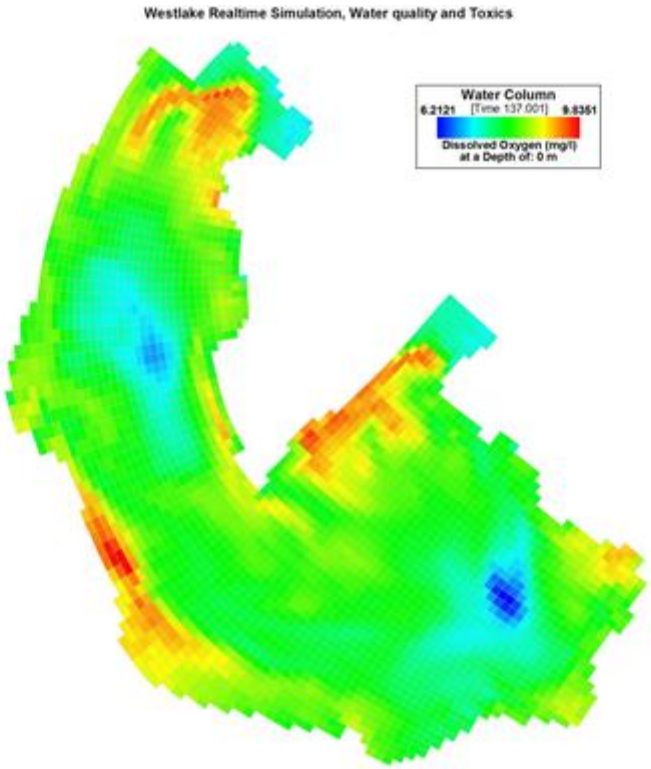


Figure 6-79 EE display of EFDC+ output – DO at 137.001.

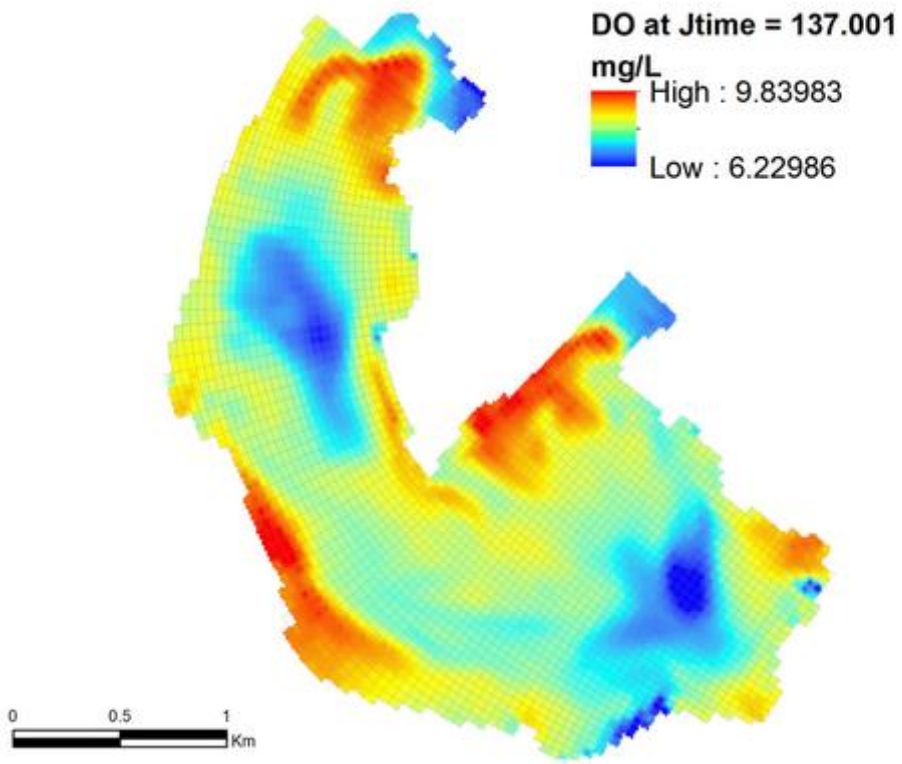


Figure 6-80 ArcGIS display of EFDC+ netCDF output – DO at 137.001

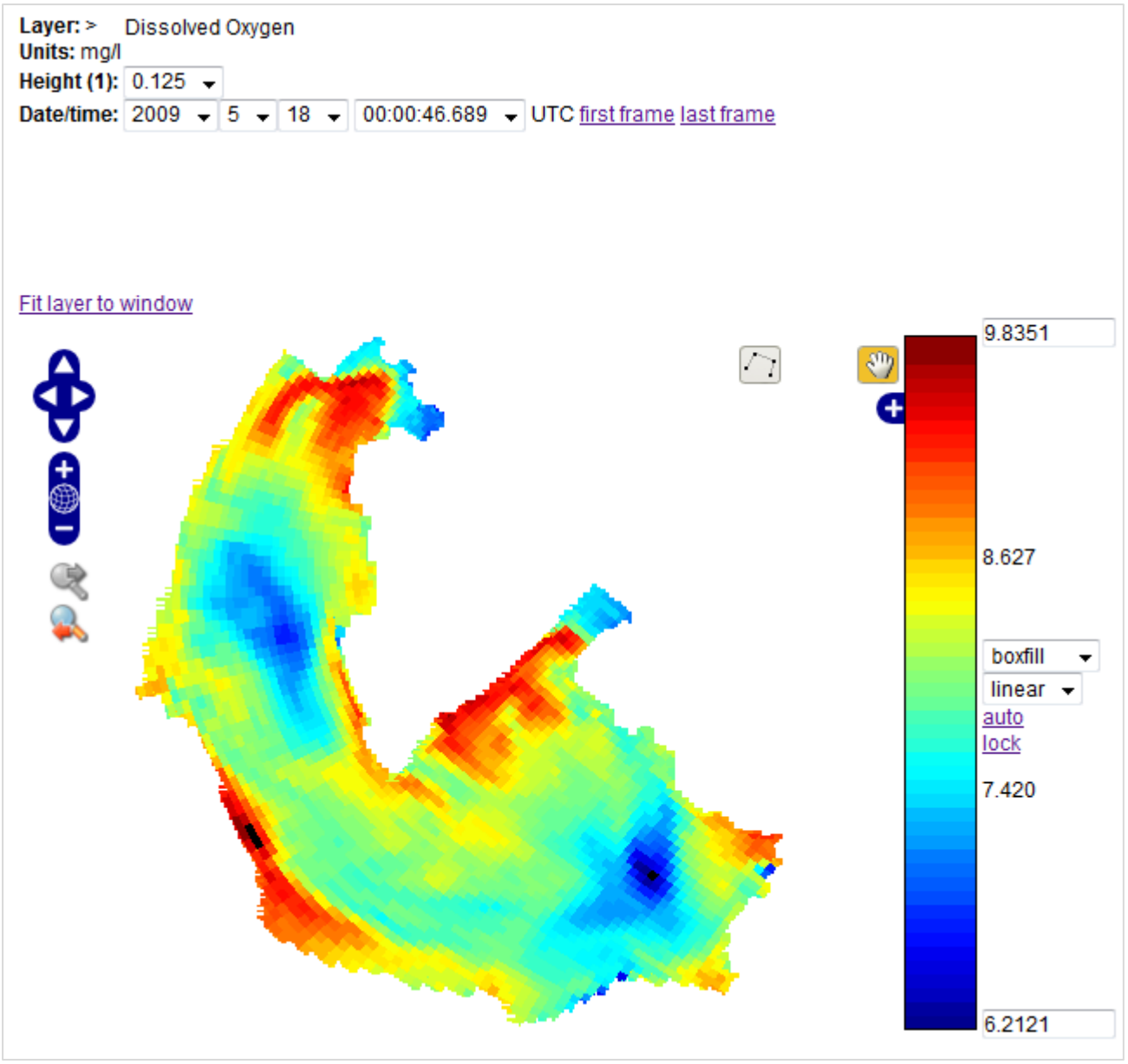


Figure 6-81 ncWMS display of EFDC+ netCDF output – DO at 137.001.

6.5.7.3 *NetCDF Parameter Description*

ID	Name	Units	positive	grid_mapping	coordinates	dim_type	fillvalue	standard_ name	long_name
1	ZBOT	m	down	crs	lat lng	2	MISSING_VALUE		Bottom Bathymetry
2	WSEL	m	up	crs	lat lng	3	MISSING_VALUE		Water Surface Elevation
3	VELX	m/s		crs	lat lng	4	MISSING_VALUE		Eastward Water Velocity
4	VELY	m/s		crs	lat lng	4	MISSING_VALUE		Northward Water Velocity
5	VELZ	m/s		crs	lat lng	4	MISSING_VALUE		Upward Water Velocity
6	TAUB	N/m2		crs	lat lng	3	MISSING_VALUE		Bottom Shear Stress
7	WINX	m/s		crs	lat lng	3	MISSING_VALUE		Eastward Wind Velocity
8	WINY	m/s		crs	lat lng	3	MISSING_VALUE		Northward Wind Velocity
9	WHEI	m		crs	lat lng	3	MISSING_VALUE		Significant Wave Height
10	WANG	degree		crs	lat lng	3	MISSING_VALUE		Wave Direction
11	WPER	s		crs	lat lng	3	MISSING_VALUE		Wave Period
12	SAL	ppt		crs	lat lng	4	MISSING_VALUE		Salinity
13	TEM	C		crs	lat lng	4	MISSING_VALUE		Temperature

14	DYE	mg/l		crs	lat lng	4	MISSING_VALUE		Dye
15	SFL	mg/l		crs	lat lng	4	MISSING_VALUE		Shellfish Larvae
16	TOX	mg/l		crs	lat lng	7	MISSING_VALUE		Toxics
17	SED	mg/l		crs	lat lng	5	MISSING_VALUE		Cohesive Sediment
18	SND	mg/l		crs	lat lng	6	MISSING_VALUE		Non-cohesive Sediment
19	CHC	ug/l		crs	lat lng	4	MISSING_VALUE		Cyanobacteria
20	CHD	ug/l		crs	lat lng	4	MISSING_VALUE		Diatom Algae
21	CHG	ug/l		crs	lat lng	4	MISSING_VALUE		Green Algae
22	ROC	mg/l		crs	lat lng	4	MISSING_VALUE		Refractory Particulate Organic Carbon
23	LOC	mg/l		crs	lat lng	4	MISSING_VALUE		Labile Particulate Organic Carbon
24	DOC	mg/l		crs	lat lng	4	MISSING_VALUE		Dissolved Organic Carbon
25	ROP	mg/l		crs	lat lng	4	MISSING_VALUE		Refractory Particulate Organic Phosphorus
26	LOP	mg/l		crs	lat lng	4	MISSING_VALUE		Labile Particulate Organic Phosphorus
27	DOP	mg/l		crs	lat lng	4	MISSING_VALUE		Dissolved Organic Phosphorus
28	P4D	mg/l		crs	lat lng	4	MISSING_VALUE		Total Phosphate

29	RON	mg/l		crs	lat lng	4	MISSING_VALUE		Refractory Particulate Organic Nitrogen
30	LON	mg/l		crs	lat lng	4	MISSING_VALUE		Labile Particulate Organic Nitrogen
31	DON	mg/l		crs	lat lng	4	MISSING_VALUE		Dissolved Organic Nitrogen
32	NHX	mg/l		crs	lat lng	4	MISSING_VALUE		Ammonia Nitrogen
33	NOX	mg/l		crs	lat lng	4	MISSING_VALUE		Nitrate Nitrogen
34	SUU	mg/l		crs	lat lng	4	MISSING_VALUE		Particulate Biogenic Silica
35	SAA	mg/l		crs	lat lng	4	MISSING_VALUE		Dissolved Available Silica
36	COD	mg/l		crs	lat lng	4	MISSING_VALUE		Chemical Oxygen Demand
37	DOX	mg/l		crs	lat lng	4	MISSING_VALUE		Dissolved Oxygen
38	TAM	mg/l		crs	lat lng	4	MISSING_VALUE		Total Active Metal
39	FCB	mpn/100ml		crs	lat lng	4	MISSING_VALUE		
40	MAC	ug/l		crs	lat lng	4	MISSING_VALUE		Macroalgae
41	MOC	kg		crs	lat lng	3	MISSING_VALUE		Oil Mass
42	TRK_LAT	degrees_north		crs		2	MISSING_VALUE		Drifter latitude
43	TRK_LON	degrees_east		crs		2	MISSING_VALUE		Drifter longitude
44	TRK_LEV	m		crs		2	MISSING_VALUE		Drifter elevation
45	TRK_VOL	m3		crs		2	MISSING_VALUE		Drifter Oil Volume

6.6 Hydrodynamics Tab

The *Hydrodynamics* tab provides the user with two sub-tabs: *General* and *Vegetation*. In the *General* sub-tab shown in Figure 6-82 the user may set the turbulence and numerical solutions options, as well as wind data, and wetting & drying as described below.

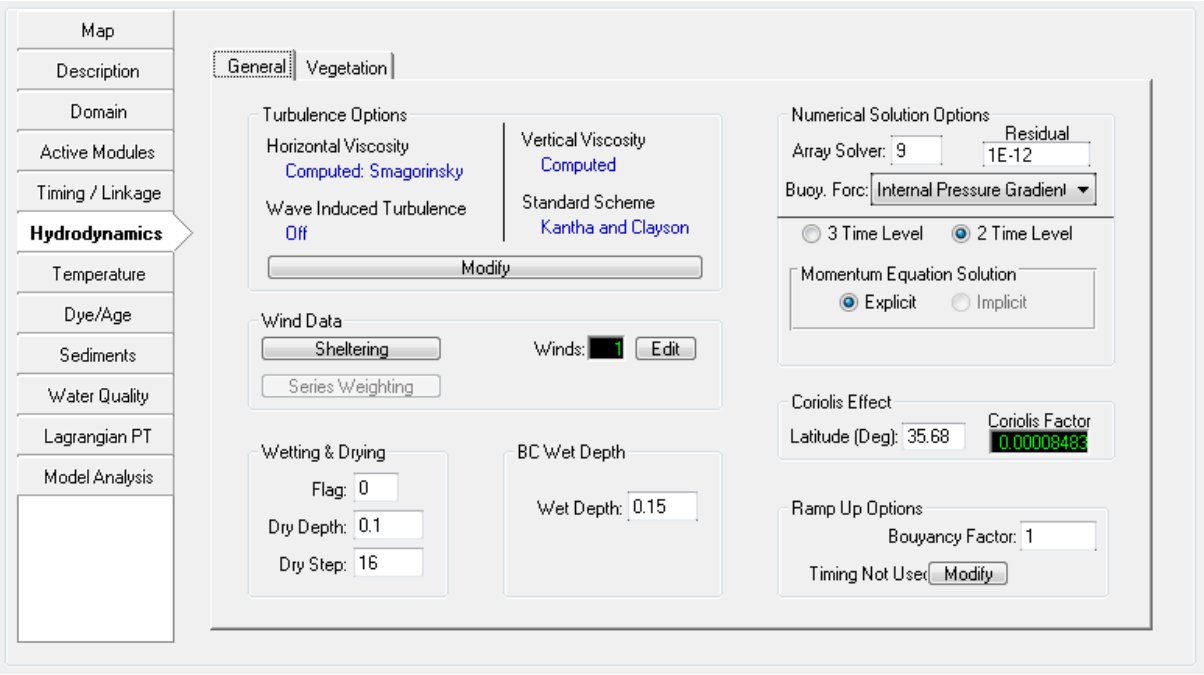


Figure 6-82 Hydrodynamics Tab: General.

6.6.1. Turbulence Options

The *Turbulence Options* frame displays the current settings for various hydrodynamic options such as *Horizontal Viscosity* and *Vertical Viscosity* and the *Standard Computational Scheme* for vertical turbulence. These options and parameter settings can be selected and modified by clicking on the *Modify* button. These options are described in the following sub-sections.

6.6.1.1 *Turbulent Diffusion*

Figure 6-83 shows the main form for the computational options for the horizontal and vertical eddy viscosities and diffusivities. The parameters and options have been grouped according to which dimension they are used for, i.e. horizontal or vertical.

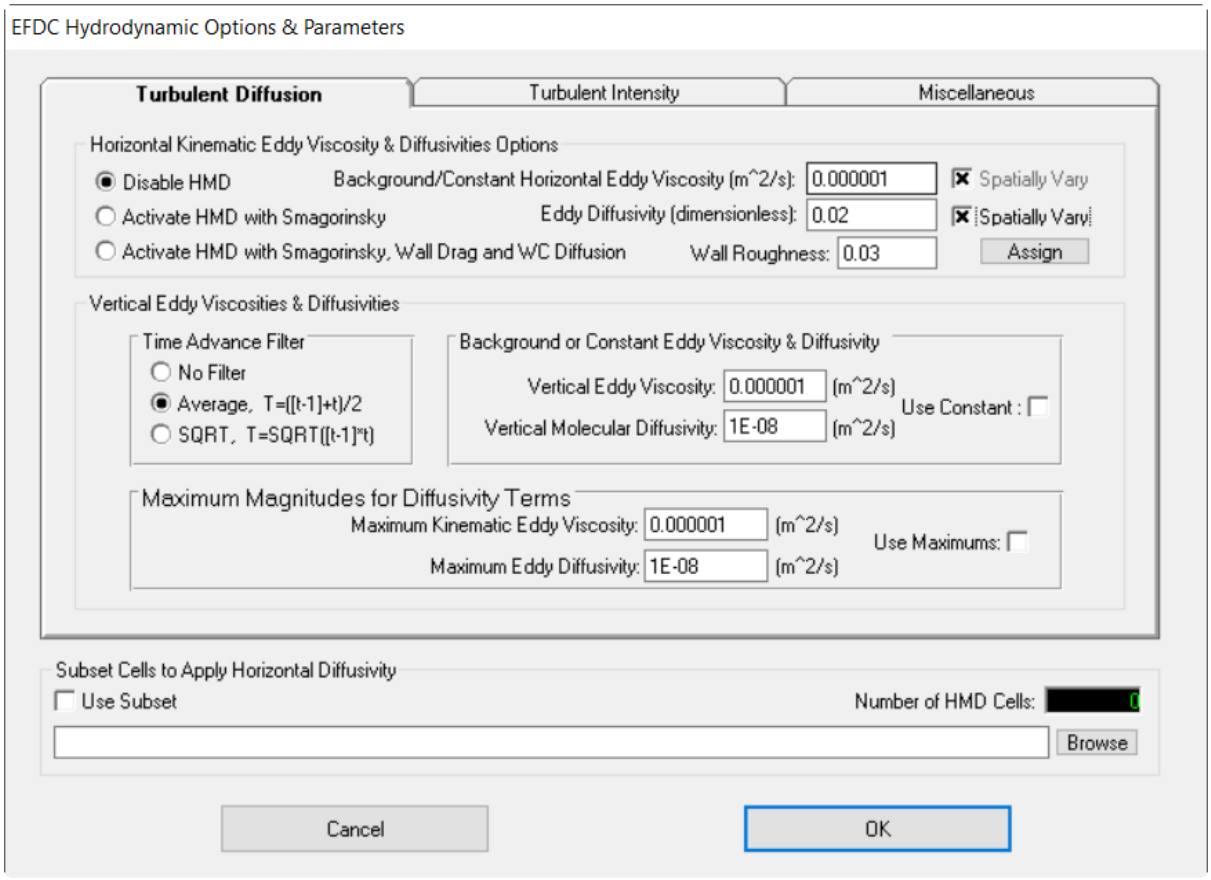


Figure 6-83 Hydrodynamics Options: Turbulent Diffusion.

The *Horizontal Kinematic Eddy Viscosity and Diffusivities Options* form provides an option box which is used to enable the Horizontal Momentum Diffusion (HMD) capability of EE. If the *Disable HMD* option is selected, the HMD is set to the *Background/constant Horizontal Eddy Viscosity* (AHO) set by the user. This is equivalent to the ISHM flag set to 0.

EEMS supports a spatially variable AHO option. When the *Spatially Vary* checkbox is selected, EFDCPlus will compute a separate AHO for each cell using the expression $ABS(AHO) * DX * DY$. This approach reduces the numerical diffusion error when the models have a large range of cell sizes.

The user should set AHD, Smagorinsky's Coefficient, in the *Eddy Diffusivity* (*Dimensionless)* dialog box when using the *Activate HMD with Smagorinsky* option. When *Activate HMD with Smagorinsky* is selected (ISHMD=1) then EE will apply the background HMD as well as Smagorinsky's subgrid if AHD>0. With this option the contaminant diffusion is off and not accounted for. If AHD=0 then the *Constant Viscosity* option will be used.

EEMS also supports a spatially variable AHD option The user needs to check both of the *Spatially Vary* checkboxes, in which case the *Assign* button is also enabled. Selecting the *Assign* button displays the form shown in Figure 6-84. EE uses a new file called AHMAP.INP to store a cell by cell list of AHO and AHD. If the AHO in the AHMAP file is a negative number, EFDCPlus will use the cell area times the input value to compute the cell's AHO according to the equation above.

Typically, a user wouldn't have data for spatially interpolated AHO and AHD, rather it is expected to use this function to assign constant values to certain regions. The AHO and AHD values can be assigned to the whole model domain using operators or constant values. They can also assign values of AHO and AHD to certain regions using polygon files. Load the polygon file, select the value for AHO or AHD, and select *Apply*. When the user saved the project EE will create an AHMAP.INP file.

To ensure that the correct values have been assigned to the correct regions, the AHO and AHD may also be visualized in *ViewPlan* using the *Fixed Params* viewing option.

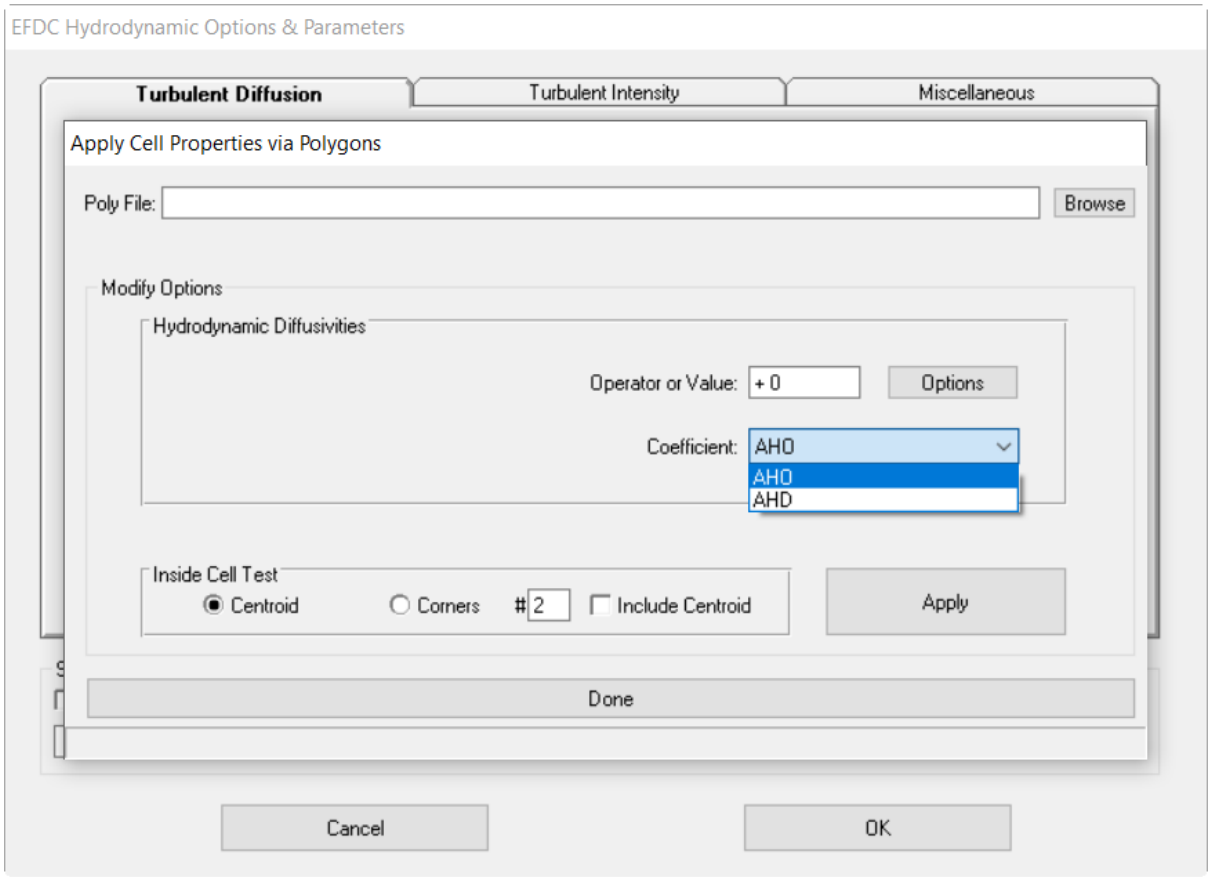


Figure 6-84 Setting spatially varying AHO and AHD.

When *Activate HMD with Smagorinsky, Wall Drag and WC Diffusion* is selected (ISHMD=2), then the full HMD is used as well as wall effects. This option also applies diffusivities to all constituent transport, which includes salinity and temperatures. This option should be used when simulating a very uniform flow system is constituent transport is needed. The resulting horizontal diffusivity from these options can be viewed from within *ViewPlan* provided velocities are also available. The vertical eddy viscosities and diffusivities (AV & AD arrays in EFDC, respectively) can also be viewed if the user configures the Internal Array Viewer (See Appendix A) to include these arrays.

The *Vertical Eddy Viscosities & Diffusivities* frame allows the user to set the Time Advance Filter and Vertical Eddy Viscosity as well as Vertical Molecular Diffusivity. The *Maximum Magnitude for Diffusivity Terms* allows the user to set the Maximum Kinetic Eddy Viscosity as well as the Maximum Eddy Diffusivity.

6.6.1.2 *Turbulent Intensity*

Figure 6-85 shows the *Turbulent Intensity* tab. Generally, only the *Vertical Turbulence Options* will be changed. At the current time, production runs should set the *Advection Scheme* to 1 and the *Sub-Option* to either Galerpin (Sub-option=1), Kantha and Clayson (Sub-option=2) or Kantha (Sub-option=3).

The *Vertical Turbulence Limiting Options* provide a drop down menu that allows the user to choose from three options: no length scale and RIQMAX limitation; to limit RIQMAX in the stability function only; and finally, to limit both the length scale and RIQMAX. When using multi-layer models and vertical turbulence is limiting, the *Limit Length Scale and Limit RIQMAX* option provides runs with less likelihood of model crashes.

The *Wall Proximity Function* also allows the user three options: no wall proximity effects on the turbulence; to be parabolic over depth wall proximity; and finally open channel wall proximity.

The *Turbulence Closure Constants* may be changed after the *Modify* check box is selected. However, these constants should not be changed without good justification.

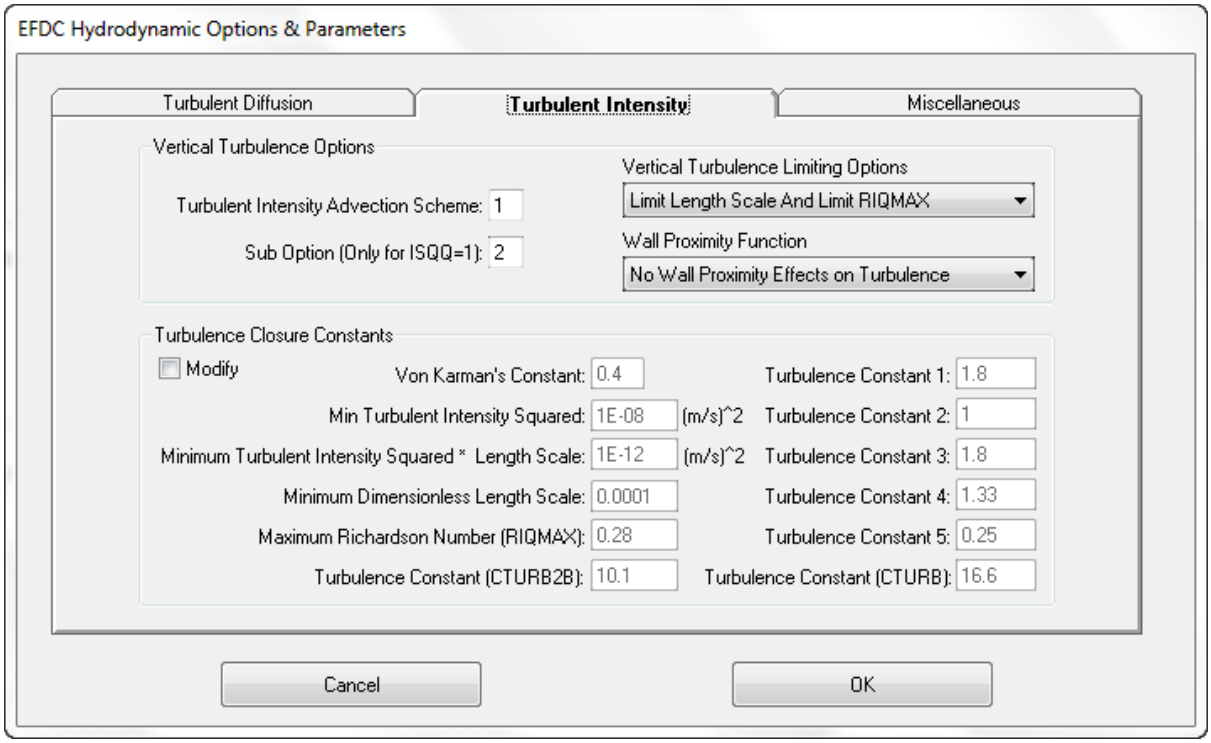


Figure 6-85 Turbulent Intensities.

6.6.1.3 *Miscellaneous Options*

Further turbulence modification options are also available under the *Miscellaneous Options* tab shown below in Figure 6-86. These options allow various momentum corrections which may be enabled with the *Momentum Correction Flag*. When the flag is set to zero momentum corrections are disabled; when set to 1 they are enabled.

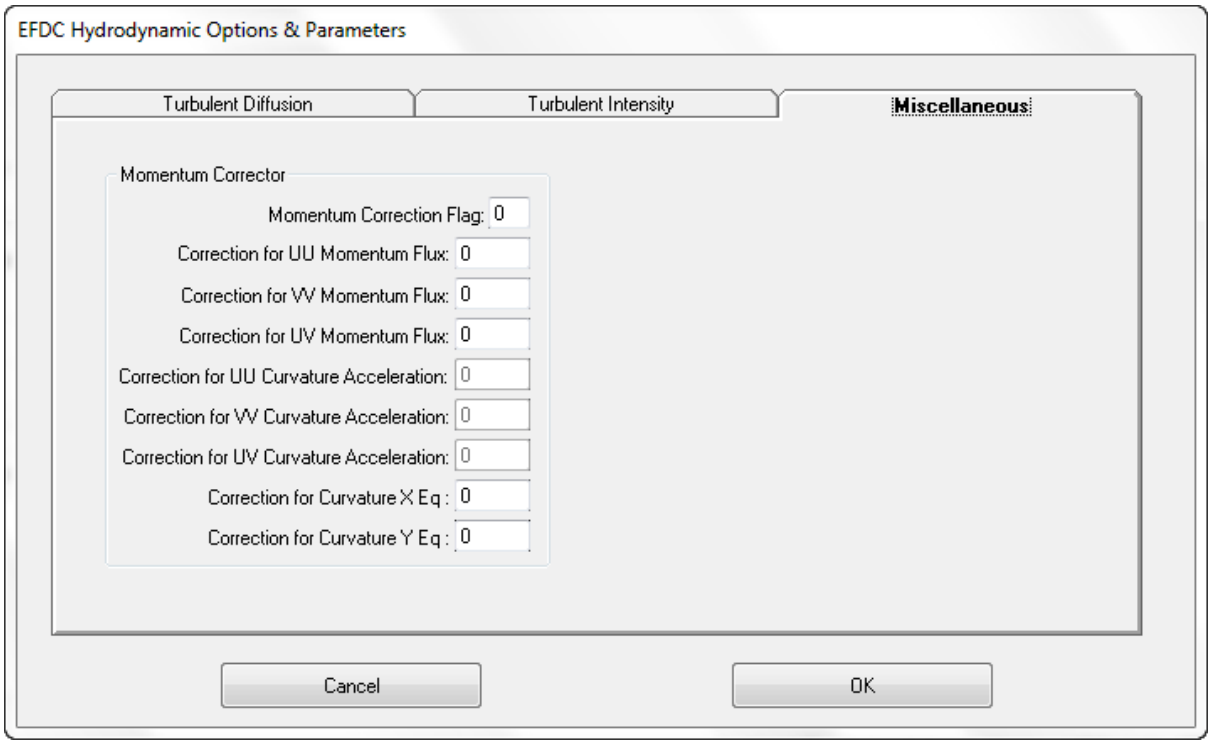


Figure 6-86 Miscellaneous Tab – Momentum Correctors

6.6.2. Wind Data

The *Wind Data* frame provides access to the WSER spatially varying options. The wind sheltering coefficient can be assigned using the *Sheltering* button and/or edited in *ViewPlan*. If more than one WSER series is being used it is necessary to must distribute the wind field assignments. This is done using the *Series Weighting* function described below.

6.6.2.1 *Wind Series*

EE has the ability to add multiple wind, atmospheric and ice stations. Stations may be defined and edited with the Edit button shown in Figure 6-82.

The user may set the coordinates of the wind series with the *Show Params* button in the *Wind BC Time Series Editing* as shown in Figure 6-87. EE will use the X, Y coordinates in the value column for display purposes. If the user has not entered the X, Y values then EE will automatically calculate X,Y coordinates based on the lat/long values provided. However, if both are entered it should be noted that EE will use the X, Y coordinates. The user can also set the anemometer height of the station.

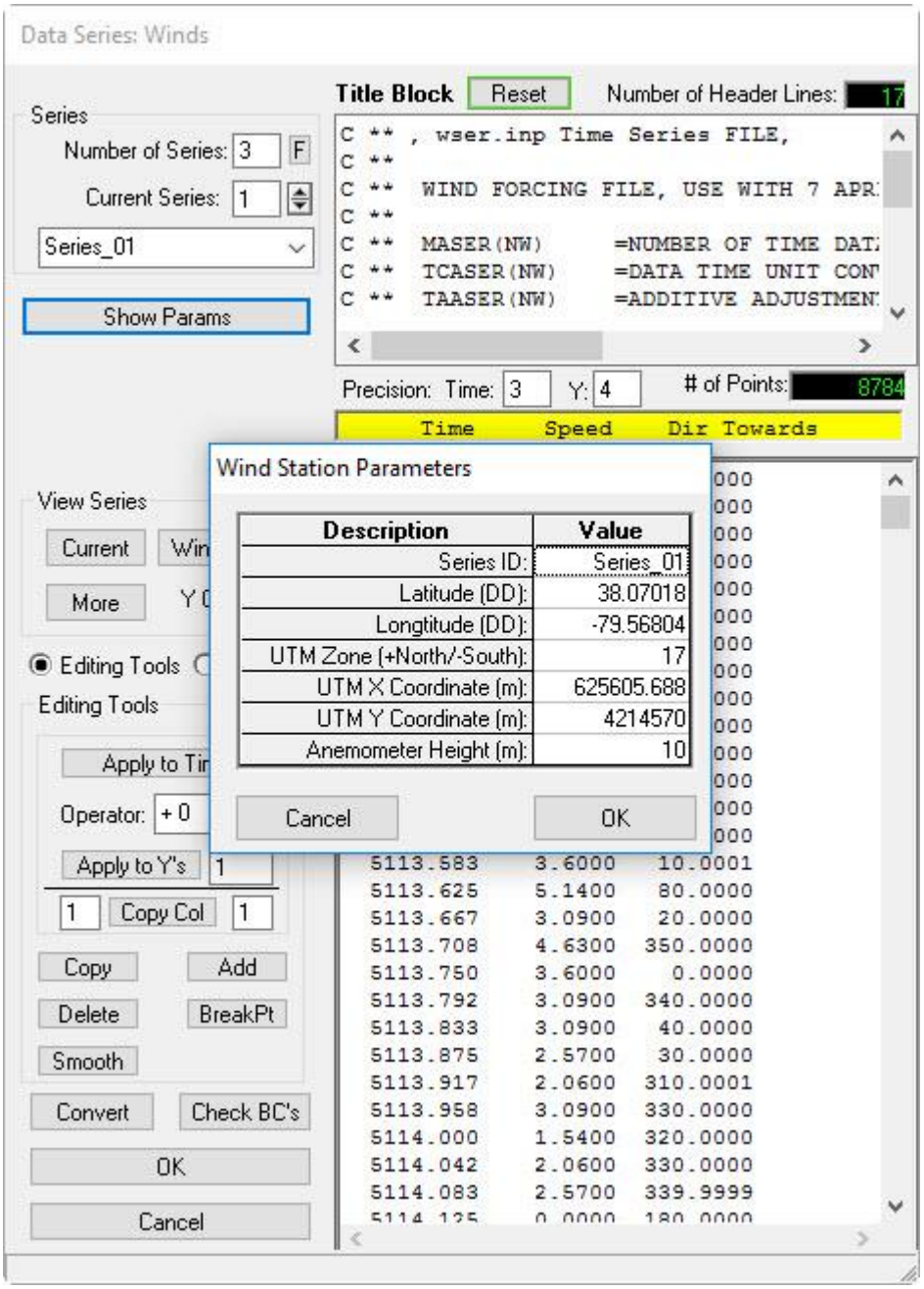


Figure 6-87 Wind Data Series: Station Coordinate Setting.

6.6.2.2 *Series Weighting*

EE and EFDC+ have been updated to provide a more flexible station weighting methodology for map files when handling multiple the atmospheric series, wind series and ice series. Each of these time series has a corresponding map files when there are multiple series. In EE8.1 a change was made to the EFDC input file for these map files so that station weighting would be handled in EFDC. While the overall approach provided users with a simpler methodology, having the station weighting calculated by EFDC removed the level of use control, and made support of legacy models more difficult. For this reason, the calculation of the station weighting is now handled by EE and the input files for EFDC have reverted to the previous format as described below.

The approach that was adopted from EE8.0 onward is to provide users with a simple tool for managing the data and stations for wind, ice and atmospheric data. This approach allows EE

to display the stations in a grid format with a quality index that indicates how the user rates each time series and time block within that series.

An example of a model domain with three wind stations is shown in Figure 6-88. This example has three stations WSER_1, WSER_2 and WSER_3. These have a total of of five time series for wind as shown in Figure 6-89.

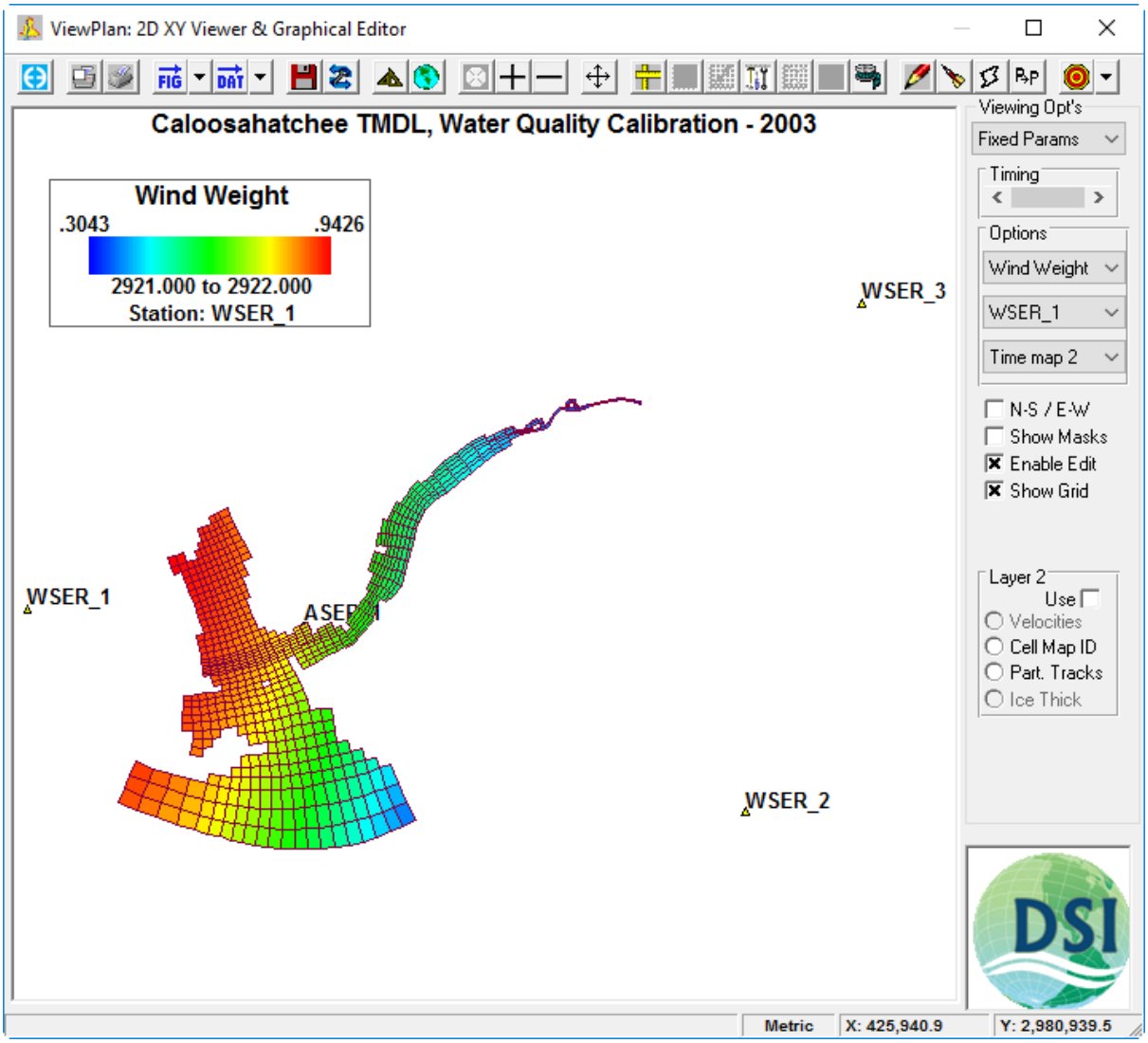


Figure 6-88 Wind stations and time series.

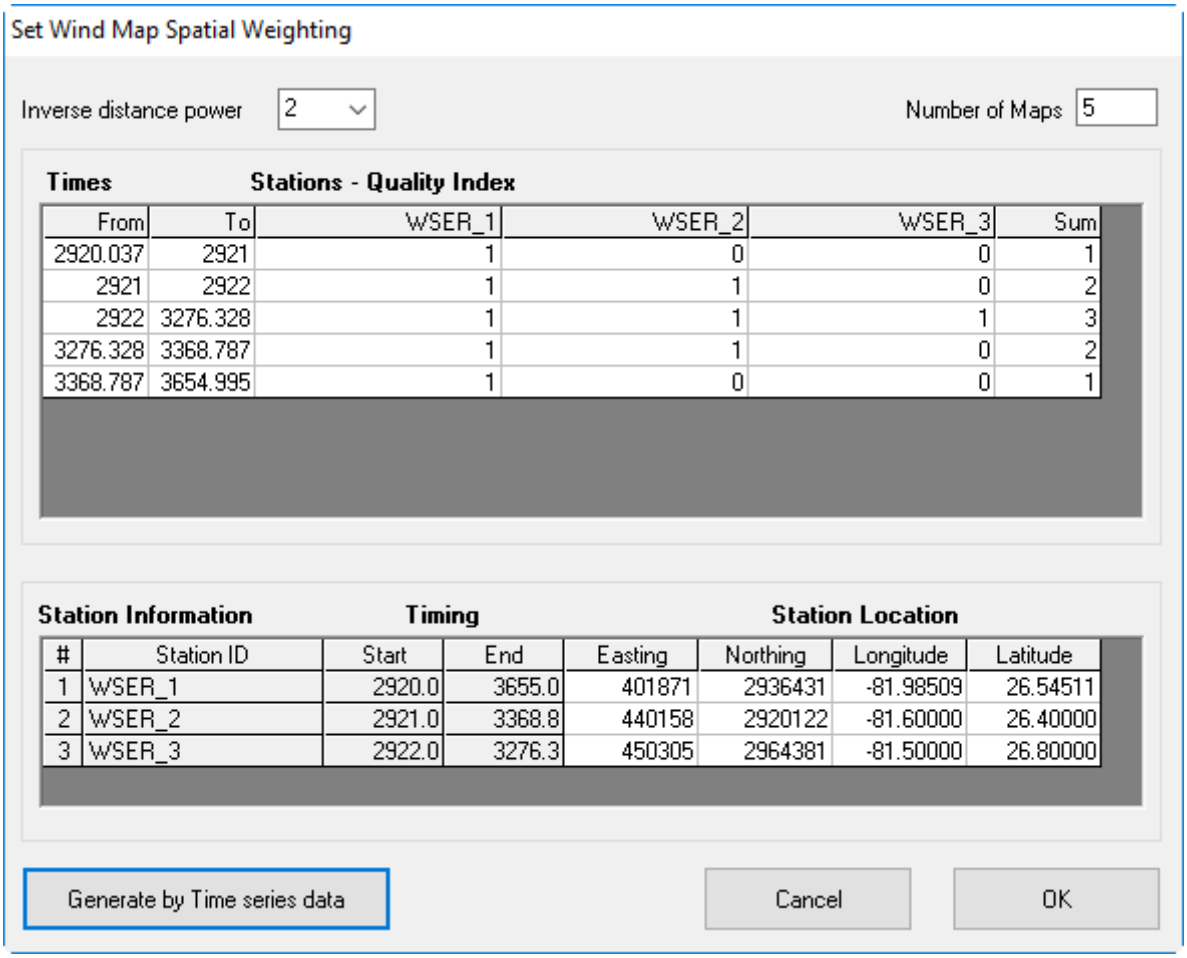


Figure 6-89 Wind Map: Default quality indexes for wind stations and time series.

In Figure 6-89, the default quality indexes have been set by EE after selecting the *Generate by Time Series Data* button. EE automatically generates the quality index with the maximum value of one for the defaults, which is 1. The user now needs to provide each time series with a quality index from 0 to 1. This weighting is based on users judgment of the quality of this data series based on various factors. If the user knows of no problems with the series then they would give it a rating of 1. However, often various factors which will influence the quality of a time series and cause the user to downgrade the series such as the distance from the model domain, or an intervening geographical feature. In this case a lower value for the quality index will be selected, and if another times series is available for that period with a higher weighting it will have a greater impact on the model.

Each time series can be separated into many parts to set different quality/influence levels if necessary. EE will use the values provided in the *Wind Map Spatial Rating* table to generate the WNDMAP.INP file, an example of which is provided in the appendix. This file contains the calendar interval (Julian day) for which the following map is valid, and then the on each row is I, J, and weighting for each time series. This makes the format similar to EFDC GVC WNDMAP.INP file, however it should be noted that GVC has certain requirements for this header files (six comment lines only). In a similar way, atmospheric series (ASER.INP), and ice series (ISER.INP) have map files which are ATMMAP.INP and ICEMAP.INP respectively.

The weighting map that EE generates is calculated based on the inverse squared distances from the cells to the stations and the qualitative coefficients for each time series. It includes the number of maps and the corresponding blocks of weighting coefficients. The first row of each block is beginning time, ending time and the qualitative coefficients. The remaining part of the block is the map of true weighting coefficients, including LA-1 data rows (L=2, LA). The true weighting coefficients are calculated as follows:

$$\beta_n = \left(\frac{\alpha_n}{R_n^2}\right) \Big/ \sum_{n=1}^{N}\left(\frac{\alpha_n}{R_n^2}\right)$$

$$0 \le \alpha_n(t) \le 1, n = 1, N$$

$$R_n^2 = \left(XC(L) - XS(n)\right)^2 + \left(YC(L) - YS(n)\right)^2, n = 1, N$$

Where

- N is the total number of time series (N the number of stations),
- N=NASER for ASER.INP; N=NWSER for WSER.INP N=NISER for ISER.INP
- XS(n), YS(n) UTM coordinates of one station to get time series,
- XC(L), YC(L) UTM coordinates of the cell L.
- L = index of cell,
- α_n is qualitative coefficient of each time series

EFDC then reads the true weighting coefficients from the map file to calculate the averaged value V(L) of each cell for every time step by:

$$V(L) = \sum_{1}^{N} \beta_n V_n$$

6.6.2.3 *Wind Rose Plotting*

A wind rose plotting function is now available within the Data series editing forms for winds. The winds data series form may be accessed in the *Domain* | *Boundary Conditions* | *Number of Input Table and Se*ries form and pressing the E button, or from Hydrodynamics tab | Wind data | Edit. Selecting the Wind Rose button shown in Figure 6-90 prompts the user the start and end time for the rose data set to be displayed. After entering these data wind rose is displayed as shown in Figure 6-91.

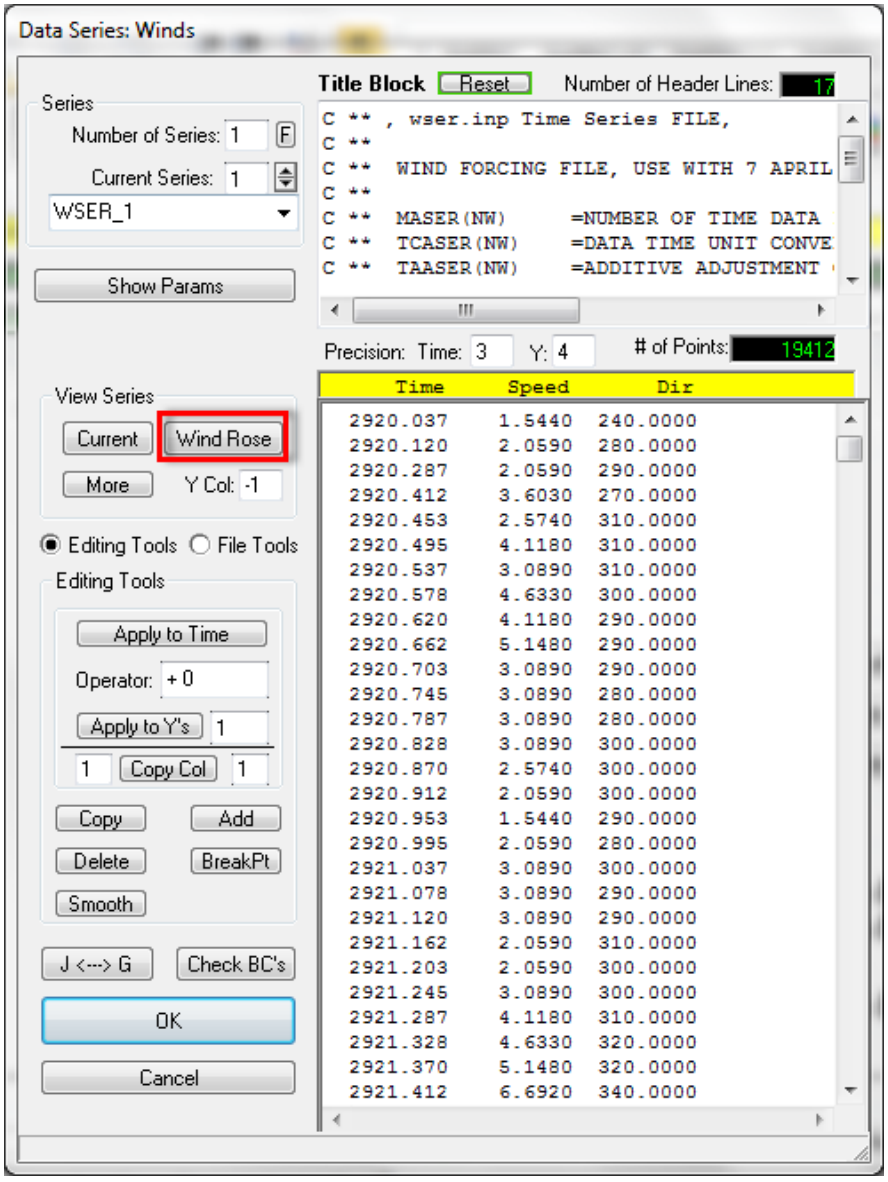


Figure 6-90 Boundary Conditions: Rose Plots for Winds.

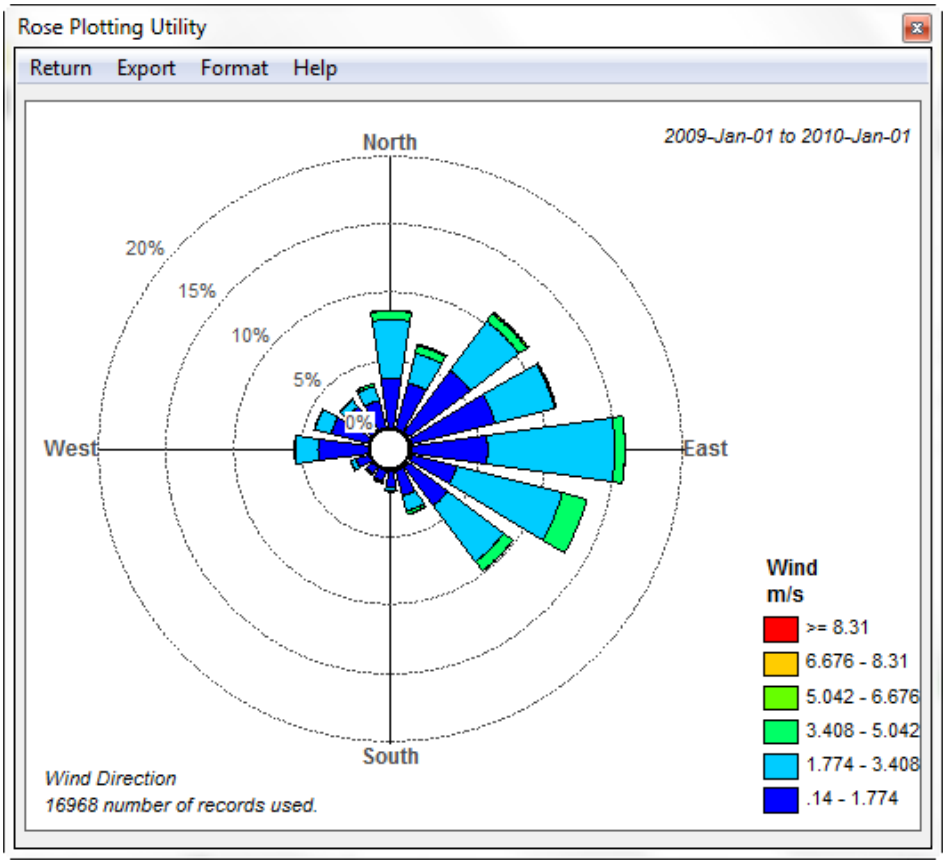


Figure 6-91 Example of a Wind Rose Plot.

A number of export functions are provided including metafile, bitmap and ASCII. The user user can also customize the rose format by RMC on the rose or selecting *Format* | *Scale* in the menu as shown in Figure 6-92.

Other options in the *Format* dropdown include editing the wind rose labels and setting the number of sectors. When setting the sector options the user can select the number of sectors, the number of degrees for the gap between sections and switch on and off the outline of the sectors.

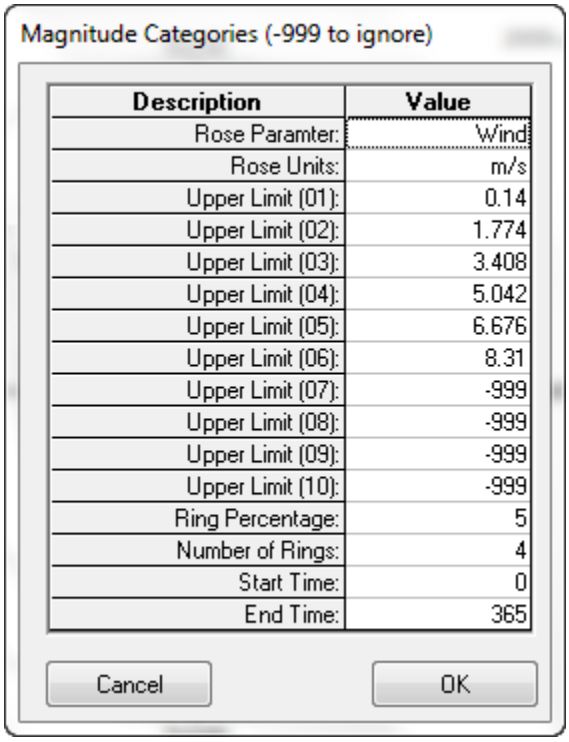


Figure 6-92 Rose Plot Category Options.

6.6.3. Wetting and Drying

The *Wetting and Drying* frame uses a flag to select various options. These options can change depending on the model settings. Selecting the *Use* check box turns on wetting and drying.

The *Dry Depth* defines at what depth a cell can be considered dry. When using wetting/drying the *Minimum Height* in the *Domain* tab | *Water Depth Setting (Legacy)* frame should be less than the *Dry Depth*. If this is not the case the initial water depths everywhere will force all of the models cells to "wet", even if they should be dry.

The *Dry Step* is used if there is a wet cell that is isolated or perched and which has a very small depth. In this case, after a certain number of time steps set by this *Dry Step* value, EFDC will waste the water. EFDC continues to keep track of this cell and the volume of water removed from the system to calculate mass balance. In EFDCPlus, if the *Dry Step* is a positive value then it doesn't have significant impact on the model. The value should be set to negative to activate dry cell wasting, eg -99. If the negative *Dry Step* is a small number then EFDC will provide more reporting of cells going dry in the run screen. If set a larger negative number (eg -1600), then it will to report less often.

In EFDC these settings are: ISDRY=0 turns off wetting and drying; abs(ISDRY)>0 turns wetting and drying on, ISDRY>1 but <99 turns wetting and drying on with zero dimension volume balance and ISDRY=99 is the algorithm for multi-face wetting and drying. If ISDRY is a negative (e.g. -99) then EFDC uses cell skipping to improve computational speed. Most applications that need wetting and drying should use ISDRY = -99.

6.6.4. Boundary Condition Wet Depth

The boundary condition *Wet Depth* is used to set the minimum depth for withdrawals i.e. it is depth below which withdrawals will be turned off from the cell.

> **Note**
> If any of the "multiplicative factor" settings for water depth or bottom elevation are 0 (or blank), EFDC will compute a zero for that parameter for the initial conditions.

6.7 Numerical Solutions Options

The *Numerical Solution Options* frame in *Hydrodynamics* | *General* (Figure 6-93) provides access to several of the numerical solutions schemes available in EFDC. From EE8.2 it has been simplified so that the *2 Time Level* and *3 Time Level* solutions are primary. The *Buoyancy Forcing Option*, *Array Solver*, and *Residual* options have been removed from the GUI as these are never set by the user (the standard case now implemented is the *Internal Pressure Gradient* option). Other options can still be set in the EFDC.INP file if the user wants to make such a change but they should understand the options and ensure the sub-models selected are correct if they make a change.

The solution schemes available with EFDC_GVC are *Standardized* (0), *Ordered R/B Conjugate gradient* (2), *Reduced R/B Conjugate gradient* (3), and *Conjugate gradient for Wetting and Drying* (9). Internal Buoyancy forcing option must be specified.

The *3 Time Level* (3TL) or *2 Time Level* (2TL) options allow the user to select the method to be used for numerical simulation. If using the EFDC_GVC model with GVC layering, the only *Time Level solution* available is the "3 Time Level".

If using the 3TL option, the user must specify the momentum solution option (ISCDMA) of Upwind, Central, and Upwind following the MOM momentum advection scheme. There are other schemes (ISCDMA>2) that are experimental.

The 2TL solution is the most robust solution technique and is recommended.

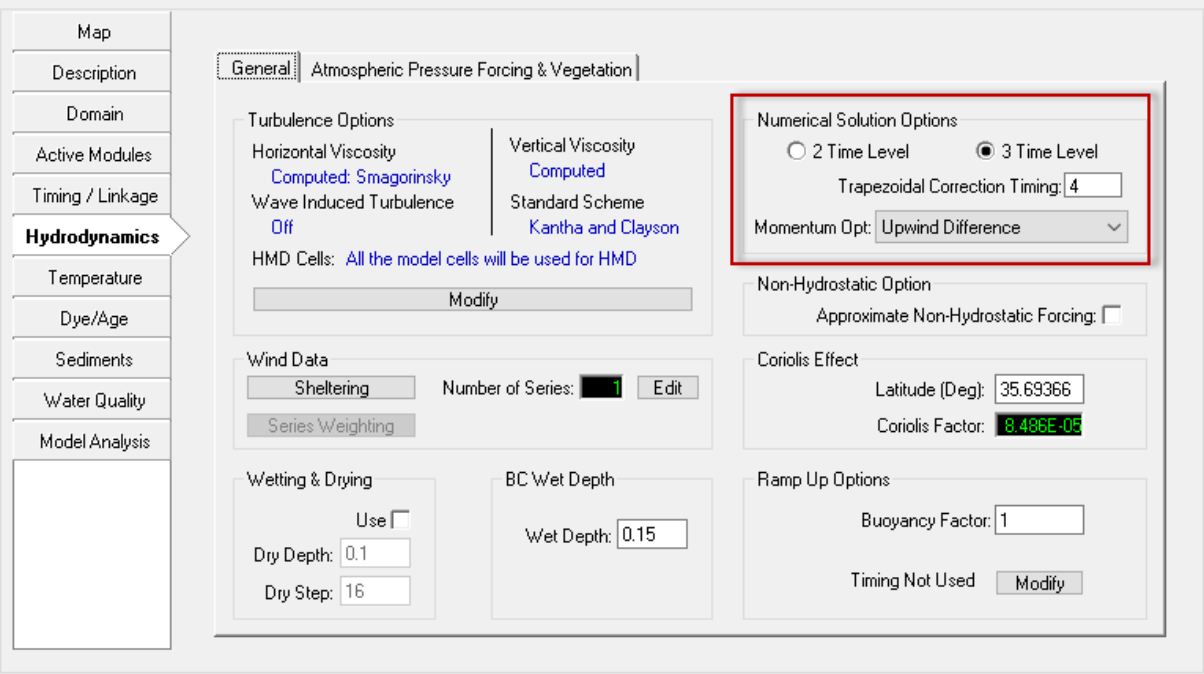


Figure 6-93 Numerical Solution Options.

6.7.1. Coriolis Effect

The *Coriolis Effect* frame allows the user to determine the Coriolis number by setting the *Latitude (Deg)*, which is the latitude of the midpoint of the model domain.

6.7.2. Ramp Up Options

The *Ramp Up Options* frame in the lower left-hand corner allows the user to modify the timing to full transition physics. Buoyancy can be adjusted with a buoyancy factor that must be between 0 and 1, with 0 being no buoyancy influence and 1 being full buoyancy influence. For some applications, the ramp up time from the ICs to natural physics require the phasing in of the buoyancy and the non-linear terms. This is accomplished using the *Modify* button and applying number of ramp up iterations for the various options.

6.7.3. Atmospheric Pressure Forcing

The EFDC_Explorer/EFDC+ Modeling System has been updated to include the atmospheric pressure horizontal gradients in the model domain on the hydrodynamics. This allows EFDC to address pressure variations in large coastal systems.

EFDC+ has been updated for the pressure solution in the CALEXP2T and CALEXP routines. The modification does not make any contributions to the hydrodynamic field when the number of pressure series (NASER) is less than two, because the horizontal gradient components of atmospheric pressure (pa) are always zero. However, if pa is not a constant in the computational domain, then gradients are computed and these will impact the external solution pressure field. This approach does not violate the hydrostatic assumption

The impact of atmospheric pressure is included in a number of hydrodynamic models including CH3D. The USACE uses this approach when they are simulating coastal systems.

Figure 6-94 shows the *Atmospheric Pressure Forcing* form in the *Hydrodynamics* tab of the main EE form. Here the user can select the *Edit* button to open each atmospheric series and edit them as shown in Figure 6-95. When there are multiple series the *Series Weighting* button is enabled and the user can set the weighting for the series as shown in Figure 6-96. Details of how to set the series weighting are described under Series Weighting. Note that atmospheric series may also be edited in the *Temperature* | *General and Data* tab when temperature is being simulated.

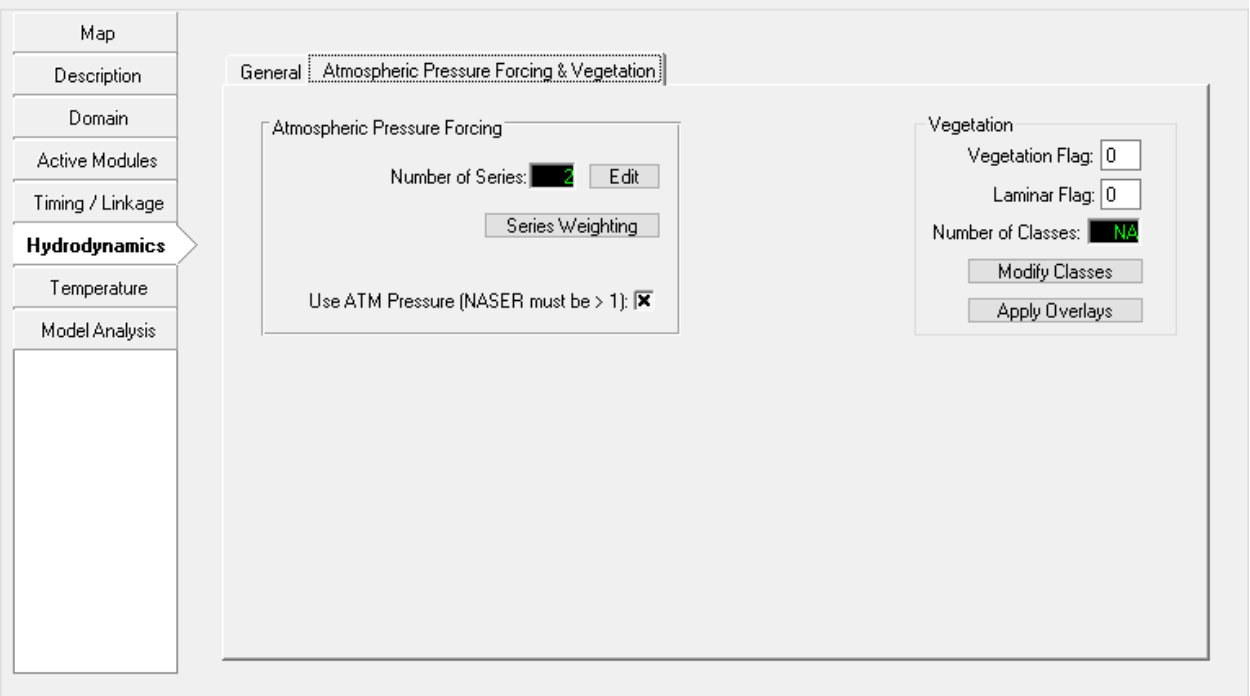


Figure 6-94 Hydrodynamic: Atmospheric Pressure Forcing frame.

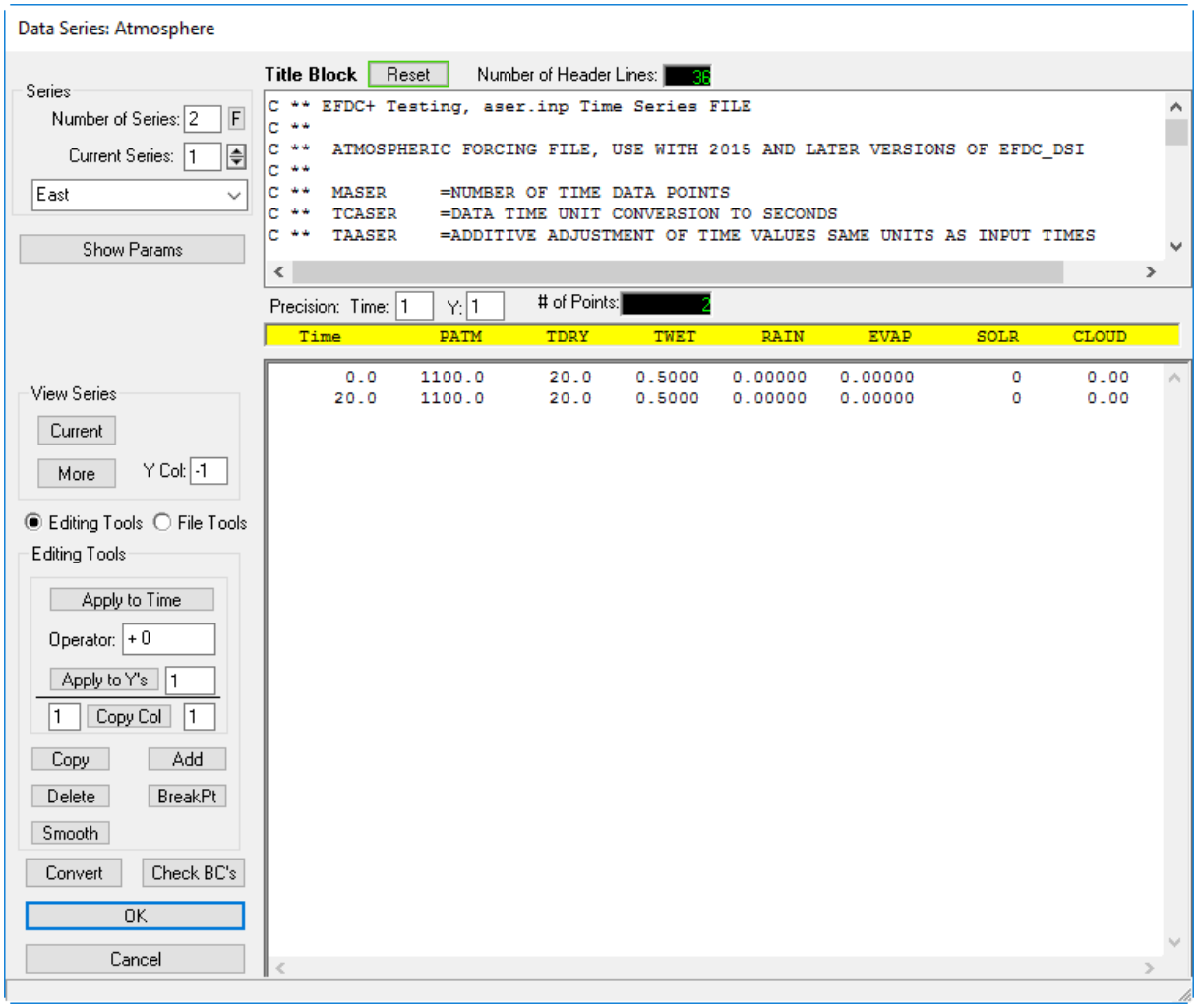


Figure 6-95 Data series for atmospheric data.

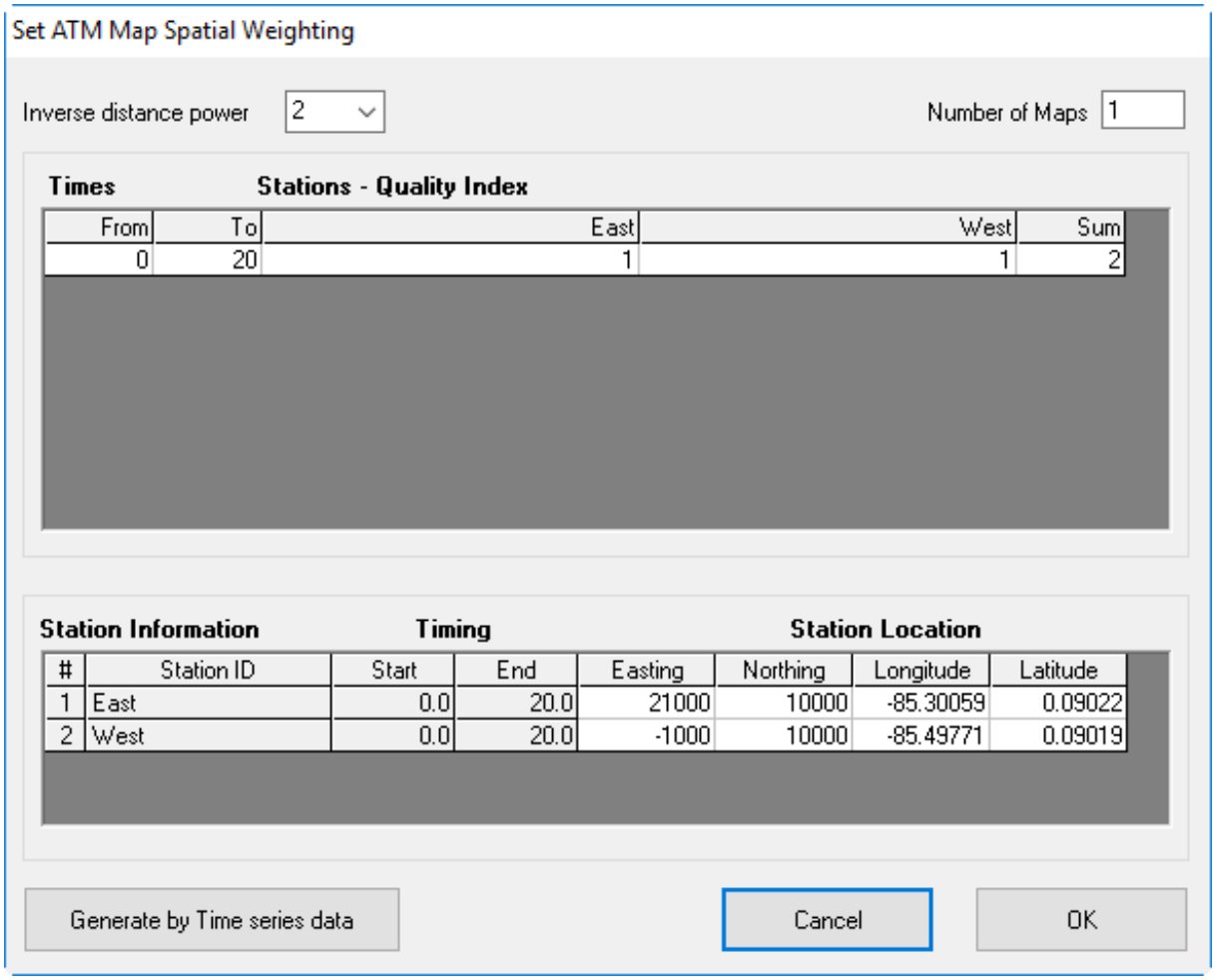
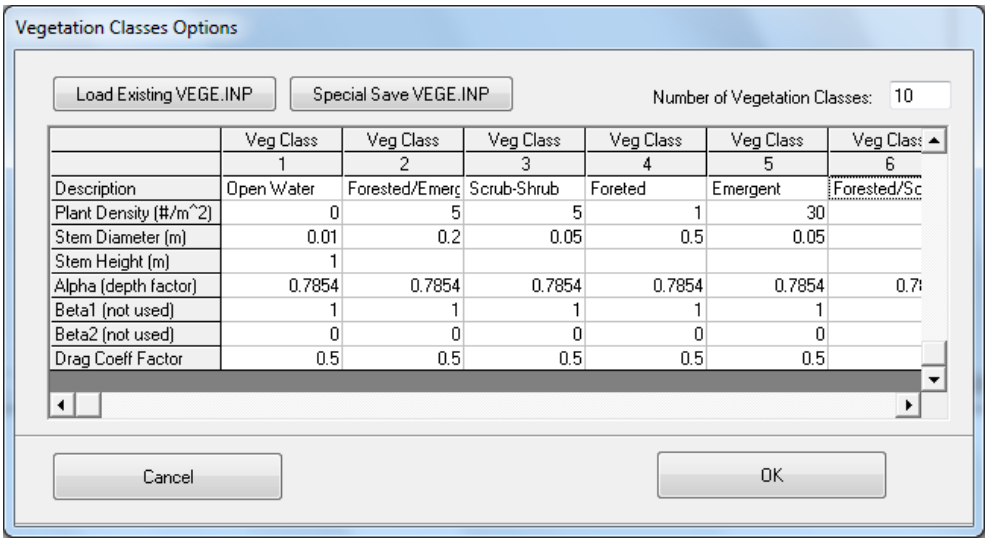


Figure 6-96 Spatial Weighting form.

6.7.4. Vegetation

In the *Vegetation* frame on the main *Hydrodynamics* tab, the user can set the primary computational option for enabling the vegetation feature and at what level (see the tooltips). The vegetation class settings are available via the *Modify Classes* button option which provides a user interface to the data that are needed for the VEGE.INP file (Figure 6-97). First, the user should specify the number of vegetation classes needed. Enter the *Number of Vegetation Classes* into the input box and press return. The grid will be expanded to accommodate the desired number of classes. All the inputs required by EFDC are shown with a couple of exceptions. The Beta1 and Beta2 variables are actually not used but are included in the list. The VEGE.INP file must have data in those columns but they are not actually used by EFDC. The other exceptions are the ID and Description fields. They are only used by EFDC_Explorer. The ID field is used to match vegetation classes to polygon ID's (see the following description of *Apply Overlays*) to automatically set the vegetation map that is needed for input via the LXLY.INP file. The *Description* field is only used for labeling.



	Veg Class 1	Veg Class 2	Veg Class 3	Veg Class 4	Veg Class 5	Veg Class 6
Description	Open Water	Forested/Emerg	Scrub-Shrub	Foreted	Emergent	Forested/Sc
Plant Density (#/m^2)	0	5	5	1	30	
Stem Diameter (m)	0.01	0.2	0.05	0.5	0.05	
Stem Height (m)	1					
Alpha (depth factor)	0.7854	0.7854	0.7854	0.7854	0.7854	0.7
Beta1 (not used)	1	1	1	1	1	
Beta2 (not used)	0	0	0	0	0	
Drag Coeff Factor	0.5	0.5	0.5	0.5	0.5	

Figure 6-97 Vegetation class parameters.

The *Apply Overlays* button uses the vegetation class ID field and matches it to input polygon ID's. Figure 6-98 shows the form for applying a vegetation map to the model cells. The file containing one or more polygons in the same file (see Appendix B for polygon formats) needs to be opened (via the *Browse* button). The polygon file will be read and the polygon ID's will be matched to the vegetation class ID's. If any vegetation classes do not have any defining polygons the user will be notified. Once the initial matching process is complete the user then must choose to perform the actual vegetation assignment. If there are unmatched vegetation classes, the user can choose to process anyway but they need to be aware that no EFDC cells will be matched to any the vegetation classes that do not have a polygon. The classes can always be edited later, if desired. Figure 6-99 shows a vegetation map that resulted from the polygon/vegetation class assignments.

A simpler approach can be used to assign a single vegetation class to a polygon region if the P2D file contains only one polygon.

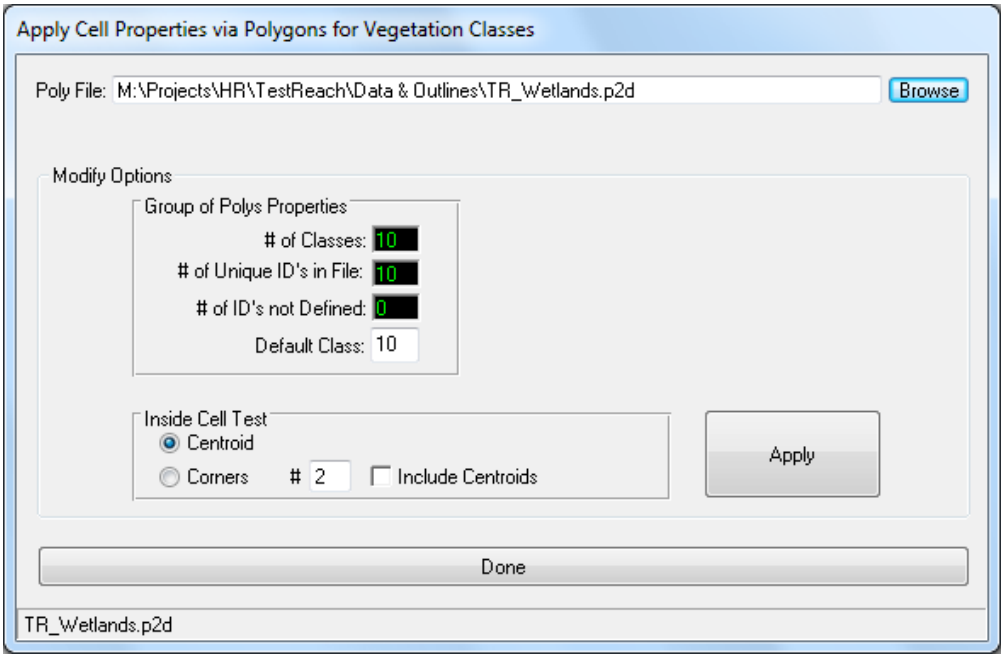


Figure 6-98 Cell property assignments: Vegetation map with IDs.

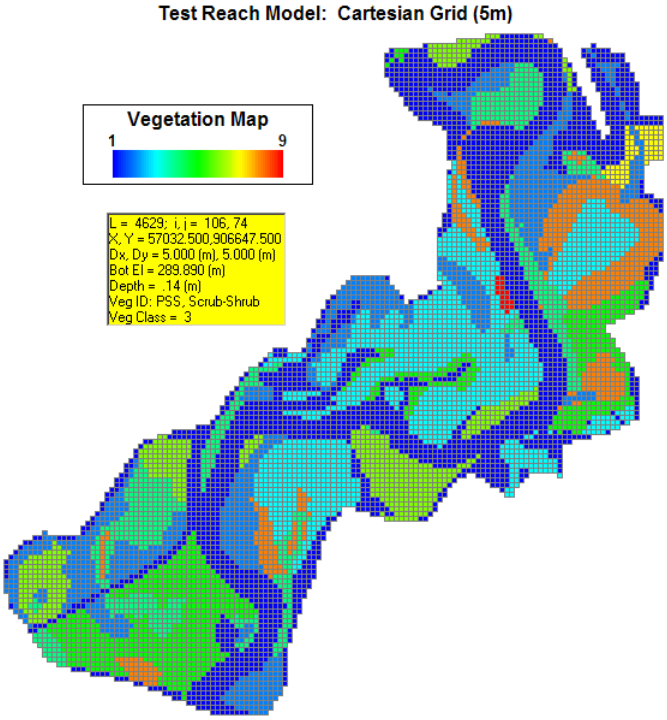


Figure 6-99 Example vegetation map assignment.

6.8 General Vertical Coordinates (GVC) Tab

When using the EPA GVC Model the GVC options are enabled as shown in Figure 6-100. The user must specify which layering option to use with the GVC model, either a Standard Sigma Vertical Grid (pre-GVC) layering option or a Generalized Vertical Grid (GVC) option that allows the number of layers to vary in the model.

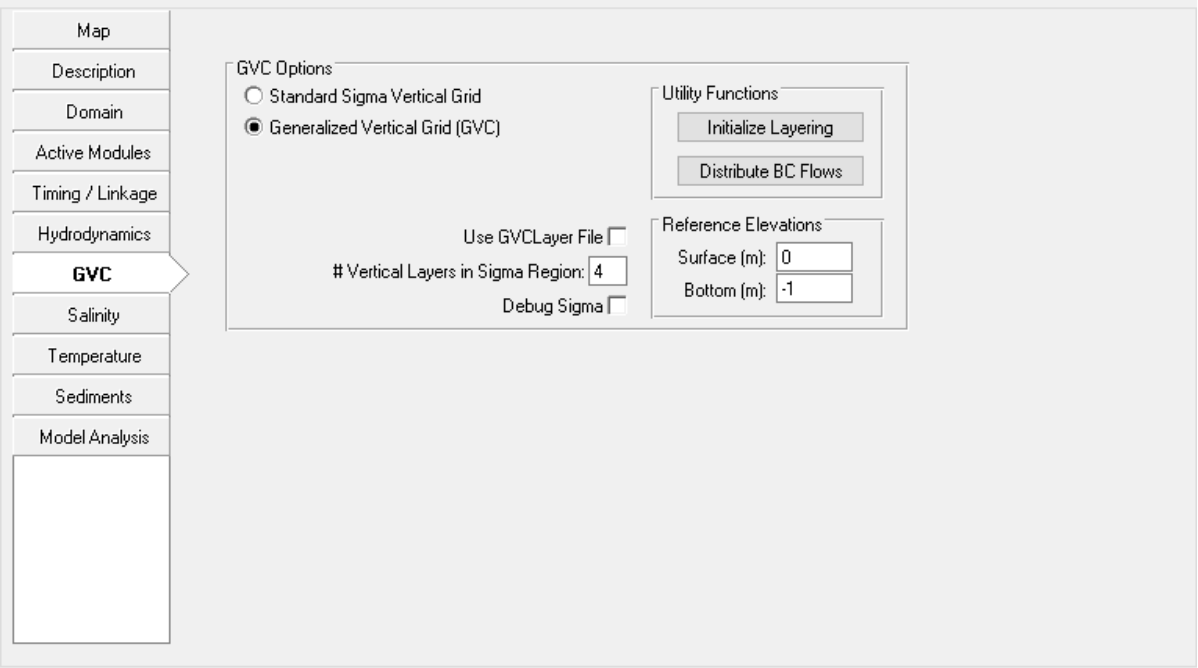


Figure 6-100 GVC Tab.

If the GVC layering option is selected the following sub-options must be specified:

- *Use GVC Layer File*: Check to manually set the cell by cell layers. When checked the GVCLAYER.INP file is written by EFDC_Explorer and then used by EFDC_GVC during the model run. If not checked, EFDC_GVC automatically assigns the GVC layers using the Surface and Bottom Reference Elevations. EFDC_Explorer can set and display the GVC layering. Use the *Initialize Layering* button to set the layering automatically, using the *Surface* and *Bottom Reference Elevations*. You can then view, modify and reset the layering in the *ViewPlan* function.
- A GVC grid has two types of cells, Local Sigma or GVC cells. Local Sigma cells use the number of layers specified in the *# of Vertical Layers in Sigma Region*. The number of layers must be less than or equal to KC.

The *Utility Options* frame allows the redistribution of the total flow into the active layers for all the flow type Boundary Conditions by pressing the *Distribute BC Flows* button. When using the GVC Grid, the *Initialize Layering* button allows the user to initialize the GVC layering using the Surface and Bottom elevations specified in the Reference Elevations form.

The Surface and Bottom Reference Elevations should initially be set to something near the maximum water surface elevation and minimum bottom elevation, respectively. These can be altered at any time and the layering recomputed using EFDC_Explorer to determine their validity. The validity of the layering can be viewed in the *ViewPlan* function, under *Fixed Params* viewing option.

6.9 Salinity Tab

Figure 6-101 shows the *Salinity* tab options. The method for assigning initial conditions for salinity is similar to that described for other water column parameters in Section 6.3.2.1. The user may also edit the time series data for this parameter with the *Edit* button in the *Time Series Data* frame.

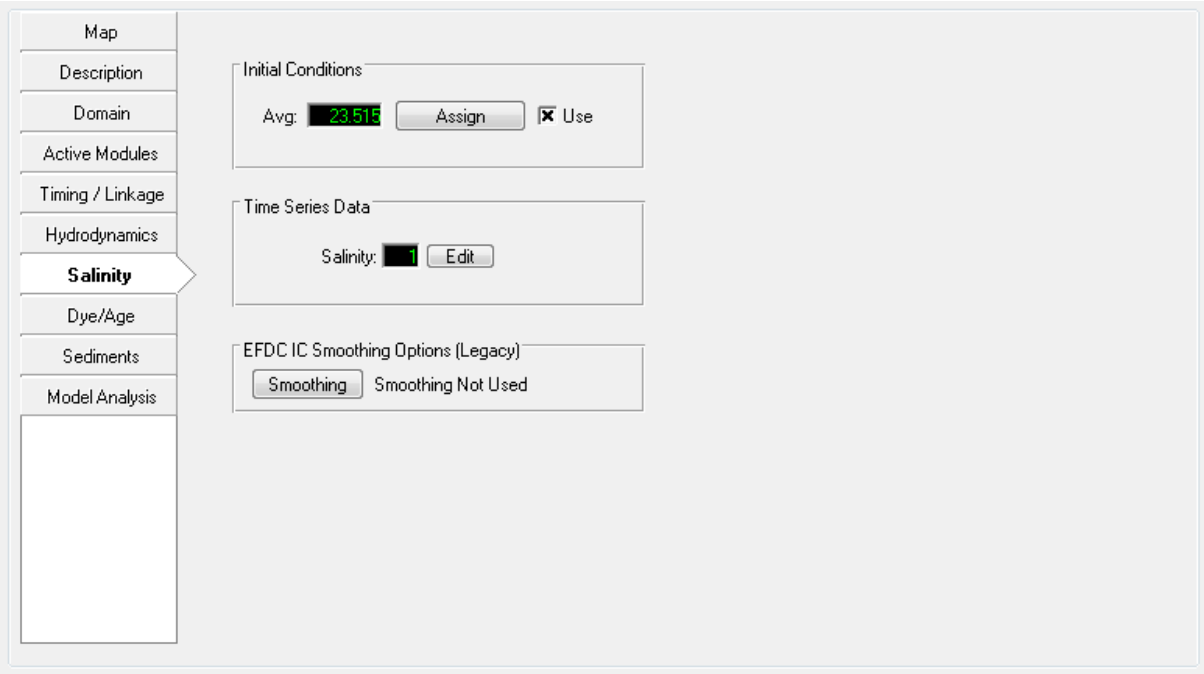


Figure 6-101 Salinity Tab

6.9.1. EFDC IC Smoothing Options

This is used to smooth initial conditions for the SAL/bottom elevation inside EFDC. However, it is recommended that smoothing be performed by EFDC_Explorer rather than the EFDC model. Pressing the *Smoothing* button brings up a form whereby the user may select the number of smoothing passes and smoothing factor for bathymetry and salinity.

6.10 Temperature Tab

If the user has selected the *Activate Temperature* checkbox in the *Active Modules* tab the *Temperature* tab will be enable as shown in Figure 6-102. There are two tabs available with for the temperature option, *General*, and *Ice Options*.

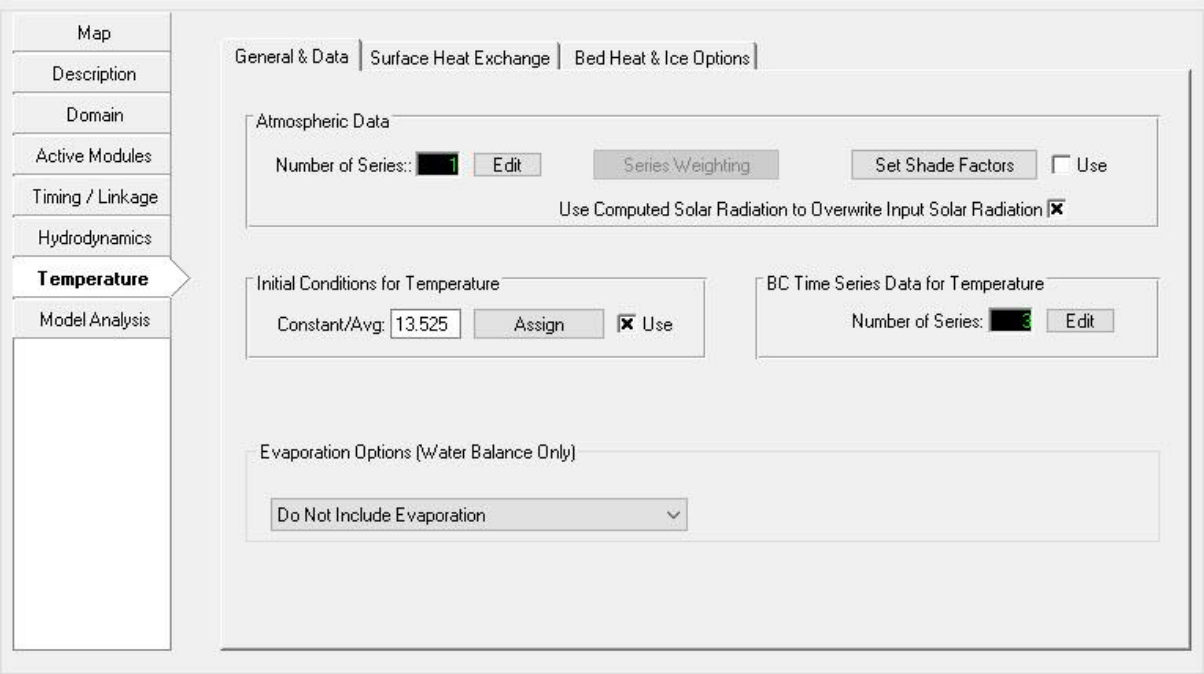


Figure 6-102 Temperature Tab.

6.10.1. General & Data

The *General & Data* tab contains the *Atmospheric Data* frame, as well as the *Initial Conditions for Temperature* frame, *BC Time Series for Temperature*, and *Evaporation Options* as shown in Figure 6-102. These options are fully described in the following sections.

6.10.1.1 *Atmospheric Data*

The *Atmospheric Data* frame provides access to creating and editing the atmospheric data series (ASER.INP) and spatial variation options. This series and weighting may also be edited from the *Hydrodynamics* | *Atmospheric Pressure Forcing and Vegetation* tab in models where temperature is not being simulated.

To set the coordinates of the atmospheric series, the user should select the *Edit* button on the *Atmospheric Data* frame which will provide the form shown in Figure 6-103. EE will use the X, Y coordinates in the value column. If the user has not entered the X, Y values then EE will automatically calculate X, Y coordinates based on the lat/long values provided. However, if both are entered it should be noted that EE will use the X, Y coordinates.

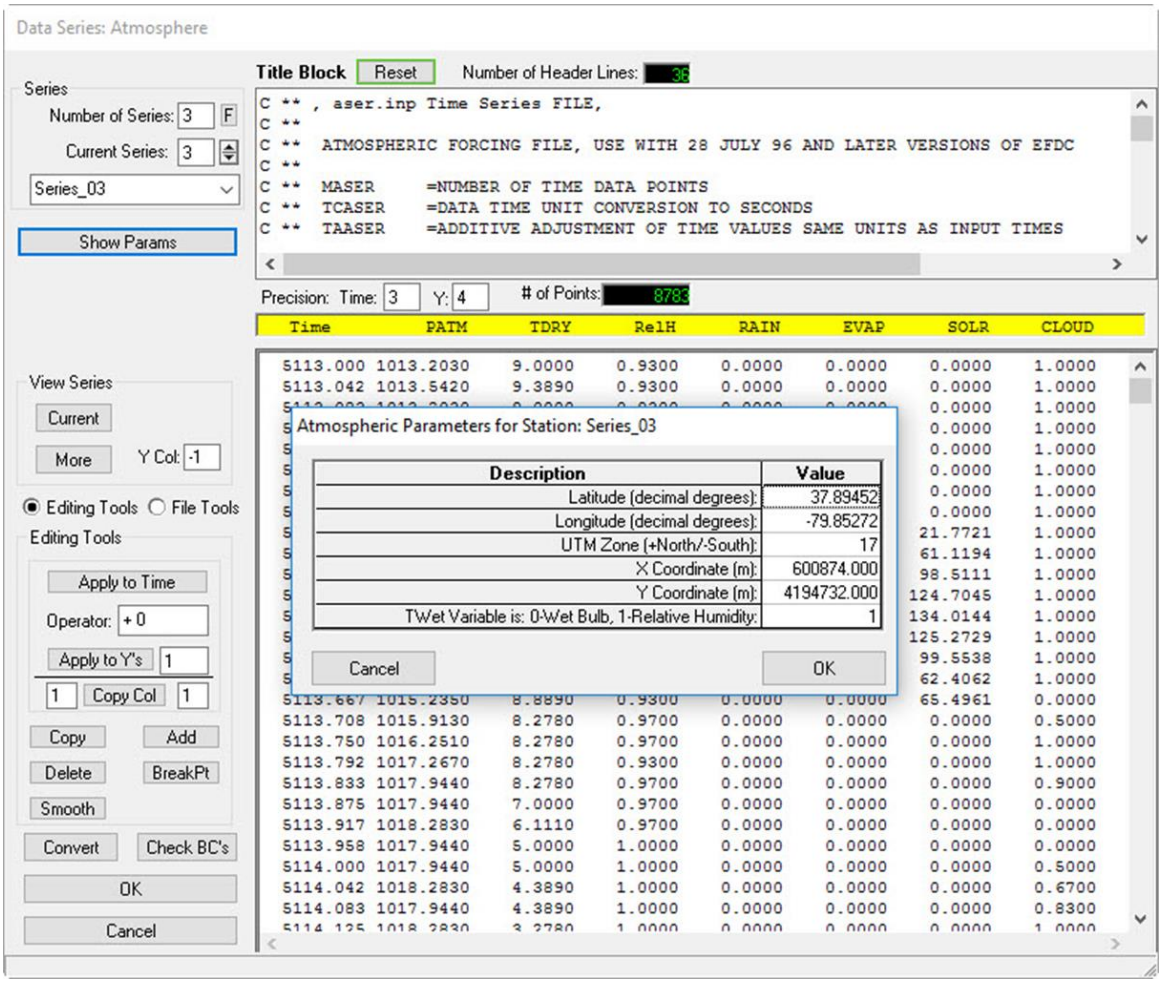


Figure 6-103 Atmospheric Data Series: Station Coordinate Setting.

The solar radiation shading factor can be assigned using the *Shade Factors* button and/or edited in *ViewPlan*.

If you are using more than one ASER series you must distribute the atmospheric data series assignments. This is done using the *Series Weighting* function shown in Figure 6-104. The same approach taken for the WSER series weighting is used for the ASER series weighting assignments (see Wind Data). The atmospheric series weight for each station used must be assigned. In *ViewPlan*, EFDC_Explorer provides an atmospheric series weighting map for each station or the wind weighting total (as a QC to make sure it adds up to 1.0).

When the user selects the *Series Weighting* button. To set the station weighting, enter the quality index in the table as shown in Interpolation options can be set in terms of the number of sectors and inverse distance power. An example of weighted stations displayed in *ViewPlan* is shown in Figure 6-105

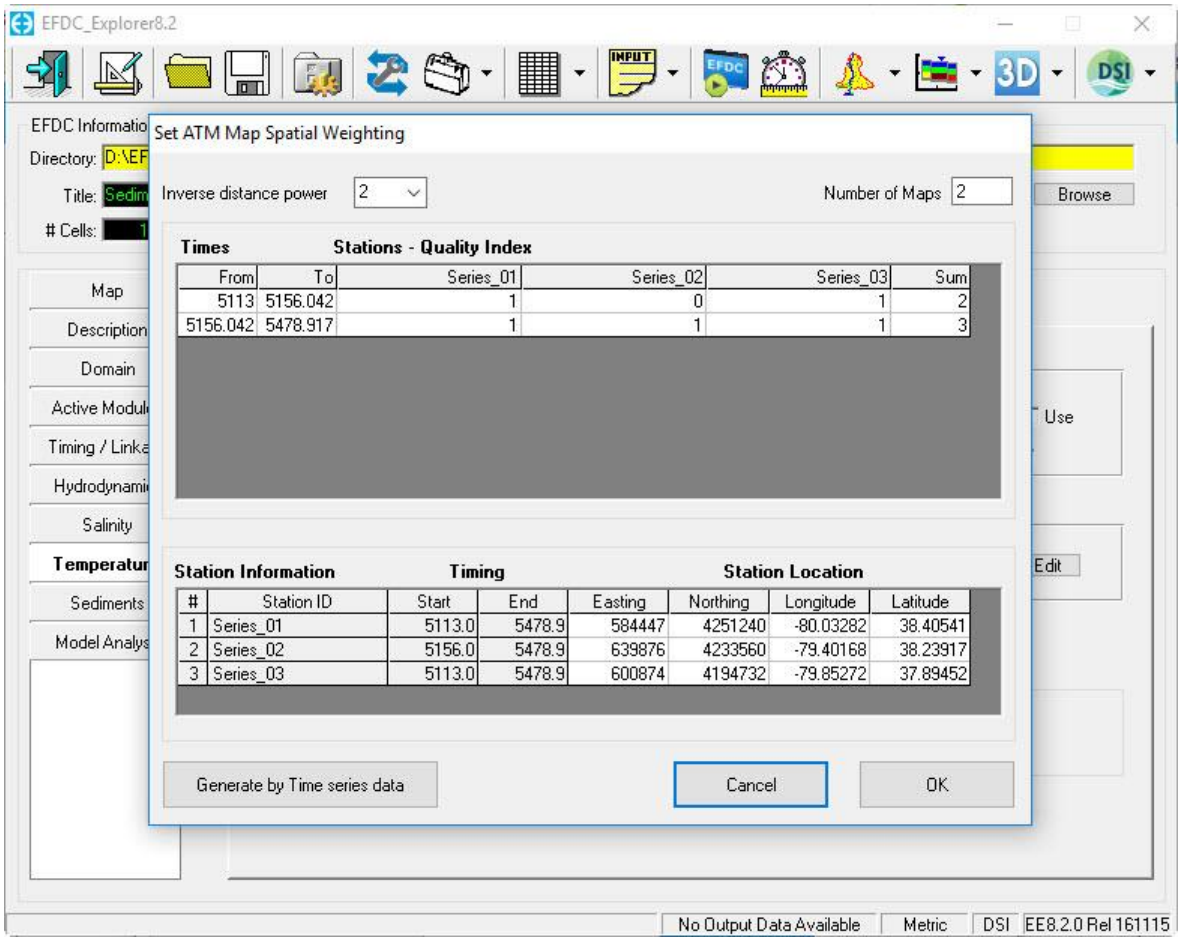


Figure 6-104 Atmospheric Series Weightings using Quality Index.

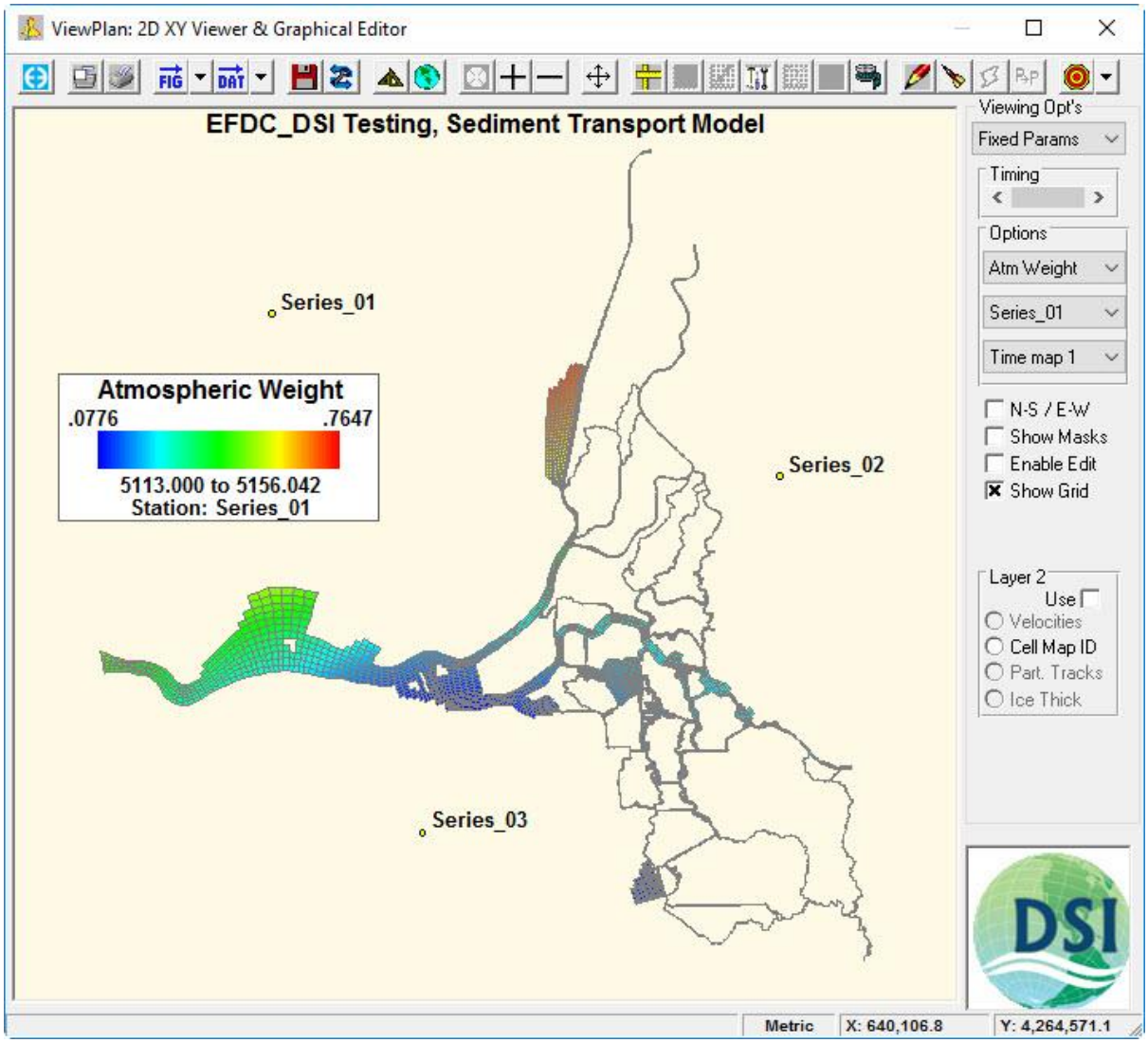


Figure 6-105 ViewPlan: Atmospheric Series Weightings.

Note that if the user manually edits the value ASER.INP to set the value for solar units conversion SOLRCVT to a value other than 1, then when EE is saved this value will be changed back to 1. SOLRCVT is to used convert solar short-wave radiation to joules/s/m^2 (watts/m^2). The reason for EE reverting this value to one is that EE will apply the conversion factor to the values when EE is saved, and after conversion is applied the conversion factor reverts to the recommended value of 1.

6.10.1.2 *Initial Conditions and Temperature*

The user sets the initial conditions for the water column as well as the time series for temperature in this frame. See Apply Cell Properties via Polygons for how to use the *Assign* button for setting initial conditions.

6.10.1.3 *Evaporation Options*

Note that the *Evaporation Options (Water Balance Only)* frame is only displayed if the appropriate surface heat exchange sub-model has been shown in the *Surface Heat Exchange* tab. *Evaporation Options* will only appear for the following surface heat exchange options:

1. Full Heal Balance (Legacy)

2. Full Heat Balance with Variable Extinction Coefficient
3. Equilibrium Temp (CE-QUAL-W2 Method)

See Section 7.4 for details on this option.

6.10.2. Surface Heat Exchange

The *Surface Heat Exchange* tab contains the *Main Heat Exchange Options* and is where the user sets the *Surface Heat Exchange Sub-model* as shown in Figure 6-106. This drop-down list now contains six options, one more than previous versions of the EFDC_Explorer Modeling System. The user will note that the *Setting*s button is no longer provided. These settings have been moved to more appropriate places, adjacent to the option that requires these setting to simplify the modeling process. Only settings which are relevant are now provided in the *Surface Heat Transfer Coefficient Frame* and in the *Light Extinction Coefficients* frame, depending on which option is selected. For example, if sediments are not being simulated the *Light Extinction for TSS* is hidden as not applicable.

The options for the *Surface Heat Exchange Sub-model* are now:

- No Atmospheric Linkage
- Full Heat Balance (Legacy)
- External Equilibrium Temperature
- Constant Equilibrium Temperature Coefficient
- Equilibrium Temperature (CE-QUAL-W2 Method)
- Full Heat Balance with Variable Extinction Coefficient

These options are described further below.

Note that to activate the heat sub-model in EFDC, the temperature constituent must be activated (*Active Modules* | *Sub-Model Computational Options* | *Details*). The ASER.INP file must be used for all of the thermal sub-models to compute the surface and bottom heat exchange processes. Also, depending on whether the current model is EFDC_GVC or EFDC+, the EFDC_Explorer form will have different options shown in the *Main Heat Exchange Options frame.*

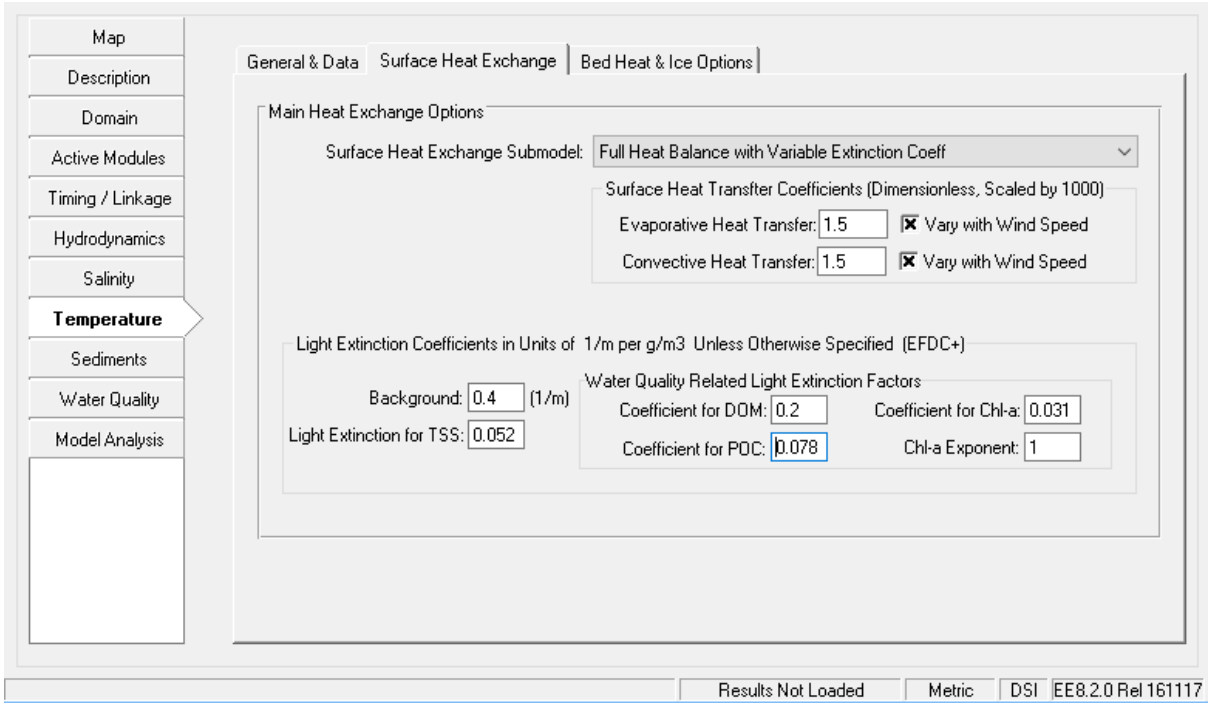


Figure 6-106 Surface Heat Exchange Options.

The *Full Heat Balance with Variable Extinction Coefficient* option was added in EE8.2 and provides the user with the option of allowing light extinction to be dependent on the total suspended solids in the water column, thereby making light extinction temporally and spatially varying. The extinction factors are determined on a cell by cell basis for each time step based on TSS, and if the water quality model is active, based on POM, DOM and Chlorophyll-a as well.

With this option the user may also set the evaporative and convective heat transfers to be impacted by the wind speed. The user should select the appropriate check box for the *Vary with Wind Speed* option, and then ensure a correct setting for the heat transfer coefficient for each option.

6.10.3. Bed Head & Ice Option

This tab contains the *Ice Simulation Options* frame, *Initial Conditions Bed Temperature* frame, and the *Bed Heat Exchange Coefficients* frame which are described in the following sections.

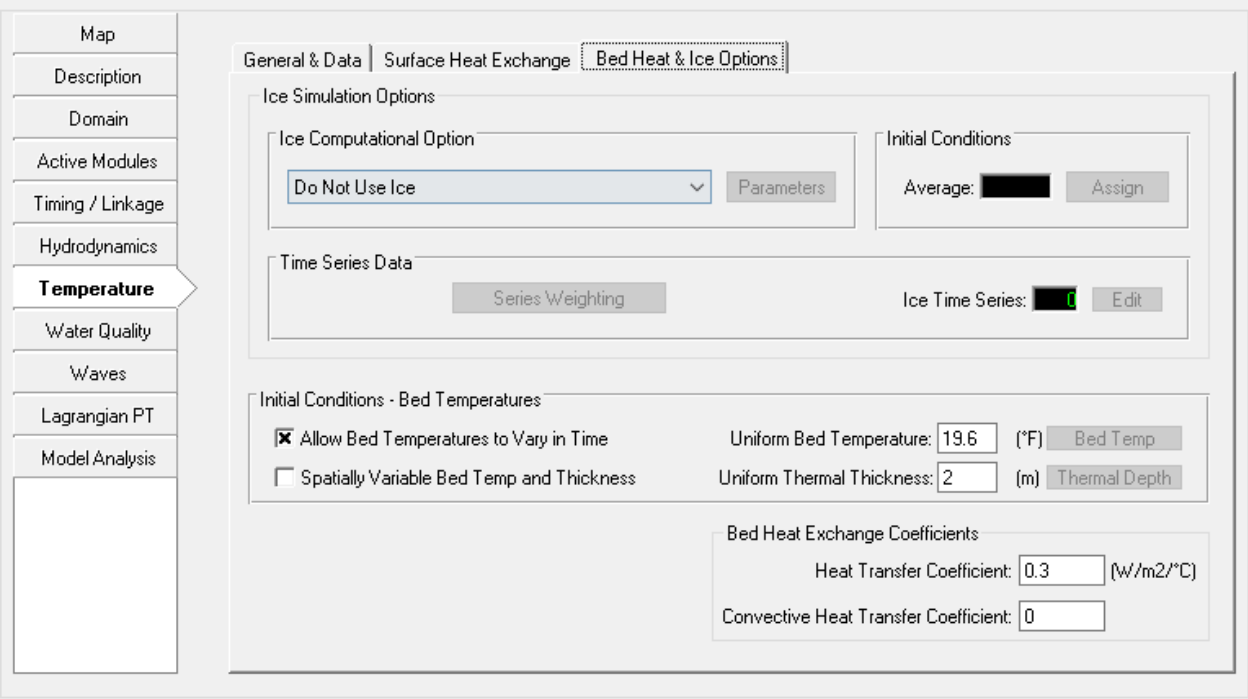


Figure 6-107 Bed Heat & Ice Options.

6.10.3.1 *Ice Simulation Options*

A robust ice sub-model is available from release of EFDC_Explorer7.3. Previous versions of EFDC had relatively limited ice modeling ability. In EE7.2 and earlier the ice conditions had to be fully specified by the user for every cell for the model simulation period. The creation of the input files was also external to EE. Now ice formation and melt is simulated by EFDC using a coupled heat model and fully handled by EFDC_Explorer.

Note that ice dynamics are not modeled at this stage. An ice dynamics sub-model would simulate the constriction of the channel by ice and the resulting bed shear caused by the transport of ice chunks. An ice dynamics sub-model is being considered for a later release of the EE modeling system.

Ice options are now available under the *Ice Options* frame under the *Temperature* tab as shown in Figure 6-107. For all the options, ice is only enabled if the user is simulating temperature.

Options for the ice sub model include:

ISICE = 0 Do not use ice

ISICE = 1 Use External Ice Time Series (ISER & ICEMAP)

ISICE = 2 Use Specified ON/OFF Ice Cover (ISTAT)

ISICE = 3 Use Heat Coupled Ice Model

ISICE = 4 Use Heat Coupled Ice Model with Frazil Transport

Application and operation of each of these options is explained in the following sections.

(1) Use External Ice Time Series (ISICE = 1)

This option is similar to the ice option available prior to EE7.3 and does not compute ice formation/melt and it is not linked to the heat balance. This option simply requires the user to provide a fraction of ice coverage and thickness of ice for every cell using formats provided in Appendix B – Data Formats. The primary impact of the ISICE =1 option is on processes that occur at the air/water interface and has no direct impact on ice melt. For those cells where ice is present the ice sub-model will:

- Limit water surface heat exchange and moderate the layer KC temperature to the specified ice temperature.
- Reduces or eliminates (based on fraction of ice coverage) reaeration of oxygen into the water column.
- Reduces or eliminates the shear stress on the surface of the water due to winds.
- Reduces or eliminates the wind speed used for all other surface exchange processes.

For the option ISICE=1, EFDC uses the ISER.INP to read time and ice thickness for externally specified ice cover, and ICEMAP.INP for the weighting coefficients of ice thickness in case of more than one-time series given in ISER.INP.

ICEMAP with the NISER weightings (the number of ice series) when NISER>1 can now be read and written.

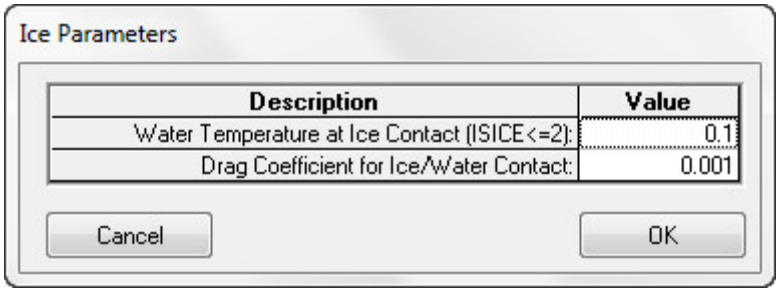


Figure 6-108 Temperature Tab: ISICE=1.

(2) Use Specified ON/OFF Ice Cover (ISICE = 2)

This option is effectively a global toggle so that ice may be turned on or off over the whole domain. The time series file that populates this option lists a date and toggle on and toggle off with the input file ISTAT.INP.

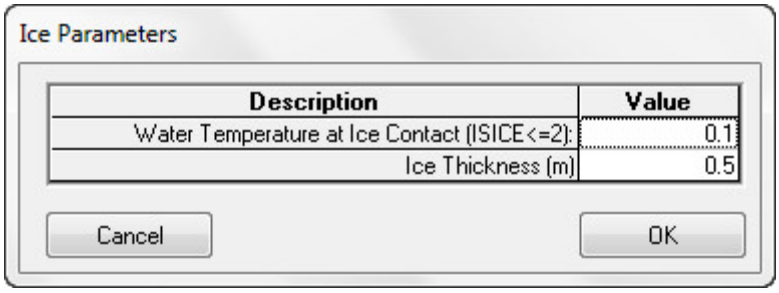


Figure 6-109 Temperature Tab: ISICE=2.

The input file ISTAT.INP file is same as old the ICECOVER.INP, with a time stamp and a status of ice cover. The convention is quite similar to that for other time series. The header contains the number of data lines, time conversion coefficients. The remaining block includes two columns: Julian time and the status of ice cover, either 0 or 1, off or on. The ice temperature and the ice thickness are stored in EFDC.INP file in C46A. Appendix B shows the format for this file. EFDC reads the file and applies it in the same way as ICECOVER.INP. All the other computations are the same as those for ISICE=1 except the initialization is whole model on or off rather than based on the ICEMAP.INP file. EFDC can also handle multiple ice series and weights based on different series like NISER.

(3) Use Heat Coupled Ice Model (ISICE = 3)

The Heat Coupled Ice Model applies mass conservation during ice growth/melt. Ice is always calculated in the heat coupled ice model, similar to CEQUAL-W2 model upon which the EFDCPlus ice sub model is based.

This option is most recommended for model simulations of lakes and reservoirs with relatively thick layers. For rivers, this option and frazil ice option are not fully representative. This is due to small layer thickness in most river models. Generally, the layers used in rivers are too thin to produce ice. Even though ice crystals form, they are not thick enough to form an ice cover.

Currently the ice sub-model in EFDCPlus is only an ice cover model and not an ice and snow cover model. The snow cover would account for snow on top of the ice and is expected to be added for an upcoming release. The ice cover model allows light to be attenuated through the ice. The solar radiation absorption is accounted for in this process. To implement this in EFDC, routines such as CALQVS and CALHEAT were modified. CALPUV was also updated so that the bed heat is handled when the elevation is below the bottom of the cell. Figure 6-110 shows the default values of the ice parameters that are required to simulate ice in EFDCPlus model. A checkbox is provided for the Use Ryan Harleman Wind Function option if desired.

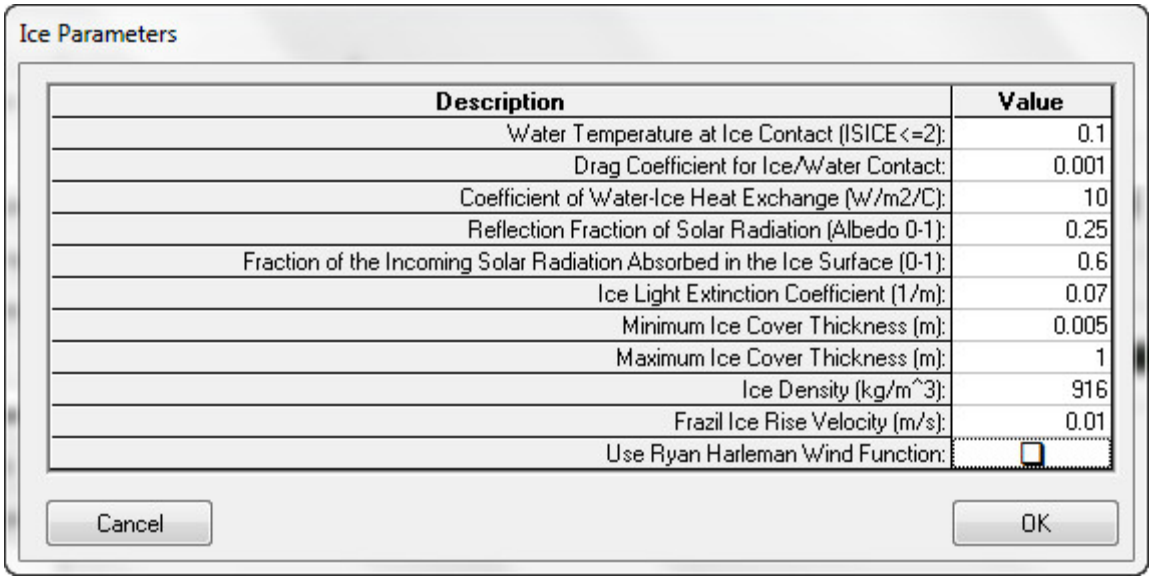


Figure 6-110 Ice Sub model: Heat Coupled Ice Parameters (ISICE =3).

ICE.INP is the initial conditions file that is only needed for ISICE = 3 & 4. The format for this file is provided in Appendix B. Note that EFDC assumes the top of ice is equal to the water surface elevation thereby allowing for higher flows in restricted depth.

(4) Frazil Ice Transport (ISICE = 4)

When calculating ice formation in a river, if the air temperature is less than freezing temperature (a value that may be lower than zero in salt water) generation of ice crystals takes place in the water. These crystals are called frazil flakes. As the frazil flakes are lighter than water they float and cause "ice pans" which may then become "ice floes". As the frazil ice rises, it is necessary to input a rising velocity into EE to account for this. In addition to the modification to the EFDC subroutines outlined for ISICE =3, a new routine, CALTRANICE is created to simulate frazil ice as a concentration.

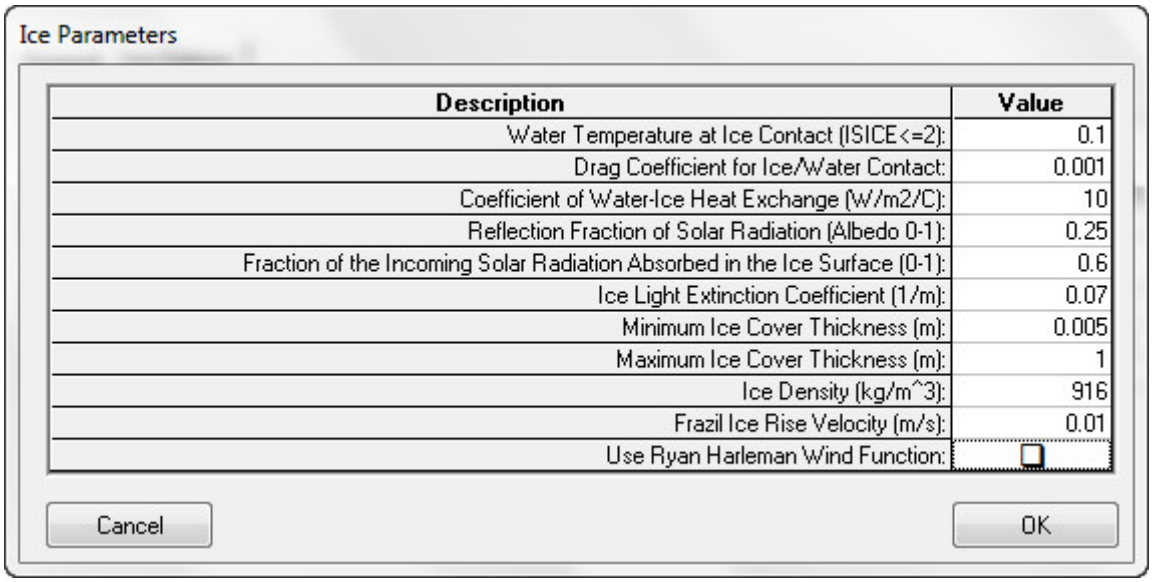


Figure 6-111 Ice Sub model: Frazil Ice Parameters (ISICE =4).

(5) Visualization of Ice

Output from the ice sub-model can be displayed in a number of ways in EE. A 2D plan view of the ice thickness or temperature may be viewed by selecting *ViewPlan* and then *Viewing Options* | *Heat: Bed/Ice*. **Figure** 6-112 shows an example of frazil ice formation in a river in Alberta, Canada. Note that initially ice forms on the river banks where depths are shallower and flow is lower.

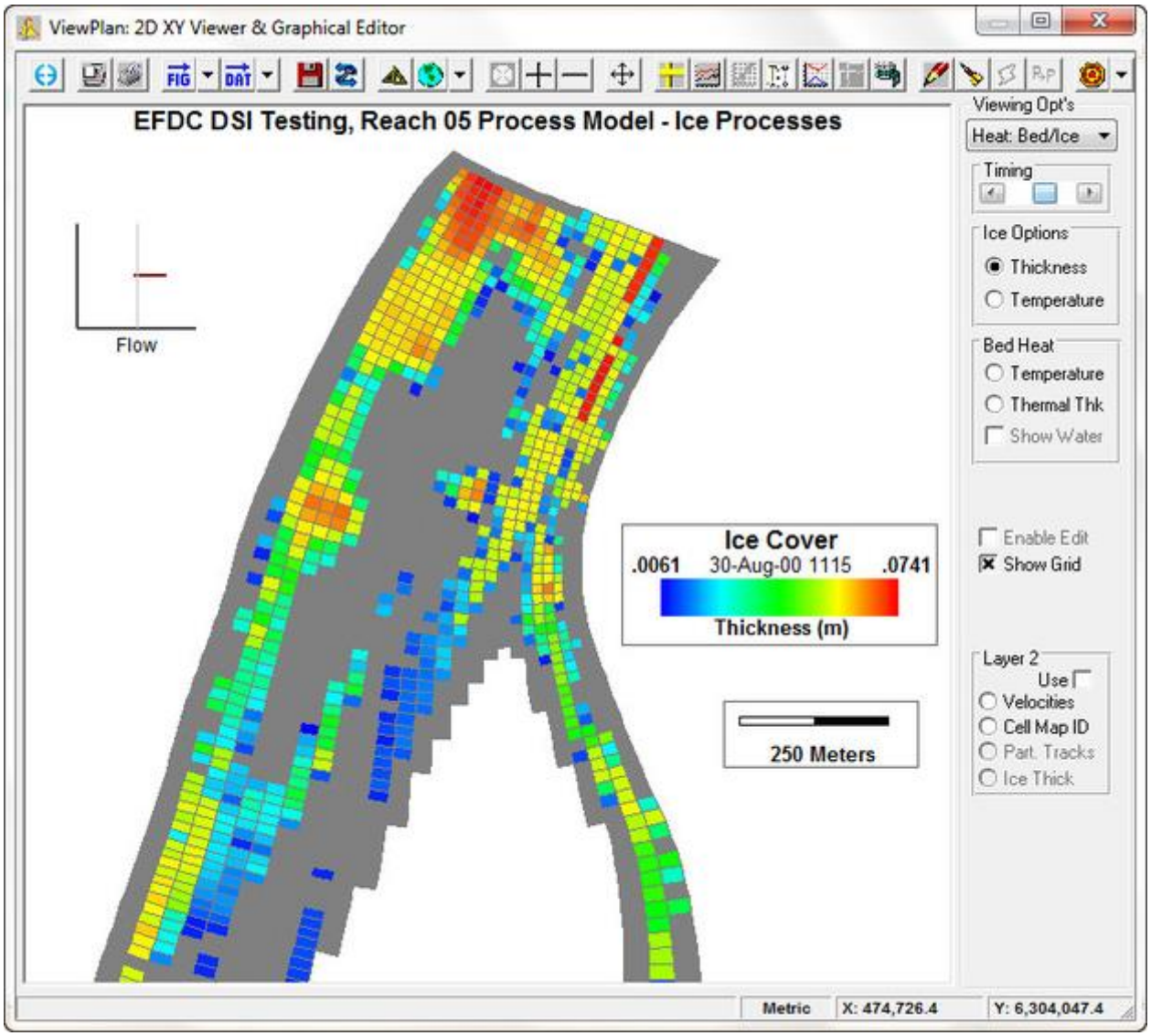


Figure 6-112 Ice Sub model: ViewPlan Frazil Ice Thickness.

The user can also visualize ice on the water column in the *ViewSlice* tool. This displays as a solid grey color as shown in Figure 6-113. The vertical exaggeration of the ice layer can be modified by the user to show it more clearly as shown in Figure 6-114.

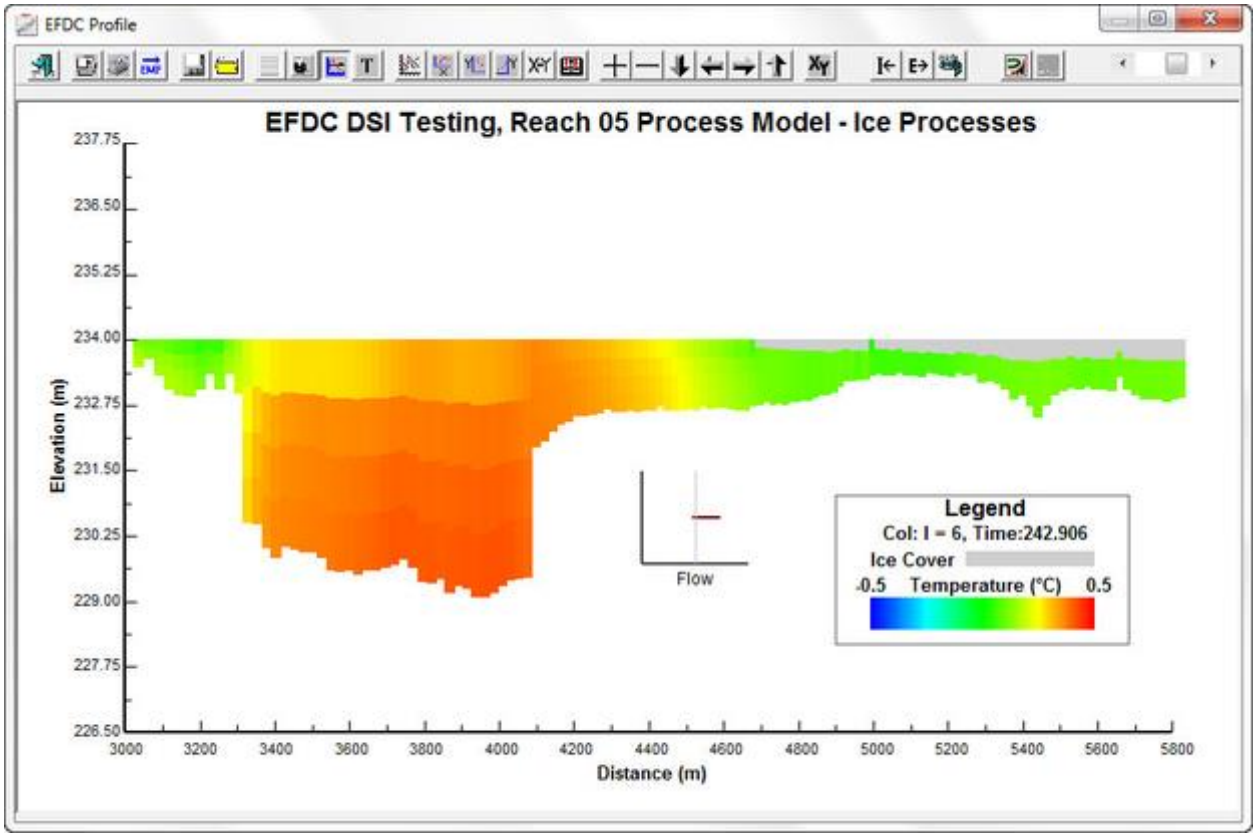


Figure 6-113 Ice Sub model: ViewSlice ice cover and WC temperature.

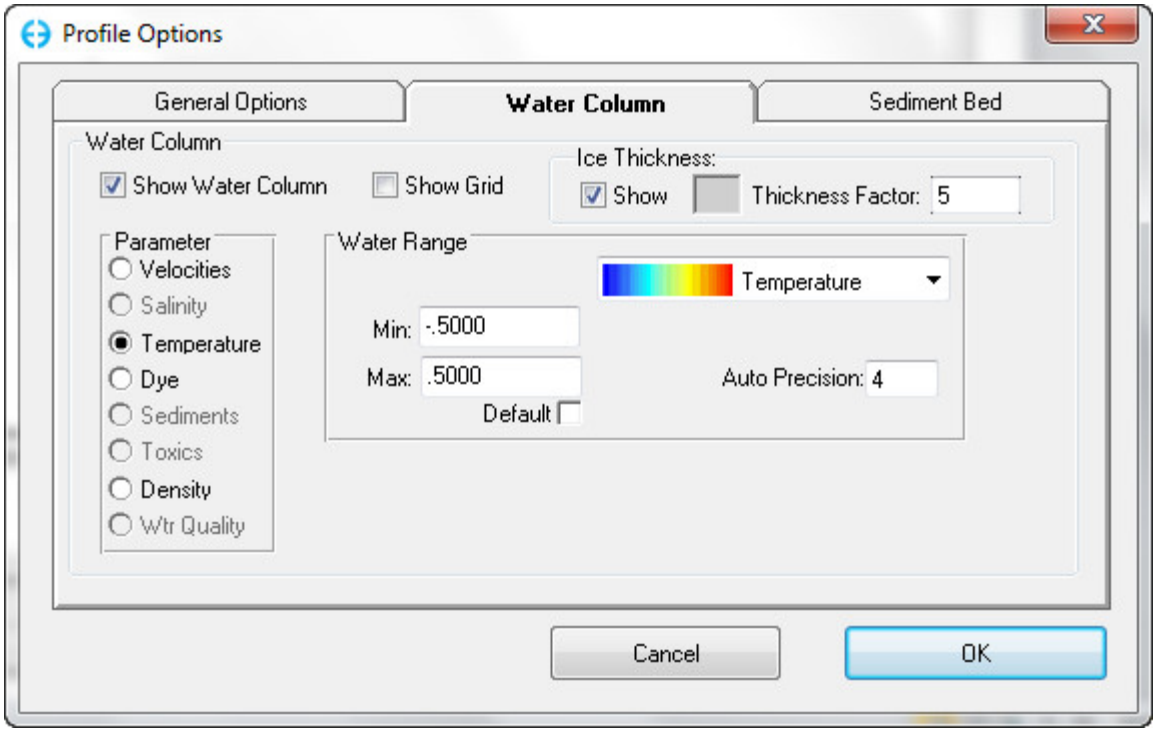


Figure 6-114 Ice Sub model: ViewSlice Options.

Ice may also be viewed in 3D with the *View3D* tool as shown in Figure 6-115.

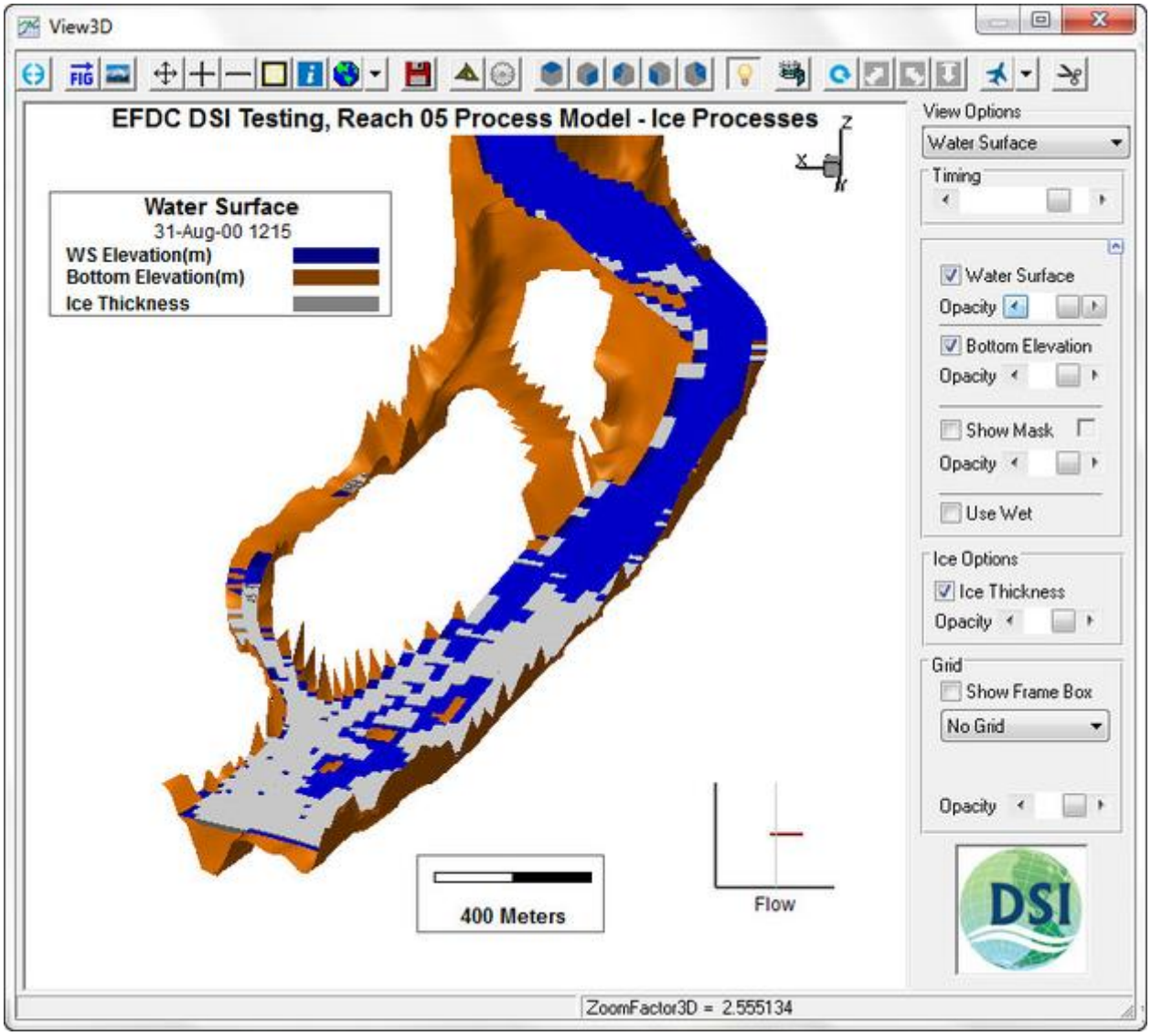


Figure 6-115 Ice Sub model: View3D ice cover and water surface elevation.

6.10.3.2 *Bed Temperature and Heat Exchange*

Initial Conditions Bed Temperature frame, and the *Bed Heat Exchange Coefficients* frame are shown below. The last of these have been moved from the *Settings* button in releases prior to EE8.2.

The EFDCPlus version of the model has two different approaches for the bed heat sub-models. If the standard EFDC full heat balance (ISTOPT(2)=1) sub-model is used, the water/sediment bed heat exchange model of the pre-GVC version is used. If the user has selected the equilibrium temperature sub-model (ISTOPT(2)=4) then the bottom heat exchange is computed using the heat exchange coefficient and the user can assign spatially varying sediment thicknesses and initial temperatures using the TEMB.INP file.

Note that the EFDC_GVC model uses a thermal bed model to determine the bed/water column heat exchange (EFDC+ Theory Guide, DSI 2017). A sediment thermal thickness is assigned for the entire model. This thickness is not related to either the sediment transport model's bed or the water quality sediment bed. This thermal layer is then divided into KBH layers (must be >2). A constant initial bed temperature can be used or alternatively the TEMPB.INP file can be used to assign a spatially variable initial temperature, by layer. Check the *Use Spatially*

Variable Bed Temp and Thickness checkbox in the *Initial Conditions - Bed Temperatures* frame for this option.

6.11 Dye/Age Tab

If the dye computational option is selected as shown in Figure 6-116 then the user may assign a “decay” rate to the otherwise conservative tracer.

The “dye” constituent may also be used as an “age of water” function to compute residence times and visualize domain/boundary water mixing if this option is selected. This option is available in EFDCPlus only.

See Section 6.3.2.1 as to how to apply cell properties via polygons with the *Dye* button in the *Initial Conditions* frame.

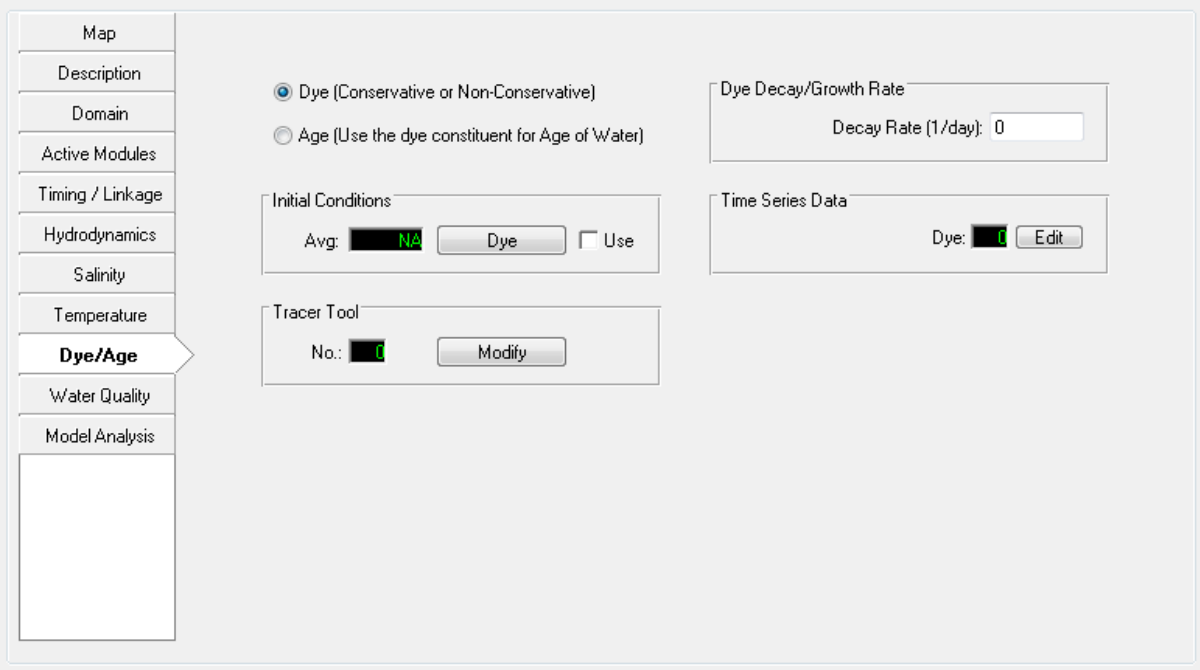


Figure 6-116 Dye/Age Tab

The “decay” rate options are shown in Table 6-1.

Table 6-1 Dye “decay” rate options.

Dye “Decay” Rate	EFDCPlus Operation
>0	First order decay rate
0	Conservative tracer
<0	First order growth rate (EFDCPlus Only)

Depending on the decay rate option, the user can view and animate the dye concentrations or age of water in *ViewPlan*.

If using the restart option, the user can initialize a dye field using the DYE.INP file by selecting the *Dye Overwrite* check box on the *Initial* tab, located in the *EFDC Restart Option* frame.

6.11.1. Tracer Tool

The purpose of the tool is to provide a quick way to setup EFDC computational options and boundary conditions for point source tracer injections. Figure 6-117 shows an example of the tracer tool. The list in the upper left of the *Dye Injection Point* frame shows the tracers currently defined. The list on the right shows all the currently defined flow type cells. The user needs to assign an existing cell or create a new dye injection point and then assign the flow and dye concentration for each point.

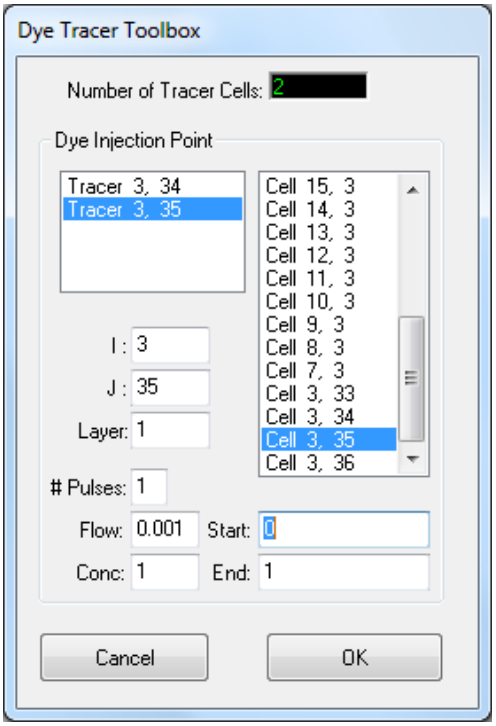


Figure 6-117 Tracer generation tool.

To define an existing boundary cell as a tracer, select the cell from the list on the right and press INS to add it to the tracer list. You must then define the flow and dye injection characteristics for each tracer cell. You can delete tracers by selecting the cell on the left and pressing DEL key. You can also create a tracer at a new location by selecting INS and entering the location for the tracer in the *I, J* and *Layer* boxes. In this example the cell I=20, J=16, has been selected to produce 5 pulses over 5 days as shown in Figure 6-118 below.

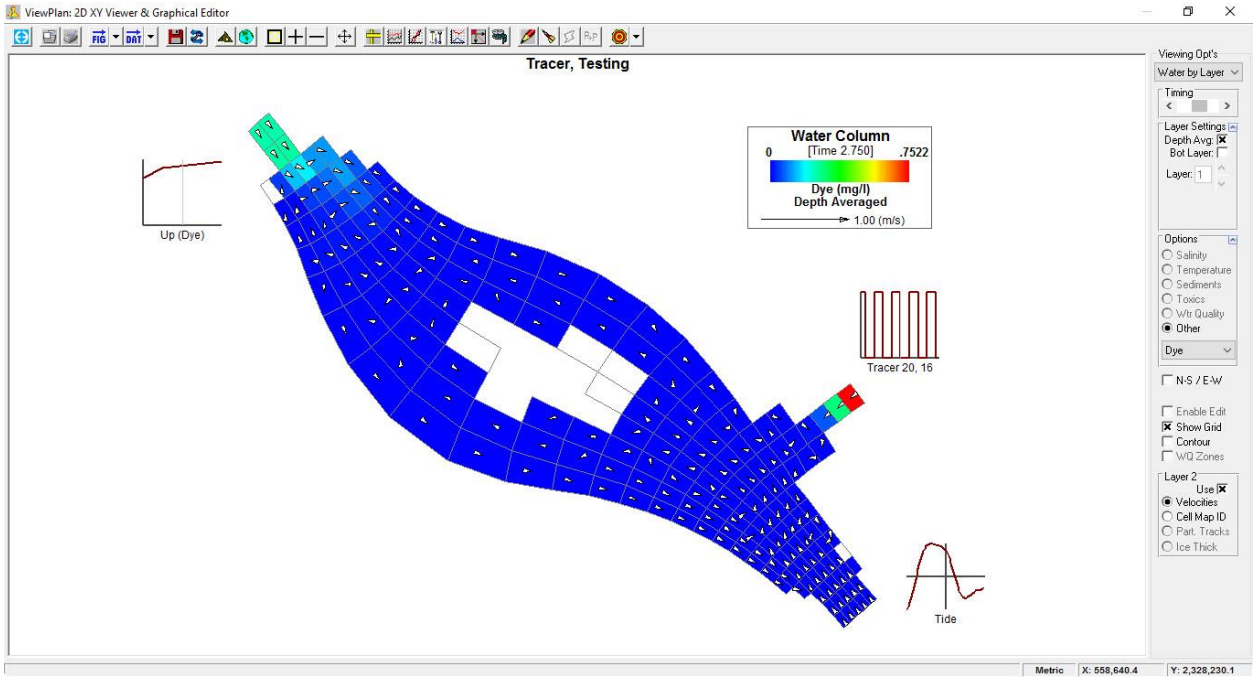


Figure 6-118 Dye Tracer Results in ViewPlan.

6.12 Sediments Tab

There are two separate sediment transport sediment sub-models implemented in EFDC. The first sub-model is original EFDC code, which is capable of simulating the transport and fate of multiple size classes of cohesive and non-cohesive suspended sediment including bed deposition and resuspension. A second sediment transport sub-model for EFDC is the SEDZLJ module that uses Sedflume data to determine erosion rates has been developed by Sandia National Laboratories. These two options for sediment are selected for in the *Active Modules* tab where the user can select either the *Use EFDC Sediment Model* or the *Use the SEDZLJ Sediment Model* option as shown in Figure 6-119.

If the user has selected EFDC Sediment Model then the *Active Cohesive Sediments* or the *Activate Non-Cohesive Sediments* checkboxes in the *Active Modules* tab the *Sediments* tab will be displayed. If the user has selected the SEDZLJ sediment model the just the *Activate Sediment*s checkbox is displayed as shown in Figure 6-120.

Once the sediment model has been turned on the Sediment tab is available as shown in Figure 2. What is displayed here will vary depending the sediment model selected.

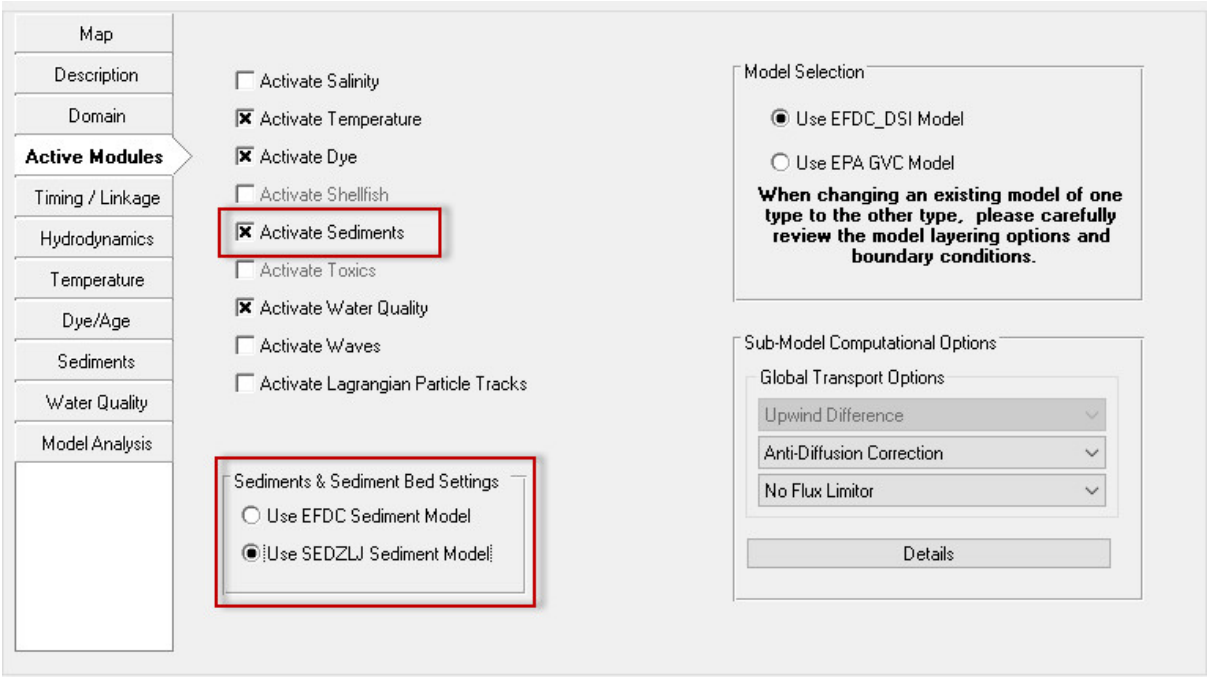


Figure 6-119 Active Modules Tab - Sediment sub-model options.

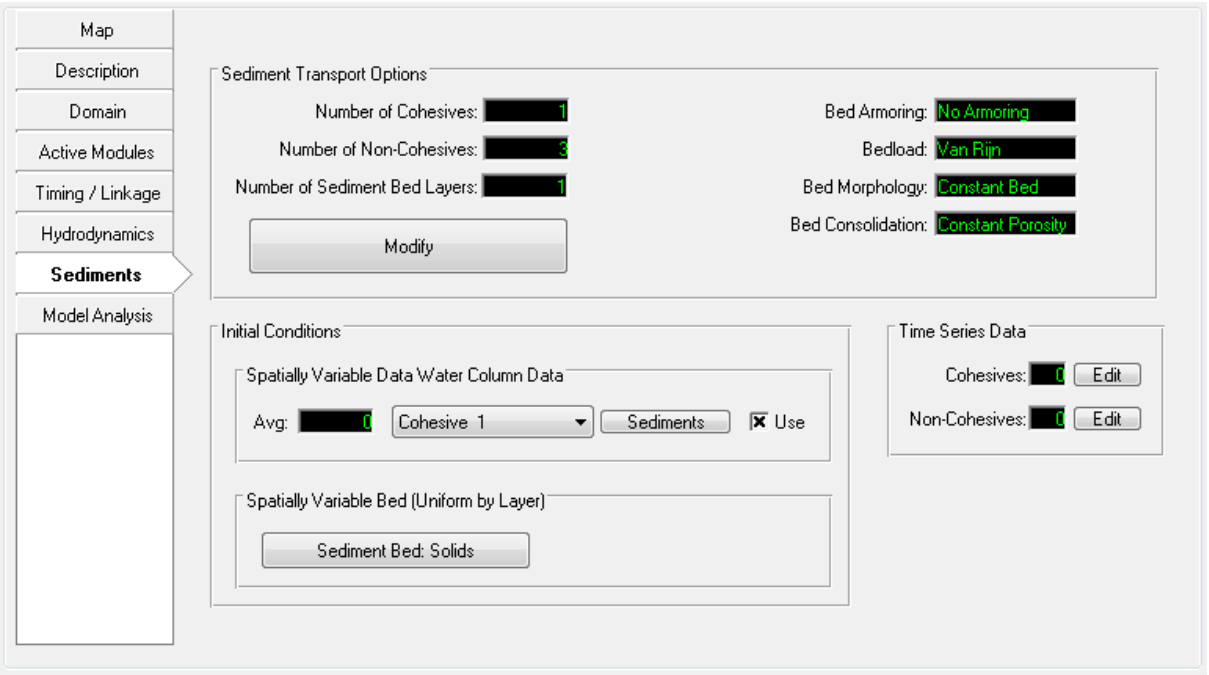


Figure 6-120 Sediments Tab for EFDC Sediment Model.

6.12.1. EFDC Sediment Model

The sediment transport parameters and options are set using the *Modify* button, which forms shown in Figure 6-121 to Figure 6-124.

The user also has further option for setting the sediment bed with EE's *Sediment Bed Initialization Tools*. These are described in detail in the *ViewPlan* discussion (Section 8.6.9.1).

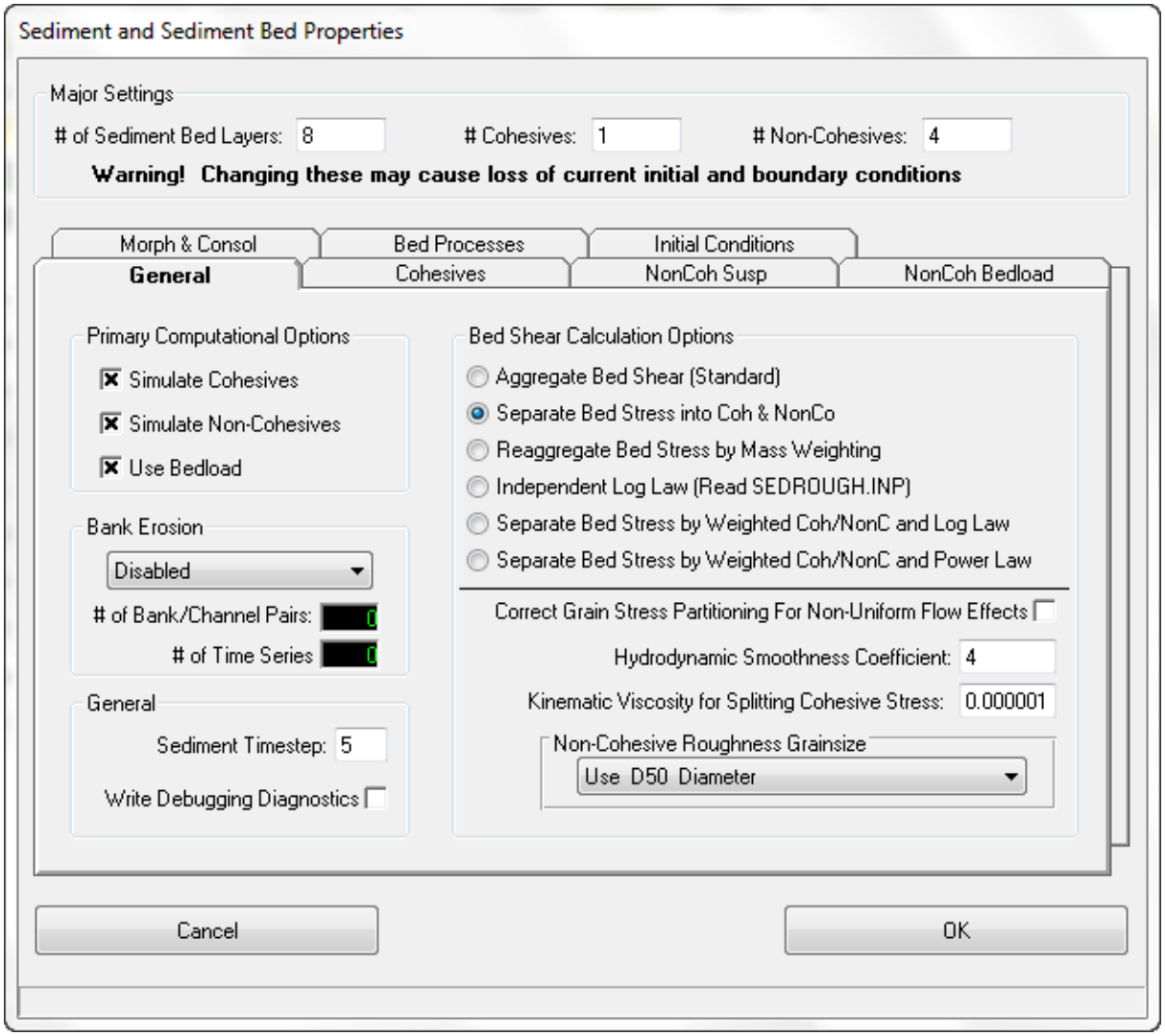


Figure 6-121 Sediment Transport – General.

6.12.1.1 *General Tab - Sediments*

The principal settings for the number of sediment classes and bed layers are specified in the top frame under *Major Settings* as shown in Figure 6-121. The user should define the number of sediment bed layers, cohesive and non-cohesive sediments. However, as with the toxics, these *Major Settings* parameters should only be changed with care and usually early in the model calibration process so as not to lose your initial and boundary conditions.

In the *General* tab the user can tell EFDC whether or not to simulate cohesives and/or non-cohesives and the approach to compute bed shear stresses for sediment transport. Un-checking the *Simulate Cohesives* and/or *Simulate Non-Cohesives* check box does not delete previously defined sediment transport parameters, it just turns off the EFDC computations for that parameter.

The *Sediment Timestep* is a multiple of the number of hydrodynamic time steps to calculate between each sediment bed process computation (i.e. deposition, erosion, consolidation, etc.). Because the bed processes are slow, relative to the hydrodynamics, this number can often be 10 or greater. The user should conduct testing by starting with a low number and increase the *Sediment Timestep* until the user detects a noticeable difference in the model results. Then the number should be reduced by some amount in order to provide a safety factor. This can then be used for the subsequent calibration and production runs.

The *Bed Shear Calculation Options* allows the user to choose from a number of different calculation options for bed shear stress computations. If non-cohesive and cohesive calculations are being performed, it is generally preferred to use the *Separate Bed Stress in Coh & NonCo* option.

6.12.1.2 *Cohesives Tab - Sediments*

Figure 6-122 shows an example of the *Cohesives* tab in the *Sediment Transport* form. The user should specify the *Cohesive Settling Flag* where: 0-Simple; 1-Huang and Metha; 2-Shresta and Orlob; 3-Ziegler and Nesbit; 98-Lick Flocculation; 99-Lick Flocculation with Advective. If the *Apply Vertical Diffusion* check box is selected then vertical diffusion will be applied to the cohesives. When the *Use Bulk (Mass) Erosion* check box is selected the bulk erosion critical stress and rate will be used in the functions CSEDTAUB and CEDRESSB (still experimental at this time).

The number of columns shown in the input parameter grid varies with the number of classes to be modeled. The Diameter setting located at the bottom of the parameter grid sets the grain size that will be used in calculating d_{50}'s for the sediment bed. It is not used for any other calculations.

Maximum and minimum *Cohesives Fluid Concentrations* may be set by the user. Checking the *Use Fluid Mud* checkbox will include cohesive fluid mud viscous effects using function CDEVIS (SEDT).

For many of the fields in the *Erosion & Deposition Parameters* grid, clicking on a cell and pressing F2 pops up helpful information for that field.

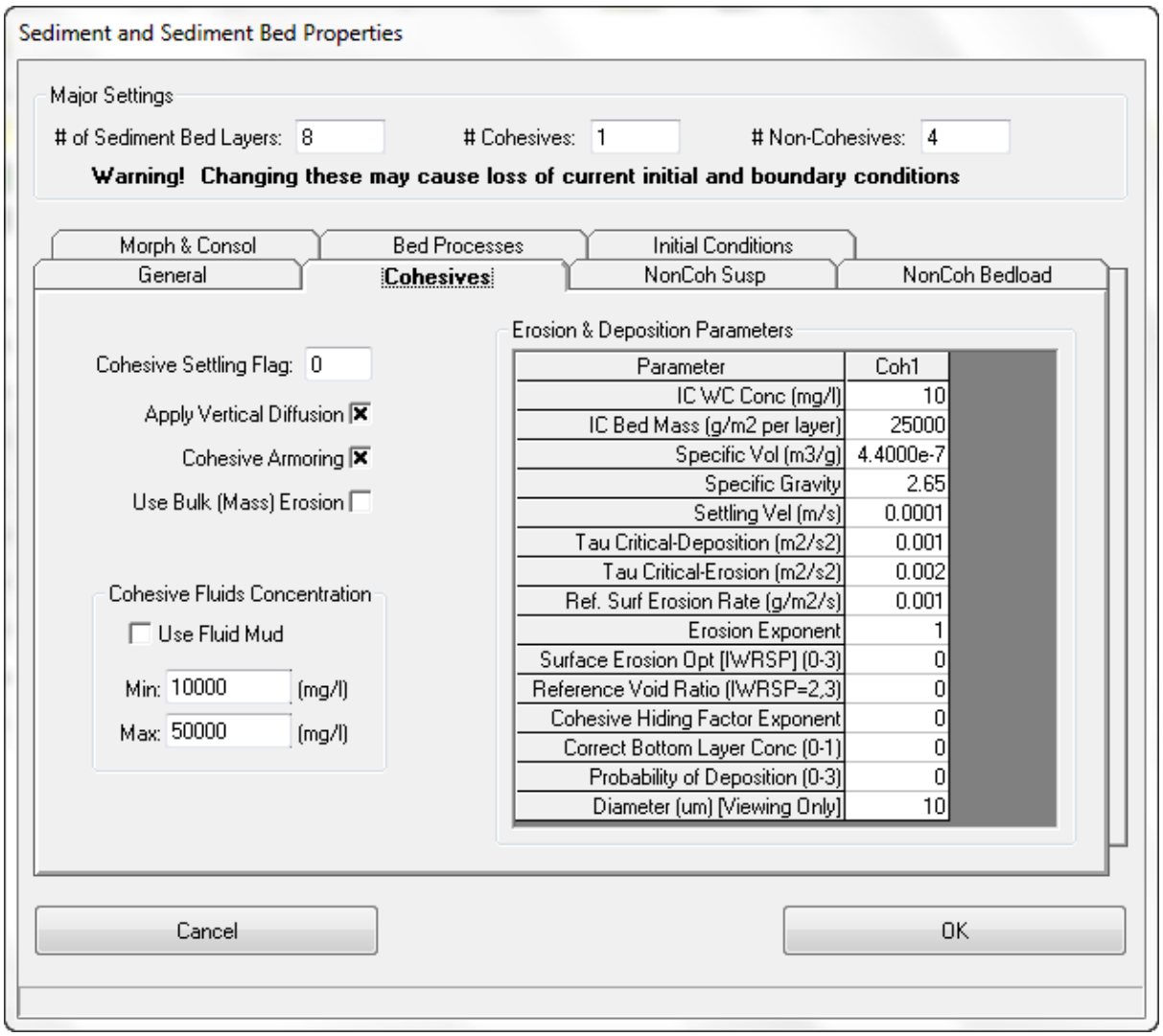


Figure 6-122 Sediment Transport – Cohesives.

6.12.1.3 *Non-Cohesives Suspended Tab - Sediments*

Figure 6-123 shows the form with the *Non-Cohesives Suspended* tab shown. It is similar in function and operation to the *Cohesives* tab. Grain size for each non-cohesive class is required for the sediment transport computations. This value is also used as the grain size for the d_{50} calculations. A number of options for computing equilibrium concentrations are available in the *Equilibrium Conc* frame. This includes the option of equilibrium concentrations calculated from Sedflume data with or without critical shear stress. Setting the settling velocity or the critical shear stress values to numbers <0 results in EFDC computing those parameters values using the Van Rijn equations (1984a, 1984b).

Pressing F2 for help pops up information relevant to the current input field.

The *Sediment Info* button provides a sediment properties calculator which uses the Van Rijn equations to compute the critical stress and setting parameters. Pressing the *Set Parameters* button will initialize the sediment properties using the Van Rijn equations if the sediment diameter is specified prior to pressing the button.

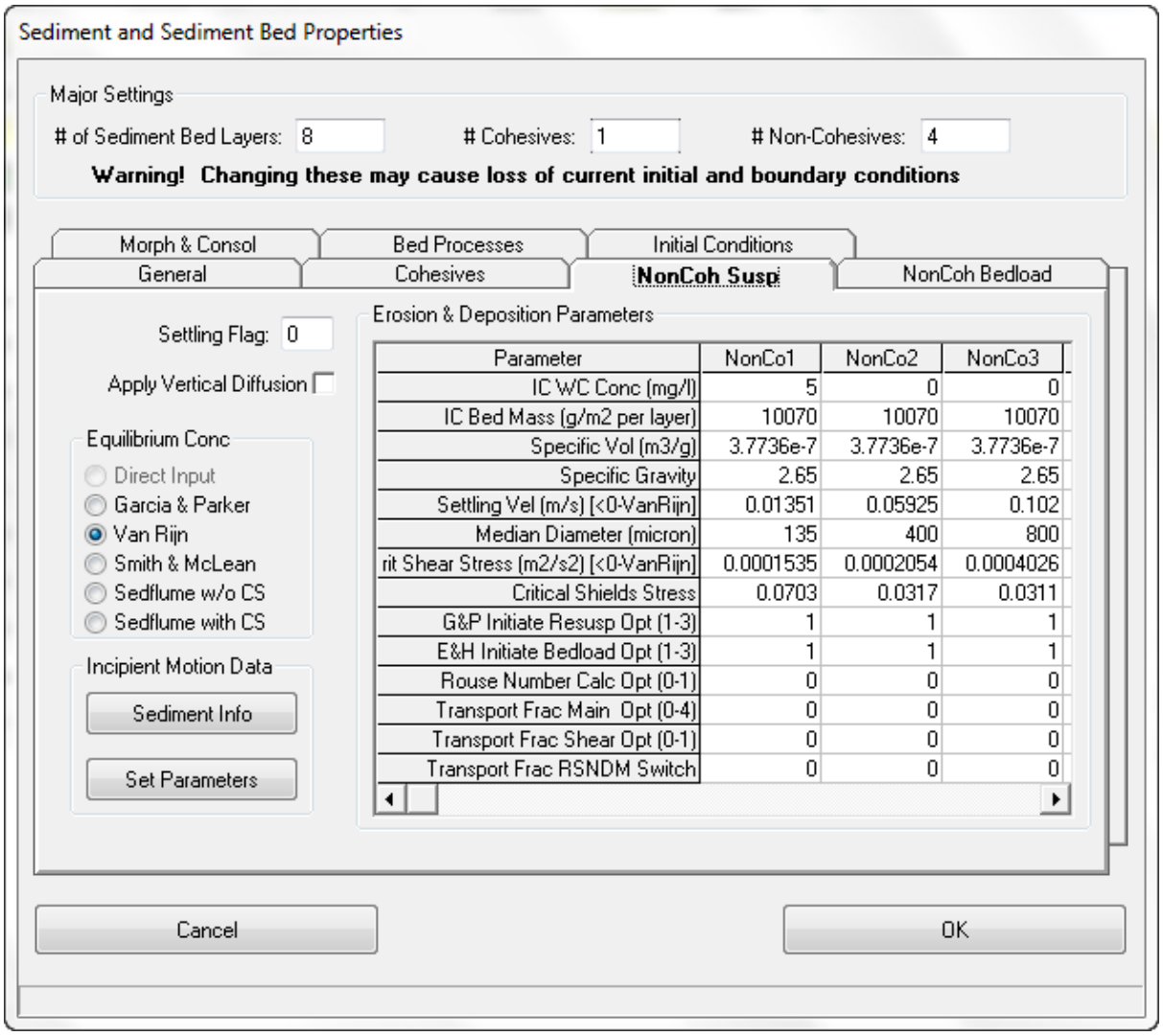


Figure 6-123 Sediment Transport – Non-Cohesives Suspended.

6.12.1.4 *Non-Cohesives Bedload - Sediments*

Figure 6-124 shows the tab for the *Non-Cohesives Bedload*. The use of the Gamma parameters varies slightly between the different EFDC codes. The user should know how the Gamma's are used by reviewing the code prior to finalizing the desired inputs.

Pressing the *Initialize Constants* button will bring up a dialog box asking the user to select the bedload approach to use. EFDC_Explorer sets the bed load transport constants to standard literature values for the computational approach selected. The user should know how the Gamma's are used by reviewing the code prior to finalizing the desired inputs.

The *Bedload Phi Options* frame allows the user to choose from the following options Constant/Manual, Van Rijn, Engelund-Hansen, and Wu, Wang &Jia.

The *Cell Face Transport Rate Option* allows calculation based on the following options: Downwind Projection; Downwind Projection with Corner Correction; and by Averaging Vector Components. The *Maximum Allowable Adverse Slope for Bedload* setting will prevent bedload movement if cell to cell slope is > BLBSNT when BLBSNT is > 0.

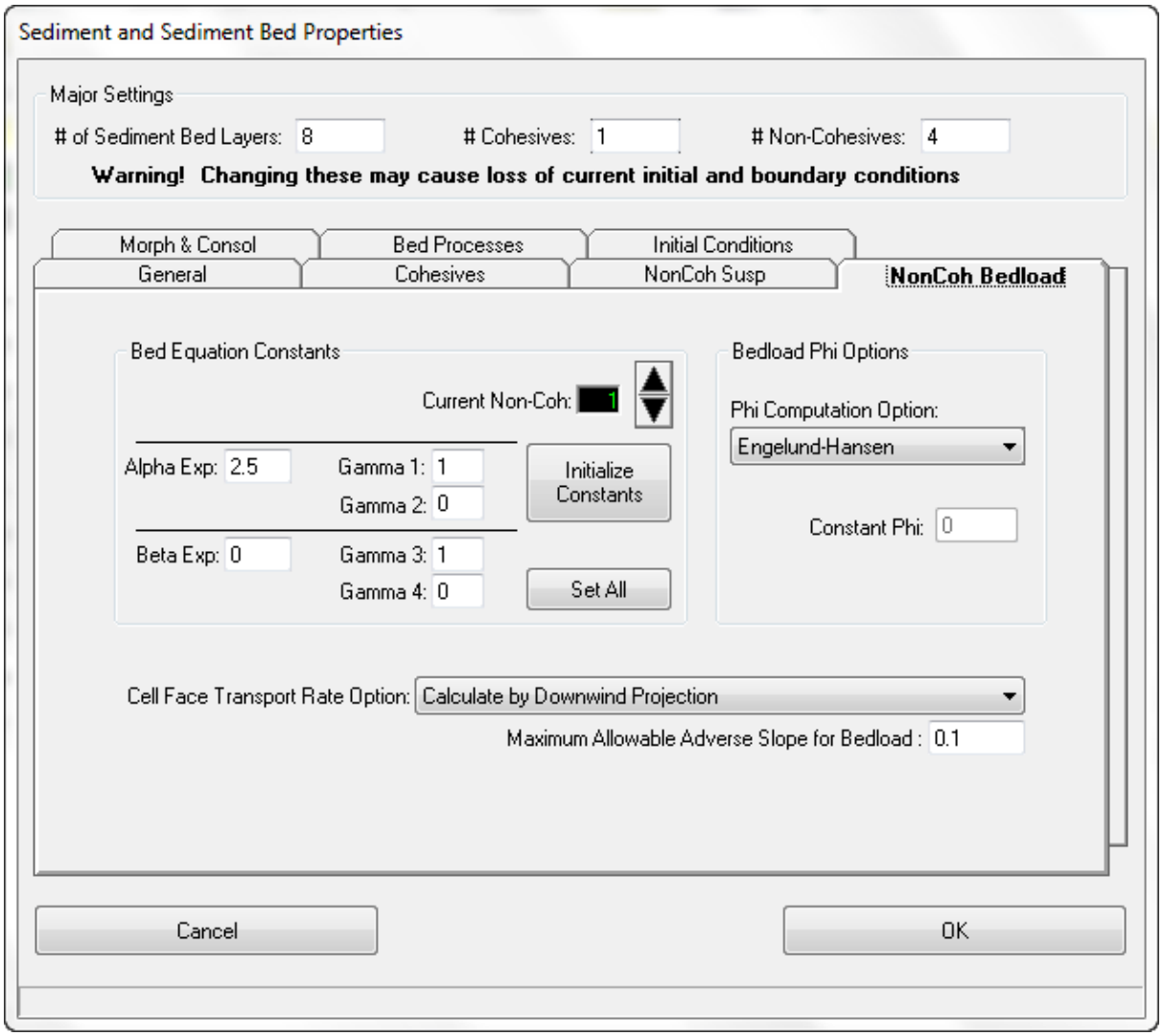


Figure 6-124 Sediment Transport – Non-Cohesives Bedload.

6.12.1.5 *Morphology & Consolidation Tab - Sediments*

Figure 6-125 shows the tab for the *Morphology & Consolidation* of the sediment properties option. In this tab the user may specify various bed consolidation and bed morphology settings.

The *Bed Morphology Options* form allows the user to choose between no bed changes (option 0) and allowing bed elevation changes (option 1). The *Max. Water Depth* is the maximum allowable depth to allow for the "lost" due to "NEG DEPTH" morphological change (EFDCPlus version only).

Bed Consolidation & Mechanics Options frame provides input to the main sediment bed consolidation options. The *Bed Mechanics* option must be 0-None, 1-Simple, or 2-Finite Strain. Option 2 is very a specialized approach and requires a detailed understanding of the of the Finite Strain implementation in the FORTRAN code. The consolidation rate is in seconds, not 1/seconds as earlier versions of EFDC.INP reported. The rate will usually be >100,000 seconds. The impact of the rate of consolidation can easily be tested using EFDC_Explorer's *Single Column Sediment Layers* post-processing function (see Sediment Bed Process Tools).

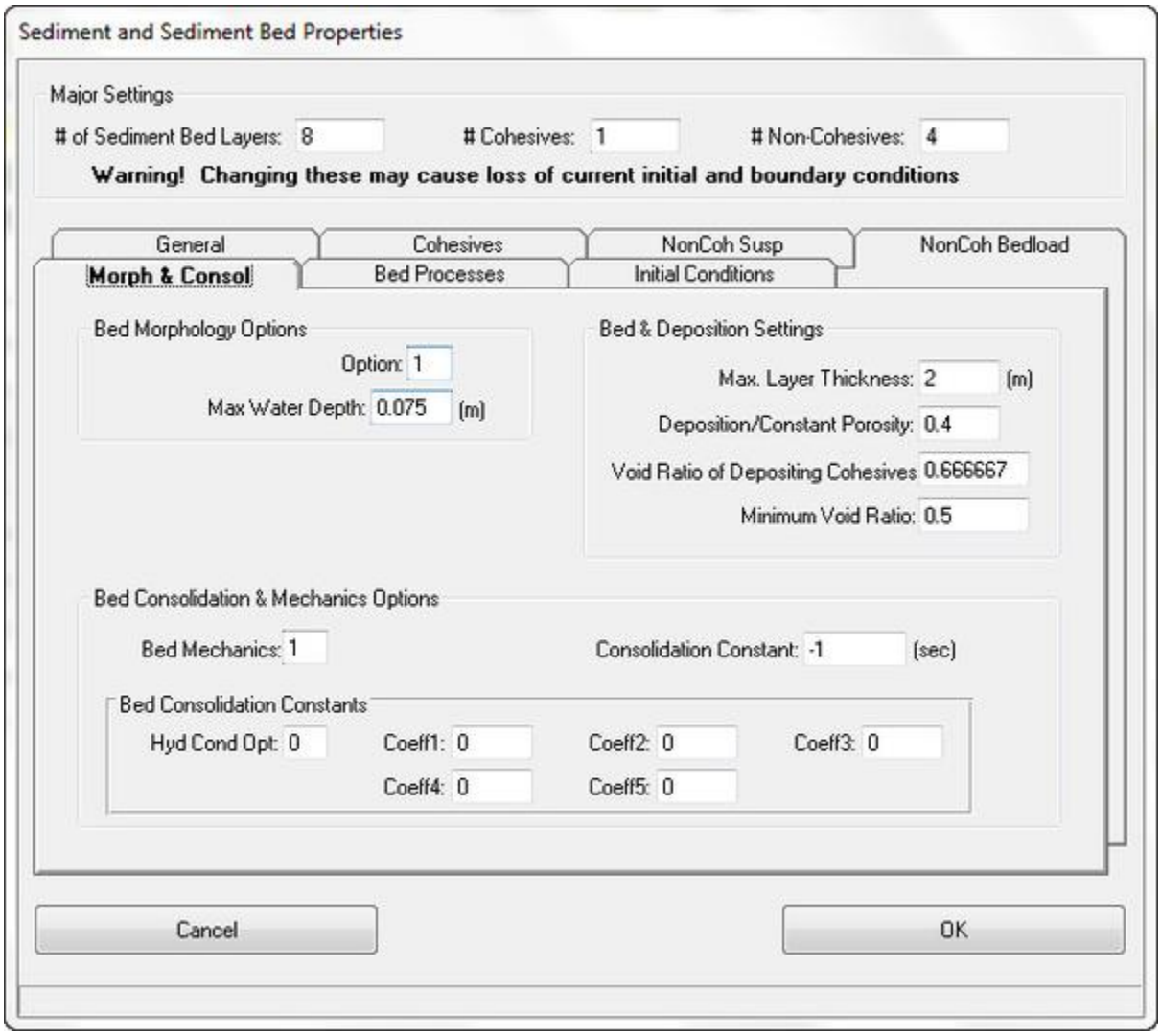


Figure 6-125 Sediment Transport – Morphology and Consolidation.

6.12.1.6 *Bed Process - Sediments*

The user options for the Bed processes tab are shown in Figure 6-126. EFDC_Explorer employs bed armoring functionality, allowing the user the options of: N*o Armoring; Non-Cohesive Armoring for Garcia-Parker;* and *Armoring with Active Parent Layering.* The layer options frame is displayed based on which armoring option is selected. For the case of armoring with *Active-Parent layering*, the user needs to specify an active layer thickness, usually <5 cm but based on the d50's, and whether the active layer is controlled by mass or thickness. EFDC can optionally initialize the active layer using the initial condition bed layering. This is a good approach for initial setup of a model when the bed configuration is not well defined.

The user may also specify the *Non-Cohesive Resuspension Options*, with a choice of modifications based on coefficients input by the user. This option modifies the critical shear stress for resuspension (i.e. TAUR) based on the fraction of cohesives.

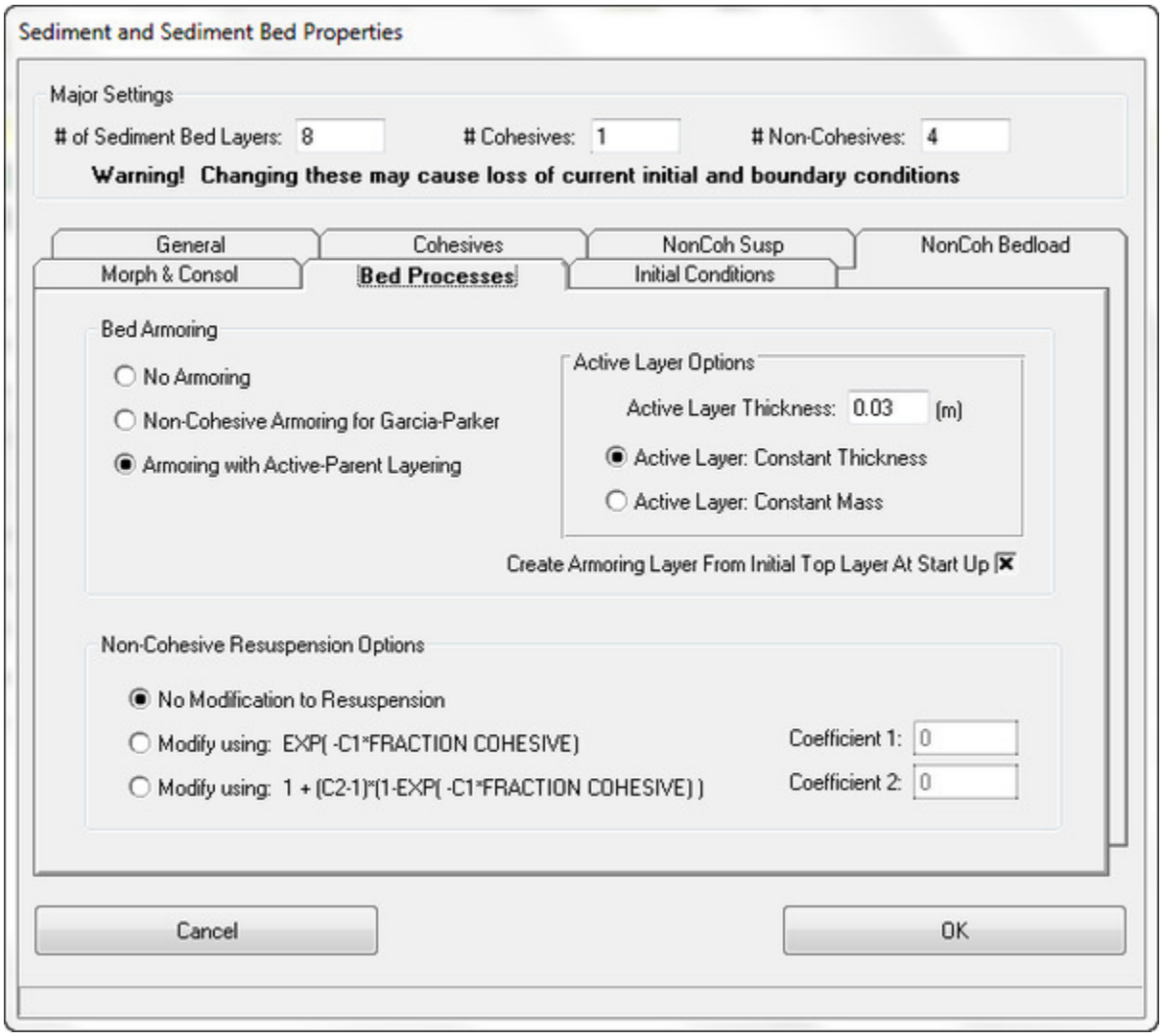


Figure 6-126 Sediment Transport – Bed Processes.

6.12.1.7 *Initial Conditions Tab - Sediments*

Figure 6-127 shows the sediment parameters form with the *Initial Conditions* tab selected. In this form the user selects how they would like to initialize the bed sediments for the model. This is done by selecting the option from the *Sediment Initial Conditions Options* frame. These options include: *Constant Water Column and Bed; Spatially Varying Water Column; Spatially Varying Bed Conditions;* and *Spatially Varying Bed and Water Column.*

If the user elects to set the *Bed Mass Specification Option* to specify the mass fraction, then and the BEDLAY.INP will required as an additional IC file.

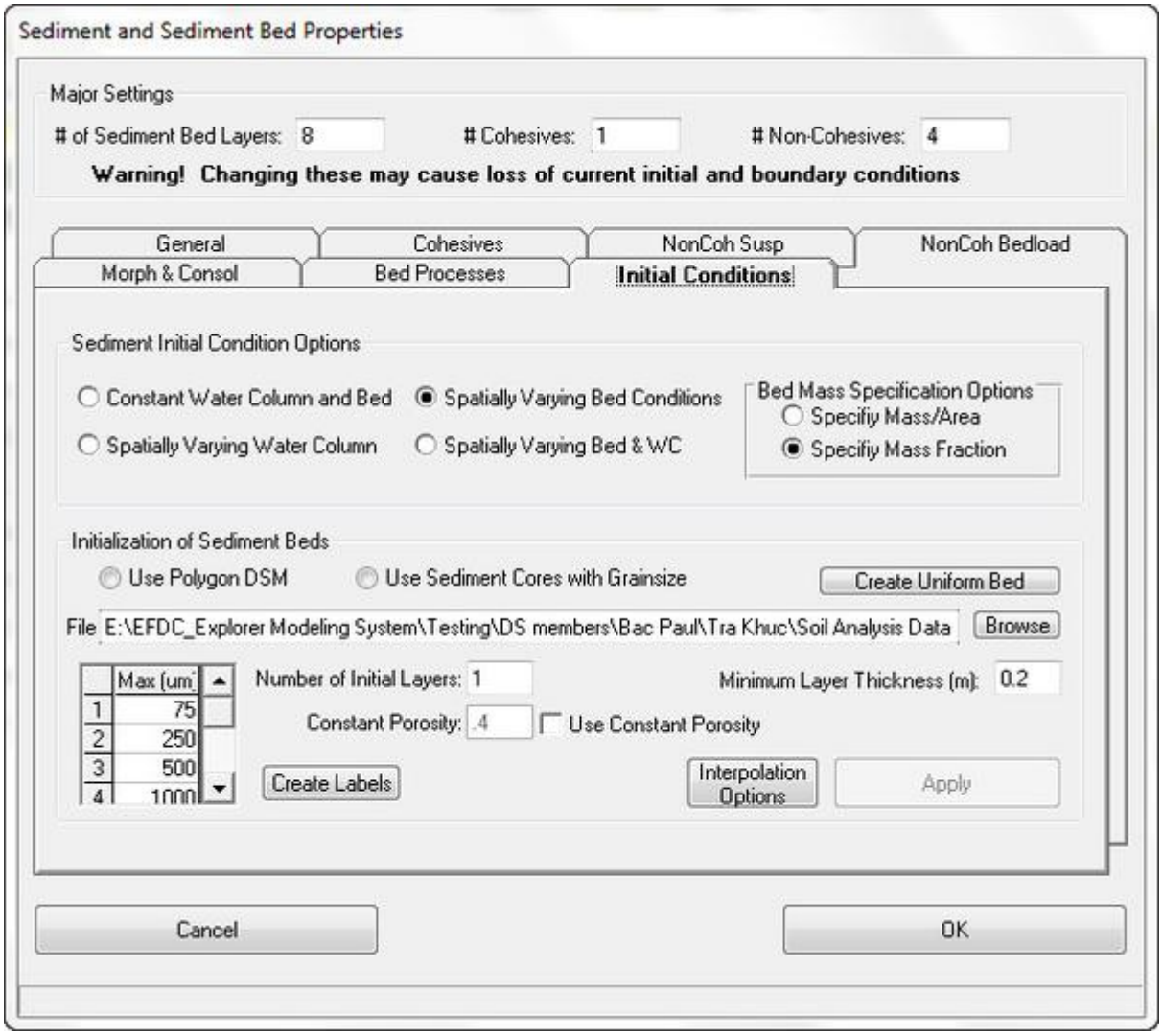


Figure 6-127 Sediment Transport – Initial Conditions.

The *Initialization of Sediment Beds* frame provides a number of options to build sediments beds which are described in more detail in Section 6.12.1.8.

6.12.1.8 *Sediment Bed Construction*

There are several ways to build sediment beds in EFDC_Explorer as shown in the *Initialization of Sediment Beds* frame in Figure 1. The simplest option is to create a uniform bed. Other options include using a polygon digital surface model, using a sediment cores with grainsize, each of which is described below.

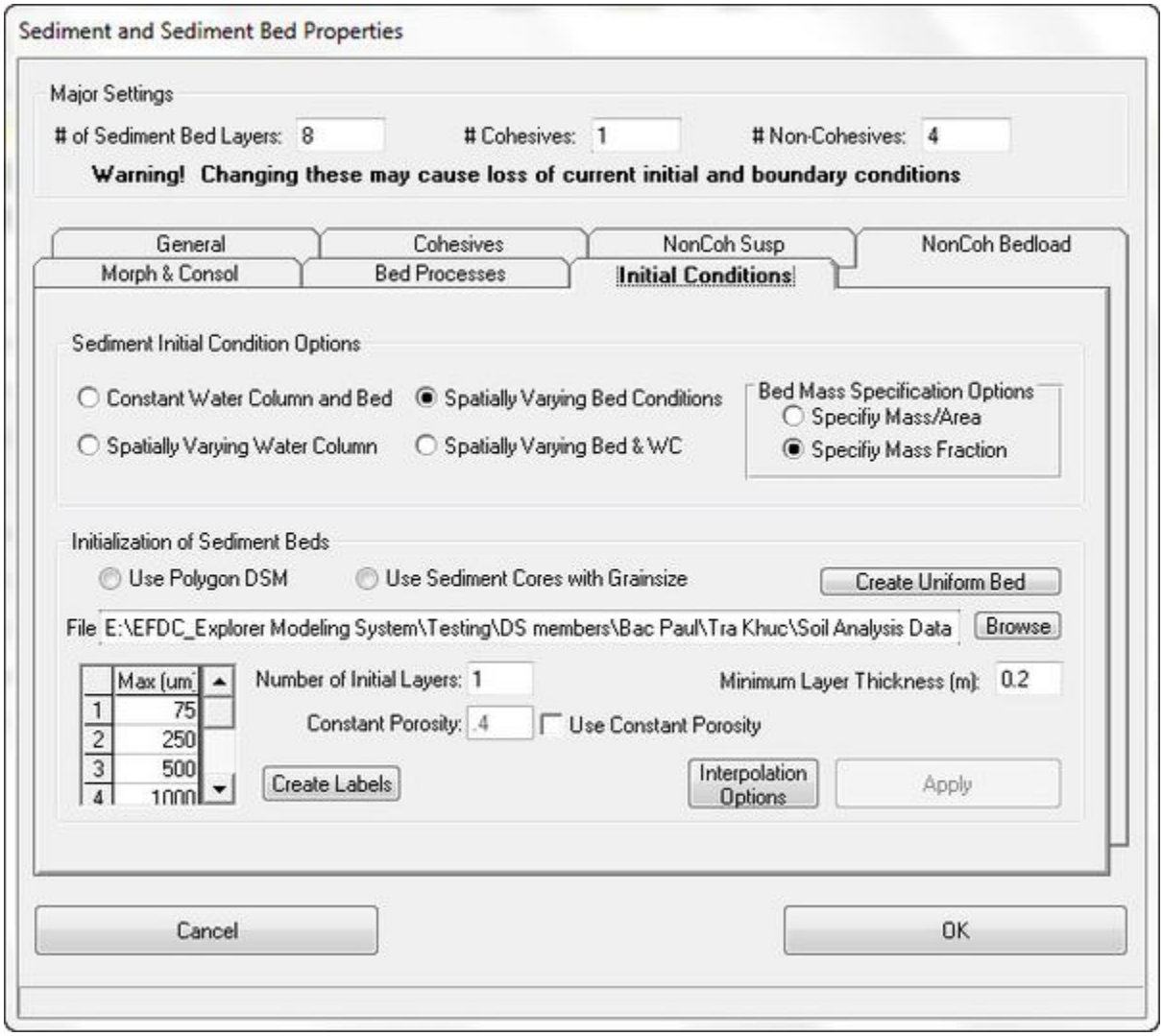


Figure 6-128 Initialization of sediment beds.

(1) Uniform Layers

To set up a simple horizontally uniform sediment bed for the model the user can use the with *Inorganic Sediment Bed Constructor: Uniform Layers* tool shown in Figure 6-129. This tool is accessed from the *Sediment Bed: Solids* button in the main *Sediments* tab. Alternatively, the user can access this option from the *Inorganic Sediment Bed Constructor: Uniform Layers* form shown in from *Sediments* tab | *Modify* button | *Initial Conditions* tab | *Create Uniform Bed* button. The *Bed Constructor* asks the user to input the number of layers, the number and types of sediments, and allows the user to specify the sediment fractions, thickness, and bulk densities for each layer. The user must still set the cohesive and non-cohesive erosion, and deposition parameters, but once the user finishes this option, the sediment bed configuration is ready for EFDC.

In the *Bed Layer Settings* frame the user may specify the number of bed layers and subsequently the thickness, porosity (thus void ratio) and mass distribution by layer. Once the sediment bed is generated the user can modify the sediment bed properties in *ViewPlan,* as needed. The layer configuration settings can be saved and later retrieved using the *Save* and the *Load* buttons in the *Bed Properties Definitions* frame.

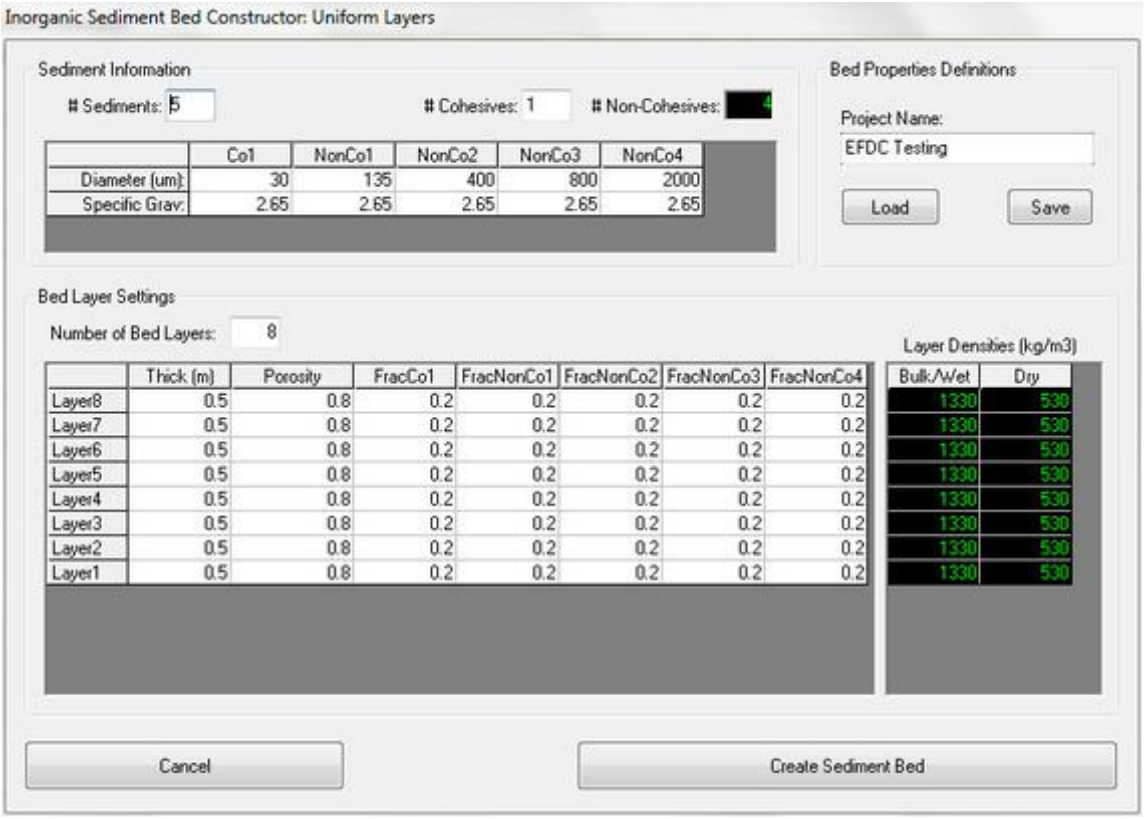


Figure 6-129 Uniform sediment bed generation tool.

It is generally expected that creating a sediment bed is done in the early phases of generating a sediment model. EE will attempt to update parameters in the *Morphology & Consolidation* tab such as the deposition porosity and cohesive void ratio to match the data entered in the *Bed Constructor.* This is not a bug and is behaving as expected.

Also, note that the *Inorganic Sediment Bed Constructor: Uniform Layer* tool adds two layers of "empty" sediment layers to allow for deposition and layer addition. The *Maximum Layer Thickness* setting in the *Morphology & Consolidation* tab controls when EFDC will create a new layer during deposition. If the top layer is already active and the thickness exceeds the maximum layer thickness, EFDC will just keep adding sediment to the top layer. This can result in a very thick layer, which is not a code problem but is a model configuration problem and should be avoided. This is expected behavior and is not a bug.

(2) Digital Sediment Model

A very useful feature in EFDC_Explorer is its ability to build the EFDC sediment files from a defined digital sediment model (DSM) that was generated by some third-party package (e.g., Spatial Explorer). This option is available from the *Sediments* tab | *Modify* button | *Initial Conditions* tab | *Initialization of Sediment Bed* form and selecting *Use Polygon DSM*. The DSM format requires a polygon followed by the layer thickness, bulk density, porosity and grain size distribution for each depth available (sediment depth intervals are based on data). More information on the data structure is available in Appendix B - Data Formats. EFDC_Explorer uses the DSM coupled with the number of size classes requested, the maximum size for each class, the number of sediment layers and the layer options (e.g., minimum layer thickness) to build the EFDC sediment files. The "Max (µm)" sediment diameters, one for each sediment class, are needed to break the sediment grainsize curves into ranges. These diameters are not the class diameters but represent the grainsize breakpoints whose geometric mean of the upper and lower limits is the corresponding sediment class' diameter.

When the user clicks on the *Apply* button after all the inputs have been provided, EFDC_Explorer generates the sediment map in memory and then writes out the files. The current sediment map (if any) will be replaced by the new one. It is recommended that the project be saved into a new sub-directory prior to implementing this option in order to save different versions of the sediment bed configurations.

(3) Sediment Cores with Grainsize Option

Another option provided in EE is the ability build a sediment bed from sediment core data. Figure 6-130 shows an example of a digital sediment model derived from sediment cores. The data contained grain size distributions by depth interval. The black diamond symbols show the locations of the cores. The plot shows the resulting depth averaged d_{50} grainsize.

The user also has many options for editing sediment cores within the *ViewPlan* form, described in detail in Section 8.6.9.

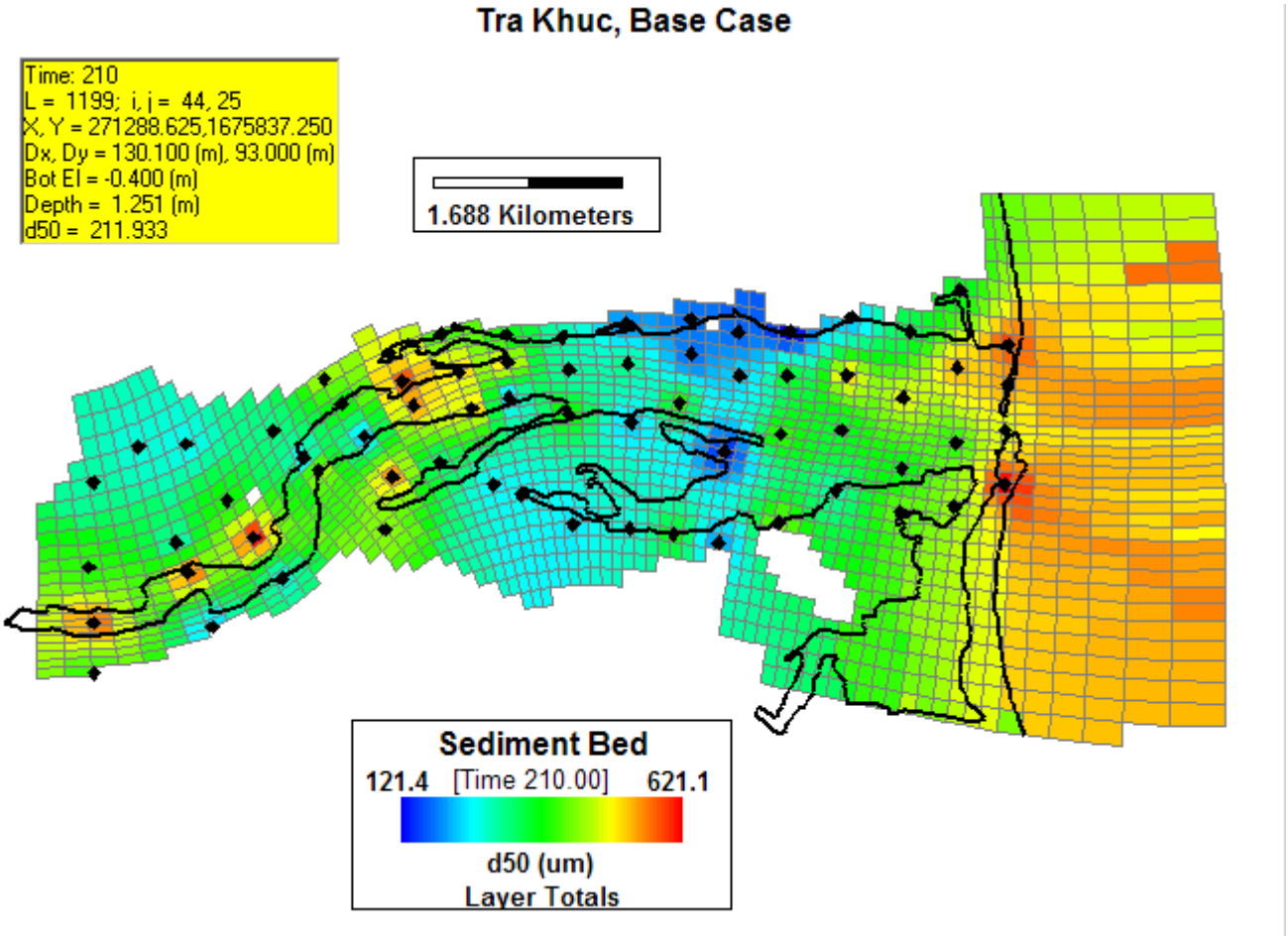


Figure 6-130 A digital sediment model generated from sediment cores with grain size.

6.12.2. SEDZLJ Sediment Model

The most state-of the-art sediment transport sub-model for EFDC is the SEDZLJ module that uses Sedflume data to determine erosion rates has been developed by Sandia National Laboratories. A detailed description of the SEDZLJ implementation in EFDC code is available in *Sandia National Laboratories Environmental Fluid Dynamics Code: Sediment Transport User Manual* (Thanh et al. 2008). The SNL approach, shown in Figure 6-131, accounts for multiple sediment size classes, has a unified treatment of suspended load and bedload, and appropriately replicates bed armoring. The resulting flow, transport, and sediment dynamics in the model is an improvement to previous models because this model directly incorporates site-specific erosion rate and shear stress data, while maintaining a physically consistent, unified treatment of bedload and suspended load.

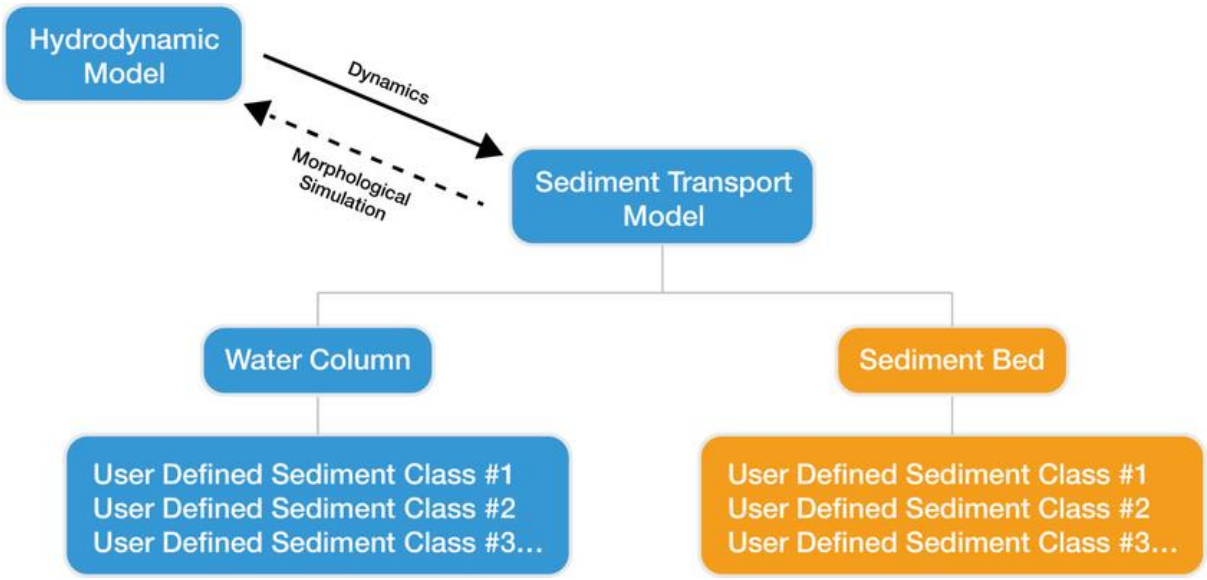


Figure 6-131 Structure of the SEDZLJ sediment transport model.

As can be seen in Figure 6-131, SEDZLJ does not distinguish between cohesive and non-cohesive groups for erosion, deposition and transport. In this sub-model, any particle size less than 200 mm is considered cohesive, and greater than 200 mm is non-cohesive for calculating the probability of deposition. Also, the size classes greater than 200 mm can be transported by bedload. The overall goal of SEDZLJ is to better represent sediment transport of typical sediments, which are a complex mixture of grain sizes with associated complex behavior. For this reason the SEDZLJ approach is now EPA's preferred standard for contaminant sites.

EE can load most existing EFDC/SEDZLJ projects. Many of these existing SEDZLJ projects use slightly different formats, including those from SNL, USACE and AnchorQEA. Most of these existing projects can now be loaded by EE and saved out to run in EFDC+. EFDC+ uses a standard DSI format of the input files that was originally developed by Earl Hayter of USACE. DSI has developed a standalone AnchorQEA to EFDC+ converter since their files are completely non-standard EFDC formats. Contact DSI for more information about this.

Examples of the SEDZLJ input files can be found in Appendix B. The primary input files for EFDC+ running the SEDZLJ sub-model are:

BED.SDF	SEDZLJ control file
ERATE.SDF	Core specific information
CORE_FIELD.SDF	Assigns a cell to a core profile where each core has certain properties)

For both the conventional EFDC and SEDZLJ sediment models, the user may set *Major Settings* for the number sediment layers and number of sediments. However, these values should be changed with caution as it may result in loss of initial conditions and boundary conditions.

In the following sections, options available in the form *SEDZLJ Sediment and Sediment Bed Properties* are described. The SEDZLJ sediment model is turned on from the *Active Modules* tab on the main form. The user should first select the *Activate Sediments* check box and then select the sediment model as shown in Figure 6-132.

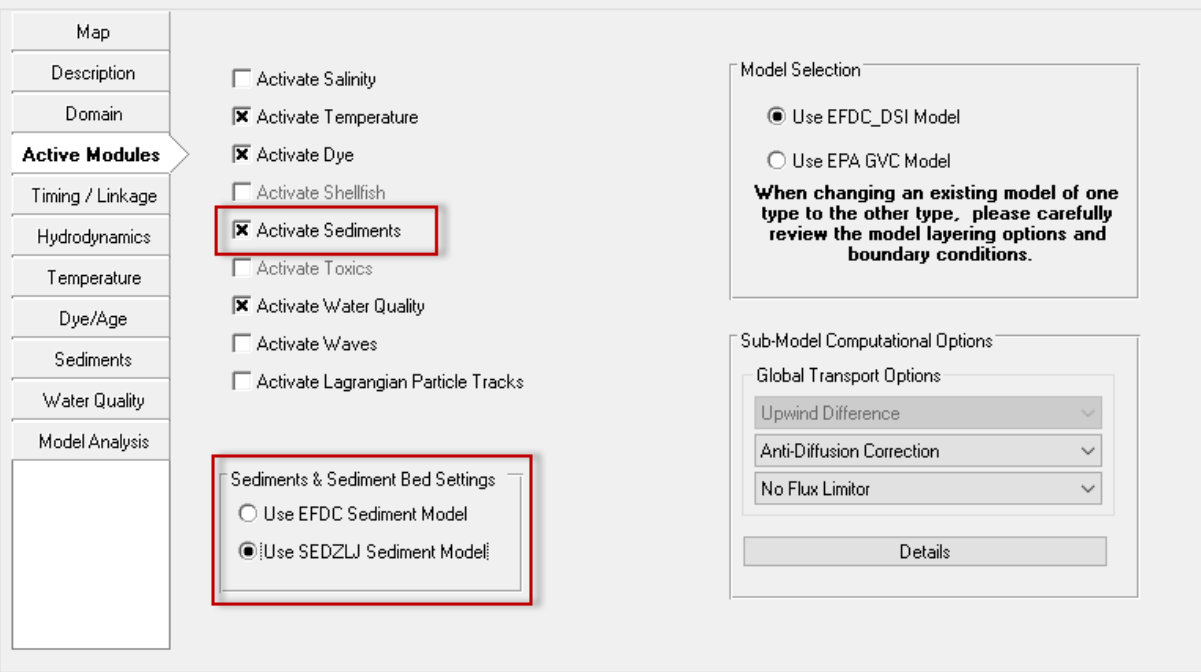


Figure 6-132 Active Module tab - SEDZLJ sediment transport sub-model.

6.12.2.1 *General Setting Tab*

The *General* tab (Figure 6-133) is used for primary computational options for the SEDZLJ sub-model. Options include allowing bed morphology changes with hydrodynamic feedback, computing bedload for coarser classes, and accounting for bed slope change.

The *Date/Time to Start Sediment Transport* allows the user to run the model without erosion and deposition active until the model is stable. After model is stable, the sediment transport is fully activated from the specified time. The *Deposition Limit of water Column Sediments* is a value from 0 to 1. This represents the minimum fraction of sediment in the bottom water layer allowed to deposit in a single time step. Setting this to zero would cause 100% of the amount to be deposited in one time step, however, this is also dependent on the settling rate specified.

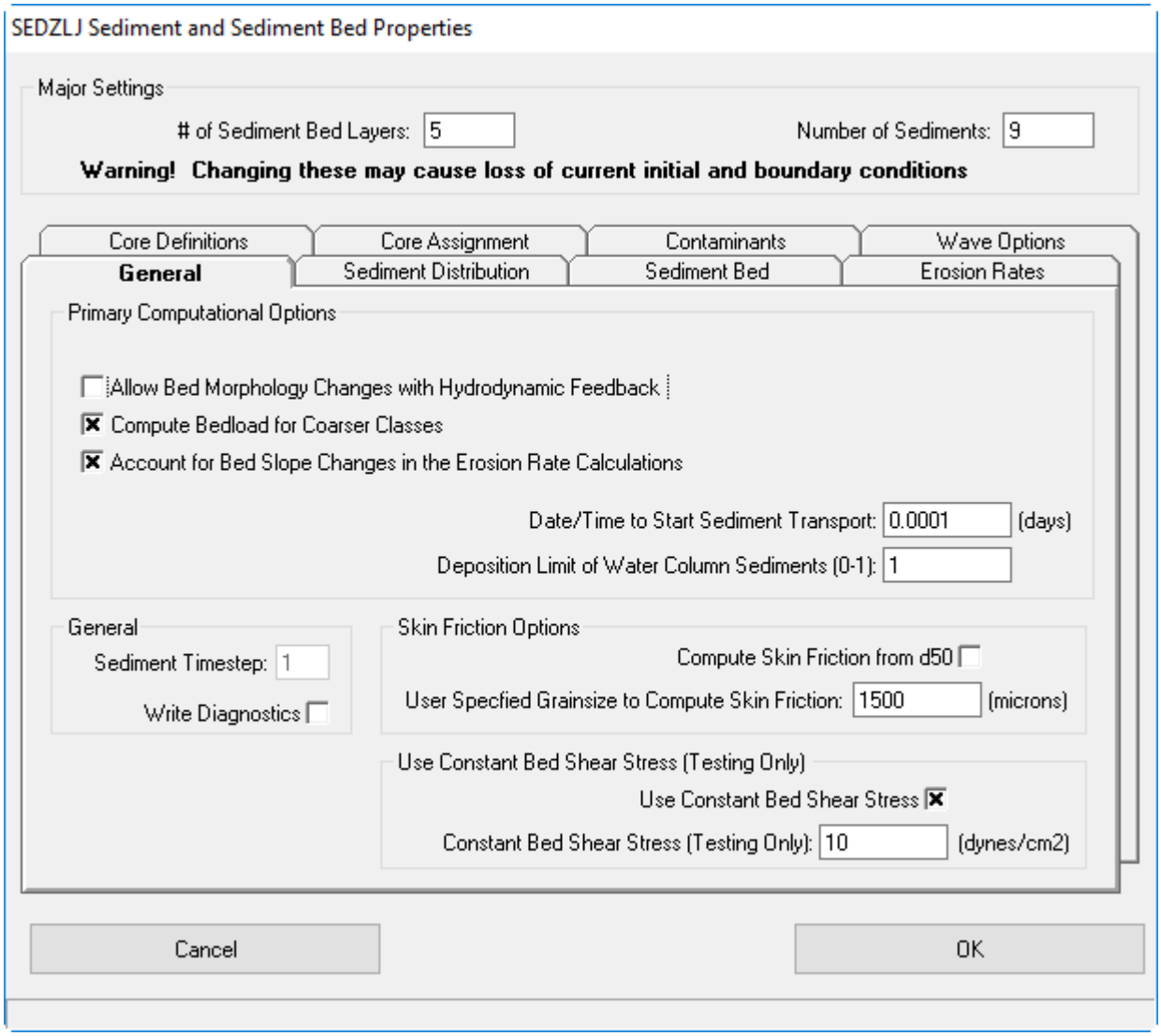


Figure 6-133 Sediment transport sub-model: General tab.

The user should refer to the SEDZLJ user manual for more information on *Skin Friction Options* and *Use Constant Bed Shear Stress* options.

6.12.2.2 *Sediment Distribution Tab*

The *Sediment Properties for Sediment Distribution* frame allows the user to set the class size, and critical shear stresses for erosion and deposition for each class. The units are in dynes/cm^2 as SEDZLJ data is generally reported in CGS units, so EE maintains these units for consistency with the reported data. It should be noted that the process of erosion and deposition are handled differently in SEDZLJ. The redeposited grain size has a different shear stress.

In the *Incipient Motion & Settling Estimates* frame shown in Figure 6-134 there is a *Sediment Info* button which reports the critical shear stress for grain sizes used in the *Sediment Properties for the Sediment Distribution* table. These calculations for the critical shear stress are based on the equations from Van Rijn (Van Rijn, et al 1984) which use uniform grain size calculations and provided for the user's information. These numbers do not have any impact in the actual model calculations. The *Initialize Defaults* button uses the values calculated here

to provide initial values for the critical shear stresses for erosion and deposition, though the user would normally have measured data for these.

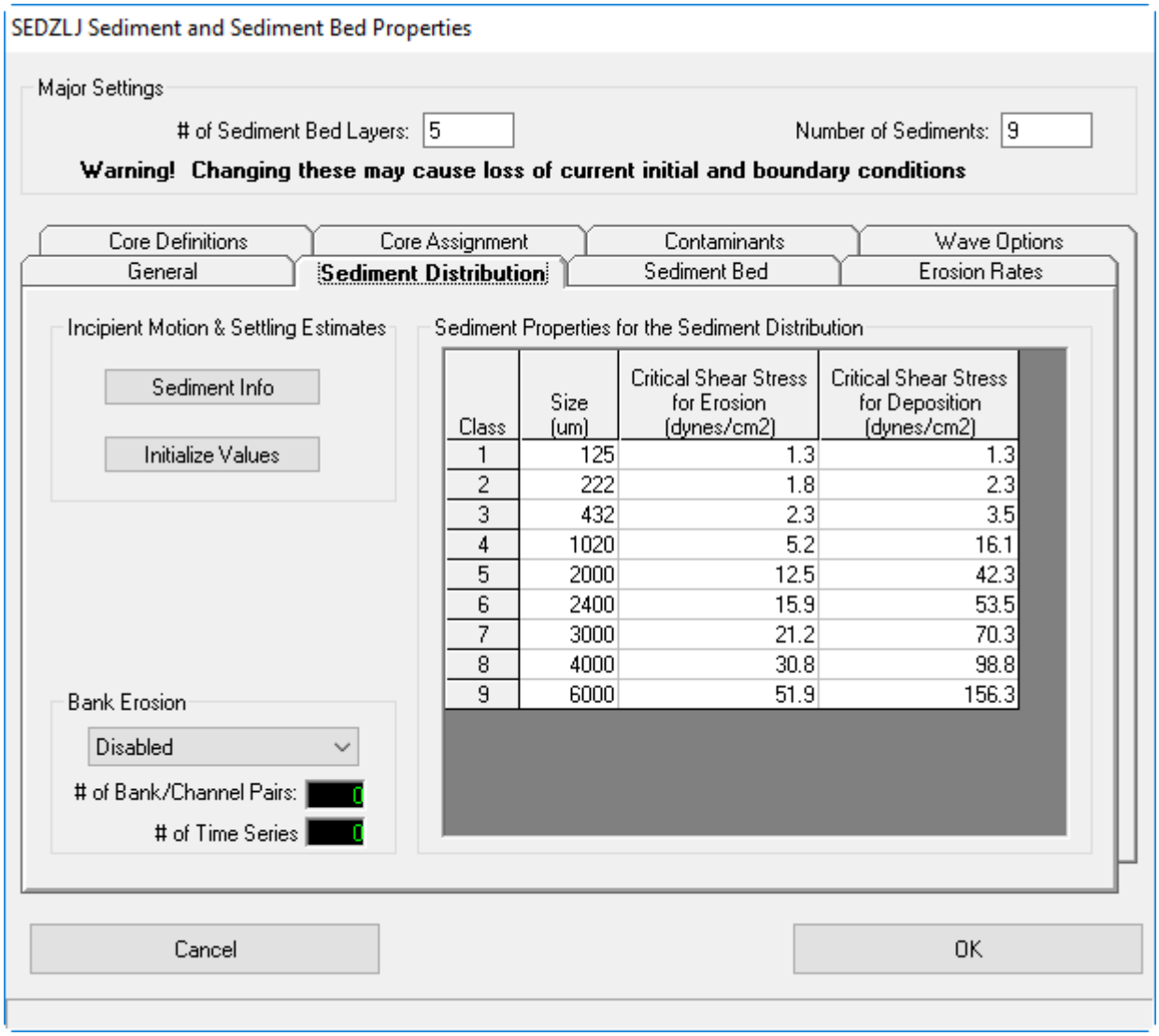


Figure 6-134 SEDZLJ Sediment transport sub-model: Sediment Distribution Tab.

6.12.2.3 *Sediment Bed Tab*

In the *Sediment Bed* tab shown in Figure 6-135, the *Sediment Bed Layering* frame allows the user to set the active layer multiplier as well as the layer thickness for all the layers.

The *Active & Deposited Sediment Characteristics* frame allows the users to specify the *Number of Deposited Sediment Classes*, as well as the *Number of Shear Stress Categories to Calculation Erosion of Newly Deposited Sediments.* Note that SEDZLJ allows for a different number of sediment classes for the sediment distribution and erosion characteristics.

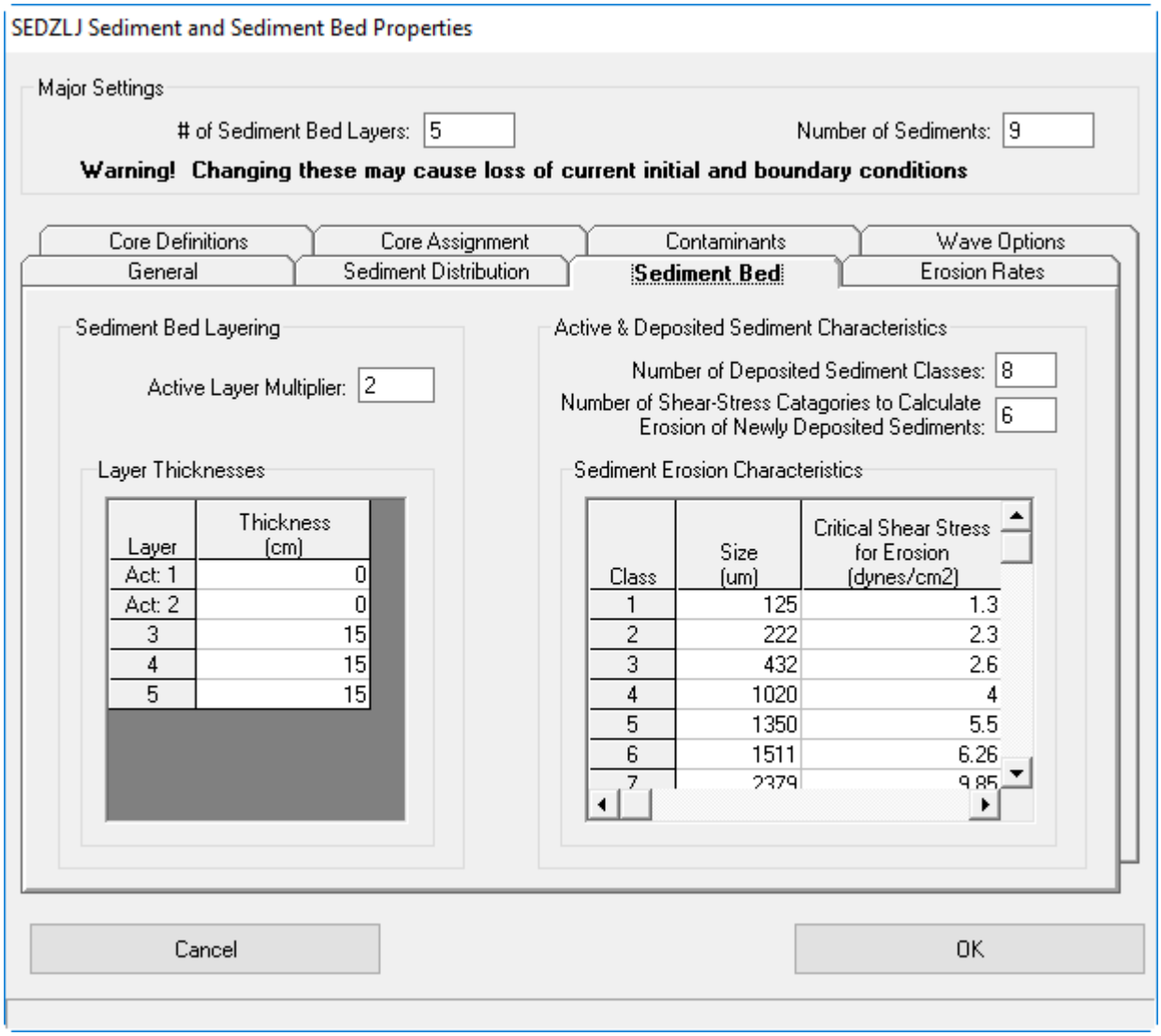


Figure 6-135 SEDZLJ Sediment transport sub-model: Sediment Bed Tab.

6.12.2.4 *Erosion Rates Tab*

The *Erosion Rates* tab provides the user with the *Active and Deposited Sediments Erosion matrix* as shown in Figure 6-136. Highlighted cells provide a warning to the user that the erosion rate is not increasing with the increasing shear category, or that erosion rates are not decreasing with the increasing grain size. This is simply a QC check for the user and EFDC+ will still run despite the inconsistent values that have been input.

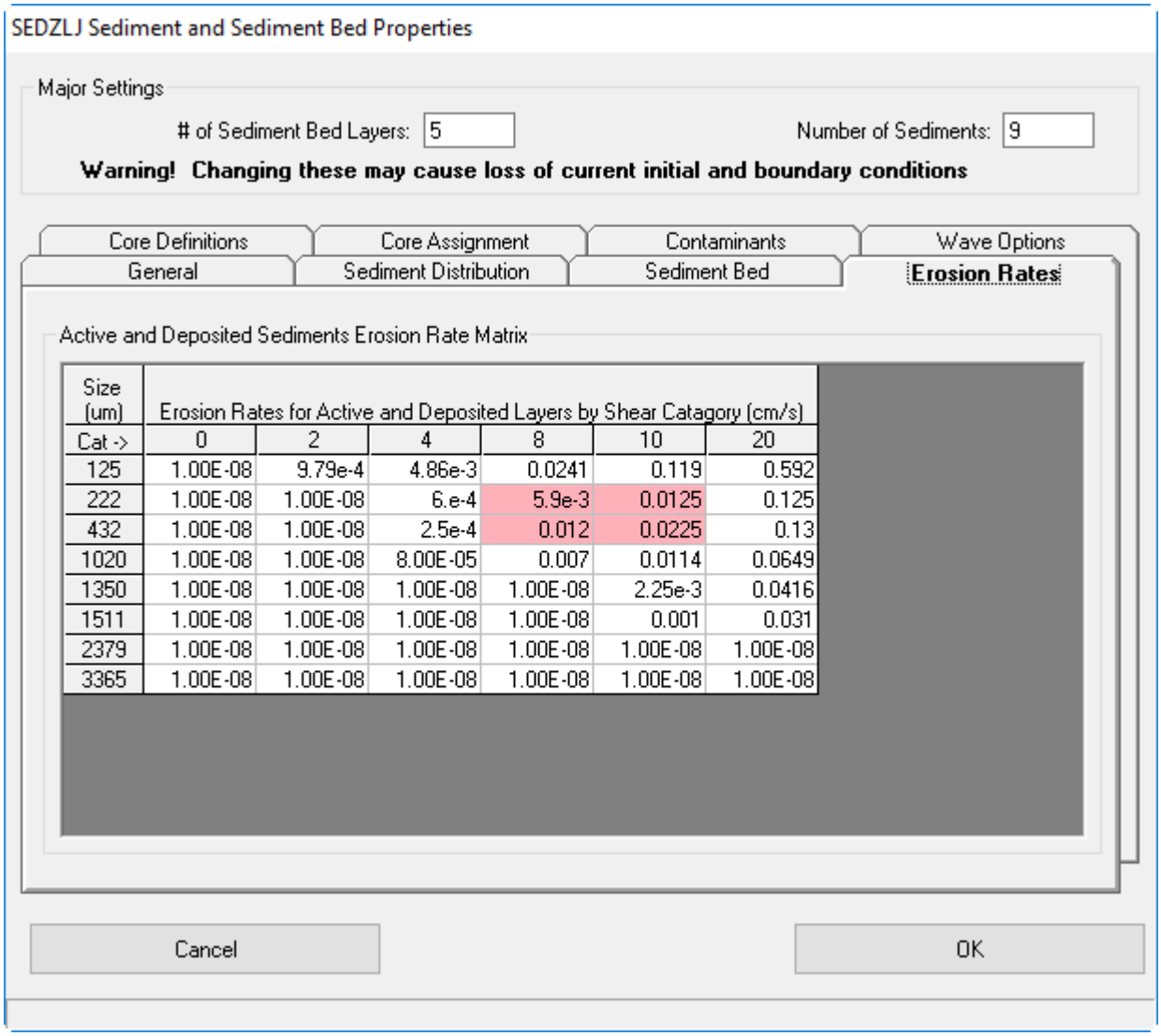


Figure 6-136 SEDZLJ Sediment transport sub-model: Erosion Rates Tab.

6.12.2.5 *Core Definition Tab*

The *Core Definitions* tab, shown in Figure 6-137 is where the user defines the *Number of SedFlume Cores,* and then assigns the key parameters associated with each core. For example, if the user sets the number Sedflume cores to two, then in the *Core Parameter* frame the drop-down will be populated with two cores. The names of these are user configurable and will appear in the information box in *ViewPlan* when looking at *SEDZLJ Options* | *Core Zone*.

For each layer in each core the user should set the critical shear stress and bulk density. The grain size distribution as a percentage also needs to be input in the first table. In the second table the user should input the initial bed erosion rates for each layer for each shear stress defined.

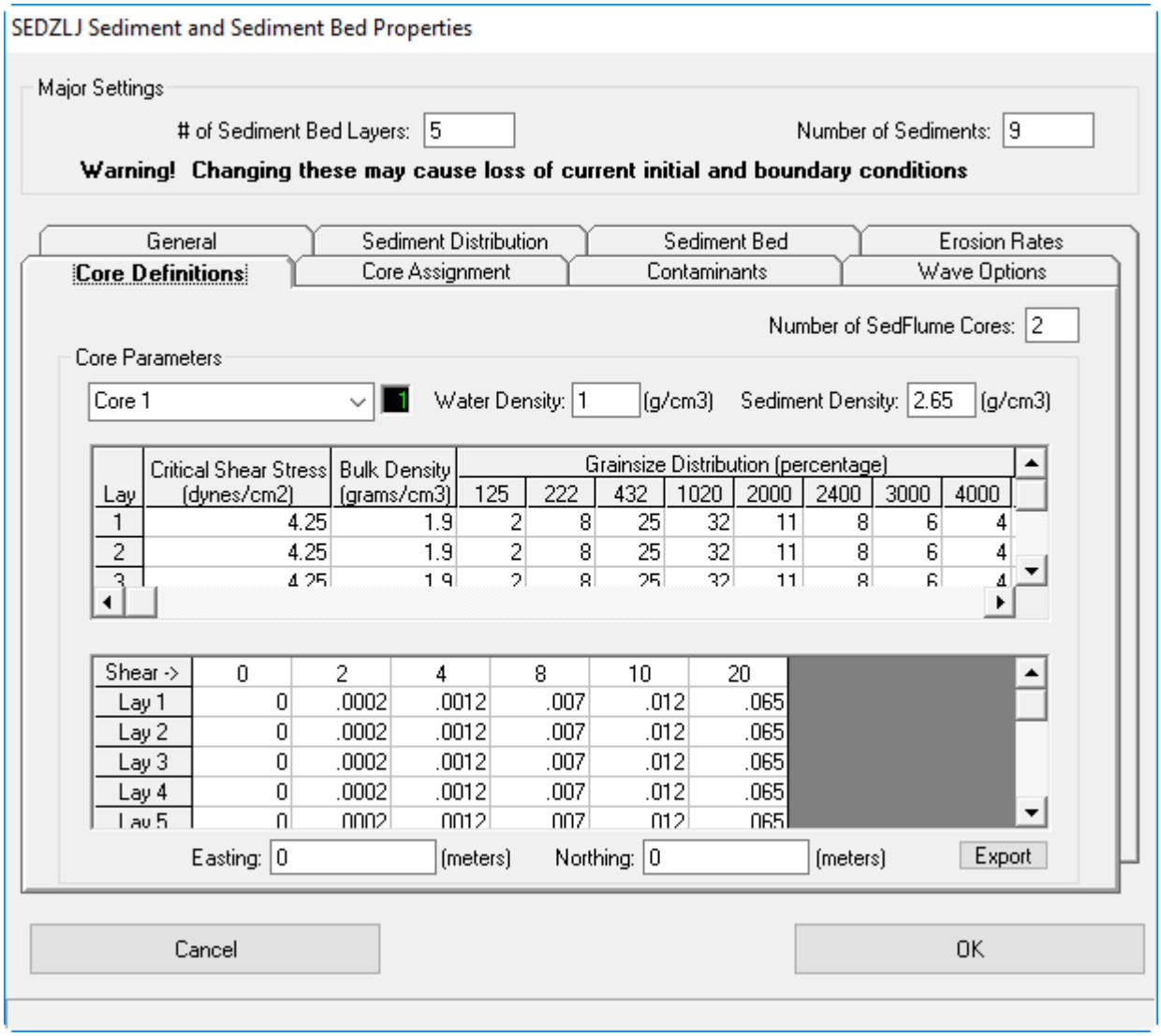


Figure 6-137 SEDZLJ Sediment transport sub-model: Core Definition Tab.

6.12.2.6 *Core Assignment Tab*

In the *Core Assignments* tab shown in Figure 6-138 the user sets the different zones in the model grid for each core type. The DSI standard for this data is just the I, J map, and the core (EFDC looks for DSI in first line). There is a different format for this file for SNL. These file format type is set in the *CORE_FIELD.SDF Format* frame.

EE allows the users to specify the default core number for all the cells in the domain or specify the core definitions on a cell by cell basis. However, users can choose the default core number value when *Allow core definition on a cell by cell basis* is checked. EE will assign default values to cells which fall outside the region where core data is not available.

The *Spatially Varying Core Assignment* frame provides options for the user to set the appropriate cores to the rest of the model domain. The *Use Nearest Neighbor Approach option uses an* interpolation method to assign core numbers using the X, Y coordinates of the core.

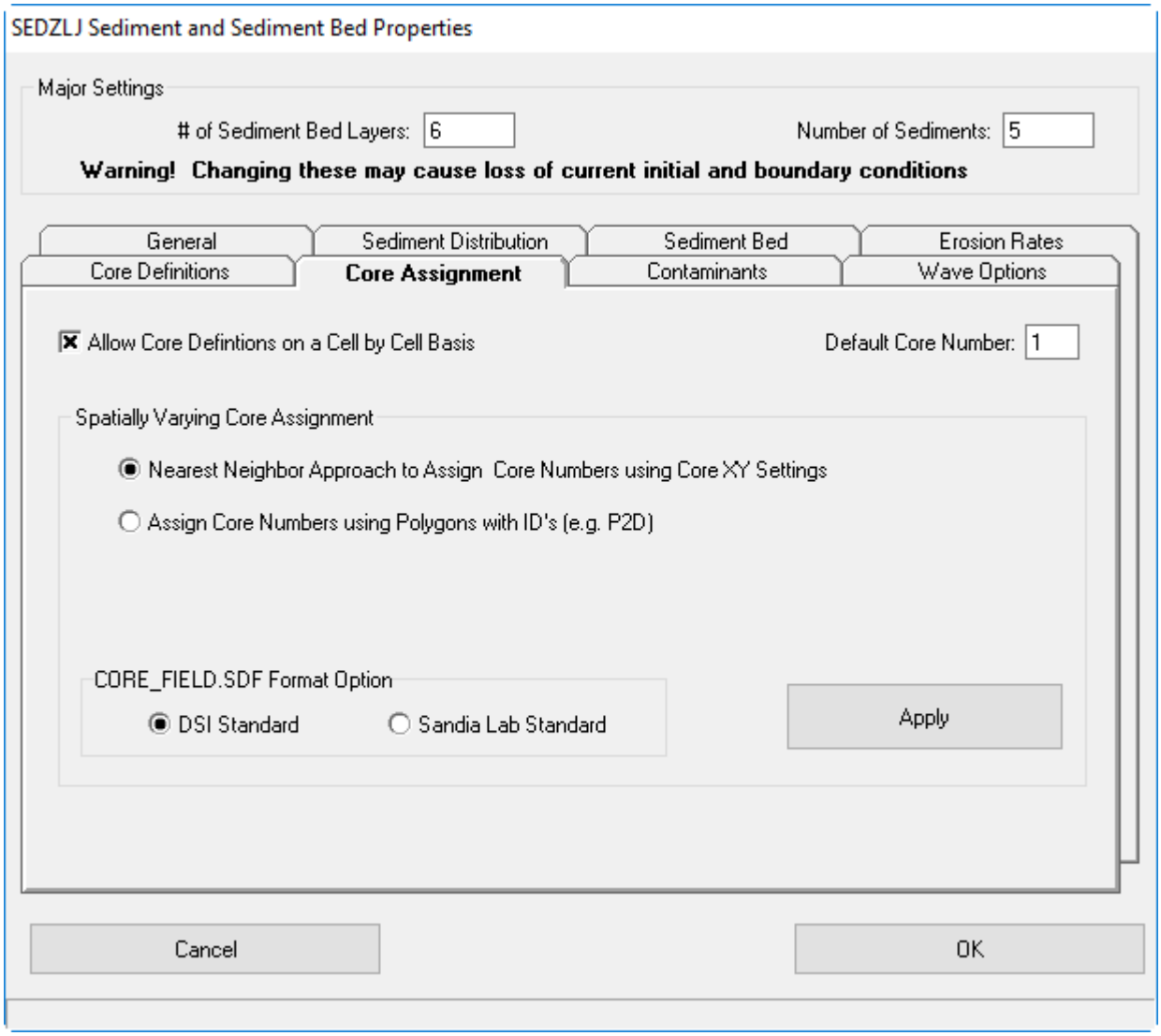


Figure 6-138 SEDZLJ Sediment transport sub-model: Core Assignment Tab.

The *Assign Core Numbers using Polygons with ID's* option is an ordered assignment based on the IDs in the P2D file. This file is begins with a single line header that is used as the ID of the polyline/polygon, which in this case is the core number. Next in this file comes the X, Y data to define the polygon. EE assigns the specified core number to the group of cells that lie inside the polygon. In some cases polygons may overlap, for example the user may make a polygon with core ID 4, which is inside that for core ID 2. In this case, core ID 4 will have precedence over core ID 2 and core ID 2 will not be used at all.

6.12.2.7 *Contaminants Tab*

The *Contaminants* tab is not functional at this stage. Contaminant transport is implemented through turning on the toxics sub-model described in the next section.

6.12.2.8 *Wave Options Tab*

The SEDZLJ wave options (Figure 6-139) can be set by the user to impact the bed shears. The *Wave Action Options* dropdown provides three options: *No Wave Action, Wind Wave (SEDZLJ),* and *STWAVE.* While these options can use the EFDC+ internal wind waves, at this stage they only impact the boundary layer. It is anticipated in the future that SEDZLJ will be updated so that it can use the internal wind waves or external wave linkages in the EFDC wave sub-model, however, at this stage this option is not available.

If the standard SEDZLJ wind wave option is used then the EE wind wave option is not used. In this case the user must ensure they have all the correct files such as FETCH.INP refer to the SEDZLJ user guide for details on this format. Conversely, if the STWAVE option is selected then that correctly formatted STWAVE files are required.

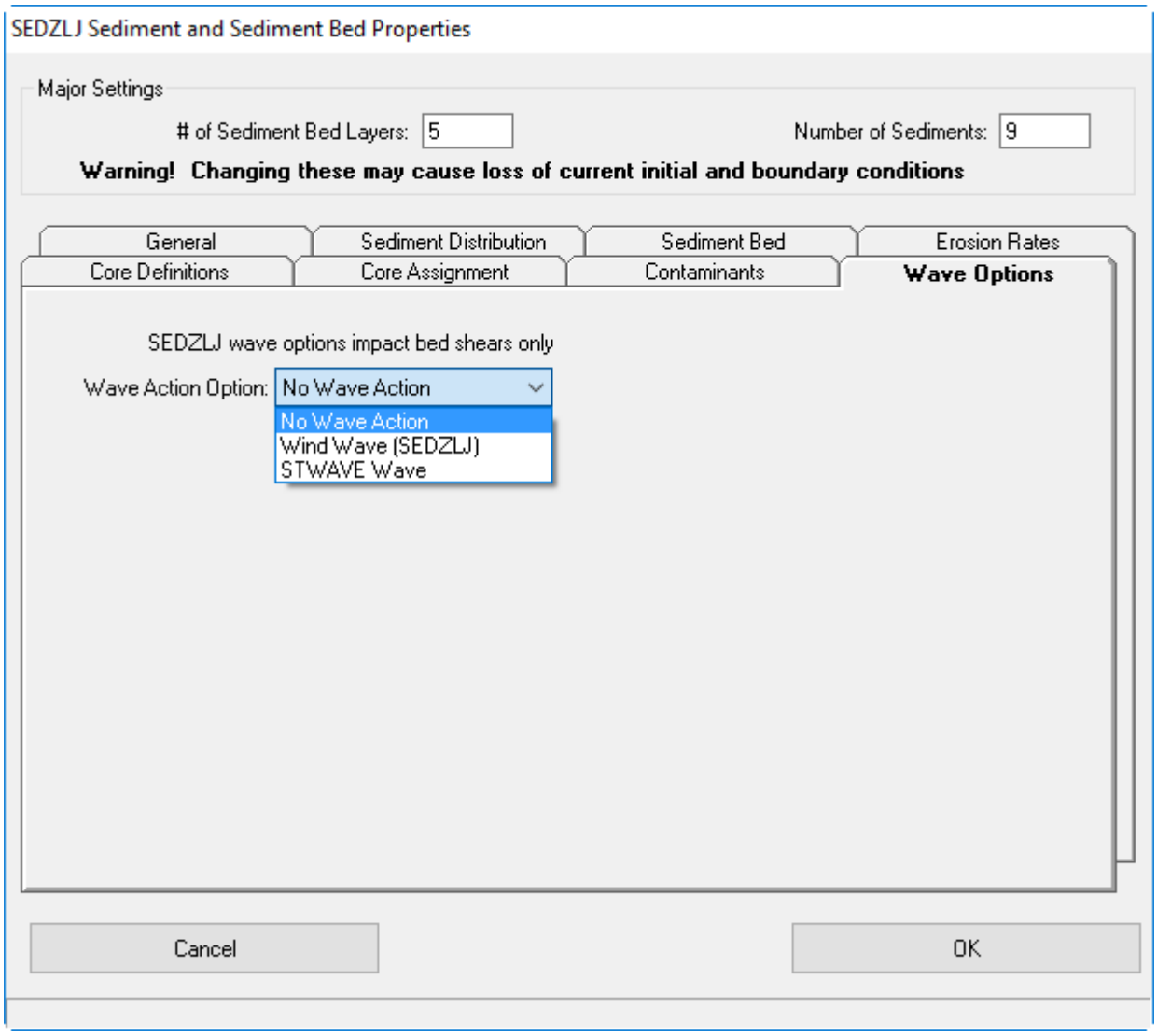


Figure 6-139 SEDZLJ Sediment transport sub-model: Wave Options tab.

6.13 Toxics Tab

When the *Activate Toxics* option has been selected in the *Active Modules* tab the *Toxics* tab will be enabled as shown in Figure 6-140. Note that the sediments module must also be selected for the toxics option to be available. The user interface for toxics when using the SEDZLJ model is the same as for when using original EFDC toxics model. However, to accommodate the new options added to toxics for EE8.3, significant changes have been made to the GUI for toxics as outlined in the following sections.

The *Toxics Options* frame informs the user of the number of toxics currently being simulated as well as the times steps for kinetics and bed processes. The user can make the initial conditions assignment for the sediment bed toxics by selecting the *Modify* button. This process is described in the next sections.

The *Initial Conditions* frame provides the *Assign* button for applying cell properties via polygons. The initial conditions set here are for the water column and sediment bed. Whether the *Use* box is selected or not will depend upon the option selected under *Toxics Options* | *Modify* | *Initial Conditions Options.* How to make the assignment to the water column is more fully described in Section 6.3.2.1 of the user manual.

Time series data for each of the toxics series may also be created and edited under the *Time Series Data* frame.

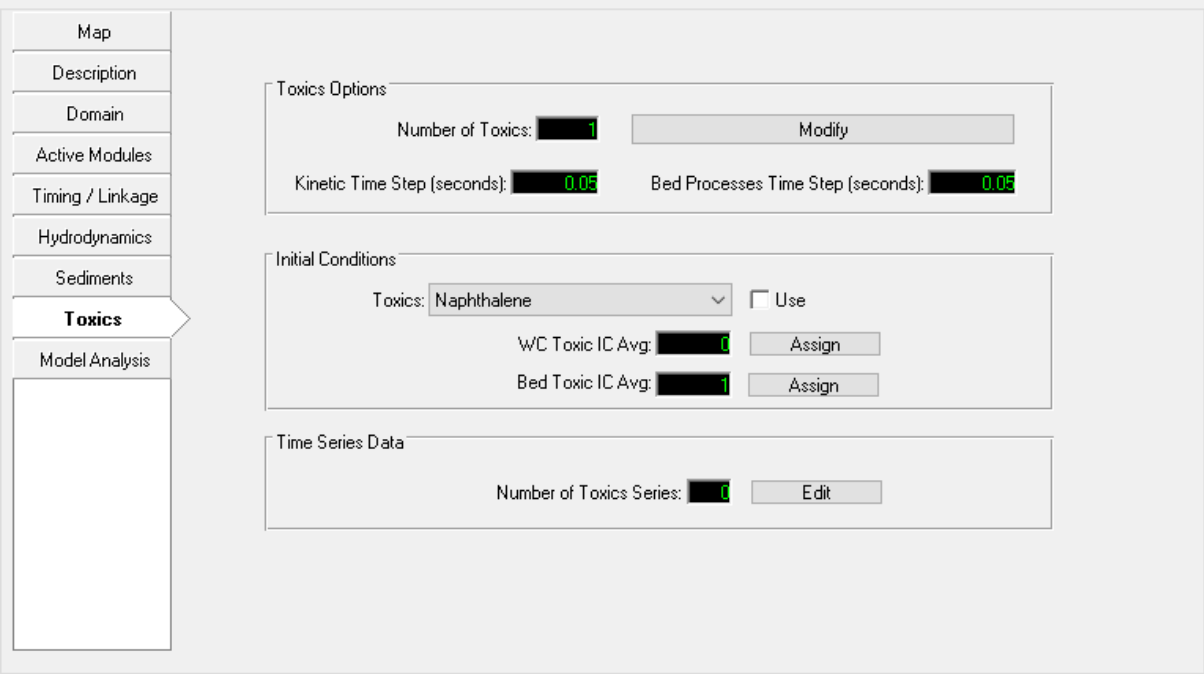


Figure 6-140 Toxics Tab.

6.13.1. Toxic and Sediment Bed Toxics Options

The toxic transport parameters and options are set using the form shown in Figure 6-141. The number of toxics included in the simulation are displayed in the top frame titled *Major Settings*. This can be changed by the user, but since boundary and initial conditions are predicated on the number of toxics, changing this value resets many of the toxics inputs. Therefore, this should only be changed when the user is prepared to reset all the initial and boundary conditions. A special case here is if the user sets the number to 0; then the toxics inputs are skipped for that project from that time forward. There are two major tabs for toxics settings, these are global options and individual toxics options described below.

6.13.1.1 *Global Options (All Toxics) Tab*

To enable the simulation of toxics, ensure the *Compute Toxics* checkbox is selected or set the simulate toxics (ISTRAN(5) = 1) under the *Active Modules* tab.

When volatilization of the toxics is being simulated the user should set the *Global Volatilization Settings*. These include velocity limit for river, and depth limit for river. If the velocities in the model are below these limits then a quiescent lake approach is used. When the values are higher than these limits then the river approach is used.

If the EFDC model is not simulating temperature then a fixed temperature approach may be set under the *Temperature if not Simulated* frame.

The *Initial Conditions Approach* frame for all toxics is set on this form. Options provided are: constant water column and bed; spatially varying water column (requires TOXW.INP file); spatially varying bed conditions (requires in TOXB.INP file); and spatially varying for both bed conditions and water column (requires both TOXW.INP and TOXB.INP).

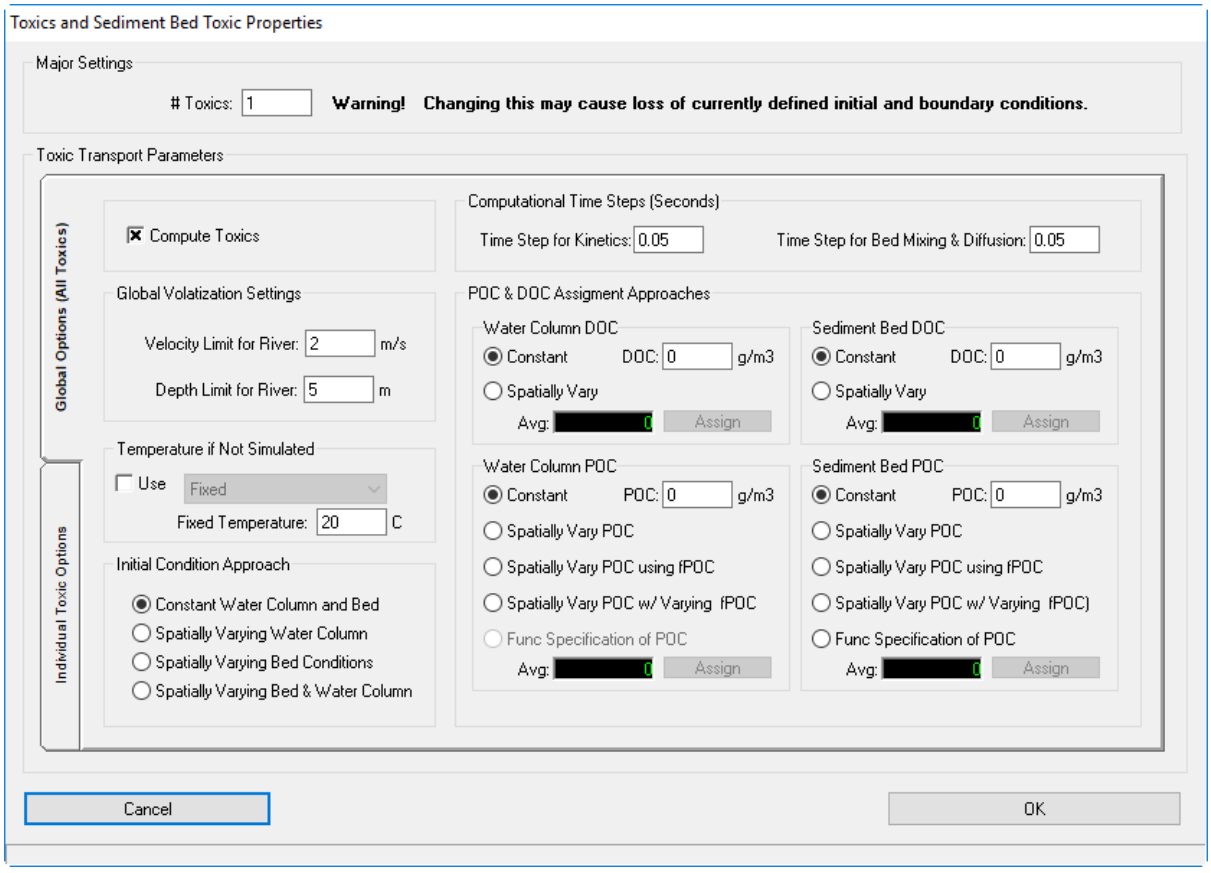


Figure 6-141 Toxics and Sediment Bed Toxic Properties options.

If a constant water column initial condition is being used then the text box under *Water Column Constant IC* frame is available. If a spatially varying water column is used then this option is greyed out. In the case when the spatially varying water column options are being used, the user should use the *Assign* button under *Initial Conditions* on the main form to set the varying water column values.

The global options tab also includes a frame for setting *Computational Time Steps* as shown in Figure 6-141. By default, the time steps for kinetics, and for bed mixing and diffusion, are set to the model time step. However, the user may adjust these as required.

The P*OC & DOC Assignment Approaches* frame (Figure 6-141) allows the user to set the particulate organic carbon and dissolved organic carbon to constant or spatially varying options for both the water column and sediment bed.

The *Water Column DOC* may be set to constant or vary spatially, in which latter case the user should use the DOCW.INP file to assign the constant time – spatially varying DOC. If the *Sediment Bed DOC* is set to be spatial varying, then the user must use the DOCB.INP file to assign the constant time – spatially varying DOC.

In the case of the *Water Column POC*, if set to *Spatial Varying POC* the user should use the POCW.INP file to assign the constant time – spatially varying POC. The spatially varying POC with fPOC option uses WC Seds and fPOC (Seds). The spatially varying POC/fPOC option uses WSeds and spatially varying fPOC from the FPOCW.INP file.

For the Sediment Bed POC, if set to *Spatial Varying POC* the user should use the POCB.INP file to assign the constant time – spatially varying POC. The *Spatially Varying POC with fPOC* option uses the bed sediments and fPOC (bed). The *Spatially Varying POC/fPOC* option uses WCSeds and spatially varying fPOC from the FPOCB.INP file. The *Functional Specification of POC* option means the user must modify the EFDC subroutine SETFPOCB to set POC in the bed.

6.13.1.2 *Individual Toxics Options Tab*

Each toxic has many of its own settings which are handled in the *Individual Toxics Options* tab shown in Figure 6-142. In this case the user must first select the toxic to be edited/entered by using the drop-down list in the upper left corner of the *Toxic Transport Parameters* frame. Once a toxic has been selected all the appropriate information is displayed on the form. The *Toxic Name* field is only used by EFDC_Explorer for labeling and information. Four sub-tabs are provided on this form, *Partition Coefficients, Initial Conditions, Diffusion & Bed Mixing,* and *Kinetic & Loss Terms*, each of which are described below.

In the *Partition Coefficients* tab the user should select the sorption model to be implemented from the radial buttons available, which include:

- *Simple Kd* which is two phase partitioning;
- *POC Only, focx TSS Dependent* which is also 2 phase partitioning, but has no DOC component. POC is calculated based on fraction of organic carbon x TSS;
- *DOC and POC*; 3 phase partitioning with POC based on input values not on fPOC.
- *DOC & foc x TSS (POC),* 3 phase partitioning, with POC based on fraction of POC per sediment class.

The contents of the grid (i.e., field) will change based on the partitioning model selected. The partitioning factors and the concentration factors are entered in the table below. Units must be consistent. Typical units for the toxic concentrations are mg/kg or µg/kg.

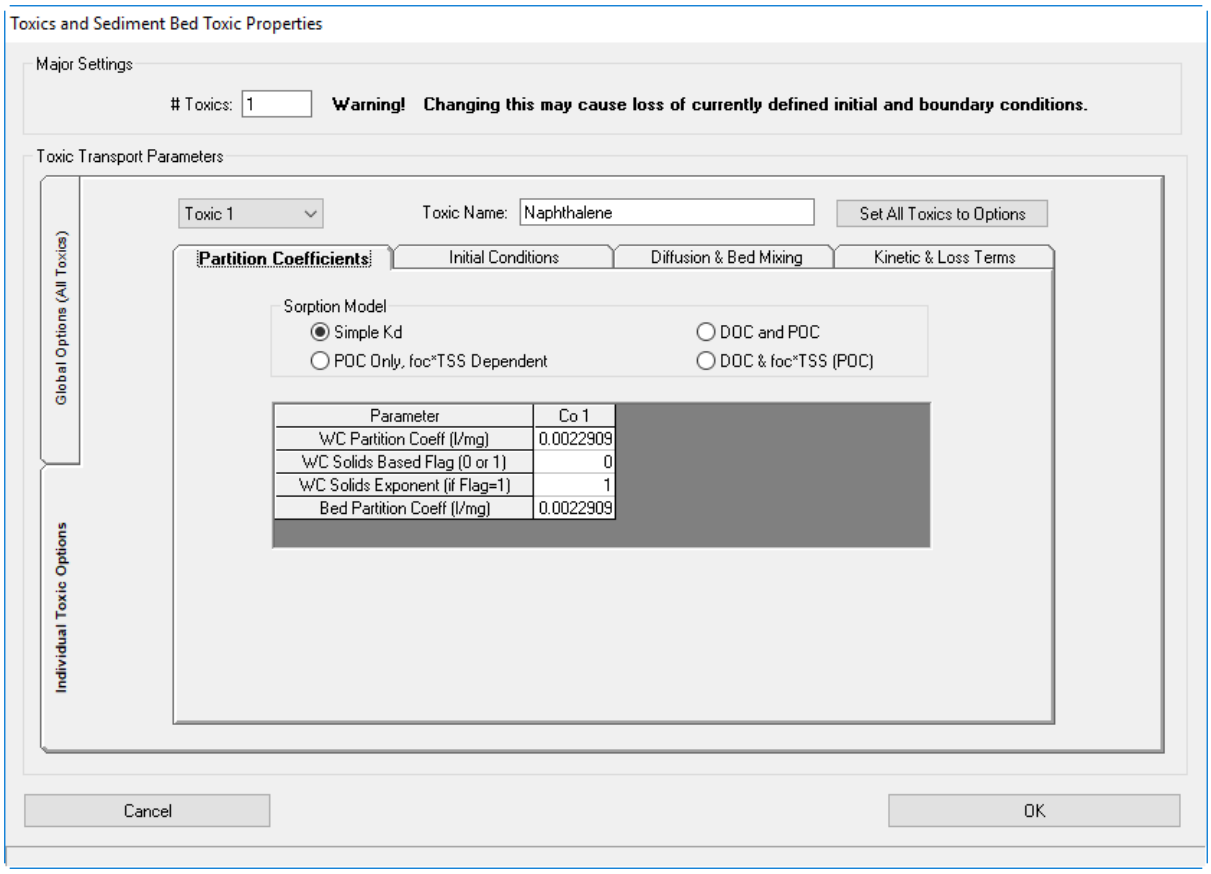


Figure 6-142 Individual Toxics: Partition Coefficients options.

The *Initial Conditions Options* tab is shown in Figure 6-143. Here the use can specify the water column toxic initial condition. This constant value is only used if a constant water column initial condition is being. If a spatially varying water column is used then this value will not be used. In the case when the spatially varying water column options are being used, the user should use the *Assign* button under *Initial Conditions* on the main form to set the varying water column values.

Bed Mass Specification field allows the user to the set the bed toxic initiation condition from two radial buttons: *Total Concentration* or *Mass Toxic/Sediment.*

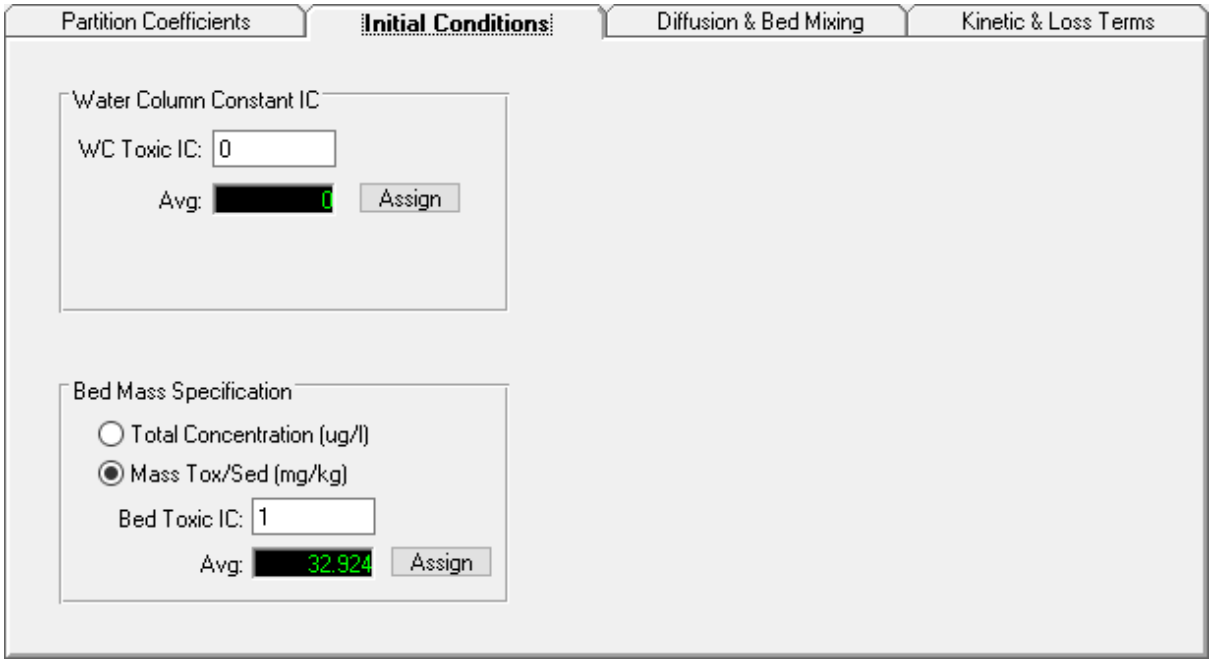


Figure 6-143 Individual Toxics: Initial Conditions options.

The *Diffusion & Bed Mixing* tab shown in Figure 6-144 allows the user to set the porewater diffusion rate and the apparent diffusion rates. For the *Porewater Diffusion Rates* frame the user specifies the porewater diffusion rate the sediment-water column interface, as well as the porewater diffusion rate between sediment layers which applies in the deeper bed. In the *Apparent Diffusion Rates* frame the user specifies the apparent diffusion rate due to particle mixing, as well as the maximum depth of the particle mixing.

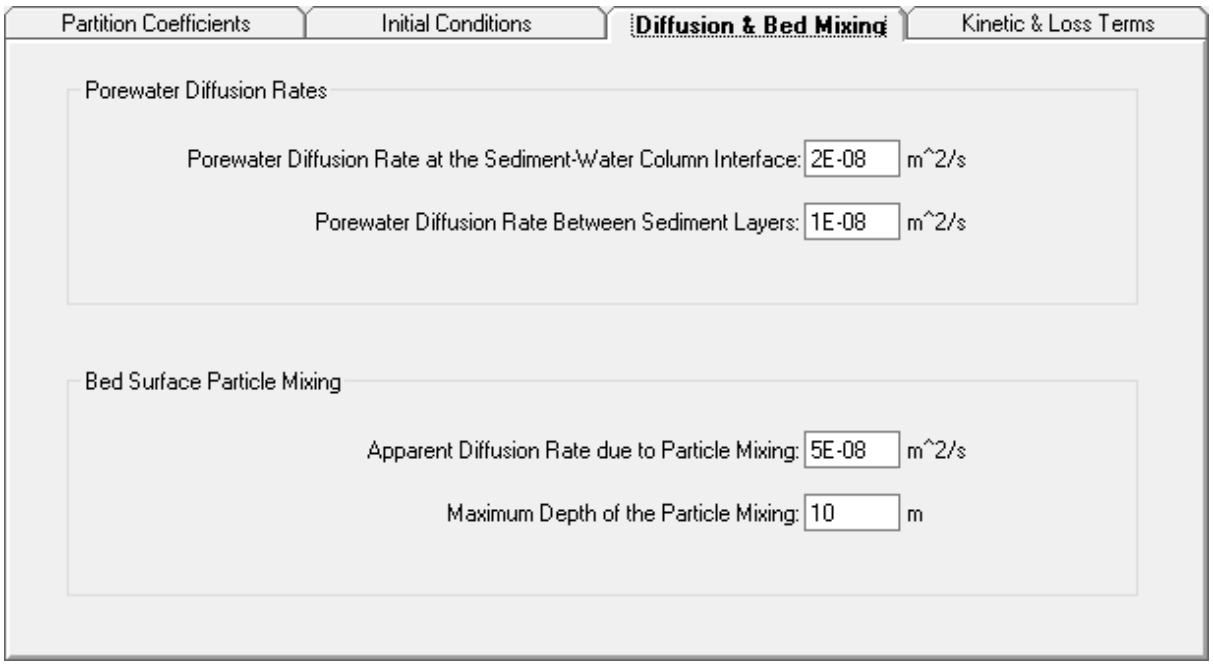


Figure 6-144 Toxics: Diffusion & Bed Mixing options.

The *Kinetic & Loss Terms* tab is shown in Figure 6-145. Here the use specifies which loss effects to implements, and then sets the decay rates for bulk decay, biodegradation, and volatilization, for the toxic class selected.

The *Bulk Decay* frame allows the user to specify the water column and sediment bed decay rates for bulk decay. The user may all set the maximum depth in the bed for which to apply bulk decay.

The *Biodegradation* frame allows the use to specify the water column and sediment bed decay rates due to bacterial degradation of the toxic. The user should all set the maximum depth in the bed for which to apply biodegradation. The biodegradation option requires the user to specify the temperature correction factor Q10 in both the water column and in the sediment bed, as well as set the reference temperature values.

The *Volatilization* frame is set to ignore by default. In the drop-down menu the user may select from the option of using the O'Connor formula, or the MacKay method for volatilization. For both of these options the user needs to specify the molecular weight of the toxic, and a temperature correction factor which is a dimensionless correction factor used to correct mass transfer rate to actual temperatures. Henry's constant for the air-water partitioning of the chemical, atmospheric concentration, and a multiplier should also be entered. This latter value is a dimensionless multiplicative factor to adjust the final volatilization rate.

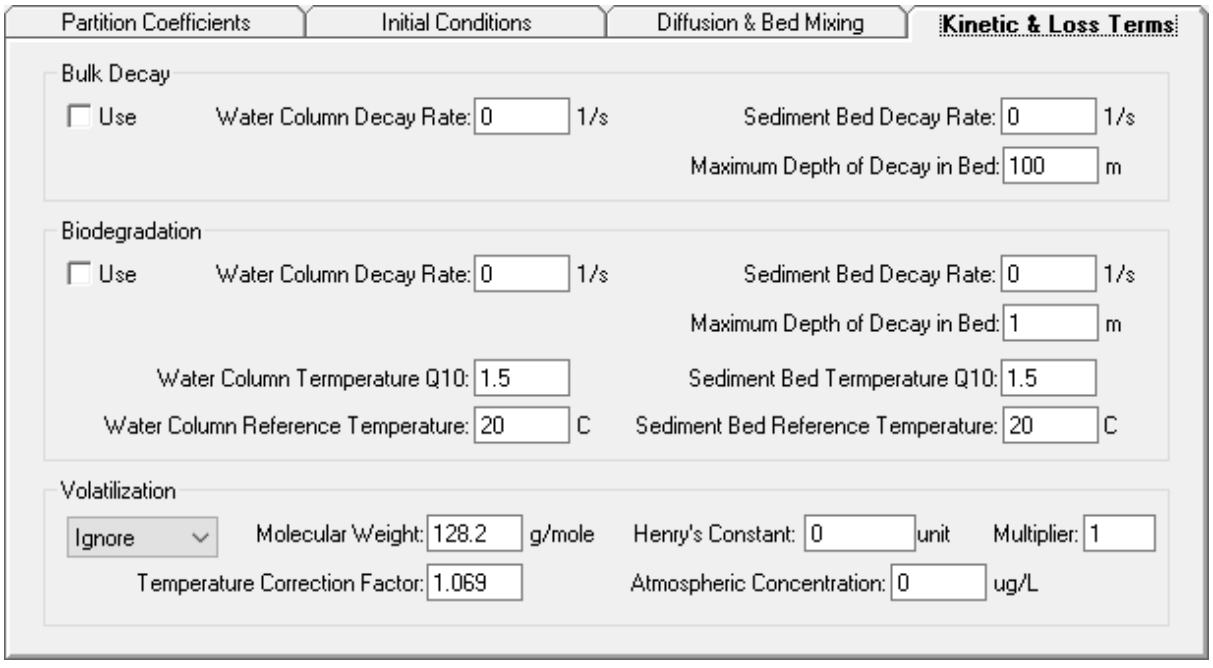


Figure 6-145 Individual Toxics: Kinetic & Loss Terms options.

6.14 Water Quality Tab

The water quality tabs of *WQ-General*, *Benth/Nutrients* and *Algae/WQ ICs* provide the user with a graphical user interface to the water quality sub-model of EFDC, sometimes called HEM3D (Park, et. al. 2000). The implementation of the water quality varies between versions of EFDC so the user should be aware of what is contained in the source code for their particular version of EFDC.

6.14.1. Water Quality Kinetics

The *WQ - Kinetics* tab is shown in Figure 6-146. Some of the user options include setting the Global Kinetic Options from one of several modules. For the EFDC_GVC version, the standard kinetic model is contained in Module 3 (ISWQLVL=3). While the other kinetic modules may be used (with care) when using the EFDC_GVC version in the standard sigma mode, if the GVC option is selected ONLY module 3 works. For the EFDCPlus version, the standard full kinetic module is Module 1 (ISWQLVL=1). Once the kinetic option has been chosen the user still needs to select which parameters will be varied/simulated within the selected kinetic option. Clicking the *Params* button brings up the list of parameters whose simulation (i.e. advection and dispersion) can be switched on or off. However, even if a parameter is not simulated, since it is still part of the kinetics, the IC concentration is used for all of the kinetic calculations.

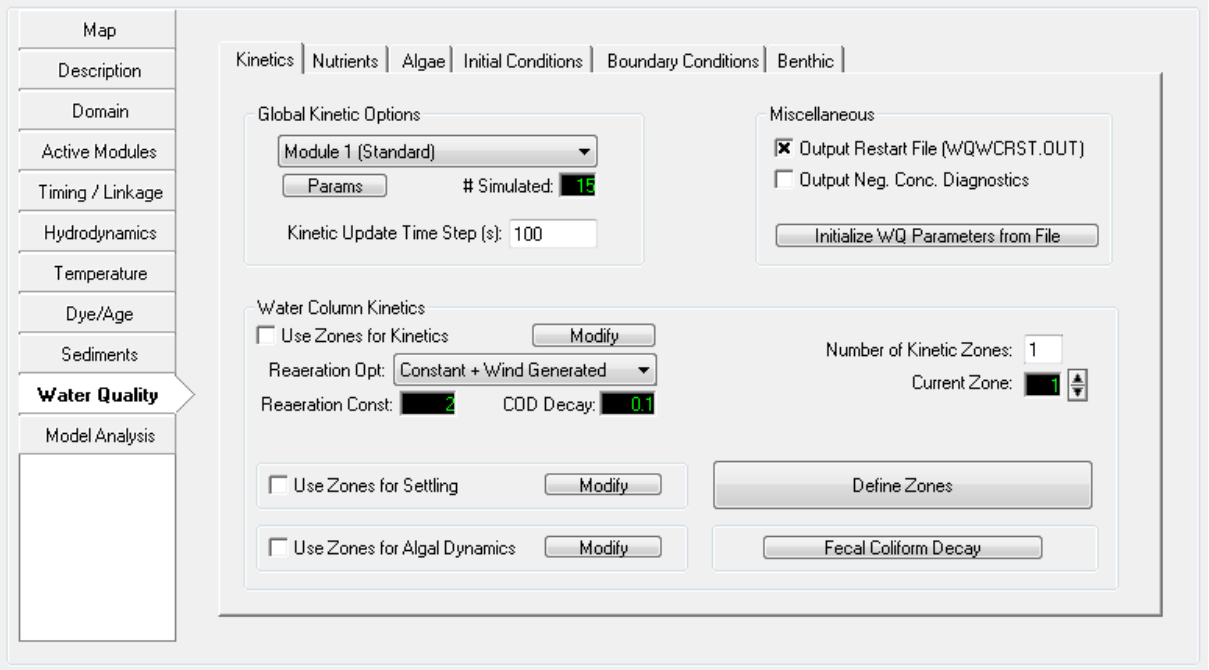


Figure 6-146 Water Quality Tab: Kinetics.

EE will automatically initialize the the water quality sub-model using the Access template file (EFDC.MDB). However, it is often convenient to initialize a new model using data from another EFDC_Explorer model with similar conditions. This is done using the "*Initialize WQ Parameters from File* in the *Miscellaneous* frame. This operation should only be conducted once at the start of the water quality model construction. This process overwrites all of the current model's settings.

EFDC allows for water column zones for the assignment of many of the water quality parameters. The WQ zones can be set using the *Define Zones* button or can be set in *ViewPlan* (*Water Column/Wtr Quality* then check *WQ Zones* checkbox). Clicking on the *Define Zones* button brings up the *Apply Cell Properties via Polygons* form (see Section

6.3.2.1). The user must first enter the maximum number of desired zones in the *Number of Kinetic Zones* box. This number can be changed at any time during the model building process.

Zones are assigned by water column layer and cell, so it is possible for the same cell to have more than one zone assigned to it, if KC>1. The parameters that vary by zone are shown in the frame *Water Quality Kinetics*. There are three groups of parameters that can vary by zone: Kinetics, Settling, and Algal Dynamics. For each group, the use of zones or not is separately controlled using the check boxes for that group. The specific parameters for the *Current Zone* for each group are modified by clicking on the *Modify* button. When the kinetic zones are enabled these values are used in preference to the default values displayed under COD parameters under the *Nutrients* tab.

The *Fecal Coliform Decay* button allows the user to set the decay rate and temperature effect constant.

6.14.2. Nutrients

Within the *Nutrient Options and Parameters* frame (Figure 6-147) settings for many of the basic nutrient parameters may be made. The nutrient parameters have been grouped by nutrient type and can be edited by clicking on the respective button. For example, to edit the nitrogen settings the user would click on the *Nitrogen* button. These are the default settings. If the user implements kinetic zones then those values will be used in place of these values, however the default values are still displayed in this form. For example, this can mean that the COD decay rate may be different in two places, however, the kinetic zone value will take precedence.

Figure 6-147 Water Quality Tab: Nutrients.

The user may also specify the *Sorption Options*. The default is “None” and parameters cannot be modified. However, if *Total Active Metal (TAM) Based* or *Cohesive Sediment Based* radial buttons are selected, the user may alter these nutrient sorption parameters with the *Modify Parameters* button.

The atmospheric wet deposition concentrations for the constituents being simulated with the kinetic module selected are edited via the *Wet Deposition* button. The dry deposition mass fluxes are edited via the *Dry Deposition* button.

6.14.3. Algae

Figure 6-148 shows the tab for *Algae/Water Quality Initial Conditions*. The *Algae Options* frame contains several buttons that, if pressed, allow the user to make adjustments to logical parameter groups. For example, the *Algal Dynamics* button brings up a form as shown in Figure 6-149. This allows the user to modify the algal default or "background" algal growth rates.

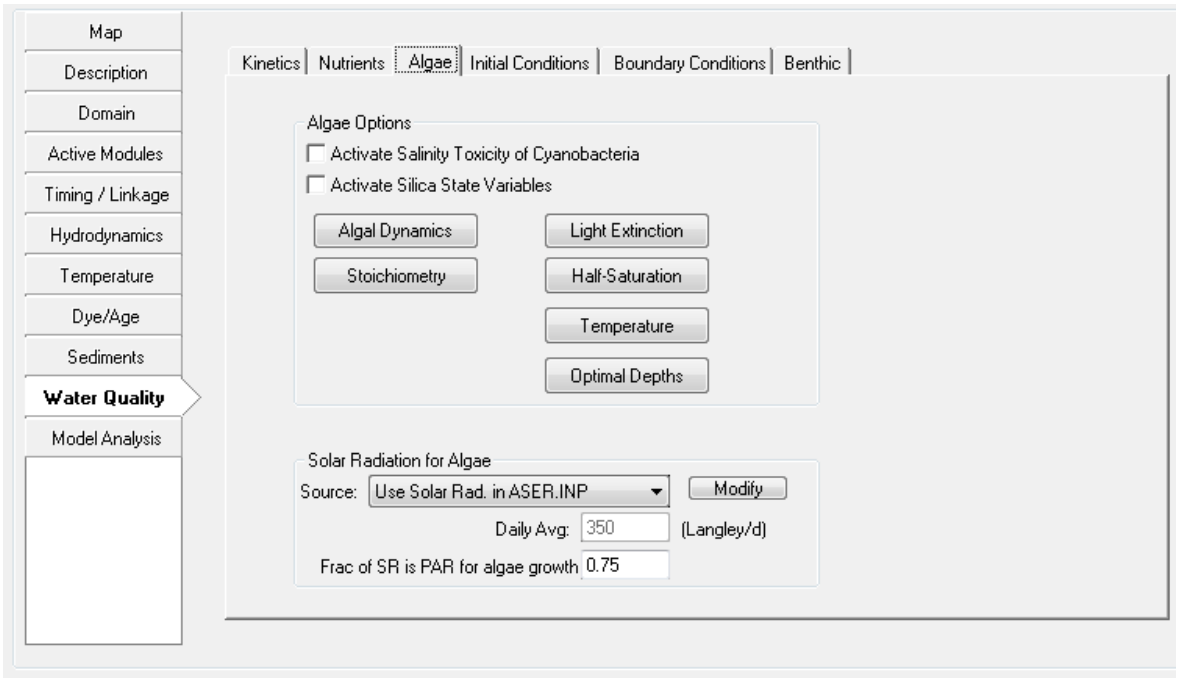


Figure 6-148 Water Quality Tab: Algae.

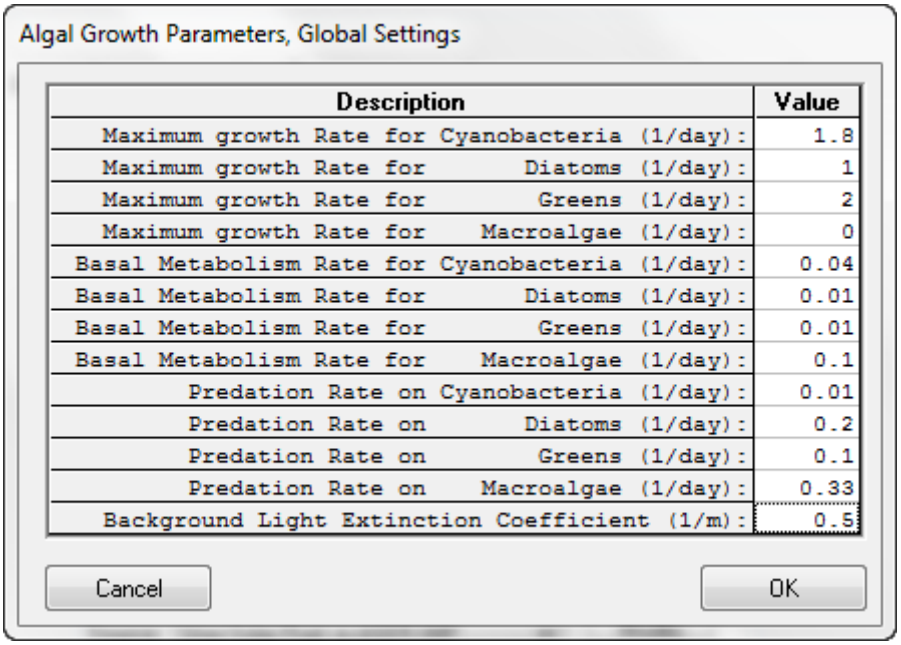


Figure 6-149 Algal dynamics parameter form.

The *Solar Radiation for Algae* frame provides the user with a drop-down menu from which the source for radiation data may be selected, these include: Constant; Read Daily Average (SUNDAY.INP); and Use Solar Radiation in ASER.INP. The *Modify* button then allows the user to set various solar radiation options including initial solar radiation at the water surface, minimum optimum solar radiation, fraction of day that is daylight amongst others.

6.14.4. Initial Conditions

The *WQ Initial Conditions* tab, shown in Figure 6-150, provides access to the sediment initial conditions settings for Water Quality. Here the user may select spatially constant or spatially varying initial conditions for each of the water quality parameters. To set spatially varying initial conditions for the water quality parameters the user must interpolate data onto the model cells using the *Apply Cell Properties via Polygons* utility. If the EFDC model uses spatially varying IC's, then the user must specify which input data format to use, i.e. WQWCRST.INP or the WQICI format. EFDC_Explorer will generate the IC's in the format specified. Pressing the *Initialize IC's* button assigns the entire domain for all parameters to the values specified as the spatially constant IC's.

The user may later edit (*Enable Edit* must be checked) any of these IC fields using ViewPlan. To access the data the user should select *Water Column* as the main Viewing Option then *Water Quality* as the sub-option, then select the parameter desired.

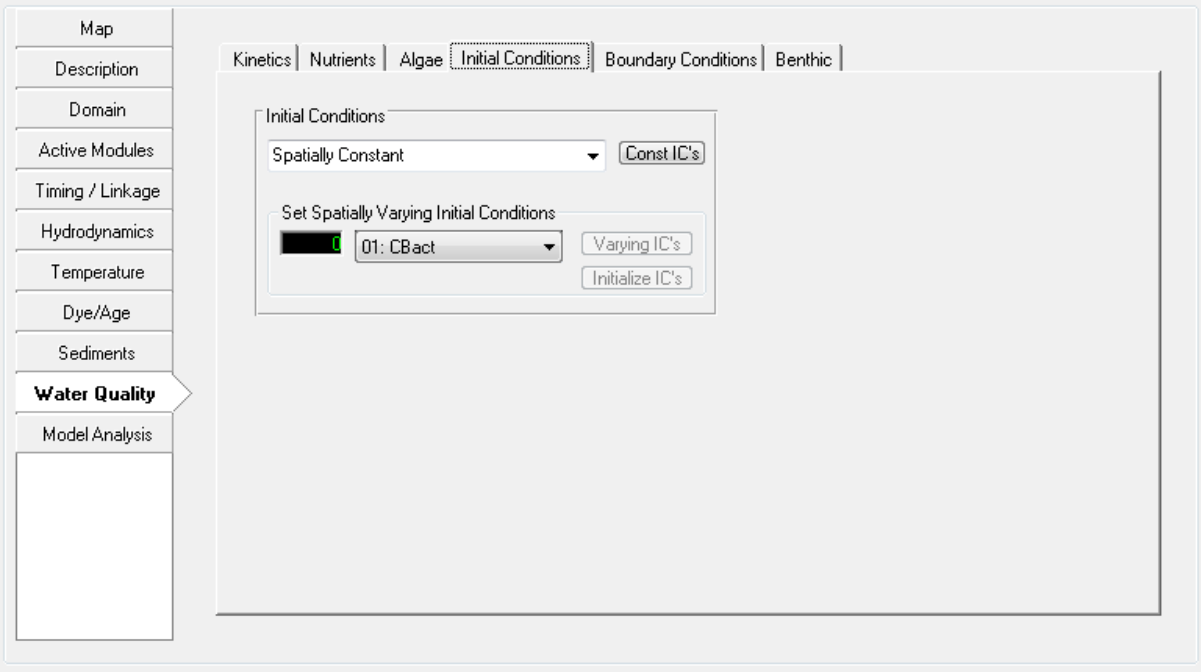


Figure 6-150 Water Quality Tab: Initial Conditions.

6.14.5. Boundary Conditions

EFDC_Explorer provides the ability to be able to define water quality boundary conditions with load concentrations rather than mass loading files. The tab that allows the user to select from various options for *Water Quality Point Source Loading Option* is shown in Figure 6-151.

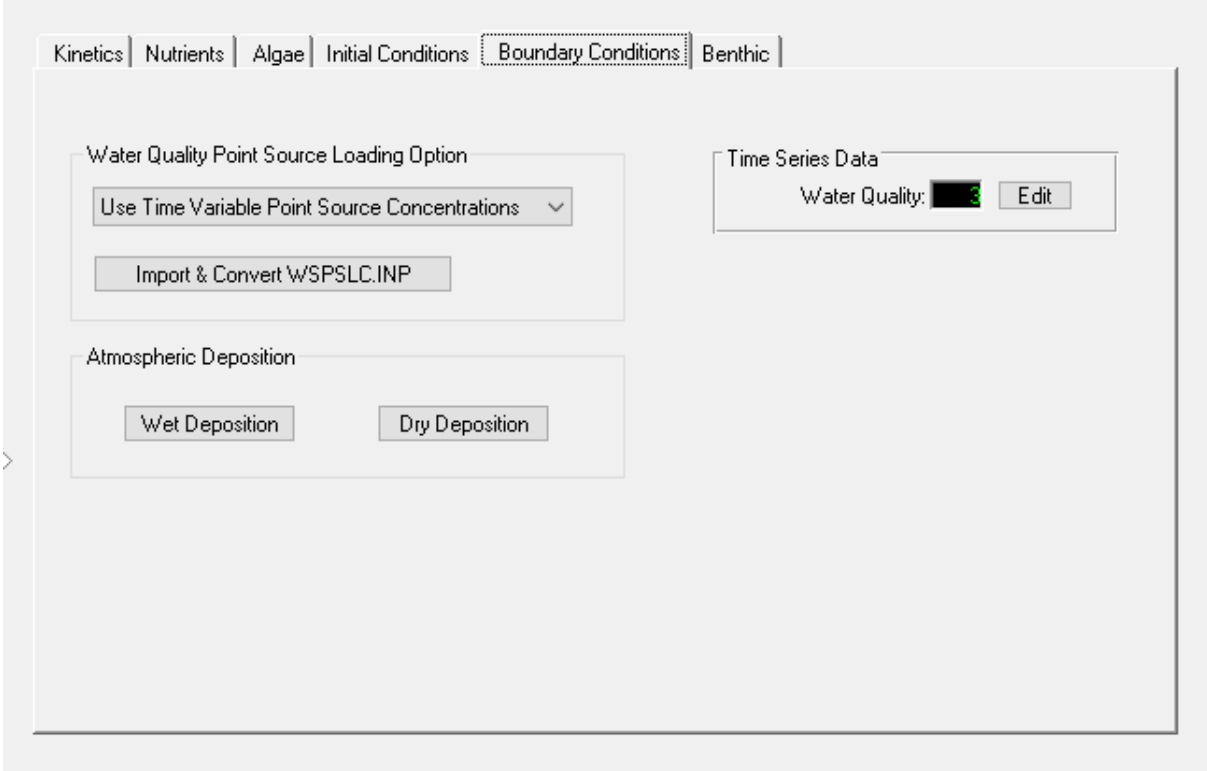


Figure 6-151 Water Quality Tab: Boundary Conditions.

The *Water Quality Point Source Loading Option* provides a drop-down table with the following three options: *Use Constant Point Source Loads, Use Time Variable Point Source MASS Loadings,* and *Use Time Variable Point Source Concentrations.*

EFDC_Explorer previously only used the larger mass loading files. The *Load Concentrations* option provides the user with smaller files and greater control. When the user selects an option different to that previously selected, EFDC_Explorer informs the user of the current BC and asks whether they want to switch to the new option. If the user responds affirmatively then EFDC_Explorer will convert and save out the option selected. With the aid of the *Import & Convert WSPSLC.INP* button the user can either overwrite or append a new water quality time series, which is a load concentration file, with concentration in kg per day. The WQ BC Load Concentrations rely on two new input files: WQPSLC.INP and WQPSL.INP. Note that WQPSL.INP is an EFDC_Explorer file, and is not used by EFDC. It is often a very large file.

When the user returns to edit the *Water Quality Tables and Series* in the *Boundary* tab EFDC_Explorer displays all 21 WQ parameters. In the *MASS Loadings* option these parameters are depth averaged and not layered, and so are vertically constant. However, after converting to concentration loadings, the user is informed that the values have been converted, and now are now stored layer by layer. Initially these values will be an average as calculated by EFDC_Explorer, but the user can now specify certain layers and assign new values as required.

6.14.6. Benthic Flux

Diagenesis/benthic flux settings are made in the *Benthic* tab shown in Figure 6-152. The nutrient benthic flux computational options are set using the *Modify Parameters* button in the *Benthic Flux* frame.

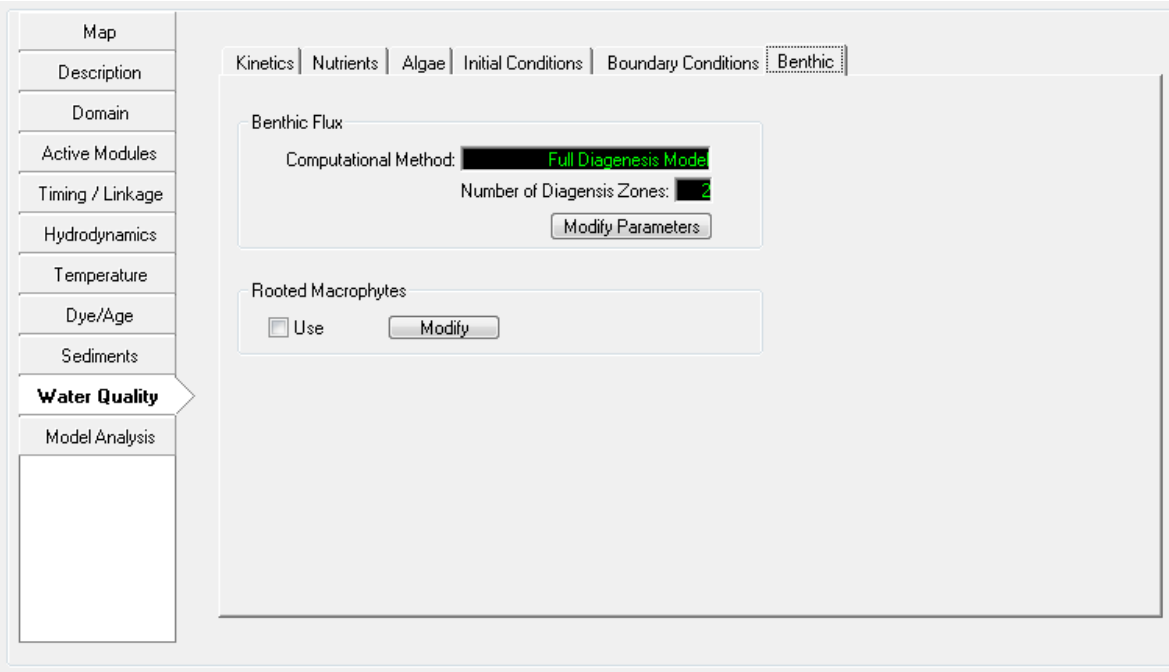


Figure 6-152 Water Quality Tab: Benthic.

Clicking the button brings up the form shown in Figure 6-153. The top frame, *Benthic Nutrient Flux Method* is the option that controls EFDC's benthic flux approach. If the nutrient mass fluxes are to be specified, either as constants or variable in time and space, then the diagenesis tabs are not used and consequently disabled. If the Full Diagenesis Model (DiToro & Fitzpatrick, 1993) option is selected then the three diagenesis tabs are available to allow the user to specify the diagenesis parameters.

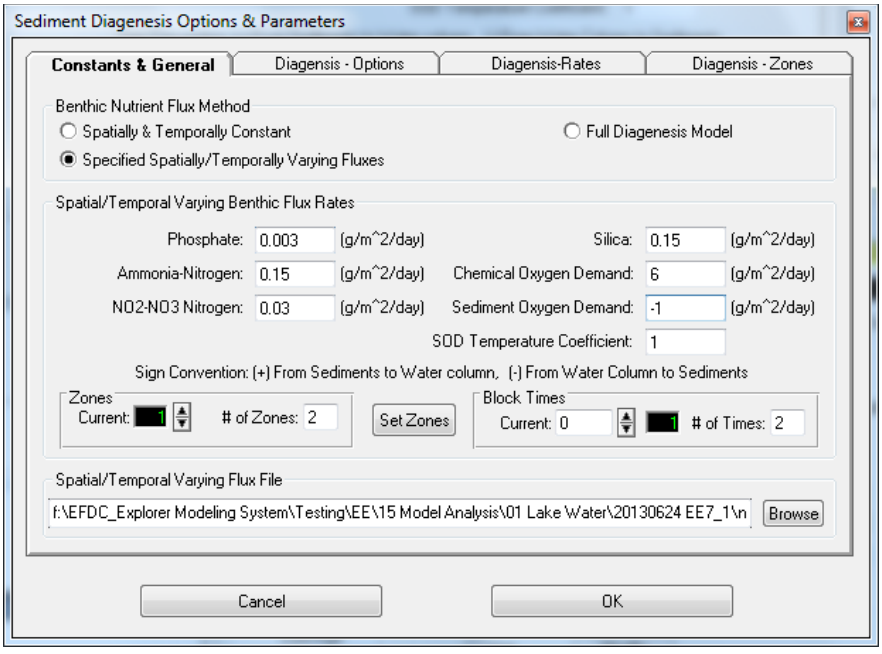


Figure 6-153 Sediment nutrient flux – Sediment Diagenesis Options and Parameters.

Figure 6-154 shows one of the diagenesis tabs for defining the diagenesis parameters. The best way to initialize all the diagenesis parameters is to use the *Load Settings* button. This will read an existing WQSD3D.INP file from another project and initialize the settings for the current project with those from the loaded project. After the user initializes the parameters they can modify the settings during the model development stages. The diagenesis values from the Chesapeake Bay ICM application (Cerco & Cole, 1995) provide a good starting point for diagenesis model parameterization.

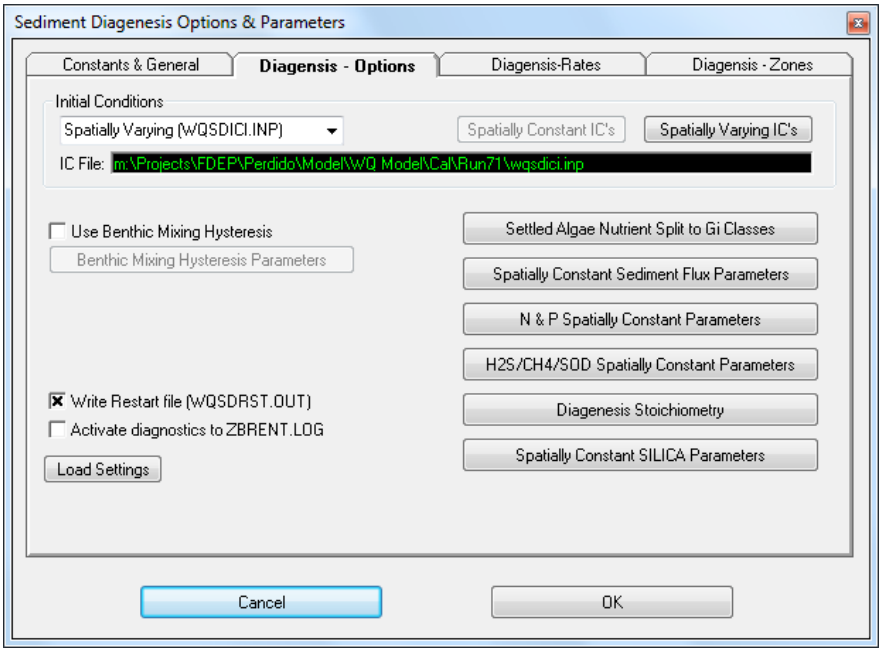


Figure 6-154 Sediment Diagenesis: Diagenesis – Options.

In addition to the diagenesis kinetic parameters, the sediment bed diagenesis initial concentrations also need to be specified. This can be accomplished by applying measured data for each parameter using the *Spatially Varying IC's* button. When pressed, the form shown in Figure 6-155 is displayed. For each of the required parameters, the user can interpolate the data onto the model. Each nutrient reaction class, i.e. G1 (most reactive), G2 and G3 (least reactive or inert) must be separately specified. The *Operator* can be used to factor or otherwise modify the interpolated field or a previously defined IC field. The data can be interpolated for the entire model in a single operation or using only selected cells by specifying a *Poly File*.

The user can initialize the spatial field to a set of constants using the *Initialize to Constants* button. It should be noted that the sediment diagenesis bed is not linked in any way to the sediment transport sub-model bed definition, scour or deposition processes.

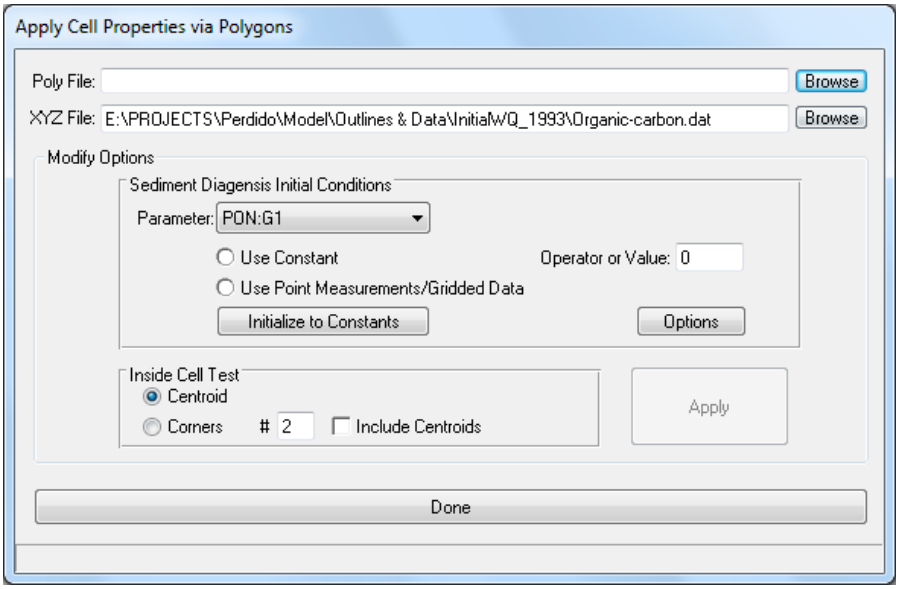


Figure 6-155 Sediment Diagenesis: setting the initial conditions.

The diagenesis kinetic parameters can be divided into zones. Figure 6-156 provides access to setting the zones and assigning the zone-specific parameters. The maximum number of zones needed is set by entering the appropriate number in the "# of Zones" field. To edit a specific zone, the user should scroll to the desired zone using the up/down arrow control.

The zones and the zone-specific parameters can be viewed and editing using the *ViewPlan* by selecting the View Option *Diagenesis*, then checking the *Show Zone* option. If the *Enable Edit* is checked, then RMC'ing on a cell allows the user to edit the cell's corresponding diagenesis zone's kinetic parameters. The cell's diagenesis zone assignments can be edited using the *Property Copy* function.

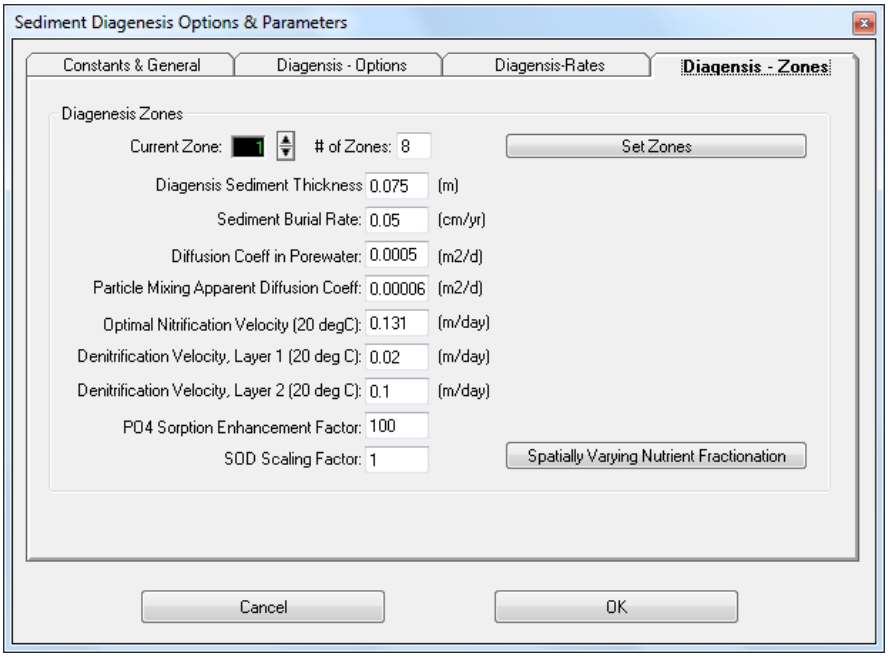


Figure 6-156 Sediment Diagenesis: Diagenesis kinetic zones.

6.14.6.1 *Rooted Macrophytes*

If the user wished to implement the *Rooted Plant and Epiphyte Model* (RPEM) the *Use* button should be selected. The *Modify* button in the *Rooted Macrophytes* frame allows the user to set various constants associated with RPEM as shown in Figure 6-157.

The *RPEM: General* tab allows the user to enable or disable a variety of combinations for RPEM, including enabling simulation of rooted plants or epiphytes; enabling epiphytes growing on rooted plants; enabling RPEM – Water Column Nutrient Interaction; enabling RPEM – Sediment Diagenesis Interaction.

For details on the theory behind this sub-model, please refer to "A Generic Rooted Aquatic Plant and Epiphyte Algae Sub-Model for EFDC" (Hamrick, 2006).

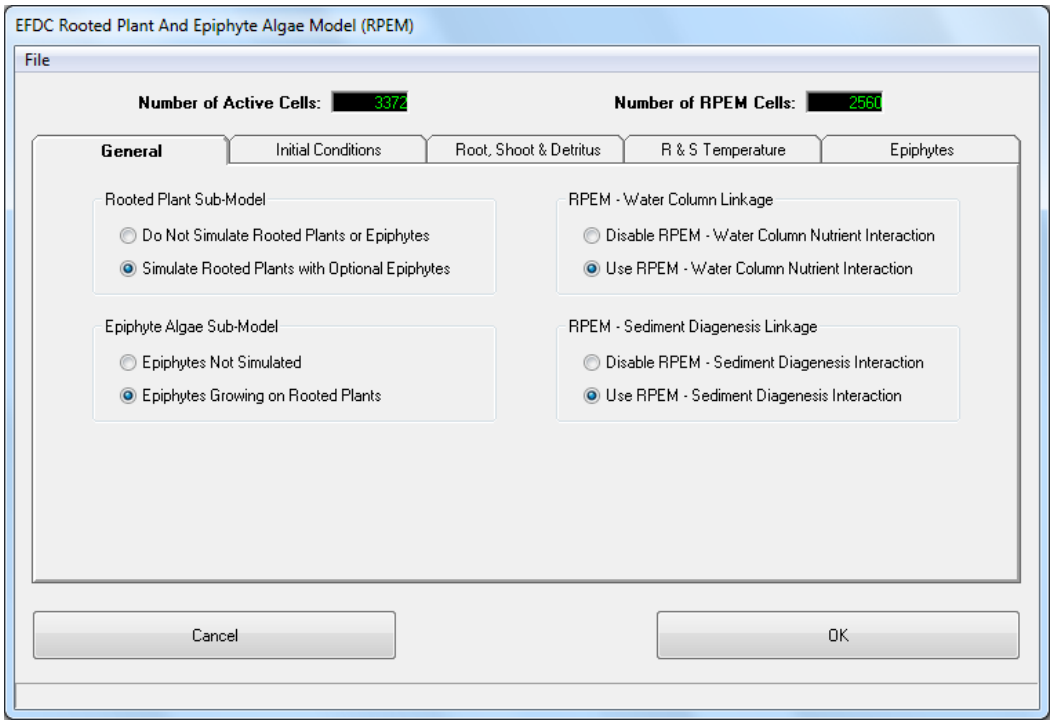


Figure 6-157 RPEM: General.

The *Initial Conditions* tab is shown Figure 6-158. This tab allows the user set various approaches for the RPEM such as: Constant IC's, Spatially varying IC's - WQRPEMSIC.INP, and Spatially varying IC's - WQRPEMRST.INP. These latter two options allow user to input variable carbon biomasses rather than setting them as constant values. Clicking on any of the buttons for setting shoot, root, epiphyte or detritus carbon ICs will bring up a form as shown in Figure 6-159. Using this form, the user may import a polyfile or datafile for the IC.

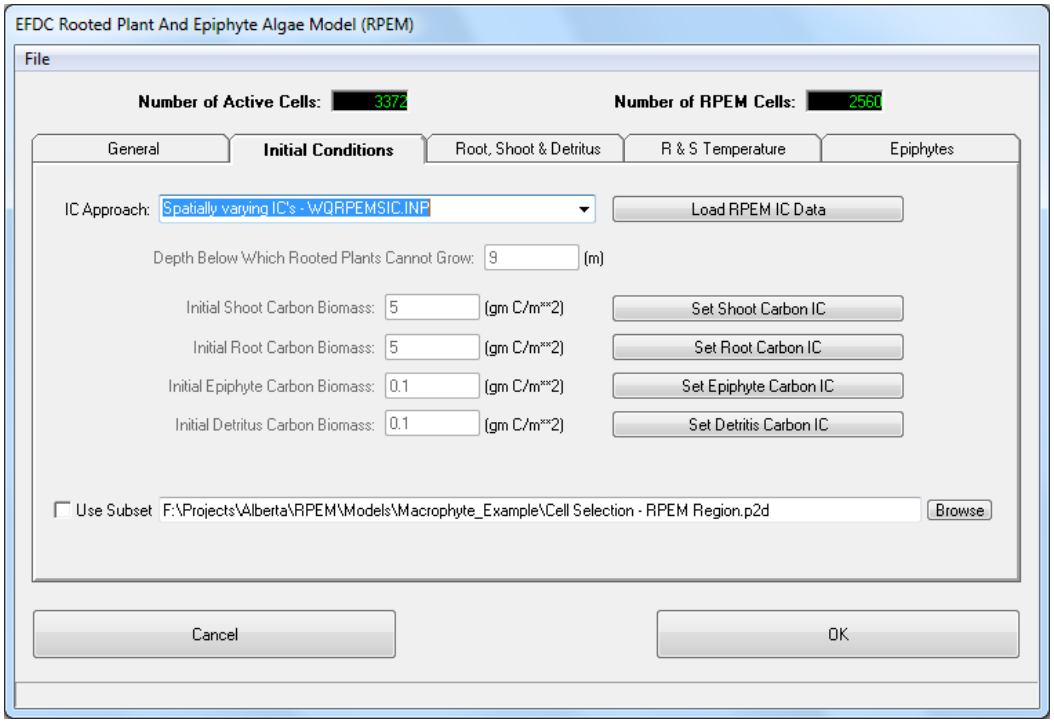


Figure 6-158 RPEM: Initial Conditions.

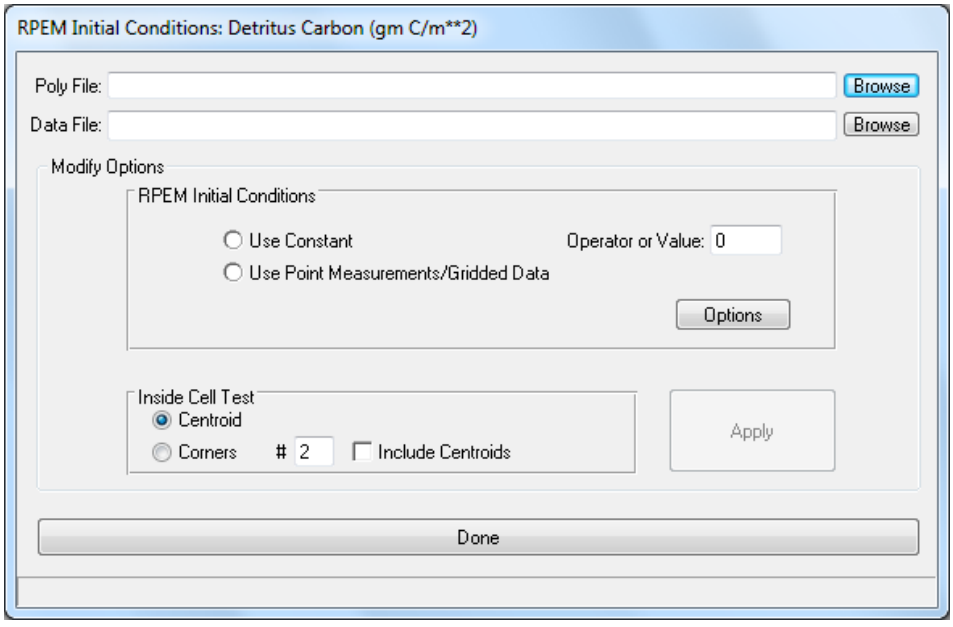


Figure 6-159 RPEM: Initial Conditions: Shoot Carbon

The *Root, Shoot & Detritus* tab sets growth and respiration rates amongst other parameters for shoots, roots and detritus as shown in Figure 6-160.

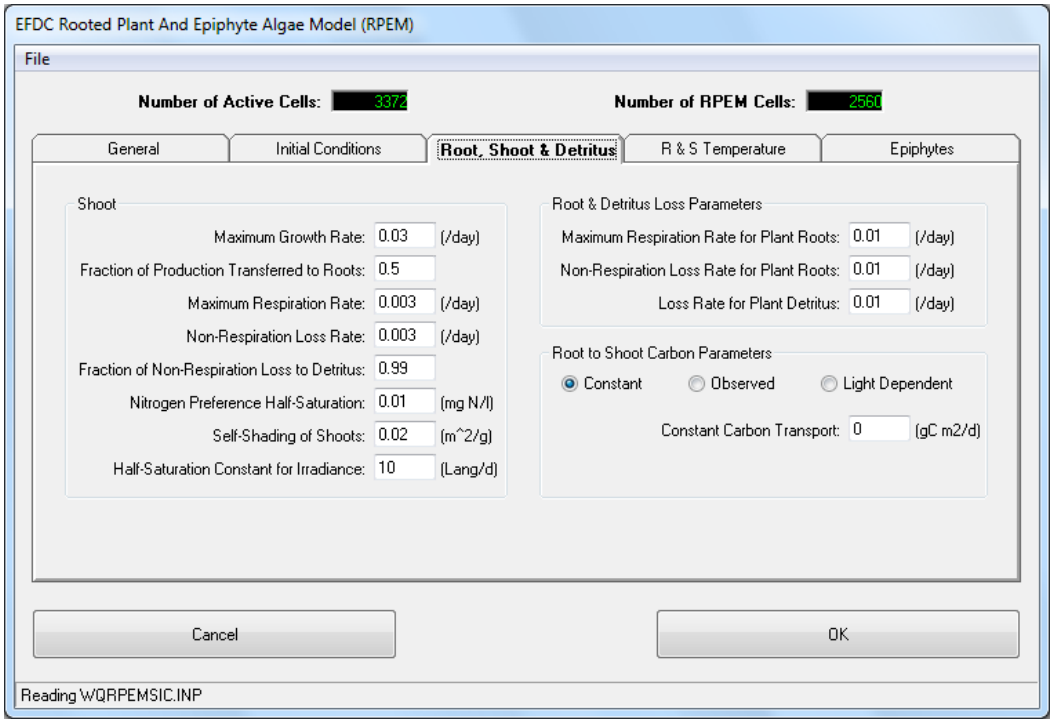


Figure 6-160 RPEM: Root, Shoot and Detritus.

Root and Shoot temperatures may be set in the *R & S Temperature* tab as shown in Figure 6-161.

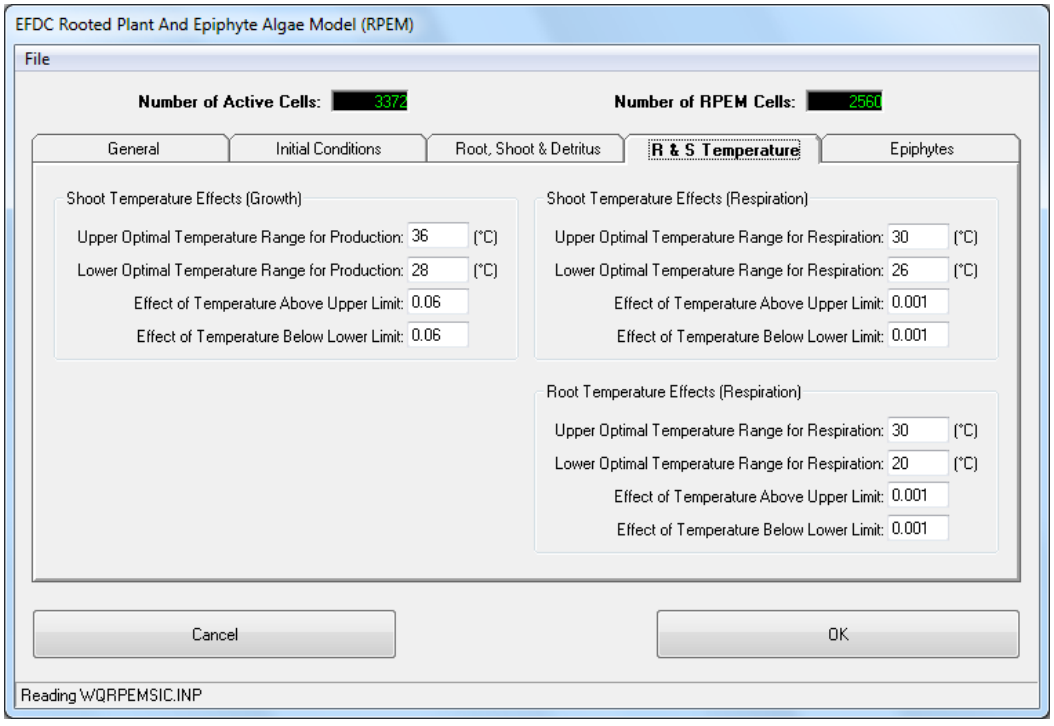


Figure 6-161 RPEM: Root &Shoot Temperature.

Constants for epiphyte growth respiration, light limitation, temperature effects for growth and respiration may be set in the *Epiphyte* tab as shown in Figure 6-162.

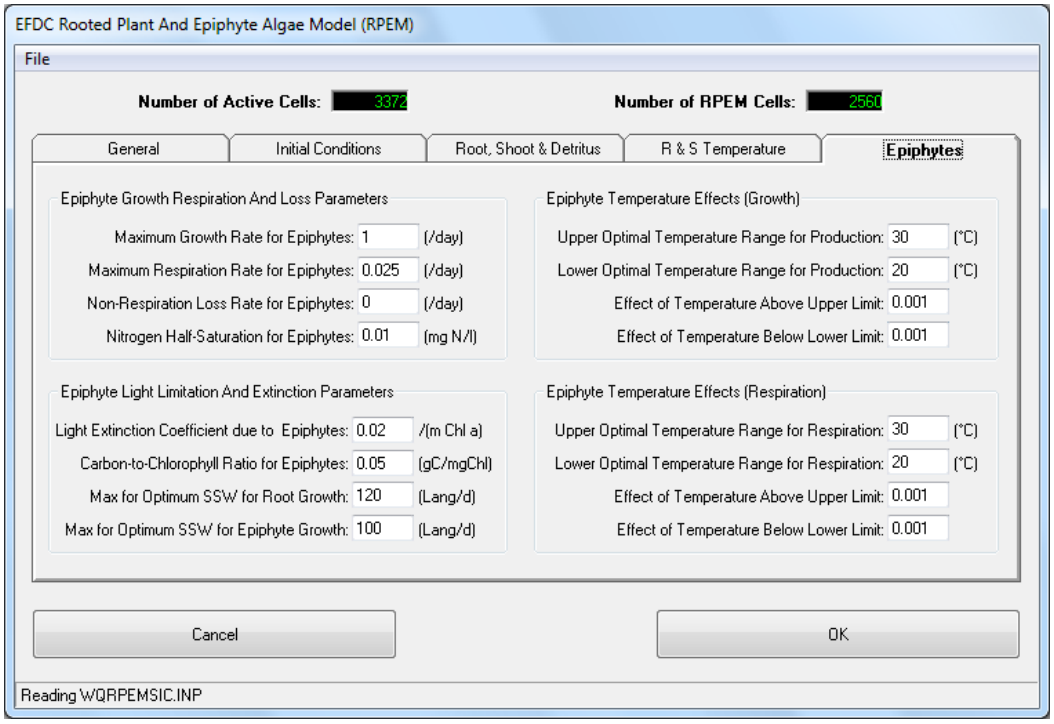


Figure 6-162 RPEM: Epiphytes.

6.15 Waves Tab

Bed shear stress associated with waves is an important parameter that contributes to sediment resuspension and transport process in coastal shallow areas and along shorelines. EFDC_Explorer has two options for incorporating waves in the flow model. These are internal wave model and external wave model as shown in Figure 6-163.

The internal wave model internally computes the wind-induced waves using wind data provided in the WSER.INP file. In contrast, the external wave model requires the output results from SWAN, REF/DIF, STWAVE and other wave models. EE will then generate the WAVE.INP and WAVETIME.INP files from output results to couple with EFDC model.

For all wave models, the user has the option of simulating radiation stress with the *Include Radiation Stress* check box as shown in Figure 6-163. Checking or unchecking this option will change the wave parameters required in the right-hand frame.

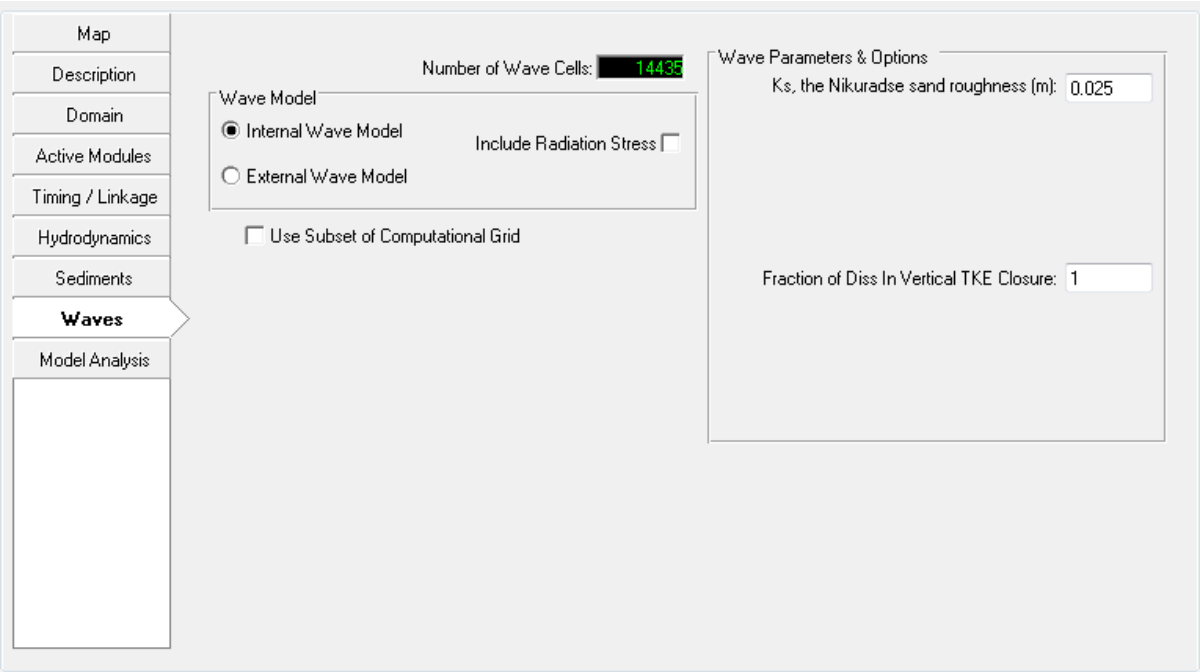


Figure 6-163 Waves Tab: Internal/External Linkage to SWAN Wave Model

6.15.1. Wave Models in EFDC

The wave model within EFDC uses a naming convention as follows:

- No Wave Effects (ISWAVE=0)
- External Linkage–Boundary Layer Only (ISWAVE=1, requires WAVEBL.INP and WV00N.INP, n=1, 2, 3… for version EE7.0, and uses a new format of WAVE.INP for EE7.1 and later. Refer to the EE7.2 guide and later for this format).
- External Linkage–Boundary Layer and Currents (ISWAVE=2, requires the new format of WAVE.INP, the same input file as for ISWAVE=1)
- Internally Generated Windwaves – Boundary Layer Only (ISWAVE=3) (EFDCPlus only)
- Internally Generated Wind-waves – Boundary Layer and Currents (ISWAVE=4) (EFDCPlus only)

6.15.2. Internal Wave Model Option

In general, the influence of wind on the flow velocity field is important while studying hydrodynamics and sediment transport in lakes, estuarine and coastal areas. Wind effects not only induce the flow current, but also generate surface waves with a wave height up to several meters. To calculate the total bed shear stress in such areas, the model must take the wave factor into account. The wave parameters such as wave height, wave direction and wave period are calculated by the SMB (Sverdrup, Munk and Bretschneider, see Zhen-Gang Ji, 2008) model. The wave direction is the same as the wind direction. This means that the effects of refraction, diffraction and reflection are not taken into account in this internal wave model.

The internal wave model option doesn't require imported external wave to simulate wind generated wave effects on bed shear stress and wave induced currents (Dang Huu Chung and Craig, P.M. 2009). The fetch for each cell by wind sector may be viewed in *ViewPlan* (see Section 8.6, *ViewPlan Main Viewing Options*).

The *Wave Parameter & Options* form allows the user to specify Ks, the Nikuradse sand roughness value as shown in Figure 6-164. This can be estimated as Ks = 2.5 x d50. The Nikuradse roughness is not the same as the hydrodynamic roughness (i.e. bottom roughness, Z0) used by EFDC to solve the hydrodynamic equations. The Nikuradse roughness is a grain roughness and represents more of a local scale phenomenon.

For the cases of ISWAVE=3 and ISWAVE=4, available only in EFDCPlus, the wind time series provided in the WSER.INP file is used to compute the instantaneous values of wave parameters with fetch calculated for each cell in sixteen directions. The effect of shoreline and EFDC internal masks are included in the fetch calculations. The resulting wave parameters are then used to calculate total bed shear stress, with bed shear stress linked to the current generated shear stress via the Grant Madsen approach.

From EE5 there has been the ability to internally generate wind-induced wave for bed shears only (ISWAVE=3). This can also include the radiation stresses for the whole water column (ISWAVE=4). These options allow the simulation of wave effects and re-suspension of sediments inside EE.

6.15.3. External Wave Model Option

This external wave model option allows the user to import wave parameter fields from other common wave models. The Steady/Unsteady option corresponds to the types of waves being imported into the EFDC model. The steady wave option means that the waves are not changing with time so the EE will not read the WAVETIME.INP file. The unsteady wave option does require the WAVETIME.INP input file. The setting in the frame reflects the way the user has imported the external waves.

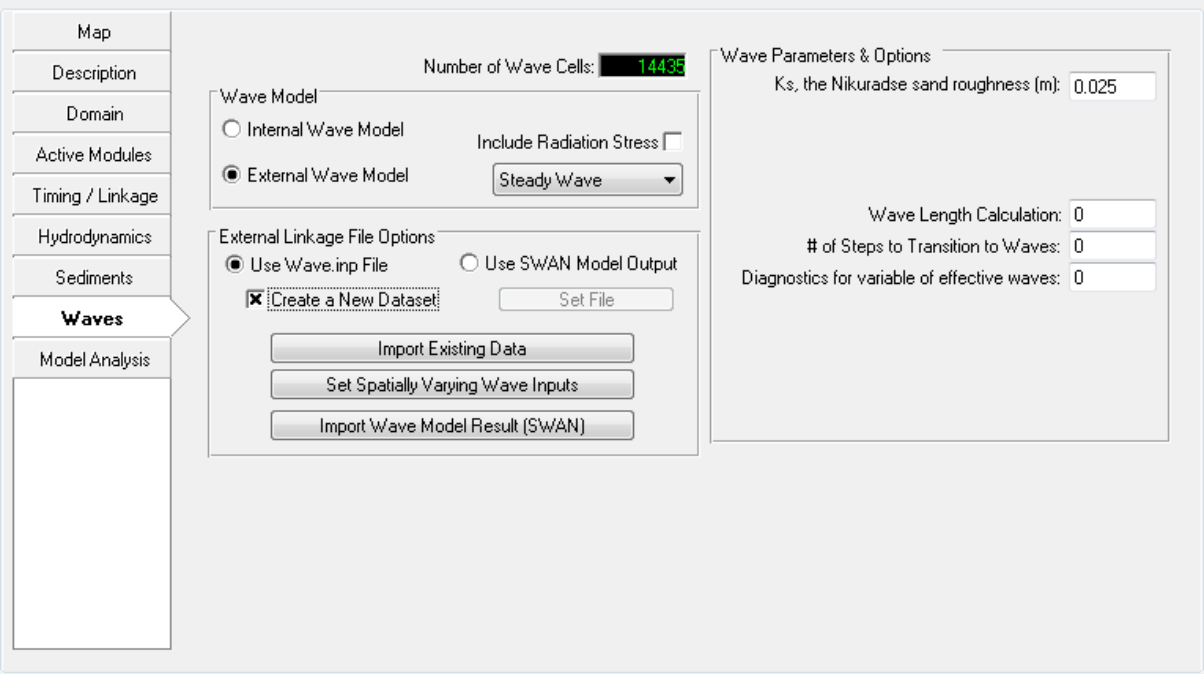


Figure 6-164 Waves Tab: External Linkage to SWAN Wave Model.

There are four different sub-options to consider while assigning the wave parameters into EFDC model as following;

a) Use Wave Input File

If the user already has a WAVE.INP file (of format shown in Appendix B 12), then the data is imported first time by checking the *New Dataset* checkbox. If a WAVE.INP has already been imported then the import options described below are greyed out until the *New dataset* box is selected.

b) Import Existing Data

EE imports an available WAVE.INP file from another project into the current project. It should be noted that two projects must have exactly the same grids.

c) Set Spatially Varying Wave Inputs

This is an option for EE7 and earlier and it is not advised be used due to the longer time required to prepare the input data. For ISWAVE=1 and ISWAVE=2 the external model results may be imported into EFDC_Explorer which will generate the required wave linkage file, depending on the ISWAVE option. Figure 6-165 shows the main import/field interpolation form for the wave parameters for ISWAVE=2. The user must match the input data file (which should be in XYZ format tab, space or comma delimited) to the parameter specified on drop down list *Wave Field Parameters* (options shown in adjacent inset). The user can either have wave height (2*wave amplitude) or wave energy. EE will compute the one from the other. The user has the option of using a polygon to select which EFDC cells will be used for the assignment. If a Poly file is not selected, then the assignment operation will be for the entire model domain. EE interpolates and converts the wave model results into the formats needed for EFDC. The interpolation process has two options, nearest neighbor interpolation or cell averaging. Cell averaging should be used when the imported data is denser than the EFDC model grid (this will usually be the case). The nearest neighbor interpolation scheme should be used if the imported data is sparser than the EFDC model grid.

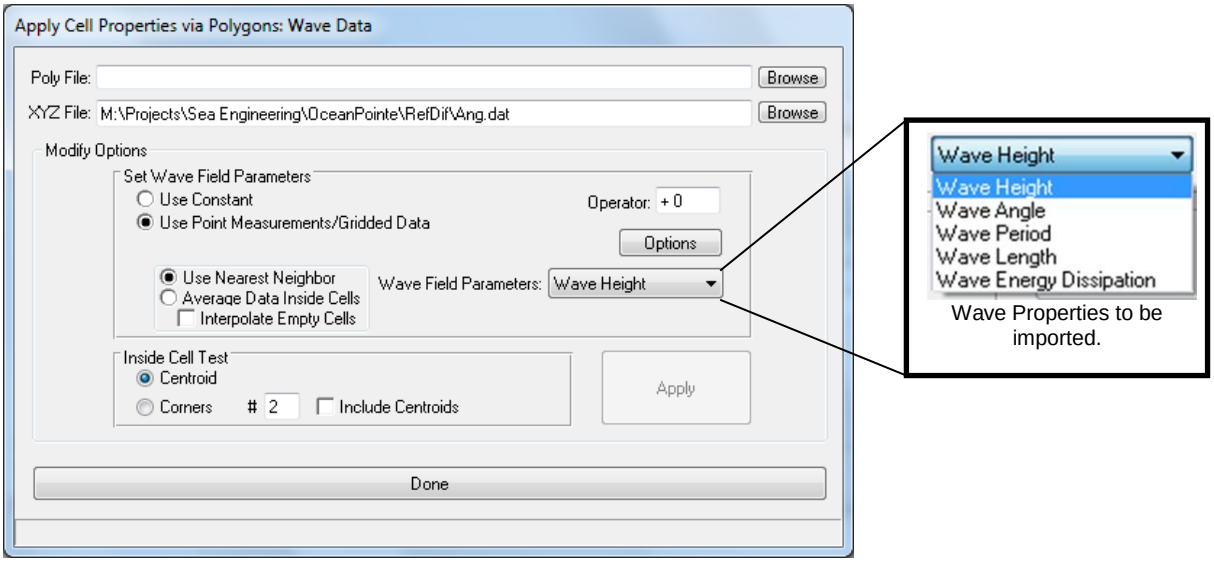


Figure 6-165 Wave generated turbulence, import data form.

DSI has extensively tested and verified the ISWAVE=2 option for both EE and EFDC using the REF/DIF / ShoreCirc modeling of rip tide currents (Svendsen, et. al., 2000). Figure 6-166 shows EFDCPlus model results for the rip tide test case. The velocity vectors (in white) have been overlaid on the bathymetry. The velocity pattern and magnitude are similar to what was computed by Svendsen.

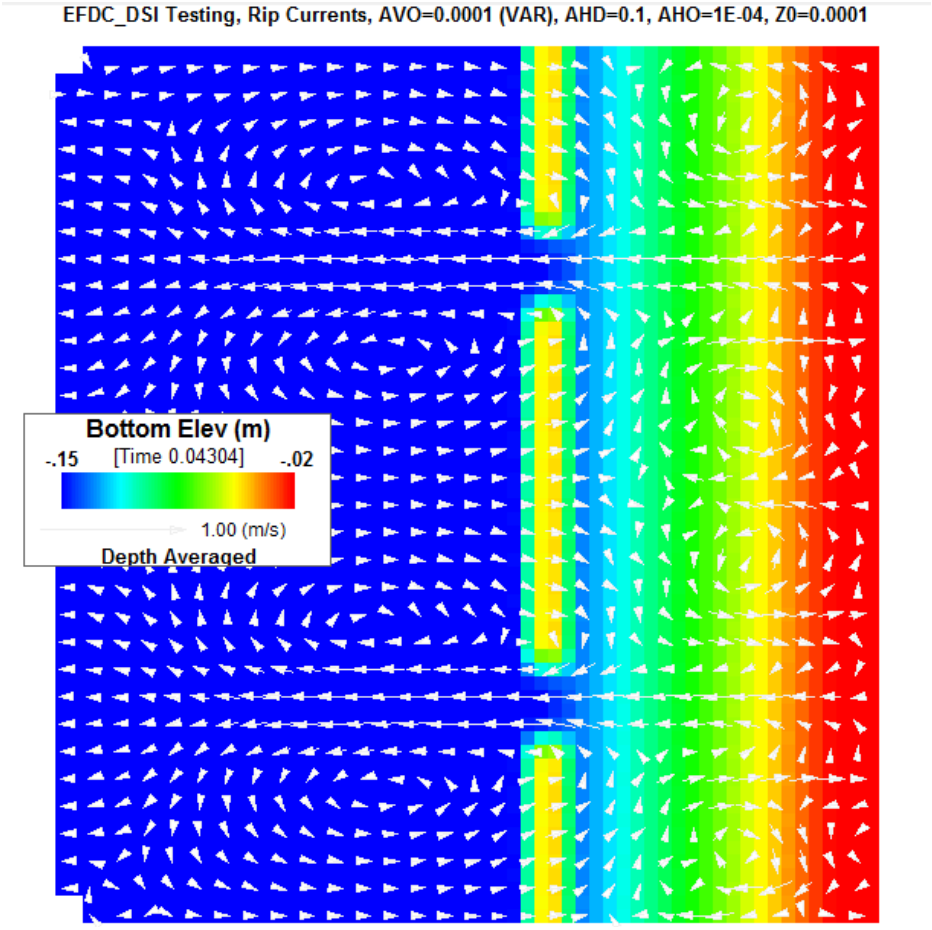


Figure 6-166 Wave generated rip tide currents overlaid on idealized beach.

d) Import Wave Model Results from SWAN

Currently EE only imports data from the SWAN wave model, however, it is anticipated that other models will be included at a later date. If the user chooses to import the SWAN output files through EE then the *Import Wave Model (SWAN)* button should be selected. This displays the form shown in Figure 6-167. The default *Work Path* is the current model directory. The user should browse to a different directory if they want to avoid saving over an existing WAVE.INP file.

In the *Import from SWAN* frame the user should first decide if they are using steady or unsteady waves with the dropdown box. If steady waves are used then then EE does not require the WAVETIME.INP file as the waves are at regular time intervals and the option to load this file is disabled. Alternatively, if unsteady waves are being used then then browse to the path for the SWAN model outputs and select the the wave time file (WAVETIME.INP). Next there are two options for SWAN input:

1. *Using output for locations* option requires the SWAN model to have same grid as the current EE model and uses the group file from SWAN, (SWAN_GRP.INP).

2. *Using output for location* option uses the data as x, y points. This option should be chosen when SWAN and EE use two different grids and wave data was exported at the locations (x, y) from the SWAN model. The latter option requires two input files: a location file and a table file. These should have been defined and saved out from the SWAN model.

Once these files have been selected the user should select the *Import* button. Two files will be created: WAVE.INP and WAVETIME.INP for the EFDC model run.

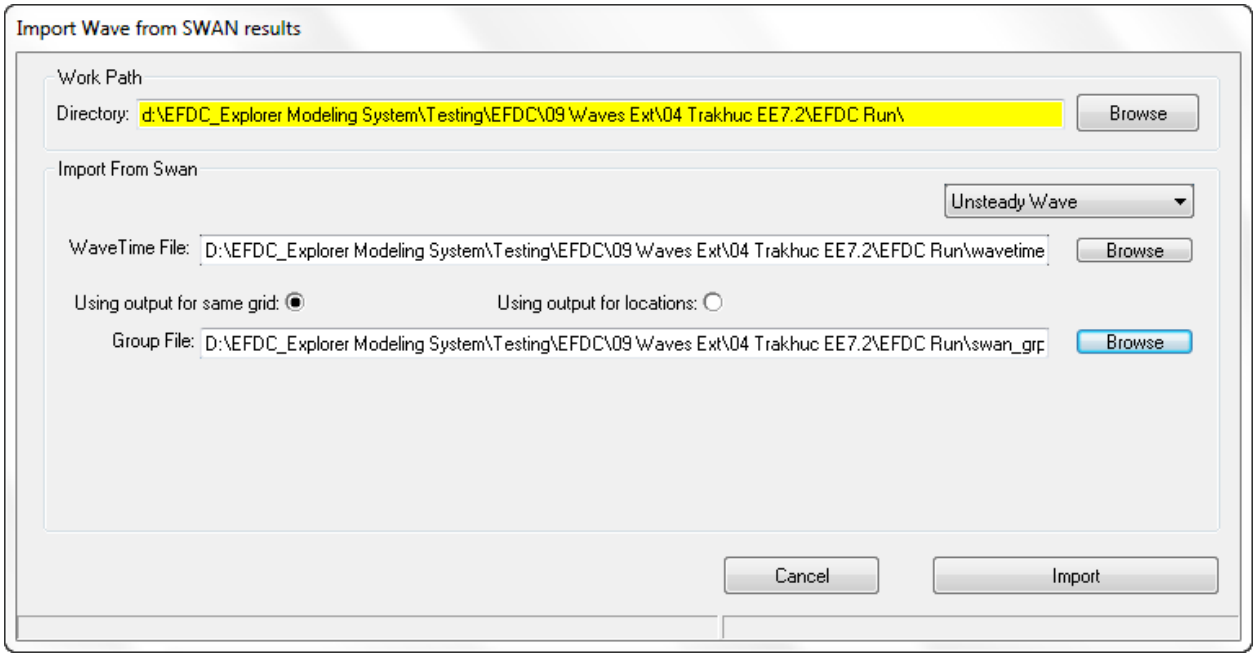


Figure 6-167 Waves Tab: SWAN Import function when using same grid as EE.

6.15.4. Creating SWAN Model Output – Running SWAN from EE

EE allows the user to build input files for SWAN and then run SWAN directly from within EE. Note that this is not a fully integrated dynamic coupling between SWAN and EFDC at this stage. Instead, EFDC will use the final SWAN output as input for the external wave model.

The steps to run SWAN from EE are as follows: First it is necessary to create a SWAN input file. This can be done from *Wave tab* | *SWAN Linkage* | *SWAN Input Settings* as shown in Figure 6-168, or from *ViewPlan* under the *Export Data* dropdown button, and selecting *Export SWAN* as shown in Figure 6-169. After selecting this option, the *Export to SWAN file* from is displayed as shown in Figure 6-170.

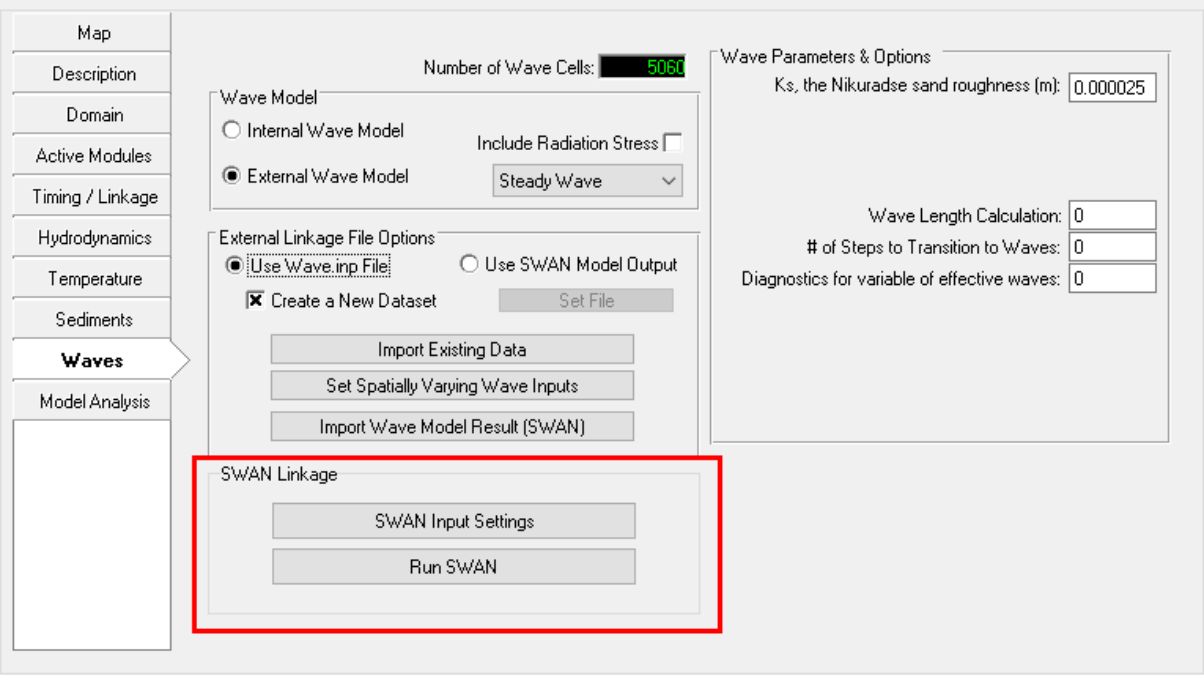


Figure 6-168 External Waves: SWAN Linkage Options.

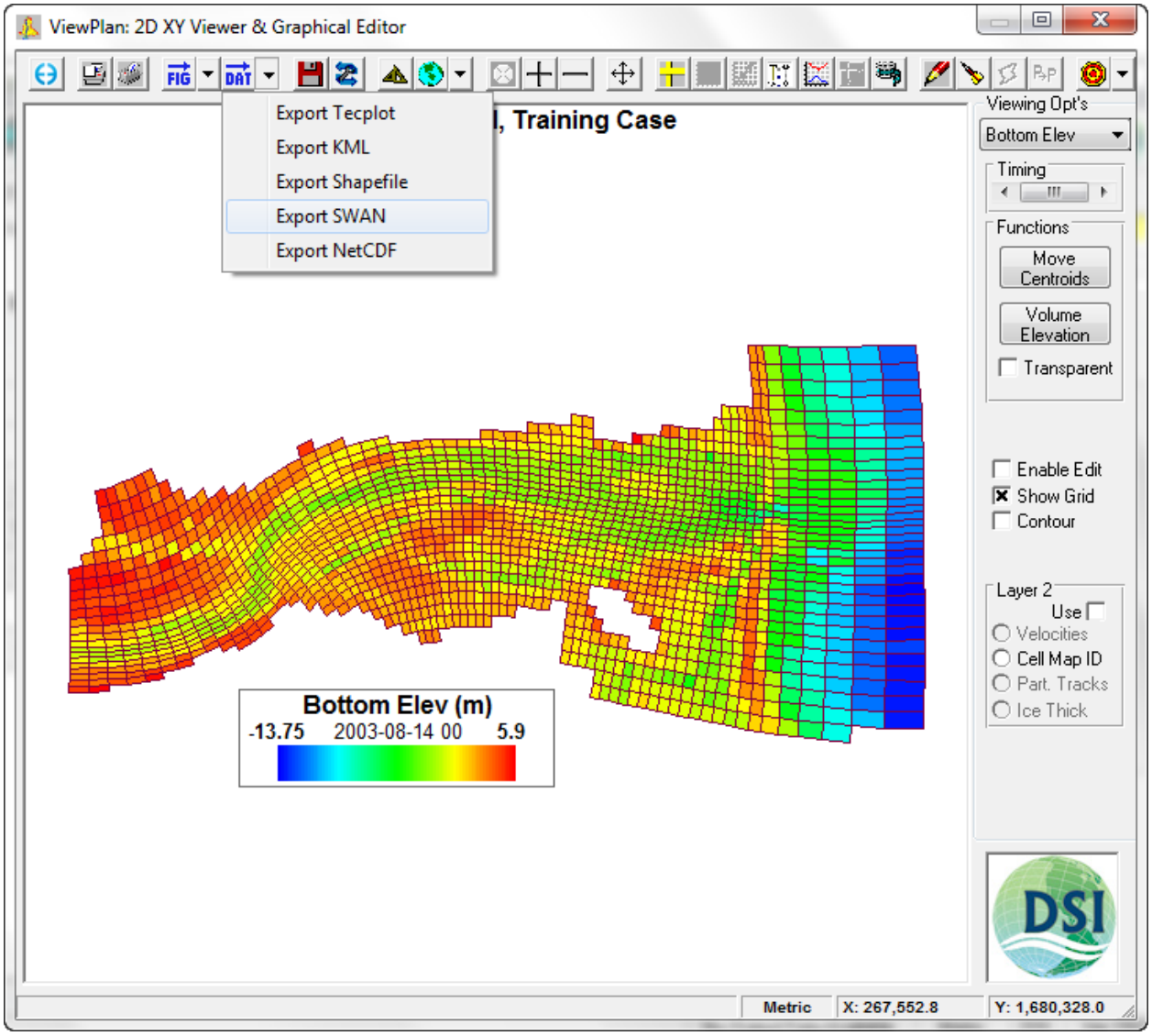


Figure 6-169 External Waves: ViewPlan generate SWAN input files (Export SWAN).

The user should select which input data is required for their SWAN model. The *Parameters* button provides further options for the creation of the SWAN inputs. The SWAN user guide should be consulted for details on how to select these parameters (http://www.swan.tudelft.nl/). If the user also intends to export the wind series for use in SWAN, the start and end times should be selected as well as the time step for this data.

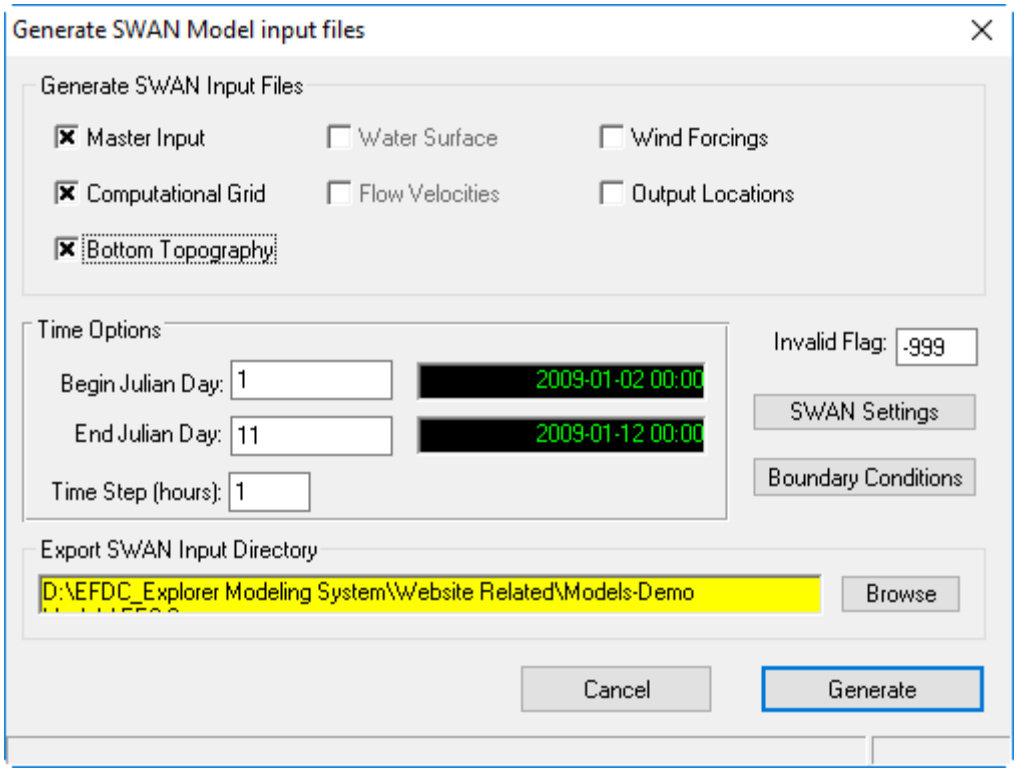


Figure 6-170 External Waves: Generate SWAN Input Files.

EE also facilitates setting the offshore boundary conditions for SWAN with the *Boundary Conditions* button. Selecting this opens the form shown in Figure 4. Beside the local wind-generated waves which are computed in the model domain by providing wind data for each model grid cell and turning on wind growth in SWAN, the user can user can also consider the influence of external waves (i.e., swells) acting on the model domain. EE detects the open boundary locations and lets the user edit each boundary condition for SWAN model. There are three options for each boundary condition group:

- No Wave, i.e. no external wave actions are considered at the selected boundary group.
- Constant: constant wave parameters in time including waves wave height, wave period, wave direction, and directional spreading are provided for the selected boundary group.
- Varying Time: wave parameters for the selected boundary group are varying in time and are provided in files for at least two points lying on the boundary section. The distance indicates the location along the boundary of the data point with respect to the first grid cell (I1,J1) of the boundary group. The total length of the boundary is provided in the text box *Length (m)*. The data file format is the TPAR file described in SWAN User's Manual. Wave parameters at every grid cell along the boundary group will be interpolated by SWAN based on the given data files.

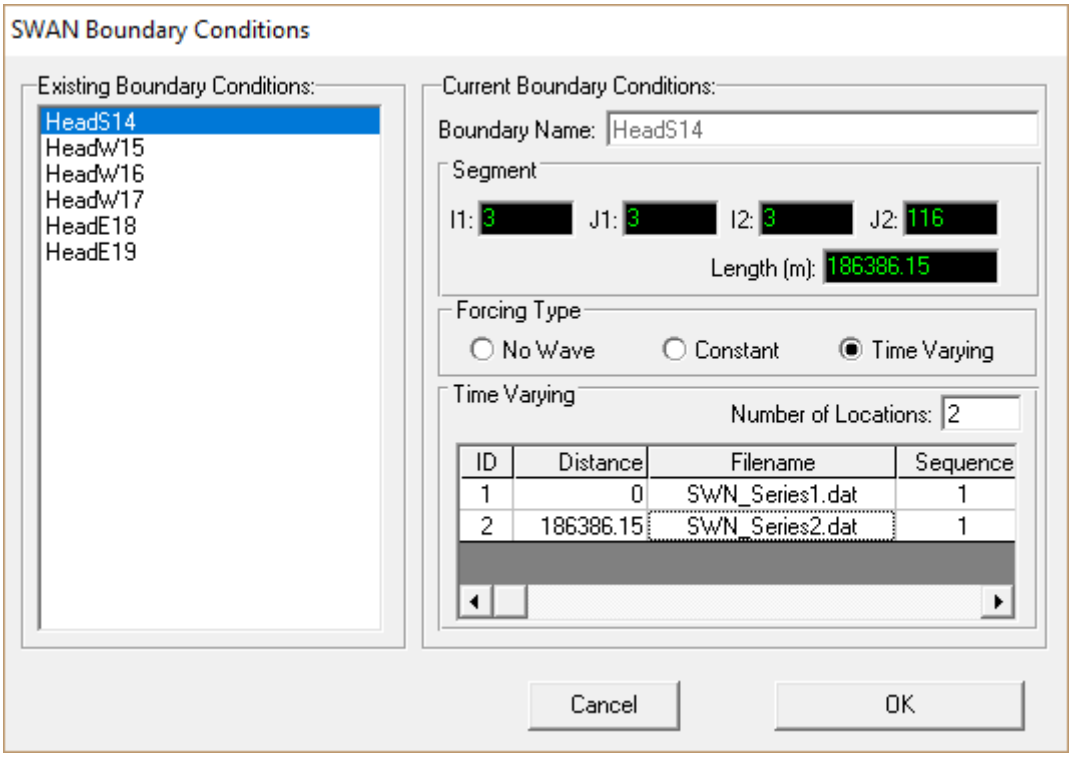


Figure 6-171 External Waves: SWAN Boundary Conditions.

The user should also ensure they have selected the correct path to generate the SWAN input files. If the path is incorrect, use the *Browse* button in the *Export SWAN Input Directory* frame (Figure 6-170) to navigate to the folder that you want to create the SWAN input files. Finally select *Generate* to create the SWAN input files.

Once the SWAN input files have been created the user should proceed to the *Waves* tab on the main EE form. Whenever *External Wave Model* option has been selected the *Run SWAN* button is displayed. As explained earlier, this provides the option of running SWAN directly from within EE (though not at the same time as EFDC is running i.e. it is not dynamically linked).

Selecting the *Run SWAN* button displays the form shown in Figure 6-172. The user should browse to the work path for the SWAN input files if necessary. They should then browse to the SWAN executable file. The SWAN exe is installed as part of the installation package for EE. SWAN is freeware under the GNU license and the executable and source code may be downloaded from the University of Technology, Delft http://www.swan.tudelft.nl/

The user should the select the *Run SWAN* button to run the SWAN model. After the SWAN model has finished running the user should select the *Create a New Data Set* checkbox and use the *Import the Wave Model Results* feature to import SWAN output into EE.

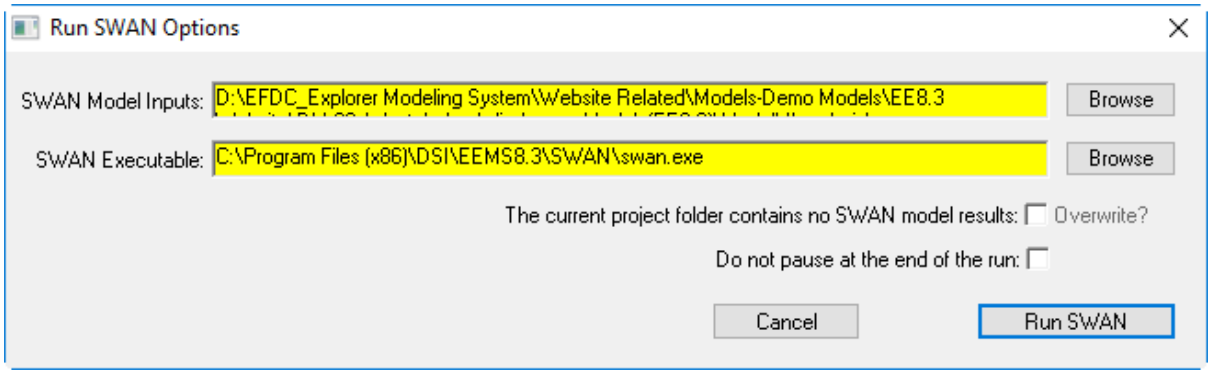


Figure 6-172 External Waves: Run SWAN internally from EE.

6.15.5. Use SWAN Model Output

The user may import SWAN model outputs directly into EFDC (Figure 6-173). This requires the SWAN user to select either a location file (table) or for a grid file (group) which can be exported from SWAN. Once again, the user should know if they are importing steady waves or unsteady waves and select the appropriate option from the dropdown box. The output files names from SWAN should be saved as:

SWAN Location file: SWAN_LOC.INP & SWAN_GRP.INP or

SWAN Table file: SWAN_TBL.INP

The radial buttons in form *USE SWAN Model Output* allow the user to select which type of input file is being input. The EFDC.INP is then updated to tell EFDC which will then look for the appropriate files in the root level of the project directory.

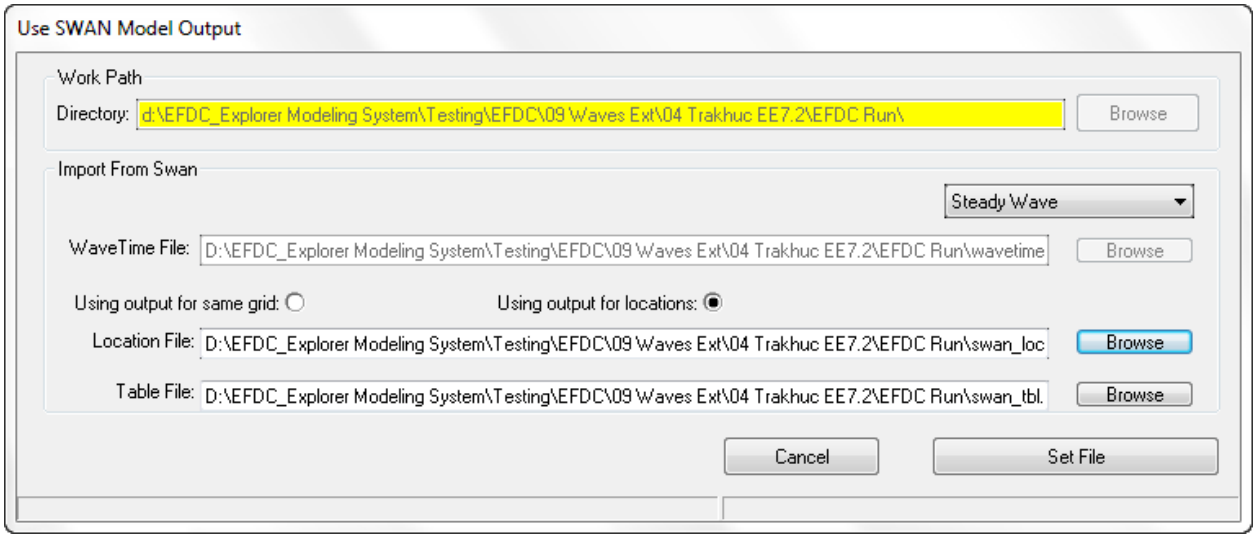


Figure 6-173 Waves Tab: Use SWAN Model output.

6.16 Lagrangian Particle Transport (LPT)

EFDC_Explorer has been updated to incorporate the pre- and post-processing of LPT's (Dang Huu Chung and P. M. Craig. 2009). Use of LPT's can be helpful when modeling oil spill tracks, emergency response, water quality applications and plume tracking. The *Lagrangian Particle Tracking* tab is shown in Figure 6-174.

Figure 6-174 Lagrangian Particle Tracking Tab.

The EE pre-processing provides full control for initial particle seeding, LPT computational option selection and plotting. The EE post-processing allows for a range of display options for the tracks, animations to the screen and or AVI files, and the ability to export any or all the particle tracks to ASCII files.

The LPT sub-model has been implemented with the following major options:

- Particles are free to move in full 3D,
- Particles can be fixed at a user specified depth, and
- A random walk component can be added to either of the two options above.

The numerical solution was separately divided into the advective transport and random components. This approach allows the user to enable (i.e. turn on random walk) or disable (advective transport only) the random components for either the horizontal and/or the vertical directions.

The method used for the solution of the differential equations is the Runge-Kutta 4 method: This method has the approximation of $O(\Delta t4)$. It has been determined that the Runge-Kutta 4 method is preferred due to its higher level of numerical accuracy. It has been shown that the computational burden of the Runge-Kutta 4 method is not significant within the overall model run times compared to other methods tested.

Within the *Lagrangian Particle Tracking Options/Drifters* frame (Figure 6-174) is reported the number of drifters set, the time for release of the drifters and the end time for the observation

of the drifters. In order to set the values related to the drifters the user should select the *Options/Particle Seeding* button. This displays the various options for setting the drifters as described below.

6.16.1. LPT Main Options Tab

Figure 6-175 shows the *LPT Main Options* tab. Within the *LPT Computational Method and Timing* frame the user must switch on drifters with the *Compute Drifters* check box before LPT can be implemented. Here the user will set the global computational period starting and ending days for the particle tracking in Julian days for all of the groups. The frequency (in minutes) of writing the particle positions to the EE_Drifters file is specified here. This frequency of output is for all groups and is independent of the normal EE output frequency.

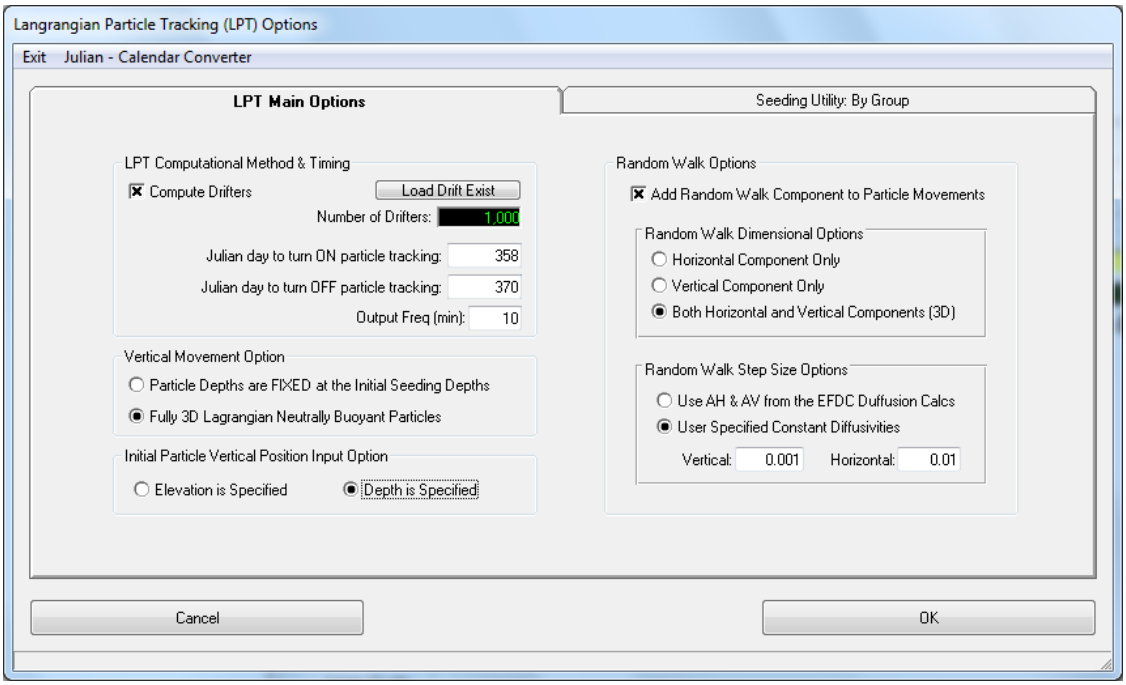


Figure 6-175 LPT Options: Main Options.

The *Vertical Movement Option* frame provides the user the option of fixing the temporal position of the particles at their initial seeding depths. If this is unchecked the particles will drift vertically as *Fully 3D Lagrangian Neutrally Buoyant Particles*. A sub-option is *Adjust Particle Depth.* This is the default option and adjusts particle depth with change in water depth. In some cases the user may wish to disallow this so that particles are not adjusted by change in water depth, and so can turn off this adjustment by unchecking this box. If the *Adjust Particle Depth* is selected, the amount of adjustment in the vertical direction depends on the ratio of particle depth and water depth. The formula of the vertical adjustment is:

$$dzp = pdep/wdep*dH$$

In which dzp is the adjustment of vertical particle depth, pdep is particle depth, wdep is water depth and dH the discrepancy of water depth of the current and previous timestep.

Within the *Initial Particle Vertical Position Input Option* the user may specify whether the vertical positions provided are elevations or depths.

If the *Random Walk* checkbox is checked the user has a number of options for the *Random Walk Dimensional Options*. The user may select the random walk in only the vertical direction, in only the horizontal direction, or in both vertical and horizontal directions at the same time.

In addition, the user may specify the *Random Walk Size Options* by selecting to allow EFDC Diffusion calculations to be used for the AH and AV. Alternatively there is the option for *Specified Constant Diffusivities* that should be inputted by the user.

6.16.2. Seeding Utility: by Group - Tab

Figure 6-176 shows the form with the various options for creating, selecting and deleting LPT groups. Any number of groups are allowed with each group having its own timing, number of particles and particle behavior. The number of particles can be added or removed from each group using the form.

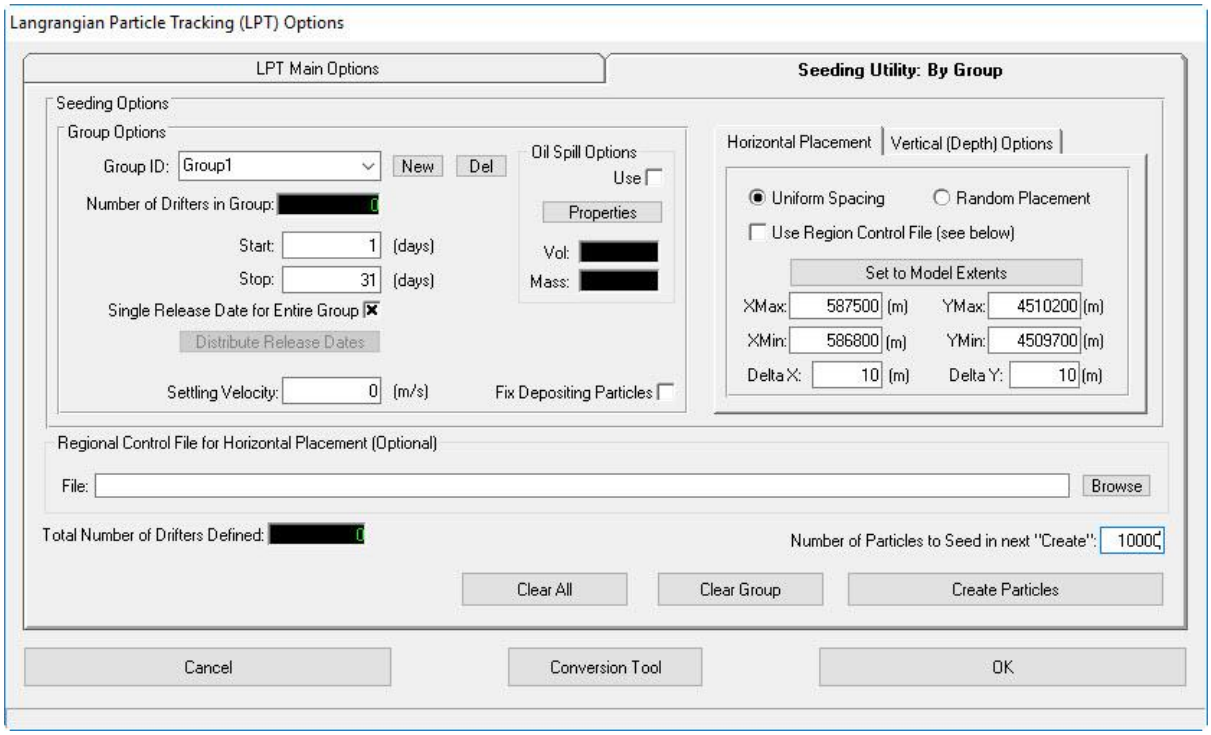


Figure 6-176 LPT Options: Seeding Utility: By Group

The *Group Options* frame allows the user to define a series of groups of drifters with a user defined ID, set with the *New* button. For each group the release and end days, as well as settling velocity, may be defined by the user. The settling velocity is always downward. When the *Clear Group* button is pressed all the user defined particle groups will be erased. The *Clear All* button will remove all the seeds for all the groups.

When the seeding options have been selected by the user, the *Create* button should be pressed which will save the selection. Another drifter may then be defined and *Create* selected until all the drifters have been defined.

The user may view the location of the drifters by selecting *OK* and then selecting *Particle Tracking* in *ViewPlan* editor. Selecting *Show IP* will display the initial position of the particle seeds. The user may create or delete drifters by right clicking on the grid.

The timing of the particle computations are set by the Julian dates shown in the *Start* and *Stop* fields for each group. EFDC will only conduct Lagrangian particle calculations for the particles in this group within the date range. The computations are also controlled by the global start and stop dates shown on the *LPT Main Options* tab.

The *Fix Depositing Particles* check box is provided for cases where the user wishes to cause particles to remain deposited on the bed after coming into contact with the bed. The default option is for this to be unchecked and particles are free to move after touching the bed.

6.16.2.1 *Horizontal Seeding Options*

The *Seeding Options* frame allows the user to select either:

(1) Uniform Spacing

This option places the initial position of the particles within a user specified region defined by the *XMin, XMax, YMin* and *YMax* at user specified intervals using the *Delta X* and *Delta Y* parameters. For example, if the user specified a region of 586800 by 587500 in the X and 4509700 by 4510200 in the Y with a spacing of 10 in the X and 10 in the Y would generate 3621 particles after the user clicks the *Create Particles* button. However, the maximum number of particles is set in the *"Number of Particles to Seed in next Create"* field. If the user specifies 1000 particles, only 1000 particles would be added, so the region would only be partially filled. However, if the user specifies 10000 particles as the maximum, only 3621 particles would be created, as limited by the seeding region and spacing.

(2) Random Placement

This option places the initial position of the particles within a user specified region defined by the *XMin, XMax, YMin* and *YMax* to randomly place the particles. The number of particles created is controlled by the *"Number of Particles to Seed in next Create"* field.

(3) Region Control File for Horizontal Placement

The *Region Control File* is an option that allows the user to specify a file (any of the EE recognized polygon files) containing one or more regions. If the *Use Region Control File* checkbox is selected then EE will look for the file is supplied in the *Regional Control File for Horizontal Placement* frame and the *Set to Model Extents* option is disabled. When the user clicks the *Create Particles* button, particles will be generated within the regions contained in the file. If the *Uniform Spacing* option is used, EE will generate a grid pattern at the spacing of the *Delta X* and *Delta Y* spacing but controlled by the region(s) contained in the file. If the *Random Placement* option is selected then the location will be assigned randomly within the region(s) in the file.

6.16.2.2 *Vertical Seeding Options*

Figure 6-177 shows the *Vertical (Depth) Options* frame which allows the user to select from one of three options: *Random Depth*, *Fixed Elevation*, and *Fixed Depth*. For the latter two options, the user should enter the elevation or depth as appropriate. *Random Depth* will randomly place the particle at a depth within the initial condition water depth. If oil spill is being modeled then *Vertical Depth Options* are disabled as oil spill can only be set at the water surface.

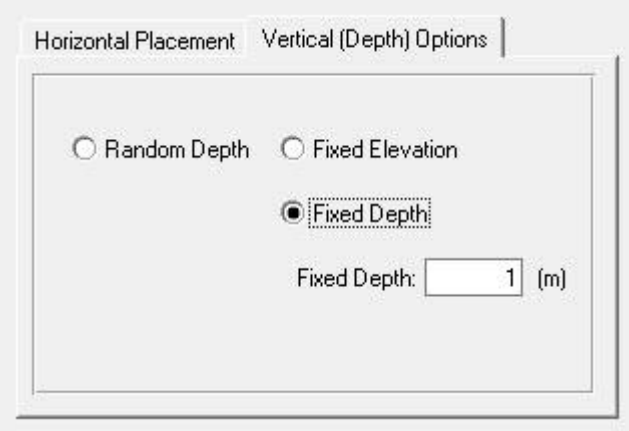


Figure 6-177 LPT Options: Seeding Utility: By Group – Vertical Placement.

6.16.2.3 *Varying the Particle Release Timing*

By default, EE will configure all the particles to be released at the *Start* time for the group that is when the *Single Release Date for Entire Group* checkbox has been selected. However, EE also allows the timed release of particles. Turning off the *Single Release Date for Entire Group* checkbox allows the user to select the *Distribute Release Dates* button which distributes the release times for the particles defined in the current group. Pressing the button displays a form with options for setting the starting and ending date. By default, these values are the same as those specified as the start and stop dates for the group, however adjusting the start and stop time will update these distributed release times. Pressing *OK* will distribute between the range entered. Even though the distributed dates are not displayed EFDC will use values the user previously selected. Pressing the *Distribute Release Dates* button again will reset the distributed release dates to the default start and stop dates for the group.

6.16.3. Examples of LPT Function

An example of the LPT function may be seen in Figure 6-178. This example consists of a rectangular domain with flat bottom, an open boundary to the east and a flow boundary along the southwest edge. U component masks were inserted to demonstrate the functionality of the Lagrangian Particle Tracks computations when masks are used.

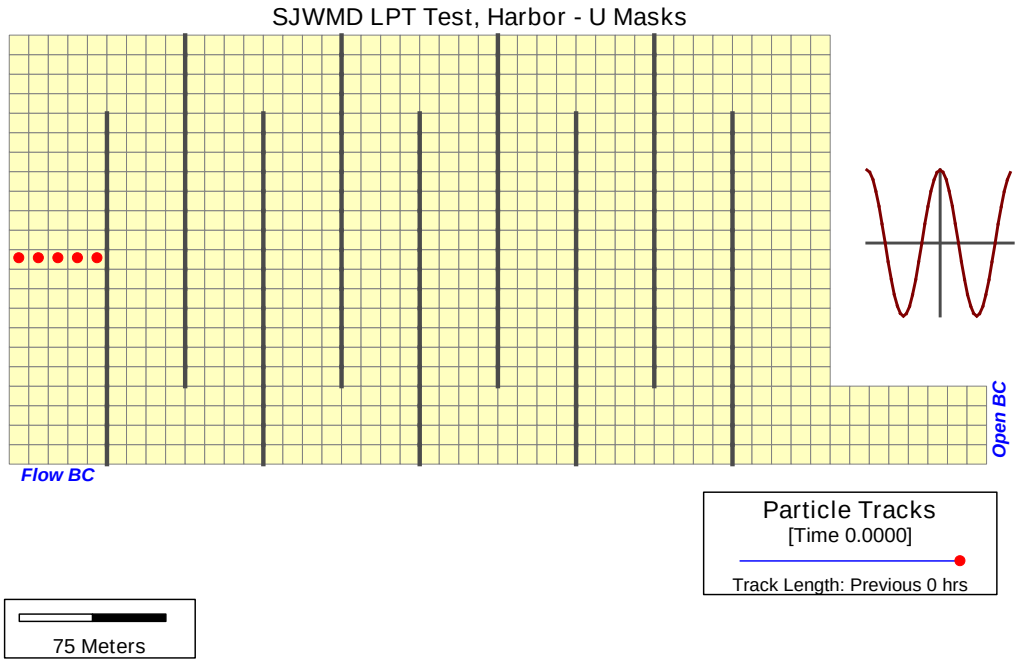


Figure 6-178 Harbor_U grid showing the masks, boundaries and initial particle locations.

The model represents the vertical component as a depth averaged system with 1 sigma layer. The depths of the 5 drifters are initialized at specified depths.

The results of the simulations are presented in Figure 6-179 showing the trajectories of the 5 drifters over one day.

Figure 6-180 shows the particle tracks colored by elevation and with random walk applied. Even though there was no vertical component, the tidal range is seen to result in changing particle elevations.

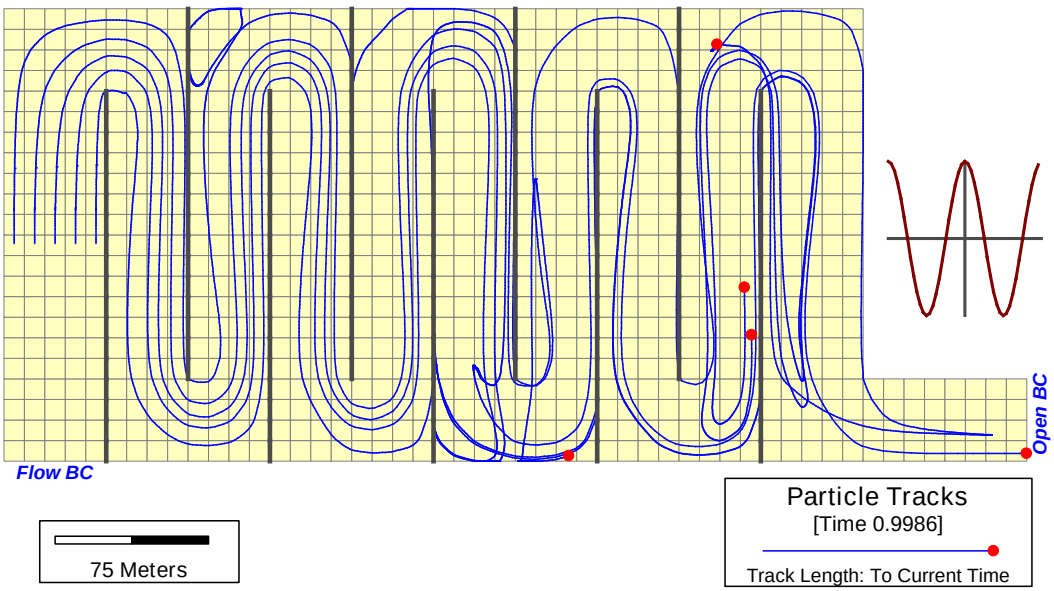


Figure 6-179 Harbor_U: Trajectories of 5 drifters over 1 day (no random walk).

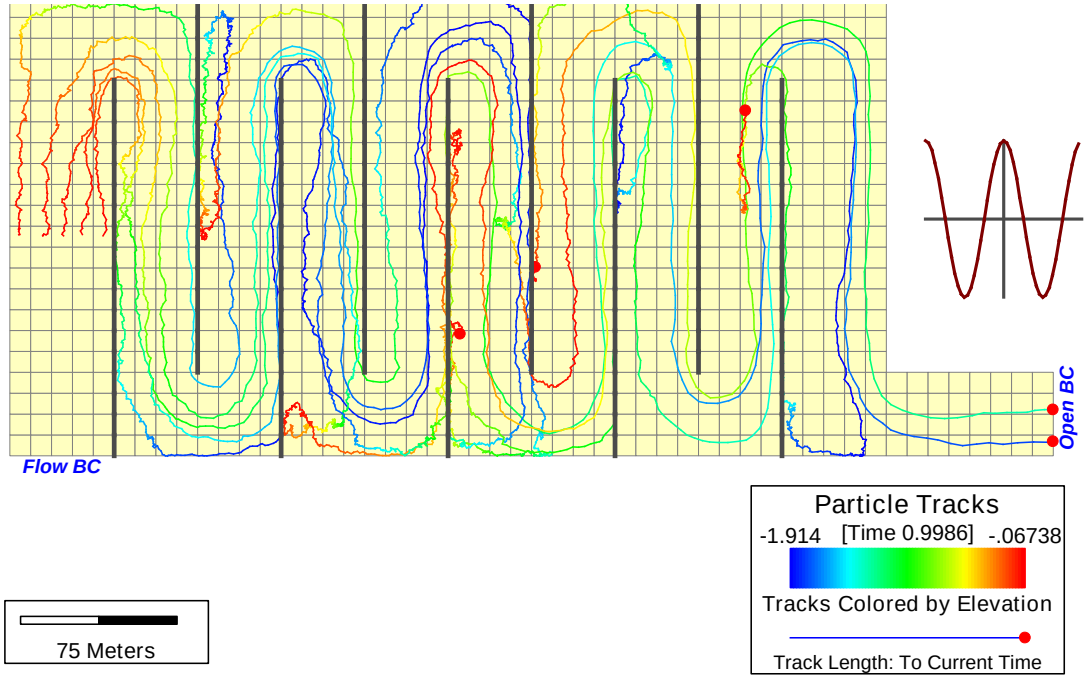


Figure 6-180 Harbor_U: Trajectories of 5 drifters over 1 day (random walk).

6.16.4. Oil Spill Modeling with Lagrangian Particle Tracking

EE7.2 and later allow the user to simulate oil spill based on drifters. Oil is maintained at the surface layer and is moved along with the hydrodynamic impacts similar to Lagrangian particles and broken down by evaporation and biodegradation.

The *Seeding Options* | *Group Options* frame is shown in Figure 6-181 below for oil spill. As there are many types of oils and different weathering of these types of oils, a group should be defined which has its own chemical property and volume per seed. When *Oil Spill Options* are enabled all the seeding options defined will apply to the group currently selected for the *Group ID* and the *Properties* button for oil spill will be enabled. In addition, below the Properties button, the volume and mass of oil for that group are displayed. As oil is always maintained at the water surface the *Vertical (Depth) Options* and *Settling Velocity* settings are disabled for oil spill groups.

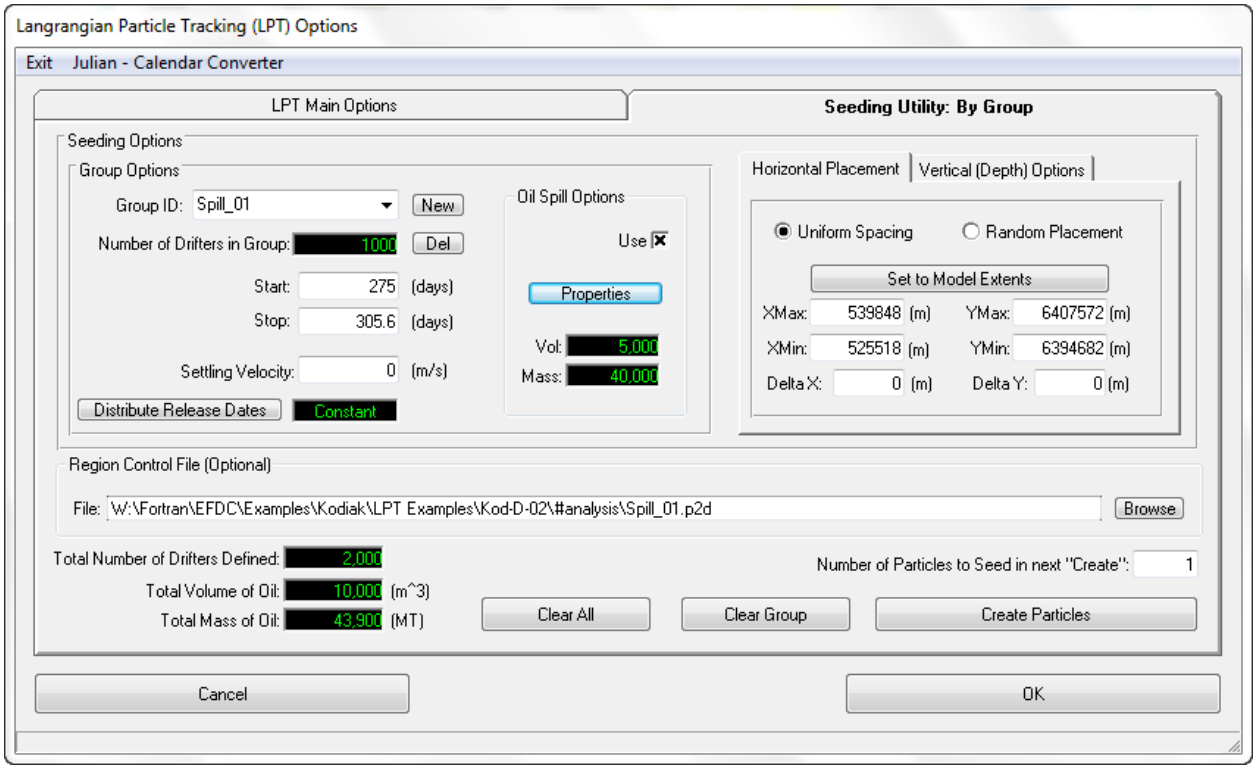


Figure 6-181 LPT: Seeding Utility by Group.

Clicking the *Properties* button will display the oil parameters settings as shown in Figure 6-182 . Each oil group requires a separate ID along with the density and the volume of oil per drifter. Biodegradation and vapor pressure are entered to provide the properties for all the drifters in the current group. Though temperature is optional, it is recommended that heat transfer be included, as the oil spill evaporation process depends on the temperature of the ambient environment. Winds also impact evaporation and it is expected that with any realistic oil spill model, winds will be included.

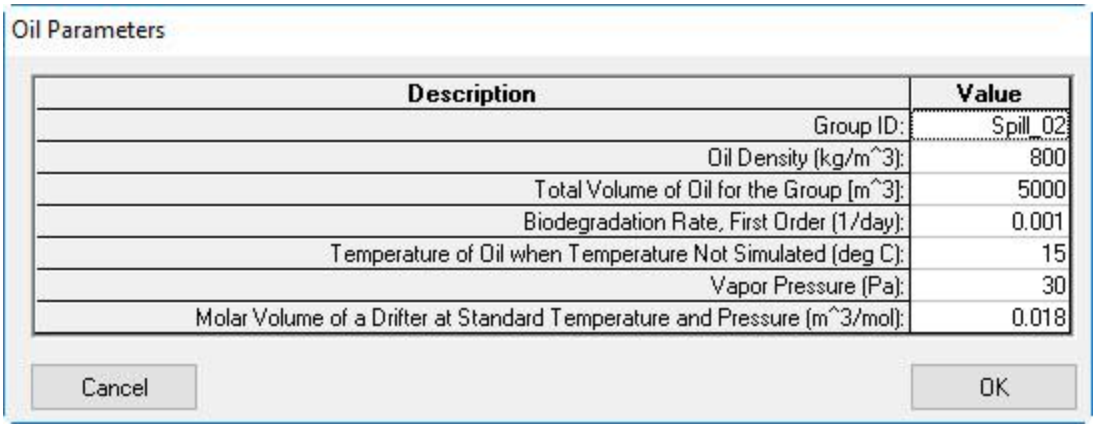


Figure 6-182 Oil Parameters Setting.

It should be noted that EFDC will cause the drifter to disappear when the oil per drifter is less than 1.0 mm^3. The user may manually choose to hide concentrations of oil higher than this by using the crop below in *ViewPlan* or clipping in *View3D*. For post-processing of the oil simulation, *ViewPlan* provides several options for displaying the oil within *Viewing Options*. Oil thickness, mass and volume may also be displayed.

When configuring an oil spill model, it should be noted that use of the *Vertical Movement Option* is ignored for groups that are designated as simulating oil. If the density of oil is less than that of water then the particles are always in the surface layer. If the oil is heavier than water, then the fully 3D option is enabled.

For simulation on the oil evaporation process, the theory of surface evaporation presented in the paper of Warren Stiver and Donald Mackay (1984) is used.

EFDCPlus allows for effects of biodegradation of the oil based on user defined biodegradation rate with a simple first order decay approach based on Stewart (1993). A rate of 0.011 per day leads to an approximately two-month half- life for oil. If temperature of an oil drifter is provided by the user then this is used as the reference rate for the optimal biodegradation.

In order to simulate a conservative oil, the user would set the degradation rate and the vapor pressure as zero.

7 Post-Processing - Form Operations

Most of the post-processing visualization features of EFDC_Explorer are located in the *ViewPlan* and *ViewSlice* functions, which are discussed in Section 8. However, there a several other features for post-processing model runs that are available from the main form. The post-processing functions are located in the *Model Analysis* tab in the main form. Three tabs provide access to these features (Figure 7-1), as discussed below.

At the top left of the form is the *Output File Loading Options* frame. This is used to show if model results have been loaded for analysis. The *Reload* checkbox forces EFDC_Explorer to reload the output files even if the files have already been loaded. This is necessary at times when an error occurs or the user is reviewing the progress of a run (see Section 5.1.6).

The user can review a run that is in process by:

- Pausing the EFDC execution (pressing any key within EFDCPlus will pause the execution).
- Checking the *Reload* checkbox.
- Using the *ViewPlan* or *ViewSlice* review the model progress.
- Restart the model, if desired, by pressing any other key than that which stopped it. Or the user may completely end the model run and make adjustments, save and re-run.
- Repeat as needed.

7.1 Model Calibration

Figure 7-1 displays the *Model Calibration* tab that provides access to some of the calibration features available in EFDC_Explorer. There are six types of calibration that may be generated:

- Time Series Comparisons,
- Correlation Plots,
- Vertical Profile Comparisons
- High Frequency Time Series
- Flux Comparisons
- Cruise Plot Comparisons

For each of these the user can provide a linkage of water column model results to measured data. Once the model/data linkage has been defined once, calibration plots and statistics can be quickly generated after each model run.

Once defined, calibration plots and statistics may be generated automatically by selecting the *Generate Now* button, or the selecting the *Generate Calibration Plots and Statistics* checkbox in *Run Options* before running a model. It should be noted that if a new model is opened while the current model is running that plots and statistics will not be automatically generated.

Pressing a *Plots* button allows the user to review the plots on screen and set each plot's style. Before viewing the plots the user will be prompted as to whether they want to *Automatically Generate Defined Plots* If the user selects this option then EE will save the plots to the project directory into a folder entitled *#calib_plots*. If the user selects *Automatically Export ASCII Data for the Defined Plots* the .DAT files are saved into a folder entitled *#calib_extraction*.

To generate the statistics, select the *Generate Now* button and check the error statistics to be reported. Selecting *Ok* will start the calculations. This can take several minutes for large models. For this reason, the user can select the *Start Run Timer* button which will show the type of plots that EE is generating at that time in the *End of Run Timer*. Once the statistics are displayed, the user can save to a text file or copy to clipboard

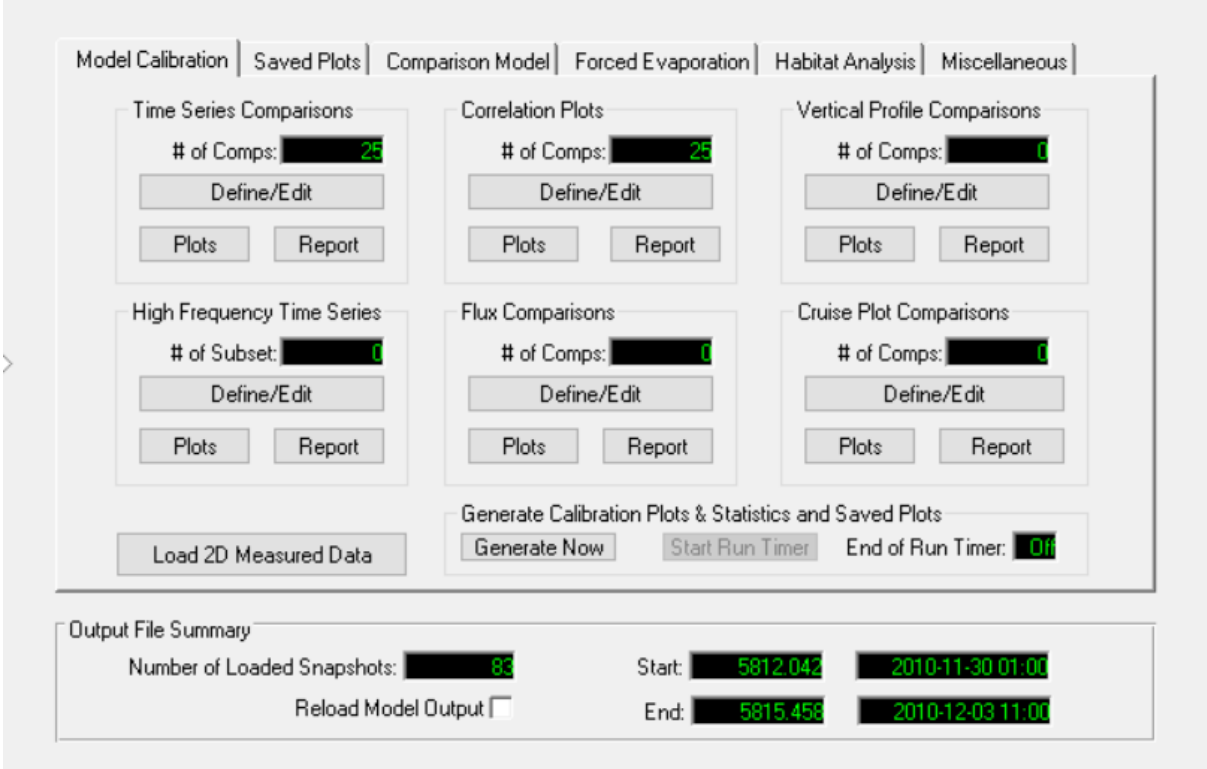


Figure 7-1 Post Processing Options - Main Form.

The plot styles are saved in the EFDC_Explorer configuration file "CalForm_TS.EE," "CalForm_CP.EE" and "CalForm_VP.EE" for the time series, correlation and vertical profile plots respectively. In addition, the file "CalForm_MMA.EE" contains plot styles for the time series option of *Min-Avg-Max* plots (see Section 10 for more details).

The time series, correlation and vertical profile model-data linkages provided in this section are only one means of conducting model-data comparisons. Almost any model parameter can be compared to data using the *Import Data* feature of the time series plotting utility. See Section 10 (Time Series Graphing) for more details.

7.1.1. Model Comparison Statistics

The error statistic is selected at the beginning of each statistical report generation process. The statistics for each parameter and station, as well as parameter composite statistics, are reported. The user may also select to automatically generate reports as shown in Figure 7-2. The reports will be automatically saved in the project directory in the "#calib_stats" folder.

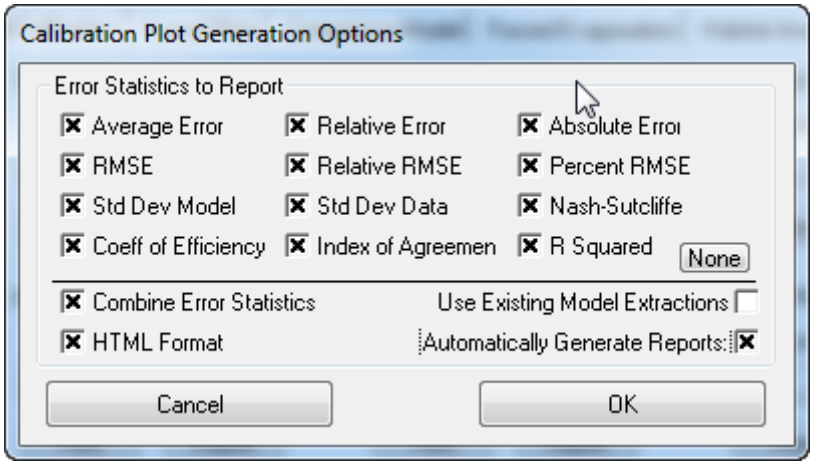


Figure 7-2 Calibration plot generation options.

The user should select the *Combine Error Statistics* checkbox to plot all the statistics on one form rather than each on a separate form. The user may select *HTML Format* so that the statistics are formatted for easy copying into a report or displaying online. To set EE so that if provides automatic summaries of statistics at the end of the model run the user should select *Automatically Generate Reports* checkbox as well as selecting *Generate Plots and Statistic at End of Model Run* in the *Run Model* form before running the model.

Extracting the data from the EFDC output can be time consuming if it is a large model run for a long period of time. To avoid having to extract the results each time the statistics are generated, the user is advised to check the *Use Existing Model Extractions* check box, which will use the extracted data still in computer memory

There are twelve pre-defined statistics available for the model-data statistical reports which are described below. The difference between the average observation and model values (Mean Error - ME) is taken to show the deviation in terms of the unit of measurement.

RMSE: The root mean squared error (or RMSE) reflects sample standard deviation of differences between simulated and observed values. Similarly, the relative RMSE (RRMSE) is the RMSE normalized by the maximum range of observed values.

The Index of Agreement (IOA) is a standardized measure of the degree of model prediction error, with values ranging from 0 (no agreement) to 1 (perfect match).

The R^2 value is a statistical measure of how close a model's predictions are to a fitted regression line. It is sometimes referred to the coefficient of determination.

Three widely used error measures are used here. "O" denotes observations and "P" denotes model predictions at the corresponding locations and times, the mean of the observed and predicted variables for "N" observations at a single or multiple observation stations is given in equations below.

Mean Observed:

$$\bar{O} = \frac{1}{N}\sum_{i=1}^{N} O_i \tag{B.1}$$

Mean Predicted:

$$\bar{P} = \frac{1}{N}\sum_{i=1}^{N} P_i \tag{B.2}$$

Standard Deviation Observed:

$$SDO = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(O_i - \bar{O}\right)^2} \tag{B.3}$$

Standard Deviation Predicted:

$$SDP = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(P_i - \bar{P}\right)^2} \tag{B.4}$$

Mean Error (ME):

$$ME = \bar{P} - \bar{O} \tag{B.5}$$

Relative Mean Error (RME):

$$RME = \frac{\bar{P} - \bar{O}}{\bar{O}} \times 100\% \tag{B.6}$$

Absolute Mean Error (MAE):

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|P_i - O_i\right| \tag{B.7}$$

Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(P_i - O_i\right)^2} \tag{B.8}$$

Relative Root Mean Square Error (RRMSE):

$$RRMSE = \frac{RMSE}{\left(O_{max} - O_{min}\right)} \times 100\% \tag{B.9}$$

Coefficient of Determination (R^2, Square of Correlation Coefficient): (B.10)

$$R^2 = \frac{\left\{\sum_{i=1}^{N}\left(O_i - \bar{O}\right)\times\left(P_i - \bar{P}\right)\right\}^2}{\left\{\sum_{i=1}^{N}\left(O_i - \bar{O}\right)^2\right\}\times\left\{\sum_{i=1}^{N}\left(P_i - \bar{P}\right)^2\right\}}$$

Nash-Sutcliffe Index of Efficiency (NSI):

$$F^2 = 1 - \frac{\sum_{i=1}^{N}\left(P_i - O_i\right)^2}{\sum_{i=1}^{N}\left(O_i - \bar{O}\right)^2} \qquad \text{(B.11)}$$

Coefficient of Efficiency (COE):

$$E = 1 - \frac{\sum_{i=1}^{N}\left|P_i - O_i\right|}{\sum_{i=1}^{N}\left|O_i - \bar{O}\right|} \qquad \text{(B.12)}$$

Index of Agreement (IOA):

$$D = 1 - \frac{\sum_{i=1}^{N}\left|P_i - O_i\right|}{\sum_{i=1}^{N}\left(\left|P_i - \bar{O}\right| + \left|O_i - \bar{O}\right|\right)} \qquad \text{(B.13)}$$

An example of a report generated by EE is shown in Figure 7-3. The bottom of the report contains a listing of the composite statistics for the entire model run. The parameters may be copied to the clipboard for pasting into Excel or some text editor. A good practice is to save these statistics in each run's directory for quick future reference.

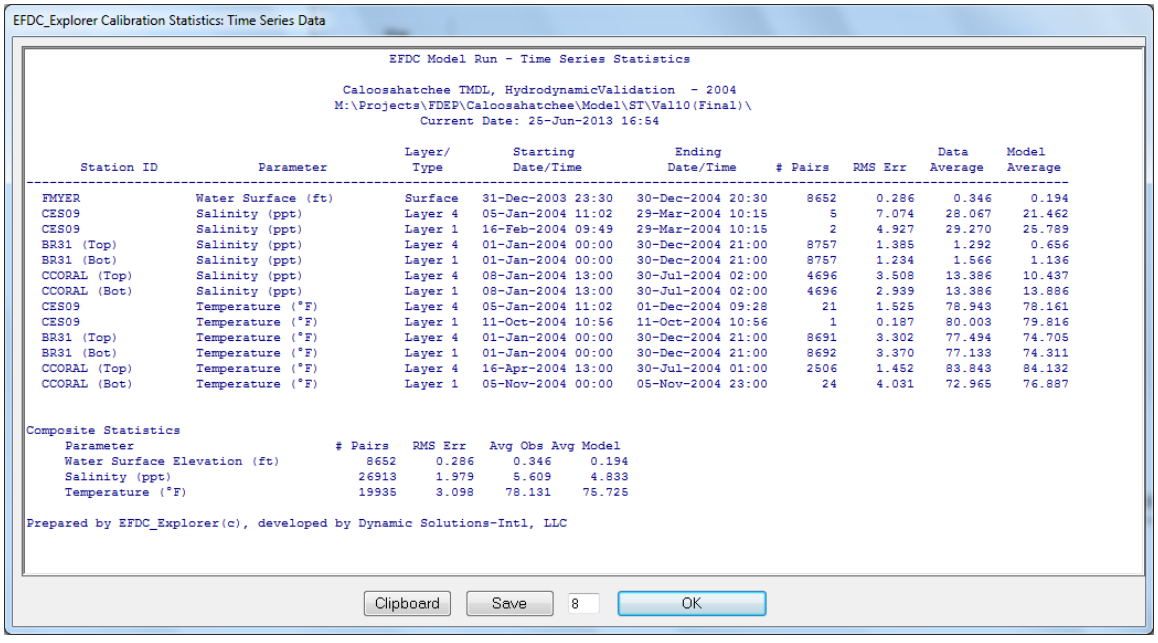


Figure 7-3 Example time series calibration statistics report.

7.1.2. Time Series Comparisons

7.1.2.1 *Model-Data Configuration*

The *Time Series Comparisons* frame contains the buttons that configure (*Define/Edit*) and plot (*Plot*) a series of EFDC cells and measured time series data. Once configured, the linkages between the EFDC cells and the data is automatically available, and the user only needs to press the *Plot* button to compare model to data for each run. Figure 7-4 displays the form used to link cells to data and other parameters. RMC on the cells to obtain additional input options or guides.

If a link is incorrectly configured and the file cannot be found at that pathname it will appear with a red highlight.

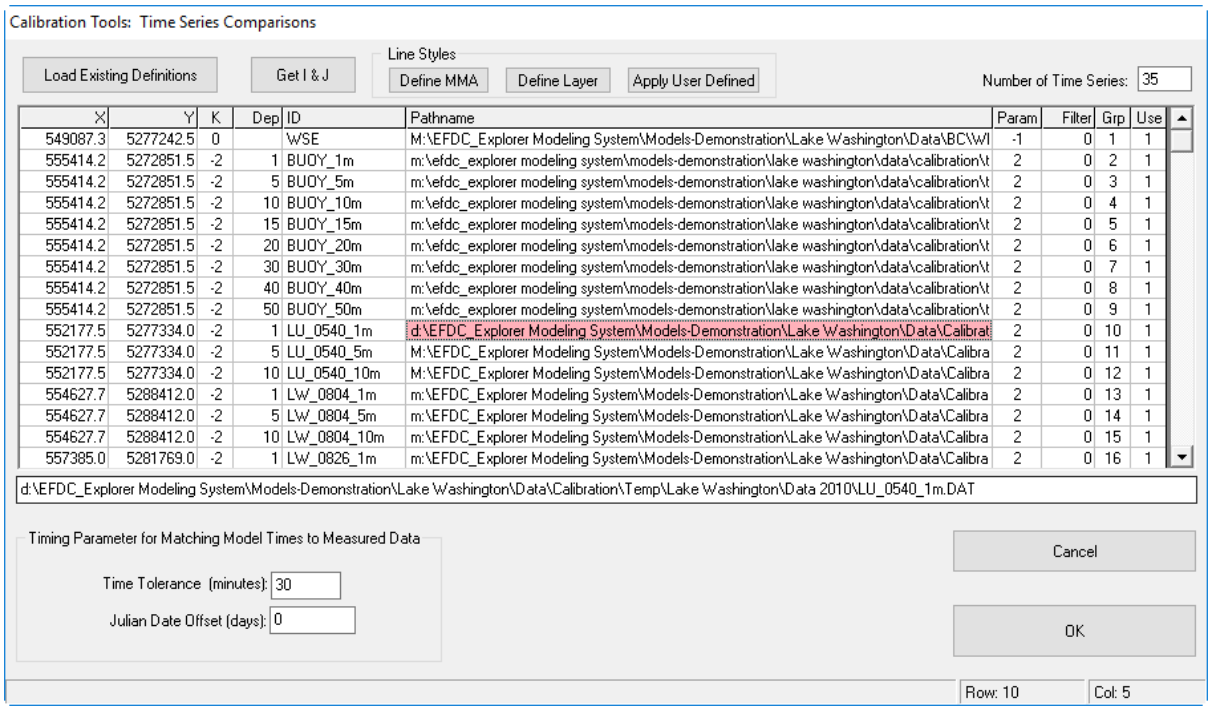
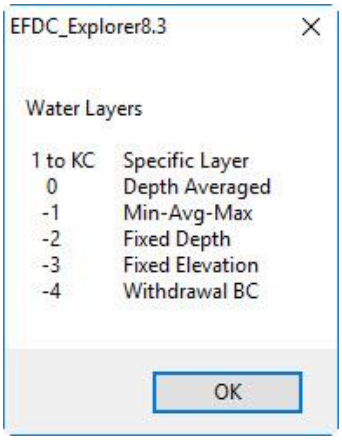


Figure 7-4 Time series calibration EFDC cell and data linkage definitions.

When first starting with the model-data linkage process, the *Number of Time Series* field will be "0". To initially define or add/remove linkages, enter the number of desired linkages. When the user presses "Enter" the data grid will be displayed with the number of lines required. The number of model-data linkages can be changed at any time. However, the definitions above the maximum number of linkages will be deleted.



The *Get I & J* button allows the user to input X and Y values in the LXLY units to determine the corresponding values of I & J. The *Line Styles* frame provides a way to define (i.e. the *Define MMA* and *Define Layer*) and then apply (via the *Apply Defaults*) common formatting for all currently defined model-data linkages.

For each model-data pair, the user must specify the information listed in each column:

X & Y: Enter the X and Y in meters of the location of the data station. The coordinate system must be the same as that used for the model. Using these coordinates the matching model cell is determined.

K: This is the layer specification option. There are three options as shown. The Min-Avg-Max (MMA) option generates three model time series for the cell, based on the water column layer results.

ID: The ID field is used for labeling the plots and statistics reports.

Pathname: This field contains the full path to the data file, in DAT or WQ formats, for the specific parameter, location and layer option. The user can enter a “None” or “Skip” in this field to only display a model time series. Rows that are shown in pink indicate pathnames that are not valid (e.g. files were deleted by the user after initial setup) and need to be redefined.

Param: Enter the parameter code that will be extracted from the model for comparison to the data contained in the corresponding data file. The list of currently available parameters is shown in Figure 7-5. RMC’ing on this field (after it has focus) pops up the browse function to allow the user to select or change the file.

Group: This is the plot number. If each line uses a unique Group # then each model-data pair will be displayed on a separate plot. If two of more model-data lines use the same Group # they will be displayed on the same plot.

Use: This is a flag to optionally turn on or off the plots and statistics computation for each model-data pair.

The *Timing Parameter for Matching Models to Measured Data* is used to set the time tolerance in minutes, such that there is a time match when the MODEL_TIME is within the DATA_TIME +/- the tolerance. The Julian date offset is used when Julian dates have been used in the data files rather than Gregorian dates. This offset allows the user to add this number of whole days to the Julian dates.

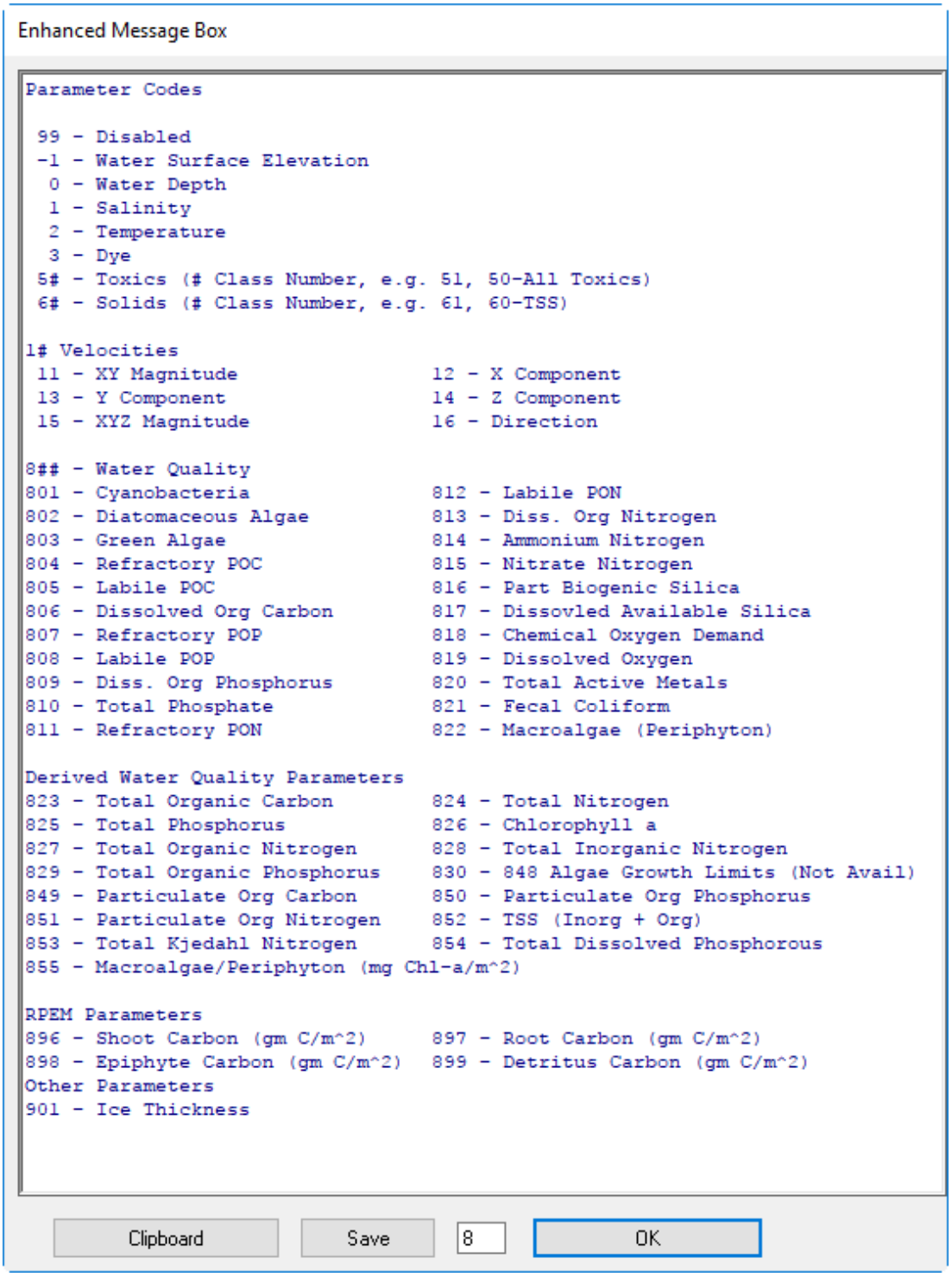


Figure 7-5 Available calibration parameter codes.

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7.1.2.2 *Time Series Plots*

The *Plots* function allows the user to view on screen or export the plots that are currently defined and enabled (i.e. Use flag = 1). The *Plots* function loads the EFDC model linkage files (e.g. EE*.out), reads the observed data files and then scans the model linkage files to build the model-data plots. The user can press ESC during the model output loading process to abort the loading and plotting. When the data has been loaded the time series plotting utility is displayed. This is the same utility used for all other time series and X-Y plots. However, when viewing the time series calibration plots three (3) additional function are available on the toolbar. The following graphic shows the time series plotting toolbar with the three calibration specific functions outlined in red.

These buttons all close the current calibration plot and, in order of the buttons shown, move to the previous plot, move to the next calibration plot or exit the time series calibration viewer. When moving from one plot to the next, when the last plot has been viewed the user is requested to either continue viewing the plots (starting over at plot 1) or exit the viewer.

Figure 7-6 and Figure 7-7 provide example plots produced by the time series calibration tool. Figure 7-6 is for a water surface elevation comparison (layer option K=0). Figure 7-7 shows a MMA time series plot (the “average” series is turned off for this example) of dissolved oxygen (layer option = -1).

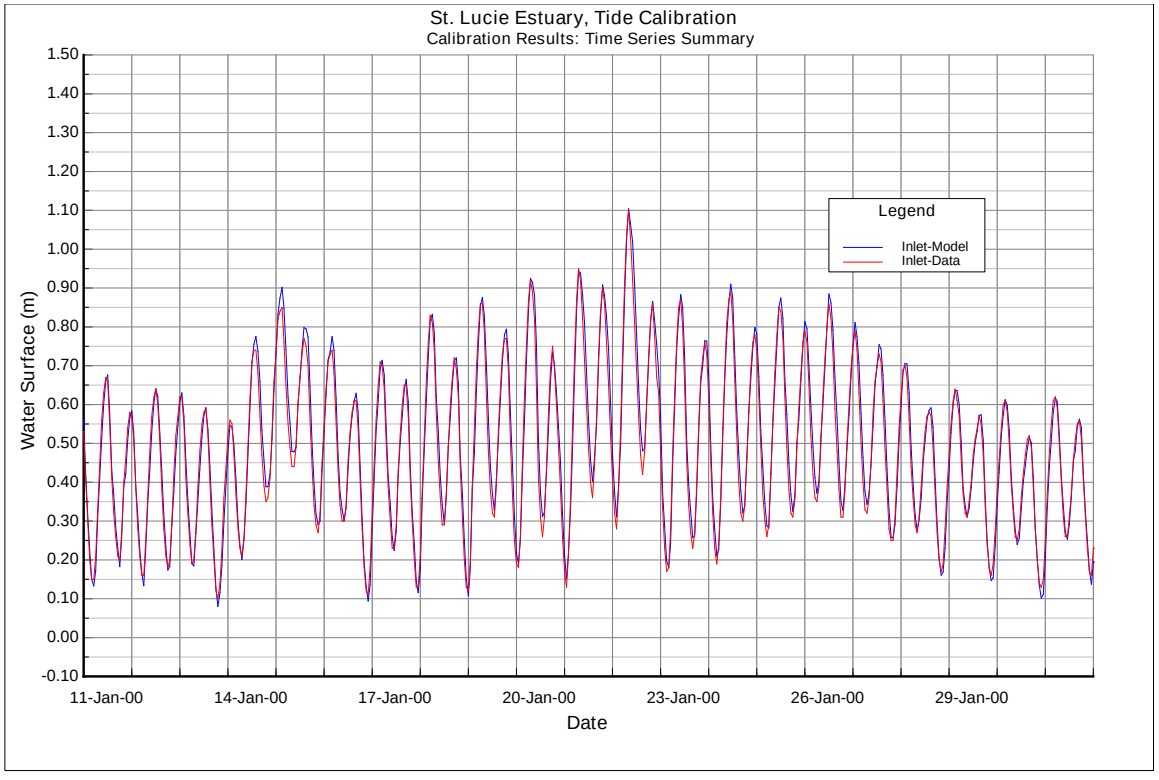

Figure 7-6 Example model-data time series comparison for water levels.

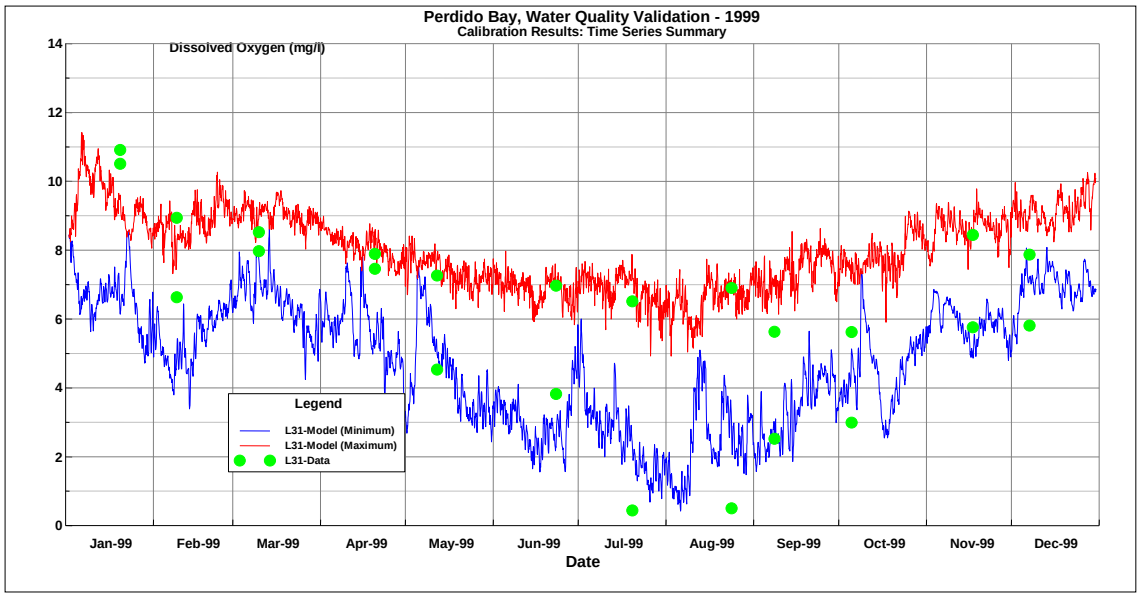

Figure 7-7 Example model-data time series comparison for dissolved oxygen.

7.1.2.3 *Low and High Pass Filter*

The *Model Calibration* | *Time Series Comparisons* tool has also been updated to allow comparison with a low and high-pass filter. To access the *Low Pass* option set the *Filter* (formerly *R Avg)* column to a number less than zero as shown in Figure 7-8. The number set is the number of hours for the low pass filter i.e. if Filter = -12.4 then the low pass filter is 12.4 hours as shown in Figure 7-9. The options for specifying the type of filter can be accessed by right mouse clicking on the Filter field as shown in Figure 7-8.

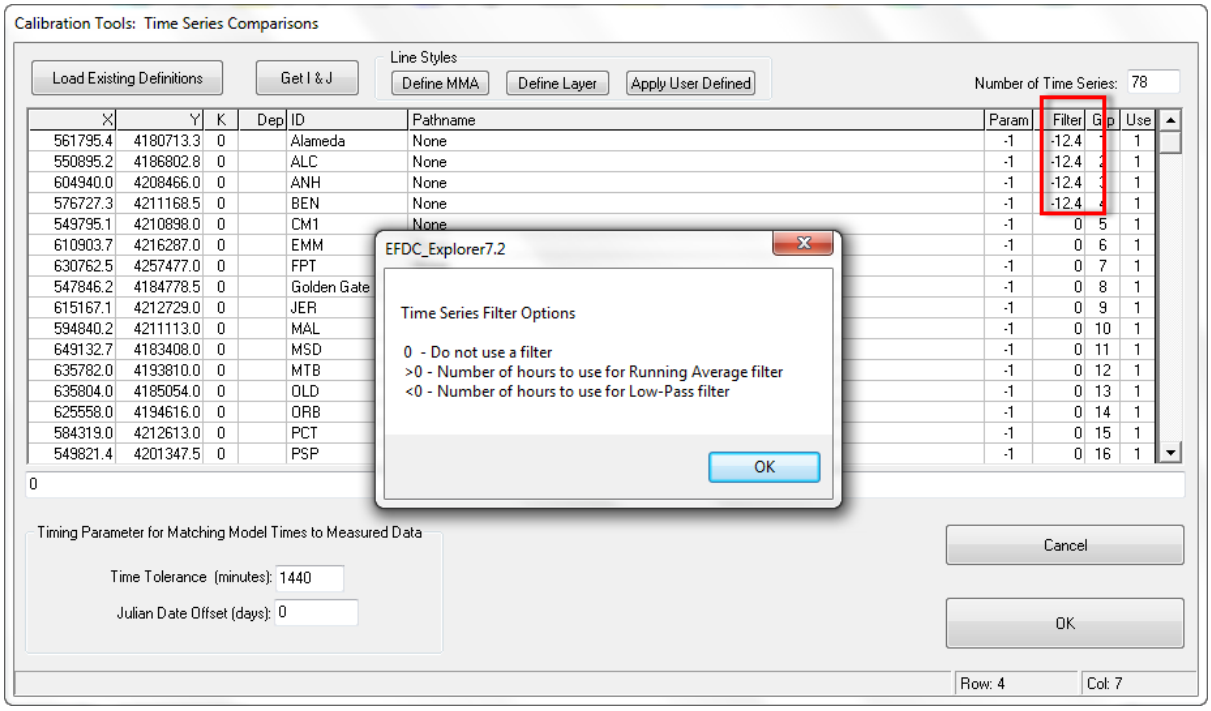

Figure 7-8 High and Low Pass Filter for Time Series Calibration Plotting.

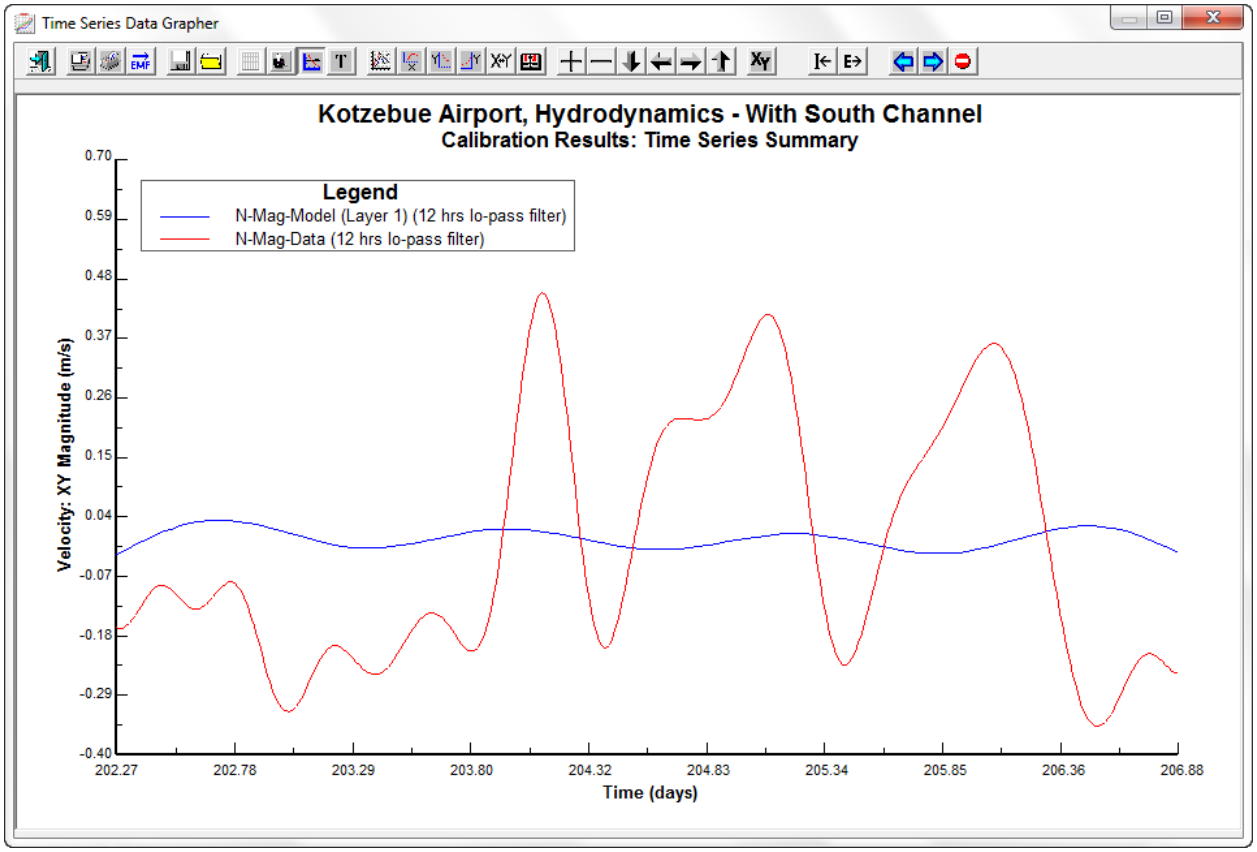

Figure 7-9 Low Pass Filter Calibration Results.

7.1.3. Correlation Plots

A recent feature of EFDC_Explorer is the ability to plot correlations between the model and existing data to assist the user in calibration. The method for setting up correlation plots is similar to that for *Model-Data Configuration* in Section 7.1.2.1. The setting up of linkages is the same as that for *Time Series Comparisons*, however, with *Correlation Plots* the user is able to select which error statistics will be displayed on the plot, as shown in Figure 7-10.

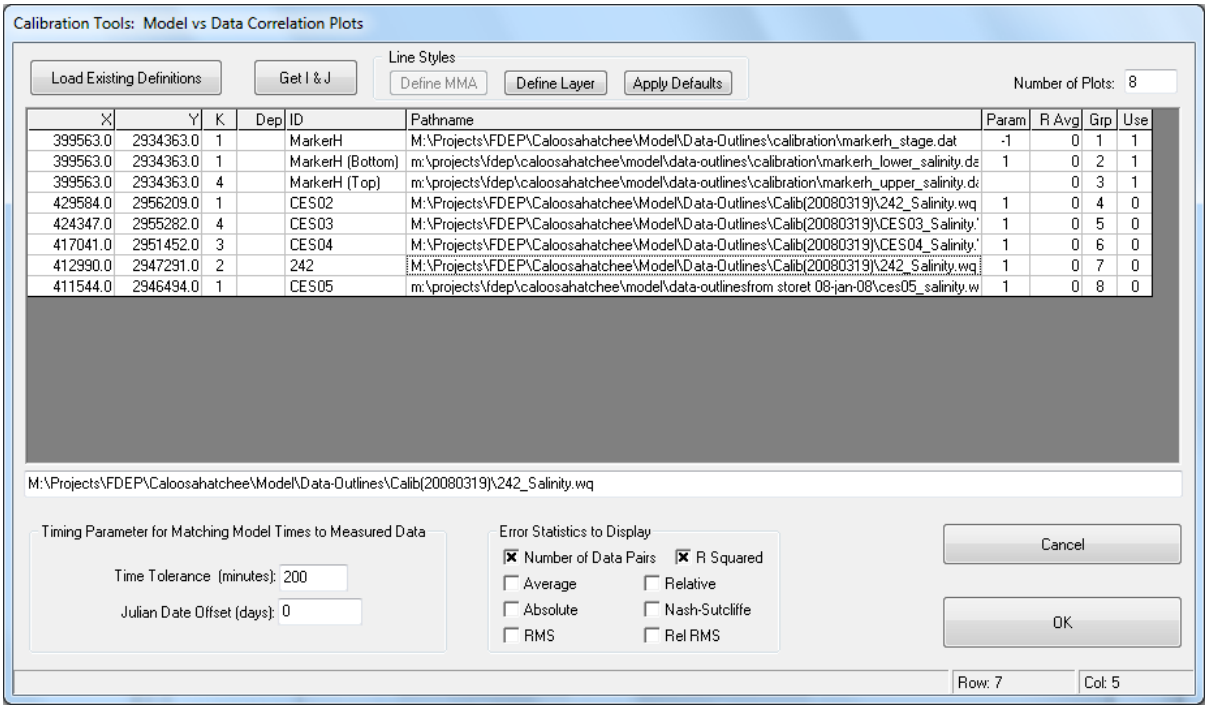

Figure 7-10 Calibration tool: Model vs Data Correlation Plots.

Error statistics which may be displayed are explained in Section 7.1.1 and include:

- R Squared
- Average
- Absolute
- RMS
- Relative
- Nash-Sutcliffe
- Relative RMS

An example of a correlation plot is shown in Figure 7-11. Note that observed data is shown on the x-axis and the model data is shown on the y-axis.

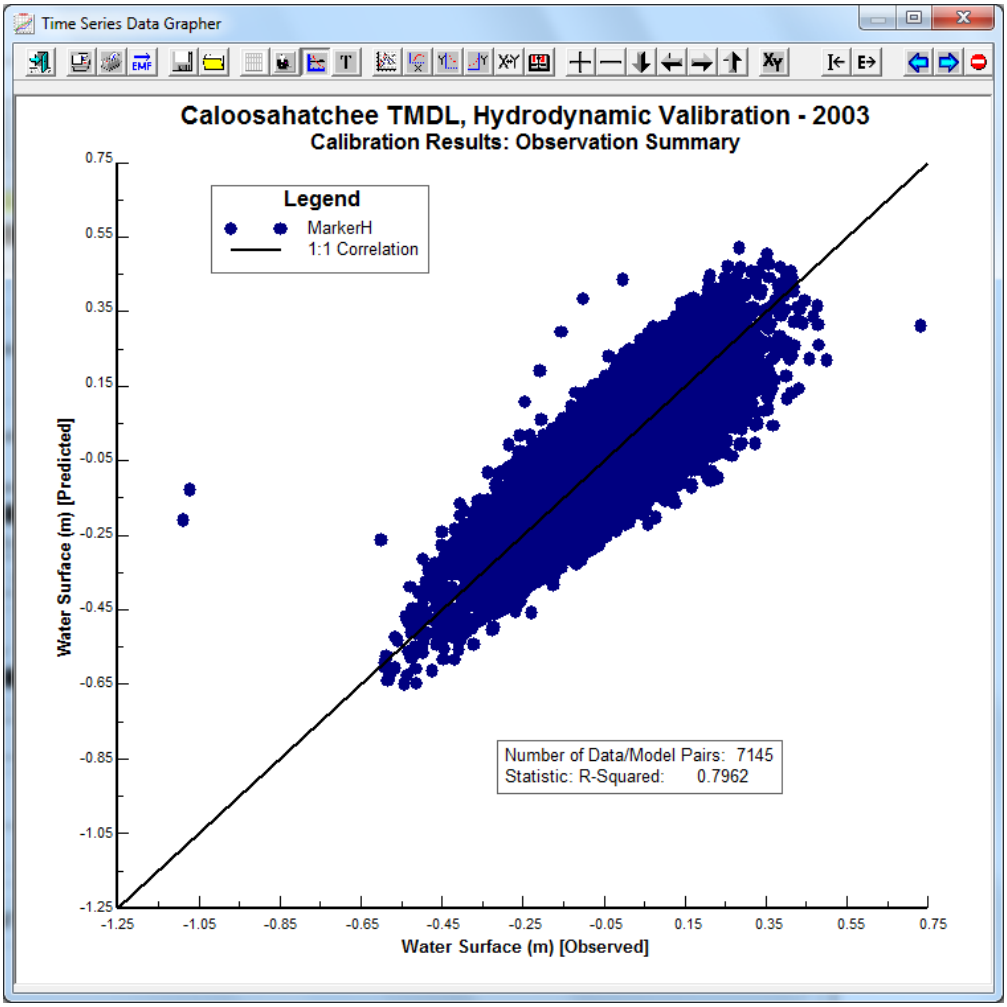

Figure 7-11 Example model data Correlation Plots comparison for water surface elevation.

An additional feature of the *Calibration plot* function is that the box size for each plot may be saved. This means that even though screen aspect ratios may vary between different computers, they will still produce the same final plots. CTRL-W sets the size of the plot box size, and when the user saves the project the size of the plot will also be saved.

7.1.4. Vertical Profile Comparisons

The *Vertical Profile Comparisons* frame contains the buttons that configure (*Define/Edit*) and plot (*Plot*) vertical profiles of EFDC cells and measured profile data. Once configured, the linkages between the EFDC cells and the data is automatically available and the user simply needs to press the *Plot* button to compare model to data for each run. Figure 7-12 displays the form used to link cells to data and other parameters. RMC on the cells to obtain additional input options or guides.

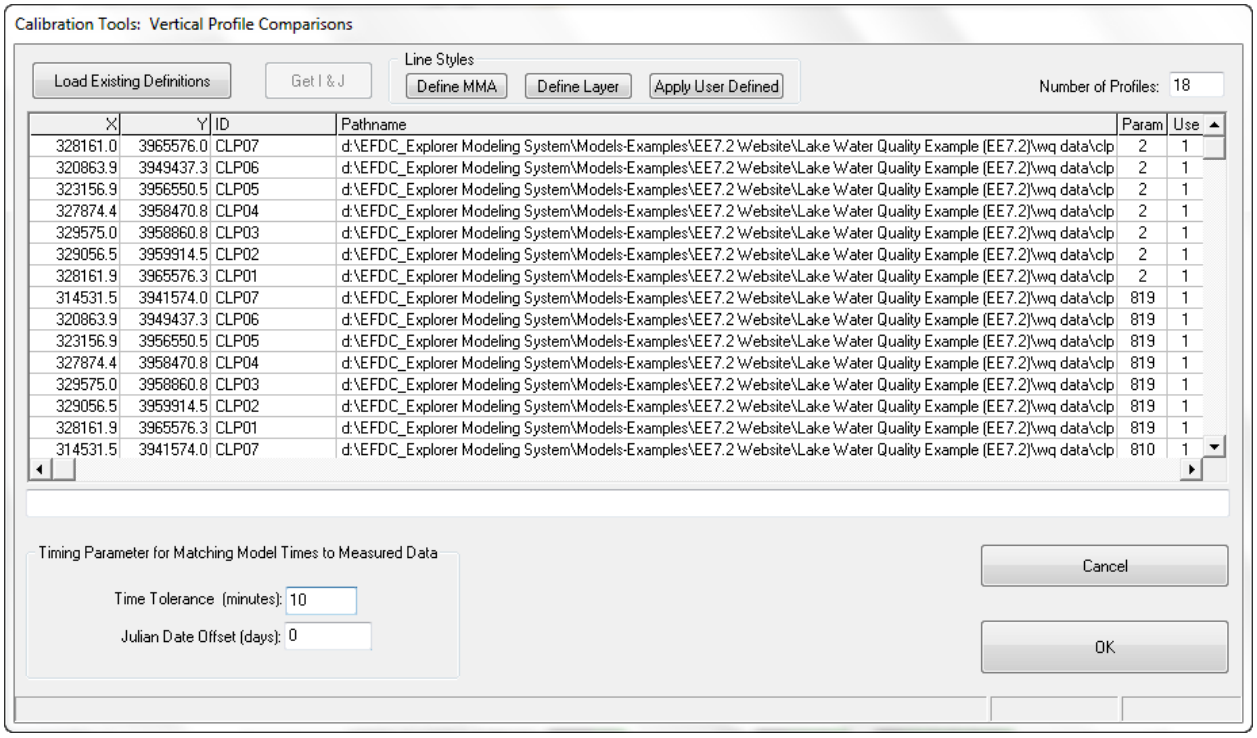

Figure 7-12 Vertical profile calibration EFDC cell and data linkage definitions.

The information is very similar to the time series model-data linkage form. However, the *Group* column is missing as the profile plot structure is pre-defined. One other major difference is the format of the data contained in the *Pathname* field. Appendix B (Format B-8) provides guidance on the format. This file can contain one or more snapshots in time for the same station and parameter.

7.1.4.1 *Vertical Profile Plots*

The *Plots* function allows the user to view on screen or export the plots that are currently defined and enabled (i.e. Use flag = 1). The *Plots* function loads the EFDC model linkage files (e.g. EE*.out), reads the observed data files and then scans the model linkage files to build the model-data plots. The user can press ESC during the model output loading process to abort the loading and plotting. When the data has been loaded the vertical profile plotting utility is displayed. This utility displays up to 8 model-data vertical profiles per page. The following graphic shows the toolbar functions.

The first group of left and right buttons move to the previous and the next vertical profile stations, respectively. The →1 and →A buttons export the current or all defined dates for the current station, respectively, to a Windows EMF file. The – and + buttons move back and forward in time for the current station. The PgUp and PgDn keys have the same functionality. The remaining buttons provide formatting control over the plots.

As an example of the vertical profile plotting function, Figure 7-13 shows a single page containing a subset of time snapshots for salinity for a data single data station for a 55 layer model.

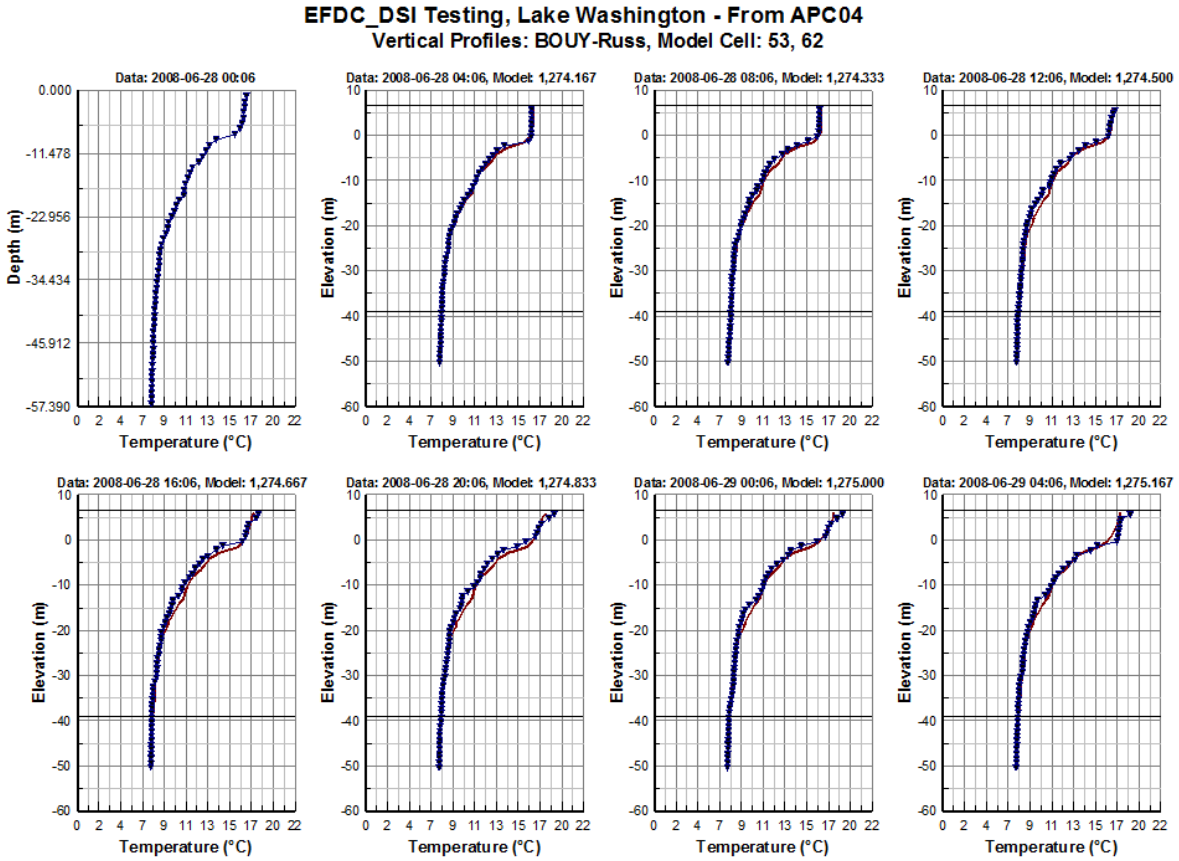

Figure 7-13 Example model-data vertical profile plot for salinity.

In order to make exporting of the Vertical Profile Plots for presentations more flexible and consistent the user can configure how many plots are displayed on each page. So instead of 8 plots (4 x 2) which is the default layout, the user may now select 4 plots (4 x 1) or 6 plots (3 x 2) etc. This option is available by selecting the *Modify Settings* button in the *Vertical Profile* tool as shown in Figure 7-14.

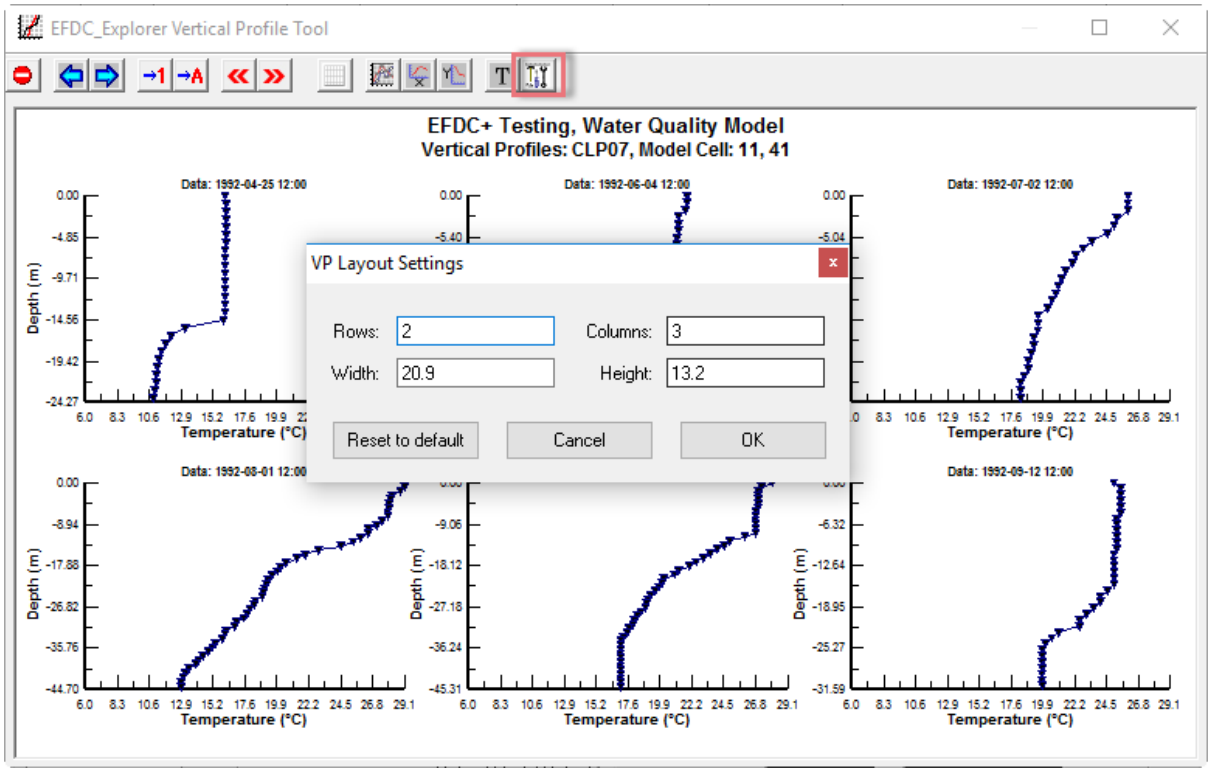

Figure 7-14 Vertical Profile plots display options.

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7.1.5. High Frequency Time Series

See Section 6.5.3.1 for details of this function.

7.1.6. Flux Comparisons

The flux comparison tool is provided to assist the user in calibration of flux or flow across a user defined line and is accessed at shown in Figure 7-1. The "# if Comps" label displays for the user the number of comparisons that have been defined. The user may *Define/Edit* these comparisons with the appropriate button which displays the form shown in Figure 7-15.

This form provides the user with option of defining the number of time series to be compared in the text box in the top right corner. The user should define an ID (usually a station name), and set the path to the predefined flux like as well as the data file for that flux line. The following five columns are used to set the parameter (based on the parameter codes shown earlier in Figure 7-5). The "R Average" column is used to filter the data with a moving average. The "Mult" column is used as multiplier and "Use" will activate or deactivate this comparison.

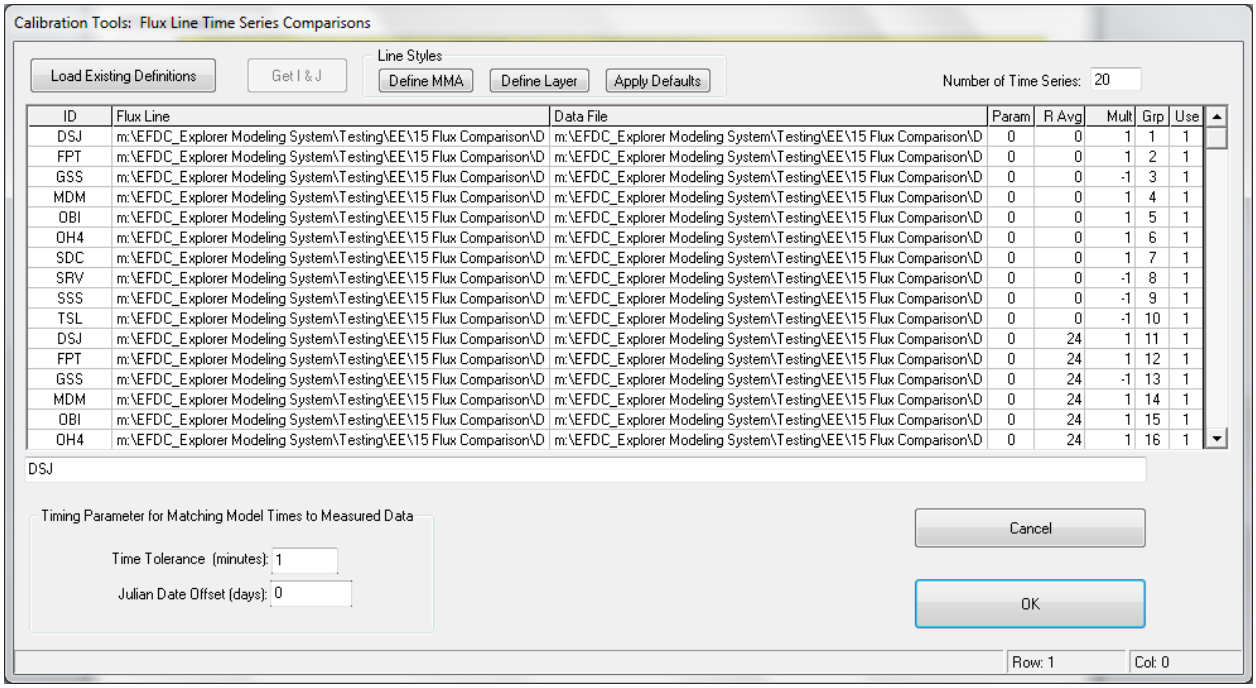


Figure 7-15 Flux Calibration

7.1.7. Cruise Plot Comparisons

Cruise plot comparisons are useful when the user wants to compare a series of continuous sampling data with model output over some distance in the domain. In *ViewPlan* mode the user may define a “cruise line”, which corresponds to the location of data sampling transects that have been undertaken on the waterbody. EE is now able to extract the data from the user defined line and compare with the measured data. This tool can be accessed in M*odel Analysis* | *Model Calibration* | *Cruise Plot Comparisons* as shown in Figure 7-16.

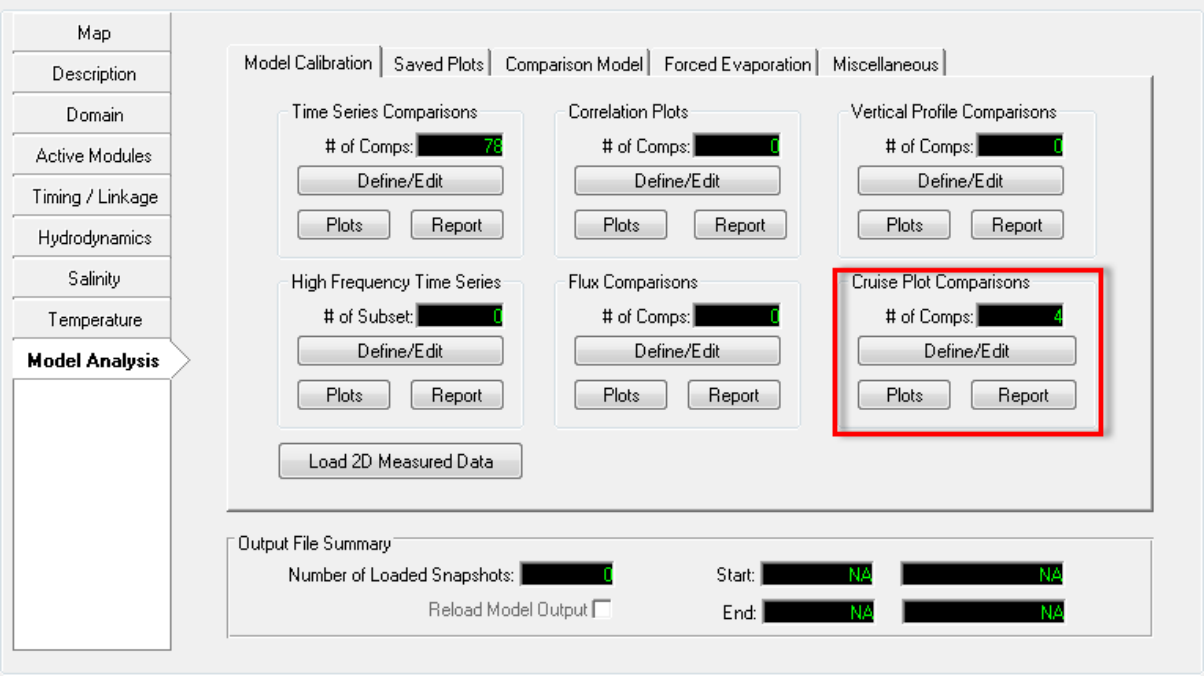


Figure 7-16 Cruise Plot Comparisons Form.

The user should click on the *Define/Edit* button to set the cruise plot data as shown in Figure 7-17. EE allows the use of two different kinds of cruise plot data. These are defined by station or by track. The user should define a cruise number ID for each data set. This ID will correspond to the USGS defined station IDs. If using native USGS data sets the user should select type 1 for the cruise data type. In this case the format is as shown in Figure 7-18. If using track data the data form should be as shown for type 2.

The data path should be set for text files that are tab delimited format required for the following parameter as explained above and shown in Figure 7-17. All the common parameter types are shown in Figure 7-18. This information box is displayed with the user RMCs on the parameter field.

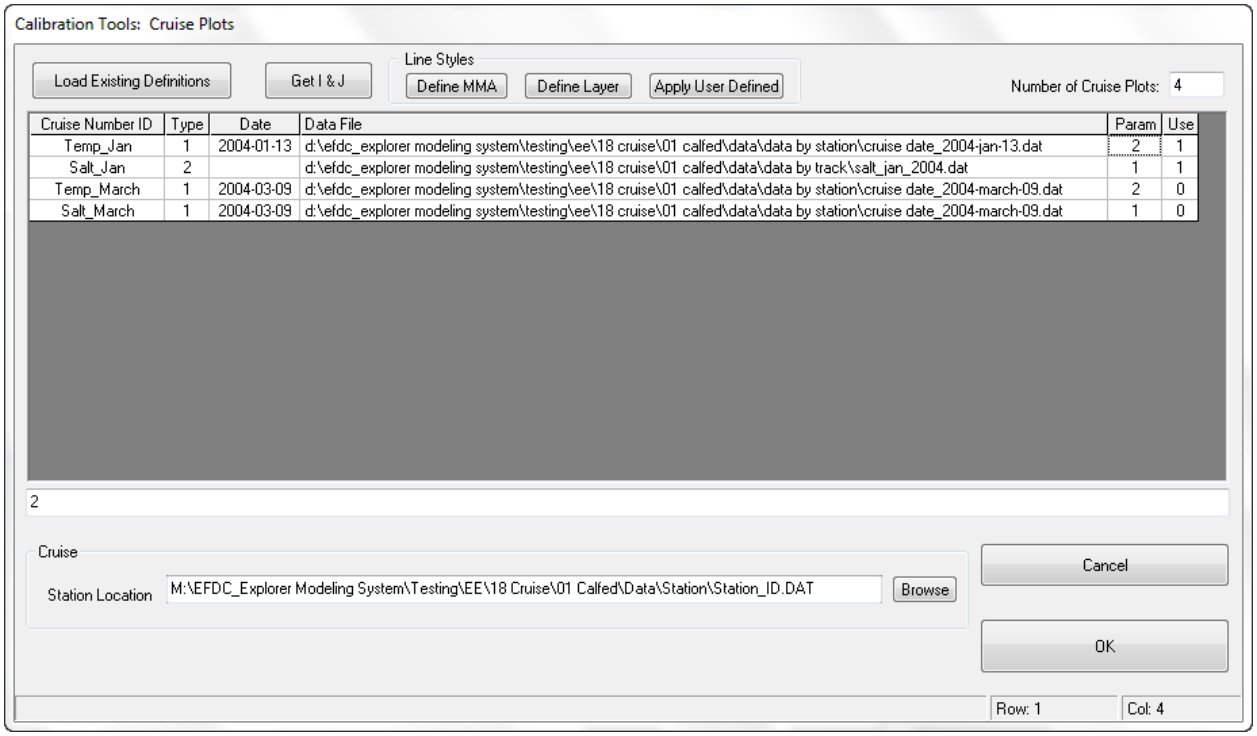


Figure 7-17 Cruise Plot Comparisons – Setting Definitions and Date File Format.

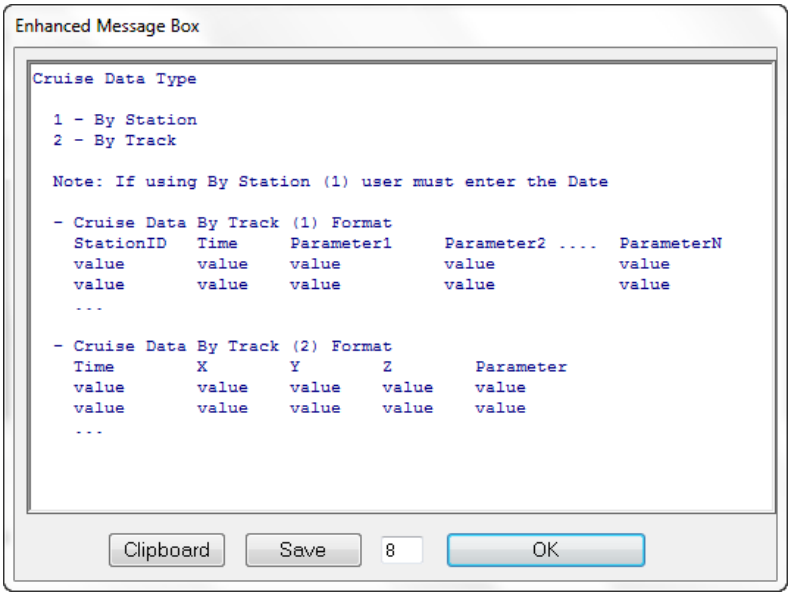


Figure 7-18 Cruise Plot Comparisons – Time Series Plot.

An example *Cruise Plot* is shown below (Figure 7-20) for two parameters: temperature (parameter 2) and salinity (ppt) (parameter 1). Figure 7-19 shows a cruise plot track in San Francisco Bay. Figure 7-20 shows a measured salinity track data from the USGS website and underneath the model output for a study on the San Joaquin – Sacramento Delta.

The colored contour represents the longitudinal and vertical distributions of salinity from the bottom to the surface of the water column. Red represents high salinity concentration, and

blue represents low salinity concentration. Color patterns from left to right represent longitudinal variability, following the cruise line. Color variations from top to bottom show the vertical variability of salinity from the water surface to the bottom. In the cruise plot, all the major locations that are located on the cruise path are displayed as the vertical dashed line.

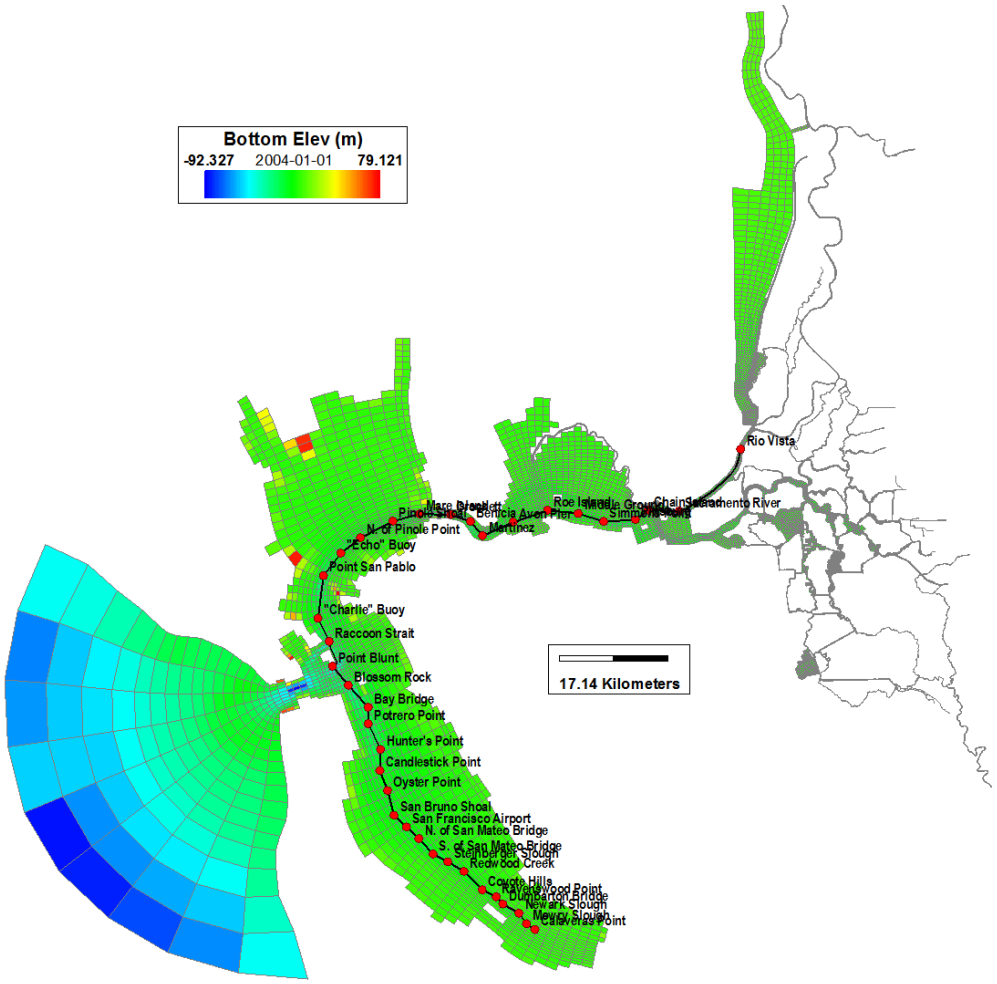


Figure 7-19 Cruise Plot Track in ViewPlan.

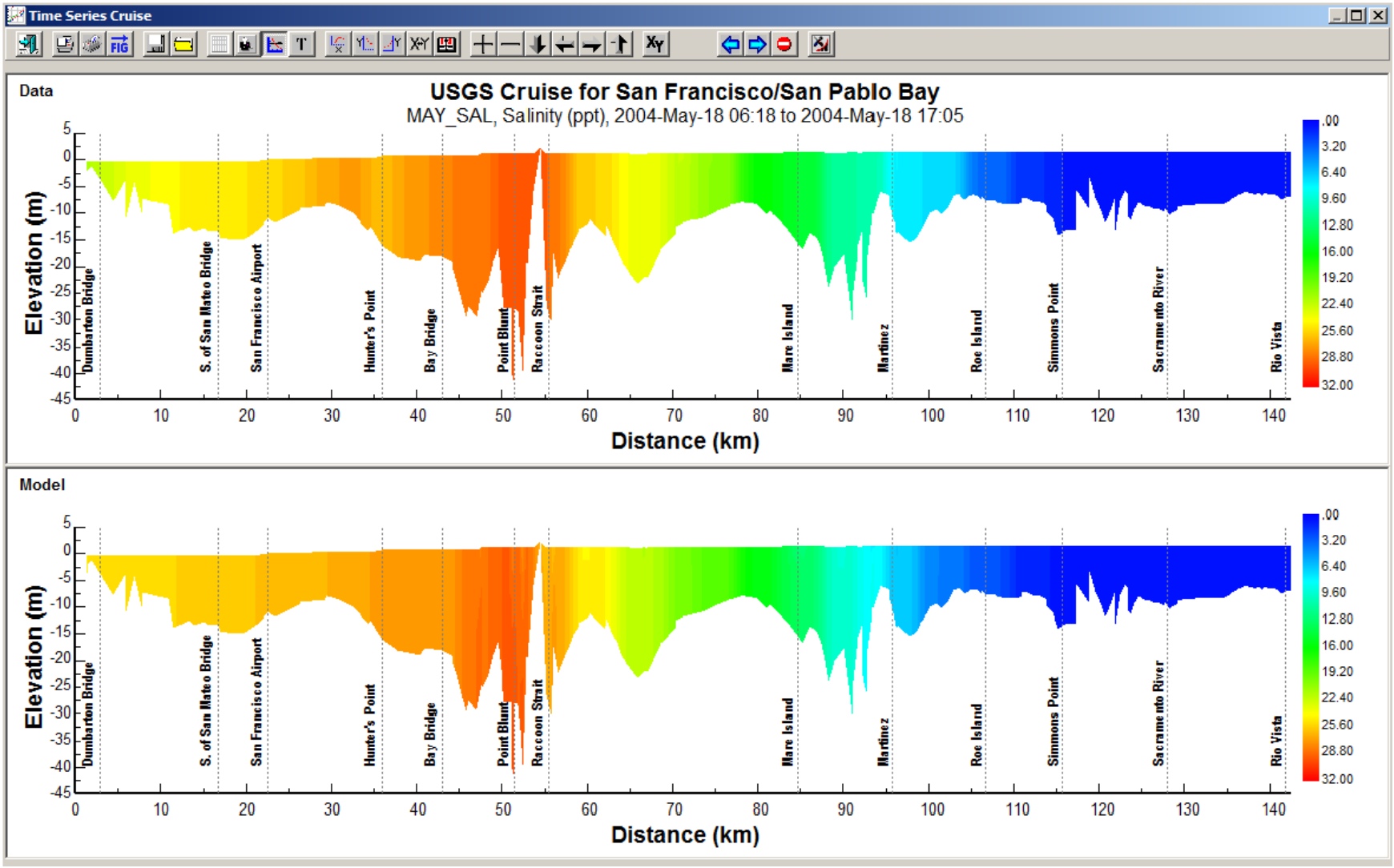


Figure 7-20 Cruise Plot Comparisons – Time Series Plot.

7.1.8. Load 2D Measured Data

The *Load 2D Measured Data* button brings up the form displayed in Figure 7-21. Currently, only velocities can be loaded here, but this feature will be expanded as resources become available. The options area is context sensitive to the option selected for importing.

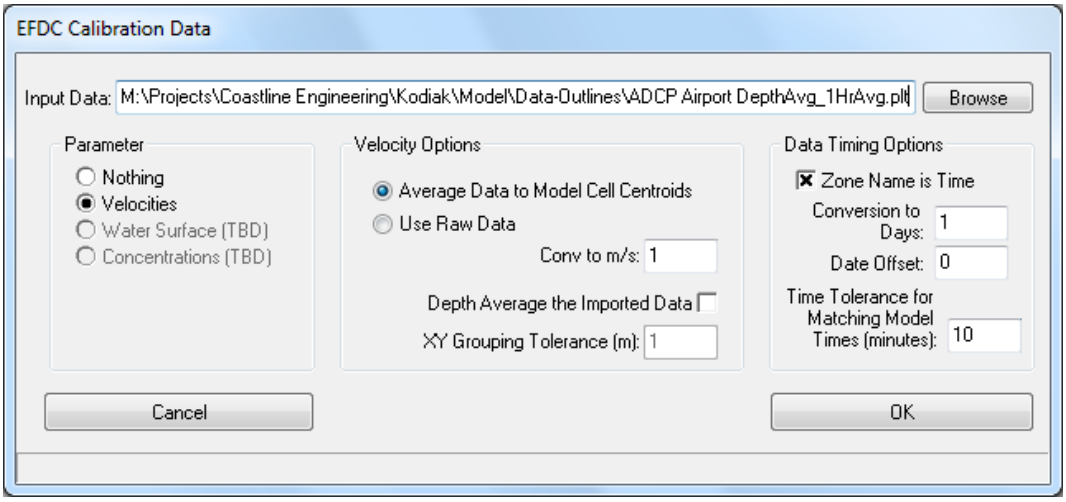


Figure 7-21 Loading measured 2D calibration data.

The data format used for the 2D data currently is the ASCII PLT format used by Tecplot®. An I, J and K 3D regular grid or a series of discrete measured 3D vectors (X, Y, Z, Vx, Vy, Vz) can be input. Pre-processing of the data is expected to be completed so that the data are ready for direct comparison to the model results. For example, if Acoustic Doppler Current Profiler (ADCP) data are being used, the spatial and temporal averaging must be done prior to importing to average the Reynolds stresses (i.e., turbulent fluctuations). One exception to this is EFDC_Explorer has the ability to average the data into the matching EFDC cells.

If the Tecplot formatted data includes multiple depths for a single location, these data can be displayed using the *ViewPlan* layering options or the user can select the *Depth Average the Imported Data* option to collapse all the data into a single X and Y component for each time. Multiple stations can be included in the same file. EFDC_Explorer reports on the data loaded. Please review this summary to ensure that it corresponds to the expected (i.e. the users understanding of the data in the Tecplot file) data.

The *Data Timing Options* frame provides some options of how EFDC_Explorer will interpret the data loaded and convert to the EFDC time reference. If varying time data is to be entered, the Tecplot Zone ID should be used to contain the data time (e.g. Zone=”15-Mar-2008 14:45”). If there is more than one zone in the Tecplot data file, during the viewing in *ViewPlan*, the data will only be displayed if the EFDC model time is within the *Time Tolerance* of the measured data.

Once imported, these data are available for plotting and statistical analysis via the *ViewPlan* | *Velocities* option. To view and calculate statistics for both the model and the data, make sure the *Show Comparison* check box is checked. Figure 7-22 shows an example of two ADCP stations (data shown in red) located in a coastal region being compare to the EFDC model results (model vectors in blue). These model-data comparisons can be animated, if desired.

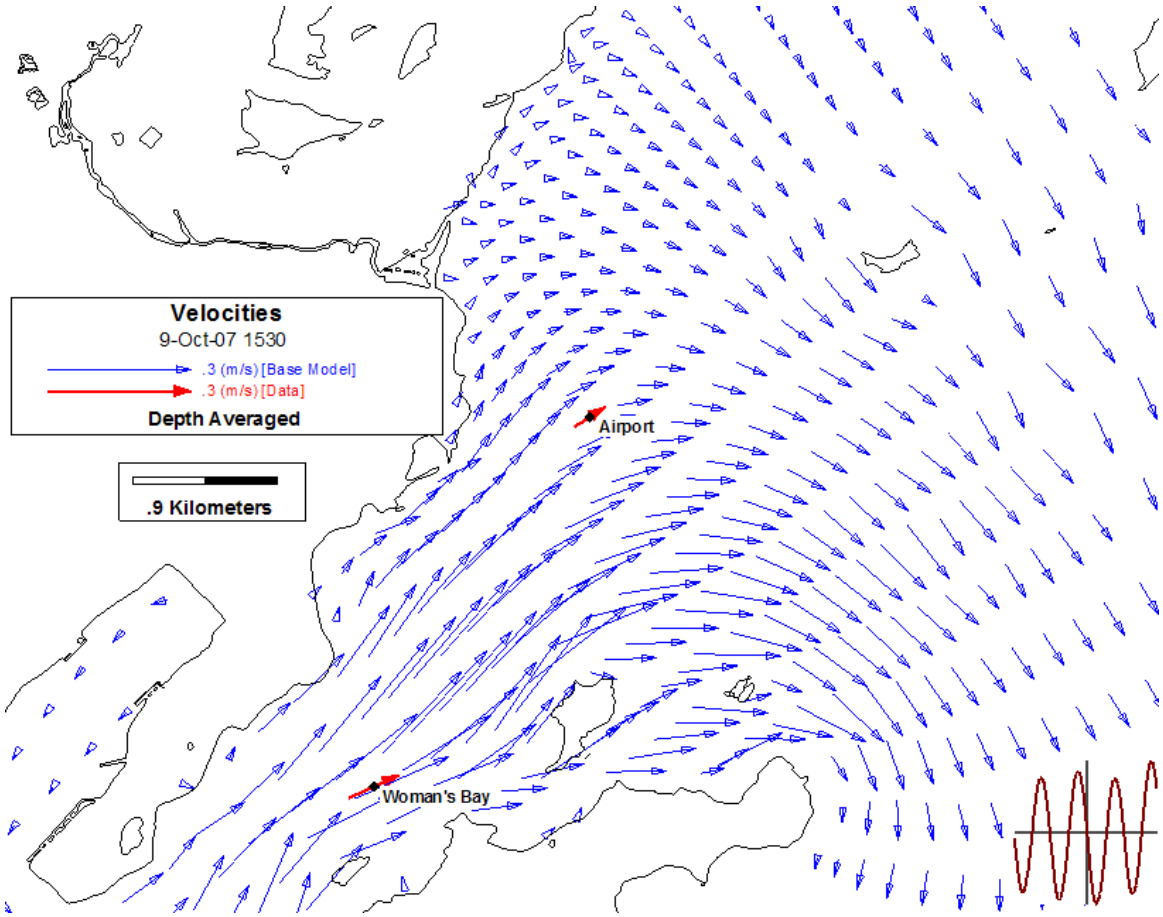


Figure 7-22 Example 2D velocity data comparison.

7.2 Saved Plots Tool

The saved plots function allows the user to apply EE in command line mode for doing batch analyses. This is accessed in *Model Analysis* | *Saved Plots* as shown in Figure 7-23. Saved plots can be generated for plots the user has created in *ViewPlan, View3D, ViewSlice*, *and 2D Longitudinal Profile Plots.*

In general, when viewing a model for which the user wants to create a standardized plot every time they view it, they first set up the desired viewing mode. For example for *ViewPlan* plots, the user may want to specify the size of the frame, the location the legend and timing frame. When this is setup to the user's satisfaction they should press *Alt-K* and save the exact layout to a file, and use *Alt-L* to reload it later. This saves a file with extension .V2D in *ViewPlan* and .V3D in *View3D* (in versions prior to EE8 this extension was .EEV). However, it should be noted these saved plots do not record the constituent viewed or the model time.

When the user wishes to use a fixed time and constituent for a whole series of plots the *Saved Plots Function* is required as shown in Figure 7-23. This is selected by pressing the *Define/Edit* button which displays the *Saved Plots* form shown in Figure 7-24. Here the user sets the ID for the constituent, the pathname for the folder containing the .V2D file, time for the snapshot and whether to use this plot or not. After the user selects *Generate Plots* all the plots will be saved in a common folder for easy comparison.

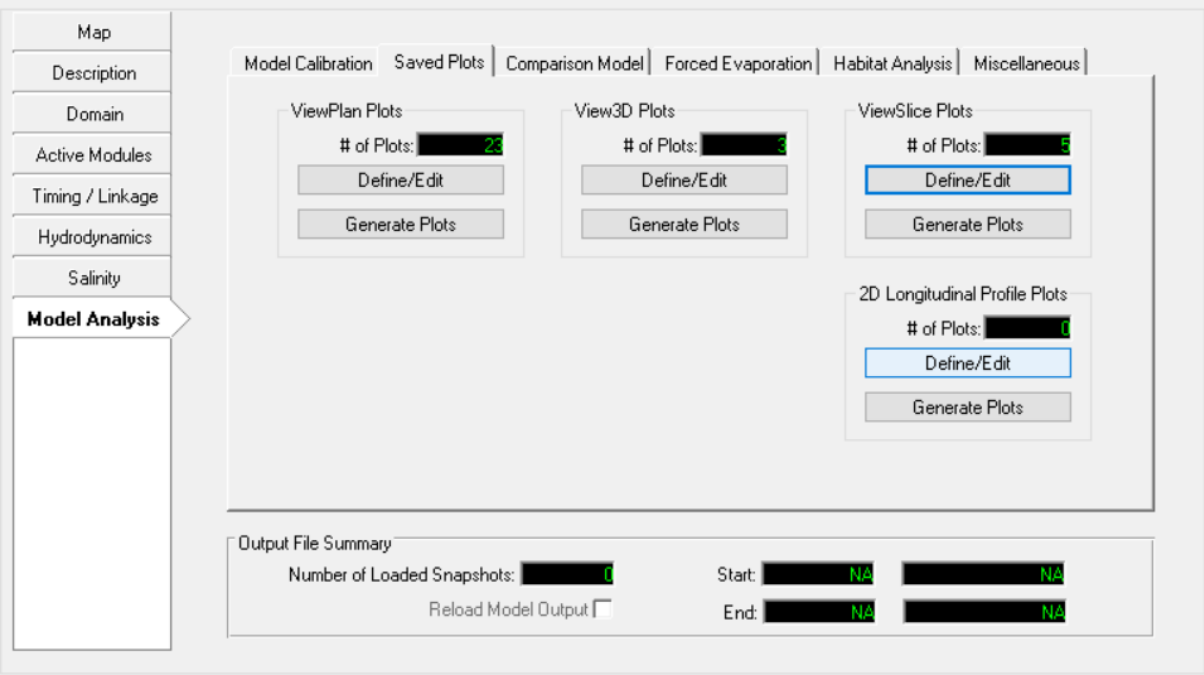


Figure 7-23 Model Analysis: Saved Plots Tool.

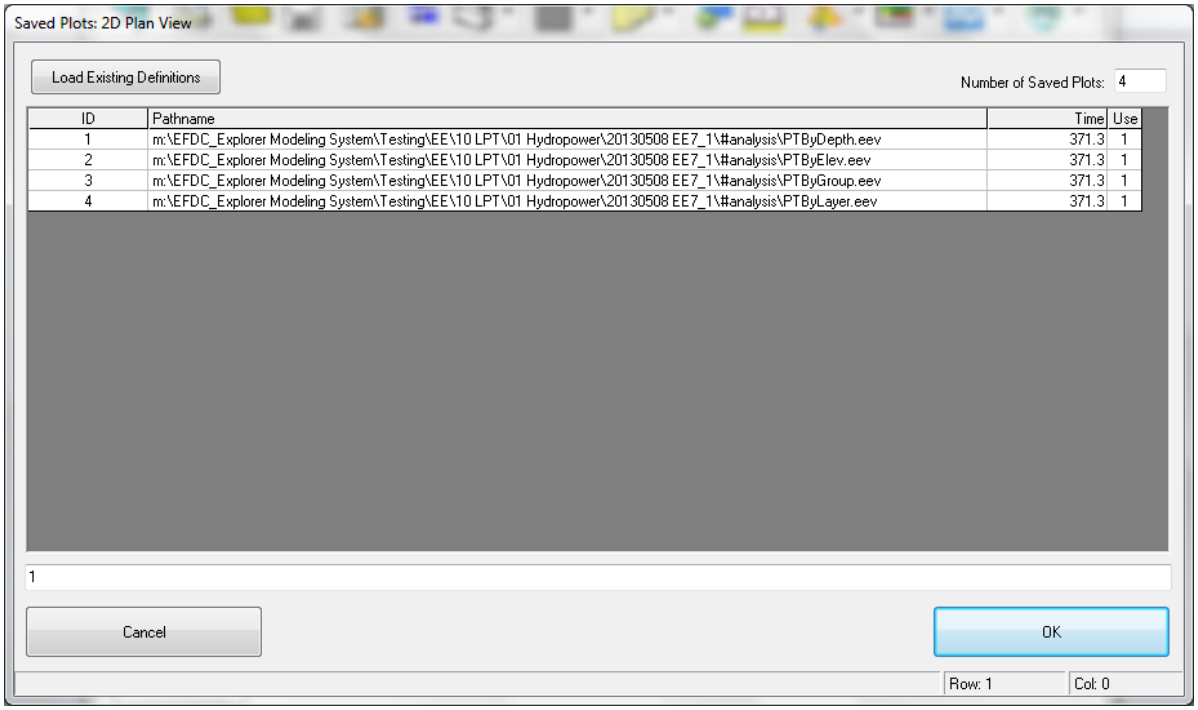


Figure 7-24 Model Analysis: Saved Plots definition form

7.3 Model Comparisons

The *Model Comparisons* tab shown in Figure 7-25 provides the user with access to the compare model input (currently limited to bathymetry) and model results (currently limited to water surface and velocities) functions.

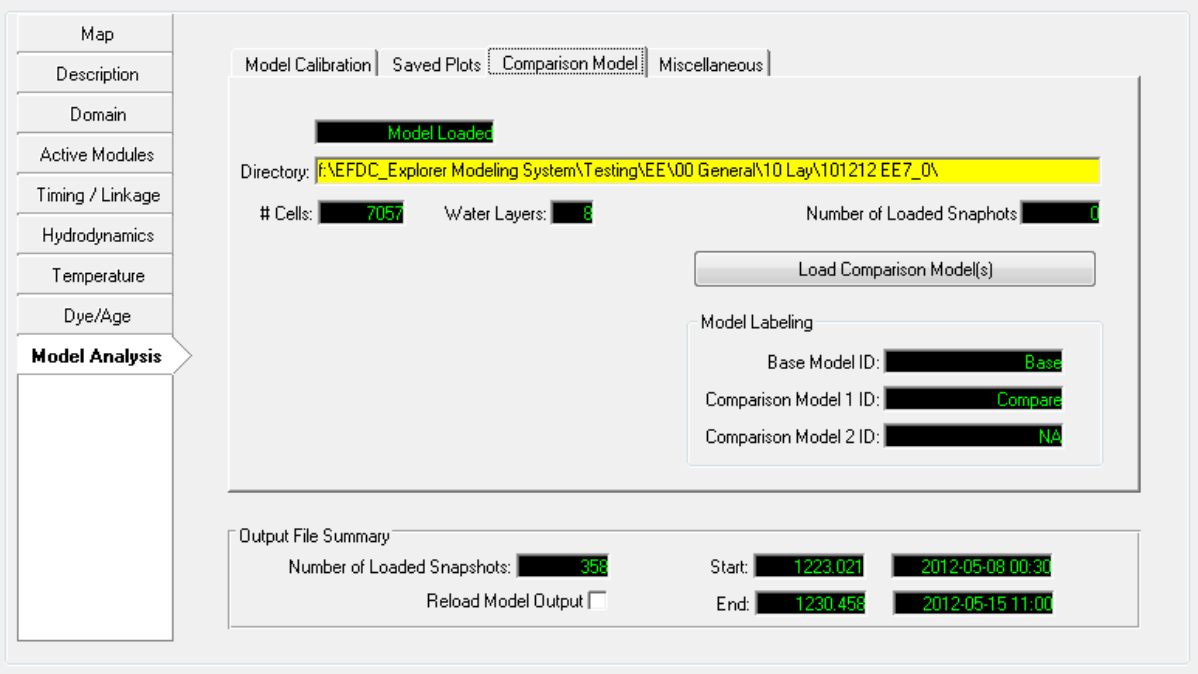


Figure 7-25 Model Analysis Tab: Comparison Data.

7.3.1. Load Comparison Model

The *Load Comparison Model* button allows the user to load a comparison model into EFDC_Explorer for plotting a range of comparisons between the two models. Figure 7-26 shows the model comparison options form. Currently, this option is functional for comparing bottom elevations, water levels, fixed parameters, velocities, sediment bed, water column and inundation extents. The results of any model comparisons are available in the appropriate viewing option of *ViewPlan* by pressing alt-M as shown in Figure 7-27.

The currently loaded model is the *Base* model. Use the browse to the project directory and then select the *Compare* model desired.

To only compare model bathymetry between models the current project's output does not need to be loaded. However, to compare the water depths and/or velocity fields between two models the output data for the current project must be loaded. If the current project's data are loaded, EFDC_Explorer allows the user to load the water depths and velocities from the *Compare* model.

The *Time Matching Option* frame provides a comparison of the snapshot frequency output between the two models. The *Time Tolerance* input box allows for some slight differences in model output times when comparing time snapshots during the simulation.

Once all the options are set, the user should press the *Load Compare* button. Until this button has been pressed, no input or output data for the Compare model is loaded.

The *Load 3rd Model* button is only used to compare water depths and elevations in *ViewPlan*. The viewing option of *Water Depths/Areal Extents* allows the user to overlay the areal extents and depths (as a function of duration and depths) for up to three scenarios. If using this option, the base model should have the largest extents with the second and/or third model having the smallest extents. A typical example would be to display the areal extents of a 100-year flood (the *Base* model), a 50-year flood *(Compare Model*) and a 10-year flood (*3rd WD* model).

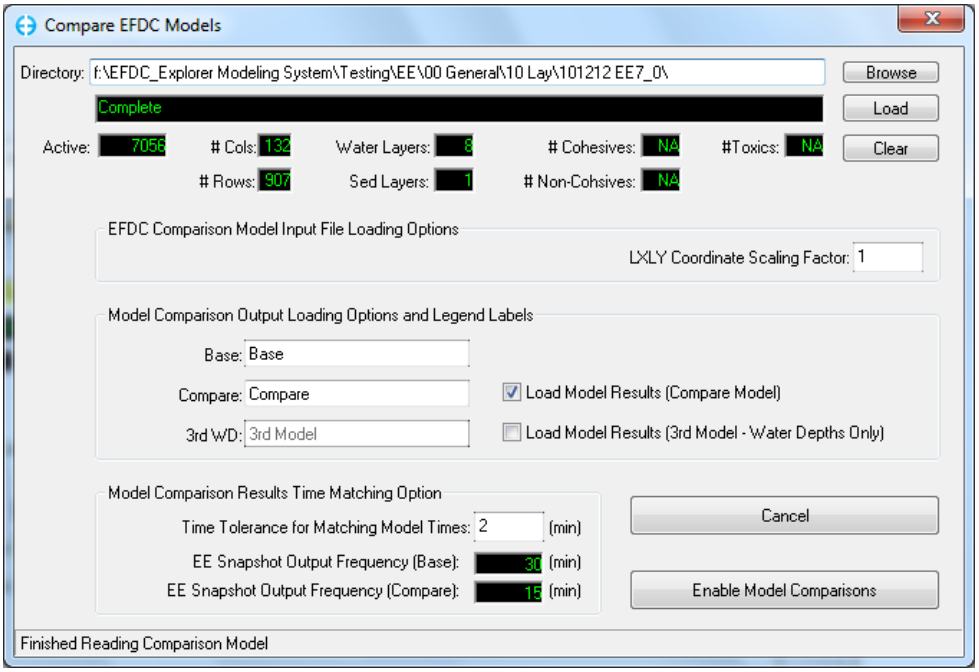


Figure 7-26 Load a comparison EFDC model.

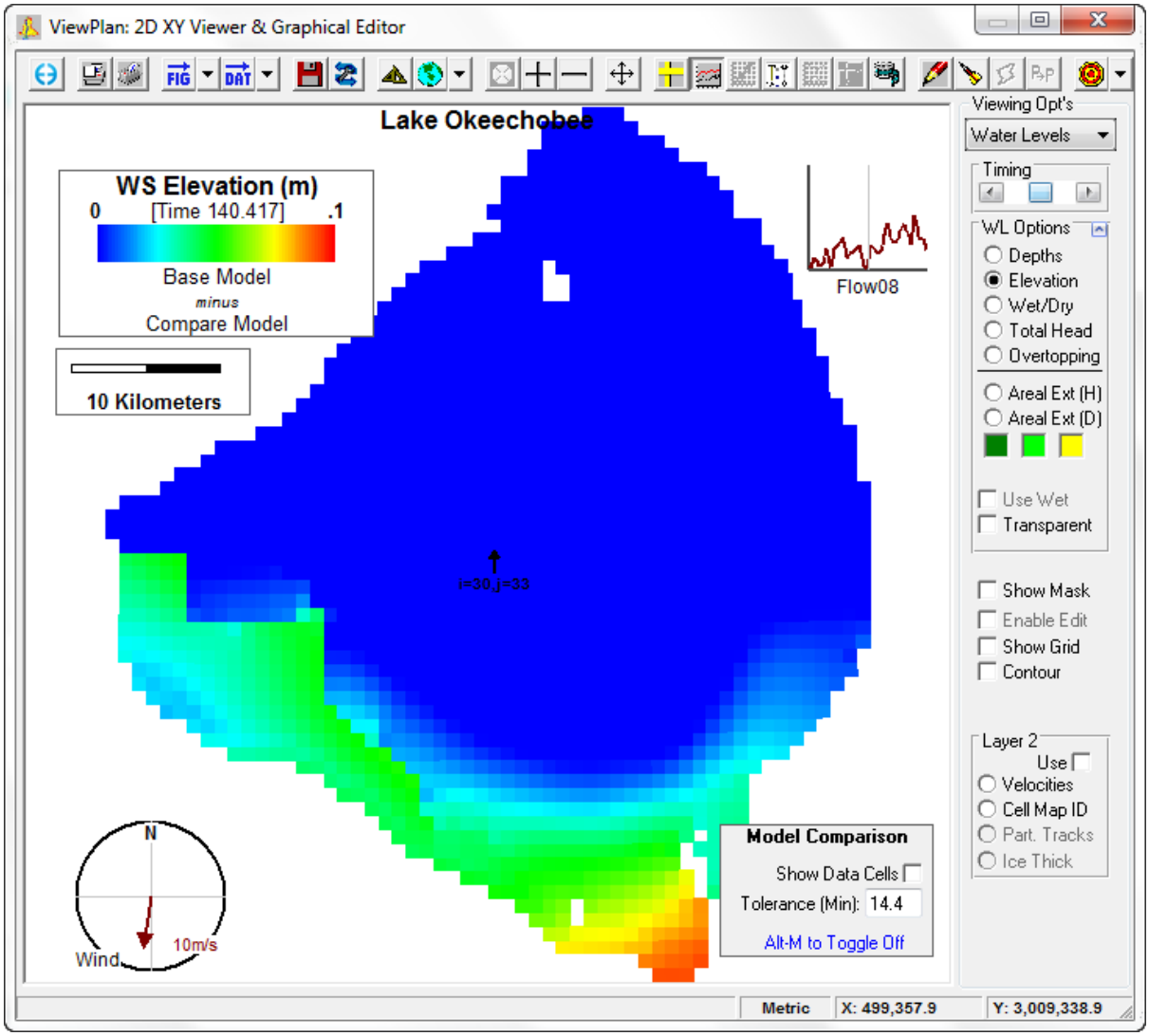


Figure 7-27 ViewPlan: Model Compare for WS Elevation.

When a user has a comparison model loaded it is also possible to view the time series extraction comparison between models. After selecting Alt-M in *ViewPlan* to compare the model results as shown in Figure 7-27, extract the time series Base – Compare models as shown in Figure 7-28.

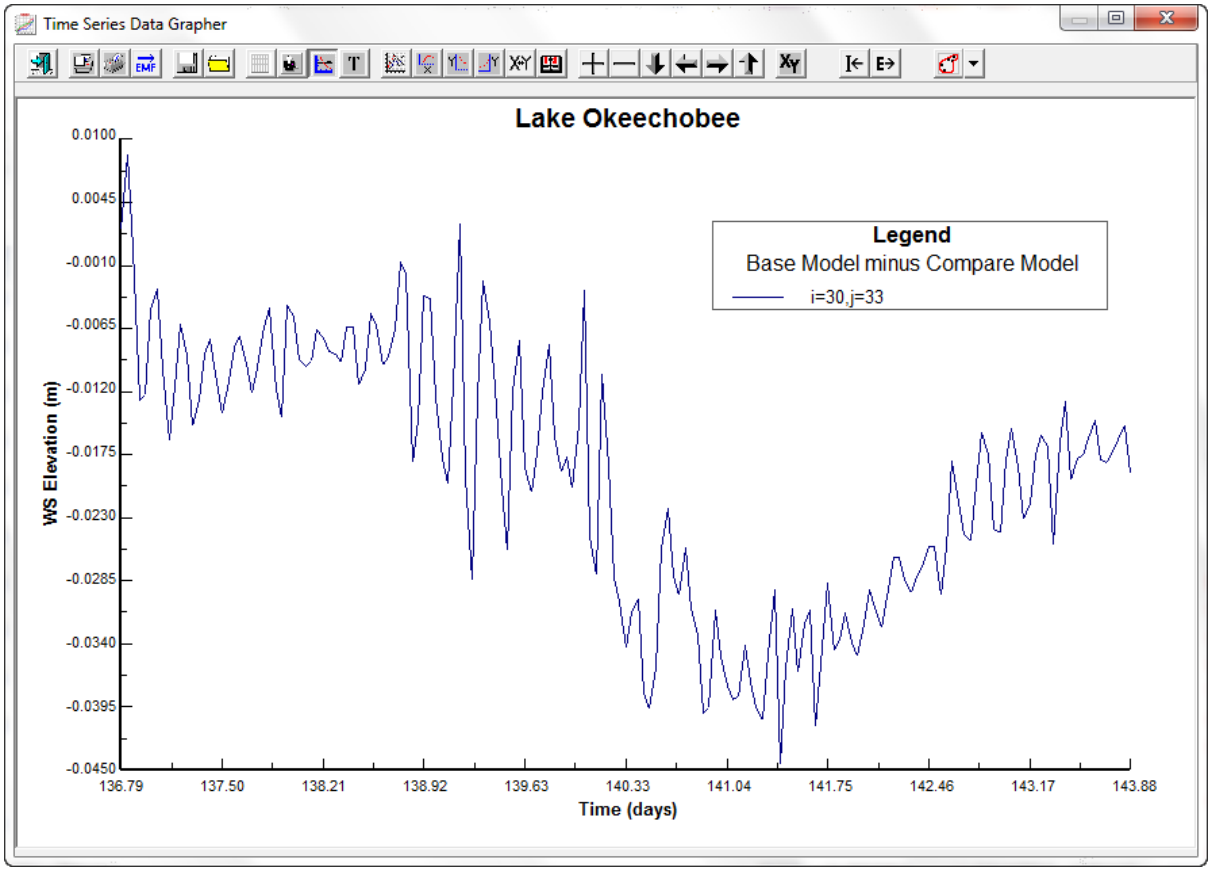


Figure 7-28 Time Series Extraction: Model Compare for WS Elevation.

7.4 Forced Evaporation

The *Forced Evaporation (FE) Analysis* capability has been developed to quantify increased evaporation induced by increased water temperatures due to releases from thermoelectric power plants. These power plants withdraw cooling waters, which once run through the plant and are returned/discharged to rivers or lakes at a higher temperature than the ambient water temperature. This higher temperature water causes additional evaporation (forced evaporation) from the river or lake. This additional evaporation is counted as water consumption by regulators as it is no longer available to downstream users.

Evaporation is dependent on wind speed, atmospheric humidity, and water temperature. There are a number of methods to compute FE using different wind functions as listed in Table 7-1. The wind functions are computed using data contained in the ASER file (ASER.INP) in EFDC.

Table 7-1 List of Evaporation Calculation Methods

IEVAP	Evaporation Approach	General Usage
0	Do Not Include Evaporation	
1	Use Evaporation from ASER	Measured or Externally Estimated
2	EFDC Original	
3	Ward, 1980	Cooling Lake
4	Harbeck, 1964	Cooling Lake
5	Brady et al, 1969	Cooling Pond
6	Anderson, 1954	Large Lake
7	Webster-Sherman, 1995	Lakes
8	Fulford-Sturm, 1984	Rivers
9	Gulliver-Stefan, 1986	Streams
10	Edinger et al., 1974	Lakes/Rivers
11	Ryan-Harleman 1974	Lakes/Rivers

Using these various evaporation methods, the model is first run with the power plant, and then run again without the power plant. EE then subtracts the output from two models and displays the difference which is the consumption of water from the power plant. The user may select "with power plant" option to calculate evaporation in the *Temperature* tab in the *Heat/Temperature* frame as shown in Figure 7-29. The evaporation options are only available if temperature is being simulated and the ASER file is used.

Once temperature is activated and the correct *Surface Heat Exchange Sub-Model* option has been selected the user can choose which evaporation approach is desired. Even if evaporative losses are not a major concern, the evaporative mass fluxes should normally be activated for most models.

Heat flux due to evaporation is always included for the *Full Heat* and the *Equilibrium Temperature (W2)* options.

EFDCPlus/EFDC_Explorer Forced Evaporation (FE) toolset results have been compared to the Electric Power Research Institute's (EPRI) FE estimates. EPRI's once through cooling FE analysis for river discharges is based on a USGS report on water consumption by thermoelectric power plants (USGS, 2013; EPRI, 2013).

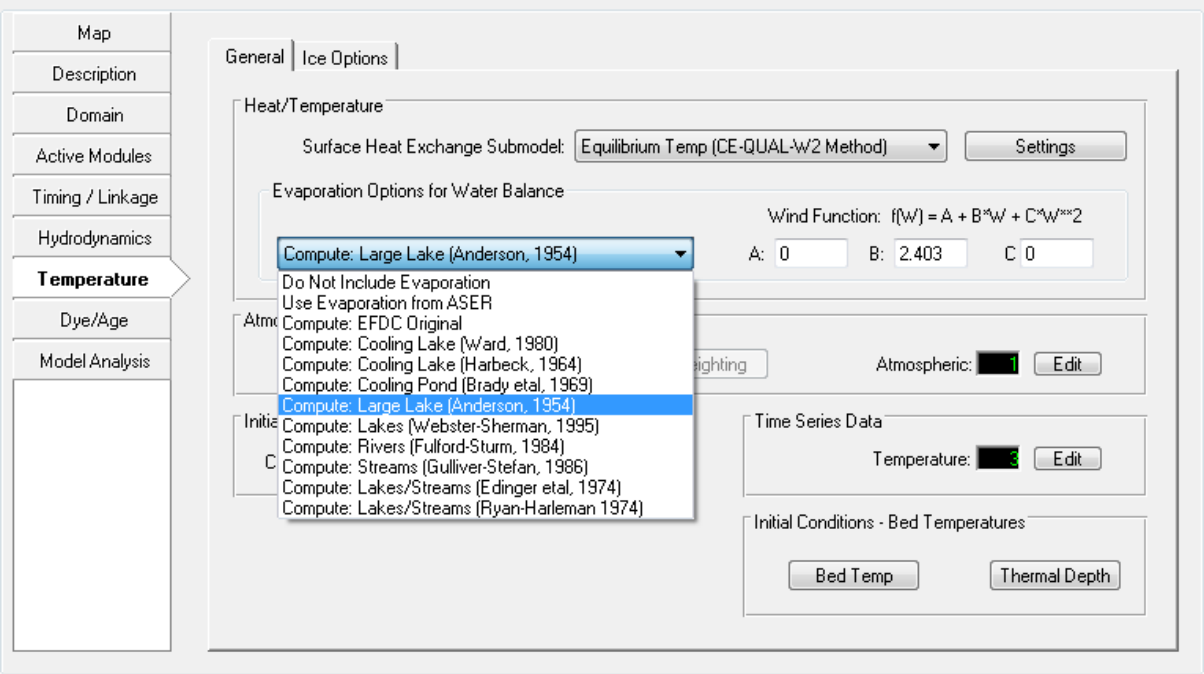


Figure 7-29 Forced Evaporation: Evaporation Options for Water Balance.

Once the models runs have been completed, the user should go to the *Model Analysis* | *Forced Evaporation* tab where the options for model comparison are provided as shown in Figure 7-30.

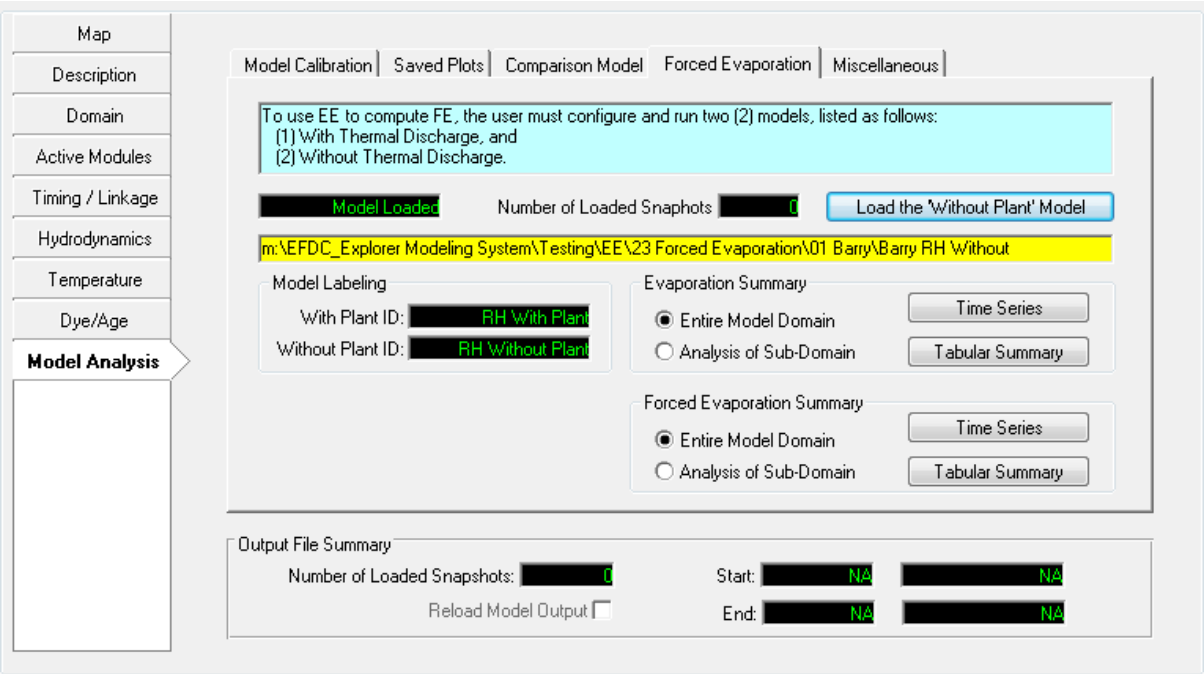


Figure 7-30 Forced Evaporation Options GUI

To display the general instructions on how to conduct a forced evaporation (FE) analysis using EE, click on the blue text box shown in Figure 7-30. EE will display the instructions shown in Figure 7-31.

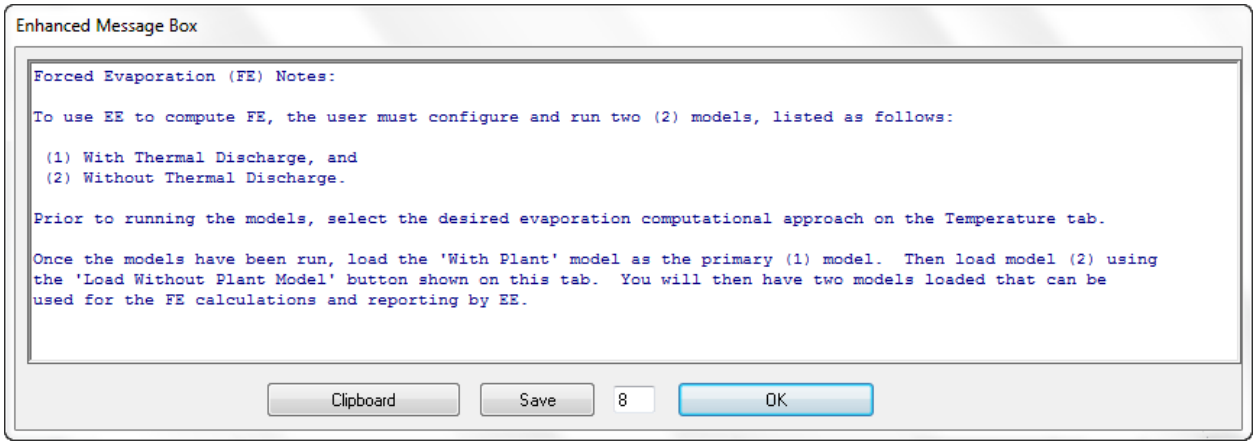


Figure 7-31 Forced Evaporation Analysis Setup Instructions.

As outlined Figure 7-31, the user should then load the *With Plant* model as the primary (1) model. Next load model (2) using the *Load Without Plant Model* button shown in Figure 7-30. This will display the form shown in Figure 7-32. After selecting *Enable Model Comparisons,* two models will have been loaded that can be used for the FE calculations and reporting by EE.

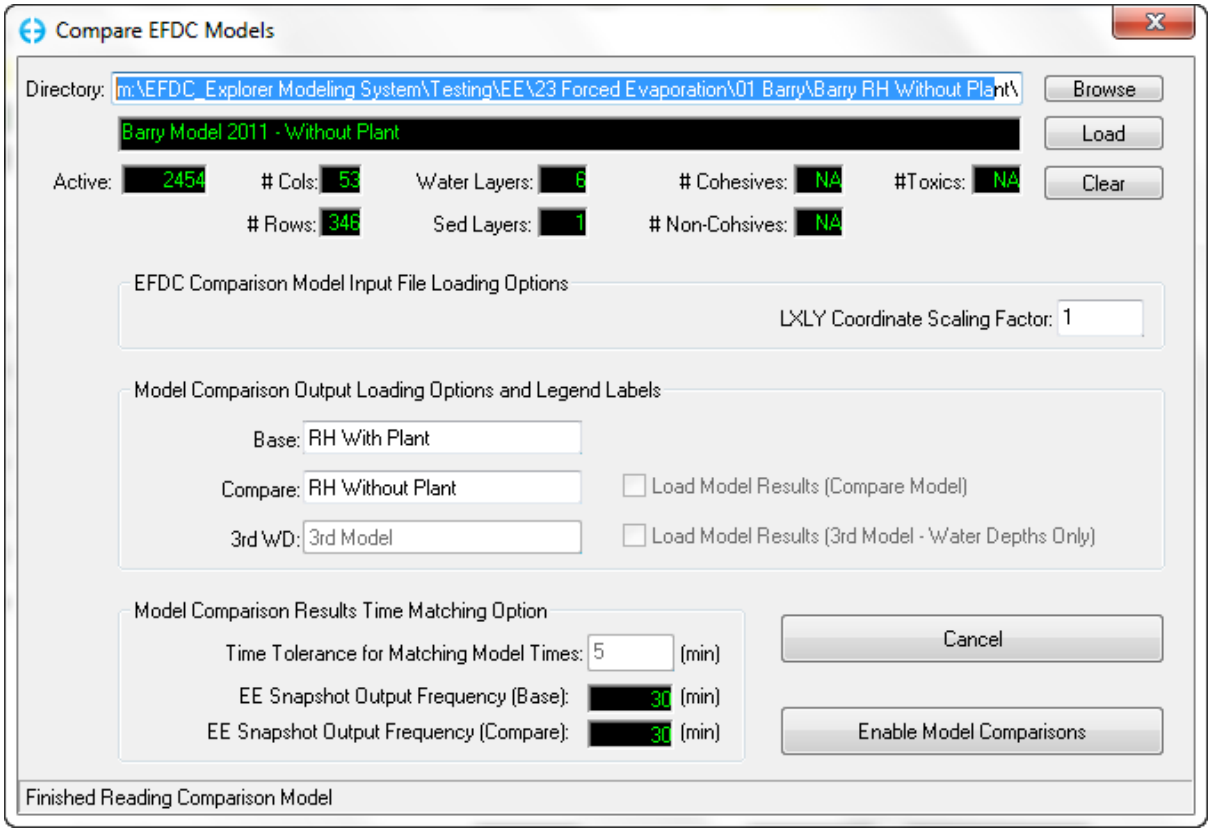


Figure 7-32 Forced Evaporation: Compare Models Options.

The *Model Labelling* frame in Figure 7-30 tells the user the type of FE option that has been selected. In Figure 3-2, “RH with Plant” refers to model run that used Ryan-Harleman approach. Time series plots may be automatically generated for either evaporation (no plant) and forced evaporation (with plant) using the buttons in the respective frames. A plot of forced evaporation (mm/day) and cumulative volume of forced evaporation is shown in Figure 7-33.

The user can also produce summaries of evaporation and forced evaporation using the *Tabular Summary* button as shown in Figure 7-30.

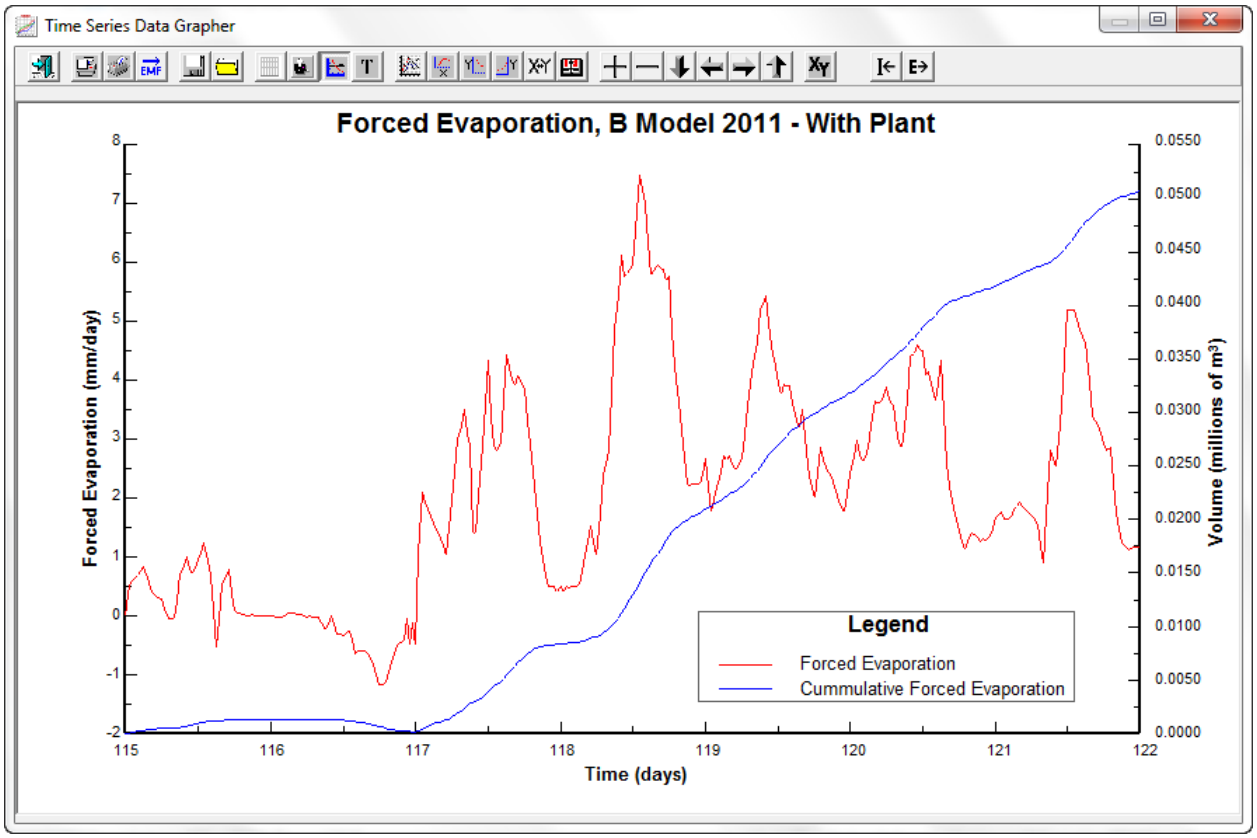


Figure 7-33 Forced Evaporation: Time series using Anderson evaporation approach.

Another way to display the impact of FE between the two models is in *ViewPlan*. Here the user should select the *Volumes* | *Evaporation* viewing option. Selecting Alt M will toggle on the *Model Comparison* tool which allows the user to visualize evaporation/rainfall as "With Plant" minus "Without Plant" models as shown in Figure 7-34.

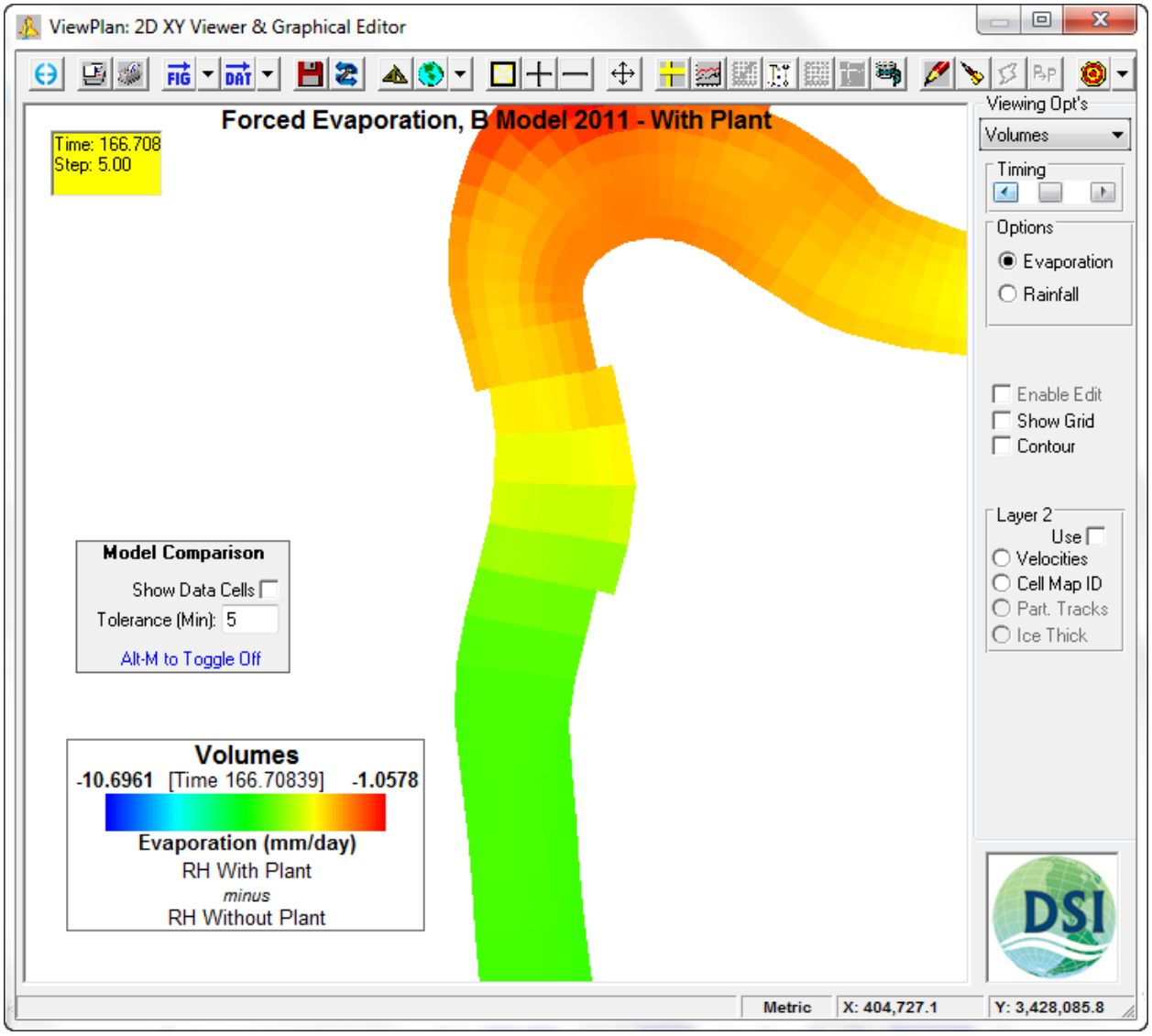


Figure 7-34 Forced Evaporation: Model Comparison using Anderson approach.

Another new and related feature in EE7.3, is the user can now generate time series of the differences in water column results as shown in Figure 7-35. DT means “delta temperature”.

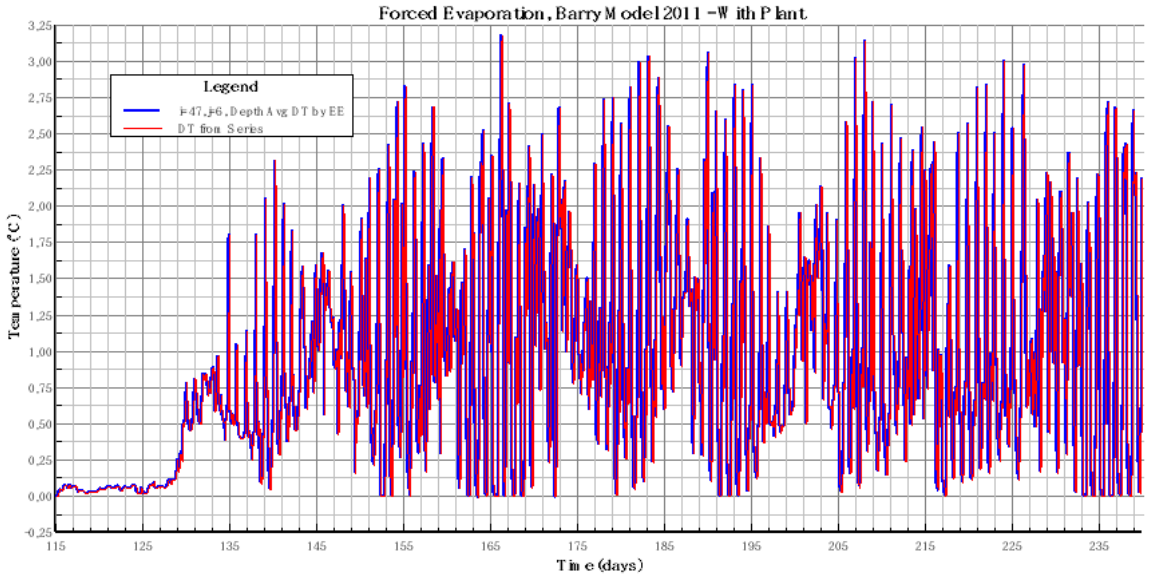


Figure 7-35 Forced Evaporation: Time series of model comparison for temperature.

7.5 Habitat Analysis

Habitat models are designed for a wide variety of planning applications where habitat information is an important consideration in the decision-making process. EEMS now supports two new types of habitat analysis tool. These are in addition to the existing temperature limit and depth criteria tool in *ViewPlan* that was initially implemented to support studies on manatee populations in the Chassahowitzka Estuary. The two new tools generate time series critical limits, as well as support the state-of-the-art Instream Flow Incremental Method. These two methods are described below.

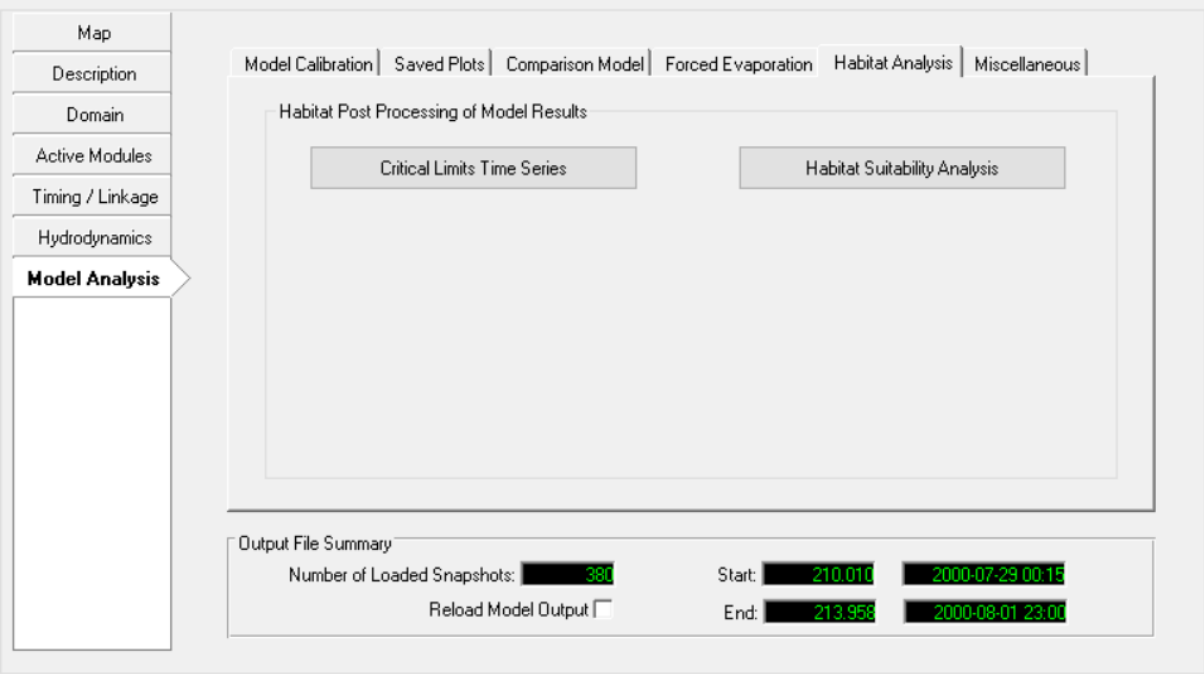


Figure 7-36 Model Analysis: Habitat Analysis form.

7.5.1. Critical Limits Time Series

The *Critical Limits Series* form is shown in Figure 7-37. The user may define up to five different parameters for the critical limits. These can be selected from any parameter that is currently activated in EE. For each dropdown list under *Parameter* the user selects the parameter of interest and then defines the minimum and maximum values for the critical limits. In the layer box a range of options are provided: single layer (enter layer number), a range of layers (eg 1-2), average of a series of layer (eg 2:6), or depth averaged (0). Once criteria for critical limits have been entered it is recommended the user save the settings. Selecting the *Save* button, the user is prompted to input the file name. EE will create a folder called #habitat to store all related files, so by default the new critical limits will be saved into this folder. Configuration files for critical limits series use ".HCL" as the extension.

The *Criteria ID* will be used to manage the different species. The *Criteria ID* is appended to the file name "Hab_Results" of the data file extracted when generating an initial time series, as described further below.

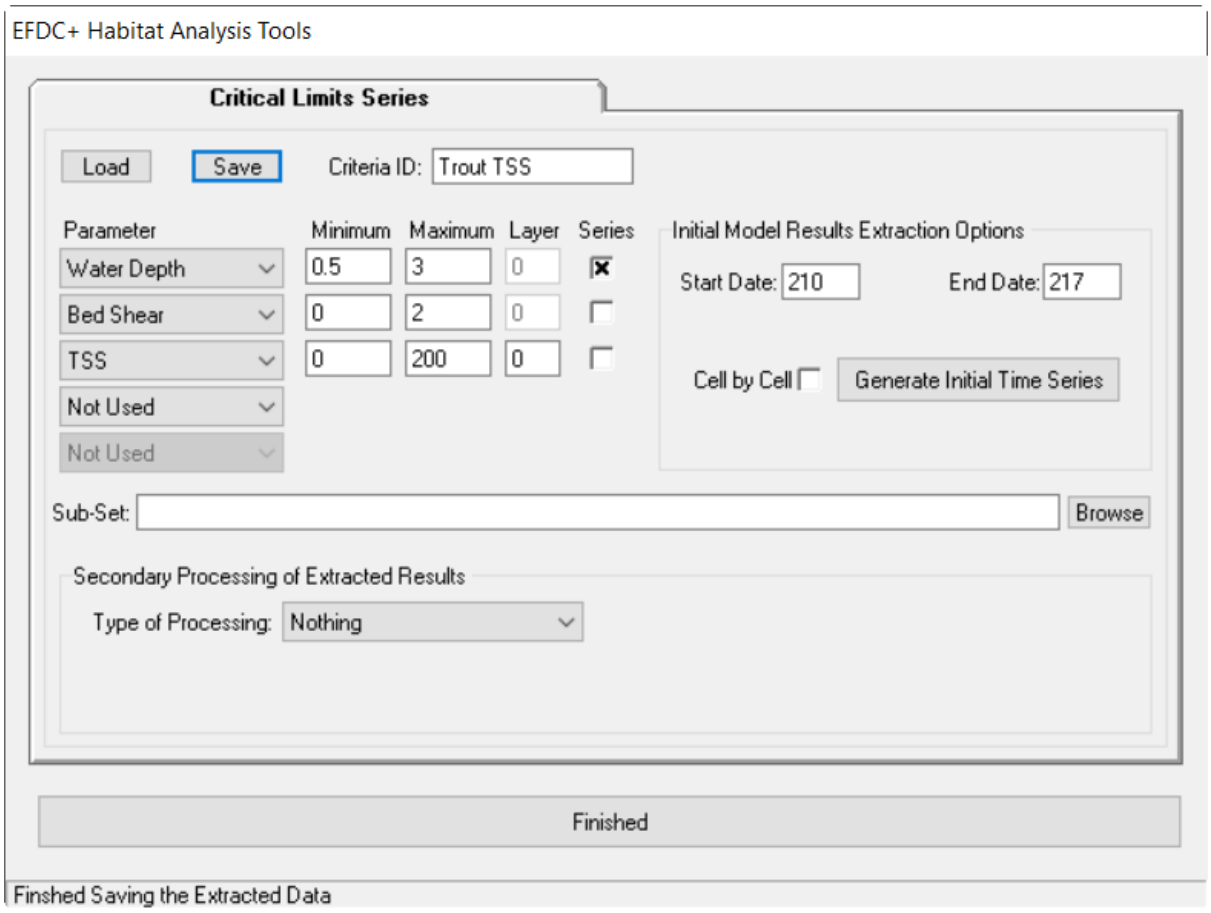


Figure 7-37 Habitat Analysis: Critical Limits Series form.

The critical limits options act as a series of toggles for each cell in each layer of the model domain. Each cell will meet either some of the criteria, or all of criteria defined. If a cell doesn't meet the all the criteria it is set to zero, whereas if a cell does meet all the criteria it is set to one. If a cell is dry it is also set to zero. Using the *Series* checkbox, the user turns on the generation of output for each parameter. Only one parameter may be output at a time.

The user should select the start and end date for the time series extraction in the *Initial Model Results Extraction Options* frame. Once this has been set the user can select the *Generate Initial Time Series* button. EE will then tell the user how many cells are in the domain and

check if the user wants to proceed with calculating the time series. Selecting the *Cell by Cell* check-box will mean that every cell is calculated for every time step. Note that it is also possible to generate the time series based on only a sub-set of the model domain by browsing to and selecting a polyline file with the *Sub-Set* box.

EE will extract all the cell information for each parameter and save it in a post processing file in the #habitat folder based on the criteria ID. In this example the file will be called Hab_Results_Trout01.dat. This file records the composite time, salinity, net weighted criteria (weighted average 0 and 1 of all the cells that meet the bathymetry criteria), area, and volume.

An example of the time series that EE displays is shown in Figure 7-38.

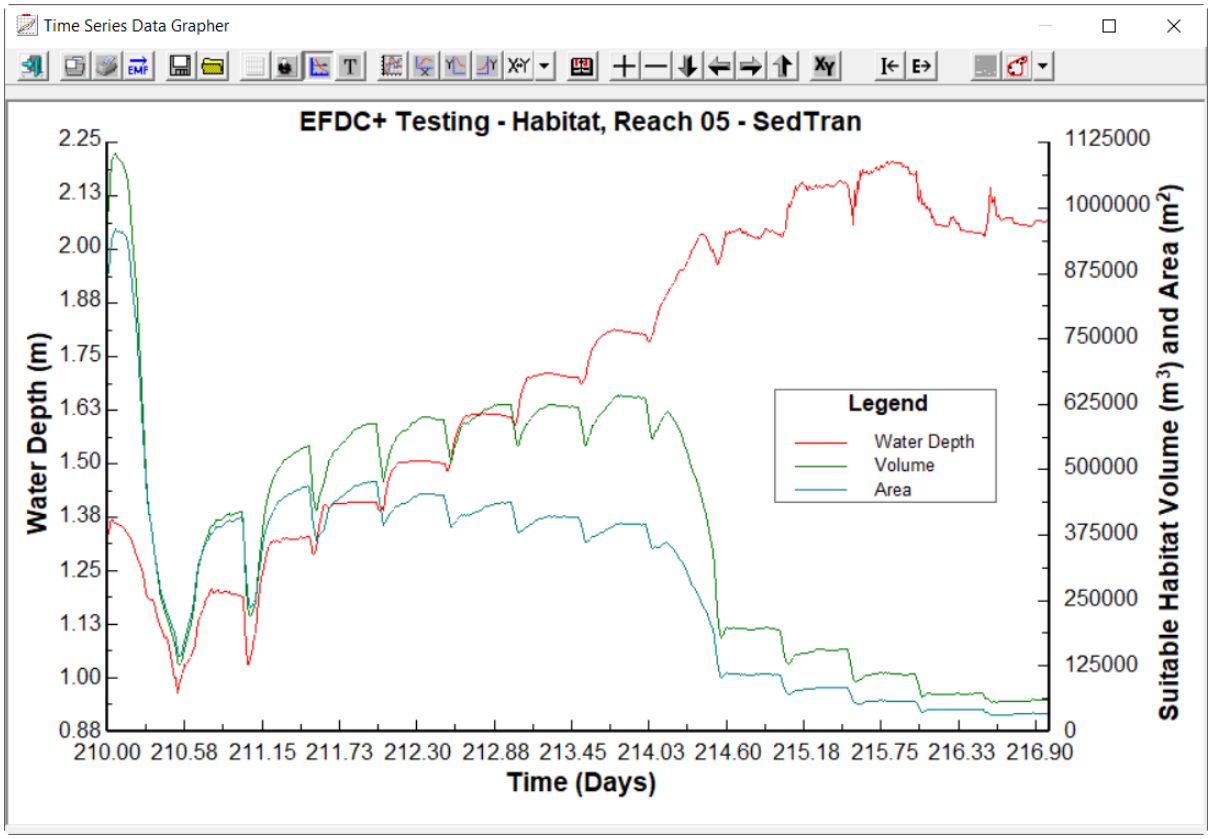


Figure 7-38 Habitat Analysis: Critical Limits Series time series.

The results in the time series plots present a composite time series of velocity and the volume and area that met these criteria. The user should export the time series as it is not automatically saved by EE.

Once a time series has been generated, and the post-processing files have been extracted, a new frame is displayed, *Secondary Processing of Extracted Results.* This option provides the user with further post-processing options. The drop-down menu has the following options: *Domain Average, Running Averages*, and *Single Cell Time Series*. For running averages, the user must specify the number of days to be applied for calculation of the running average.

7.5.2. Habitat Suitability Criteria Analysis

The Habitat Suitability Criteria (HSC) models provide an objective, quantifiable method of assessing the existing habitat conditions for a given aquatic species within a study area by measuring how well each habitat variable meets the habitat requirements of the species by life stage. This serves as an objective basis for predicting probable project impacts, documenting post-project impacts, and guiding habitat protection, mitigation, enhancement, and management decision.

Instream Flow Incremental Method (IFIM) is a method of determining the HSC, and is one of the most widely used instruments in the world for assessing the effect of flow manipulation on river habitats. Originally developed by the US Fish and Wildlife Service (Bovee, 1982), USA EPA describes IFIM as integrating "concepts of water-supply planning, analytical hydraulic engineering models, and empirically derived habitat-versus-flow functions to address water-use and instream-flow issues and questions concerning life-stage-specific effects on selected species and the general well-being of aquatic biological populations. The method has undergone continual refinement by the Fort Collins Science Center scientists and has remained a state-of-the-art tool. A major component of IFIM is a collection of computer models called the Physical Habitat Simulation Model (PHABSIM). This model incorporates hydrology, stream morphology, and microhabitat preferences to determine relations between streamflow and habitat availability" (EPA, 2018).

At this stage, however, the PHABSIM models to not simulate water quality. PHABSIM concentrates on simulation of physical habitat based on depth, velocity, and a channel index. In their manual on PHABSIM, Milhous and Waddle (2012) refer users to other sets of models, such as the QUAL-2 model for approaches to simulate water quality constituents. Furthermore, PHABSIM, River2D and similar IFIM models are restricted to 2D simulation. With EEMS now enhanced to perform the IFIM analysis, it serves as a fully highly advanced 3D PHABSIM, incorporating water quality, sediment, toxics and many other highly relevant sub-models.

Habitat availability with IFIM is measured using an index called the weighted useable area (WUA). This is the wetted area of a stream weighted by its suitability for use by an organism. In EE WUA is calculated using the product of the habitat suitability indices, which are further described below:

$$WUA = \sum_L A_L C_L \prod_1^n s_i$$

Where

- A_L is the area of the cell
- C_L is the habitat suitability index for the channel index associated with the cell substrate and/or vegetation, if not used or needed a value of 1.0 is used,
- L is the cell index
- S_i is the habitat suitability index for parameter i for up to n parameters.

Another common measure is weighted usable volume (WUV),

$$WUV = \sum_L V_L C_L \prod_1^n s_i$$

Where

- V_L is the volume of the cell

The interface for the IFIM approach to configuring habitat suitability criteria in EE is shown in Figure 7-39. When saving and loading the configuration files for habitat suitability criteria EE uses ".HSC" as the extension.

The *Criteria ID* will be used to manage the different species. The *Criteria ID* is appended to the file name "HSC_Results" of the data file extracted when generating an initial time series. For the example shown in Figure 7-39 the extracted file will be called "HSC_Results_Walleye – Open Water.dat".

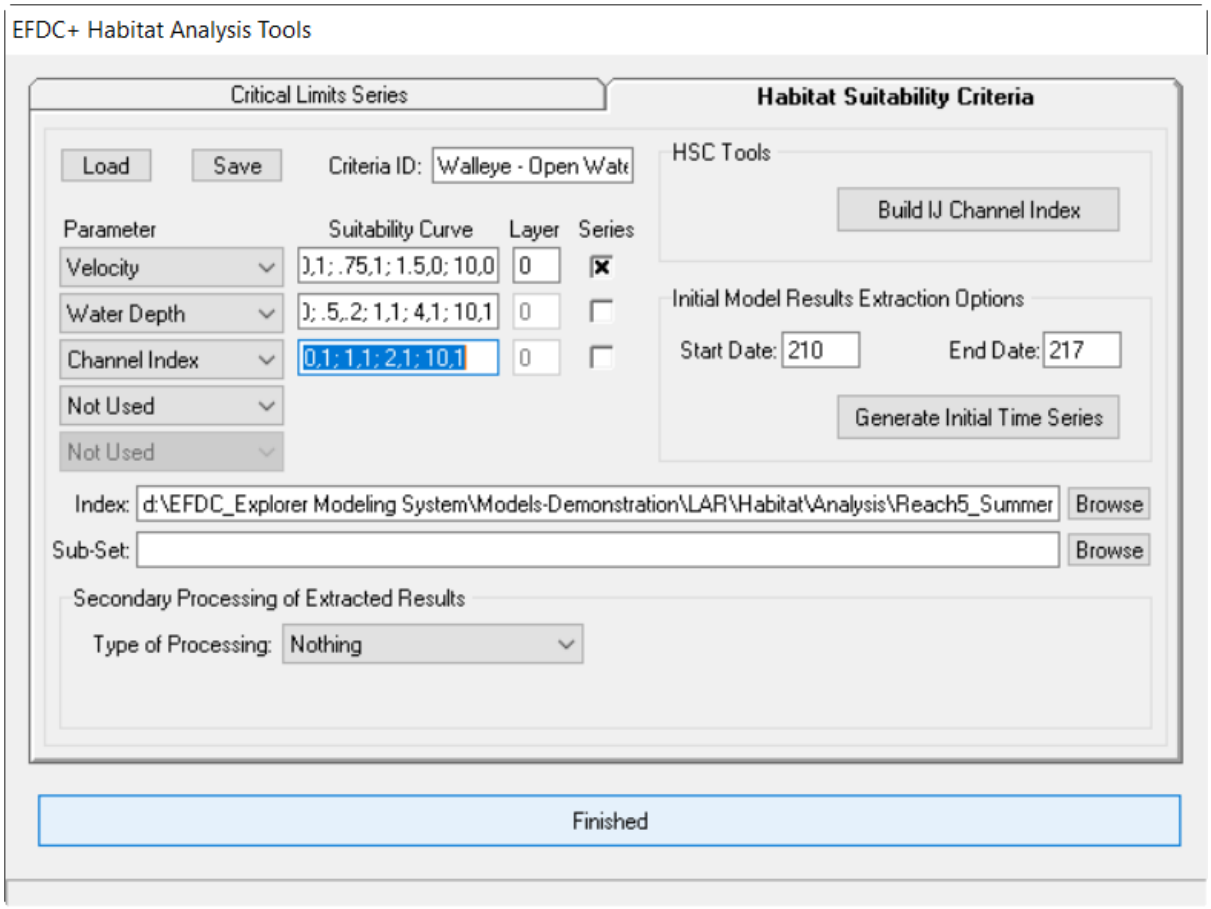


Figure 7-39 Model Analysis: Habitat Suitability Criteria form.

Parameters that may be defined include bottom elevation, water depth, bed shear, channel index, velocity, fraction of light, cohesives, and TSS. The general approach for calculating the weighted usable area (WUA) for habitat is quite similar to that for the critical limits approach. One significant difference in the model setup is that the inflows are incremental, or stepped in IFIM approach. There are two main differences in the post-processing of the data: defining suitability index curves, and specifying a channel index. These are described below.

7.5.2.1 *Suitability Index Curves*

Rather than using simple maximum and minimum limits for a parameter, as was the case for critical limits, the HSI approach uses Suitability Index (SI) curves. These SI curves are constructed by quantifying field and laboratory information on the effect of each habitat variable, such as temperature, dissolved oxygen, spawning gravel size, and siltation, on the growth, survival, or biomass of a species by life stage. The SI curves assume that increments of growth, survival, or biomass plotted on the y-axis can be directly converted into an index of suitability on the x-axis. The index of suitability ranges from 0.0 to 1.0 with 0.0 indicating unsuitable conditions and 1.0 optimal conditions. Measurements of each habitat variable taken at the proper time in the field can be applied to the SI curves to assess the suitability of the variable in meeting the habitat requirements of the species by life stage

To facilitate the input of these curves, EE provides a tool to create and view SI curves. After selecting a parameter from the dropdown menu, selecting the *Suitability Curves* text box and pressing *Enter* will open the *XY Data Editing Tool*, which is a spreadsheet type form with a corresponding graph as shown in Figure 7-40. The user may choose to initially develop the curves in a spreadsheet and copy them into the *Suitability Curve* form where they will be stored and plotted. Alternatively, they can be typed directly into the form as (parameter, suitability). These x, y values are comma delimited, with each pair then separated by a semi-colon.

After defining the SI curves the user should select the layer range as described in the *Critical Limits* section. It is required to select the checkbox to assign which series is to be plotted, and the start and end date of extraction from the model, before selecting *Generate Initial Time Series.*

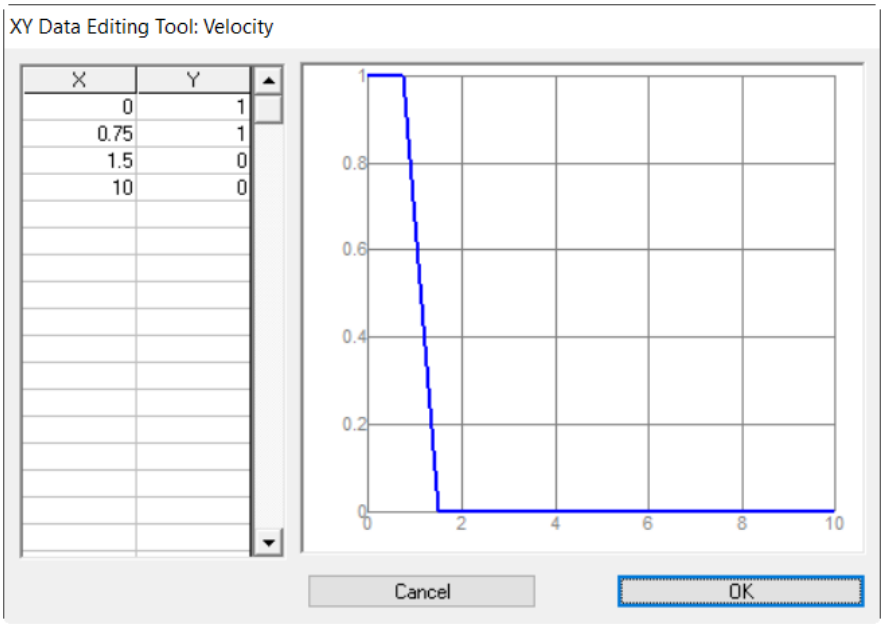


Figure 7-40 Editing tool for suitability index curves.

7.5.2.2 *Channel Index*

Another option available in the HSC approach, which is not in the *Critical Limits* option, is the defining of a channel index. The purpose of the channel index is to allow the user to evaluate the habitat suitability based on nature of the channel bed, which may vary in different regions of the stream, for example with sand, silt or gravel. This index can be set based on different locations in the channel that the species may be occupying. The channel index is a numeric value that has been assigned to each cell. EE allows the user to assign any value to a cell, though typically they range from 0 to 1, or 0 to 100. *Channel index* is one of the parameter options, and should be selected from the dropdown menu if required.

If the user doesn't have a channel index file already, they should first build the channel index with the *Build IJ Index* button. Selecting this will open the tool shown in Figure 7-41. In this form the user should browse to a channel index file from another tool such as River2D (.CHI) and import this file. River2D format is: "X, Y, Elevation, Substrate Code", where X and Y should be in UTM. However, EE will prompt the user for the column containing the channel index so other formats can also be imported. Alternatively, the user can generate this externally in a spreadsheet. EE will interpolate the channel index to the I, J grid. The user selects the name of the output file, which will be saved in the EE file format with extension .CI and of the type: I, J, Channel Index.

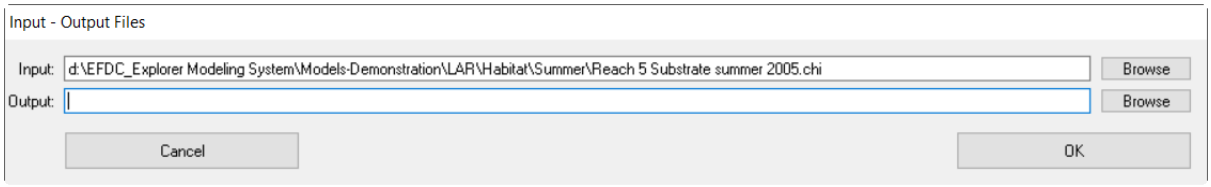


Figure 7-41 Habitat Suitability Criteria: Channel index tool.

Once the user has a channel index file already created they should browse to it in the *Index* box. As described in the critical limits section, it is also possible to generate the time series based on only a sub-set of the model domain by browsing to and selecting a polyline file with the *Sub-Set* box. If no file is selected in the *Index* option EE will effectively ignore this option by assuming the index is 1.0 everywhere.

The user should select the start and end date for the time series extraction in the *Initial Model Results Extraction Options* frame. Once this has been set the user can select the *Generate Initial Time Series* button. EE will then confirm the number of cells in the model domain and check if the user wants to proceed with calculating the time series.

To create time series EE will extract all of the cell information and save it in a post processing file in the #habitat folder with a name based on the *Criteria ID*. If the criteria ID is "Salmon01" then the file will be called HSC_Results_Salmon01.dat. This file records the composite time, salinity, net weighted criteria (weighted average 0 and 1 of all the cells that meet the bathymetry criteria), area, and volume.

Once a time series has been generated, and the post-processing files have been extracted, a new frame is displayed, *Secondary Processing of Extracted Results as shown in* Figure 7-42*.*

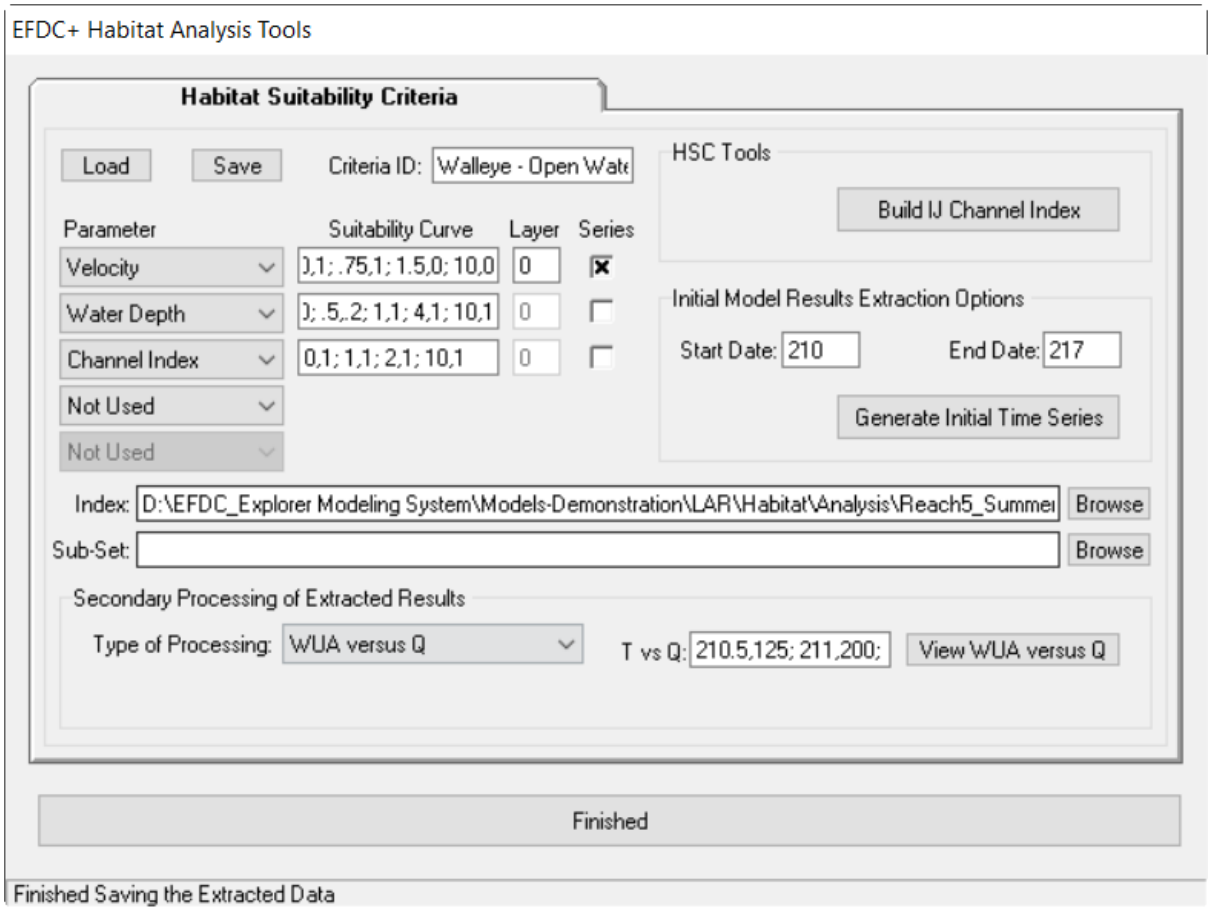


Figure 7-42 Suitability Criteria: Secondary processing.

The drop-down menu for *Type of Processing* has the following options:

- *WUA Time Series:* This is a time series of the weighted usable area
- *WUA vs Q:* For this option the user must create a time series versus flow curve. This curve should match the initial boundary conditions that are being applied to the model. The user should enter the time versus flow (T vs Q) as x, y points in the table. EE will extract the WUA for the flow at this time and plot WUA vs flow.
- *HSC Time Series:* This is a number from 0 to 1 that is weighted over the entire domain.
- *WUV Time Series:* time series of weighted usable volume
- *Volume Time Series:* time series of volume
- *Parameter Time Series:* Depending on which option with a check box has been selected (only one at a time may be selected), EE will extract the time series for that parameter

After selecting one of these options EE will display a button to the right of the menu. Selecting this button will display the time series such as that for the WUA time series shown in Figure 7-43. If the user has extracted the results previously in this session, and then wants to view another secondary processing type, EE will prompt whether to use the data in memory, or whether to extract the data again.

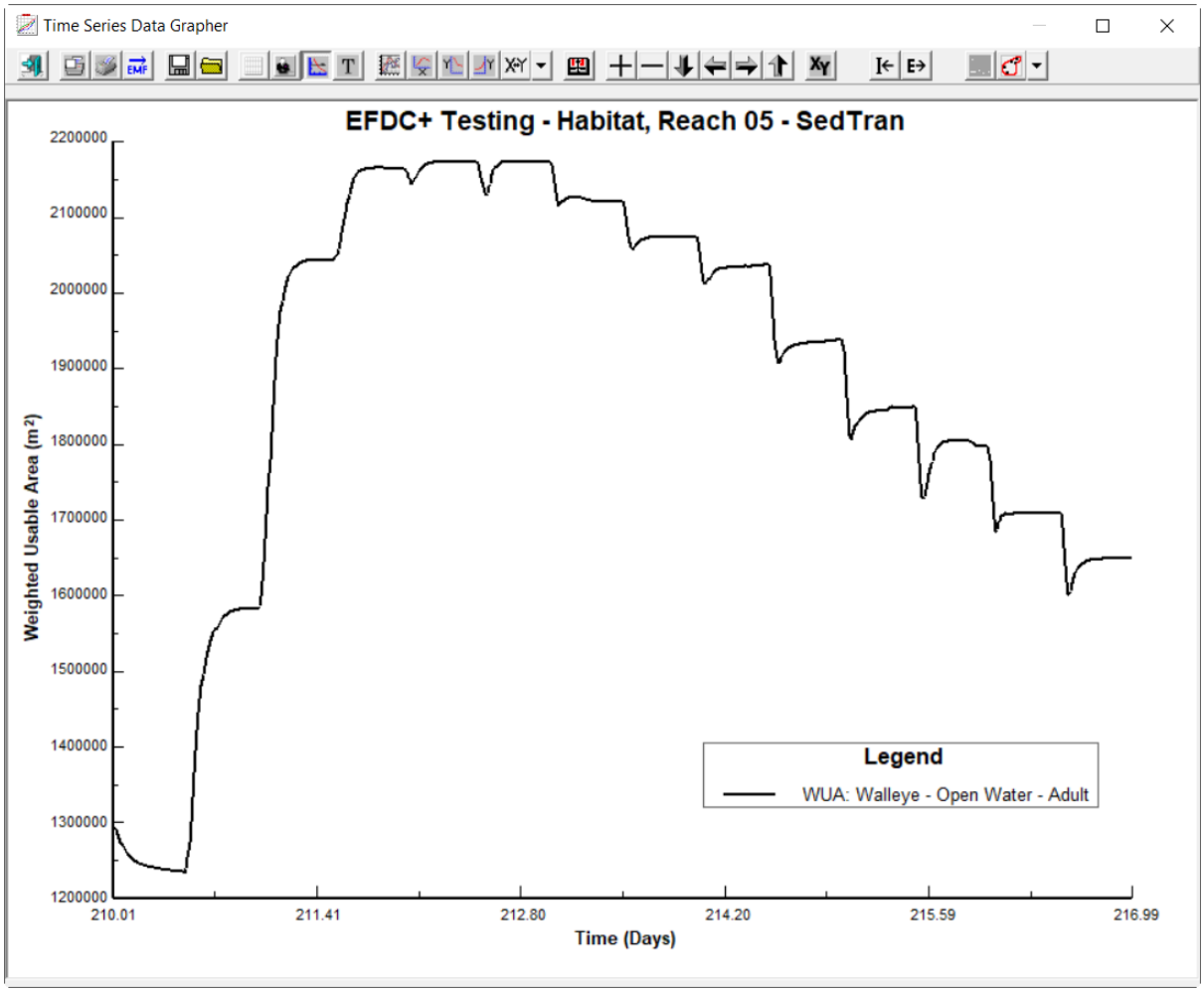


Figure 7-43 Weighted Usable Area time series.

7.6 Miscellaneous/Profile/Series

Figure 7-44 show the Miscellaneous Tab for Model Analysis. The various features are described under each frame below.

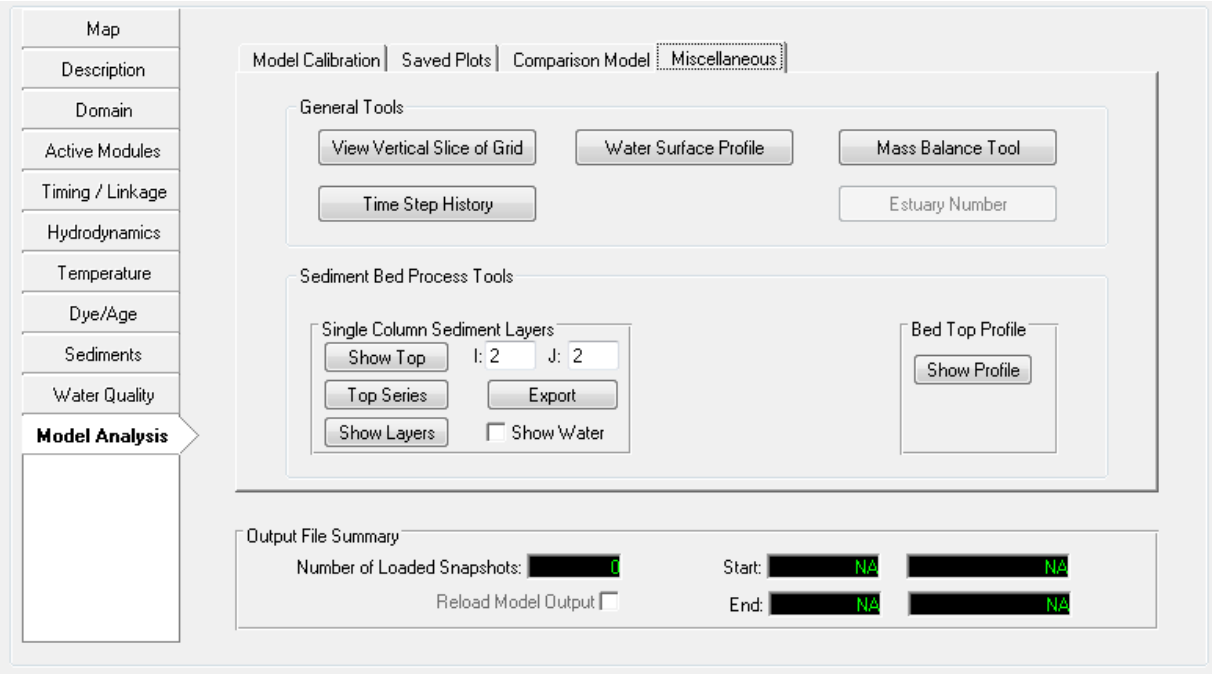


Figure 7-44 Model Analysis Tab: Miscellaneous.

7.6.1. General Tools

Access to some of the profiling and time series options are available on the *Profile/Series* tab (Figure 7-45). Each major option or feature will be discussed below.

The *Profile Location Option* frame needs to be set first before a profile can be extracted from EFDC *ViewSlice* or any of the profiles shown in the *Post Processing* form. There are three profile extraction options. The user may either select a value of I to extract the active J cells along that I, or select a value of J to extract the active I cells along that J. The third option is to use a *Drape Line*, which is a polyline in the same coordinate system as the LXLY data. The I & J's along the line will be assembled and the profile will be output along that slice.

7.6.1.1 *View Vertical Slice of Grid*

The *View Vertical Slice of Grid* button simply extracts a profile using the settings in the *Profile Location Option* frame and displays the water column and sediment bed layering (if KB>0). Figure 7-45 shows an example slice. If the drape line or the I/J extraction falls across inactive cells a gap will be displayed. The water surface displayed is based on the initial condition depths.

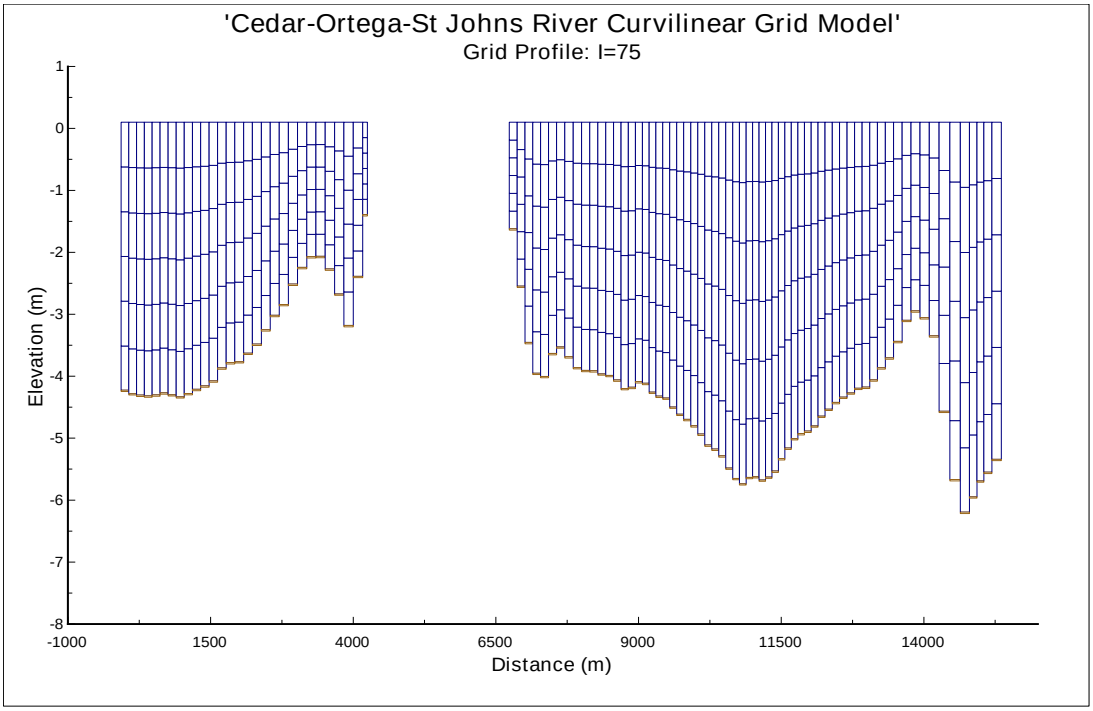


Figure 7-45 Example grid profile plot.

The *Time Step History* button provides a time series plot of the internal time step that EFDC used for the simulation being reviewed. This is of particular interest only when dynamic time stepping was used for the simulation.

7.6.1.2 *Water Surface Profile*

The *Water Surface Profile* button extracts a profile using the settings in the *Profile Location Option* frame and then displays the results. Two lines are shown, one for the water surface elevation and the other for the bottom. The user can "scroll" up and down along all the snapshot times by pressing the PgUp and PgDn keys using a snapshot increment of 1. Larger snapshot increments are available using the Shift key for increments of 10 and the Ctrl key for increments of 100. Many other functions are available and may be by pressing F2. See Section 10 for a detailed explanation of how to format and use *XY Times Series* graphs. An example water surface profile is shown in Figure 7-46.

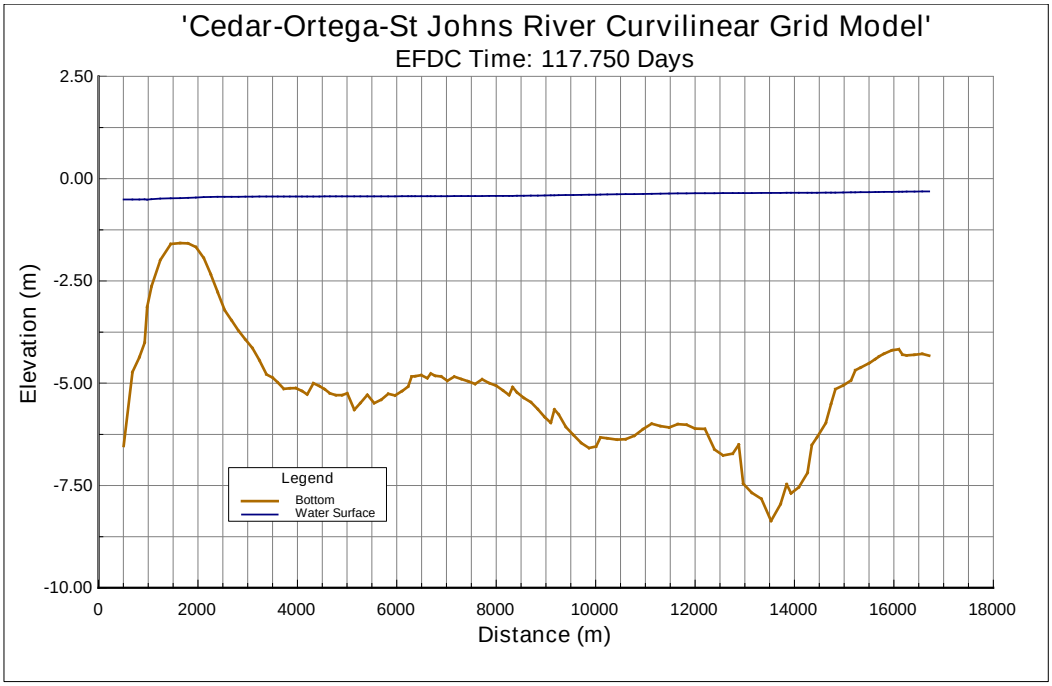


Figure 7-46 Example water surface elevation profile with bathymetry.

7.6.1.3 *Mass Balance Tool*

The *Mass Balance Tool* (

Figure 7-47) allows the user to evaluate the total model's sediment and/or toxic balance as well as determining the mass fluxes through each boundary. Time series plots of the mass loading through the flow type boundaries can also be obtained (see Figure 6-19). The mass loading plots can be obtained and used without computing the mass balance.

This tool is computing the mass balance based on model output snapshots. The smaller the output snapshot interval the more accurate the reported results will be.

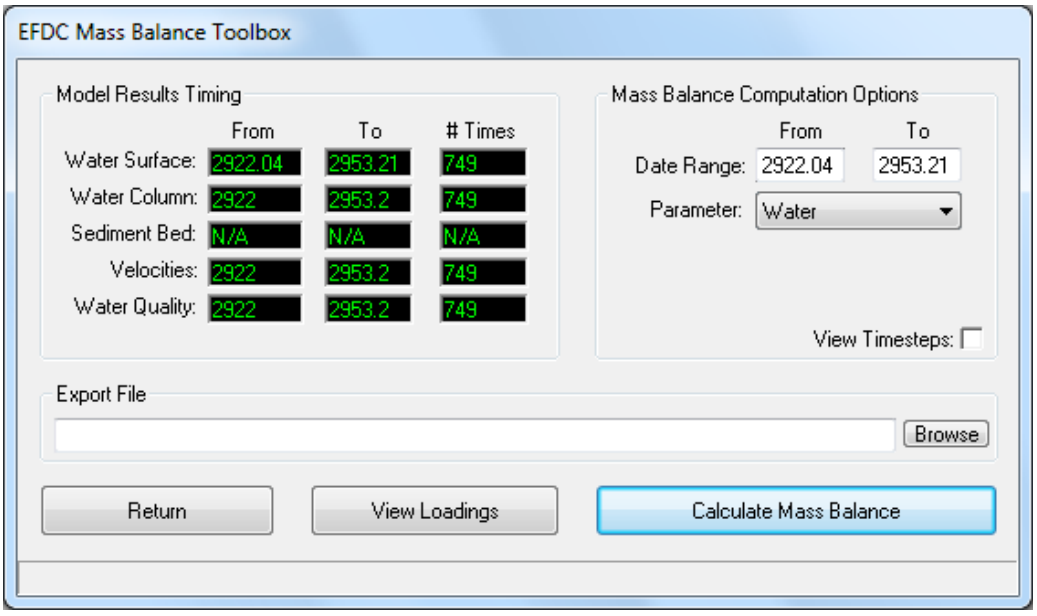


Figure 7-47 Mass Balance Tool Options Form.

7.6.2. Sediment Bed Process Tools

The features located in the *Single Column Sediment Layers* frame all deal with analyzing in detail a single cell and its sediments with time. These options produce single cell plots similar in style to the profile plots, but for only one cell. You can use the PgUp and PgDn keys to scroll up and down in time or press the animate toolbar button to view the consolidation in time.

Figure 7-48 provides examples of results using the *Show Layers* option with *Show Water* checkbox checked. This example used simple bed consolidation (IBMECH=1) with bed elevation changes (IMORPH=1). The consolidation rate (SEDVRDT) was set to 100,000 sec. Note that as the sediment consolidates (Figure 7-48(a) to Figure 7-48(b)), the water depth increases, reflecting the extrusion of water from the sediment porosity.

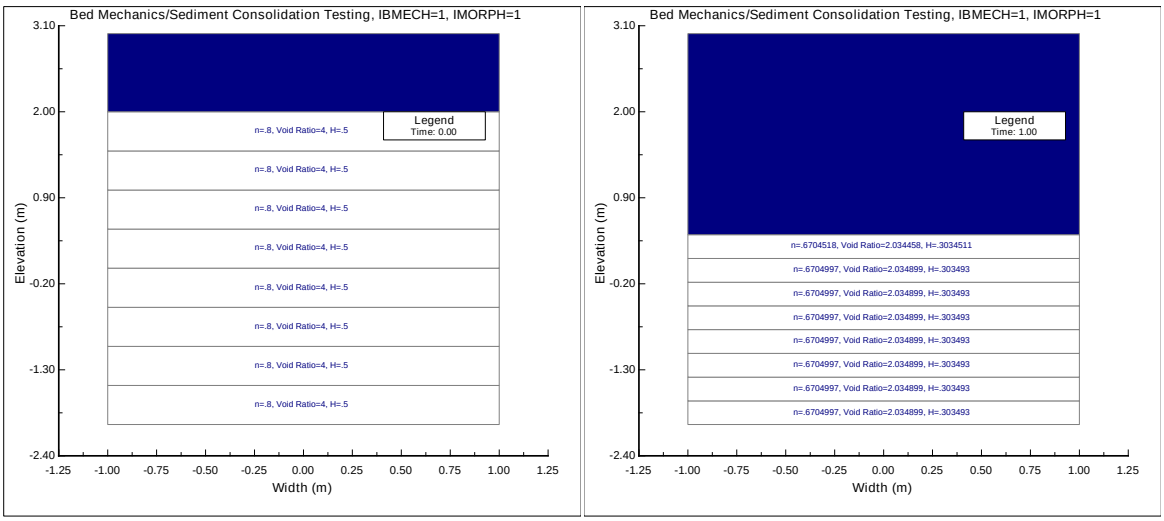


Figure 7-48 Sediment column consolidation: (a) Initial conditions (b) End of Day 1.

7.6.3. Bed Top Profile

The *Bed Top Profile* function provides a 2D plot of a vertical slice through the model showing the water column and the sediment bed. Figure 7-49 provides an example for the Pa Vinh (Son La Province, Vietnam) dam's tailwater sediment transport analysis. Each bar represents a model cell. The thicker top bar represents the water column concentrations of total suspended solids (TSS) and the bottom, horizontally grouped, bars represent the sediment bed.

The height of the water column bar represents the water depth. The color of the water column bar represents the TSS concentration of the cell at the time shown in the legend. The color range corresponds to the TSS values shown on the key, above the color bar.

The height of the sediment bars represents the thickness of the upper most sediment layer that is interacting with the water column. There is a separate sediment bar for each grain size class being simulated. The color range corresponds to the mass fraction shown on the key, below the color bar. If the cell contains no sediments in a given class, that class' bar is not filled. The sediment class grain size is labeled in the key for each sediment bar.

RMC'ing on the legend or pressing the toolbar legend button will provide access to adjustments to the plot settings.

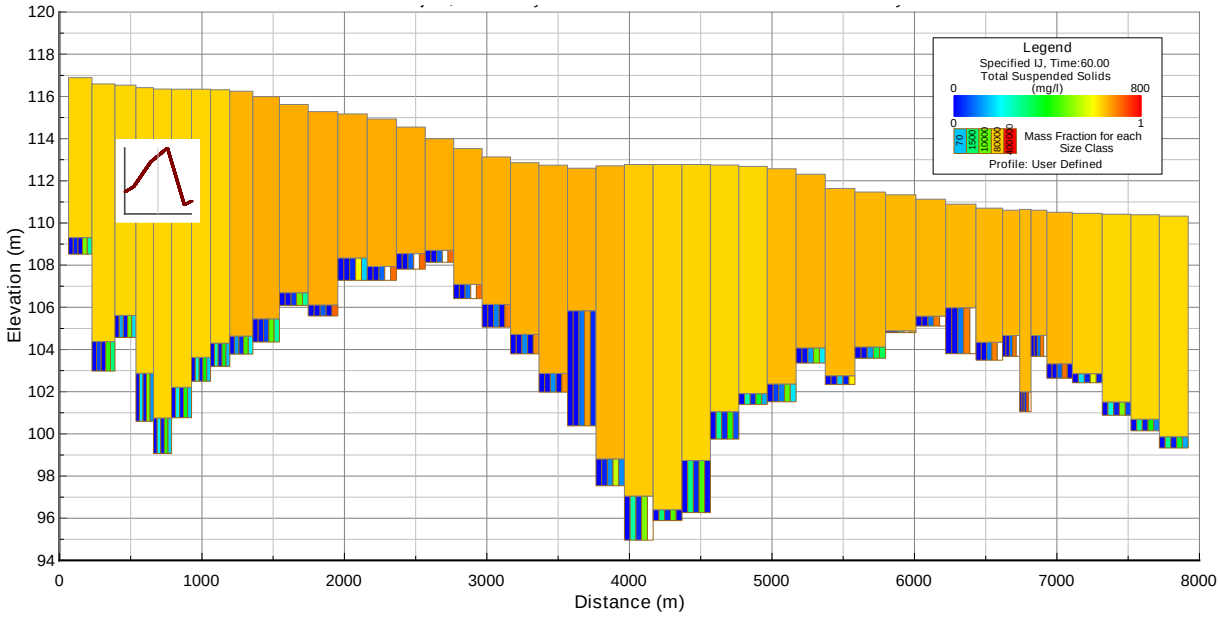
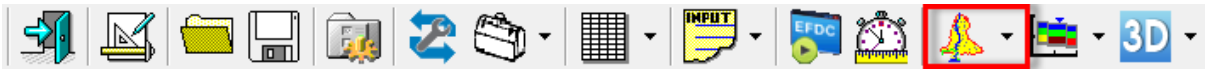


Figure 7-49 Example "Bed Top Profile" for water column and sediment bed.

8 ViewPlan



The *ViewPlan* button of the main toolbar provides access to the primary utility for the pre-processor visualization and map based interface functions as well as the primary post-processing utility.

ViewPlan button provides the user with a dropdown menu comprising the following options:

View Initial Conditions: This option may be used with the user is setting up the model or wants to view the ICs prior to the model run

View Model Results: This option is used after the model has been run and the user wished to analyze the model results.

View Model Results (Re-scan Output): This option should be used if the user has merged two models or has continued a model run and want EE to go back to the beginning of the output data set.

Table 8-1 contains a list of the main parameters that can be displayed using *ViewPlan*. The Sub-Options column lists only the major sub-options. Almost every function has a number of sub-options and features to combine and split the data in a range of different ways. These sub-options are explored in more detail in the following sub-sections.

Table 8-1 Main Functions of ViewPlan.

ViewPlan Option	Description	Sub-Options	Time Variable
Cell Indices	Displays EFDC model number schemes for I & J indexes and the linear L index.	NA	No
Cell Map	Displays the computational cell map that is saved in the CELL.INP file	NA	No
Bottom Elevation	Displays the bottom elevations	NA	Yes
Water Levels	Displays various views using the water levels	Depth Elevation Wet/Dry Total Head, Overtopping Areal Extents (FEMA) Areal Extents (Depth)	Yes
Boundary C's	Displays the model domain with the boundary cells shown by type	NA	Yes
Fixed Parameters	Displays the model input parameters that are (generally) fixed in time.	Roughness Cell Angles Wind Shelter Shading Factors GVC Layers Groundwater Map Wind Weightings Atm Weightings	No
Model Metrics	Displays some model metrics. Some of the sub-options are time variable and others are fixed.	CFL Time Step Courant # Orthogonal Deviation Celerity	Yes

ViewPlan Option	Description	Sub-Options	Time Variable
		Froude # Richardson # Densimetric Froude # Reynolds #	
Velocities	Displays velocity vector plots and flux tool. Velocity vectors can be overlaid many of the other *Viewing Opt's*	2D Velocity Vectors Magnitudes, Vertical Velocities Velocity Anchor Point Regular Grid Extraction Flux Tool	Yes
Bed Shear	Displays bed shear stress.	Shear – Total Shear – Cohesives Shear – Non-cohesives Wave Only Current Only	Yes
Sediment Bed	Displays the sediment bed parameter values for the selected sup-option.	Top layer Layer Thickness Sediment mass Sediment fraction Porosity Computed d_{50} Delta (scour/deposition) Bed Shear Stress Edit Sediment Cores	Yes
Heat: Bed/Ice	Displays the heat in the sediment bed or in the ice layer	Ice Temperature Ice Thickness Bed Heat Temperature Bed Heat Thermal Thickness	Yes
Volumes	Displays rate of changes in water column volume	Evaporation Rainfall	Yes
Water Column *Water by Layer* *or* *Water by Depth*	Displays the water column parameters values for the selected sup-option	Salinity Temperature Sediments Toxics Dissolved, POC bound, DOC Complexed Water Quality 21 EFDC Parameters Trophic State Index Other Derived Params WQ Kinetic Zones Age Water Density Secchi Depth % Irradiance Habitat Analysis Volumetric Analysis	Yes
Diagenesis	Displays organic and nutrient sediment concentrations (by G class) and nutrient fluxes.	Concentrations of PON, POP, POC, NH4HN, NO3-N, PO4-P, H2S, Silica Benthic Stress Sediment Temperature	Yes

ViewPlan Option	Description	Sub-Options	Time Variable
		Flux Rates PON, POP, POC, SOD, COD, NH4, NO3, PO4, Silica	
Vegetation Map	Displays a map of the vegetation classes	NA	Yes
Modify Channel	Displays the channel modifier showing connectivity	NA	No
Wave Parameter	Displays the wave parameter values for the selected sub option	Energy Height Radiation Shear Stress XX, YY, XY Dissipation Angle Period Total Depth (Derived) Water Surface Elevation (Derived)	No
Computed Variable	Displays the internal EFDC for the selected sub option	Horizontal diffusivities Other User Defined	Yes

8.1 Simulation Results Loading

As a post-processor, *ViewPlan* requires the output data, as well as the input data, to be loaded. The user can control when output data is loaded using the *Output Loading Frame* as shown in Figure 8-1. When a file is initially loaded the *Output File Loading* frame will appear as *No Results Loaded* until the *ViewPlan* button is pressed. Once *ViewPlan* has been used the *Output File Loading* will appear as *Model Results Loaded* After output files have been loaded, they will not be reloaded for that project unless the user selects the *Re-Scan* option on the dropdown, checks the *Reload* check box in the calibration section (see Section 7.1), loads a new project or exits EFDC_Explorer and then restarts it.

The user should note that this is a redesign from previous versions of EE. The user can now choose from the drop-down menu to load model results or not (IC only). The default option, if the main toolbar button is selected is *View Model Results*. If no model results are available, only the IC's will be available for display.

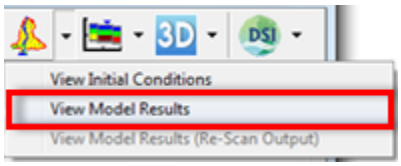


Figure 8-1 Model results loading options.

The *Re-Scan* menu option is useful when the user is loading the results while the EFDC model is running (see Section 5.1.6). It is better to pause EFDC before you load the data to prevent file handling errors but it is not required, especially if EFDC is writing to the EFDC_Explorer files infrequently. If the *Re-Scan* menu option is used, it will cause EFDC_Explorer to reload the model results requested at that time.

The water depths are automatically loaded as they are necessary for any post processing to be performed. The depths are stored in the EE_WS.OUT file. This file contains the depths for all the cells for every snapshot time. This file is completely loaded into memory when post-processing. If the water surface file is not loaded, no post processing is available.

For all other model results, only the current time is loaded. This approach allows for quicker file loading and viewing and allows for very large model results files (>4 gigabytes) to be saved and used. However, when generating any time series type processing, each snapshot has to be loaded for all the times requested. Many hard disks have caches to allow frequently accessed files to load quicker. Therefore, the first time a time series is processed the data may load slowly. However, the subsequent processing may be much faster.

During the file loading process, pressing the ESC key aborts the file loading process and returns to the main EFDC_Explorer form.

8.2 Introduction ViewPlan Form

The *ViewPlan* form viewing options is adjusted based on what parameters were simulated, what data has been loaded, and what option has been requested by the user. Figure 8-2 shows an example of the *ViewPlan* form with the *Water Column* option selected. This plot shows several standard features of *ViewPlan* including the *Timing Frame*, *Legend* and the *Horizontal Scale*. With the exception of the *Legend*, these features can be turned on or off.

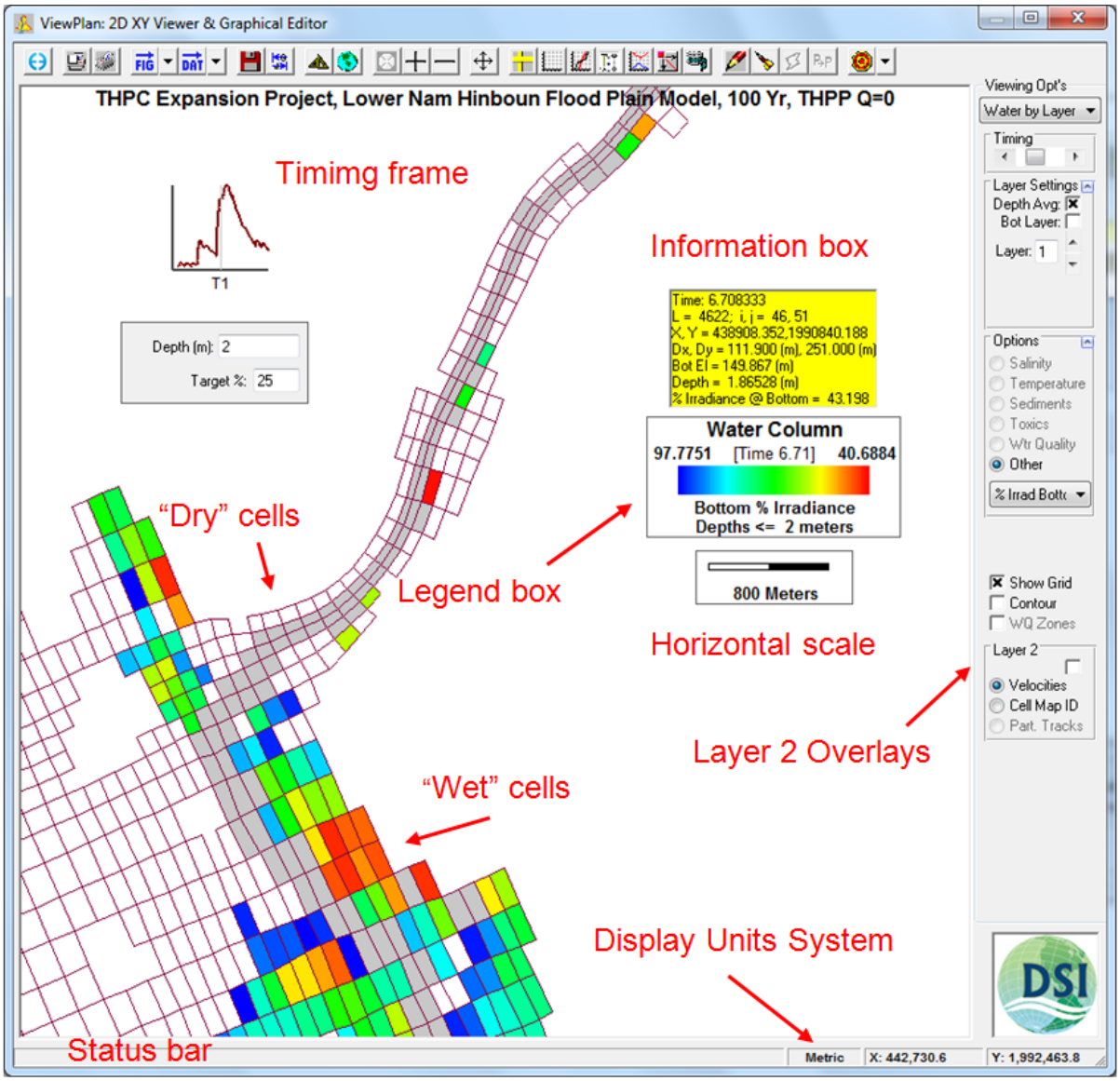


Figure 8-2 ViewPlan main form.

The *Viewing Opt's* frame contains a dropdown list of all the available viewing, editing, and/or post-processing major topics. The list of items in this list may change each time the user changes parameters in the pre-processor function or loads a different model results file (see Section 7.1). Figure 8-3 shows the *ViewPlan* context sensitive area for alternative *Viewing Opt's*. It can be seen that having selected the *Water Column* major option different options are available. Within the context area only the appropriate options will be shown.

The toolbar located at the top of the form provides functions that operate on/with the current *Viewing Opt's* selected. For example, clicking on the animate button animates the variable that is being displayed. Table 8-2 provides details of the various functions available from the toolbar. Section 8.6 provides more details on the various *Viewing Opt's* available.

The *Timing* frame provides a scroll bar that provides direct access to the model output snap shots. When the slider bar is scrolled completely to the left (Timer = 0) the data displayed are the initial conditions specified in the model input. The current time is displayed in Julian date in the legend. The resolution of the time display is controlled by the EFDC_Explorer settings for time resolution.

The form may be resized (shrinking it too small will cause EFDC_Explorer to limit the size), maximized and minimized. However, if minimized, it must be restored prior to doing anything with EFDC_Explorer.

8.2.1. Mouse Functions

The toolbar changes the function of some of the mouse clicks, but in general the following summarize the basic mouse click functions.

8.2.1.1 *Repositioning Legend & Other Objects*

To reposition EFDC_Explorer pop-up's, the legend, labels, notes, dialog windows and frames the user may left mouse click, hold it down and drag it to move the object to another location on the plot. If the legend is moved off the display and the form resized so that the legend is no longer visible, it will be repositioned to the center of the current view.

8.2.1.2 *Cell Information*

Using the mouse, point to a specific cell and then left mouse click (LMC) to display that cell's general information along with the data of the currently selected parameter, with any sub-options. Right mouse click (RMC) copies this information to the Windows clipboard. Figure 8-3 shows an example for salinity.

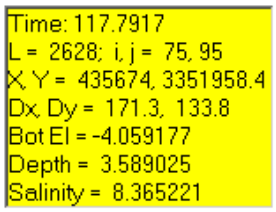


Figure 8-3 Cell Information example.

8.2.1.3 *Right Mouse Click*

Right mouse click (RMC) on the object to perform the following:

The Legend Brings up the Display Options form.

A Cell (BC) Displays the pop-up shown which allows the options of editing the cell, deactivating the cell, or *Set as show I,J*. This sets the I and J in *Run Time Status* form within the *Grid & General* tab. Depending on the BC type clicked, the view cell will display the primary boundary condition parameter as well as the corresponding concentrations. The view group displays total group flows or fluxes.

A Cell (Edit) When displaying the domain with the *Enable Edit* check box selected, except for boundary conditions, a *Modify Cell* form (Figure 8-4) is displayed that allows the user to modify many of the cells properties in one location.

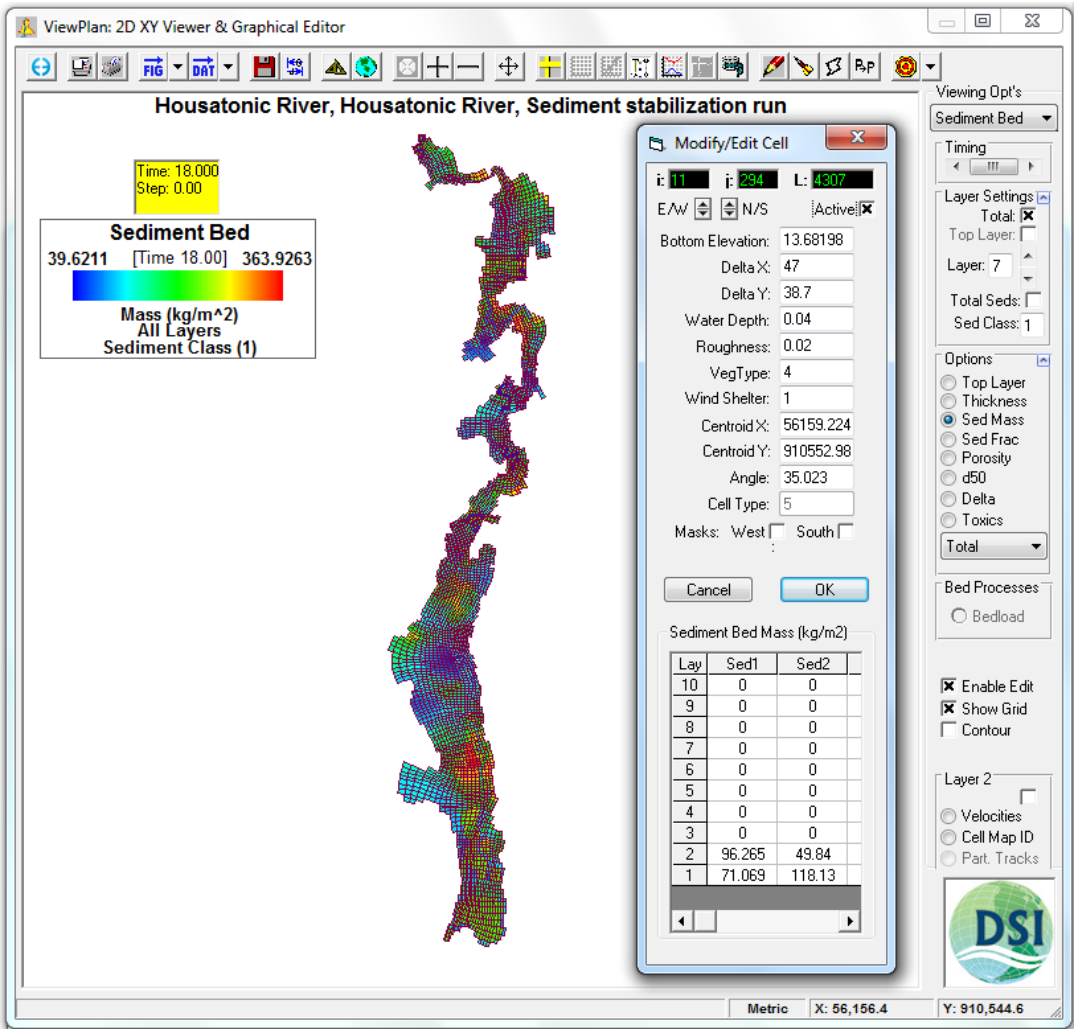


Figure 8-4 Modify Cell form with bed layer-sediment mass sub-option.

In addition to these functions, RMC'ing on zones for the water quality, diagenesis and vegetation zones allows the user to edit the data for the entire zone.

During post-processing water quality, if viewing dissolved oxygen with the time series toggle on, the user can select from a series of pre-defined "bundles" of nutrients.

8.2.2. Shortcut Keys

To obtain help on the keystroke functions press, F2. What is actually shown in the message box pop-up depends on what data/model results are loaded and the currently selected view. Table 8-2 provides a listing of the keystroke functions in *ViewPlan*.

Table 8-2 ViewPlan keystroke function summary.

Panning & Zooming	
Space LMC	Pan with cursor
Mouse Scroll	Zoom in and out
F3	Modify Pan & Zoom Increments. The size of the panning and zooming steps are adjusted here.
F5	Refresh Current View
left	Pan East.
right	Pan West
up	Pan North
down	Pan South
+	Zoom In
-	Zoom Out
Ctrl-E	Zoom Model Extents
Cell Property Copy Editing	
Alt-LMC	Get Property. Alt-LMC'ing on a cell will copy that cell's current property to the operator box.
Ctrl-RMC	Set Property. Ctrl-RMC'ing on a cell will apply the value or operator in the operator box to that cell (Copy Cell Property must be enabled)
Smoothing of Initial Conditions Field	
Ctrl-S	Smooth the Currently Selected IC Field. This applies only to bottom elevations and the water column.
Rectangular Cell Selection (for Editing Cell Properties)	
	To Select Cells in a Rectangle hold the ALT key down and Right Mouse drag. The 'Enable Edit' checkbox must be selected.
Post Processing – Time Selection	
PgDn	Show Next Snapshot
PgUp	Show Previous Snapshot
Shift-PgDn	Show Next Snapshot – Increment:12
Shift-PgUp	Show Previous Snapshot – Increment:12
Ctrl–PgDn	Show Previous Snapshot – Increment:120
Ctrl-PgUp	Show Previous Snapshot – Increment:120
Ctrl-G	Go to a Specific Date. Format of date to enter depends on whether the user is displaying Julian or Gregorian dates.
Specific View: Water Levels, Inundation Extents	

Ctrl-O	Export the displayed inundation outline as a P2D file.
Specific View: Sediment Bed, Bed Shear Stress	
Alt-P	Toggle Display of Bed Shear or Stream Power
F7 or Ctrl-T	Toggle Sediment Core Editing
Specific View: Water Column	
Alt-B	Conduct a Bottom Irradiance Analysis
Alt-C	Conduct a Volumetric Analysis
Alt-H	Conduct a Habitat Analysis
Specific View: Comparison Model	
Alt-M	Toggle the display of model comparison
Alt-V	Compute and display the volume differences between two models in bottom elevation view
Specific View: Velocities	
RMC	Define Velocity Profile (Velocity View Only)
Calibration Data Display	
Ctrl-<	Go to previous data snapshot. If calibration data has been configured and the current view is the water column, this keystroke will cause EFDC_Explorer to jump forward to the next measured data point and display the model results with the data or residuals labeled.
Ctrl->	Go to next data snapshot.
Series/Profile Locations	
Alt-I	Load the Series/Profile Locations from a File
Alt-O	Save the Series/Profile Locations to a File
Layout and Viewing Options	
Ctrl-W	Set the View Port to a Specified Size
Alt-W	Set the View Port to a Specified Scale
F6	Toggle Grid Display
F7	Toggle Julian Date as Date Format
F8	Toggle Display of Masks
F9	Toggle Transparency for Current View
F11	Toggle the Display of Titles
F12	Toggle On/Off Greyscale Color Ramps
Alt-K	Save the Layout (form size, scale, legend location, etc.) to a File
Alt-L	Load a Layout from a File and apply it to the current view.
Miscellaneous Functions	
Ctrl-A	Animate the Current View
Alt-E	Export the Current View as a Metafile

Ctrl-M	Toggle Metric/English Display Units
Alt-R	Compute and Display time series plot of River Kilometer versus time for a target value.
Ctrl-V	View a Specified Cell using L or I,J.
Ctrl-P	Photo Viewer – load an image to display on the *ViewPlan* workspace

8.2.3. Toolbar Summary

The *ViewPlan* toolbar provides access to a range of different functions and utilities (Figure 7-5). Some of these functions are dependent on the current context while some are not. Table 7.3 contains a summary of each function.

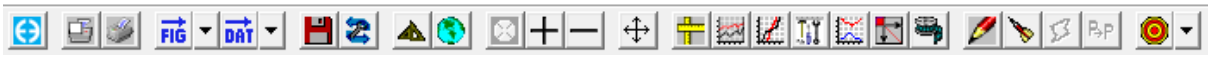


Figure 8-5 ViewPlan Toolbar.

Table 8-3 Summary of ViewPlan toolbar.

	Exit *ViewPlan.*	General
	Printer setup options.	
	Print to current printer the current view.	
	Export the current view to a Windows enhanced metafile (.EMF) or as a Bitmap (.JPG)	
	Export the current major option to Tecplot, KML, Shapefile or to SWAN	
	Create a new EFDC model using the current time to assign the IC's.	
	Julian date ←→ Gregorian calendar calculator.	
	Display options for color ramp, vectors, grid lines, overlays, etc.	Display
	Toggles the display dereferenced bitmap backgrounds.	
	Zoom extents.	Navigation
	Zoom in at fixed increments.	
	Zoom out at fixed increments.	
	Pan Left.	
	Pan Right.	

	Pan Up.	
	Pan Down.	
	Distance tool. Distances are displayed in the status bar.	
	Time series tool. Point and click on cells to build a group of cells.	Post-Processing
	Vertical profile tool. Available in Water Column option of ViewSlice.	
	General statistics tool. Results are copied into the clipboard.	
	Longitudinal profile tool.	
	Water and Mass flux (see below for details)	
	Animation tool. Output animations to the screen and/or AVI files.	
	Polyline/polygon creation/edit tool (see below for details).	
	Model Results Extraction Tool - a point & click data extraction tool.	
	Multi-cell selection tool for general purpose editing using Modify Cell form.	Pre-Pro
	Copy cell property button. A rapid point & click method of cell by cell adjustments using operators.	
	View calibration stations, data and/or Model-Data residuals.	Post-Pro

8.2.3.1 *Printer Setup*

The current default printer is automatically used by EFDC_Explorer. Figure 8-6 shows an example printer setup form that appears when the highlighted toolbar button is clicked. If no printers are available during the startup of EFDC_Explorer, it will display a warning but will continue. Besides being used for printing, the settings from this form also impact certain exported graphics. The primary setting used is the portrait versus landscape option to set page orientation.

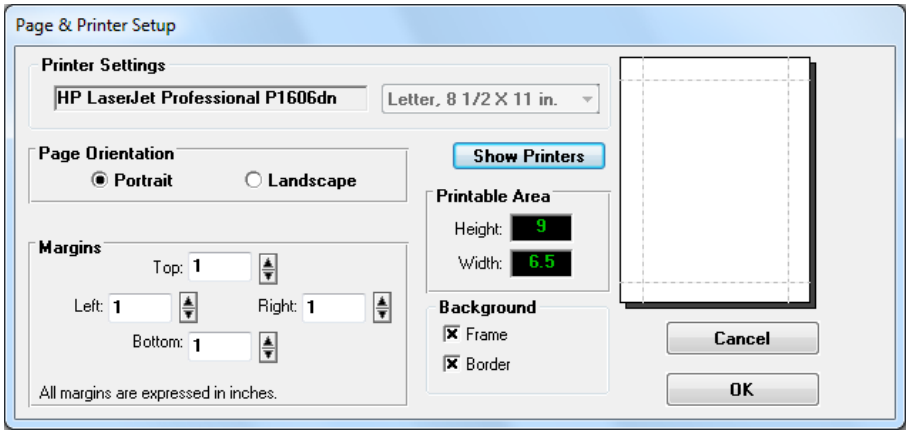


Figure 8-6 Printer Setup.

8.2.3.2 *Export EMF and Bitmap Files*

As most of the modeling efforts end up in engineering and scientific reports, it is very important for EFDC_Explorer to have the ability to produce high quality graphics that can go directly into all of the most popular word processing packages. The ability to export enhanced metafiles (EMF's) meets this need. EMF's are a native Windows based graphic that easily imports into almost all word processing and presentation packages.

The size and shape of the metafile depends on two things in EFDC_Explorer. If you are exporting the results using the *Show Border* check box, then the size and shape will reflect the printer settings. If the check box is not displayed then the size and shape will reflect the current *ViewPlan* window.

Another useful format for images for reports and presentations is bitmap (.BMP). EFDC_Explorer has the ability to export bitmaps at user defined resolutions as shown in Figure 8-7. Here the user may set the resolution in dots per inch and the size of the bitmap in pixels in the X and Y directions.

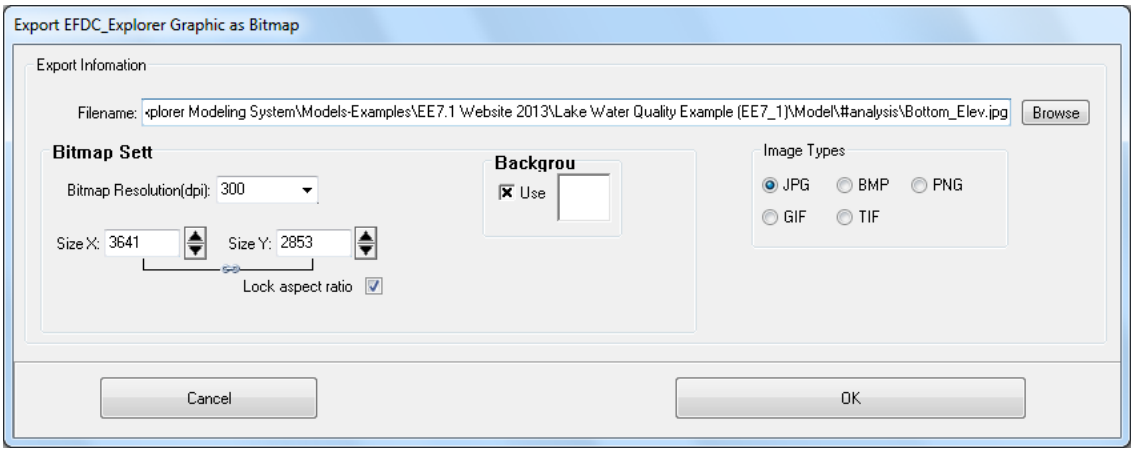


Figure 8-7 Export EMF file as Bitmap.

8.2.3.3 *Export Tecplot, KML, Shapefile Files*

It is recognized that EFDC_Explorer does not and will never handle all the post-processing desires/needs of every project. Therefore, the capability to export the data to third party packages via an ASCII file is provided by the inclusion of the TP function from the toolbar whereby the user may export Tecplot® files. When selected, *ViewPlan* displays the form shown in Figure 8-8. Here the user can select the beginning and ending times to export as well as the skip interval. From EE7.3 users can export the water column parameters in addition to bed parameters directly to Tecplot. The Tecplot export function in *Water By Layer* and *Water By Depth* option has also been implemented. When exporting the sediment bed toxics EE exports the real values and does not truncate the output.

The user must select the times to be exported. The user may manually select the times using the Windows standard LMC, Ctrl-LMC and Shift-LMC to select one or more times. Another time selection option is to press *Select All* (the button then toggles to *Select None*) and set the *Output Interval*. An Output Interval of 1 will output all selected times, a value of 2 will output every other snapshot, etc. The number of snapshots that will be exported is shown in the # *Selected* box.

The *Horizontal Skip* is used to set a skip count for exporting the model cells. A skip count of 1 means every cell will be exported.

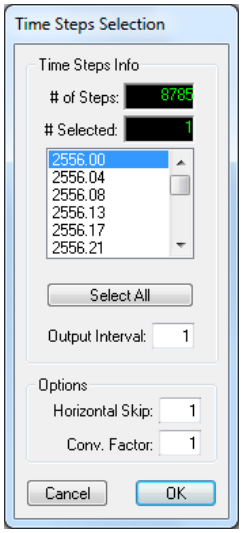


Figure 8-8 Tecplot export timing options.

Also on the *ViewPlan* | *DAT* button drop down menu is the ability the export KML (Keyhole Markup Language) files. KML are native Google Earth© files, using a type of XML language used for expressing geographic annotation and visualization within web based, 2-D and 3-D maps. After selecting this option and entering a file name the user is prompted for the UTM Time Zone (1 to 60) and Opacity (%). After creating a KML the file may be loaded into Google Earth© where each I,J cell is displayed as a place (see Figure 8-9).

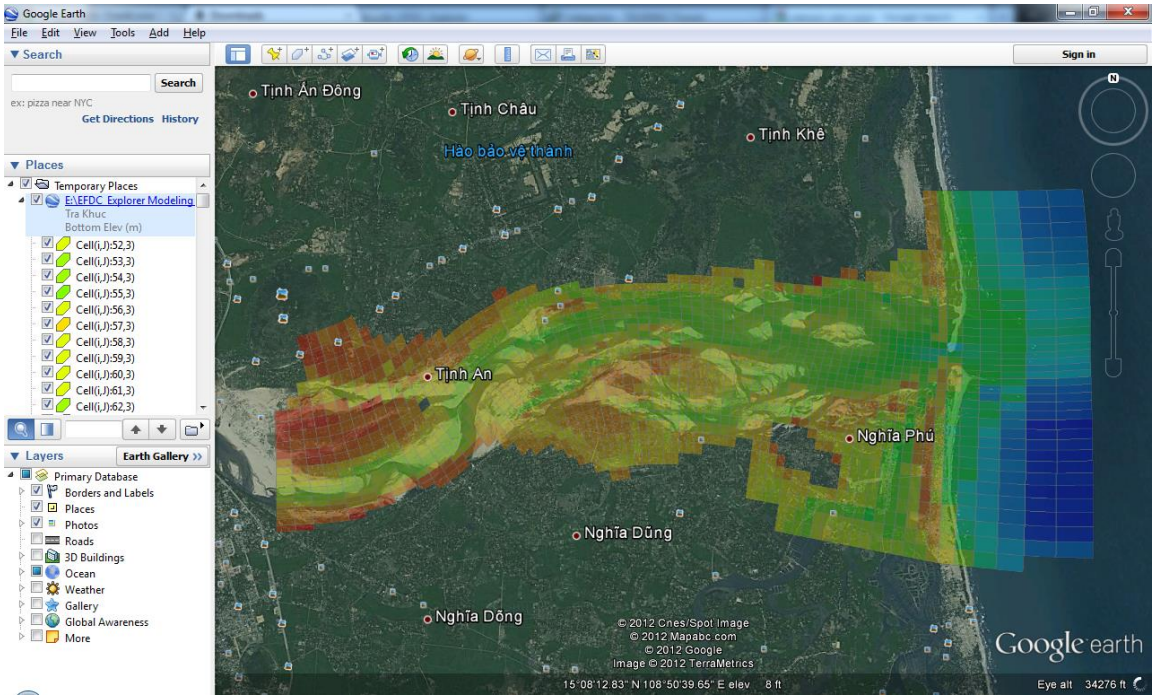


Figure 8-9 KML file exported from EFDC_Explorer to Google Earth©.

As of release of EE7.2 there is an option to export whichever *Viewing Option* is selected in ESRI © shapefile format. In addition to the shapefile, the projection file is also provided and the sample shapefile loaded in ArcGIS is shown in Figure 8-10.

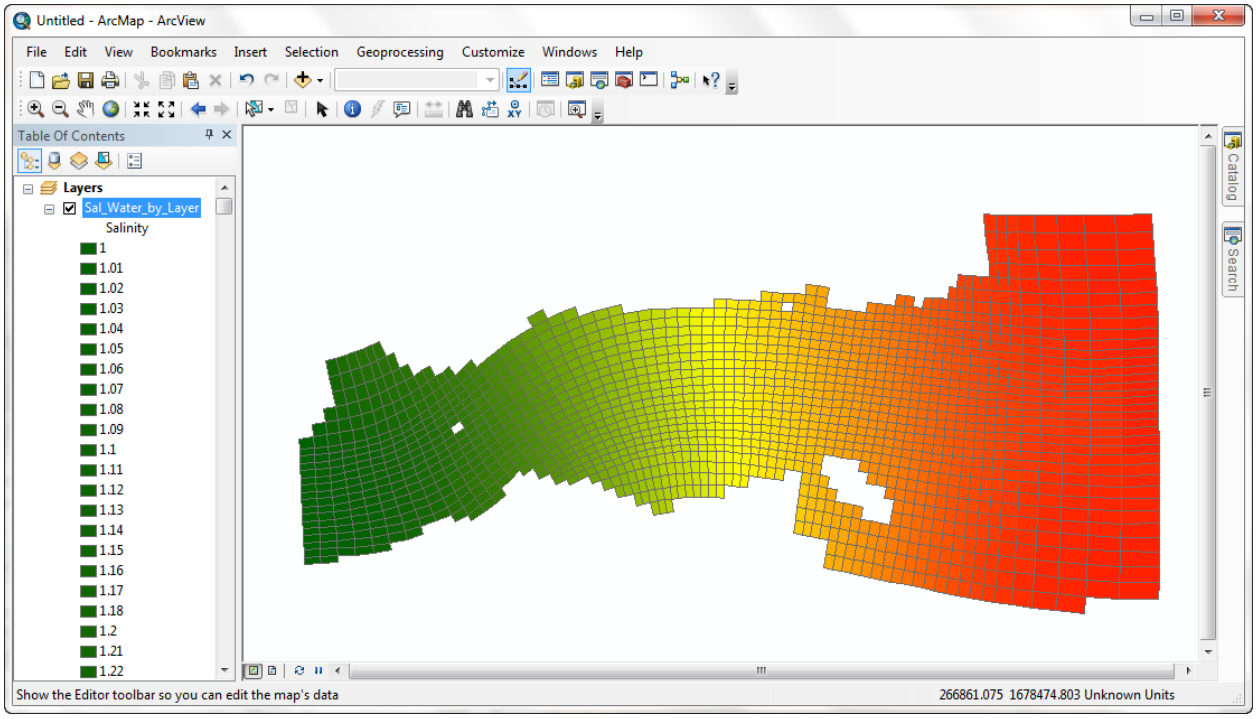


Figure 8-10 Shapefile showing Salinity viewed in ArcView.

8.2.3.4 *Export NetCDF Files*

From the release of EE7 it is possible to export model output in NetCDF format. NetCDF (Network Common Data Form) is a community standard for sharing scientific data. Developed by Unidata, it is a set of software libraries and machine-independent data formats that support the creation, access, and sharing of array-oriented scientific data.

EE uses the CF (Climate and Forecast) conventions which Unidata describes as "designed to promote the processing and sharing of files created with the NetCDF API. The CF conventions are increasingly gaining acceptance and have been adopted by a number of projects and groups as a primary standard. The conventions define metadata that provide a definitive description of what the data in each variable represents, and the spatial and temporal properties of the data. This enables users of data from various sources to decide which quantities are comparable, and facilitates building applications with powerful extraction, re-gridding, and display capabilities." http://www.unidata.ucar.edu/software/netcdf/

Figure 8-11 shows the dropdown menu for exporting data from *ViewPlan.* When the user selects the *Export NetCDF* option then form shown in Figure 8-12 is displayed.

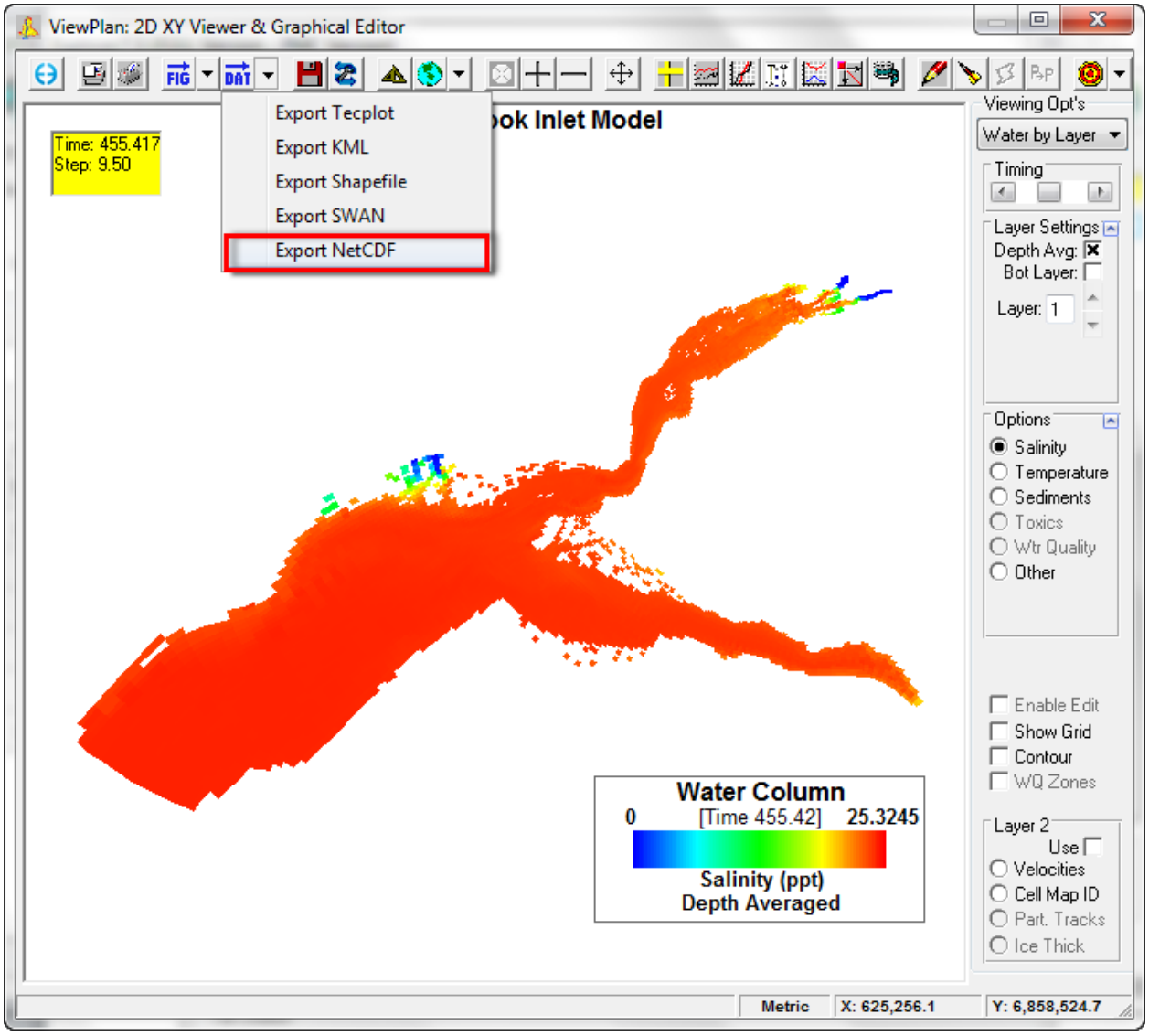


Figure 8-11 ViewPlan: Export NetCDF Option.

Here the user may select which data is to be exported. The *Export to NetCDF files* form allows the user to select static data such as the model grid and initial bottom elevations to be exported. Dynamic data from hydrodynamics and constituent transport may also be exported. The user should select the begin and end time for the export of the data.

In the *File Creation* frame the user may select from exporting all the NetCDF data in one file, or separate it into a series of files, one for each day.

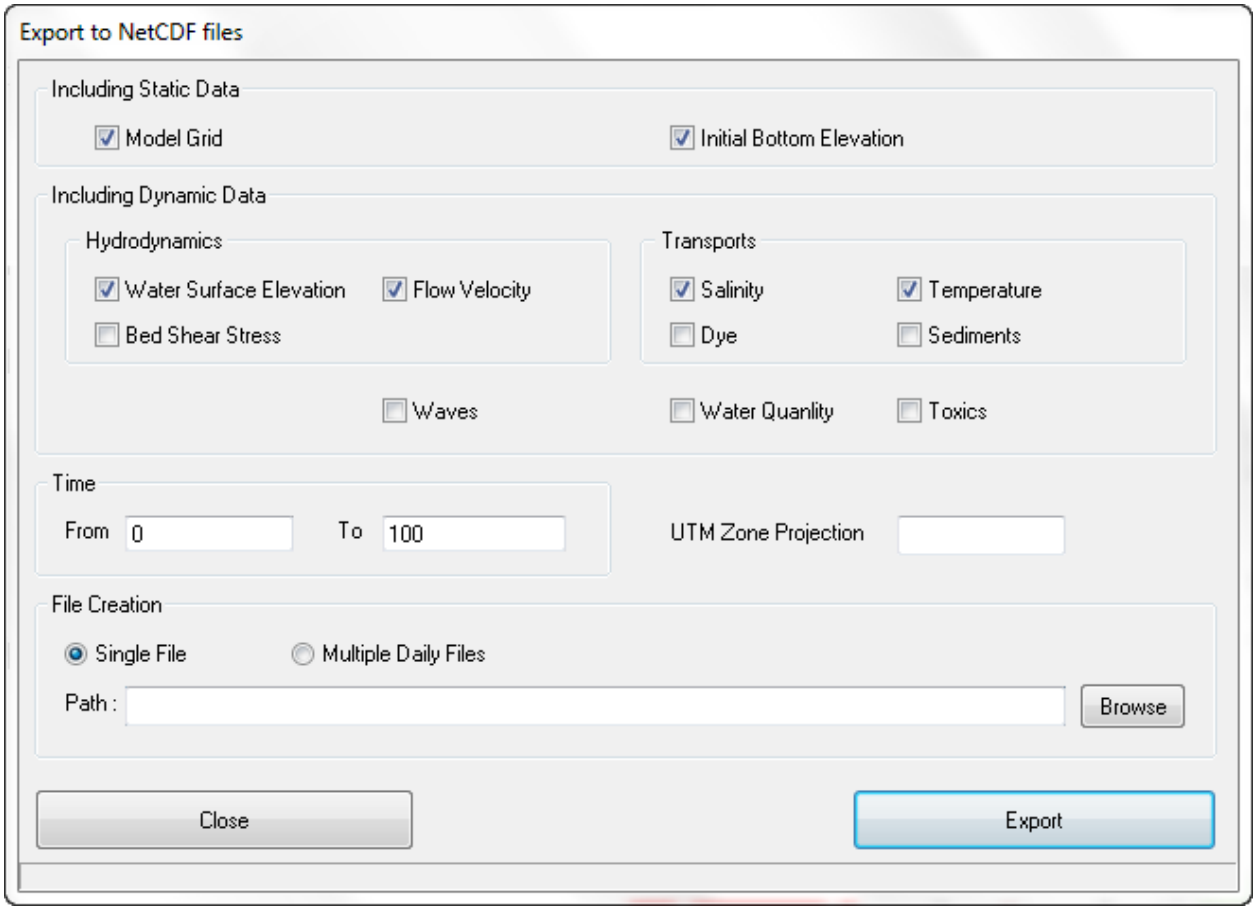


Figure 8-12 Export NetCDF Files Options.

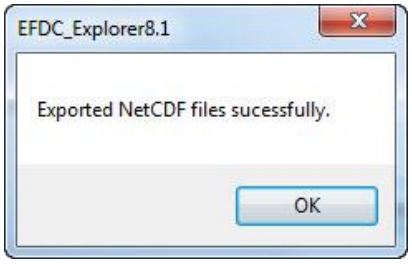


Figure 8-13 NetCDF Files Exported

8.2.3.5 *Create New EFDC Model*

This function saves to disk a new project using the model results at the time of current view (as indicated in the legend) as the initial conditions. When selected the user is prompted to select a new project directory to save the new model to. All appropriate flags and settings are adjusted in the current model to allow the user to load the newly saved model (the original model is still the current project) and run EFDC. All parameters being simulated use a spatially variable initial conditions file.

This feature is useful for testing various adjustments or corrections to the model inputs after a particular date. An example would be to test why a model crashed after a certain time by adjusting boundary or wetting//drying options and running a "continuation" model. If appropriate, the model results from a continuation model can be merged with the original model to produce a single model output for plotting and analysis. See Section 6.5.6 for more detail.

8.2.3.6 *Polyline/Polygon Creation Tool*

EE makes extensive use of polygons when applying cell properties and the *Polygon Tool* is the primary utility in EFDC_Explorer to create and edit the various polylines and polygons that the user may need to help define boundary cells, flux lines, model annotation, velocity profiles, etc. In these cases, cells that are inside the polygon(s), using the *Inside Cell Test* options, are adjusted according to the options specified in the *Modify Options* frame. When the toolbar button is pressed, EFDC_Explorer asks the user for a file to load in order to edit existing data. The user can either select the file to edit or press cancel to start with no existing lines.

From the release of EE7.2 this tool is greatly simplified and provides the user with separate toolbar on the workspace. The Polyline button is shown in Figure 8-14 using a small red rectangle on the top menu. When selected, it provides the user with a number of options for creating and editing polylines and polygons which are described in Table 8-4. Note that LMC commences a new line and RMC ends the line. After RMC the user must select an ID String for the polyline (Figure 8-14). However, it is important to note that entering a value here does not save the polyline and if left blank will delete the polyline. A .P2D file may contain multiple polylines each with different ID strings. After entering the ID the user must select the *Save* button for the polyline/s to be actually saved for later use. Though the default format is P2D, DX files can also be written by explicitly specifying the file extension as DX.

When the polygon/polyline editing tool is active many of the other standard mouse operations are disabled.

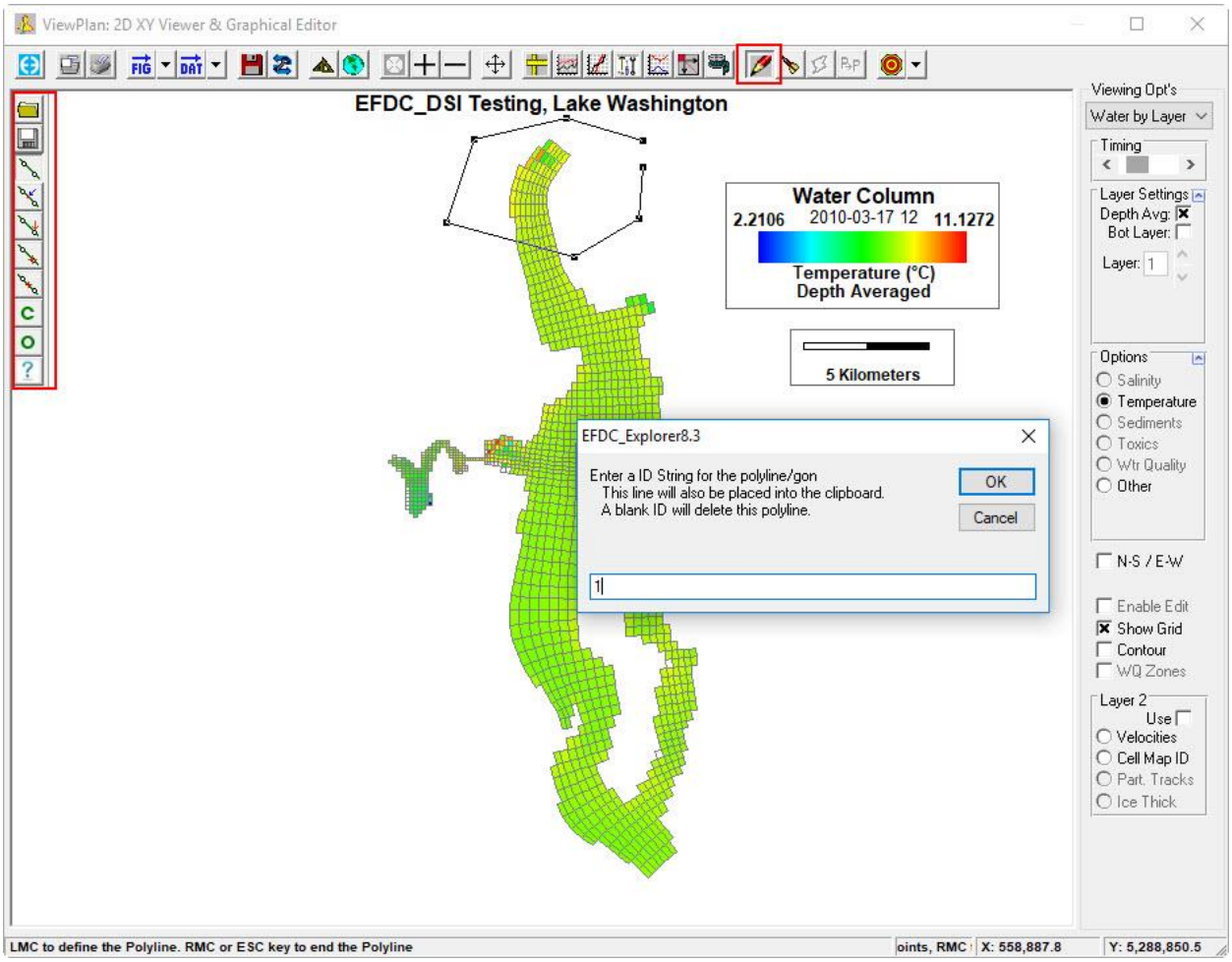


Figure 8-14 ViewPlan: Polygon Tool.

Table 8-4 Polyline Tool Buttons.

	Open an existing polyline for editing.
	Save a polyline. It is necessary to select this icon for the polyline to saved.
	Draw a new line. LMC in the workspace creates a point. Moving the mouse and LMC in another location creates another point with a line joining the two points. RMC to end drawing the polyline.
	Delete previously created line. When this icon is selected, clicking on a line will delete it.
	Insert points on an already created line. Once a point is inserted it can be moved to reshape the line.
	Move points on a polyline.
	Delete points on a polyline.
	Close the current polyline to create a polygon
	Open the current polygon to make a polyline.
	Help for using polyline tool

8.2.3.7 *Water and Mass flux*

This tool is used to calculate the water and mass flux for WQ parameters such as salinity or DO and is most useful when wanting to compute pollutant loadings

To use this tool the user must first have a flux line, or create a flux line with the polyline tool in *ViewPlan*. The user should then select the Water and Mass flux button and select polyline over which the flux is to be calculated as shown in Figure 8-15. From this dialog box the user may also select from various flow options, including total flow, the I,J direction for which flux will be calculated, or the dominant flow.

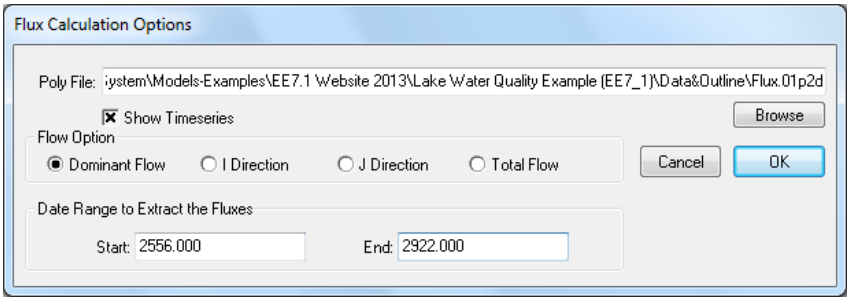


Figure 8-15 Flux calculation options.

EFDC_Explorer will then calculate a time series showing the flow, mass flux (g/s) and average concentration. If the user switches off time series check box then a table of EW and NS total and absolute fluxes is reported for the mass flux past that line for the time period specified.

8.2.3.8 *View Calibration Data*

When calibration data in the Time Series (Section. 7.1.1) or Vertical Profile (Section. 7.1.4) has been configured the user can use the *View Calibration Data* function to display calibration information on selected plots. The symbols, fonts and time tolerance use the data posting options in the Display Options form (Section. 8.3). The data components of this function only apply to the *Water Column* view. When viewing the water column results EFDC_Explorer matches the viewing parameter to the data parameter. If there is no data currently configured data for the current viewing parameter a message will be displayed

The View Calibration Data toolbar function toggles on and off the display of the following:

Show Data Stations: Label the current plot with Station IDs. This information will be displayed on most views even if there is no matching parameter in the calibration data.

Show Data Values: Labels the current plot with calibration data if there is a parameter match to the data and the data is within the tolerance time of the current view.

Show Data Residuals: Labels the current plot with residual (Model – Data) if there is a parameter match to the data and the data is within the tolerance time of the current view.

Use Layer Options/Show All Layers: This option enforces the calibration data layer options to the current view. If not checked the data and residuals displayed will be based on depth averages.

Set Model/Data Time Tolerance: Allows the user to set the number of minutes allowed between the model snapshot time and the data time. If the absolute time differences are less than or equal then EFDC_Explorer will consider the two time the same and display the appropriate data or residuals.

Show Previous Data: This function jumps the current view backwards in time to the previous model-data match for the current parameter. If a match is not found, or the current model time is less than the first data record, a message will be displayed in the status bar indicating that there was no match. Ctrl-< keystroke performs the same function.

Show Next Data: This function jumps the current view up to the next model-data match for the current parameter. If a match is not found, or the current model time is greater than the last data record, a message will be displayed in the status bar indicating that there was no match. Ctrl-> keystroke performs the same function.

8.2.3.9 *Model Results Extraction Tool - Export XY data file as I,J*

The *Model Results Extraction Tool* (for extraction of x, y and current value using the mouse or pointer) in *ViewPlan* has been updated from release of EE7.2 with a function to extract/load model results as I,J. Previously EE only allowed exporting as X,Y points. The new GUI is shown in Figure 8-16.

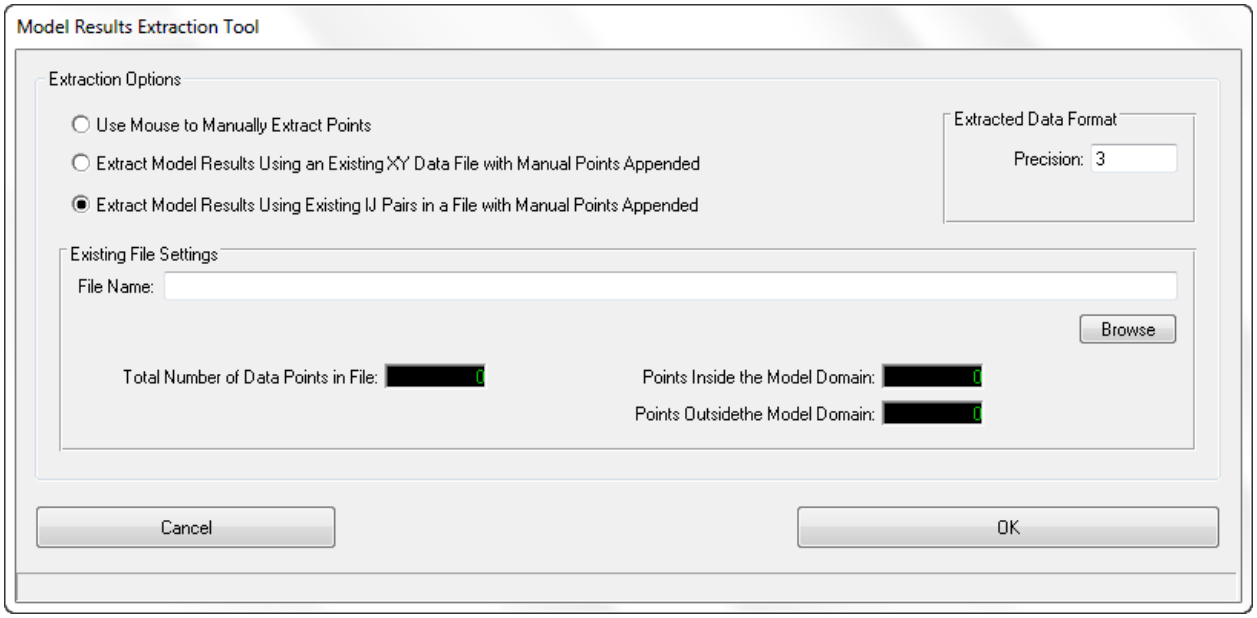


Figure 8-16 Extraction Data Tool.

8.2.4. Navigating the View

The *ViewPlan* toolbar has a range of different viewing navigation functions. However, the mouse and mouse – keyboard combination also have some very useful navigation capabilities. The following summarize the two main navigation features not on the toolbar.

Zooming: The user may zoom to any region of the view simply by RMC, holding down and dragging the mouse. When the user releases the mouse button the screen will be zoomed, centered on the area selected.

Panning: The user may pan in any direction by holding the Shift key down and RMC, holding down and dragging the mouse in the direction and distance desired.

The other zooming and panning functions provided on the toolbar are also available. At times the zoom extents button is grayed out due to the view already being zoomed to full.

8.2.5. Reporting Units

EFDC_Explorer can generate the plots and tables in either metric (the default) or English (Imperial) units. This feature is toggled on and off using the Alt-M keystroke. The current reporting units system is shown in the status bar at the bottom of the form. The user can also LMC the units label on the status bar to toggle the units.

EFDC and EFDC_Explorer require the model generation and inputs to be in the metric system. The use of the English units for display is only for reporting. In order to edit or change the model inputs the reporting units must be set to metric.

8.2.6. Second Layer

Many of the *Viewing Opt's* can be overlaid with either velocity vectors or cell indexes using the options in the *2nd Layer* frame. The display format of the 2nd Layer parameter is controlled by the current settings for that parameter. For example, if velocity vectors are to be overlaid water column TSS and the user wants to change the format and/or plotting locations of the velocities, the user must go to the Velocity view, make the desired changes and then go back to the Water Column view and redisplay the TSS concentrations. The velocity vectors will then be overlaid on the TSS plot in the new format.

8.3 ViewPlan Display Options

The primary display options form for *ViewPlan* is displayed by one of several access methods. They include pressing the button on the toolbar or RMC'ing on either the legend, horizontal scale or timing frame. The *Display Options* form consists of five tabs shown in Figure 8-17 to Figure 8-24.

8.3.1. General Options Tab

The *General Options* tab is shown in Figure 8-17. In the *Model Grid Characteristics* frame the user may change a line format or color for a particular option. This is done by LMC on the picture box displaying the format. A format form will be displayed allowing the user to make the desired adjustments. The background color of the plot may similarly be adjusted in the corresponding frame.

The values that represent the blue and red ends of the color ramp can be specified. The direction of the color ramp depends whether the blue value is > or < the red value. If the *Autoscale with View* option is selected then the blue and red range will be reset to the minimum and maximum values for the current parameter that are within the current viewport.

From release of EE7.3 the user has a number of options for displaying *Color Ramps* in *ViewPlan* and *View3D*. The typical choice is the temperature color ramp which shows blue for low values and red for high values. Many other color ramps are now available from the dropdown list. If the user selects *Single Color* from the dropdown then the color ramp will use a gradient based on the *Single Color* option selected beneath the drop-down list.

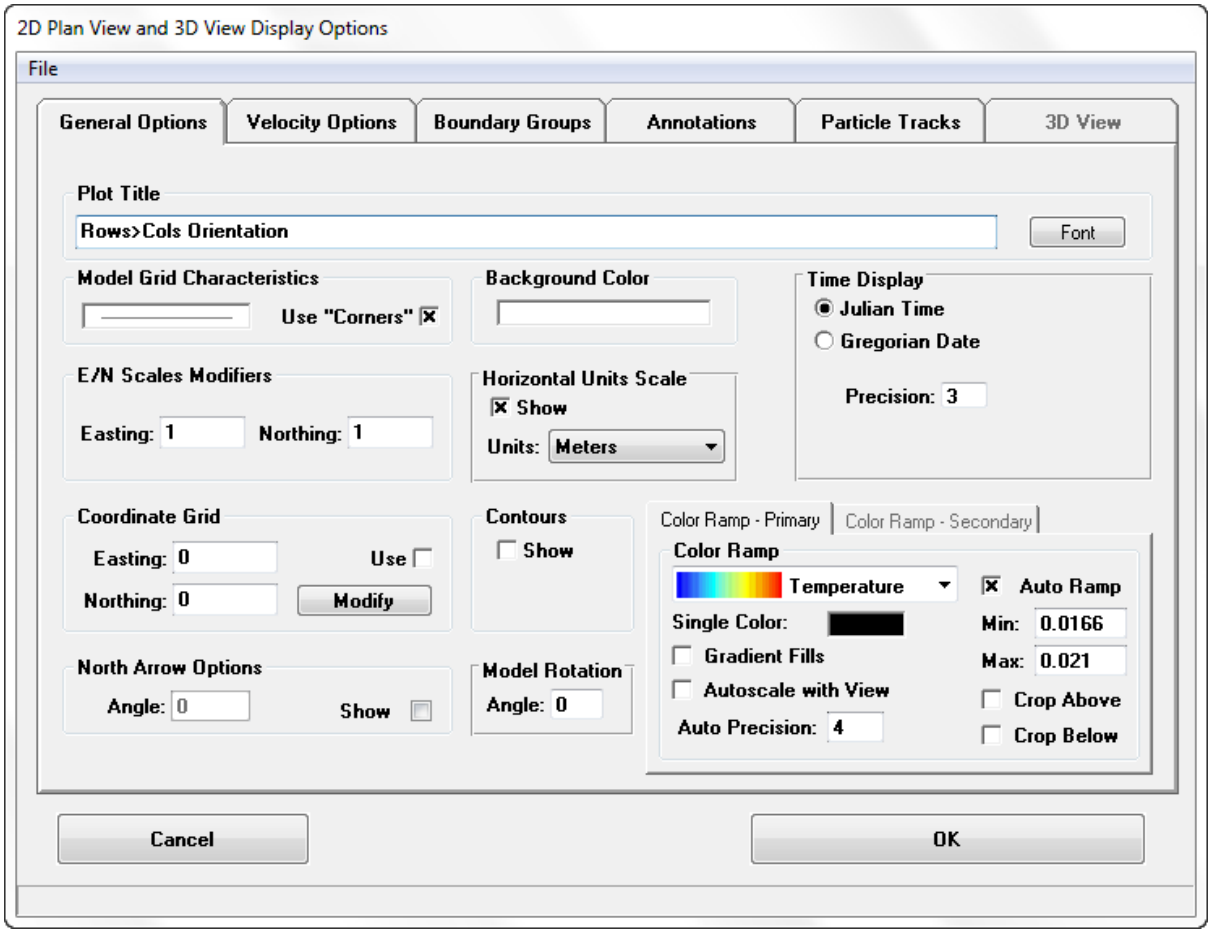


Figure 8-17 ViewPlan Display Options: General Options.

The *E/N Scale Modifiers* is used to scale up one dimension in order to better view a model. This should only be used for models that only have one active cell across the dimension needing scaling. For example, viewing a test case model of a straight river 100 km long (I component) and 100 meters wide (J component). If viewed with a 1:1 horizontal scale it would be difficult to see the model cell colors. Using a scaling factor of 5 in this case would produce better visualizations.

The display of the horizontal scale is optional. The units are set in the *Units* dropdown list. The scale units are set independently of the current reporting units (Section 8.2.5).

A grid can be overlaid on the model using *Coordinate Grid* options. The desired delta X and delta Y spacing should be entered in the Easting and Northing fields, respectively. The label and line styles can be adjusted by pressing the *Modify* button. On the formatting form, the user can select the *Link* check box to assign the same format for both the X and Y.

ViewPlan will display a north arrow when the *Show* box is selected in the *North Arrow Options* frame. The user may also rotate the arrow to align with the actual northerly direction by rotation the arrow clockwise by the number of degrees input. Furthermore, a *Model Rotation* option enables the user to rotate the model for display, thereby allowing it to fit better on the screen or printed page and improve visual alignment. As the model rotates, the north arrow automatically is updated based on the new rotation selected. While EE could previously rotate the model domain about a centroid, this new feature from release of EE7.2 changes the real co-ordinates in order to rotate the model.

Rotation is in the clockwise direction; however, the user may also enter a negative angle for counter-clockwise rotation. Note that by default the north arrow rotates with the model and may be manually adjusted if required. Figure 8-21 shows an example of a rotated model.

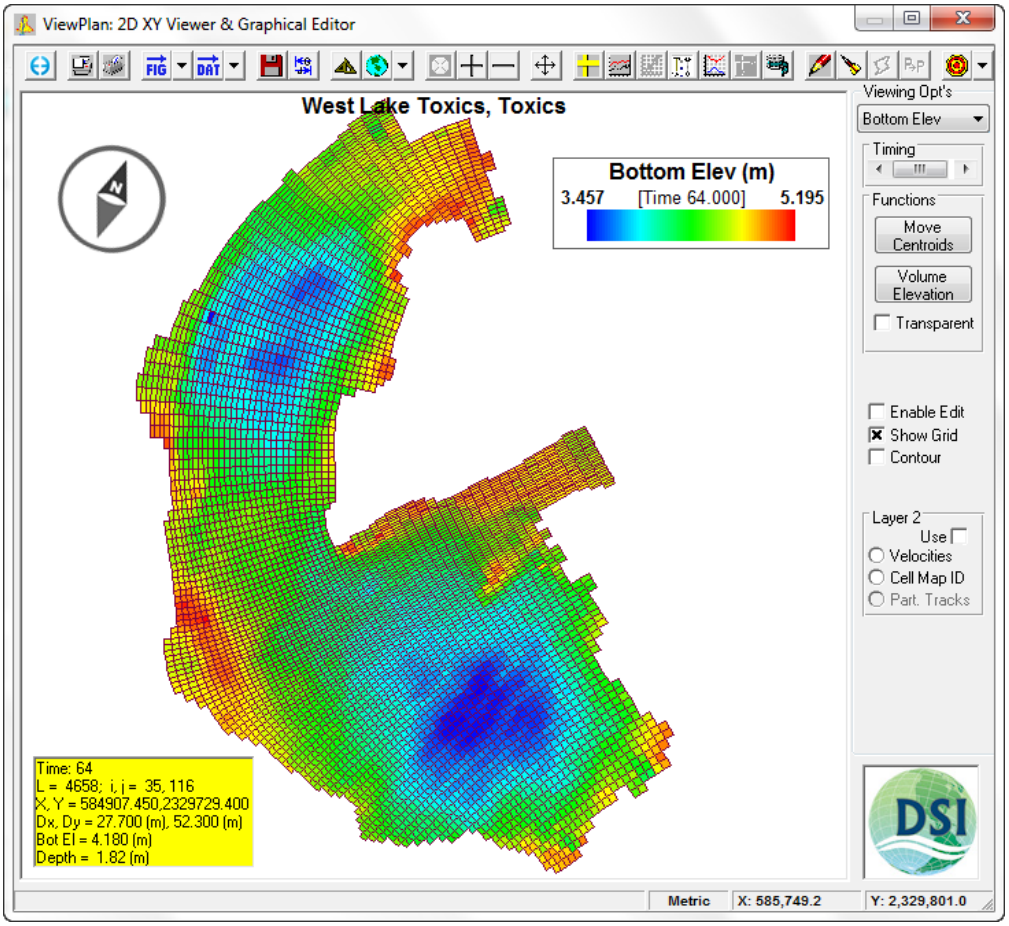


Figure 8-18 ViewPlan: Model Rotation.

8.3.2. Velocity Options Tab

Figure 8-19 shows the *Velocity Options* tab of the *Display Options* form. This tab provides access to the velocity vector formatting options including scaling, display units, and many other formatting options. If the *Scale Vectors* check box is checked then the length of the velocity vectors will be proportional to the *Scale* factor. If this checkbox is not selected then a constant length vector with the correct rotation will be shown and the magnitude will be labeled next to the vector.

The *Primary* vector controls the vector display format of the currently loaded EFDC model's velocities. The *Secondary* vector controls either a second model's velocities (see Section 7.3.1) or data (see Section. 7.1.8). To change the style of the vectors, LMC on the displayed vector.

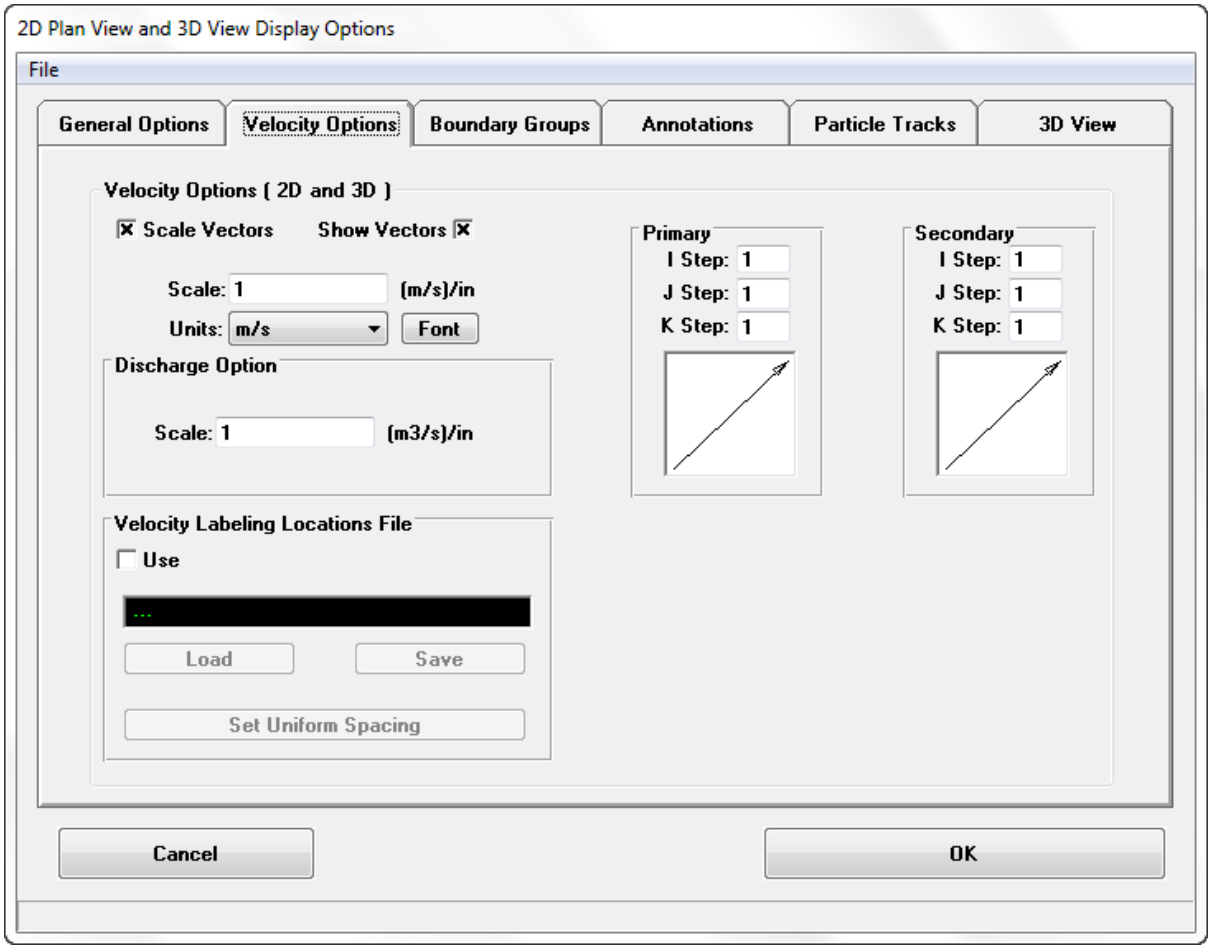


Figure 8-19 ViewPlan Display Options: Velocity Options Tab.

When plotting the velocity vectors, EFDC_Explorer generally loops over the I and J components with the steps shown (e.g. “i Step”). If more control is needed for a project, the user may select the *Velocity Labeling Locations* file. This file is an ASCII file (DAT or P2D format) containing a list of XY coordinates. If selected, EFDC_Explorer loads the XY’s and determines the corresponding cell list. If points are outside the model domain they are ignored. Then when plotting velocity vectors, only those cells in the list will be plotted.

Another feature of EE is the ability to be able to set velocity vectors at regular user-defined intervals independent of the grid and cell size. This is useful when a model varies from fine to coarse grid resolutions. In this case it is not helpful to have velocity arrows sparse in some areas and crowded in others. Selecting the *Set Uniform Spacing* button prompts the user for the delta X and delta Y for the velocity posting file, and subsequently prompts the user for the X and Y point of the lower left corner for the calculation of the spacing. By default this is the lower left corner. EE the creates a velocity map of the posting locations. The user may also load and save the labels and grid for

The user may also set the anchor point of vector: as start, midpoint or endpoint. This is done by clicking on vector arrow in *Velocity Options* which displays a form shown in Figure 8-20. Various other formatting options are available such as the color, taper and anchor point symbol.

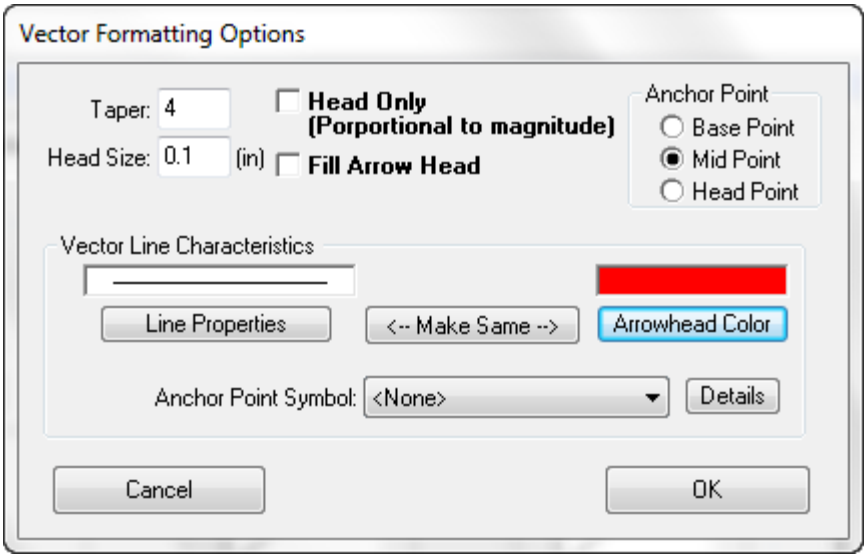


Figure 8-20 ViewPlan Display Options: Vector Formatting Options

8.3.3. Boundary Groups Tab

The way boundary conditions are displayed may specified by the user as shown in Figure 8-21. The boundary conditions are listed in the *Boundary Condition Display* form. The *Color Cells by Groups* check box will allow all boundary conditions of a similar type to be of the same color.

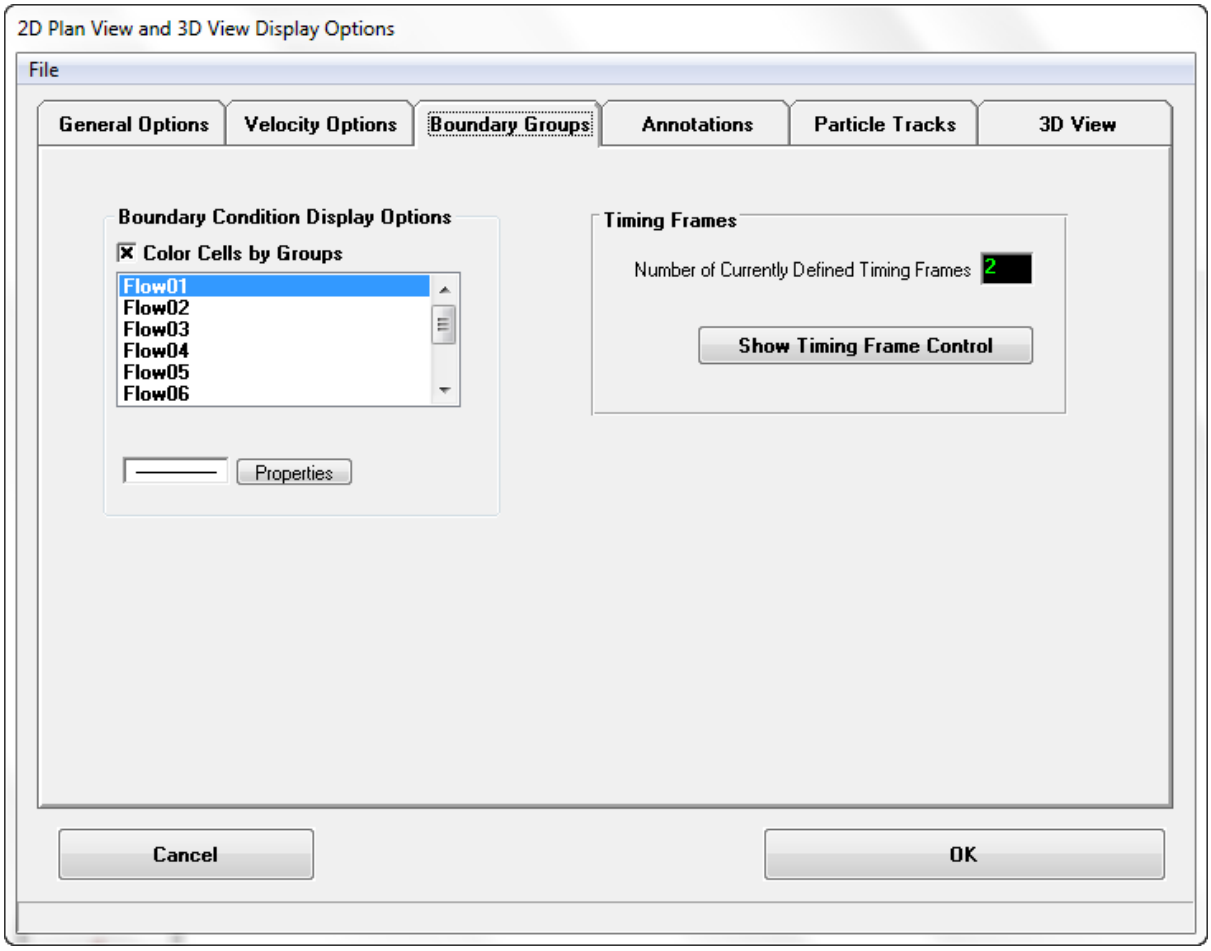


Figure 8-21 ViewPlan Display Options: Boundary Conditions.

Clicking on one of the boundary conditions followed by the *Properties* button allows the user to select the polyline properties such as color, like thickness and line style as shown in Figure 8-22.

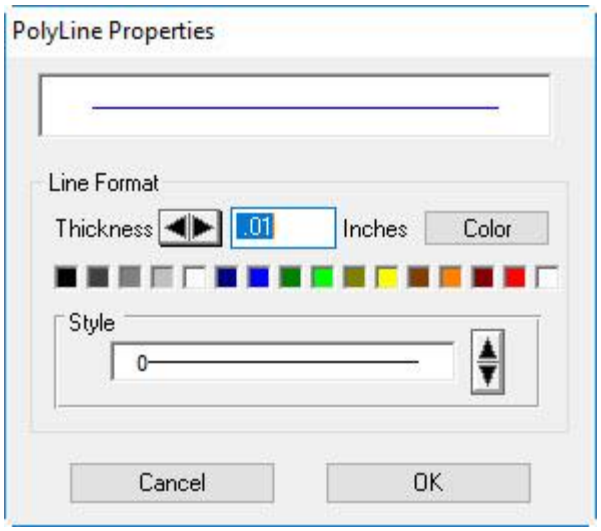


Figure 8-22 Typical Polyline Properties Options.

The *Timing Frames* frame provides the user with the number of currently defined timing frames that are displayed for each boundary condition on the *ViewPlan*. To edit this select the *Show Timing Frame Control* button. The form shown in Figure 8-23 allows the user to define new frames by selecting the *New* button. Each defined frame corresponds to boundary condition listing in the *Boundary Group to Show* frame with the selected BC highlighted in blue. The scale of the timing frame may be adjusted up or down to set the Y-axis to most suitable level.

The value of the scale field is dependent on what type of boundary forcing time series is selected. For winds the scale is m/s, for flows and pressure the scale is days. RMC on the *Timing Frame* allows the user to jump to the time selected on the frame.

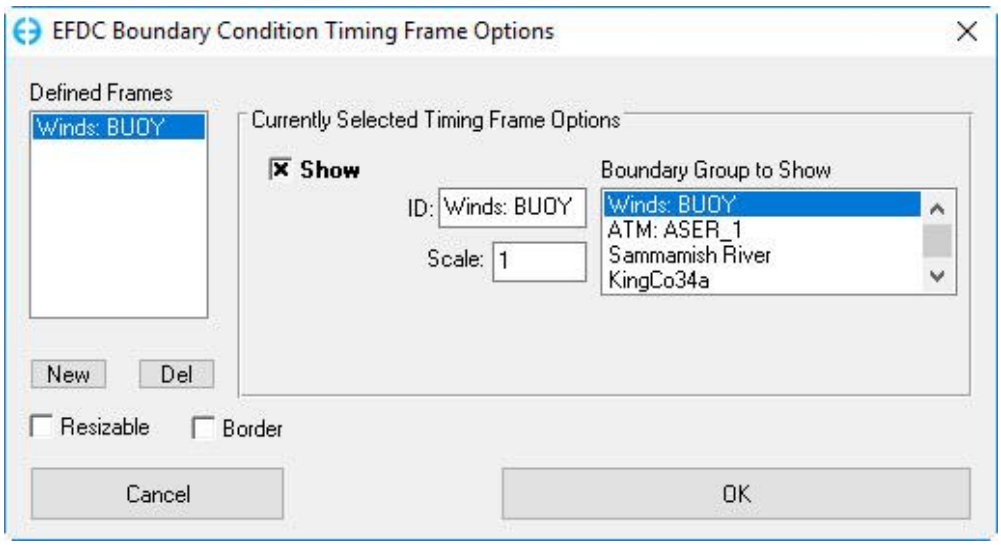


Figure 8-23 Timing Frame Options Control.

8.3.4. Annotations Tab

The main user labeling and overlays are controlled in the fourth tab of the *Display Options* form (Figure 8-24). A labels file and a data posting file are essentially the same format (X Y Label). The main difference is the labels in the labels file can be individually formatted while those in the data posting file all use a single format. For labels, EFDC_Explorer uses a file with the same name as the file with the labels but with a LBF extension to save the individual label formats.

Google Maps © standard .KML files may also be imported as a data posting or labels file.

When calibration data in the *Time Series* (Section 10) or *ViewSlice* (Section 9) has been configured the user can use the *View Calibration Data* (Section 8.2.3.8) toolbar function to display calibration information on selected plots. The symbols, fonts and time tolerance in the *Data Posting File* frame control the display format of the calibration data.

The *Global View* checkboxes allow the user to turn on or off the entire corresponding feature. For both labels and overlays, individual items can be turned on or off also.

To change the display format of an overlay line, LMC on the line display.

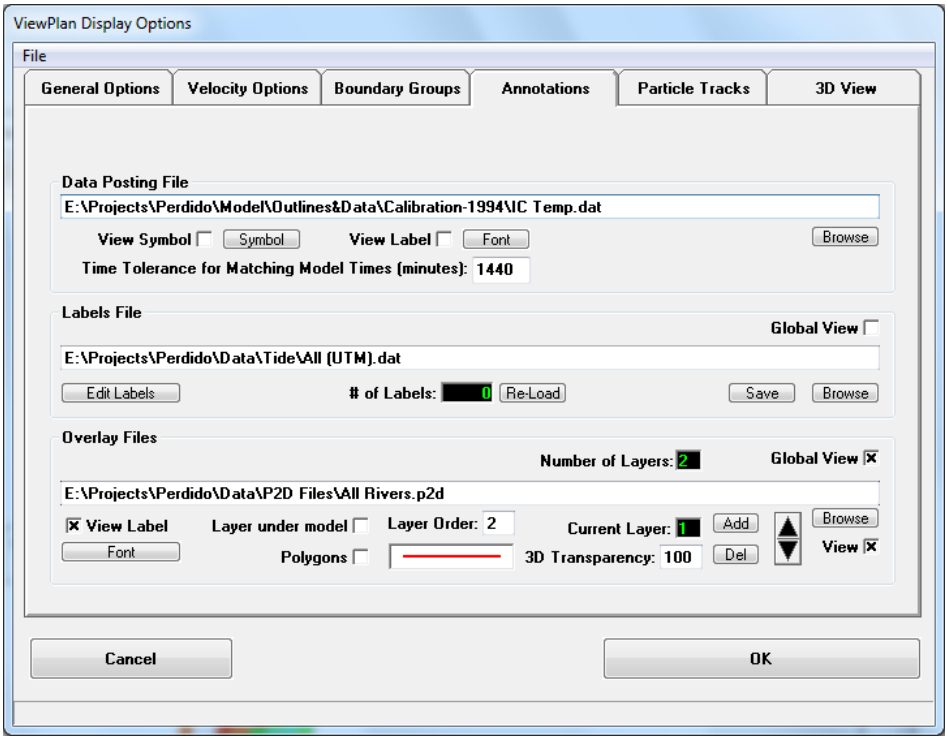


Figure 8-24 ViewPlan Display Options: Annotations.

The *Layer under model* checkbox tells EFDC_Explorer to plot the specified layer first and overlay model results on this.

The *Polygons* check box will display the XY data as closed polygons.

8.3.5. Particle Tracking

When the using the Particle Tracking functions, the user is able to configure the display characteristics of the particles in the *Particle Tracks* tab as shown in Figure 8-25. The user is able to set the *Time History* of the particles, such as whether to display:

- the entire track to the current time,
- the entire track (for the whole time),
- the recent history of the track, up to a time specified by the user,
- current position of the particle only.

In addition to this, the user may also specify the width and colors of the tracking lines, and set the symbols for the particles.

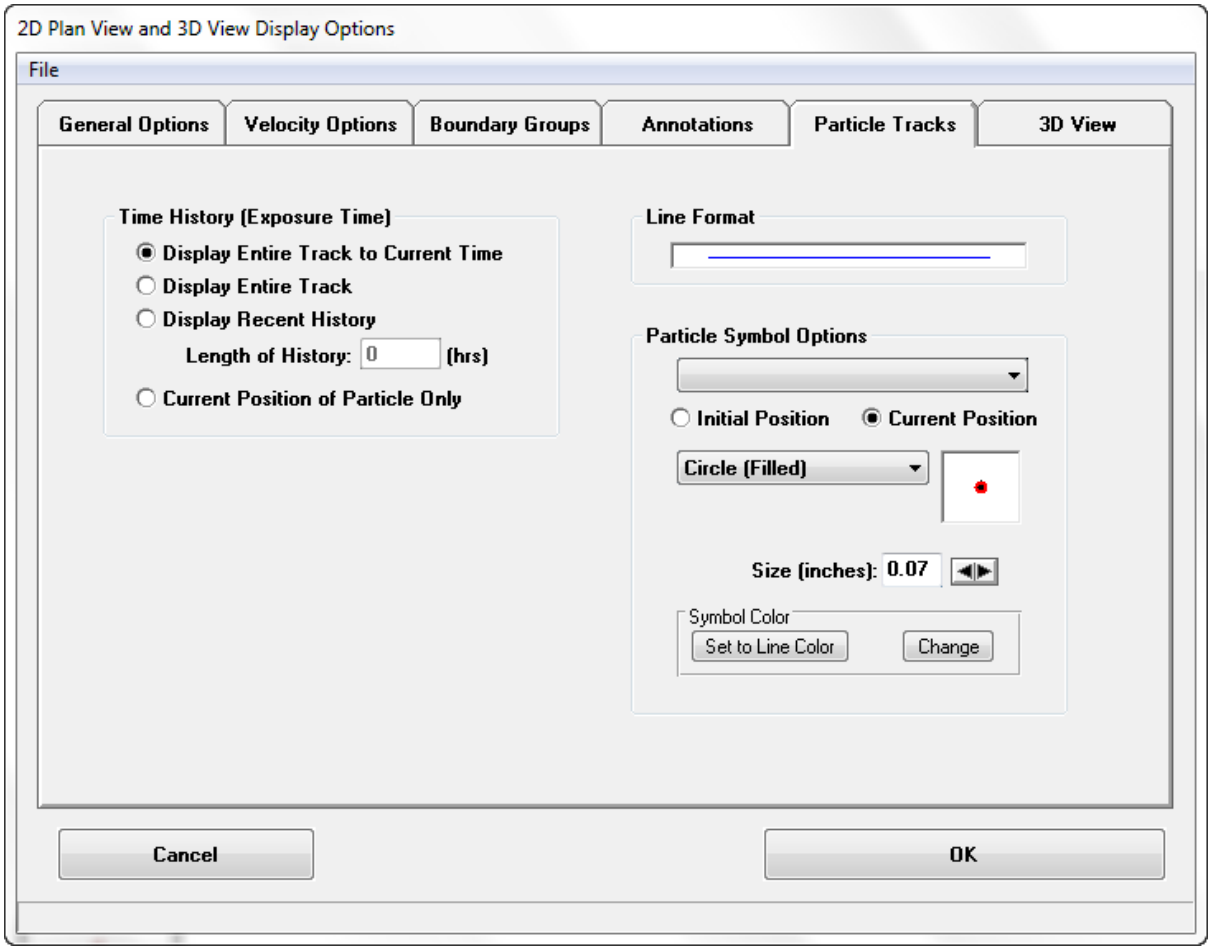


Figure 8-25 ViewPlan Display Options: Particle Tracks.

8.4 General Pre-Processing Functions

ViewPlan provides access to the visual/point & click editing features of EFDC_Explorer. The features and functions discussed in this section relate to more than one Viewing Opt. View specific pre-processing functions are discussed in the respective sections below.

ViewPlan is used for both pre- and post-processing. Therefore, if model results are loaded, the user needs to make sure the current time is set to the 0^{th} time or initial conditions. Of course, only initial conditions data can be edited, the model results cannot be edited. When the user desires to edit the initial conditions, make sure the timing scroll bar is moved to the far left. The *Enable Edit* check box should then be displayed and enabled. As a safeguard to prevent inadvertent data modification, you can only edit data if the *Enable Edit* check box is checked. The following discussions assume that the above two conditions have been met.

8.4.1. Single Cell Edits

To edit the cell properties of a single cell, RMC on the desired cell. Depending on the *Viewing Option* either the cell editing form will be displayed or a popup menu will be displayed. If a menu is displayed select *Edit*. The *Modify/Edit Cell* form is then displayed (see Figure 8-4). You may enter a new value for any of the parameters displayed or use an operator (see Section. 1.3.4).

8.4.2. Multiple Cell Edits

A group of cells may be edited at the same time using the same *Modify/Edit Cell* form shown in Figure 8-26.. The groups of cells may be selected in one of two ways. The simplest is to use the Alt-RMC and drag method to select a group of cells within a rectangular box. All the cells whose centroids are within the box will be included in the group edit. The second method is to use the polygon based *Multi-cell selection tool* from the *ViewPlan* toolbar ( ). Click on the toolbar button and you will be asked for a polygon file. This option allows you to have more control over which cells will be selected than a simple rectangle. Select your file and the selection method and then press *Apply*. A list of cells that match the criteria will be included in the cell group to edit.

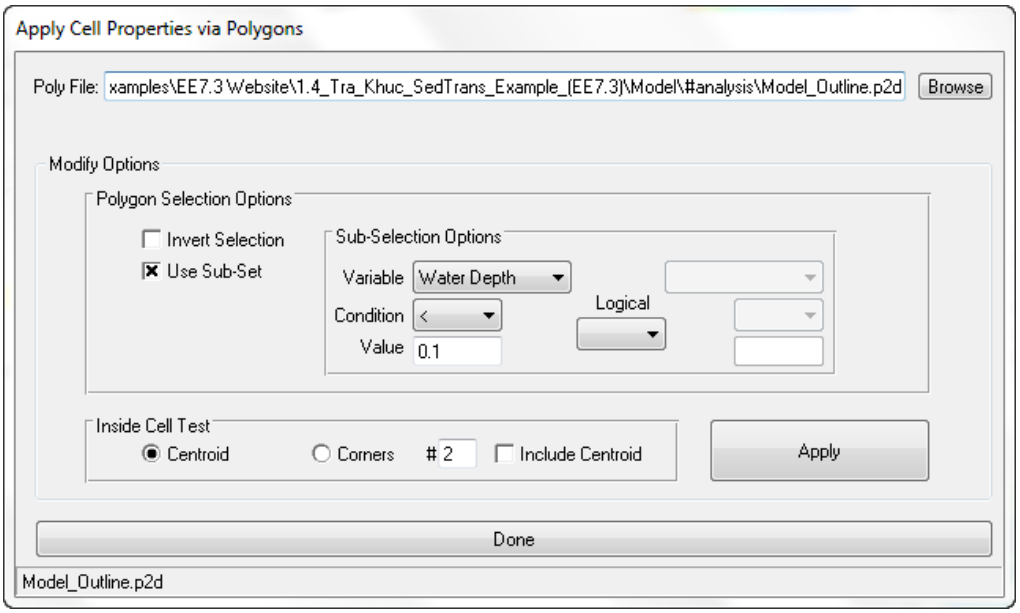


Figure 8-26 ViewPlan: Apply cell properties with a polygon.

The *Edit/Modify* form will come up allowing the user to make group changes to the parameters. Some properties do not have a value in the input box because that property varies between the selected cells. If a fixed value is shown, then that value is constant for the selected cells. To replace a property with a new value, enter the new value into the appropriate input box. To apply the *Operator* function to offset or otherwise adjust all the properties in the selected region, put a "+", "-", "*", or a "/" as the first character followed by a space and then the value to apply. For example, say the user wants to lower a region's bottom elevations by 0.5 meters. Select the region desired, then type in the *Bottom Elev*: input box "- 0.5". EFDC_Explorer will then lower all the bottom elevations of the cells in the group by the amount specified. The method can be used for any of the cell properties.

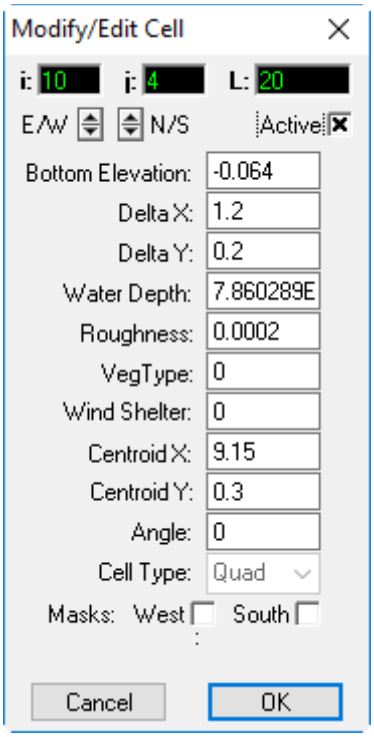


Figure 8-27 ViewPlan: Editing cell parameters.

8.4.3. Cell to Cell Copy/Assign

This feature allows the user to select a "Source Cell" and then copy the current property into subsequent cells ("Target Cell"). First the user must turn on the *Copy Cell Properties* button on the toolbar. The "Source Cell" is then selected using Ctrl-LMC. An input box is then displayed and filled with the Source Cell's property. The user can change the value or use the operator method noted above. The user can then apply the "Source Cell" property (or whatever is in the property input box) by Ctrl-RMC. The user can keep applying the value or operator by continuing to Ctrl-RMC'ing on the desired cells.

The user can also apply Operators using this function. For example, to apply a rise of 50% over the current value of whatever parameter is being edited, follow these steps:

- Select the desired *Viewing Opt's*.
- If output data is currently loaded, set the current time back to the start. The *Enabled Edit* will be enabled if the user is at the IC timing position.
- Check *Enable Edit*.
- Press the P→P button on the toolbar. This is a toggle button. This function will be active until the user toggles it off by pressing the button again or changing the "Viewing Opt's".
- Enter "* 1.5" in the *Cur Value* field (note the space between the * and the 1.5 is required).
- RMC'ing on the cells, while holding down the Control key, will apply a 150% factor to the value of the cell. For example, if bottom roughness is being edited and the initial value was 0.02 m, after Ctrl-RMC'ing the updated value would be 0.03 m. The cell color would be updated.

8.4.4. Data Field Smoothing

The bathymetry, salinity and temperature data can be smoothed over the whole domain or using a polygon to subset the cells. Pressing Ctrl-S displays the smoothing control form. The *Load* button in the *Polygon* frame can be used to input the polygon file, otherwise the entire domain will be used. Enter a weight in the *Smoothing Factor* input box and then click *Apply* as many times as desired. Each click performs a single pass through the data. After each pass the data are redisplayed so that the user can view the results prior to applying another smoothing pass.

8.5 General Post-Processing Functions

The basic use of EFDC_Explorer's ViewPlan post-processing function is 2D maps of various parameters at user specified times. The viewing options have many variations depending on the context. For example, for water column information the user may specify depth averaged results or review the results for each layer of the water column. Another example is for the sediment bed, for which the user may view the results as either mass weighted depth averaged results (e.g. d_{50}'s), total mass summed over all the layers, or layer by layer.

The desired time is scrolled to using the *Timing* scroll bar, using the PgUp/PgDn keystroke combinations or Ctrl-G to jump directly to a specific date. The output settings are set using the *Display Options* form accessible from either the toolbar or RMC on the legend, timing frame or horizontal scale.

Some of the general features of the post-processing will be discussed in the following subsections.

8.5.1. Time Series

A family of time series curves of the same parameter can quickly be displayed. Press the *Time Series* button on the toolbar and then LMC on each cell to be displayed. For all parameters except for the water depth derived parameters, each time the user selects a new cell the *Cell Selection List* is updated with the I, J and K of that cell and the cell location is shown on the model domain as shown in Figure 8-28. A number is shown next to each cell selected that corresponds to the cell # in the list. There is no limit to the number of time series can be shown at once.

The user may manually adjust the I, J and K value in the list, as well as turn on or off the cell by setting *Use* to 1 or 0 respectively. Selecting *Clear* clears the whole list. The user may save and reload a list of time series location for use at a later time using the *Save* and *Load* buttons.

Selecting *Show* causes the time series data to be loaded and displayed for the selected parameter and cells in the list. Section 10 provides a discussion of the use and options for the *Time Series* utility.

Toggle the *Time Series* button to reset the series list or change the view option. The previously used list of cells is remembered after the user returns the time series option.

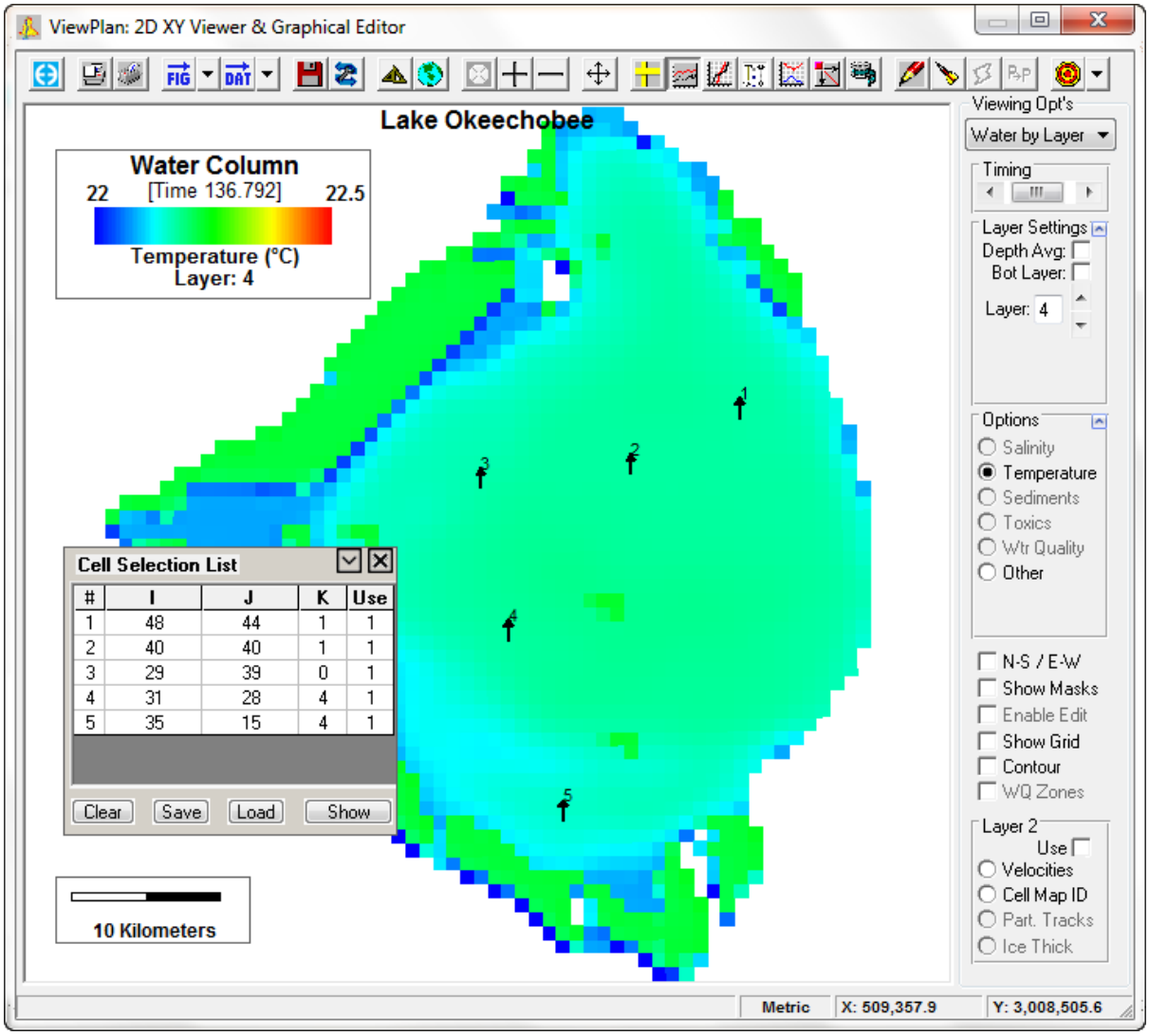


Figure 8-28 ViewPlan: Cell selection for times series plots.

8.5.2. Vertical Profile Plots

Vertical profile plots maybe be displayed by selecting the highlighted button in Figure 8-29. The user can then click on the cells for which vertical profiles will be displayed and an I, J, K list will be displayed for each cell selected. After the user selects Show a vertical profile will be displayed as shown in Figure 8-30.

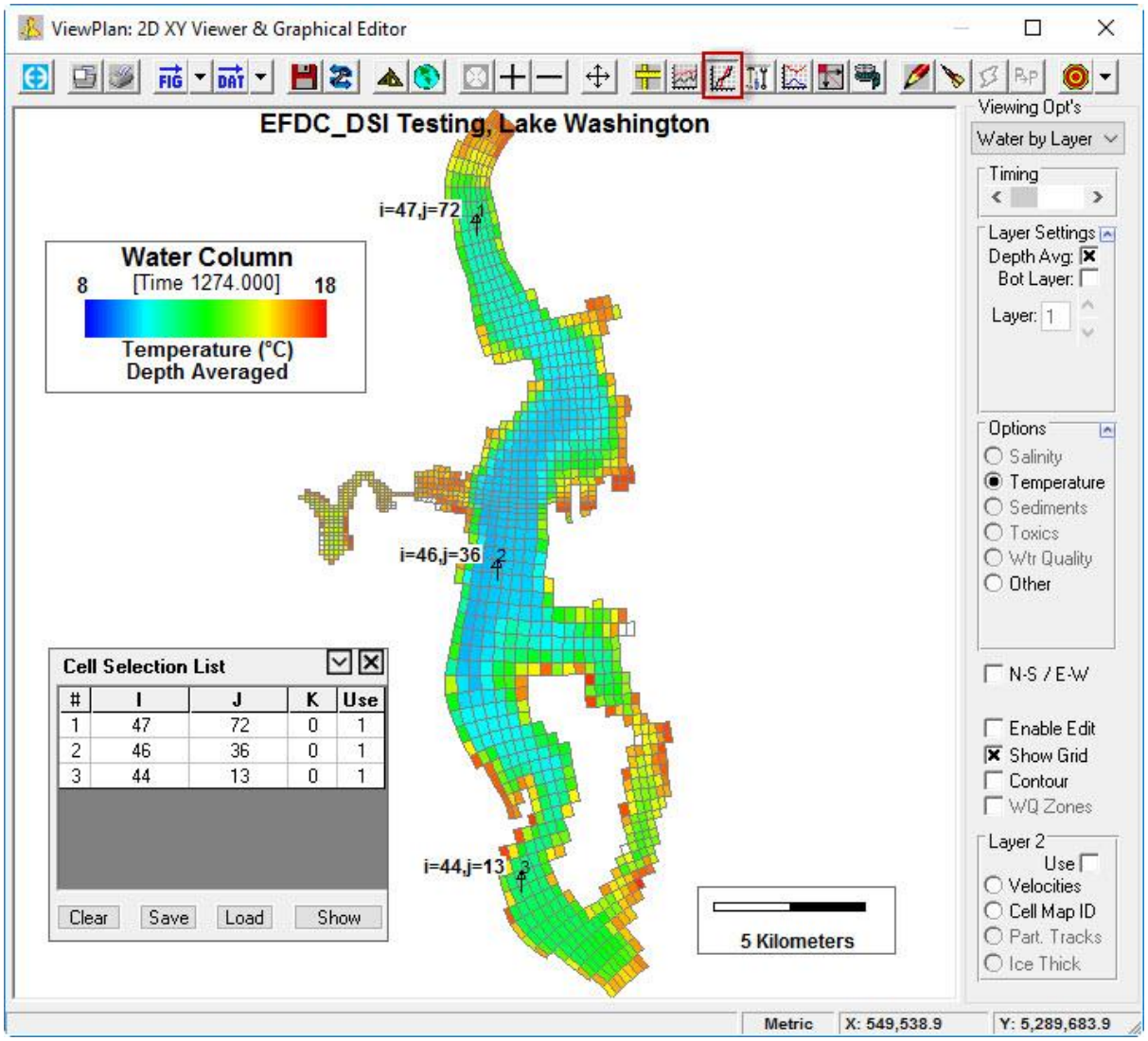


Figure 8-29 ViewPlan: Vertical Profile plotting tools.

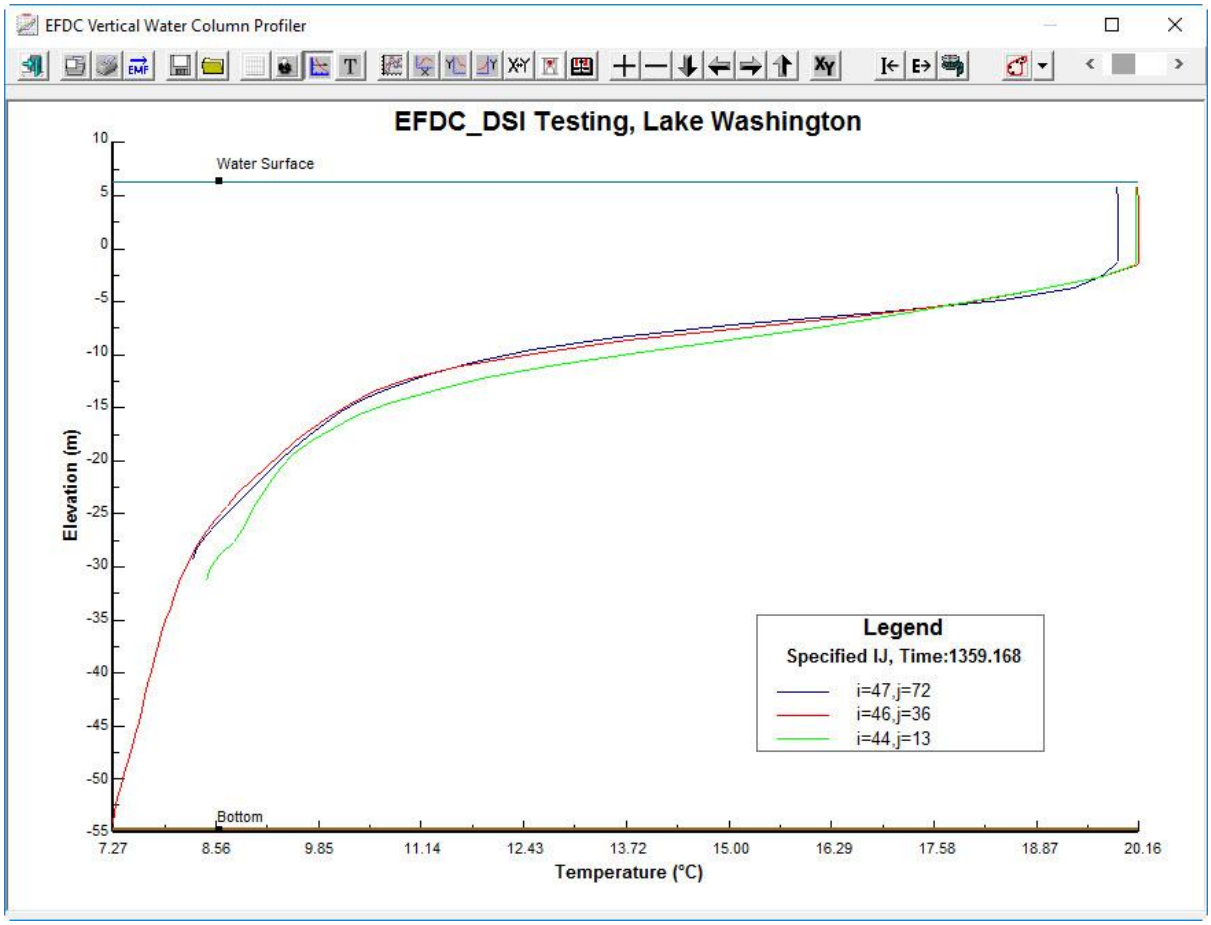


Figure 8-30 Vertical Profile plot.

8.5.3. General Statistics

ViewPlan has the ability to calculate some general statistics of the current parameter and snapshot time. Click the *General Statistics* button and the user will be prompted with a browse window to select a polygon file. If the user clicks *Cancel*, EFDC_Explorer will use the current window (not the entire domain). If the user selects a polygon file, then the cells inside that polygon will be used for the calculation.

From the selected cell list a set of general statistics of the currently displayed snapshot in time will be computed. The exact statistics vary with the parameter.

For example, a drifter counting tool can be accessed using this tool. Here the user can load a polyline or use the current plan view of the model to determine the number of drifters for either a snap-shot in time or generate a time series for the whole model. This can be useful for users who want to know the quantitative distribution of particles and generate density maps of particles.

The *General Statistics* button is shown in Figure 8-31. When selected the pop up will ask if the user wants to user a polygon. If the user selects “Yes” then EE will prompt for the polyfile. A summary of statistics for that polyfile or a time series will be generated.

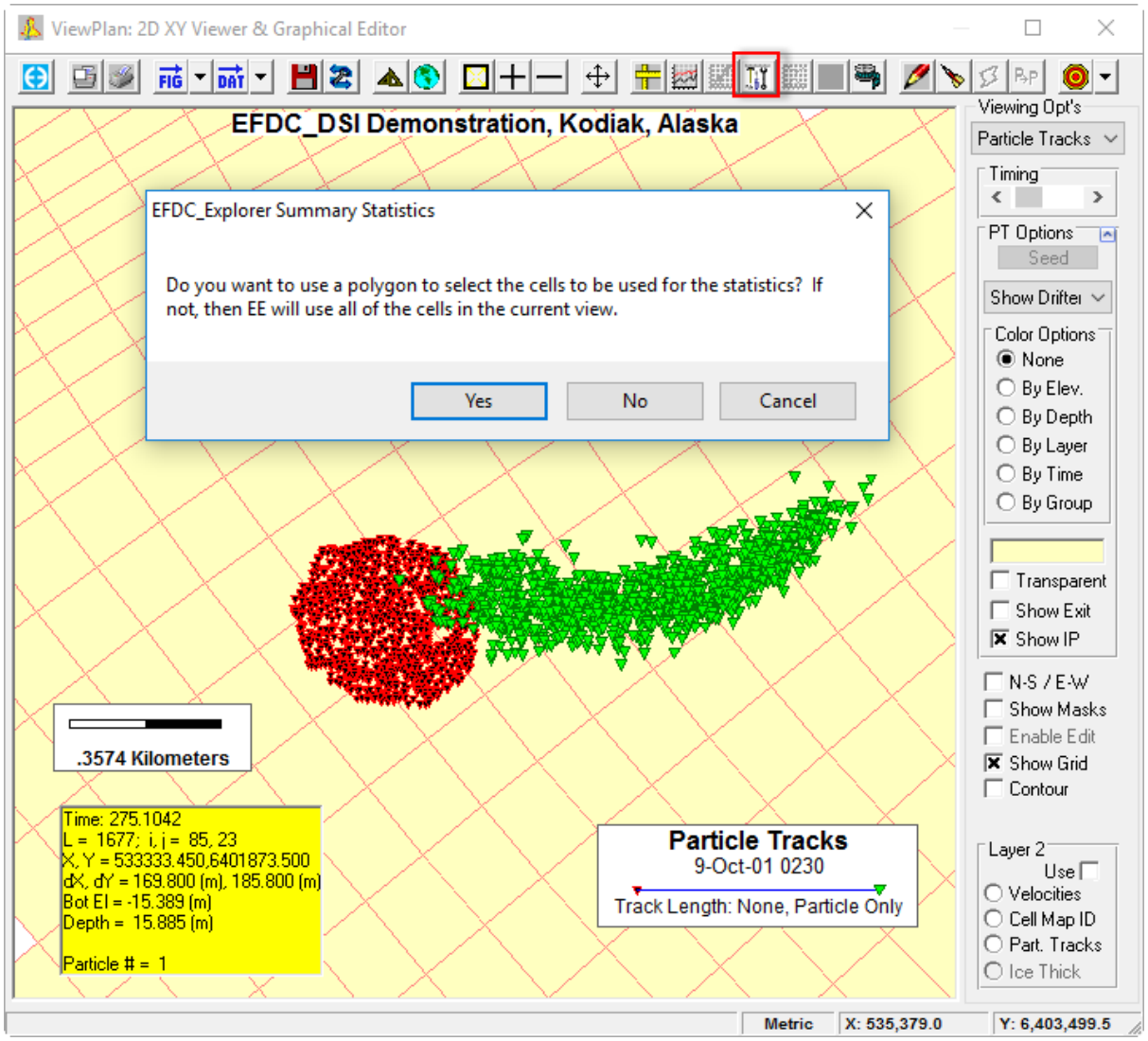
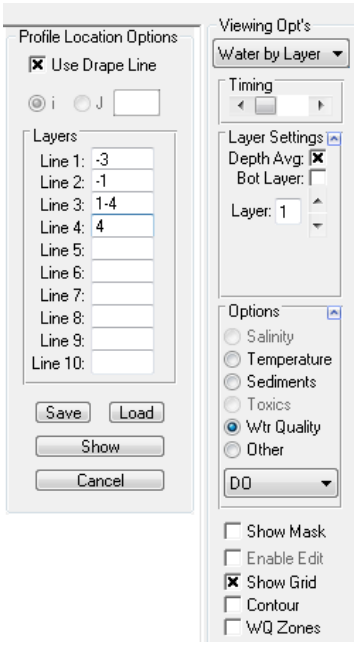


Figure 8-31 General Statistics: Concentration of drifters.

8.5.4. Longitudinal Profile Plots

When viewing the water column in many of the *ViewPlan | Viewing Options* the drop-down the Longitudinal Profile tool is available on the toolbar (the button is greyed when not available). This function is similar to the 2D profile *ViewSlice* but provides an XY longitudinal plot of the current parameter along some profile. The approach to define the data extraction location is identical to the 2D *Slice/Extraction Options* used by *ViewSlice* (see Section 9-2). If a *Use Drape Line* is selected the user should define the layers required, and after selecting the *Show* button will be prompted for the drape line.



Up to ten lines are allowed per plot. The water column data selected is the main parameter to be plotted. However, the layering options are controlled by the *Longitudinal Profile* tool, not the settings in the *Layer Settings* frame. There are some special cases available for display. The following list provides the options available for each line.

Layer # A specific layer number or a range of layers to average. In the adjacent graphic the "Line 4" field has "1-4" entered. Therefore for that line DO being plotted is the average DO for those layers.

0 If a "0" is entered then the value extracted will be the depth averaged value.

-1 Special Case: Water surface elevation.

-2 Special Case: Water depth.

-3 Special Case: Bottom elevation.

-4 Special Case: Bed shear stress.

If the layer field is blank for a certain line, it will be skipped.

The *Profile Locations Options* graphic above shows five lines being defined with three of the lines extracting dissolved oxygen (DO) for layer 1, the average of layers 1-4 and layer 4. Once the options have all been set, the user must press *Show* to view the profile. The user can save and later reload a set of line definitions using the *Save* and *Load* buttons, respectively. Figure 8-32 shows the longitudinal profile plot resulting from the selections. These profiles can be animated to the screen or AVI file using the animate button from the toolbar.

Users may display graph/plot labels in metric or British imperial units using the CTRL M keystroke to toggle the units. Selecting ALT M will toggle the x axis from feet to miles.

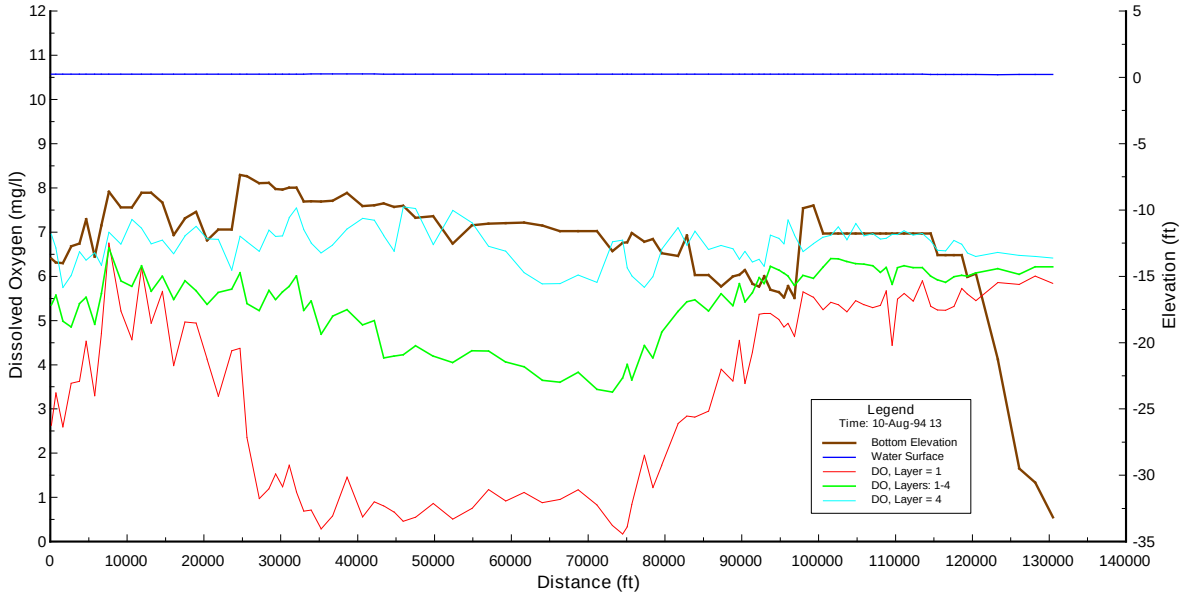


Figure 8-32 Water Column longitudinal profile of dissolved oxygen.

8.5.4.1 *Timing Frame*

The *Model Timing Options* button provides the user with access to the *Profile Options* form wherein is a *Timing Frames* frame. Selecting the *Show Timing Frame Control* allows the user to toggle on and off the timing frame as shown in Figure 8-33 below.

Multiple timing frames may be defined and displayed by the user similar to the use of this tool in the *ViewPlan* mode as shown in Figure 8-34.

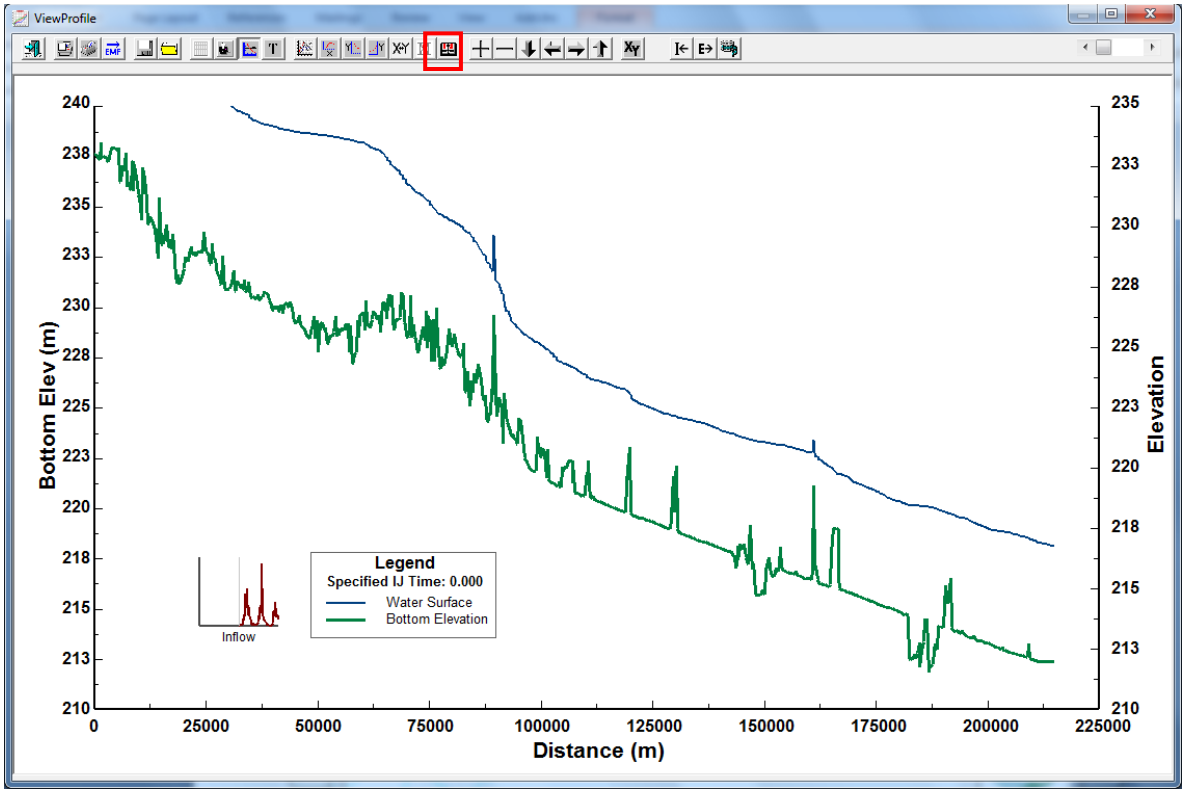


Figure 8-33 ViewPlan: Longitudinal Profile Plots Timing Frame

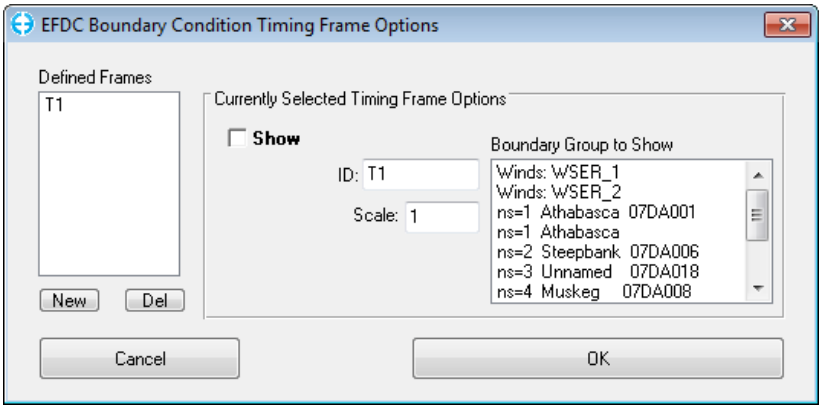


Figure 8-34 ViewPlan: Profile Plots Timing Frame Options.

8.5.5. Water and Mass Flux Tool

The flux comparison tool is provided to assist the user in calibration of flux or flow across a user defined line as shown in Figure 8-35. The user should browse to a poly line that specifies the line across which the flux will be calculated. A number of *Flow Options* are provided by selecting the radial buttons. Options include *Dominant Flow, I Direction, J Direction* and *Total Flow.* The user may also specify a time range over which the flux will be calculated in the *Date Range to Extract the Fluxes* frame.

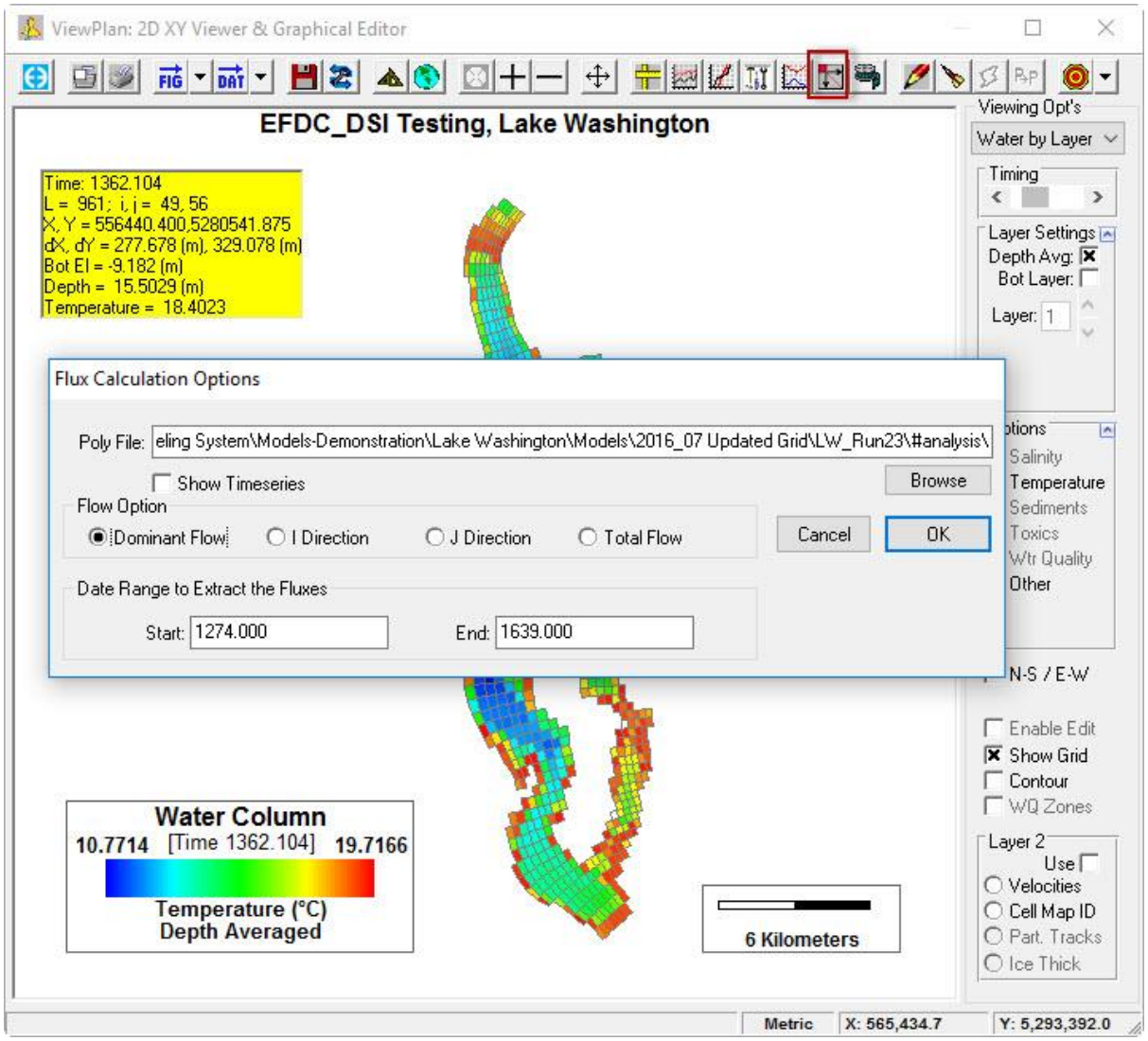


Figure 8-35 ViewPlan: Water and Mass Flux Tool.

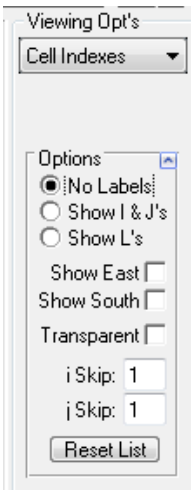
8.5.6. Animation of Results

The results of any of the time dependent results can be output as an animation to either the screen or an AVI file. If the animation is to be saved to an AVI file, the user is asked for the number of frames per second to be output to the file. This will be application specific, but a number of 4 frames per second generally provides a fairly smooth, but not too fast animation. The user should be aware that these files can get quite large quickly. The frame size of the animation will match the frame of the *ViewPlan*. Press any key to stop an animation.

8.6 ViewPlan Main Viewing Options

As previously mentioned the *ViewPlan* | *Viewing Opt's* is adjusted based on what parameters are being modeled, what data has been loaded, and what option has been requested at a particular time by the user. All of the available options are described below.

8.6.1. Cell Indexes



This *Viewing Options* displays the EFDC indexes of I & J and the linear index "L" on a cell map based on LXLY coordinates. The data posting label format is used to display the indexes.

The *i Skip* and *j Skip* fields can be used to reduce the frequency of labeling, if the labels are too crowded. To only label certain cells do the following:

- Enter very large skip values for both the i and j indexes.
- RMC on the cells to be labeled.
- Refresh the view. Until the *Reset List* is pressed, each time the user views the Cell Indexes only the user defined cells will be labeled.

The *Transparent* check box causes the grid displayed to use no fill so only the grid outline is displayed. This feature is also applied for the water column cells, if the cells are "dry". This feature is needed when using geo-referenced bitmap backgrounds to reveal the underlying background.

8.6.2. Cell Map

The Cell Map Viewing Opt displays the cell as stored in the computer memory. Figure 8-36 shows an example cell map using the *Uniform Grid* option. The I and J axes can be labeled, as they are in this example.

Cells can be deactivated or activated using this view. RMC'ing on a cell pops up a menu that allows the user to *Edit, Activate, Deactivate,* or *Set as Show IJ.* The user can deactivate cells from several *Viewing Opt* views. Deactivated cells are still defined and are in memory. The number of active cells does not change until the user saves the project. When the user reloads the project the deactivated cells will be completely removed.

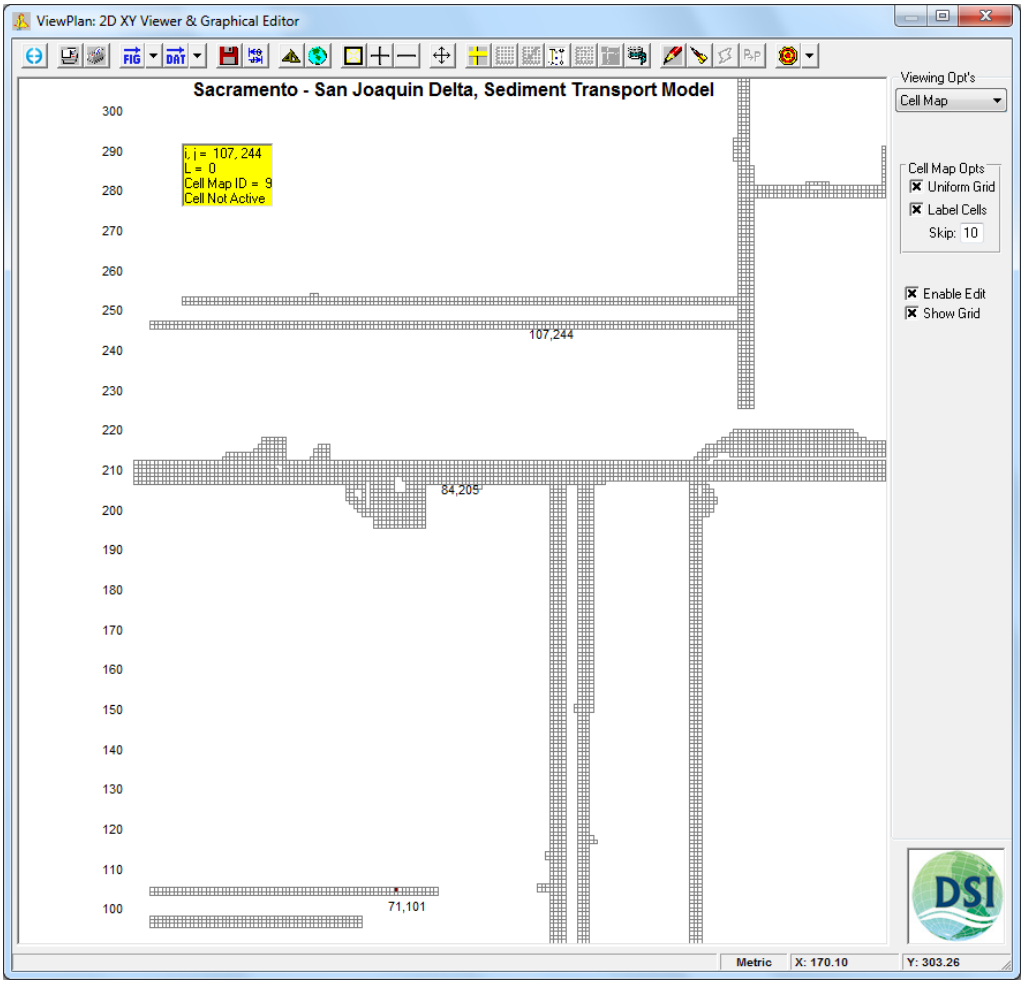
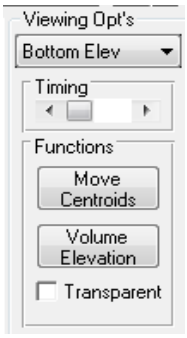


Figure 8-36 ViewPlan: Example Cell Map.

Retaining deactivated cells in memory allows the user to reactivate a cell without losing information, if needed, during the cell editing process. To reactivate a deactivated cell or an originally deactivated cell, RMC on the cell and select activate. If the cell was previously activated it will just change the active cell flag to on. On the other hand, if the cell was not previously activated EFDC_Explorer looks to the immediately adjacent active cells to try to set the appropriate cell dimensions, depth, rotation, etc. Because EFDC_Explorer is using adjacent cells to assign properties, the user should add cells in a manner that has the most active cells adjacent to the cell being activated. The cell properties are displayed to allow the user the opportunity to adjust the cell properties before accepting the activation.

If cells are activated near the edge of the IJ space, EFDC_Explorer adds to the appropriate dimension and shifts the model to reflect the new IJ's.

8.6.3. Bottom Elev



The *Viewing Opt* | *Bottom Elev* displays a plan view of the model bathymetry. The scales can be changed in the *Display Option* form to show resolution, if desired.

An option to shift the cell centroids is available by pressing the "Move Centroids". The *Volume Evaluation* button displays a XY plot of the volume-area relationship and the area-elevation relationship.

Use the *Transparent* function to view an underlying georeferenced bitmap. If the *Show Grid* option is not checked the grid outlines will be colored but the cells will not be filled.

8.6.3.1 *Bathymetry Comparison*

If a *Comparison* model has been loaded (see Section 7.3.1) alt-M allows the user to compare the bathymetry between two models, the current model and the Comparison model. Alt-M toggles the display of *Model Comparison* allowing the bathymetry differences (Base - Compare) to be displayed instead of elevations. This feature is also available for other Viewing Options (see Section 8.2.6). The volume differences can be computed with the Alt-V keystroke. The total cut/fill/net volumes will be displayed and then placed onto the clipboard.

8.6.4. Water Levels

Water Levels option provides for a range of water depth derived parameters. The two main uses of this option are to plot water depths and water surface elevations. Some of the functions only use the water depths and bathymetry, which is always available, but some of the functions also require velocities to have been loaded. The graphic to the right shows some of the water level options grayed out because the velocities were not available. The following summarizes the other options:

Wet/Dry: Provides a two-color display for wet cells (blue) and dry (gray). The wet/dry determination is made using the dry depth unless the *Use Wet* option is checked, in which case the wetting depth will be used. The areas and the numbers of wet/dry cells are reported.

Total Head: Displays the total head, water surface elevation + velocity head ($v^2/2g$). Only available if the velocities have been loaded.

Overtopping: This displays the FEMA defined overtopping depth which is defined as: depth + velocity head ($v^2/2g$). Only available if the velocities have been loaded.

Areal Ext (H): This option displays the FEMA velocity hazard level defined as depth * velocity head ($v^2/2g$). The areas displayed and the areas computed are based on the computed hazard level over a specified minimum. The user can change the minimum level and EFDC_Explorer will recompute the areas and display the results.

Areal Ext (D): This option displays an inundation map and computes areas using a specified minimum depth and duration. The user can change the minimum depths and durations and

EFDC_Explorer will recompute the areas and display the results. If a Compare model or a Compare model and a 3rd model have been loaded (Section 7.3.1) then each model's results will be plotted in the order of Base, Compare then 3rd. The colors displayed for each model can be changed by LMC'ing on the colored boxes beneath the *Areal Ext* checkboxes.

Figure 8-37 provides an example using a Base and a Compare model. Note the areas computed and displayed in the yellow information box.

Delta: Displays the differences in water surface elevations between the Base model and the Compare (i.e. Delta = Base-Compare). A Compare model, with depths, must be loaded for this option to be available.

The default is to use the "dry depth" as the cutoff between "wet" and "dry" cells. If *Use Wet* is checked then the "wet depth" water level will be used. Using the wet depth results in less cells being considered wet. This option is only applicable for models using wetting/drying options.

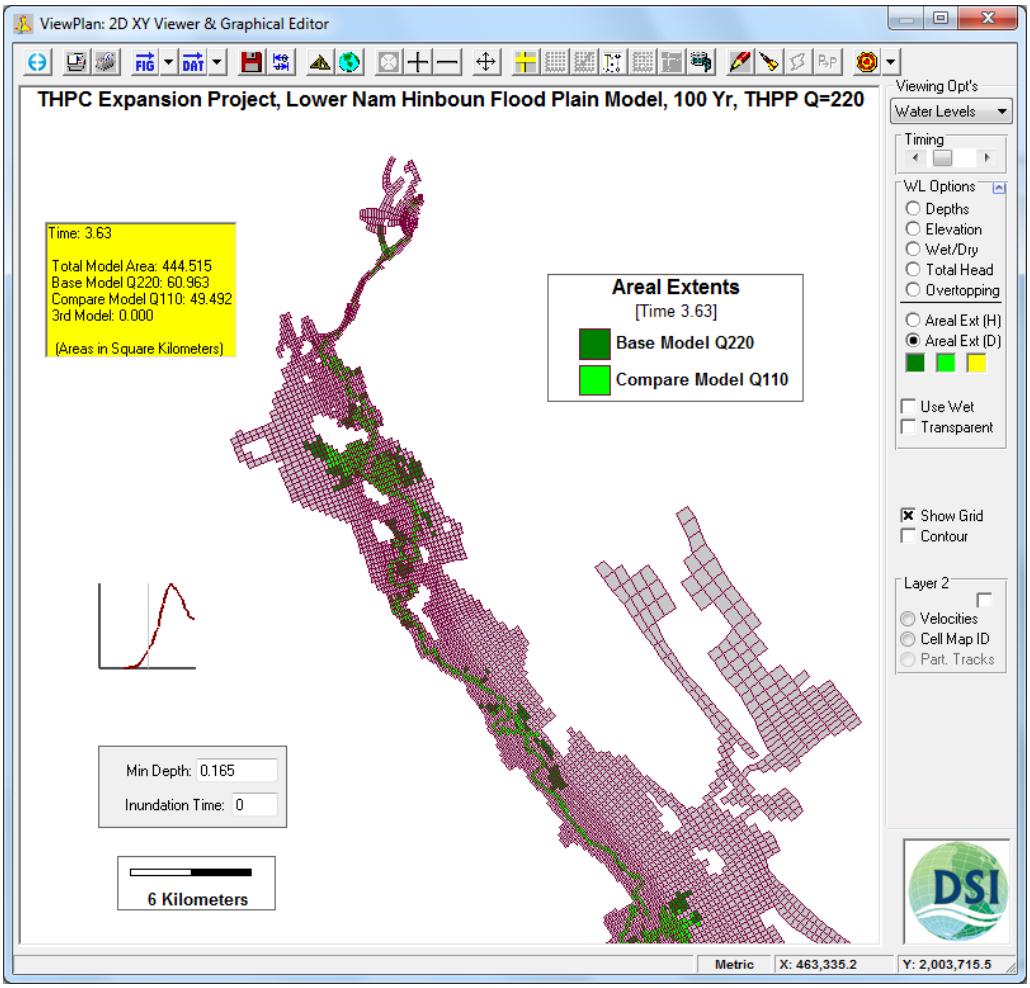


Figure 8-37 View Plan: Water Level showing Areal Extents based on depth-durations.

8.6.5. Boundary Conditions

The *Viewing Opt* | *Boundary C's* allows the user to display the location and type of boundary conditions currently defined. The user may also view the associated time series forcings. These depend on the type of time series defined: whether lookup table or equation based. The *View Cell Lookup Table* and *View Group Lookup Table* are used when the HS uses a lookup table. If the HS uses an equation then the *View Cell Time Series* and *View Group Time Series* options are enabled as shown in Figure 8-38. An example of how to use these options is provided below.

Options Area

If editing is enabled when the user RMC's on a cell, the menu (shown adjacent) is displayed. The user can choose:

Edit To edit the boundary group.

New To add a new boundary condition group. The new group is not required use the cell RMC'd.

Delete To delete the boundary group.

Add to Adjacent Adds the cell RMC'd to an existing boundary group that has as one of its current cell as an adjacent cell.

LMC'ing on a cell that has one or more boundary conditions currently assigned to that cell lists the boundary group(s) and reports on the boundary forcing at the time shown in the Options Area. The user can enter new times between the *Min* and *Max* shown and the forcings will be displayed.

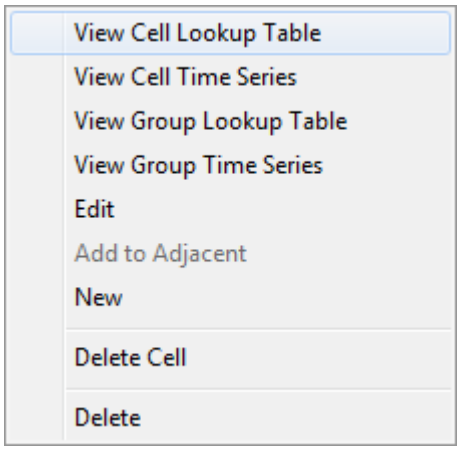


Figure 8-38 View Plan: Boundary conditions RMC popup menu.

EE8 has the capability to plot the flow time series of an actual boundary condition including hydraulic structures defined by equations. EFDC+ creates a new output file, EE_BC.OUT that stores the flows for each boundary condition / cells at each time step. The user can now RMC on any BC in a model that has model output loaded and see the time series of flow.

An example of how to view the time series is shown in Figure 8-39. This model has an open (tidal) boundary at the western boundary and an open (tidal) boundary in the eastern boundary. A culvert has been configured in the middle of the domain, with the upstream cell of the culvert marked with a "U" and the downstream cell marked with a "D". The downstream (D) cell for the culvert has been RMC'd and the user can now select a look-up table or time

series. If the culvert does not use a look-up table this option is not available. In this case the culvert has a time series for flow which will be displayed as shown in Figure 8-40.

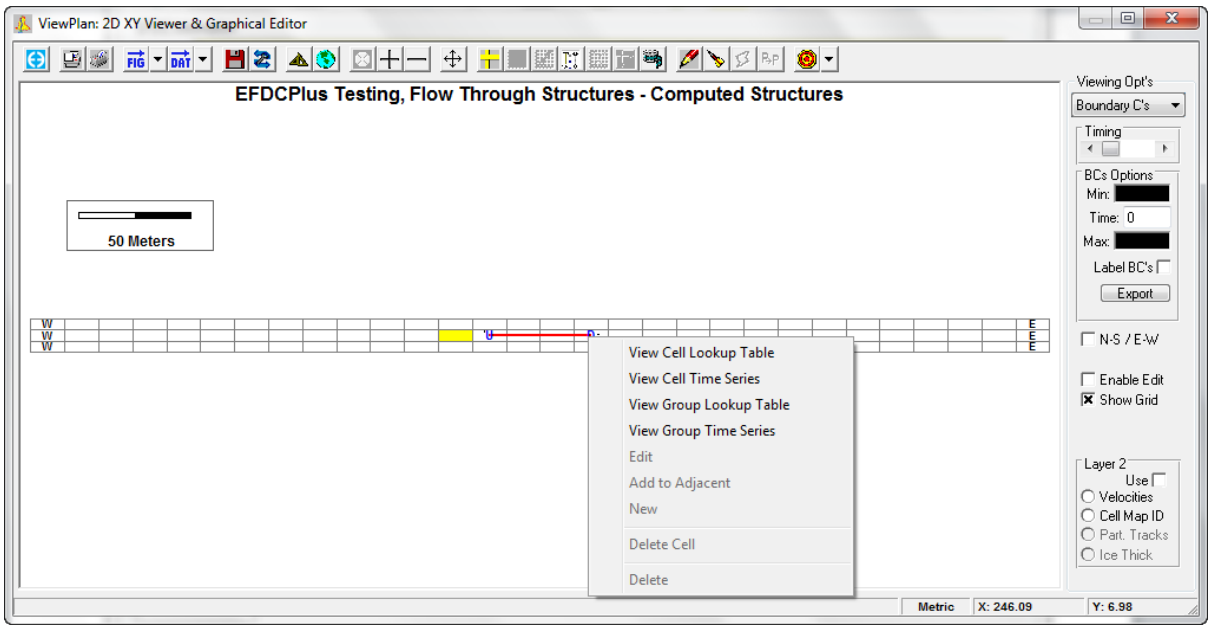


Figure 8-39 Viewing flow time series in a culvert.

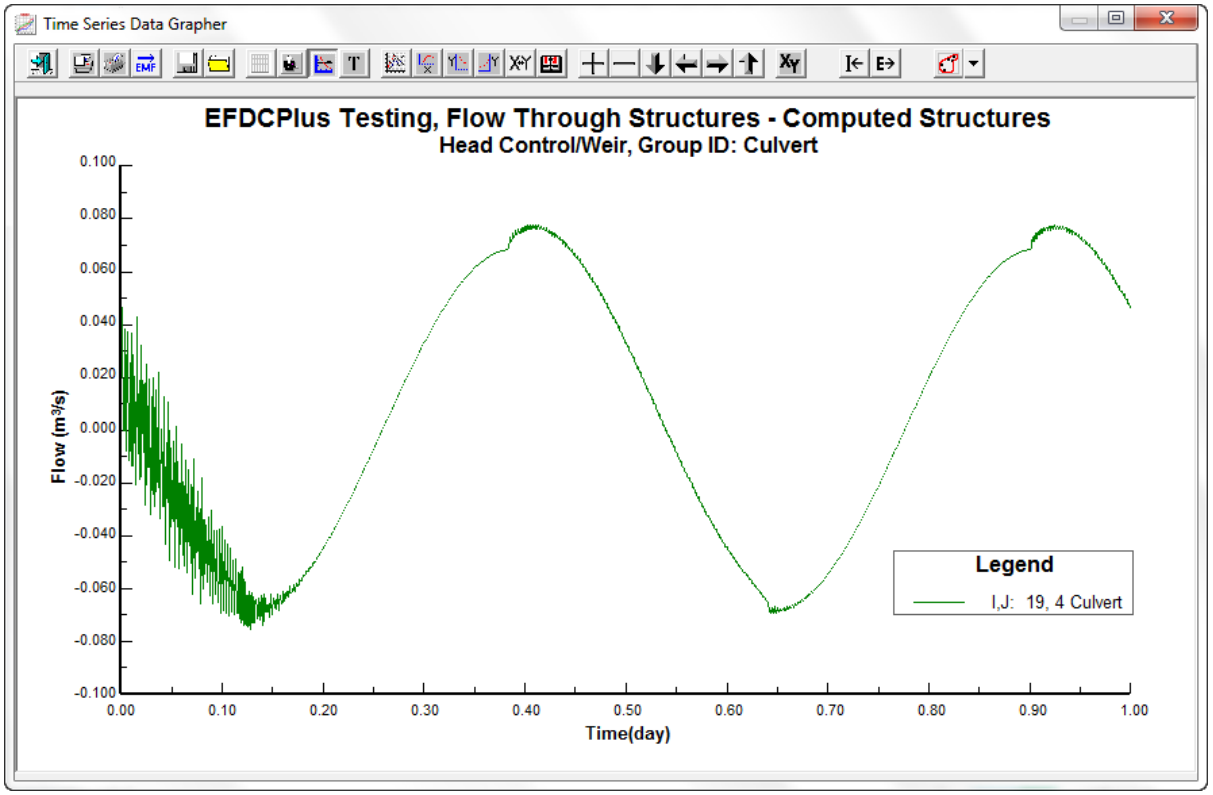


Figure 8-40 Time series for a culvert.

8.6.6. Fixed Parameters

The *Fixed Params* viewing option provides access to viewing and editing parameters that are invariant in time. These parameters include:

- *Roughness*
- *Cell Angles*
- *Cell Connectors* (N-S and E-W cell mapping)
- *Wind Shelter* (shown only if WSER>0)
- *Shading Factors* (shown only if shading is used)
- *GVC Layers* (shown only if GVC Model is used)
- *Groundwater* Map (shown only if groundwater is activated)
- *Wind Series Weightings* (shown only if NWSER>1)
- *Atmospheric Series Weightings* (shown only if NASER>1)
- *AHO* (eddy viscosity)
- *AHD* (Smagorinsky's coefficient)
- *Generic IJ Value*

GVC Layers

Most of the *Fixed Params* do not have sub-options. Only the *GVC Layers* option has sub-options. The adjacent graphic shows the options area when the *GVC Layers* was selected.

When viewing the *GVC Layers*, RMC'ing on the model provides the user with several options to reset the layering or compute the layering using specified surface and bottom reference elevations. The GVC layering type impacts the assignment process. The vertical layering type needs to have been set to 1 for GVC, or 2 for local Sigma. *Bad layers* will display the cells whose bottom elevation, relative to SELVREF, is bad when using automatic layering.

The Generic IJ Value option is used to display any data overlay on the model on a cell by cell basis. The is especially useful when wanting to view the channel index for habitat analysis or display a particular value on every cell to compare with model output or for reporting purposes.. On selecting this option EE prompts for an input file. This file should be of the format: I, J, value. This file may be comma or tab delimited. The options button allows the user to enter the parameter name and units. If this is left blank then EE will prompt the user for this information. EE retains the link to this data when the model is closed or saved for convenience when reopening the *Generic I J value*. It may be changed by browsing to another file.

8.6.6.1 *Cell Connectors*

In EE8 and EFDC+ users can connect disconnected sub-domains via N-S and E-W connectors. See Section 4.1.4 on how to import multiple domains. To connect the sub-domains, the user should navigate to *ViewPlan* | *Viewing Options: Fixed Parameters* | *Cell Connectors* or *Bottom Elevation* and select *Enable Edit*. In order to view any existing connection the *N-S / E-W* checkbox should also be selected. When the user RMCs on the workspace a number of option for creating, deleting and editing cells connections is displayed as shown in Figure 8-41. If no cell connections are currently configure then only the New Cell Connections option is accessible, and *Delete Cell Connection*, *Delete All Cell Connections*, and *Edit Cell Connections* are greyed out.

Selecting *New Cell Connections* provides three options, including the *New Single Cell Connection*. This allows the user to simply LMC on a border cell on each subdomain and create a connection. The user can also draw polylines for the connections to be made with the *New Group Connection Across 2 Polylines*. Alternatively, the user may have existing polylines

created that can be loaded and used to make the connections with the *New Group Connection in existing Polyline Files* option.

If the user chooses the option of *New Group Connection Across 2 Polylines* shown in Figure 8-41, they can then draw a new polyline on each subdomain.

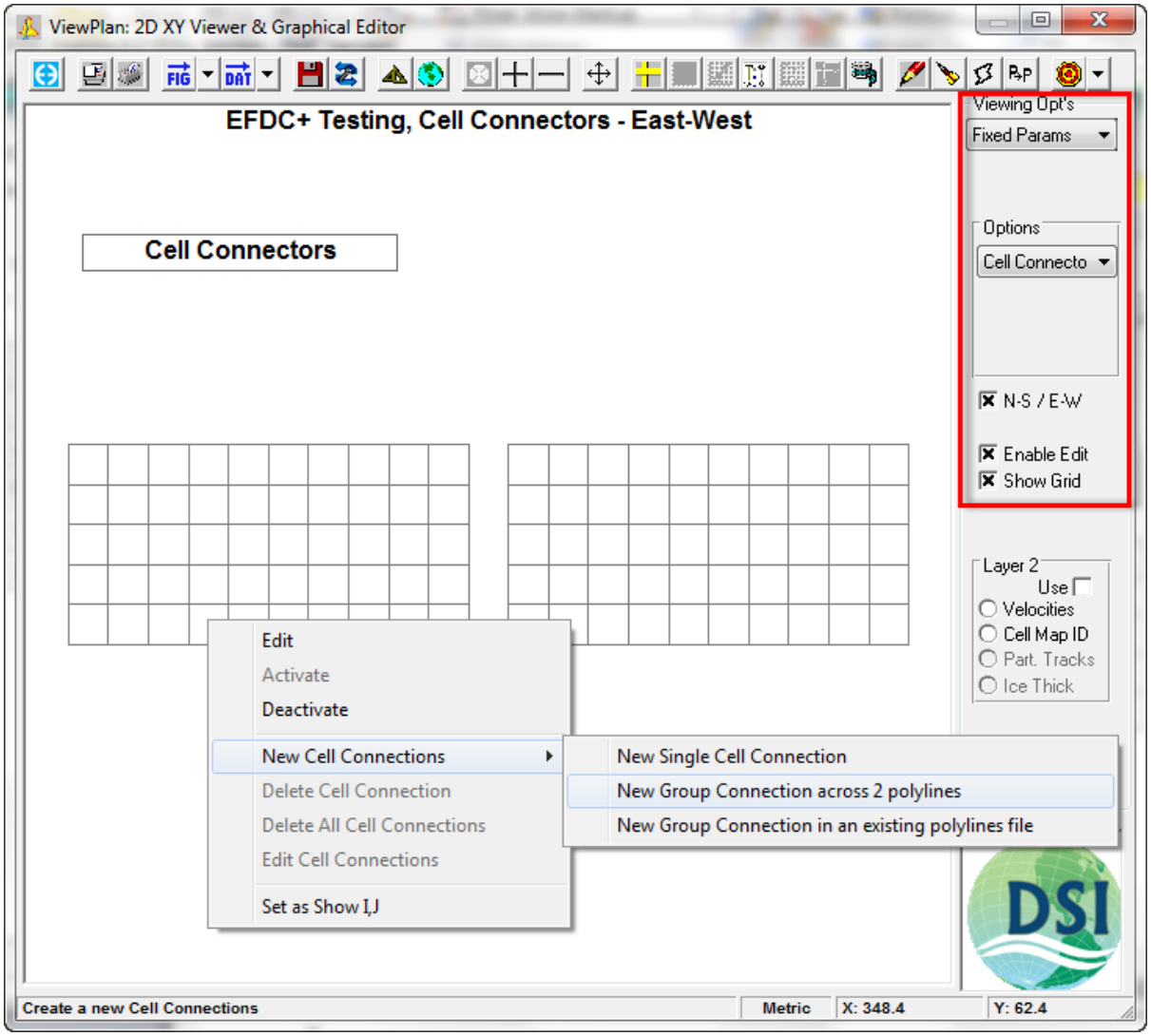


Figure 8-41 ViewPlan: Sub-domain connection options.

The user should LMC to start a polyline, and then make LMC each point required to create a polyline on the subdomain border, and then RMC to end the polyline. Once the two polygons have been created, one for each subdomain, EE will prompt the user for N-S, E-W or both kinds of connections by entering the number 1, 2 or 3 respectively as shown in Figure 8-42. EE will then automatically make the connections for the user as shown in Figure 8-43.

Note that in some cases some connections may be made that are not wanted by the user and these should be deleted with the *Delete Cell Connection* option in the RMC form.

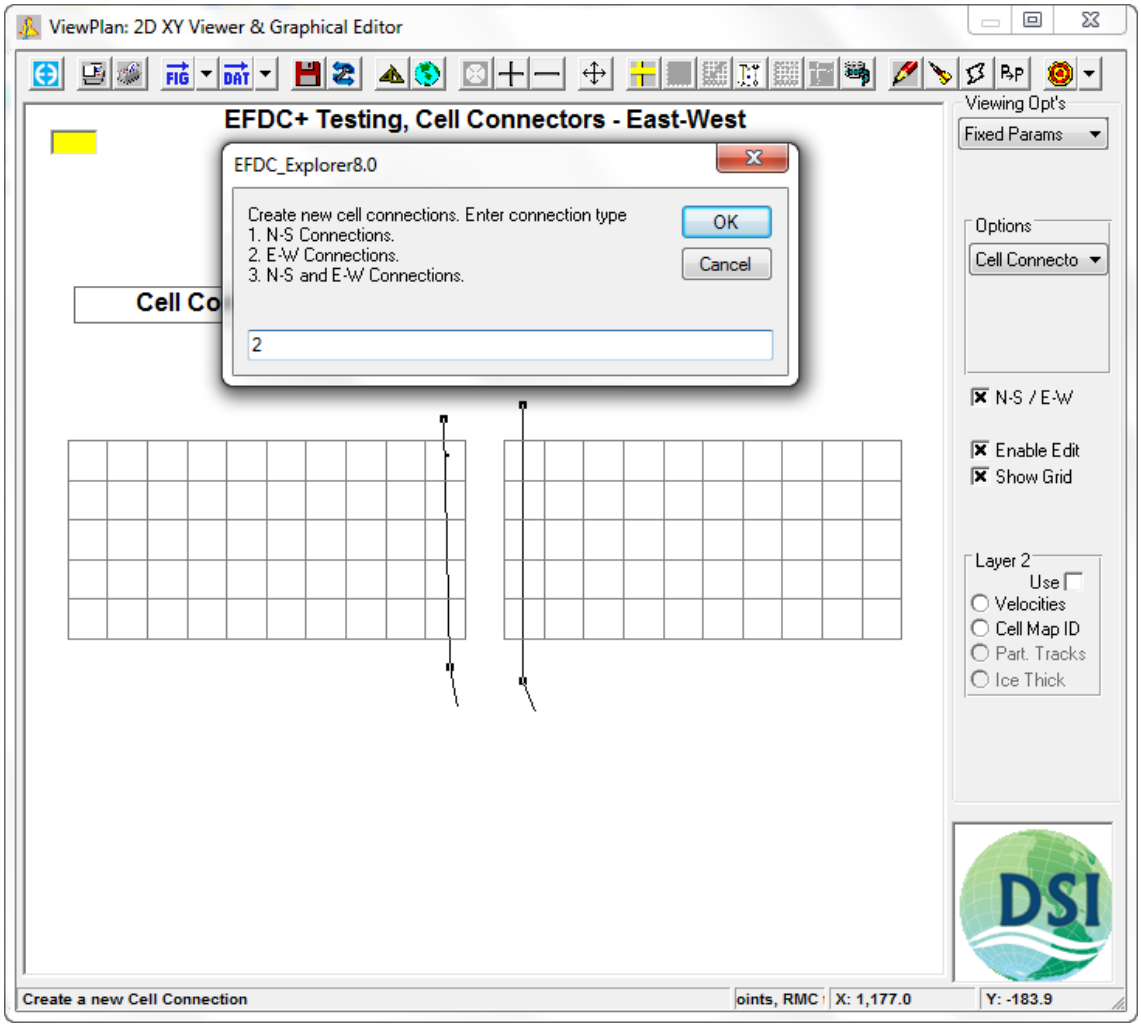


Figure 8-42 ViewPlan: E-W Connections defined by two polylines.

If the connectors line up in a way that does not match with the requirement of the user, it is possible to select the option *Edit Cell Connections* from the RMC on the connectors. This provides the user with a form to move the connector in a more convenient way with the L index list of both East and West cells. The user should select one cell with a LMC or multiple cells with “Control + LMC” and then use the *Up* and *Down* buttons to shift these connections as shown in Figure 8-44.

An example of the MAPPGEW.INP file that EFDC uses to map the connectors is provided in the appendix for the user’s reference.

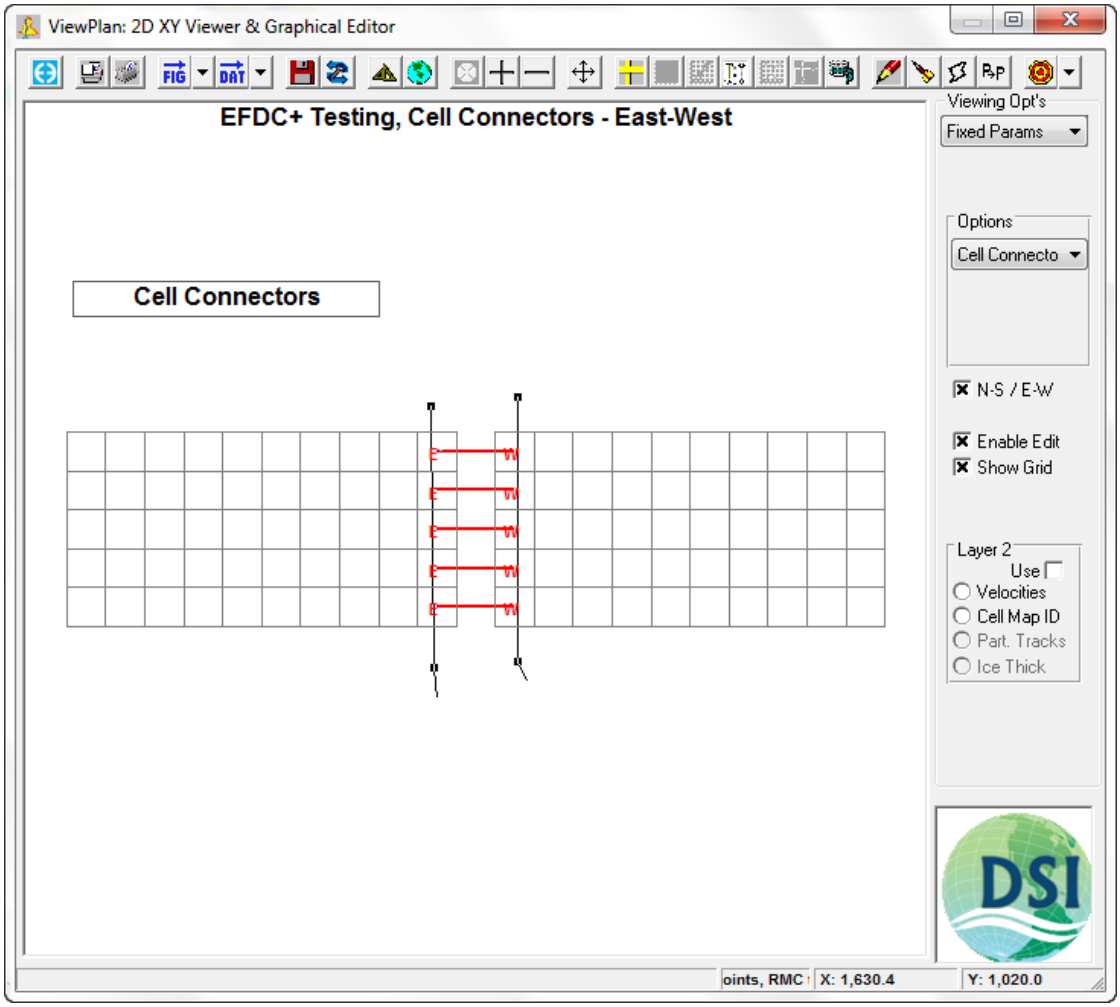


Figure 8-43 ViewPlan: E-W Connections created via a polyline.

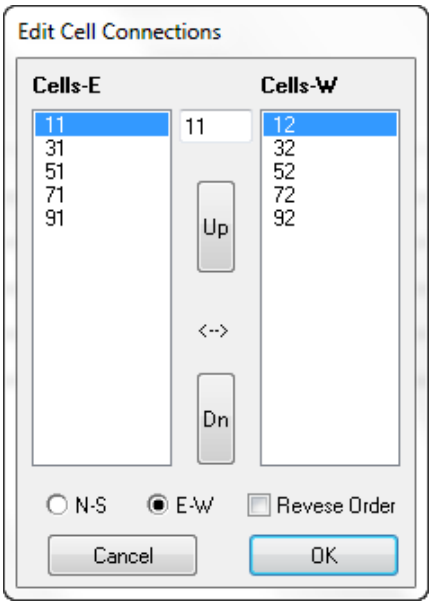


Figure 8-44 ViewPlan: N-S & E-W connection editor.

8.6.7. Model Metrics

The options available to the user in *Model Metrics* drop down menu are:

CFL Time Step Displays the Courant-Friedrichs-Levy (CFL) time step computed for each cell. This is a good guide to the appropriate time step, especially if using the two-time level solution.

Courant # Displays the Courant numbers for the model based on the time step settings of the model. If adaptive time stepping is specified, the time step used for the initial display of the Courant numbers will be 0 seconds. A *Courant Calculator* is available by setting the focus to the plot and then pressing "T". This allows the user to specify a new time step to be used for the Courant numbers.

Orthogonal Deviation Displays the orthogonal deviation for the model grid. The computations require four cells; therefore, the top and right edge cells are skipped. The average of the absolute deviation is reported in the legend.

Celerity Displays the computed celerity for each cell.

Velocities must be loaded for the following options to be available:

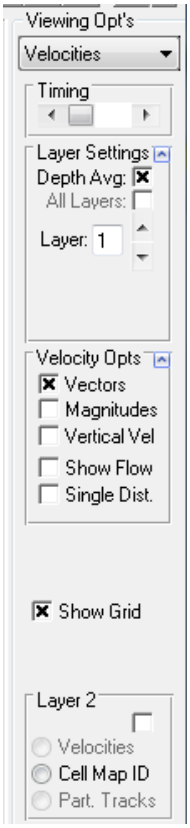
Froude # Displays the computed Froude Number for each cell.

Richardson # Displays the computed Richardson Number for each cell.

Densimetric Froude # Displays the computed Densimetric Froude Number for each cell.

Reynolds # Displays the computed Reynolds Number for each cell at the selected time step for each layer. The kinematic viscosity is used for the computations of the Reynolds number.

8.6.8. Velocities



Viewing Opt | *Velocities* displays the velocity vectors and magnitudes for the model. The velocities from a run are stored in the EFDC_Explorer linkage file EE_VEL.OUT file. The first option the user needs to select is the layering option. There are three ways to handle the layers; depth averaged, a specified layer or display all layers. For the *All Layers* option, the vectors for each cell are to correspond to the layer, blue for layer 1 and red for layer KC.

The *Velocity Opts* frame controls the type of velocity plot to display. These options can be combined. For example, the user can plot the 2D vectors over cells that have been colored base on vertical velocity magnitudes. The details of the formats of the color ramps and vector styles is adjusted using the Display Options form.

The *Show Flow* checkbox, when checked, causes EFDC_Explorer to report flows instead of velocities when LMC'ing on the form and when cutting a profile (Section 8.6.8.1)

EFDC computes the velocities on the south and west faces of a cell. These are the velocities saved in the linkage file. When displaying the velocity field, EFDC_Explorer transforms the cell face velocities in IJ space to cell centered velocities in XY space. When computing fluxes, EFDC_Explorer uses the IJ space cell face velocities.

8.6.8.1 *Profile Tool*

EFDC_Explorer provides a quick method for displaying a velocity profile or a flow profile, if the option "Show Flows" is checked. While in the velocity view, RMC on a cell located on one side of the desired profile location. A popup menu will ask the user if they want to generate a "Profile". When clicked the message "Starting i, j = ?,?". Now select the end point with the Right Button". Press OK to continue. If the next mouse click is a RMC, EFDC_Explorer will extract the velocities or flows from the cells between the two cells selected, inclusive. Multiple profiles can be extracted, However, they will all be for the current time.

8.6.8.2 *Water Flux Tool*

Another method to extract flows across certain cell boundaries is provided via the *Flux Tool*. This method calculates the flow across one or more *flux lines* contained in the P2D file. Selecting the *Flux Tool* button display the option form shown in Figure 8-45.

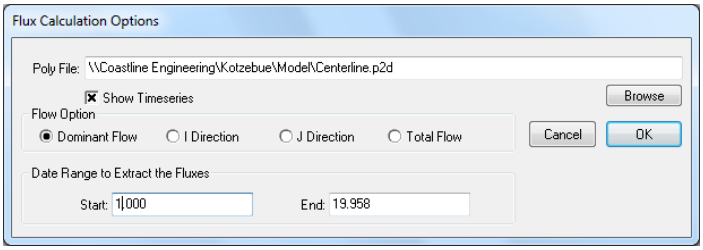


Figure 8-45 Water flux tool control options.

The water flux tool computes either total discharge across the section or on a layer by layer basis, based on the options selected on the *ViewPlan* form. If a single snapshot is selected

then the results will be displayed in a dialog box and also placed onto the clipboard. If *Show Time-series* is selected then the user must specify which flow component desired. The following is a description of each type:

Total Flow: The absolute value of the total flux across the section.

EW Flow: The sum of the flows in the EW (based on the cell map, i.e. EW=I component flow)

NS Flow: The sum of the flows in the NS (based on the cell map, i.e. NS=J component flow)

Dominant Flow: Computes the total flow across the section and assigns a sign based on dominance.

The *Dominant Flow* option is the most useful for most applications.

If the "Poly File" has more than one section/flux line defined, EFDC_Explorer will compute the fluxes for all the defined lines (up to 10 max). Figure 8-46 shows an example of total discharge using the dominant flow option for the San Francisco Bay application.

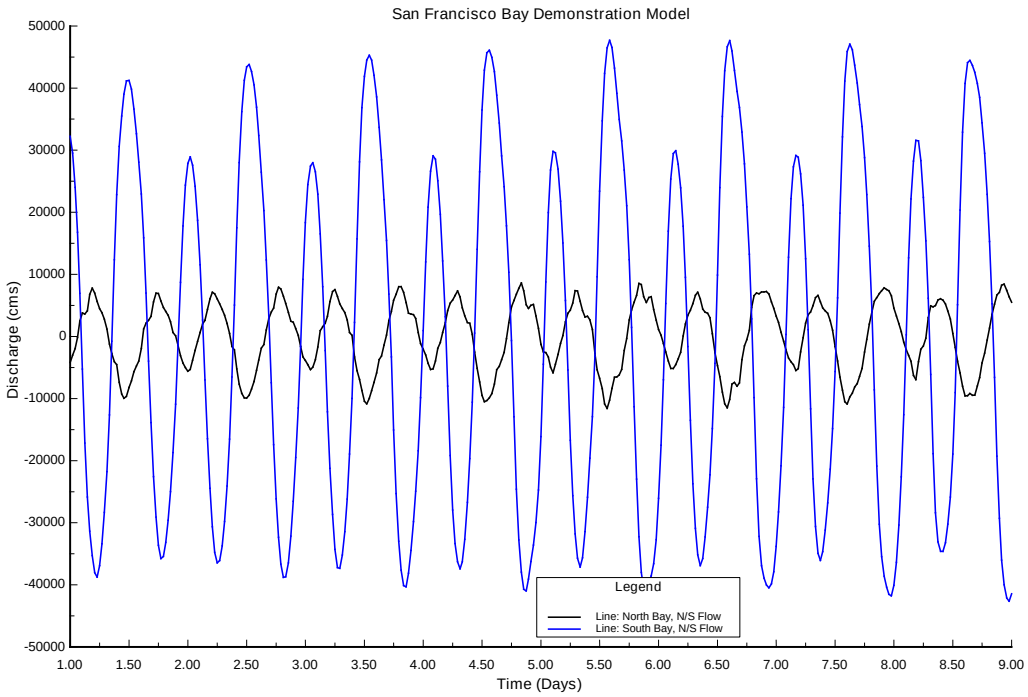


Figure 8-46 Water Flux tool example results using Dominant Flow.

8.6.8.3 *Velocity Profile*

The *Velocity Profile* feature allows the user to pass a straight line over the model domain for the cells in which a 2-D velocity profile is to be displayed. This feature is different from the longitudinal profile that is located on the top menu on View Plan. The user may access this function with RMC within the model domain to select the *Profile* option as shown in Figure 8-47. The location where the user RMCs, that would be the starting point for the new profile. The user is then prompted to the end the line with another RMC. Multiple lines may be created and plotted together by starting a new profile where the user is asked whether to add, reset or abort the new profile. If the user selects “add”, then the multiple profiles would be plotted on the same plot.

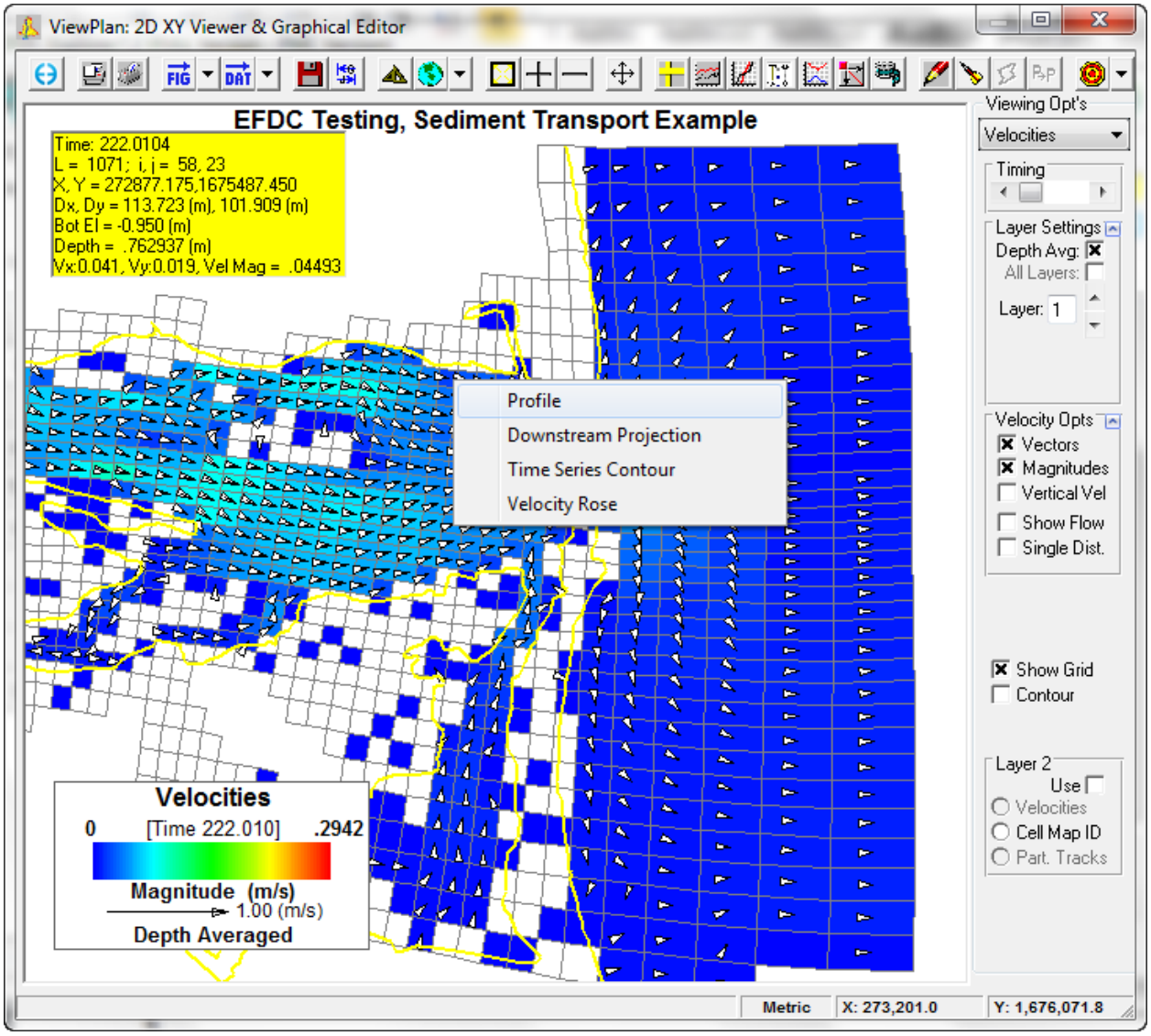


Figure 8-47 ViewPlan: Velocity Profile Selection.

An example of the profile is shown in Figure 8-48.

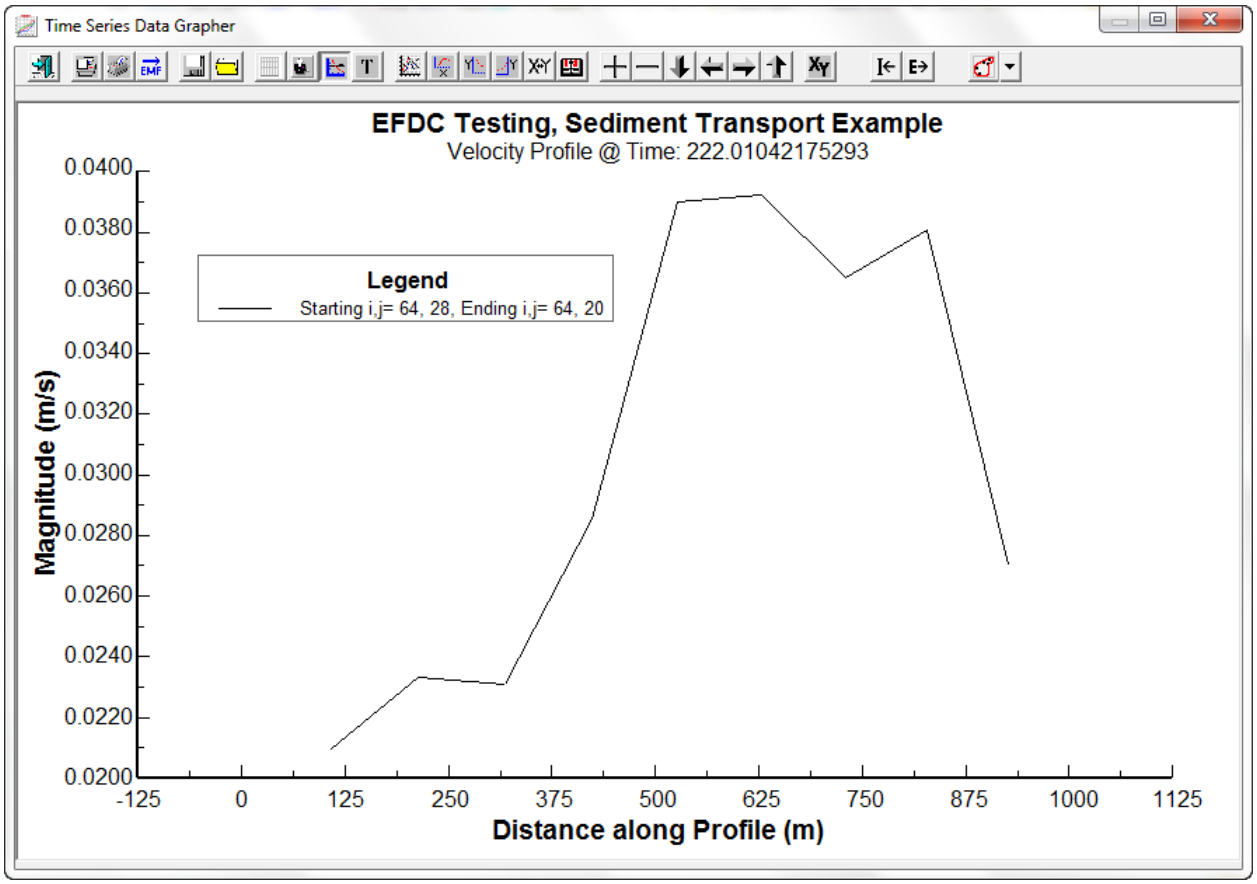


Figure 8-48 ViewPlan: Velocity Profile.

8.6.8.4 *Downstream Projection for Velocities:*

This feature allows the user to specify an orientation angle (degrees clockwise from north, looking downstream) in order to plot velocity magnitude of upstream and downstream flow during the EE data extraction. It is accessed by RMC when in the *Velocities* viewing option as shown in Figure 8-47.

If the user has selected *All Layers* in the *Layer Settings* then the user can optionally plot the average current velocity for the water column entirely or partly. This feature is primarily used when an exact comparison of model results with measured data is required in cases where the depth below the surface of the measuring device is known. For the settings shown in Figure 8-49 EE will produce the plot shown in Figure 8-50 for all layers. *From (%)* is used to indicate the percentage of the water column to be displayed from the bottom of the water column up to the *To (%)*. For example, in a 10 m water column, from 10% to 90% means the velocity magnitude it will be plotted from 1 m above the bottom and up to 9 m from the bottom.

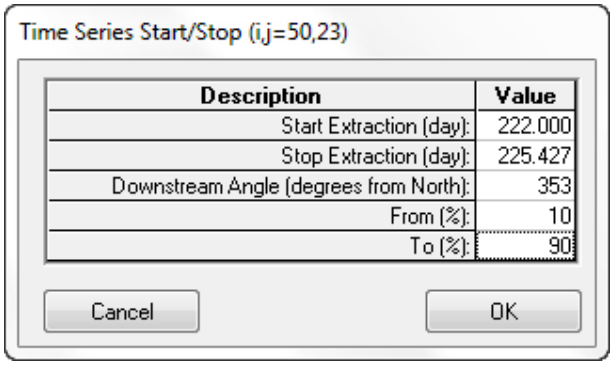


Figure 8-49 ViewPlan: Time Series – Velocity Orientation Output.

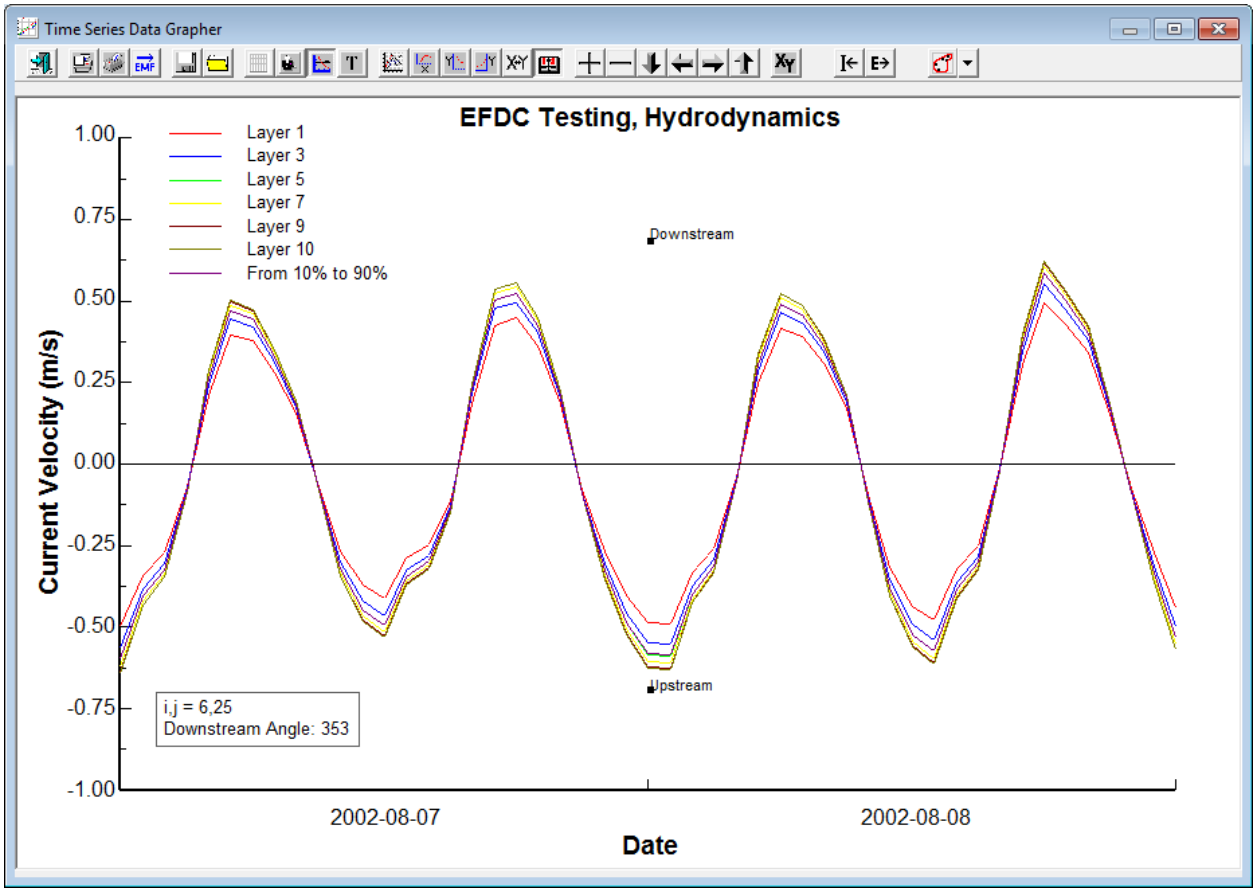


Figure 8-50 Velocity Projection Plot for all layers (layers 2, 4, 6, 8 hidden for clarity).

8.6.8.5 *Velocity Rose*

This feature allows the user to view the speed and direction distribution at a given cell over a user defined period of time. To plot the velocity rose, *velocities* viewing option should be selected in the *ViewPlan* and the user can select the location of interest with RMC. This tool is available for all layers as well as depth averaged and displays a rose as shown in Figure 8-51. The location of the cell being displayed is shown in the bottom left hand corner of the form. The velocity roses can be attached to the *ViewPlan* or can be floating as a separated window. If desired, the velocity rose can be attached to plan view by clicking on the *Show rose on the ViewPlan* under the Export drop down menu in Figure 25. The user may export the rose plot as a metafile, bitmap or ASCII file from the *Export* menu.

The *Format* menu provides options for editing the labels, setting the scale, and adjusting the sectors. The user may also save the defined rose plot format and reload it at a later time, or use it for another cell to apply the same formatting style with the *Save* and *Load* options.

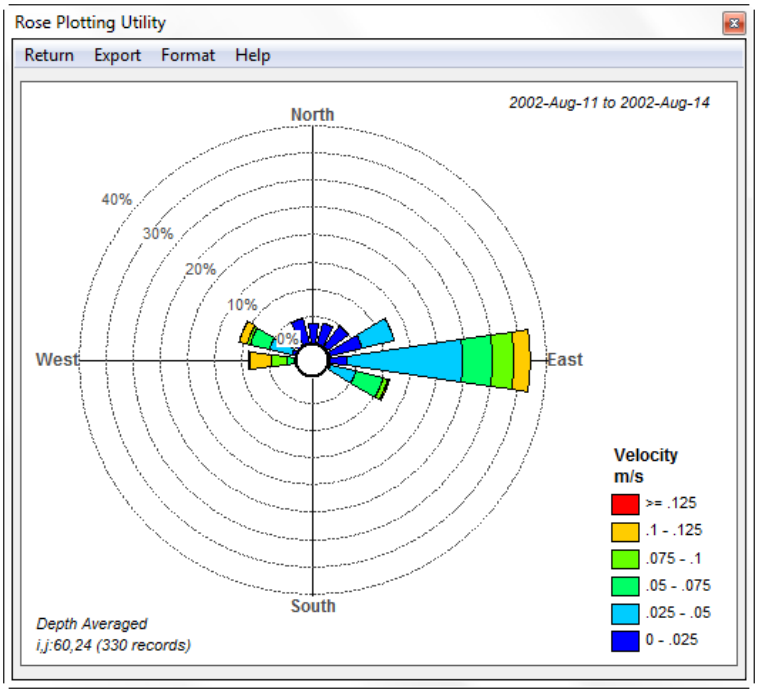


Figure 8-51 Time Series: Rose Plots for Velocity.

Multiple velocity roses may be displayed in *ViewPlan* by selecting the *Show on ViewPlan* option and change the value to 1 as result shown in Figure 8-52. Selecting *Export* | *Save* will save the layout and magnitude categories as set by the user and later loaded again to maintain the format.

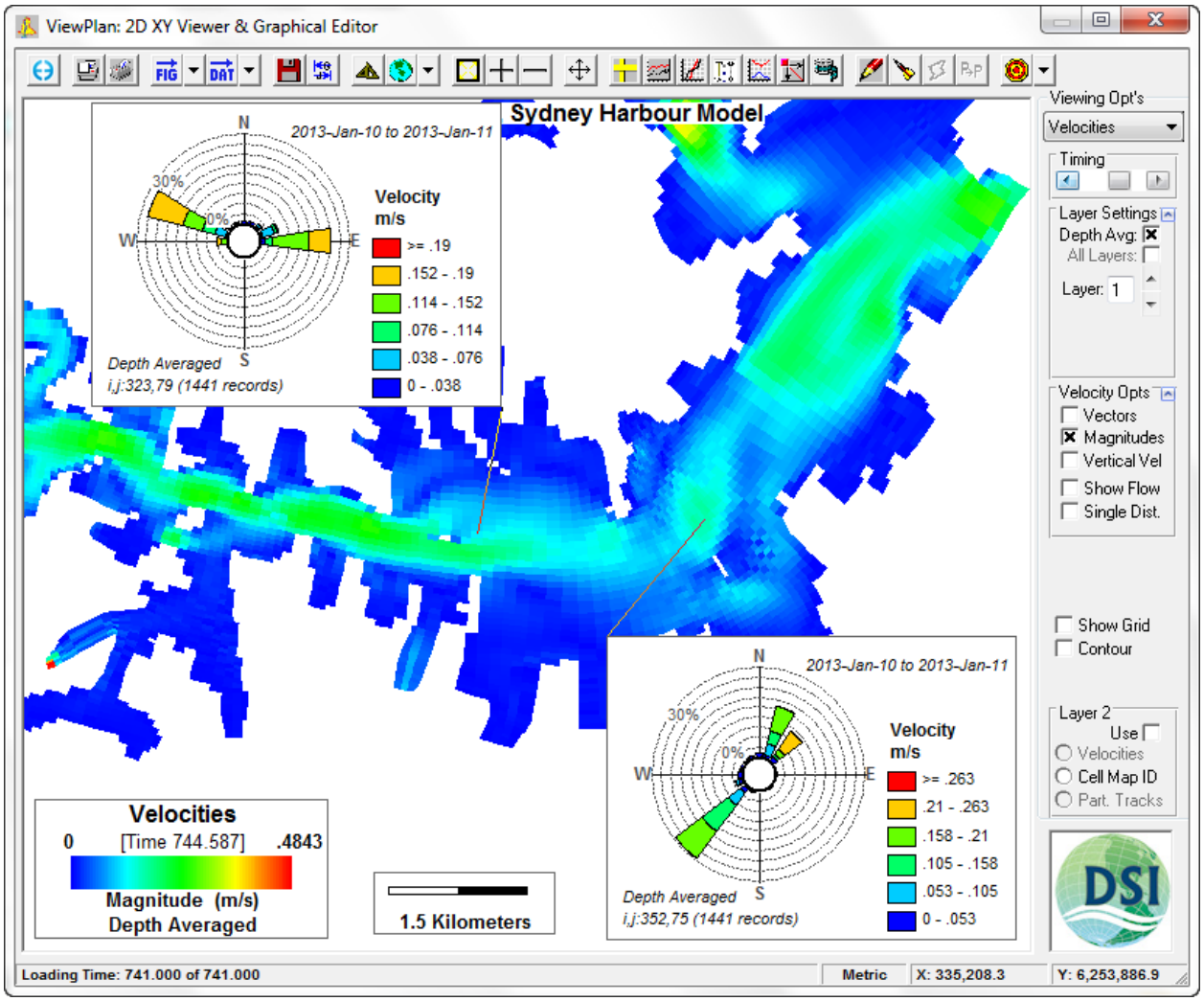


Figure 8-52 Multiple Velocity Rose Plots.

8.6.9. Sediment Bed

The *Sediment Bed* Viewing Opt provides access to the cohesive and non-cohesive sediment transport bed properties. The suspended sediment plotting and analysis capabilities are available in the *Water Column Viewing Opt*. Figure 8-53 shows the *ViewPlan* with Sediment Bed Viewing Opt selected. The cell edit form also shown in Figure 8-53 is a typical view when editing sediment mass (this model used KB=10). The *Sediment Bed* option is always available, even if sediment transport is not being simulated. This option is always available because the bed shear stress is always computed and available to post-process.

When viewing sediment data, the first option to set is the *Layer Settings*. EFDC uses a sediment bed layering scheme where the sediment-water column interface layer number can vary from cell to cell. The maximum number of layers (i.e. KB) is set in the Sediment parameter form (Section 6.12). The user has three options for viewing sediment data: *Total* (i.e. sum of all the layers), a specified layer or the *Top Layer*. The layer number that is in contact with the water column can vary so the *Top Layer* option provides an easy way to view the sediment layer that is in contact with the water column.

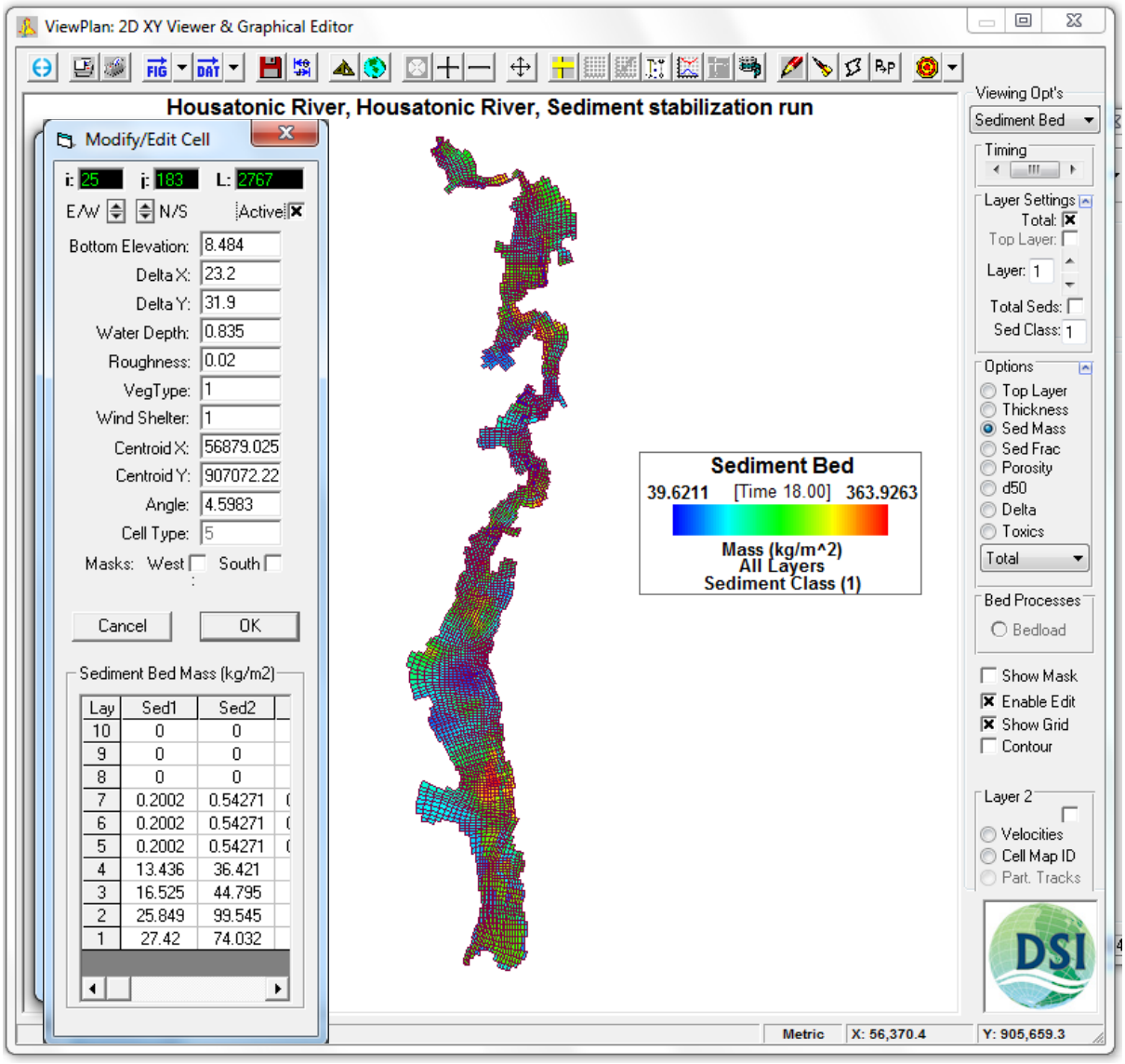


Figure 8-53 ViewPlan: Sediment Bed with Cell Editing.

The sediments can be viewed a single class at a time or all together as total sediment. The user needs to specify either the *Total Seds* checkbox to select all classes. If only one class is desired, uncheck the *Total Seds* option and specify the class in the *Sed Class* field.

The *Delta* function displays the differences in bottom elevation between the initial condition elevation and the bottom elevation at the current time. The computation is $BotEl_{cur}$-$BotEl_{IC}$ therefore, delta's < 0 indicate scour and delta's > 0 indicate deposition. In order for the *Delta* option to function the bed morphology option needs to be set to 1 (*Bed Morphology Options* in *Bed & Consol* tab of the *Sediments* form), which allows the bottom elevations to change with sediment transport.

Note

When drawing the Flux Lines be aware of the fact that the velocities in EFDC are computed on the South and West faces. Try to draw the Flux Lines along cell faces to obtain the most accurate fluxes.
Cell selection is based on the Flux Line crossing through any part of a cell. The entire cell face will be used to compute the flux for that cell.

8.6.9.1 *Sediment Bed Initialization Tools*

To facilitate rapid building of sediment bed models EE implements a powerful sediment core initialization bed tools within *ViewPlan*. If the model is configured for sediments then F7 toggles on and off the sediment bed tool. RMC and selecting *Load Cores* allows the user to the bring in the sediment core file (.DAT file). This produces a list of all the cores, number of cores and sets the sediment bed. The user is then prompted to plot the cores labeled with d50's or the label with core ID. The top layer of the sediment cores is then plotted in *ViewPlan*.

RMC and selecting *Edit Core* core displays the *Sediment Core Editing* tool as shown in Figure 8-54. This tool allows the sediment core data and sample interval data to be edited.

RMC on an existing sediment core provides several options including defining the core as "Current", deleting the core, and editing the core. Defining the core as "current" allows the user to then RMC in another location and copy the "current" core to that location by RMC and selecting *Add a new [core no.]*. This will add a new core to end of the file, and duplicates the core number.

RMC anywhere else in the viewing area allows the user to define a new core at that point as well as load and save the sediment core data.

Note that if the editing window is closed that editing is switched off, but data is not saved until the user RMCs and selects *Save Cores*.

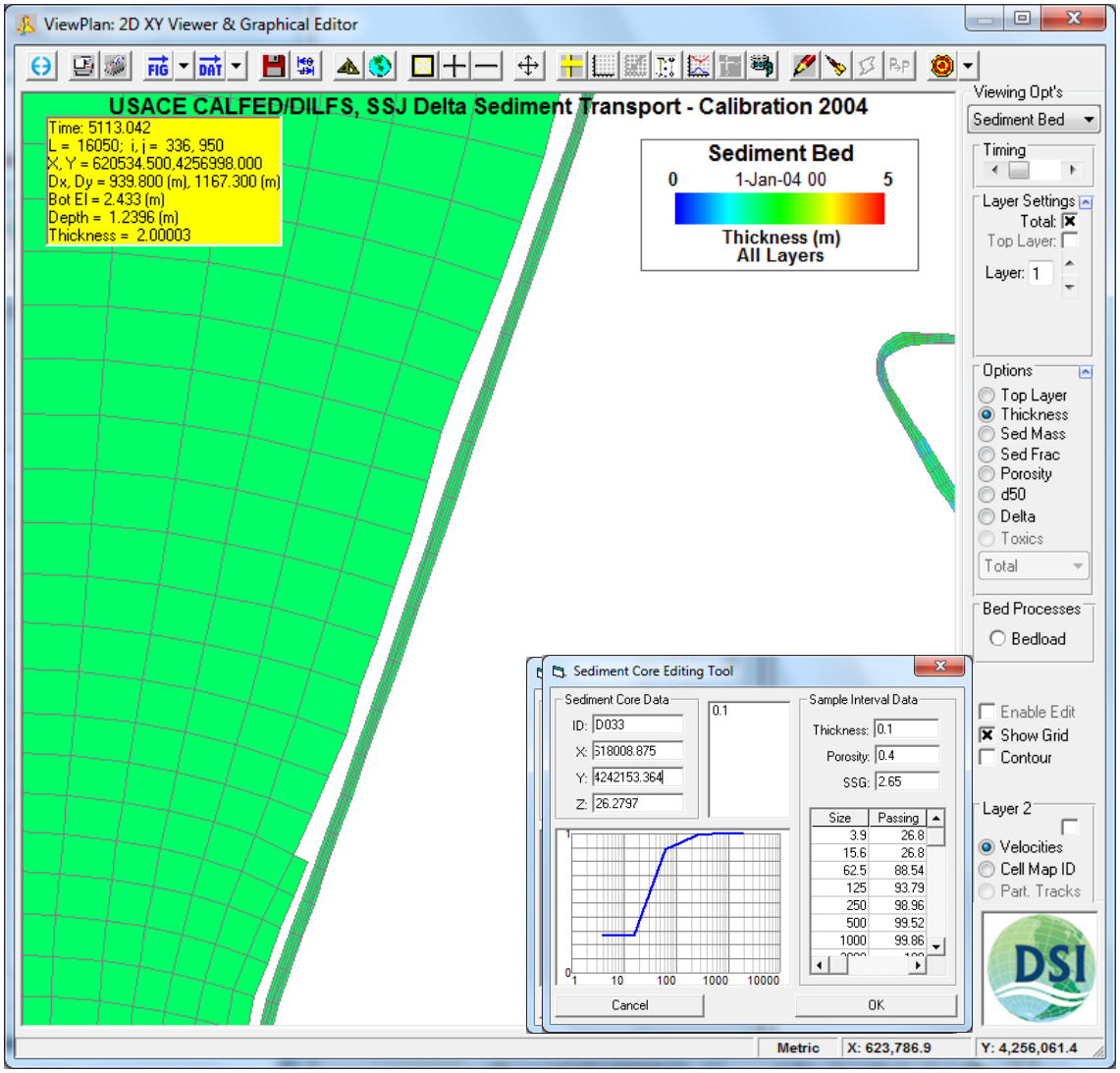


Figure 8-54 ViewPlan: Sediment core editing.

8.6.10. SEDZLJ Option

An alternative to the standard sediment transport approach is the the start-of-the-art SEDZLJ sub-model. The SEDZLJ Sediment sub-model is implemented in EFDC_Explorer for display of SEDZLJ input data and results. However, at this time, the user cannot create or edit input data.

If the user has created a valid SEDZLJ model then the option for SEDZLJ may be displayed. The options available include display of the *Core Zone, Erosion Rate, Deposition Rate, Net Dep/Erosion Rate, Bedload Concentration, d50 (Active Layer).*

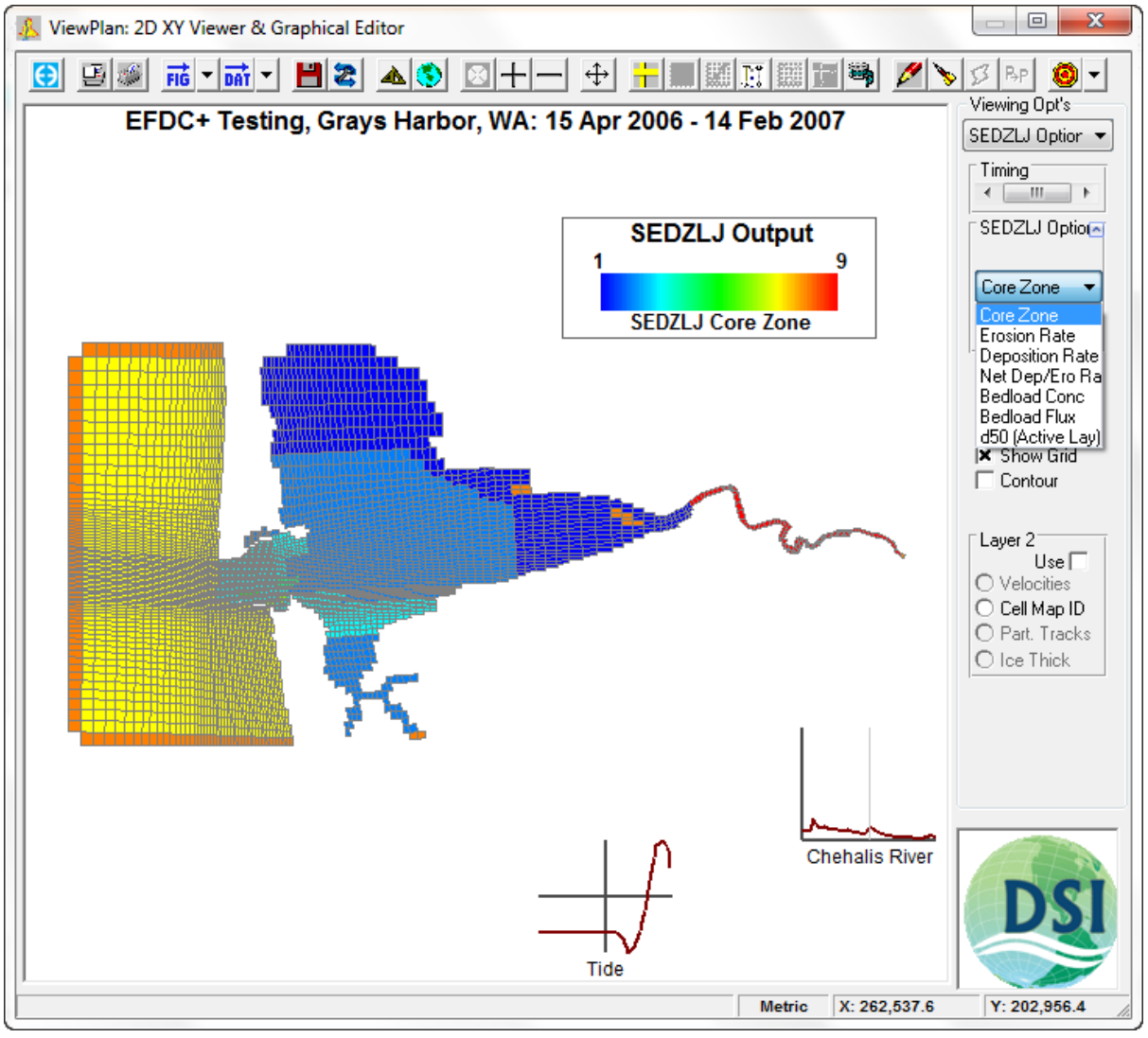


Figure 8-55 ViewPlan: SEDZLJ viewing options.

8.6.11. Toxics

The *Toxics* option is enabled only if toxics are being simulated. When selected, the labels in the Layer Settings frame changes. The *Total Seds* becomes *Total Tox* and the *Sed Class* becomes *Tox Class*. Their function is the same but they apply to toxics classes instead of sediment classes. In addition the user can select the state of the toxics, either:

Total is the total toxic concentration for the class option selected.

Tox (Diss) is the dissolved phase concentrations for the class option selected.

Tox (DOC) is the toxic concentration of the dissolved organic carbon (DOC) complexed phase for the class option selected.

Tox (POC) is the toxic concentration of the particulate organic carbon (POC) adsorbed phase for the class option selected.

POC is the POC concentrations assigned to the sediments.

DOC is the DOC concentrations assigned to the sediments.

8.6.12. Bed Processes

In the *Bed Processes* frame several options are available to display bed shear stress: total bed shear, cohesive fraction bed shear and non-cohesive fraction bed shear. The availability of the bed shear stress option is influenced by the bed shear stress computational option specified in Section 6.12.2.3.

When viewing bed shear stress, the user can toggle between stream power (Watts/m^2) and bed shear (Newton/m^2) by pressing Alt-P. Set the focus to the plot by LMC on the plot.

The bedload flux rate, if simulated, can be displayed by selecting the *Bedload* option.

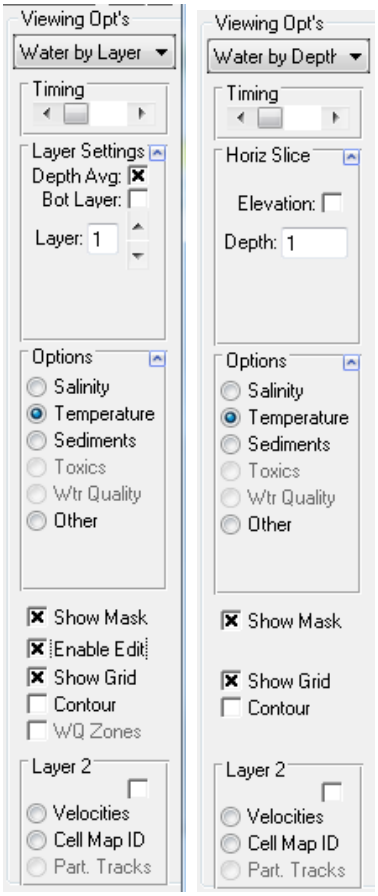
8.6.13. Heat: Bed/Ice

Viewing Opt | *Heat: Bed/Ice* allows the user to view bed heat in terms of temperature or thermal thickness, as well as for. The nature of the information displayed depends on whether the model is EFDC_GVC or EFDCPlus.

When viewing the Temperature option only the top layer (i.e. the layer in contact with the water column) is displayed. If the *Time Series* toolbar function is on then the *Show Water* checkbox is enabled and the user can view the time series of the sediment temperature and the bottom water layer (which varies for the GVC model) together.

8.6.14. Water Column – by Layer and by Depth

The *Water Column Viewing Option* is the primary 2D viewer for all of the water column related parameters. The water column may be viewed by layer or by depth. For the former, the first options to set are the *Layer Settings*. There are three layering options for processing the water column data. They are *Depth Avg* for depth averaged results, *Bot Layer* for viewing the bottom active water column layer (this varies in the GVC model, standard sigma model this is layer 1) and a specified layer.



When viewing sediments or toxics the *Total Seds/Total Tox* and *Sed Class/Tox Class* are used to decide how to treat the display results, i.e. as totals or by class.

The data displayed in the 2D plot can be edited using the editing tools available if the current time is set to the initial conditions.

The options listed in the *Options* frame will only be enabled if the correct computational flags are set. The *Density* option is only available if salinity and/or temperature are being simulated. If only salinity is simulated the temperature is taken as a constant 20°C.

For most *Viewing Opt's* the *Statistics* tool on the toolbar is available to compute a range of statistics on the current view. For the *Water Column* option the *Statistics* tool provides additional capability. The Statistics tool can compute a time series of the mass weighted averages for any parameter for all or any subset of the model.

8.6.14.1 *Water Quality*

If water quality is being simulated the *Wtr Quality* option button is enabled as is the parameter dropdown list. In addition to the 22 EFDC water quality parameters, EFDC_Explorer has 26 derived parameters that can also be displayed. Table 8-5 provides a complete list of the EFDC and EFDC_Explorer derived parameters.

The kinetic subroutines in EFDC allow the parameters to be turned on and off, depending on the level of detail in a model. However, when a parameter is not turned on it still has a value (i.e. the initial condition) and may be used in the kinetic equations of parameters that are being simulated. Therefore, EFDC_Explorer always allows the display of all EFDC parameters. However, to make it clear as to which parameters were simulated and which were not, a "(NS)" is placed at the end of the parameter abbreviation shown in the dropdown list for those parameters not simulated.

For the EFDC_Explorer derived parameters a "(DP)" is placed on the end of the abbreviation to clearly indicate that the parameter was derived.

8.6.14.2 *Irradiance*

The ability to determine the % Bottom Irradiance can useful when the user is interested in cases where the amount of light penetration must be determined, such as where there are macrophytes or aquatic vegetation. Note that this tool only determines the potential for irradiance, and does not allow for day or night or factor in cloud cover etc. Pressing ALT-B accesses the Bottom Irradiance Tool. The user is first prompted for analysis of a polygon, which must then be loaded, or the whole model.

The subsequent option form is shown in Figure 8-56. The user should enter the *Maximum Bottom Depth* in meters. Cells deeper than this depth are displayed as white. As depths will vary due to tides it is usually most useful to calculate an average for the *Depth Criteria Approach.* The user should finally enter the *Irradiance Target* as a percentage. EFDC_Explorer will then calculate all the cells that fit the criteria and calculate statistics for those cells.

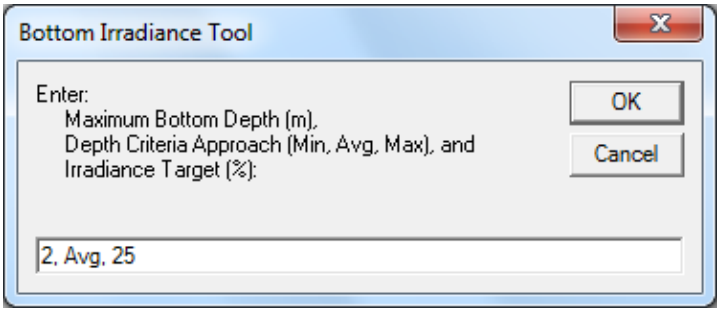


Figure 8-56 ViewPlan: Water Column, % Irradiance Tool.

Table 8-5 Water quality parameter list available for display.

EFDC/HEM3D Parameters	
Abbrev	**Name**
CBact	Cyanobacteria
Alg-D	Diatom Algae
Alg-G	Green Algae
RPOrg C	Refractory Particulate Organic Carbon
LPOrg C	Labile Particulate Organic Carbon
DOrg C	Dissolved Organic Carbon
RPOrg P	Refractory Particulate Organic Phosphorus
LPOrg P	Labile Particulate Organic Phosphorus
DOrg P	Dissolved Organic Phosphorus
TPO4-P	Total Phosphate
RPOrg N	Refractory Particulate Organic Nitrogen
LPOrg N	Labile Particulate Organic Nitrogen
DOrg N	Dissolved Organic Nitrogen
NH4-N	Ammonia Nitrogen
NO3-N	Nitrate Nitrogen
PBioSi	Particulate Biogenic Silica
AvailSi	Dissolved Available Silica
COD	Chemical Oxygen Demand
DO	Dissolved Oxygen
TActM	Total Active Metals
FColi	Fecal Coliform
MacAlg	Macroalgae
EFDC_Explorer Derived Parameters	
Tot C	Total Organic Carbon
Tot P	Total Phosphorus
TORN	Total Organic Nitrogen
TORP	Total Organic Phosphorus

EFDC_Explorer Derived Parameters (cont.)	
Abbrev	**Name**
POrg C	Particulate Organic Carbon
POrg N	Particulate Organic Nitrogen
TKN	Total Kjedahl Nitrogen
Tot N	Total Nitrogen
Chl a	Chlorophyll a
TION	Total Inorganic Nitrogen
LimP-C	Algae Limit: Phosphorus-Cyanobacteria
LimP-D	Algae Limit: Phosphorus-Diatom Algae
LimP-G	Algae Limit: Phosphorus-Green Algae
LimN-C	Algae Limit: Nitrogen-Cyanobacteria
LimN-D	Algae Limit: Nitrogen-Diatom Algae
LimN-G	Algae Limit: Nitrogen-Green Algae
LimNP-C	Algae Limit: N and P-Cyanobacteria
LimNP-D	Algae Limit: N and P-Diatom Algae
LimNP-G	Algae Limit: N and P-Green Algae
LimL-C	Algae Limit: Light-Cyanobacteria
LimL-D	Algae Limit: Light-Diatom Algae
LimL-G	Algae Limit: Light-Green Algae
LimT-C	Algae Limit: Temperature-Cyanobacteria
LimT-D	Algae Limit: Temperature-Diatom Algae
LimT-G	Algae Limit: Temperature-Green Algae
LimA-C	Algae Limit: All Factors-Cyanobacteria
LimA-D	Algae Limit: All Factors-Diatom Algae
LimA-G	Algae Limit: All Factors-Green Algae
TSI	Carlson's Trophic State Index
TSS	Total Suspended Solids (Inorg & Org)
POrg P	Particulate Organic Phosphorus

8.6.14.3 *Habitat Analysis*

The user may undertake several kinds of habitat analysis from *ViewPlan* by selecting Alt-H. This will open the *EEMS Habitat Tools* option form as shown in Figure 8-57. Select 1, 2, or 3, for the *Thermal and Depth Criteria Limits, Critical Limits,* or *HSC* options respectively. The latter two options are only available if they have been correctly configured as described in Section 7.5.

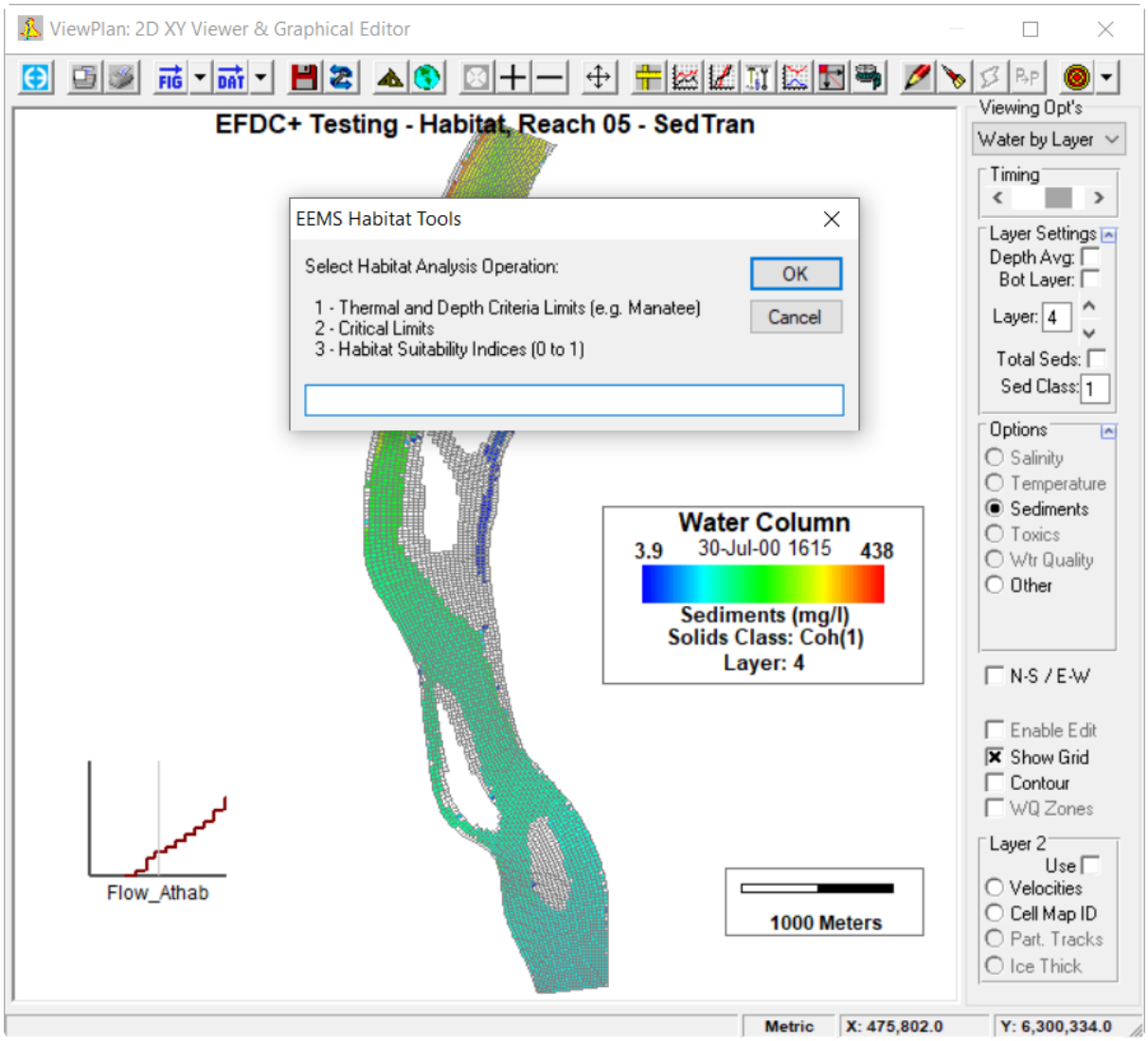


Figure 8-57 ViewPlan: Habitat tools.

The most basic option is option 1, *Thermal and Depth Criteria Limits.* This analysis tool was initially implemented to support studies on manatee populations in the Chassahowitzka Estuary. This kind of analysis of the water column is available when the *Water by Layer* drop down is selected as shown in Figure 8-58. The user may analyze any of the parameters displayed in the *Viewing Options* frame. In the *Critical Habitat Analysis* form the user should set critical levels (in this case temperature, ^{o}C), critical depths and critical duration (in hours) and press *Display*. EE will then calculate the number of cells for which these criteria apply, displaying the area and volume in the yellow *Habitat Criteria* box, as well displaying the critical cells. A second critical duration may also be displayed concurrently.

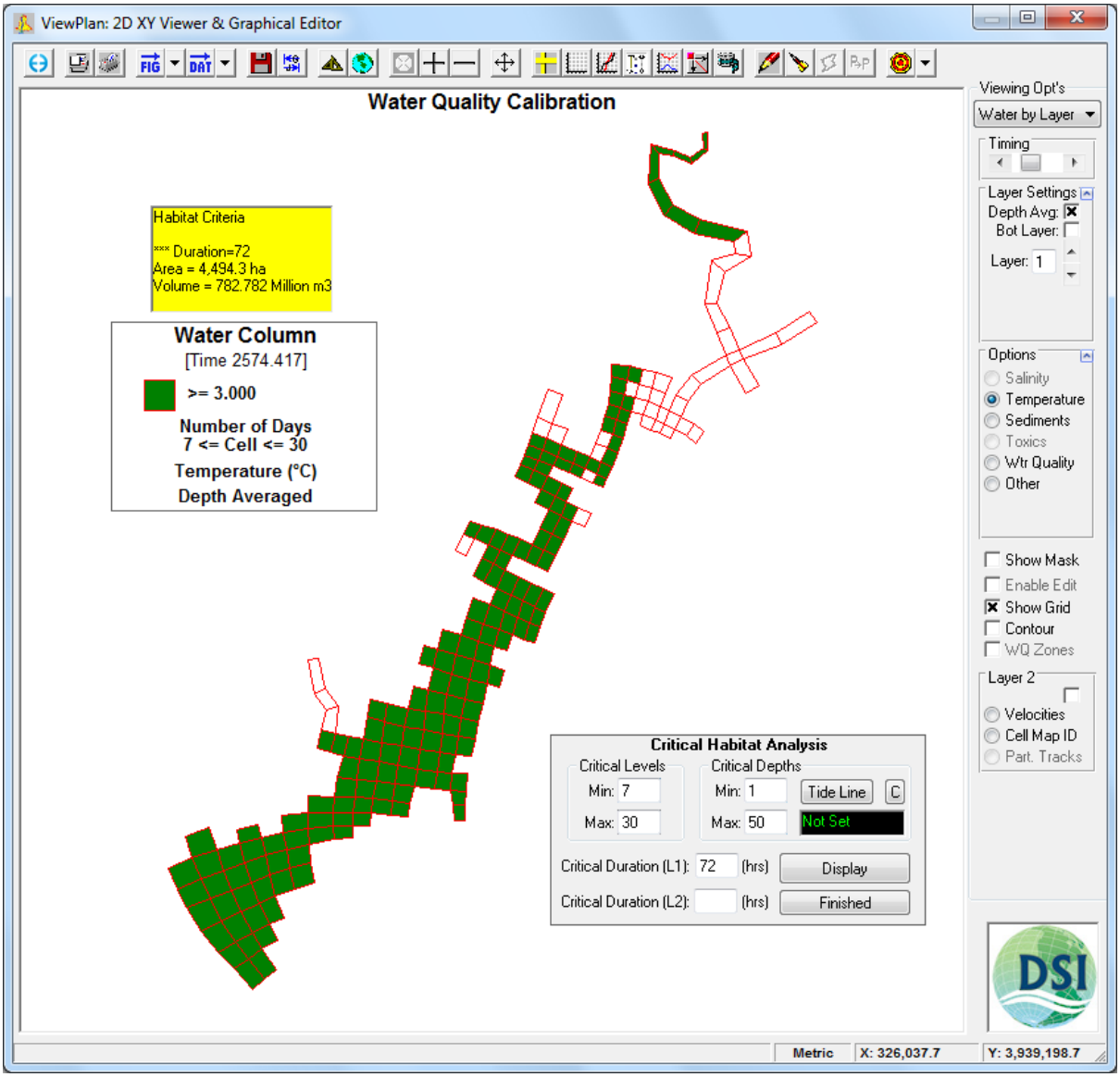


Figure 8-58 ViewPlan: Water by Layer, Habitat Analysis tool.

Selecting option 2 or 3 will cause EE to prompt the user for the habitat limits or criteria file. EE will then display areas satisfied by the criteria, as shown in Figure 8-59. This option is available for all the parameters that may be set in the habitat critical limits, or suitability criteria. If a cell is dry it is shown as grey. For the critical limits the default color ramp is white (0) to green (1). For habitat suitability criteria the same principle applies, however shades of green from 0 to 1 are displayed.

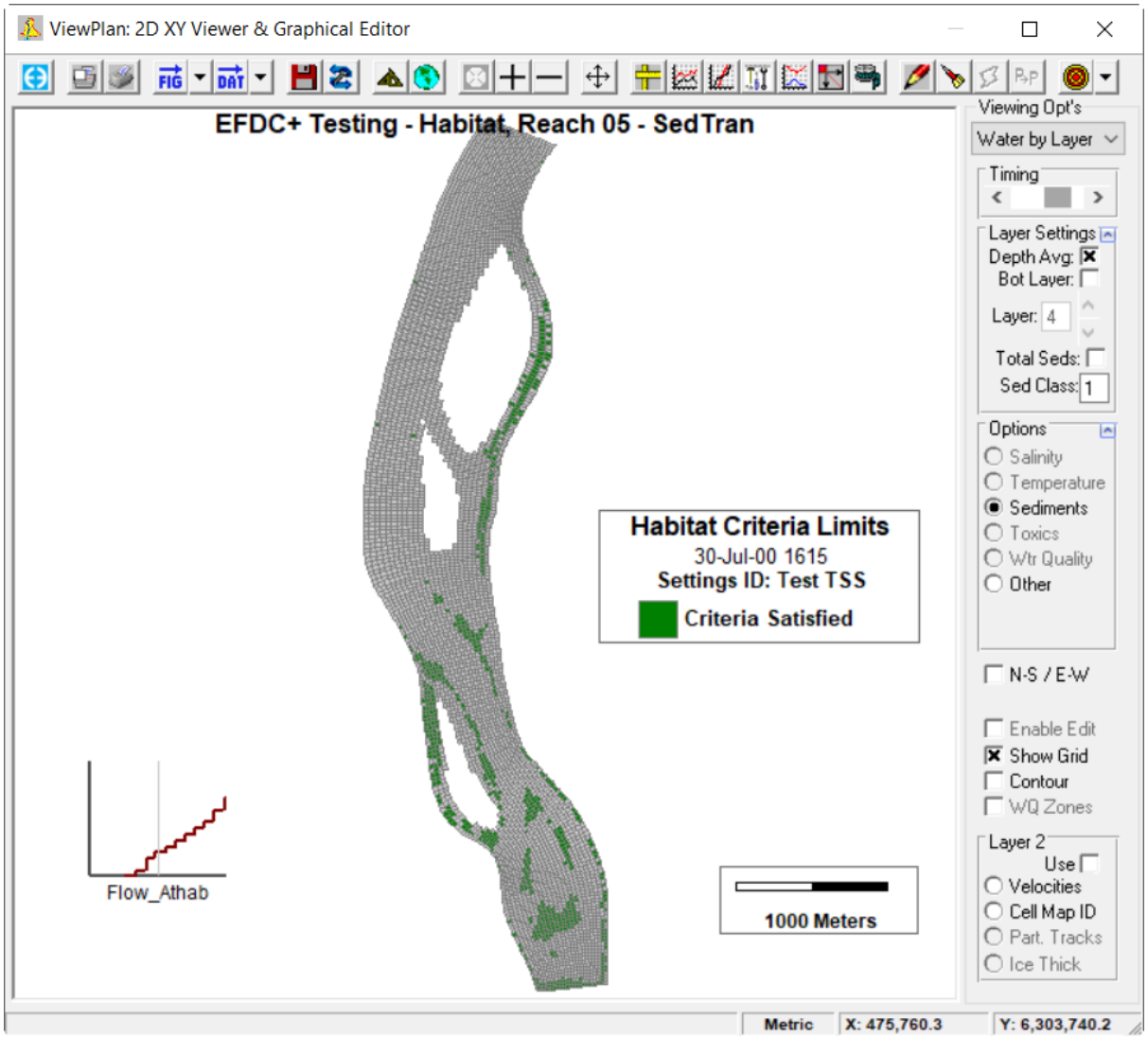
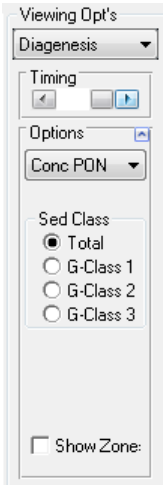


Figure 8-59 ViewPlan: Habitat criteria limits.

8.6.15. Sediment Diagenesis/Specified Fluxes

If the full sediment diagenesis sub-model is being used then the user has the ability to view the concentrations and/or resulting nutrient fluxes using *Viewing Opt* | *Diagenesis*. If the user is applying a constant or time variable specified nutrient flux then the user can view the nutrient flux zones and flux rates using *Viewing Opt* | *Sediment Flux*.



8.6.15.1 *Sediment Diagenesis*

When the full diagenesis option has been selected the *Viewing Opt* | *Diagenesis* is available to view/edit the initial concentrations and, after a model run, post-process the concentrations and nutrient fluxes. An example of the *Options* frame is shown in the adjacent graphic. Below the dropdown list for the diagenesis parameter is a sub-option area that changes with the parameter selected.

The *Show Zone* checkbox overrides the current parameter and shows the diagenesis zones instead. For pre-processing, the zones can be edited using the cell property copy function and the diagenesis parameters can be edited by RMC'ing on a cell (if *Enable Edit* is checked).

The diagenesis parameters that are available for viewing are provided in Table 8-6. The sub-options that are available for each parameter are also listed.

Table 8-6 List of sediment diagenesis parameters and sub-options available.

Abbrev	Name	Sub-Options
Conc PON	Concentration of PON	Total or by G Class
Conc POP	Concentration of PON	Total or by G Class
Conc POC	Concentration of PON	Total or by G Class
Conc NH4-N	Concentration of Ammonium	Total or by Layer
Conc NO3-N	Concentration of Nitrate	Total or by Layer
Conc PO4-P	Concentration of Phosphate	Total or by Layer
Conc H2S	Concentration of Hydrogen Sulfide	Total or by Layer
Conc Silica	Concentration of Silica	Particulate Biogenic or by Layer
Benthic Stress	Benthic Stress	
Sed Temp	Sediment Temperature (not linked to heat sub-model)	
PON Flux	PON Flux between Sediment-Water	Total or by G Class
POP Flux	POP Flux between Sediment-Water	Total or by G Class
POC Flux	POC Flux between Sediment-Water	Total or by G Class
SOD	Sediment Oxygen Demand	Total or Only Carbonaceous or Only Nitrogenous
COD	Chemical Oxygen Demand Flux	
NH4 Flux	Ammonium Flux between Sediment-Water	
NO3 Flux	Nitrate Flux between Sediment-Water	
PO4 Flux	Phosphate Flux between Sediment-Water	
Silica Flux	Silica Flux between Sediment-Water	

8.6.15.2 *Sediment Flux*

There are two specified flux options, one for invariant fluxes horizontally and temporally, and another that varies by zones in conjunction with mud/sand distributions (not linked to the sediment transport sub-model).

The parameters that are available for display are:

- Mud Zone
- Sand Zone
- Percent Mud
- Sediment Oxygen Demand (SOD)
- Chemical Oxygen Demand (COD)
- Dissolved Available Silica (SAD)
- Nitrate (NO3)
- Ammonium (NH4)
- Phosphate (PO4)

If the temporally varying option was selected the *Time (days):* field is displayed to view the nutrient fluxes assigned for that day. These are input flux rates, not computed flux rates by EFDC.

8.6.16. Vegetation Map

If vegetation has been activated then *Viewing Opt* | *Vegetation Map* is available. There are no sub-options available. When LMC'ing on a cell, the vegetation class information for that cell is displayed. The cell property copy feature is available to quickly edit vegetation class assignments.

8.6.17. Internal Variables (EFDCPlus Only)

After the model has been run and the EFDCPlus source code has been properly configured, the EFDC_Explorer linkage file EE_ARRAYS.OUT will have been generated. When post-processing a model run, if this file exists, EFDC_Explorer builds a parameter list from the data contained in the linkage file. Any of these variables contained in the linkage file can then be displayed, time series'd, etc.

8.6.18. ModChannel

If the channel modifier feature of EFDC has been activated then *Viewing Opt* | *ModChannel* is available. The *Channel Modifier Flag* must be set in the *Domain* tab in the the main form (see Section 6.3.16.3.1).

When selected, the model grid will be displayed with gray lines representing the “pipes” or channel modifiers shown. This function enables the user to create and edit channel modifiers (i.e. pipes). The user should then enable edit and use the right mouse click to choose the start point for the channel modifier. The user will then be prompted for the end point and edit form shown in Figure 8-60. In this way the user can add, delete or modify channel modifiers. Care should be exercised in using channel modifiers as model instability is sometimes increased and mass balance errors can occur.

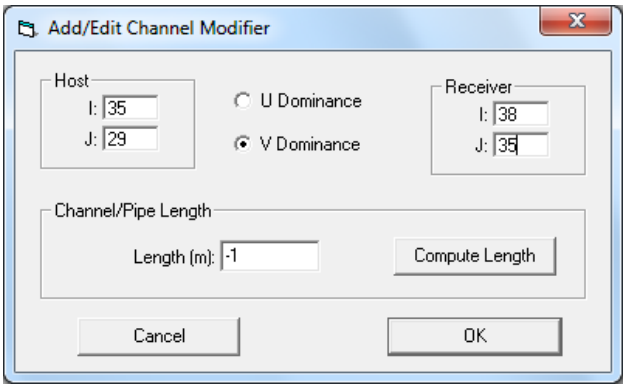


Figure 8-60 Add/Edit Channel Modifier option form.

8.6.19. Wave Parameters

If the wave options have been activated (Active Modules tab) then *Viewing Opt* | *Wave Parameter* will be available. This feature displays the 2D assignment of the wave parameters. The input parameters that can be displayed are:

- Wave Cells
- Wave Energy.
- Wave Height.
- Radiation Shear Stress: XX Component.
- Radiation Shear Stress: YY Component.
- Radiation Shear Stress: XY Component.
- Dissipation Losses.
- Wave Angle.
- Wave Period

In addition, the derived parameters listed below are available:

- Total Depth (with wave heights).
- Water Surface (with wave heights).

8.6.20. Volumetric Analysis

A Volumetric analysis may be undertaken to determine the volume of water influenced by the selected water column parameter. The volumetric analysis is accessed by the Alt-C keystroke which displays the input table shown in Figure 8-61.

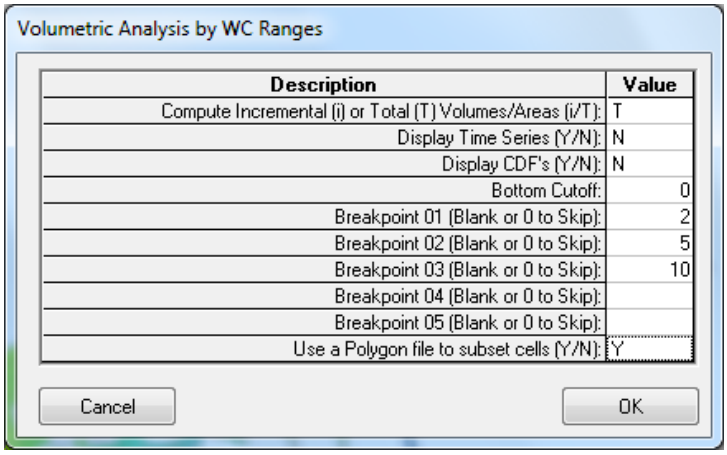


Figure 8-61 ViewPlan: Volumetric Analysis Tool.

The user may compute the volume, area and shoreline length either incrementally or as a total. If *Incremental* is selected EE will calculate from the *Bottom Cutoff* to *Breakpoint 01*, then from *Breakpoint 01* to *Breakpoint 02*, and so on. If *Total* is selected then EE will calculate from the *Bottom Cutoff* to each *Breakpoint* separately. The user should also select whether to calculate a subset of the model with a polygon or the whole model area.

Figure 8-62 displays the output of the *Volumetric Analysis* for salinity with the breakpoints as shown in Figure 8-61.

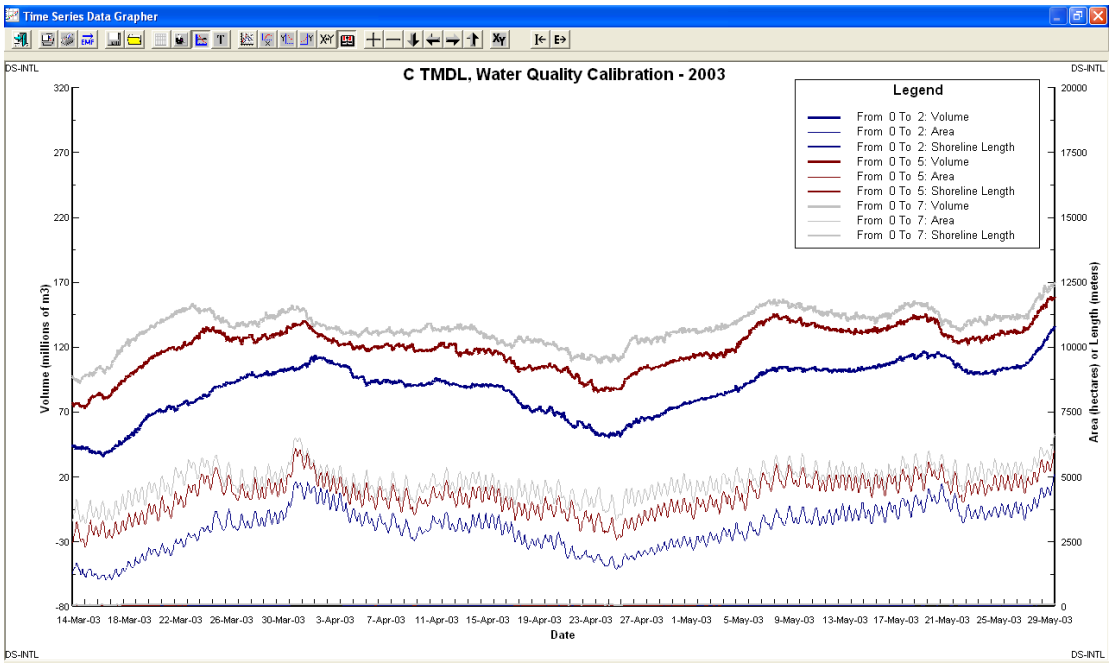


Figure 8-62 ViewPlan: Volumetric Analysis Time Series.

8.6.21. 2D Contour Lines

Within the *ViewPlan* tool, EE has the capability to plot 2D contour lines for any simulated parameters at any time steps. The contour option is enabled by selecting the *Contour* check box in the Viewing Options form. Beside the check box is a button to open the *Contour Control Form* which contains two tabs such as *Formatting* and *Contour Intervals* as shown in Figure 8-63

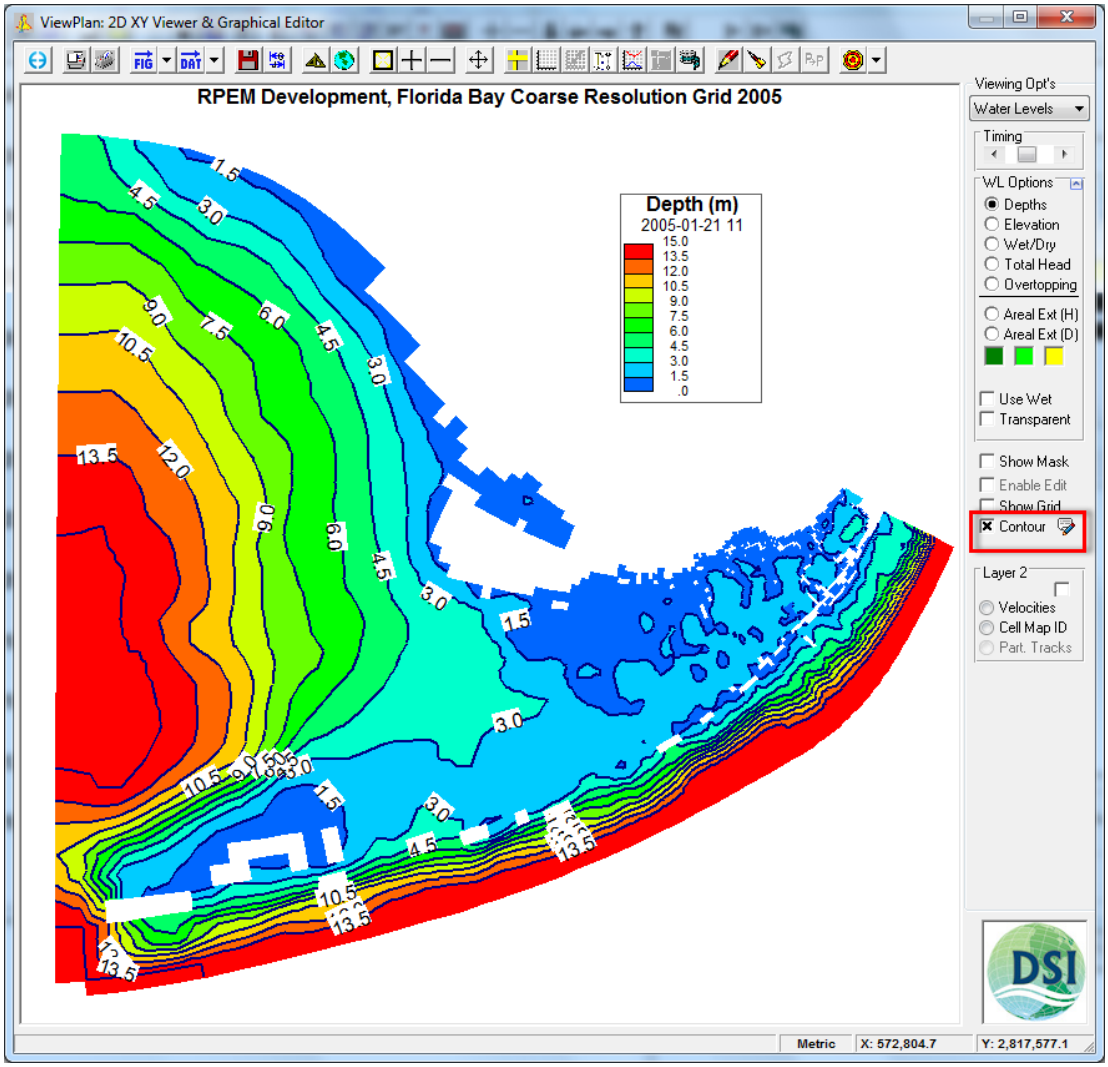


Figure 8-63 ViewPlan: Contour view.

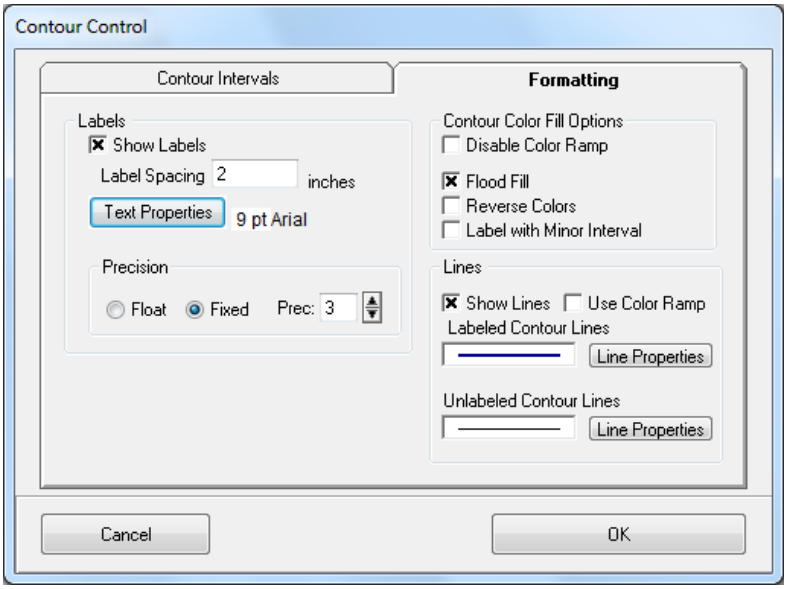


Figure 8-64 ViewPlan: Contour Control form – Formatting.

8.6.21.1 *Formatting Tab*

Formatting tab provides numerous formatting options as shown in Figure 8-64. The *Labels* frame provides a check box to switch on or off the labels and a text box for entering the spacing between the labels. The style and font of the labels may also be edited with the *Text Properties* button. In addition the background and foreground colors, label position and label precision may also be modified.

The *Data Points* frame allows display of symbols at the cell corners and editing of the symbol properties with the adjacent button. The user may also display values of the parameter currently being viewed with the *Show Values* check box. The font and position of the labels may be edited with the adjacent button.

The *Precision* frame provides two options for the numerical precision of the contour labels, floating or fixed. If *Float* is selected the precision value is ignored. Only if the *Fixed* button is selected is the label displayed to the numerical precision entered in the text box.

Contour Color Fill Options provide the capacity to “flood fill,” that is to fill the space between contours with the color based on the legend color ramp. It is recommended that when using contours the user should apply flood fill as it provides a constant color over the midpoint of the contour intervals. Note however, that this option overrides the *Outline Only*” option in the *Labels* tab. The *Reverse Colors* check box reverses the color ramp in the legend.

The *Lines* frame provides various options for editing the contour lines. Lines may be toggled on and off with the *Show Lines* check box. The color of the contour from the color ramp is toggled on and off with the *Contour Color* check box. For both the major (labeled) and minor (unlabeled) lines, line thickness, color and style may be edited with the *Line Properties* button.

8.6.21.2 *Contour Intervals Tab*

The *contour internals* tab is shown in Figure 8-65 below. The *From the Data* frame displays upper and lower limits for the parameter selected for the user's information. The contour tool assumes the minimum and maximum contour interval from this data and automatically sets the number of contours to 11, basing the major interval on this. The minor interval is set by default as one third of the major interval. Adjusting the minor interval will recalculate the major interval and number of contours accordingly. Modifying the number of contours will recalculate the major and minor interval values.

The *Clip Range* button will clip all contours outside the max and min contour values defined by the user.

The *View Options* frame provides two options for methods of calculating the contours: based on the corners of the cell, or on the cell centroid. The *Searching Radius* text box tells EE what radius of cells should be used in the calculation of the contours. The *Use Spline* check box will employ a spline function for calculation of the contour line. The number of points to be used in the spline calculation is user defined.

The *Outline Only* checkbox will turn off the legend and background colors and display only the contour lines. Note that this will not be visible if *Flood Fill* has been selected.

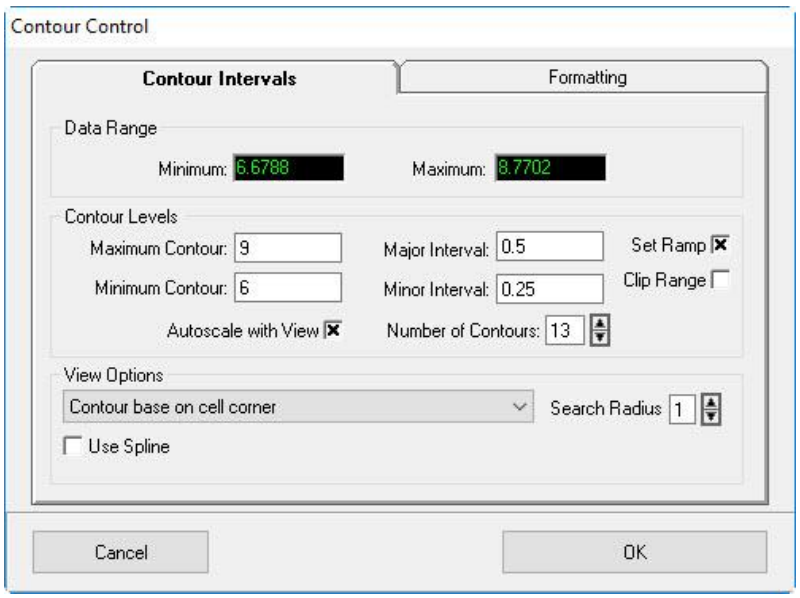


Figure 8-65 ViewPlan: Contour Control – Limits

8.6.22. Time Series Contours

Time series of velocity contour profile is a feature developed from release of EE7.2 to plot the vertical constituent contours over the water depth column, somewhat like a “heatmap”. Effectively they are the I,J Component on a time-depth background.

This feature is accessed by RMC from *Velocities* or *Water by Layer* or *Water by Depth* viewing options (See Figure 8-66) and provides a time series of that constituent’s contour profiles as shown in Figure 8-67. This is useful for analyzing the velocity speed of river or tidal discharge from the bottom to the top of the water column. Other transport parameters such as temperature and salinity are also available with this option. Figure 8-68 shows the time series contour plot control which is accessed by RMC on legend. This allows the user to define the maximum and minimum contours and to set the precision of the values as either floating or fixed.

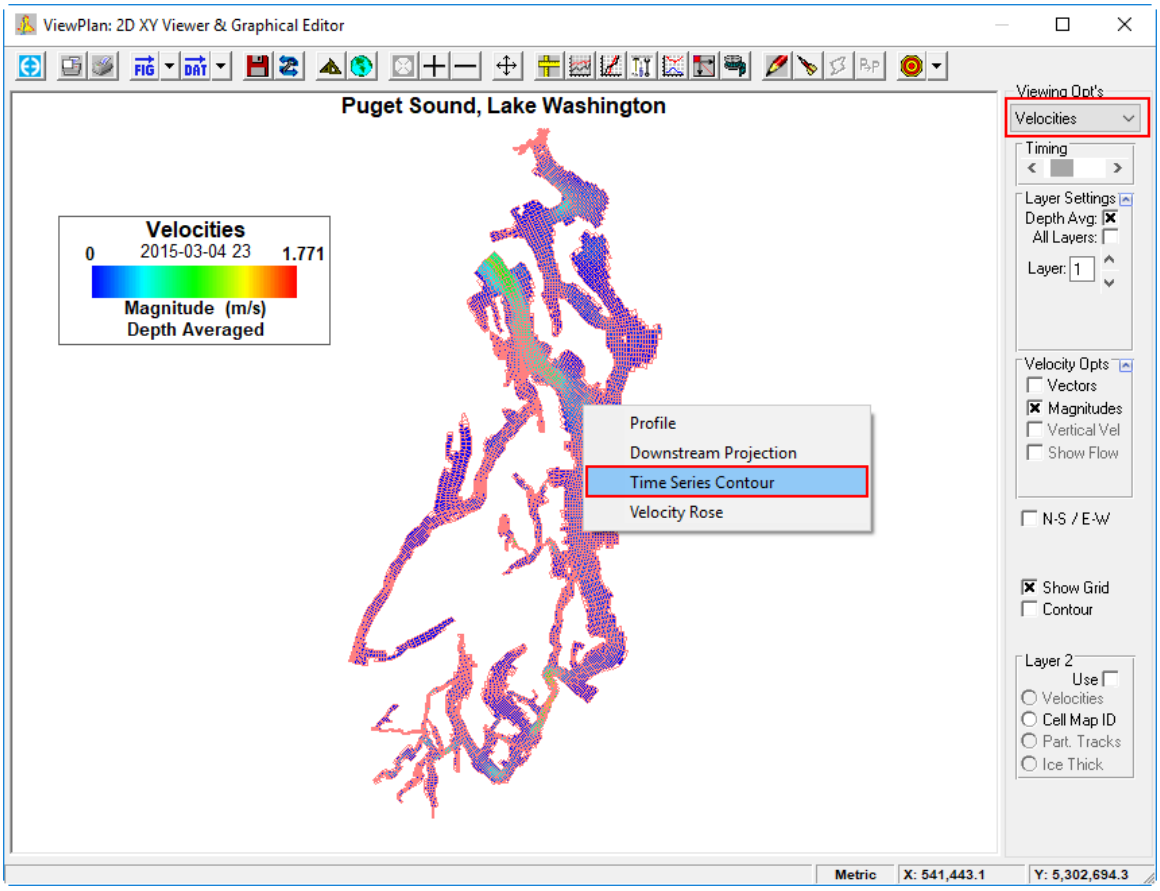


Figure 8-66 Time Series Contour option.

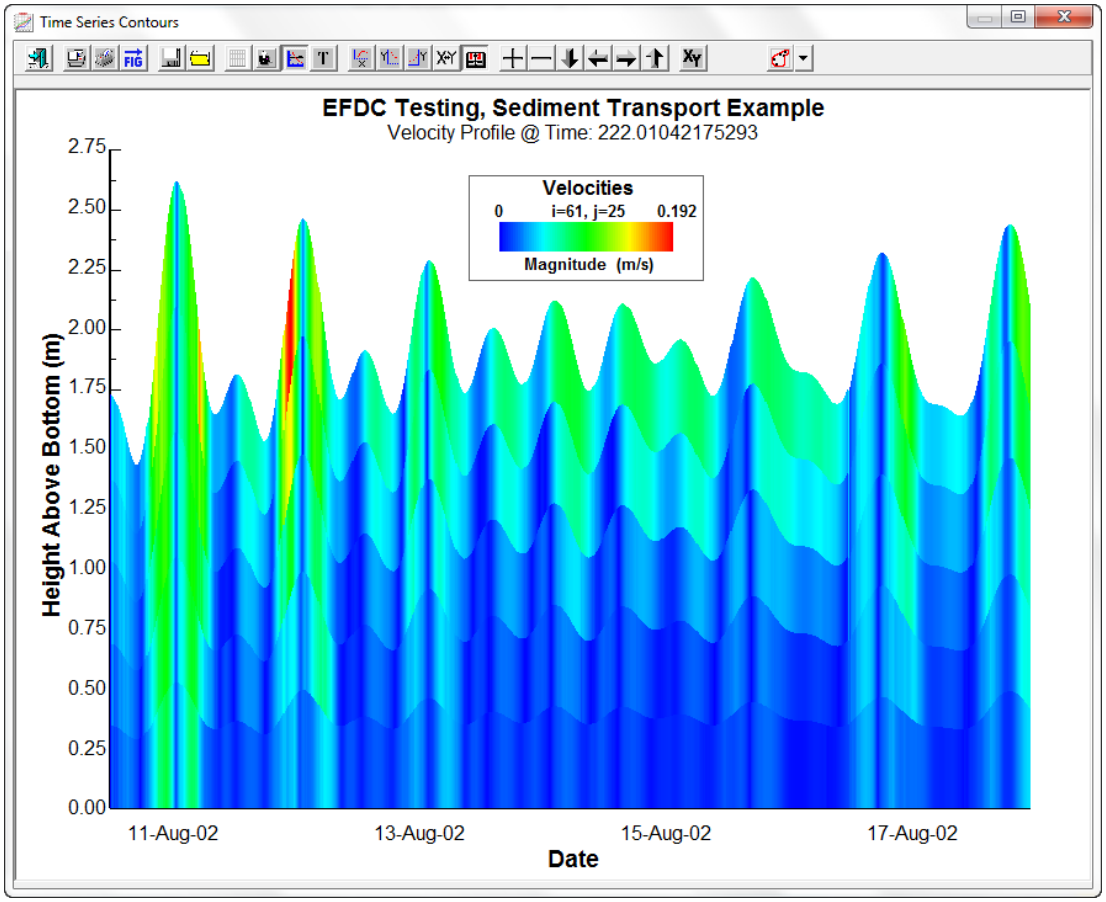


Figure 8-67 Time Series: Velocity Contour Plot.

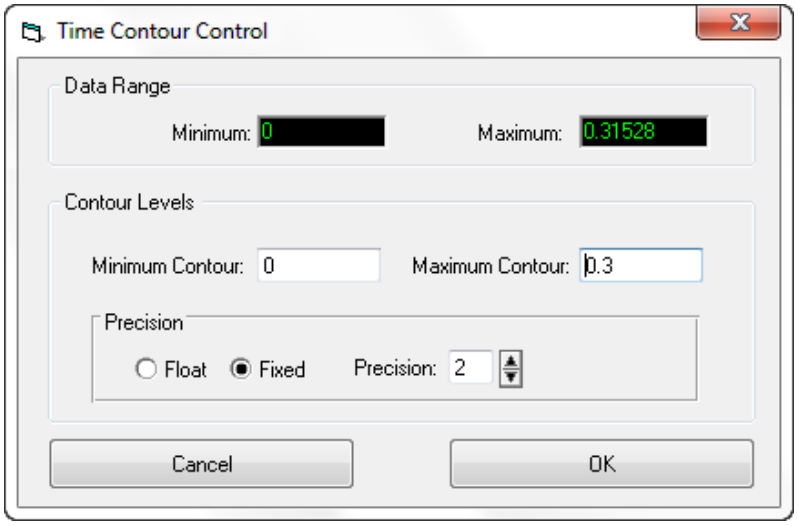


Figure 8-68 Time Series: Velocity Contour Plot Control.

The user can also select to plot a *Time Series Contour* by each layer, depth averaged or for all layers.

8.6.23. High and Low Pass Filter for Time Series

EE has filtering capabilities which are applicable for any time series plots. This involves the integration of a low pass and a high pass filtering to the time series plotting utility. Two types of filters (Lanczos and Fast Fourier Transforms) are available. This feature has also been added to the *Time series Comparison* form and *Statistics tools* in the *Model Analysis* tab as described in the next section.

The method of High- and Low-pass Filtering is that the instantaneous current velocity can be represented as the sum of time-averaged value and fluctuations of high and low frequencies as follows:

$$v_i = \bar{v} + \widetilde{V_h} + \widetilde{V_l} \qquad (1)$$

where v_i is instantaneous current velocity; $\bar{v}$ is time-averaged velocity; and $\widetilde{V_h}, \widetilde{V_l}$ are fluctuations due to the high and low frequencies respectively.

Based on the Fast Fourier Transform (FFT) method for energy spectra in frequency domain, these fluctuations were obtained dependent on the defined cut-off frequency i.e. at 1, 3, and 5 filtering hours. After that the inverse FFT was used to plot the high or low frequencies results.

High-Pass option: $v_i = \bar{v} + \widetilde{V_h}$ (2)

Low-Pass option: $v_i = \bar{v} + \widetilde{V_l}$ (3)

In the *Time Series Plotting* utility, the user can choose any time series to filter. There is an option to *Create to new line* in order to add this filtered line into the plot as shown in Figure 8-69. If *Apply to this Line* is selected the time series will be replaced with its filtered series, an example of which is shown in Figure 8-70.

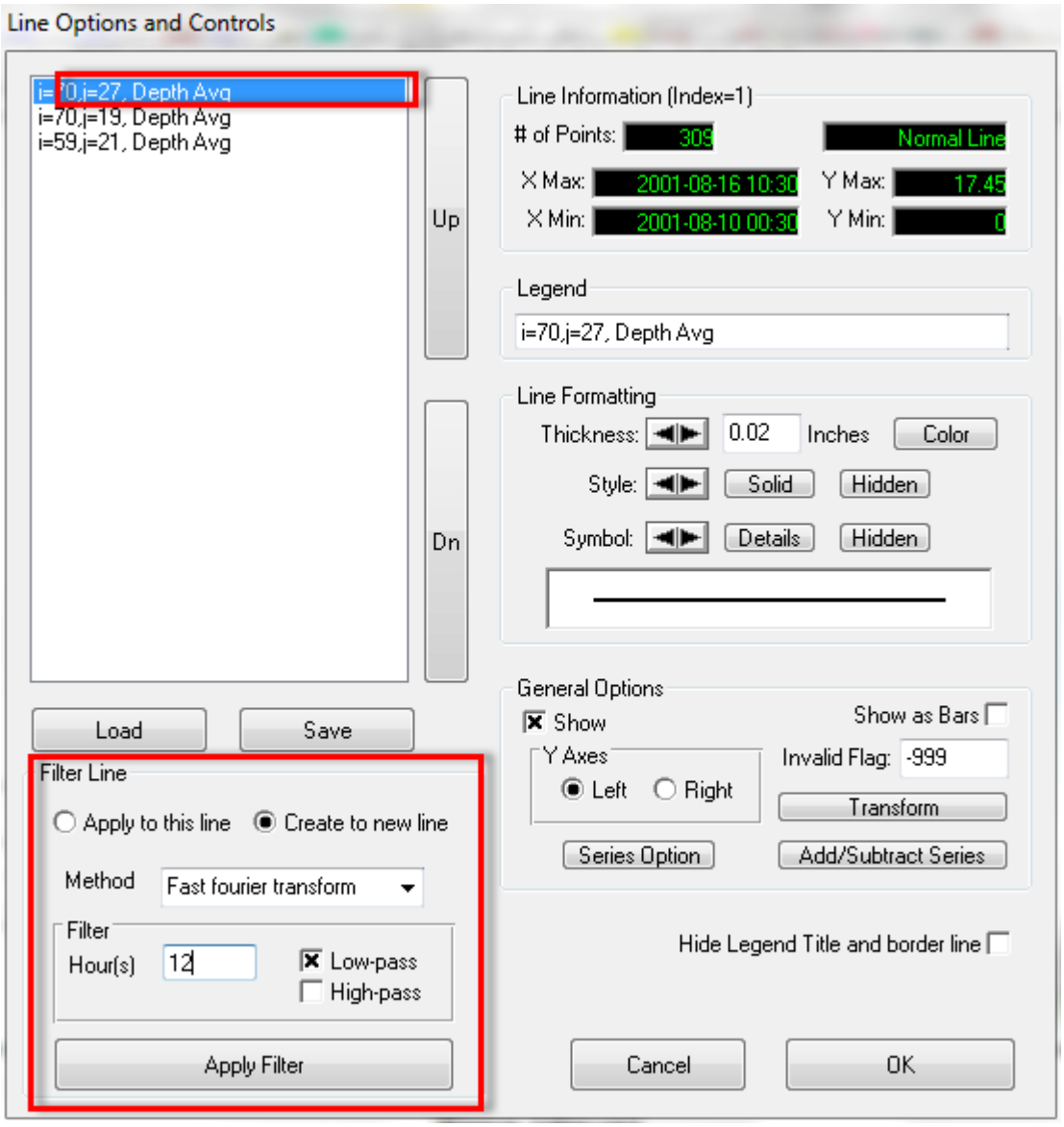


Figure 8-69 High and Low Pass Filter for Time Series Plotting.

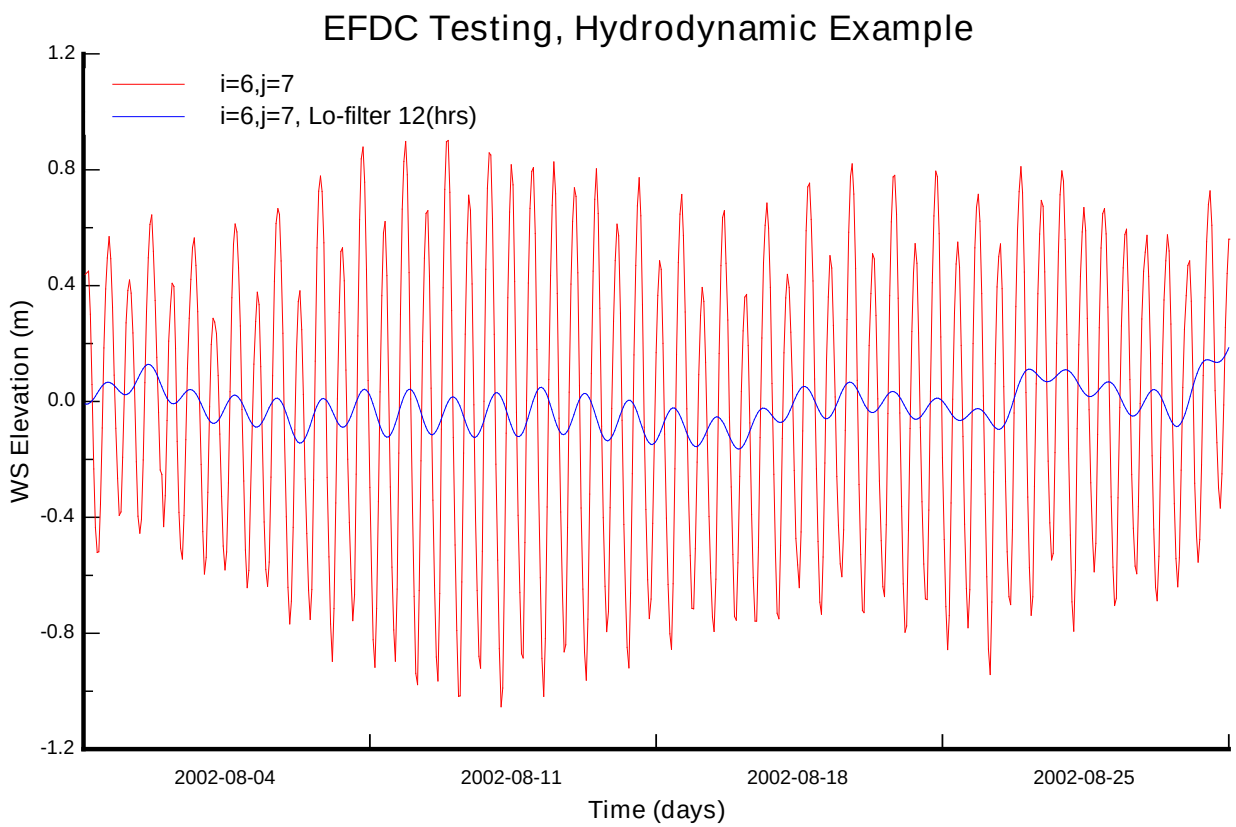


Figure 8-70 Time Series plot of WSEL (red) and low pass filter (blue) at location i=6, j=7.

8.6.24. Mean Mass Transport - Averaging of 2D/3D fields

The mean mass transport (MMT) tool is used to remove tidal series or environmental time series by providing a running average of 2-D and 3-D fields. The MMT tool essentially displays a single snap shot calculated for an averaging period of a given number of hours. The averaging period is set from the start time to end time of the residual field and displayed for every cell. The tool can be accessed from *ViewPlan* using the CTRL-R keystroke. The user is then prompted for the *Start Time* and *End Time* in Julian or Calendar time. Pressing *Apply* will display the MMT for that time period as shown in Figure 8-71 for this estuary model.

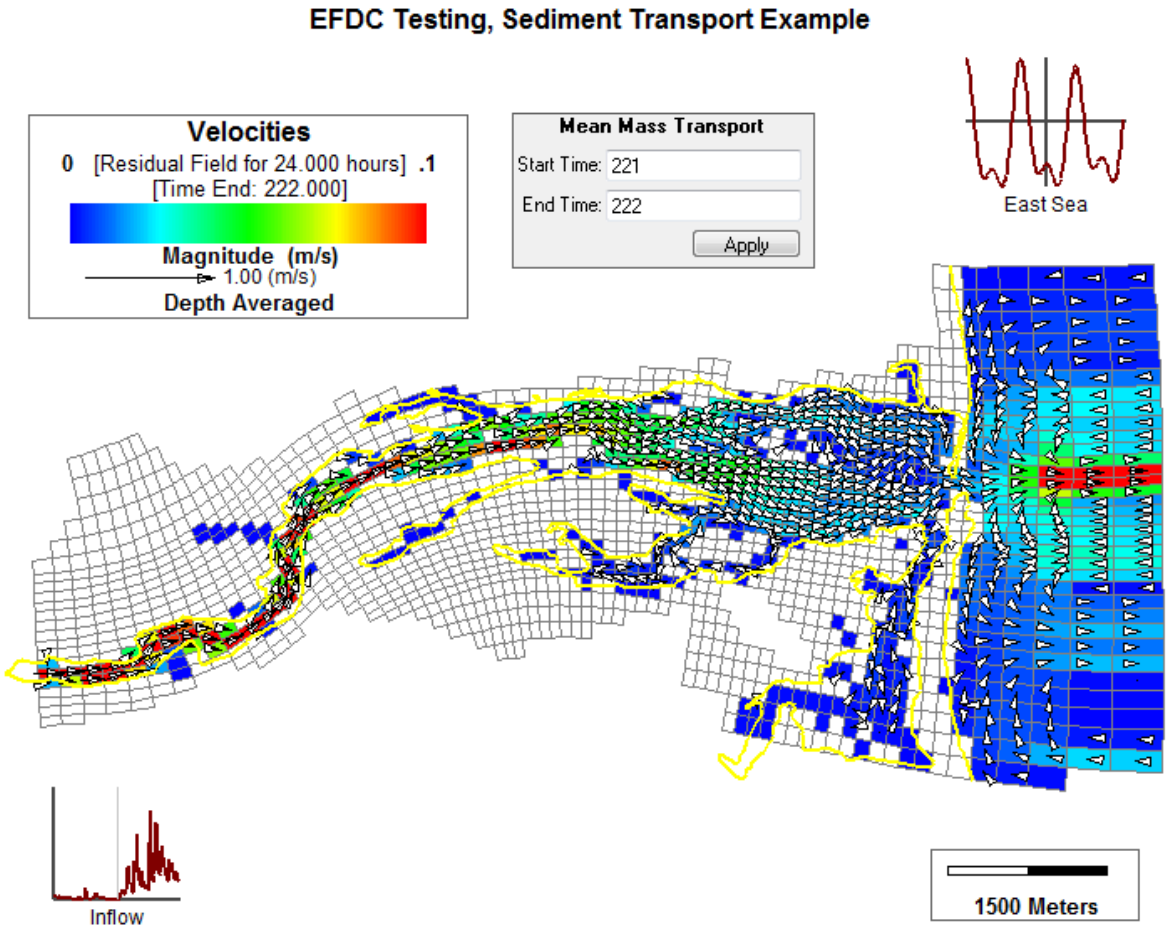


Figure 8-71 ViewPlan: Mean Mass Transport.

8.6.25. Particle Tracks and Oil Spill

The Viewing Option for Particle Tracks provides the user with a number of options for display of particle tracks and oil spill. The drop-down box *Show Drifter* also allows options for *Oil thickness*, *Oil mass* or *Oil volume* of oil is being simulated. If the drifters are displayed instead of oil then various *Color Options* may be selected, including color by elevation, by depth, by layer, by time or by group. The latter option is shown in Figure 8-72.

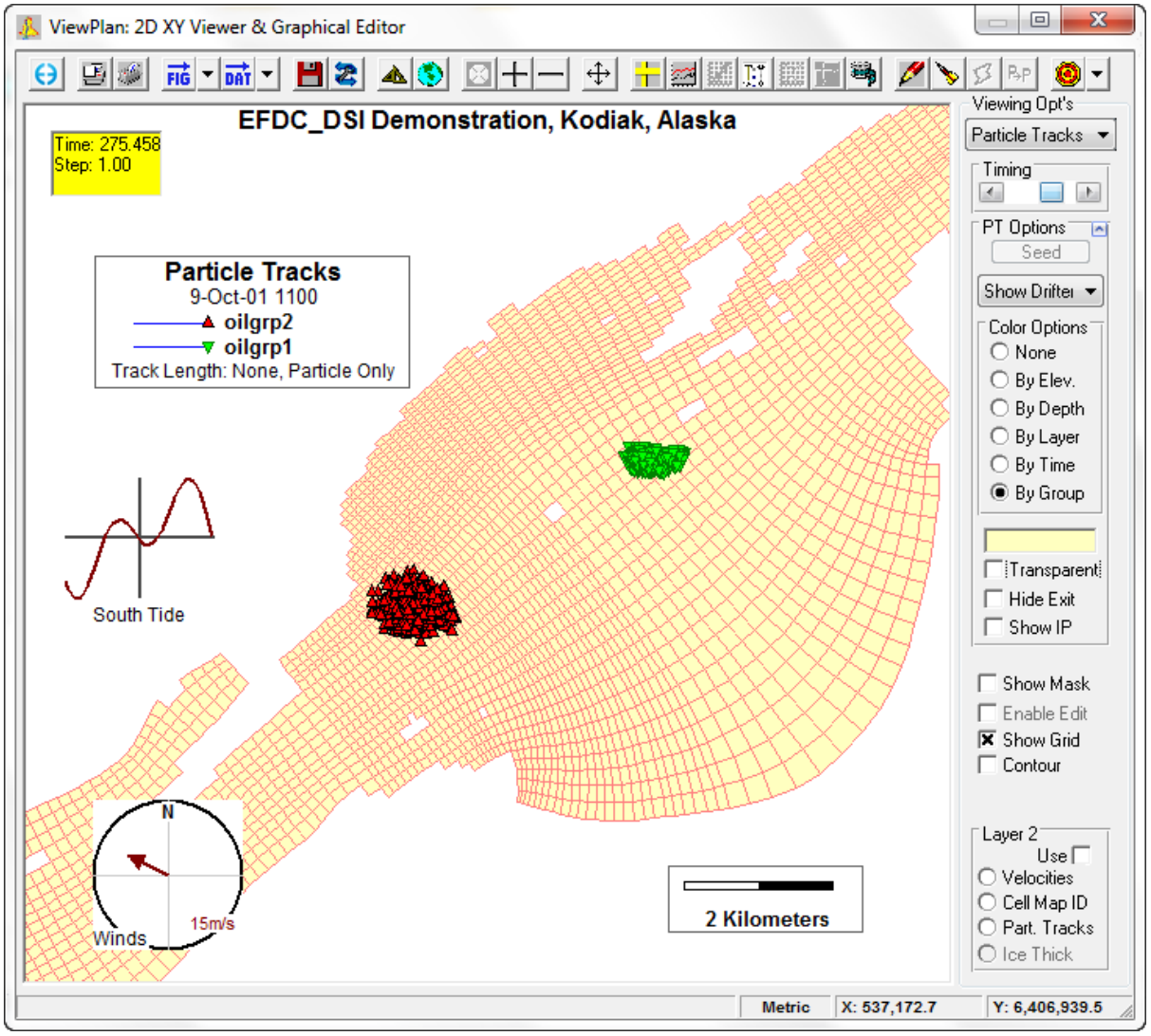


Figure 8-72 ViewPlan: Particles displayed by groups.

In the particle tracking sub-model of earlier versions of EE, particles that reach the model boundary stick to the edge of the boundary. This is to show that the particles have left the model domain and are no longer being calculated. From EE7.3 a new method has been added which provides the user an option to allow the particles to disappear if the the *Hide After Exit* check box is selected (see section 11.1.7 for an example).

As each group has its own chemical property and volume per seed to allow for different oil type, EE calculated the summation of all the groups to display total thickness or mass. These summaries of total oil volume and area are provided in the *General Statistics* button for the current time and as a time series.

If oil is being simulated by any of the drifter groups, then the total oil volume and total mass of oil is displayed at bottom of the form as shown in Figure 8-73.

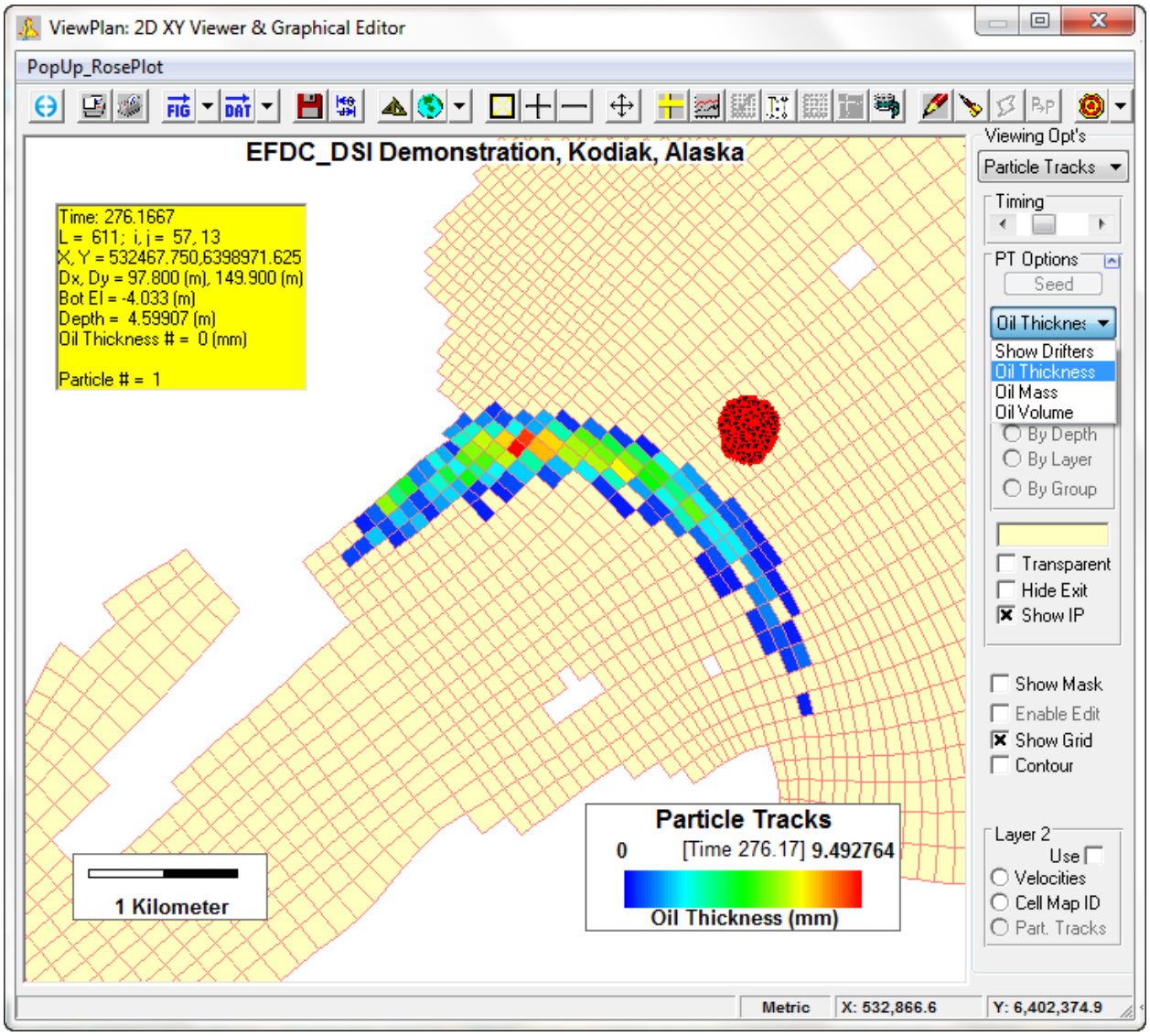


Figure 8-73 ViewPlan: Oil Spill viewing options.

8.7 Other Features

8.7.1. Photo Viewer

A new function in *ViewPlan* is the ability to add images to work space to illustrate the model domain. This is accessed via the *CTRL-P* keystroke which turns on the image viewer. RMC on a cell to *Add* and then browse to an image that location for it to be added to the workspace as shown in Figure 8-74.

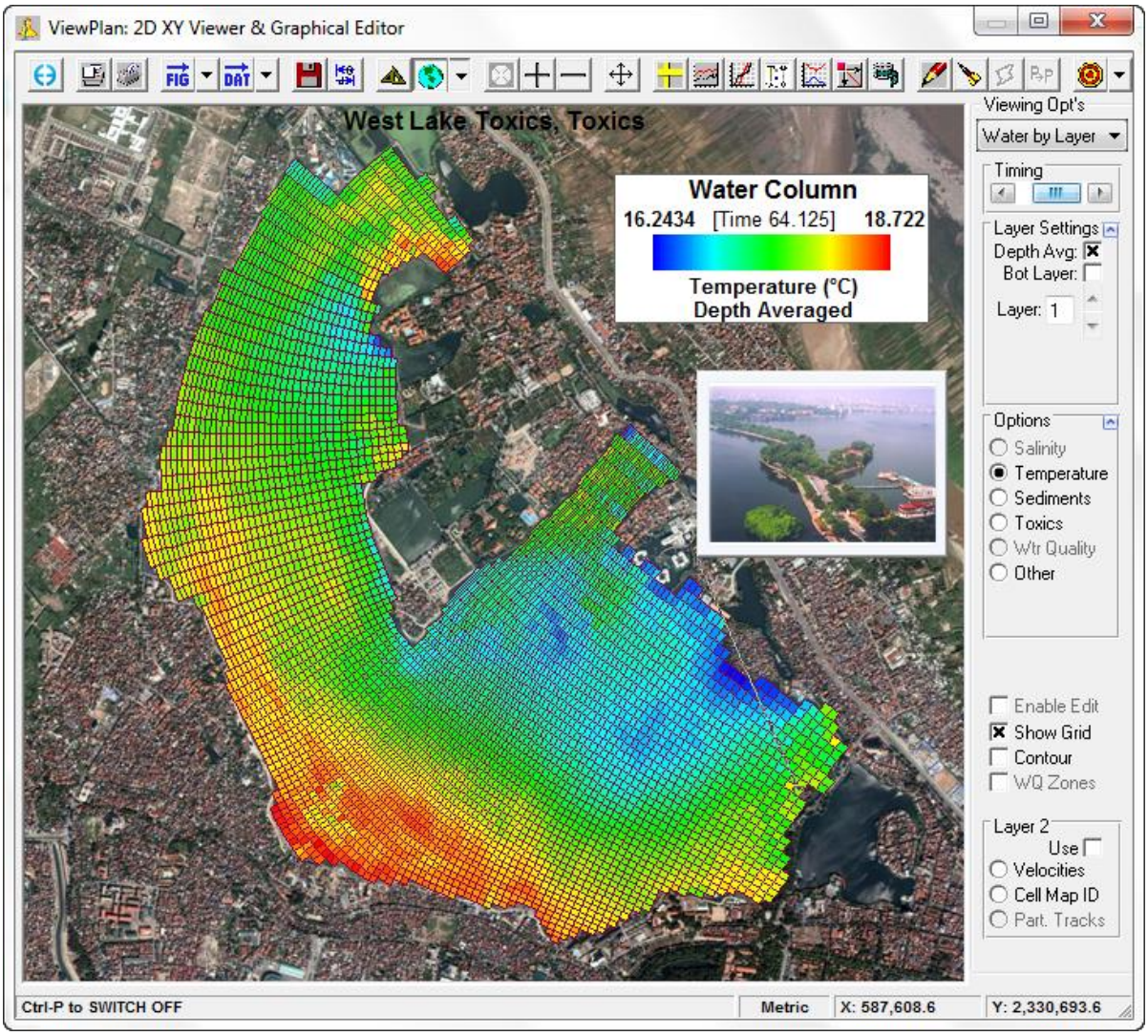


Figure 8-74 ViewPlan: Image Viewer.

8.7.2. Change Legend Font Sizes

EE has the option of adjusting the font, color and size of the text label in the legend in *ViewPlan.* This can be achieved by using ALT- F and selecting options as shown in Figure 8-75

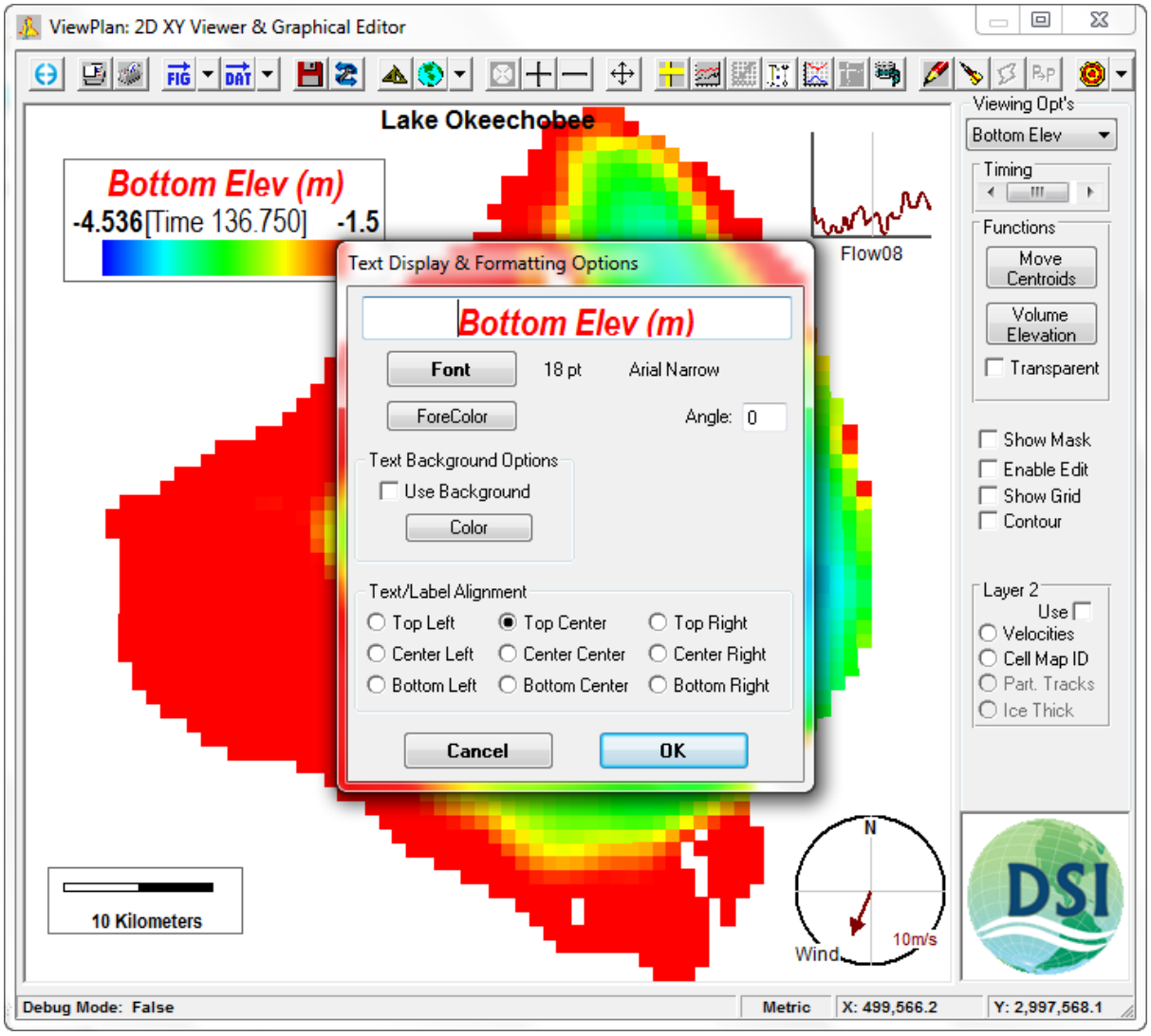


Figure 8-75 ViewPlan: Adjustable legend front sizes.

8.7.3. Display Symbols in Titles and Legends

EE8 allows the user to include symbols and alternative fonts in the titles and legends. This is primarily designed to provide the option to include Greek symbols in the plot axis. The user should select the “Control + Insert” key when editing a legend or title for the font option and table of characters to be displayed as shown in Figure 8-76.

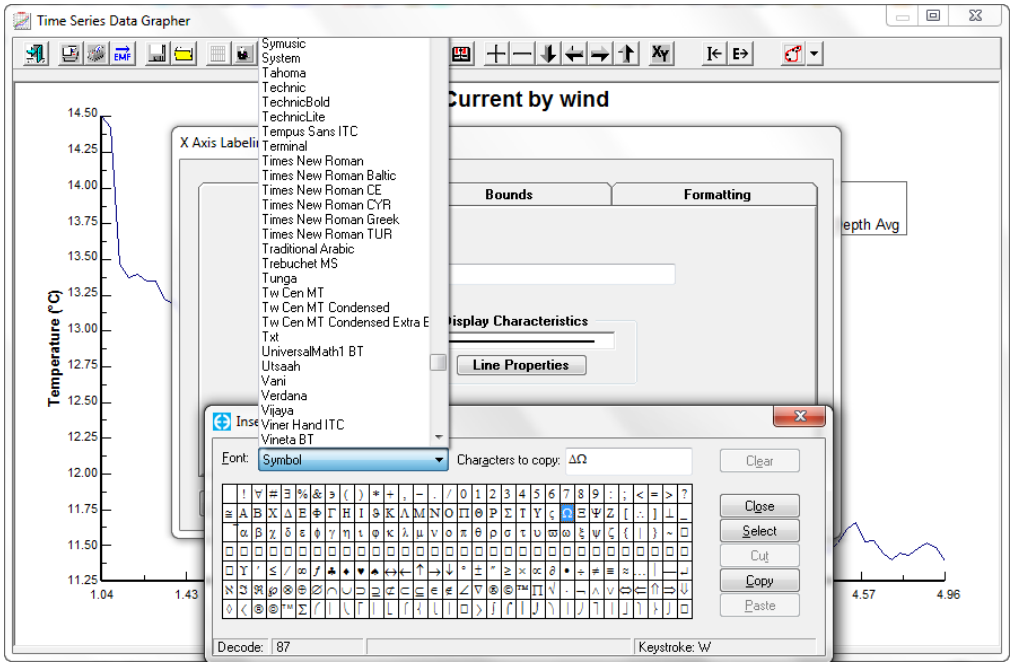


Figure 8-76 Selecting Greek symbols for Legend and Y-axis label.

If the user wishes to mix Latin and Greek fonts together in one line it is recommended to use the Greek Arial font and select the appropriate characters as both fonts can be combined with this option as shown in Figure 8-77.

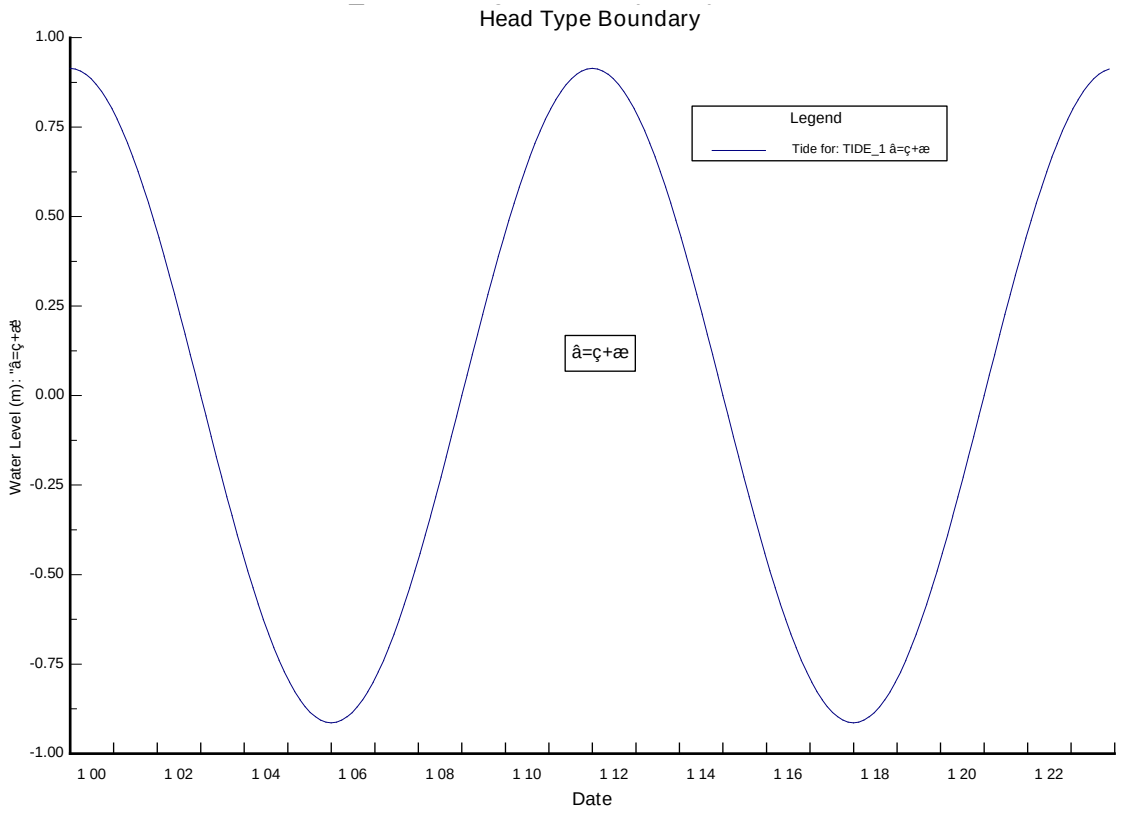
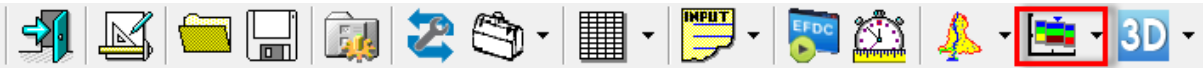


Figure 8-77 Greek symbols in legend and Y-axis label.

9 ViewSlice



The *ViewSlice* button of the main toolbar provides access to the profile/cross section post-processing utility of EFDC_Explorer. An example of the type of plot available is shown in Figure 9-1. The contents of the current cell can be displayed, as in *ViewPlan*, by LMC'ing on the cell. Many of the operations of this feature are similar to *ViewPlan*.

Similar to *ViewPlan*, the *ViewSlice* button provides the user with a dropdown menu comprising the following options:

View Initial Conditions: This option may be used with the user is setting up the model or wants to view the ICs prior to the model run

View Model Results: This option is used after the model has been run and the user wished to analyze the model results.

View Model Results (Re-scan Output): This option should be used if the user has merged two models or has continued a model run and want EE to go back to the beginning of the output data set.

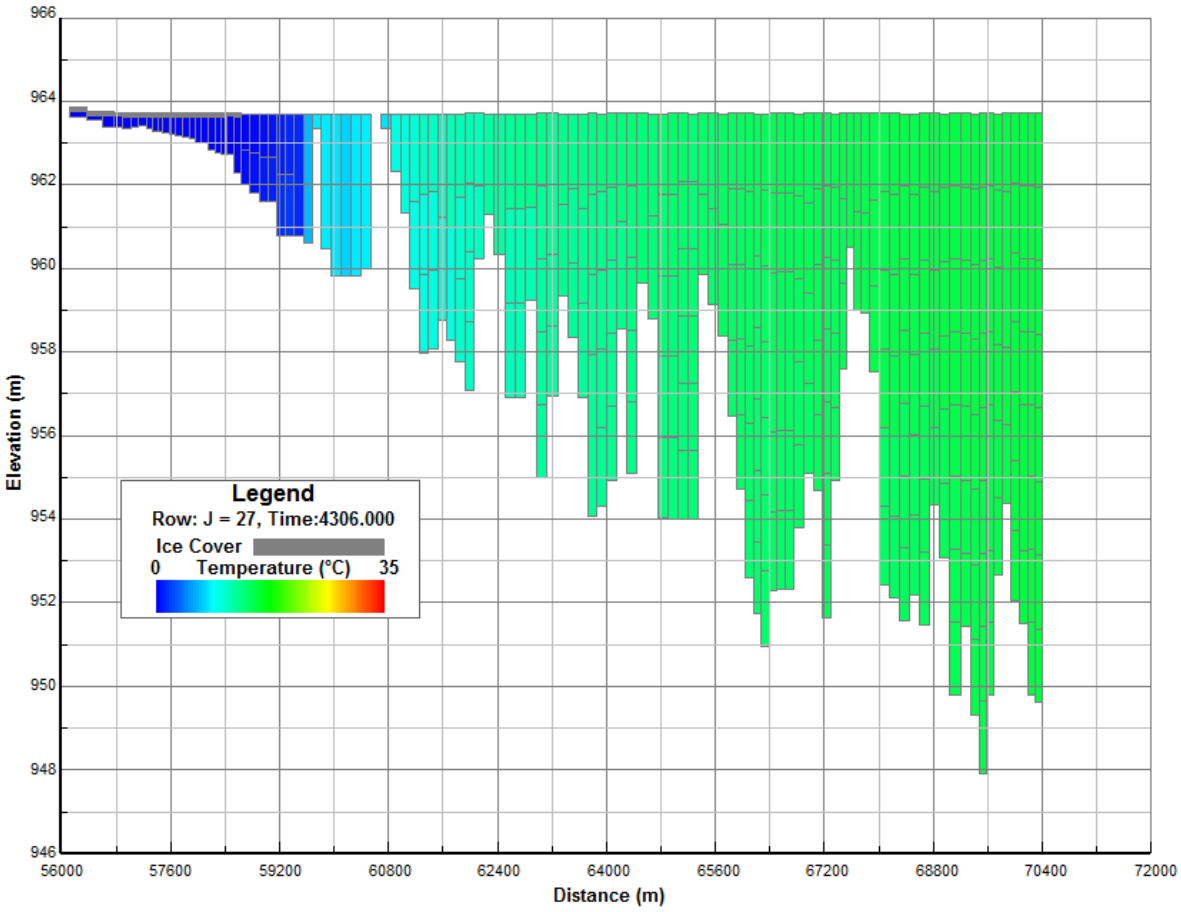


Figure 9-1 ViewSlice example for temperature and ice at one snapshot.

9.1 Slice/Profile Selection

The *Slice/Extraction Options* frame needs to be set first before a profile can be extracted from EFDC. There are three options shown in Figure 9-2. The user may either select a value of I to extract the active J cells along that I, or select a value of J to extract the active I cells along that J. The third option is to use a "Drape Line", which is a polyline in the same coordinate system as the LXLY data. The I & J's from along the line will be assembled and the profile will be output along that slice.

If an I or J extraction is used, the user can use the + /- keys to scroll up and down the selected coordinate. For example, if the I5 was selected, the + keystroke would display the I6 section, the – keystroke would display the I4 section. This feature is not applicable to the "Drape Line".

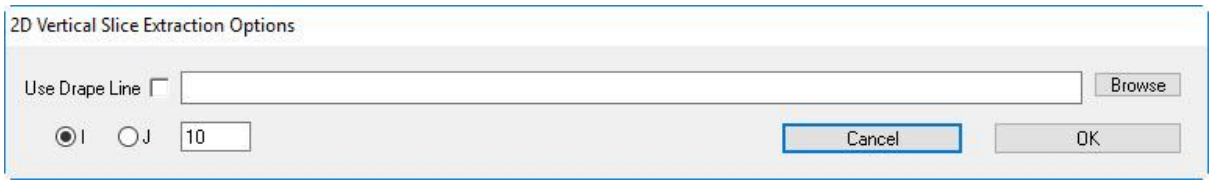


Figure 9-2 2D Vertical Slice Extraction Options.

9.2 Toolbar Summary

The *ViewSlice* toolbar provides access to a range of different functions and utilities. Some of these functions are dependent on the current context while some are not. Table 9-1 shows a summary of each function.

During animations it is usually better to fix the Y axis (elevations). The default is automatic scaling. During an animation the water surface elevation changes due to tides and/or storm events. If the axis is set to automatic scaling the plot appears to jump around due to different scaling at different times. This problem is solved by setting the elevation minimum and maximums to an appropriate range.

Table 9-1 Summary of ViewSlice toolbar.

General Functions	
	Exit ViewSlice.
	Printer setup options.
	Print to current printer the current view.
	Export the current view to a Windows enhanced metafile (EMF).
Toggles & Options	
	Toggle the grid.
	Show/Edit comment boxes
	Show/Hide markers
	Toggle between Julian date/Gregorian date formats
Plot Format and Labels	
	Edit text and fonts for the titles.

	Parameter display options/settings.
	Access the X axis options form.
	Access the left Y axis options form.
	Access the right Y axis options form.
	Set all the axes formats to the one last edited.
Navigation Functions	
	Zoom in at fixed increments.
	Zoom out at fixed increments.
	Pan Left.
	Pan Right.
	Pan Up.
	Pan Down.
Utility Functions	
	Toggle the display of the coordinates.
	Animation tool. Output animations to the screen and/or AVI files.

9.3 Primary Display Options

The user may select which parameters are to be displayed by accessing the *Display Options* form by RMC'ing on the legend, pressing Ctrl-O or using the toolbar. There are four sub-tabs *General Options, Water Column, Ice and sediment Bed*. An example of the form is shown in Figure 9-3.

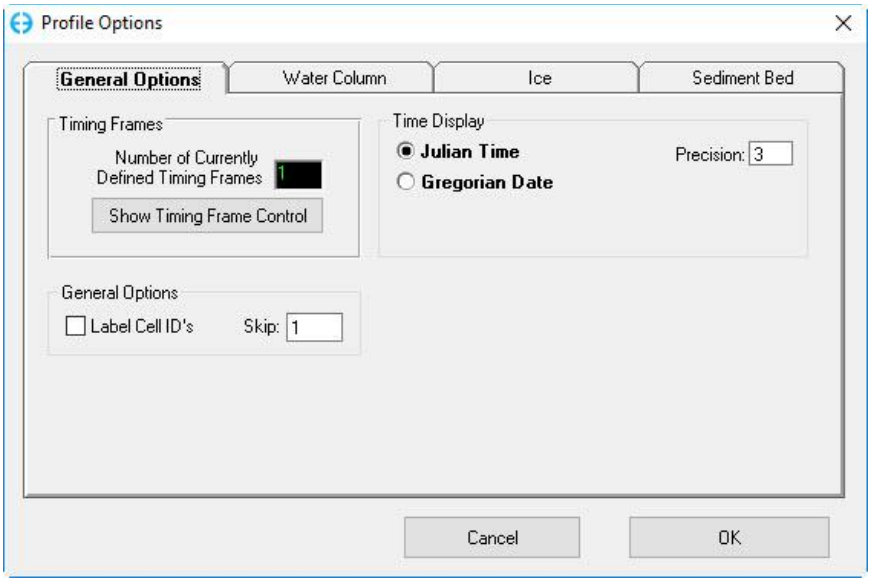


Figure 9-3 ViewSlice display options.

9.3.1. Velocity Magnitude by Color in ViewSlice

EE can display the velocity magnitude in the *ViewSlice* screen in much the same way as it does in the *ViewPlan* screen. This may be seen in Figure 9-4 below where arrows indicate direction of flow and the color.

Note that the user may go to any date with the *Go to Date* function (Ctrl-G) and scroll along the i or j axis with "+" and "-"keys on the keyboard.

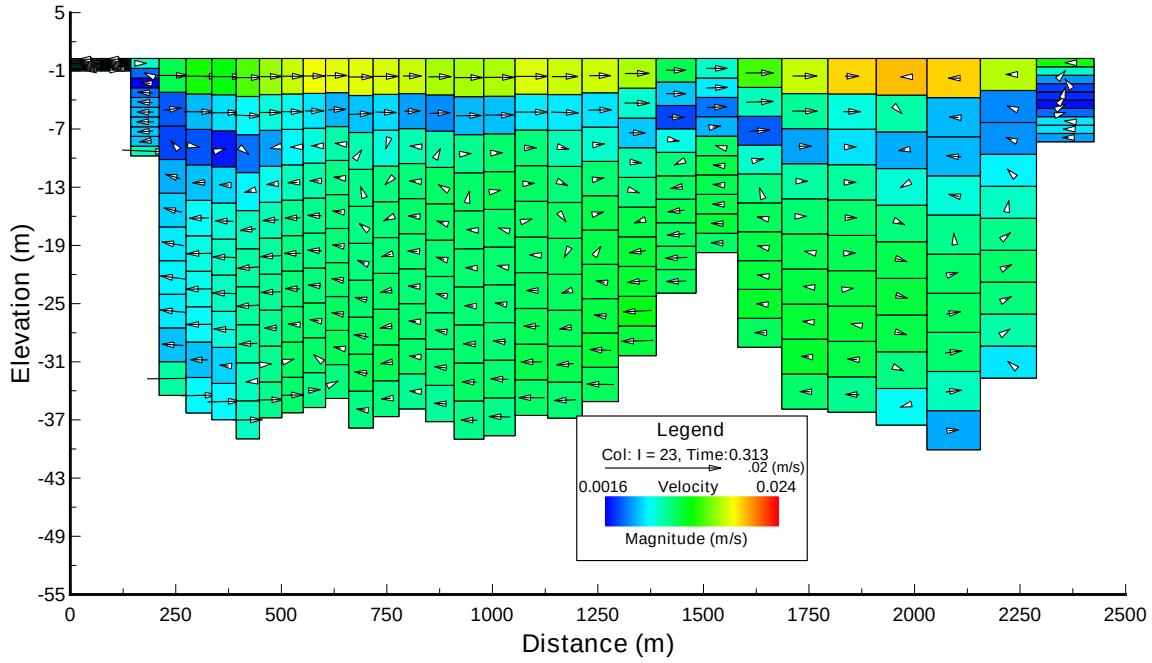


Figure 9-4 ViewSlice: Velocity color by magnitude.

9.3.2. Contour Lines in ViewSlice

EE now has the capability to produce contour lines in the *ViewSlice*. The contour option is enabled by selecting the *Show Contour* button in the *ViewSlice* form.

Next to the *Show Contour* button is the button *Show Contour Control*, which is used to open and modify the contours with two tabs: *Contour Intervals* and *Formatting* as shown in Figure 9-5. The function of these forms is the same as those for the *ViewSlice* contour features explained earlier. One different feature is the *Contour display ratio* which controls the algorithm to divide the cells before drawing the contour. If the height of cells are quite different then a higher number may be required to ensure correct display, however this will impact calculation time.

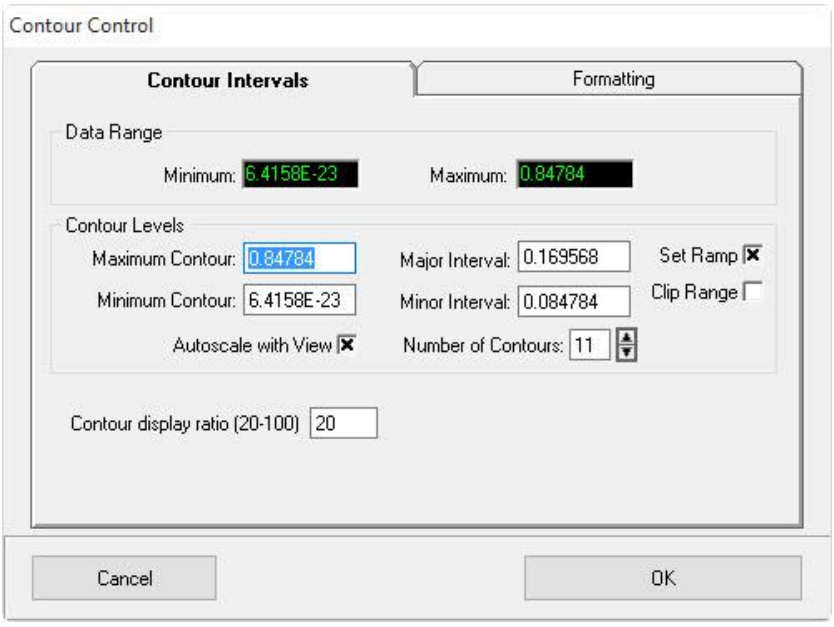


Figure 9-5 ViewSlice: Contour Control.

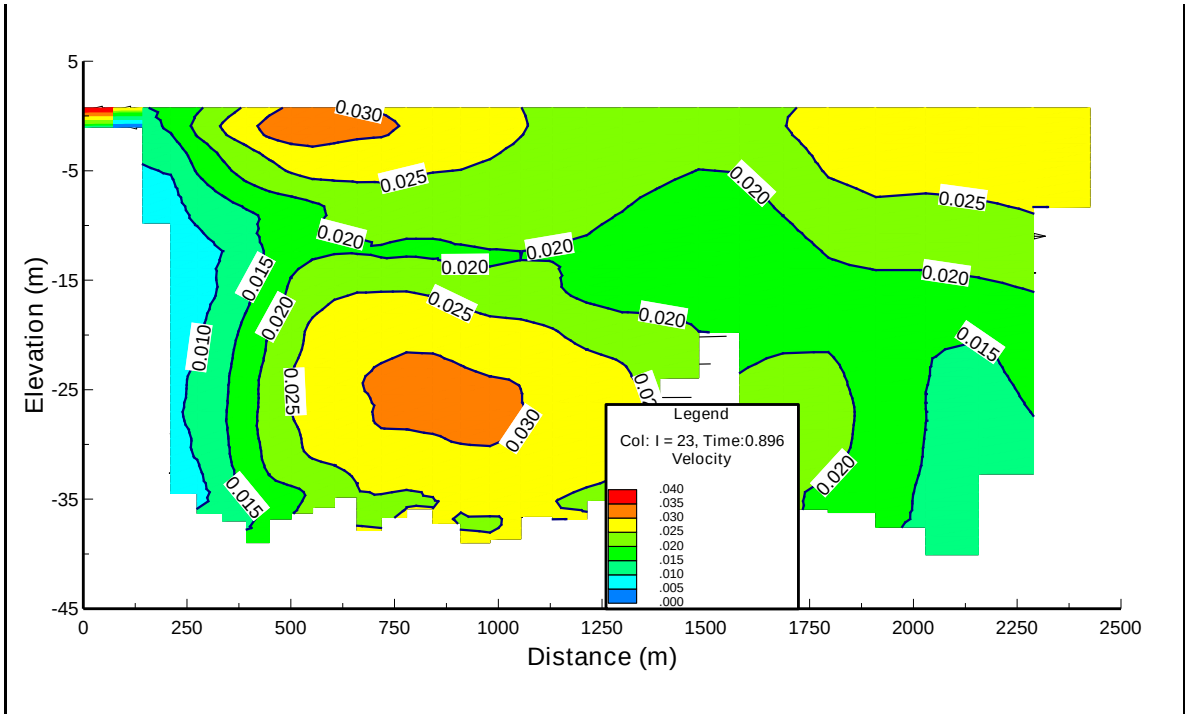


Figure 9-6 ViewSlice: Contour view.

9.4 Function Keys

During both pre- & post processing of the model the shortcut keys can be used to edit the graphs as shown in Figure 9-7. This form can be accessed via function key F2.

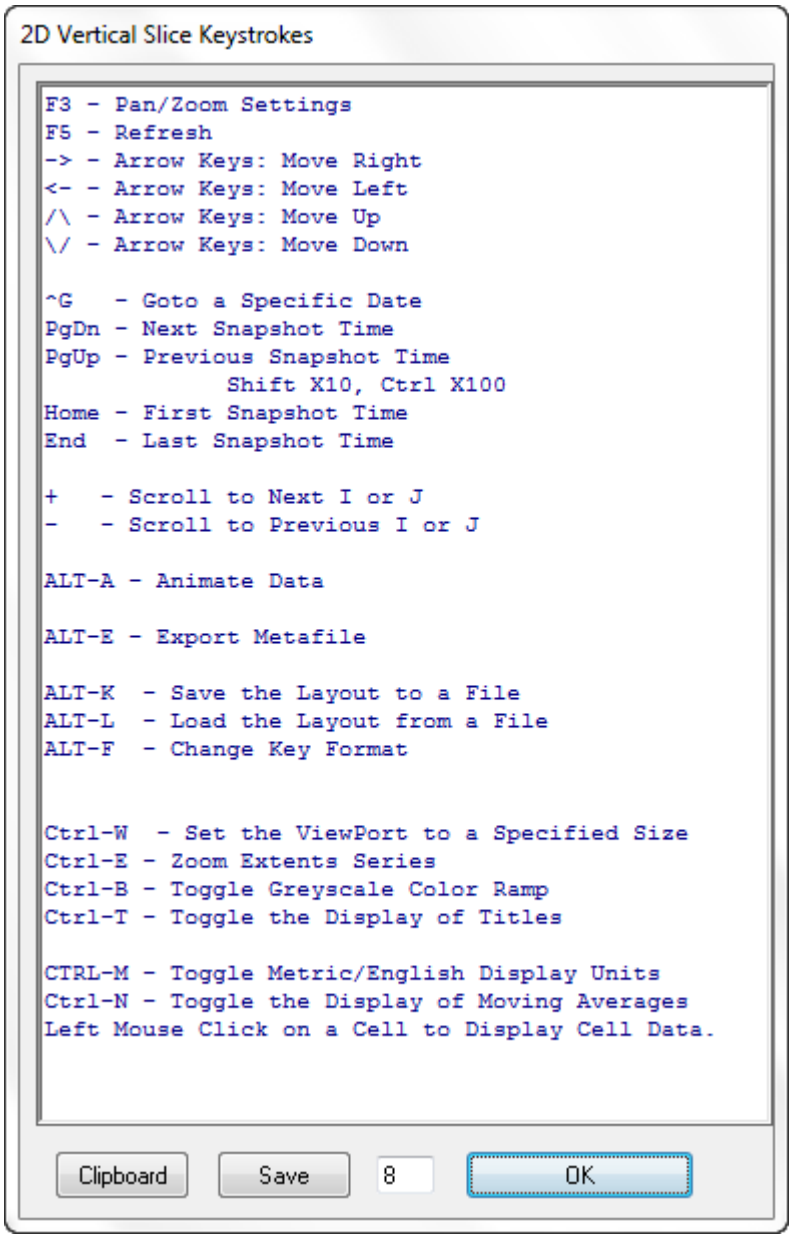


Figure 9-7 ViewSlice keystroke functions.

10 Time Series Graphing

Many of the plotting features of EFDC_Explorer use the *Time Series Graphing* (*TSG*) utility. In addition to time series plotting, the *TSG* is also used for any XY plot as well as *ViewSlice* (but with a completely different plotting subroutine). Figure 10-1 shows a screen capture of the *TSG*. This figure demonstrates the capability of the *TSG* to import multiple data files for comparison with model results.

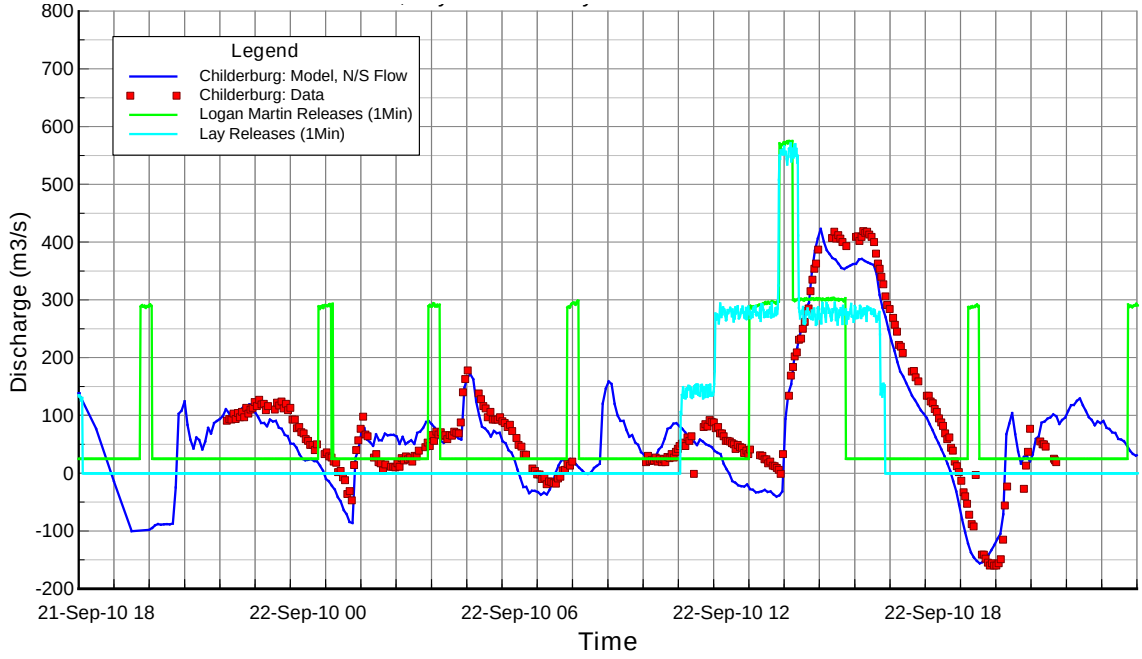


Figure 10-1 Time Series Plot.

Most of the functionality of the *TSP* is provided in the toolbar at the top of the utility but some of the functionality is accessed via keystrokes by press F2. (see Figure 10-2).

Table 10-1 provides a brief summary of the toolbar functions. This is very similar to the *ViewSlice* toolbar. The main difference is the import (I←) and export (E→) data series features. These are extremely useful to save a data series out to a file, and then later, when viewing another model or other series, load that series in for comparison. The format of these files is shown in Appendix B. A "DAT" file (the default extension) can contain an almost unlimited series. Since the TSP is used to view every time series from water column to sediments and boundary forcing series, the user can develop their own series from data for calibration or any other comparison purposes.

Exporting: When exporting, the *TSP* exports only the data in the current view. If data has been loaded but "turned off" that data will not be exported. The legend is exported with each series.

Importing: The *TSP* utility first requests a file to import. It then loads all the lines in the file and displays them with the legend and asks the user which series to import. Once the series to be imported is selected, the user is then asked which series to insert the imported series into. The *TSP* makes an initial series assignment but the user can select any series from 1 to 10. Existing series can be overwritten during the import process, if desired.

The left (primary) and right (secondary) Y axes can be edited by using the toolbar buttons shown in Table 9-1 or they can be edited by RMC'ing on the areas of the axis labels. The same method applies to access to the X-axis editor.

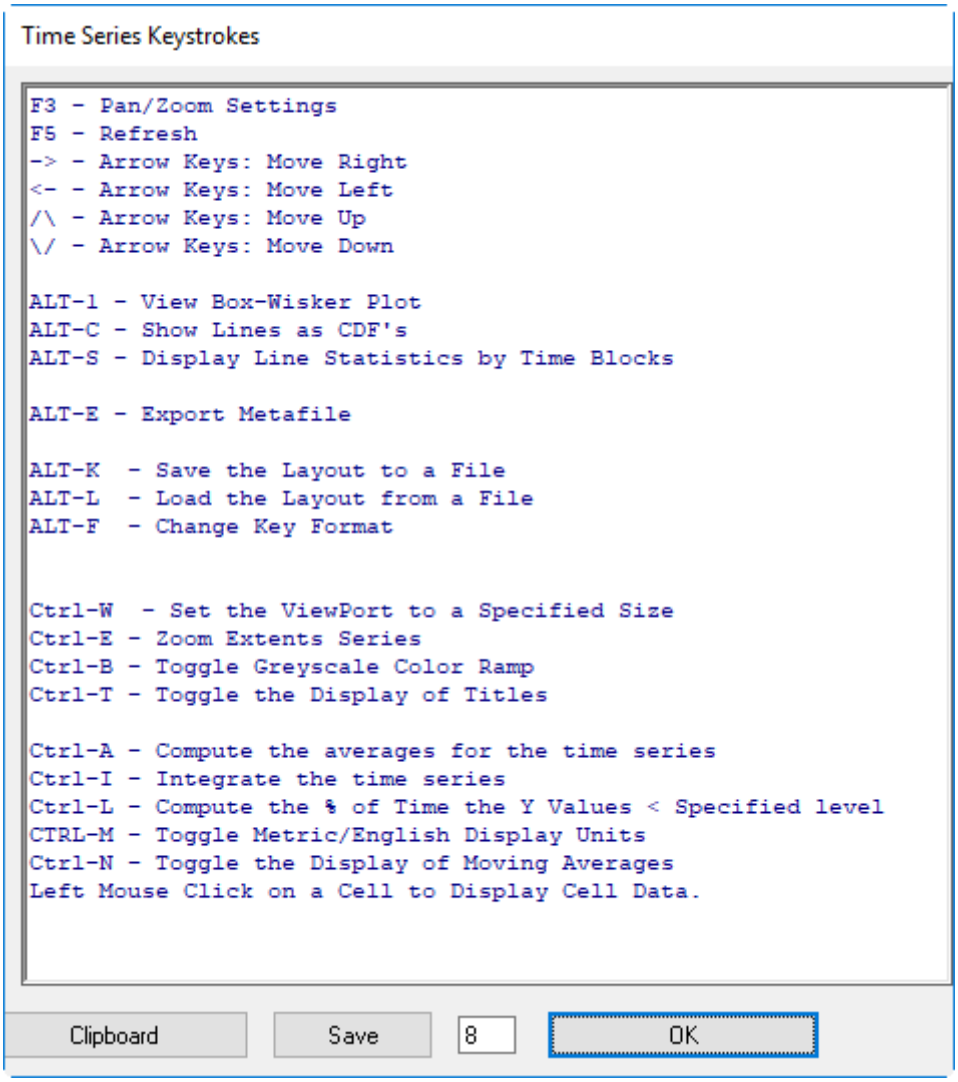


Figure 10-2 Time Series Plotting Utility keystroke functions.

Table 10-1 Summary of Time Series Plotting Utility toolbar.

General Functions	
	Exit ViewSlice.
	Printer setup options.
	Print to current printer the current view.
	Export the current view to a Windows enhanced metafile (EMF).
	Save settings
	Read settings
Toggles & Options	
	Toggle the grid.

	Show/Edit comment boxes
	Show/Hide markers
Plot Format and Labels	
	Edit text and fonts for the titles.
	Display options/settings.
	Access the X axis options form.
	Access the left Y axis options form.
	Access the right Y axis options form.
	Set all the axes formats to the one last edited.
	Toggle between Julian date/Gregorian date formats
Navigation Functions	
	Zoom in at fixed increments.
	Zoom out at fixed increments.
	Pan Left.
	Pan Right.
	Pan Up.
	Pan Down.
Utility Functions	
	Toggle the display of the coordinates.
	Import Series from a file
	Export Data to a file
	Animation tool. Output animations to the screen and/or AVI files.

10.1 Analysis and Statistics

For any defined time series the user can use the keystroke functions (F2) to toggle the display for box & whisker plots (Alt-1) or show the lines as *Cumulative Distribution Functions* (CDF's) (Alt-C). The user can also compute average values for each series for the data within the x axis minimum and maximum (Ctrl-A). These values are also copied to the clipboard to be easily copied into Excel© integrate the series (X*Y*User specified conversion factor).

Ctrl-I allows the user to integrate curves using a simple multiplication, where x is time, and y is Q in m^3/s as shown in Figure 10-3. However, as the time is in days the user must adjust the units between days and seconds when prompted.

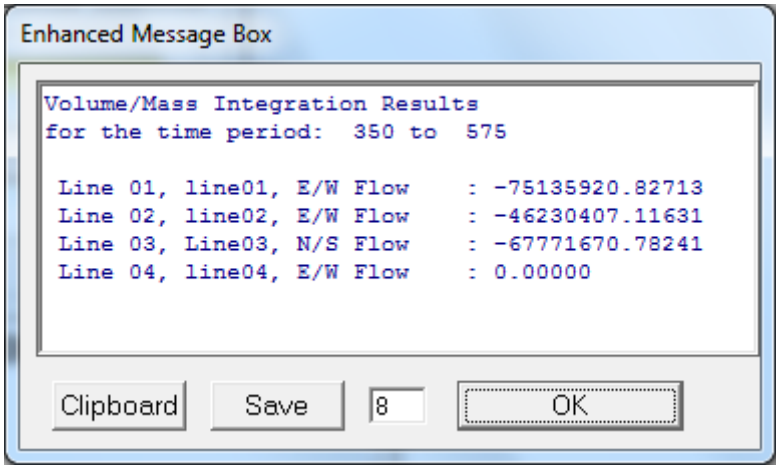


Figure 10-3 TSP: Integration of Time Series.

Line statistics by time block may also be displayed with the keystroke Alt-S. This is similar to a box and whisker plot data, but it generates a set of statistics. When the user wants to analyze each line by month they should select “1 M”. All the data is tab delimited and copied to the clipboard. The statistics generated are the average, standard deviation, minimum, maximum, lower quartile, median, upper quartile, standard error 95% and confidence interval. (See Figure 10-4)

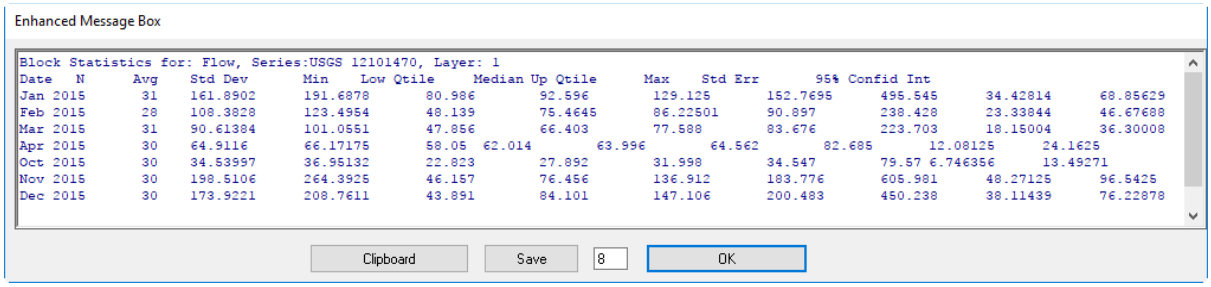


Figure 10-4 Time Block Statistics of Time Series.

10.2 Series Options

The user may select which parameters are to be displayed by accessing the *Line Options and Control* form by RMC'ing on the legend or using the toolbar. An example of the form is shown in Figure 10-5. The lines available for viewing are displayed on the left in the selection box. On the right of the form is the *Line Information* which for the line selected tells the user the number of points in the line as well as the maximum and minimum values of the line. The user may change the legend text for the selected line. In the *Line Formatting* frame the use may set the thickness, color and style with drop-down menus.

In the *General Options* frame the user can toggle the display of each line by checking *Show* for the series selected. An extra Y-axes can be added to the right of the graph by setting the *Right* tab in the "Y–axis" sub frame. Other features include changing lines to bar graphs with the "Show as Bars" check box.

In *Options* form provides options for the user to adjust the data for plotting purposes. The edited data does not update the EFDC data directly. If desired an edited series can be exported from the TSP and used to update the model input data.

In early versions of EE only 10 lines were allowed on a plot at any one time, but this limitation has now been removed. LMC on one of the series in the right-hand box will highlight that series. Holding down the control key allows the user to select multiple series. The user can then change line/point styles in the *Line Formatting Frame*. The *Legend* frame modifies what title will appear on the legend for the last series selected.

The series formatting and axes formatting can be saved for later retrieval using the save and load settings buttons on the toolbar.

A series can be adjusted by pressing the *Transform* button having selected the series. This displays a form with additive and multiplicative adjustment factors for both the X and Y components. Clicking OK to this form immediately applies the adjustments.

The form also provides a means to add or subtract one series from another and place the resultant series into a specified series # with the *Add/Subtract Series* button.

The user also has the options to delete (rather than just turn off or hide) lines from the *Time Series Grapher* tool. This is shown in Figure 10-6 where the user has selected two lines. Then by selecting the *Delete* button the lines are removed from the legend after the user has confirmed they want to remove these lines (Figure 10-7). The lines are also removed from the time series display and from the *Line Options and Controls* menu as shown in Figure 10-8.

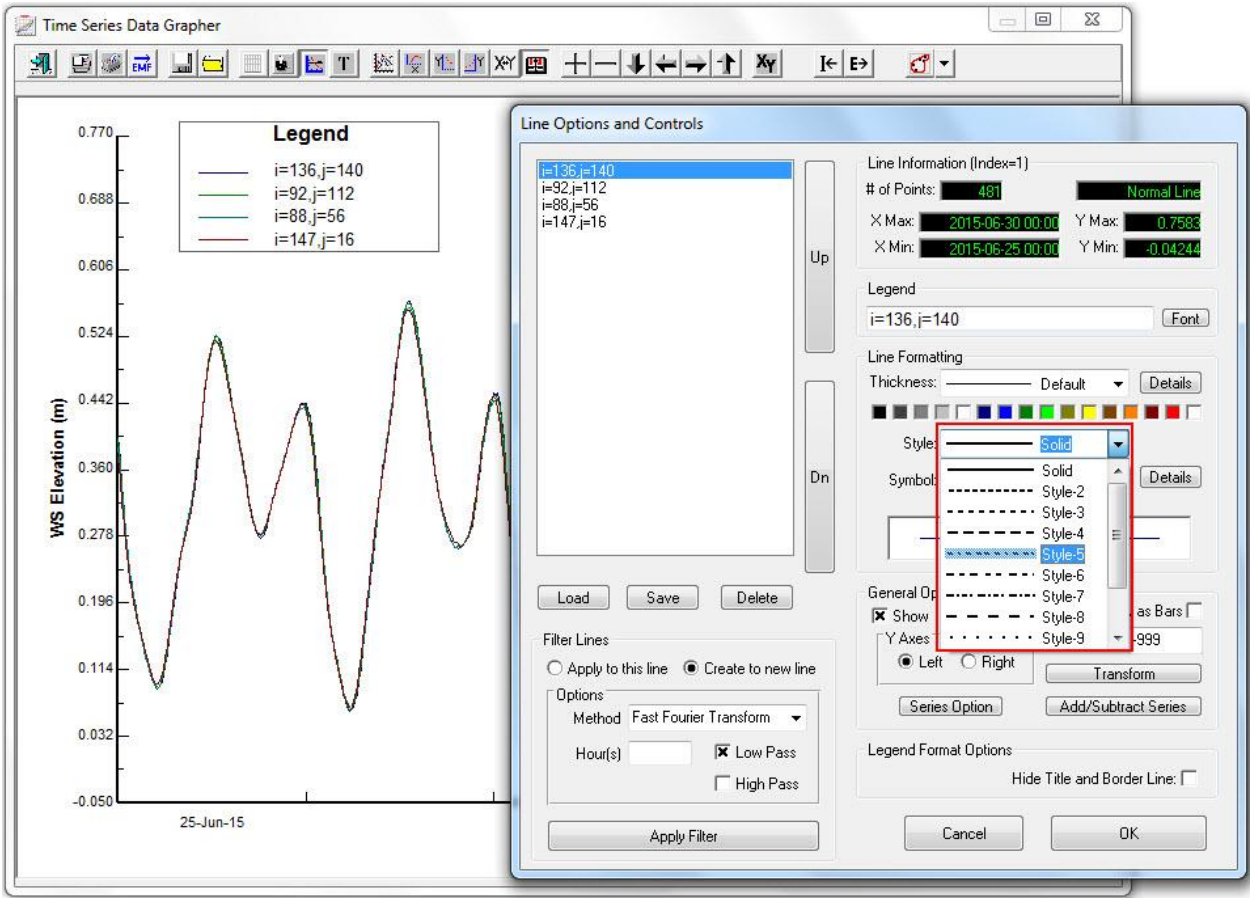


Figure 10-5 Time Series Grapher: Line Formatting

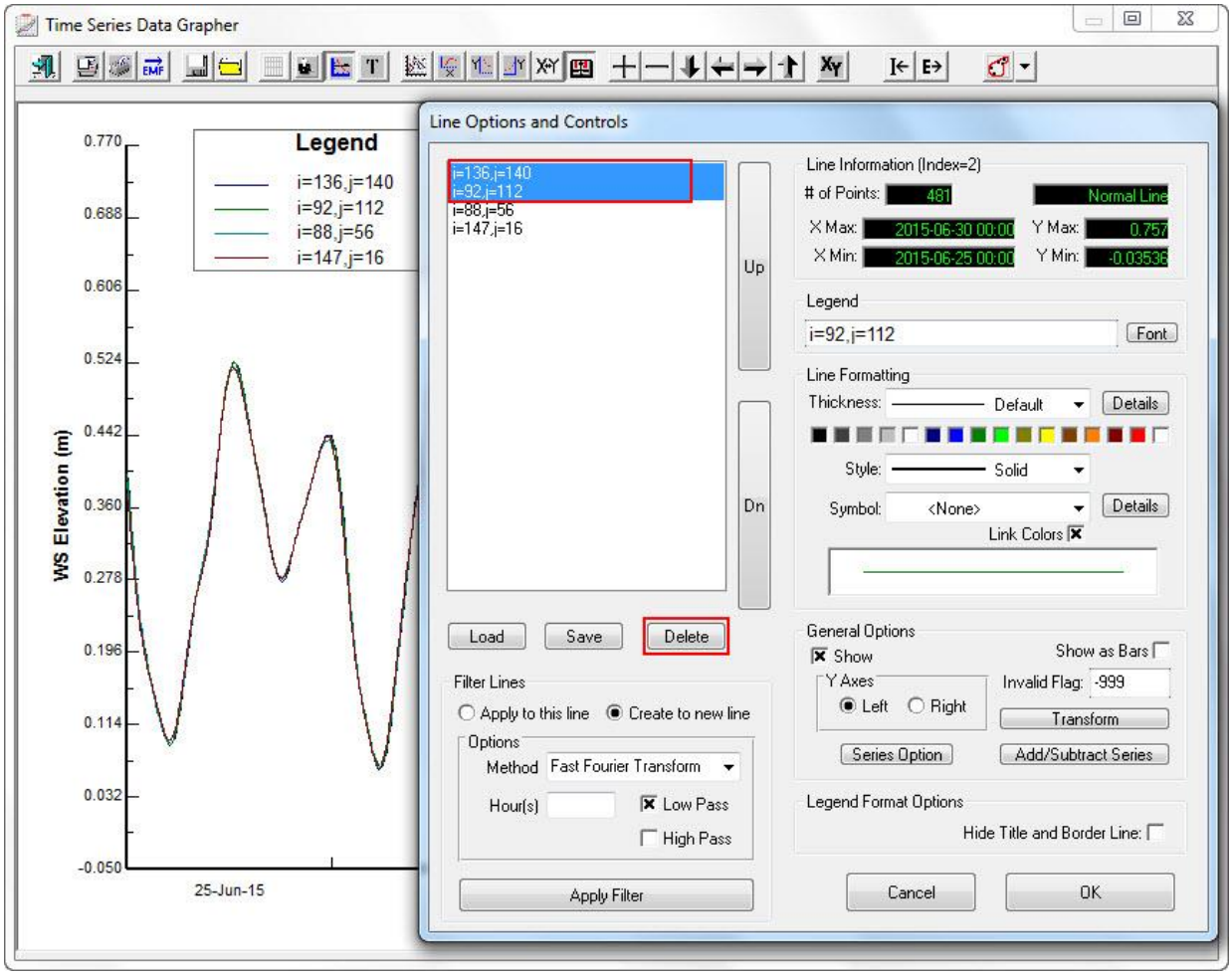


Figure 10-6 Time Series Grapher: Deleting lines.

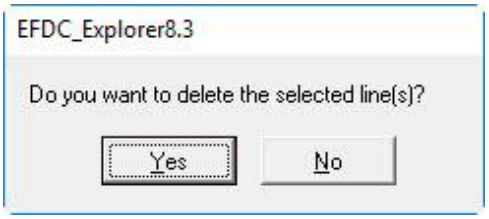


Figure 10-7 Confirmation to delete selected lines

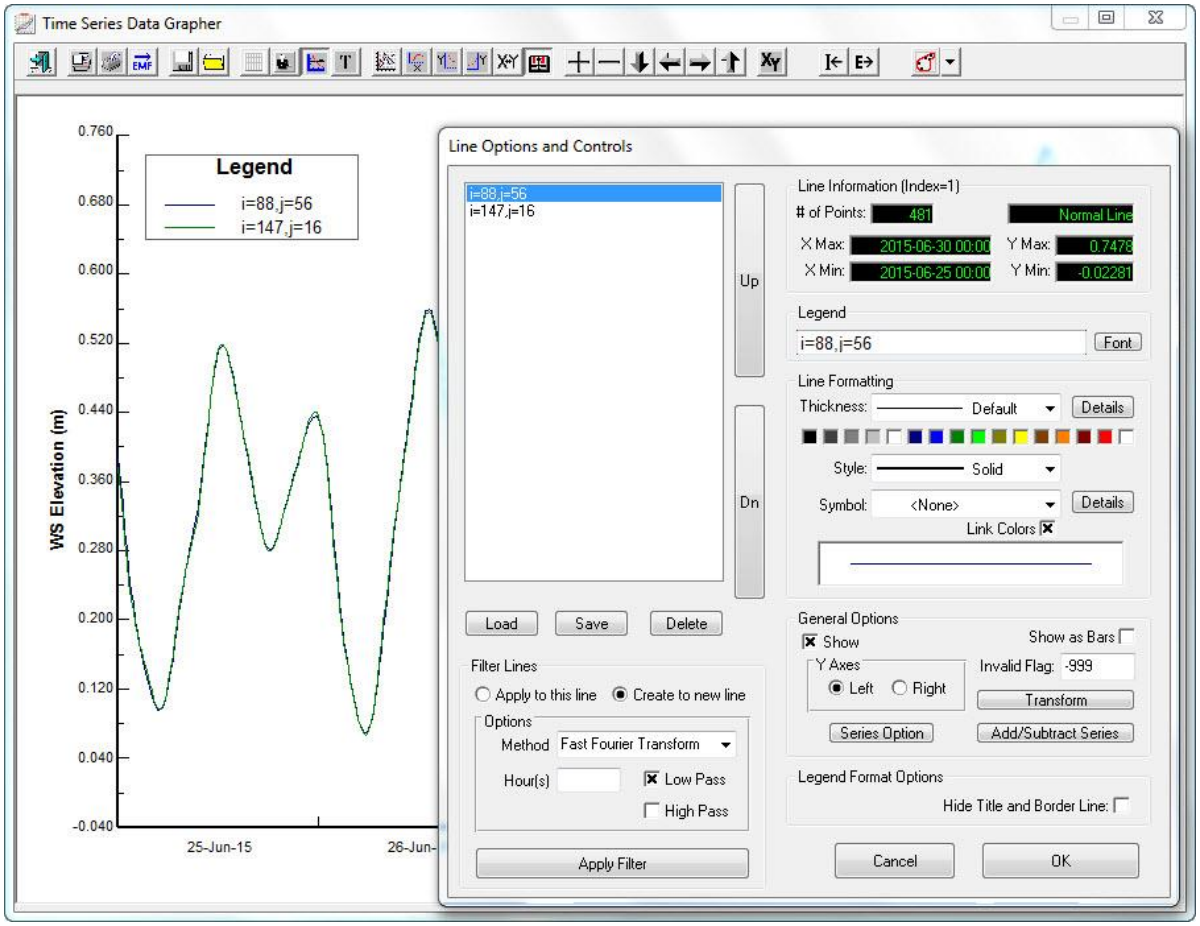


Figure 10-8 Time Series Grapher: Deleted lines.

10.2.1. Axis Labeling Options

The standard value Y-axis formatting form is accessed by RMC on the axis and is shown in Figure 10-9 and Figure 10-10. The use of these forms is intuitive. Axis labels may be entered in the *Main Label* box, however the formatting for the label is set in the *Formatting* tab. The user may specify the color and thickness of the Y axis line by pressing the *Line Properties* box in the *Axis Display Characteristics* frame.

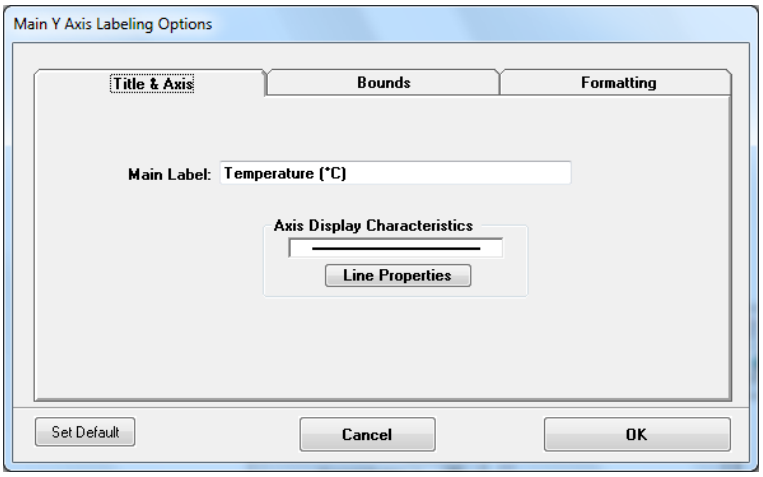


Figure 10-9 TSP Utility – Titles and Axis tab.

Scaling options include use of logarithmic axis, automatic scaling, or user specified max and min for both major and minor tick marks, as shown in Figure 10-10. Other options include reversing the order of the series (applicable only for the X axis) and showing logarithmic ticks.

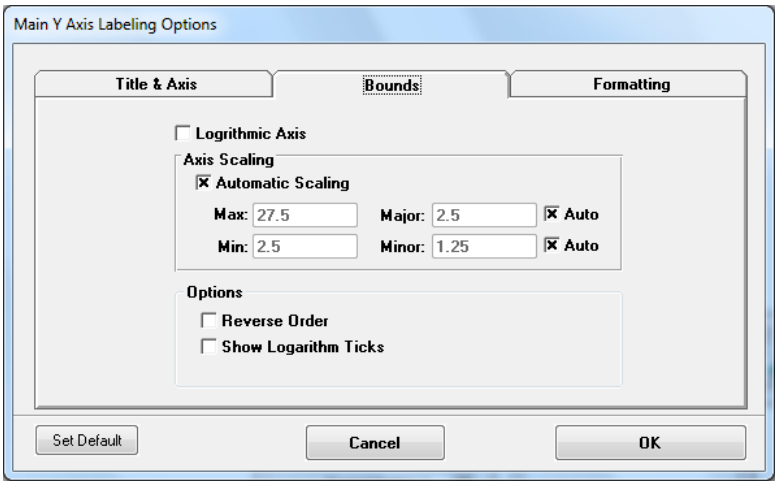


Figure 10-10 TSG Utility – Axis bounds options tab.

Formatting options available are shown in Figure 10-11. Here the user may set font types, colors, alignment and size for both the title and the labels. Grid line styles and colors for major and minor grid lines may also be set. The number of significant figures may be set in the *Label Precision* frame.

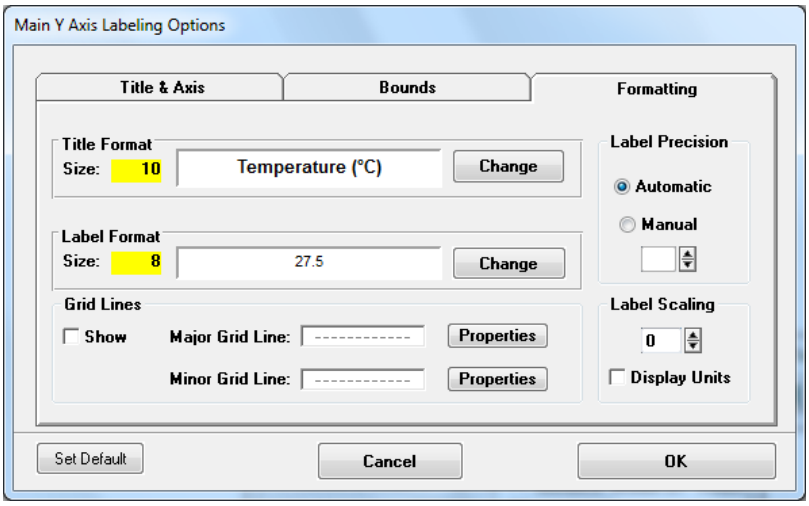


Figure 10-11 TSG Utility – Axis Formatting options tab.

The axis labeling method for the X axis is the same as that for the Y axis when using Julian dates. However, when a Gregorian date format is used on the X axis an alternative date axis form is provided as shown in Figure 10-12. Most of the options in this form are intuitive. However, one shortcut to selecting a calendar year is to enter the year number (e.g. 2003) in either the min or max field and then hit enter or tab and the dates will be set, along with appropriate tick marks.

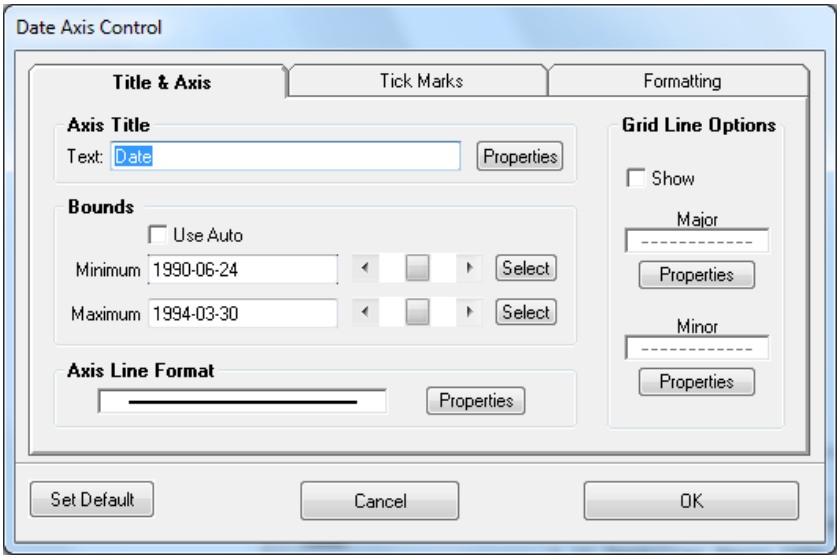


Figure 10-12 TSG Utility date axis options form.

10.2.1.1 *Subscript and Superscript in Graph Legends*

The *Time Series Grapher* and *EFDC Profile* tools have been updated to display user defined subscripts and superscripts in the legends. This is done using two underscore key strokes ("__") and a single caret ("^") for subscripts and superscripts respectively. An example of implementing subscripts and superscripts is shown in Figure 10-13. If several characters are part of a group they should be placed in curly brackets ("{ }"). For example to display: $x^2 + \log_2 x + \log_{10} x$, the user should write x^2 + log__2x + log__{10}x as the label as shown in Figure 10-13.

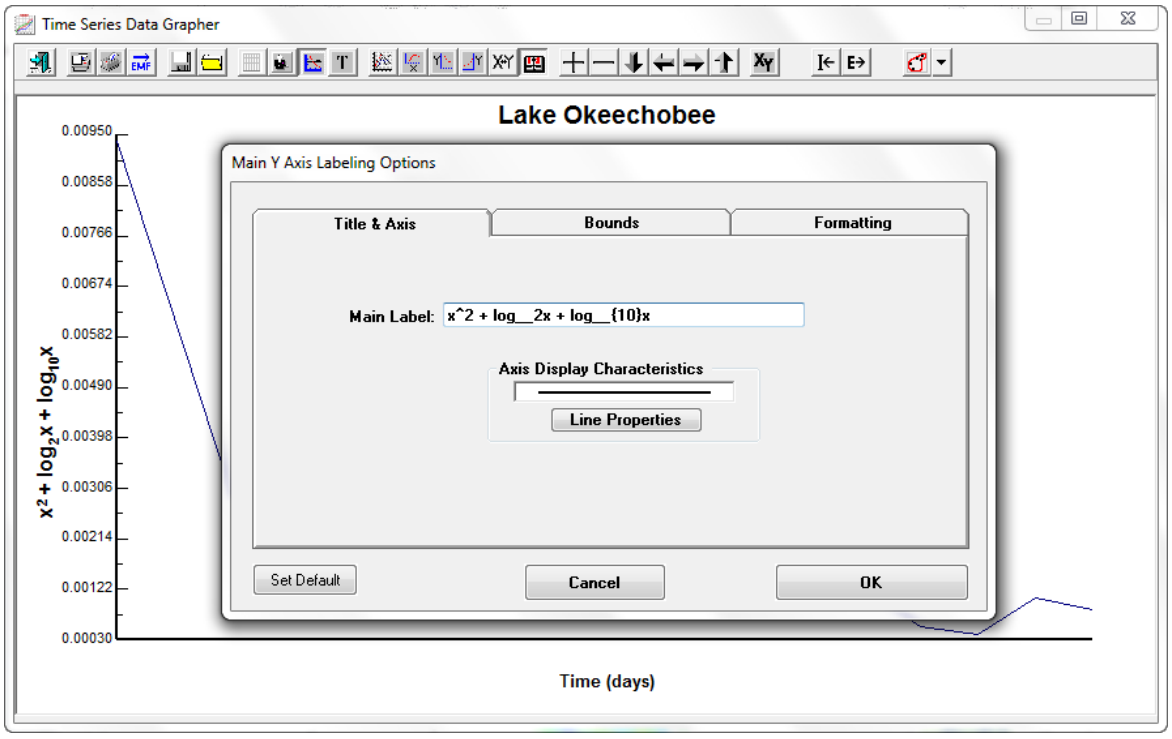


Figure 10-13 Time Series Grapher: adding subscripts and superscripts.

10.2.2. Zoom Pan Inset

A new option available within the TSP Utility (and also available in *Model Timing Options* of the *Longitudinal Profiler* tool) is that of displaying a *Zoom Inset Frame* so the user can zoom in to areas of interest in the time series and show the context of the zoomed area. This option is shown in Figure 10-14 below. The user may zoom in and then move about the plot with the zoom window in the inset frame (note that if the zoom or pan buttons are used the user must manually refresh the inset box by moving the zoom window).

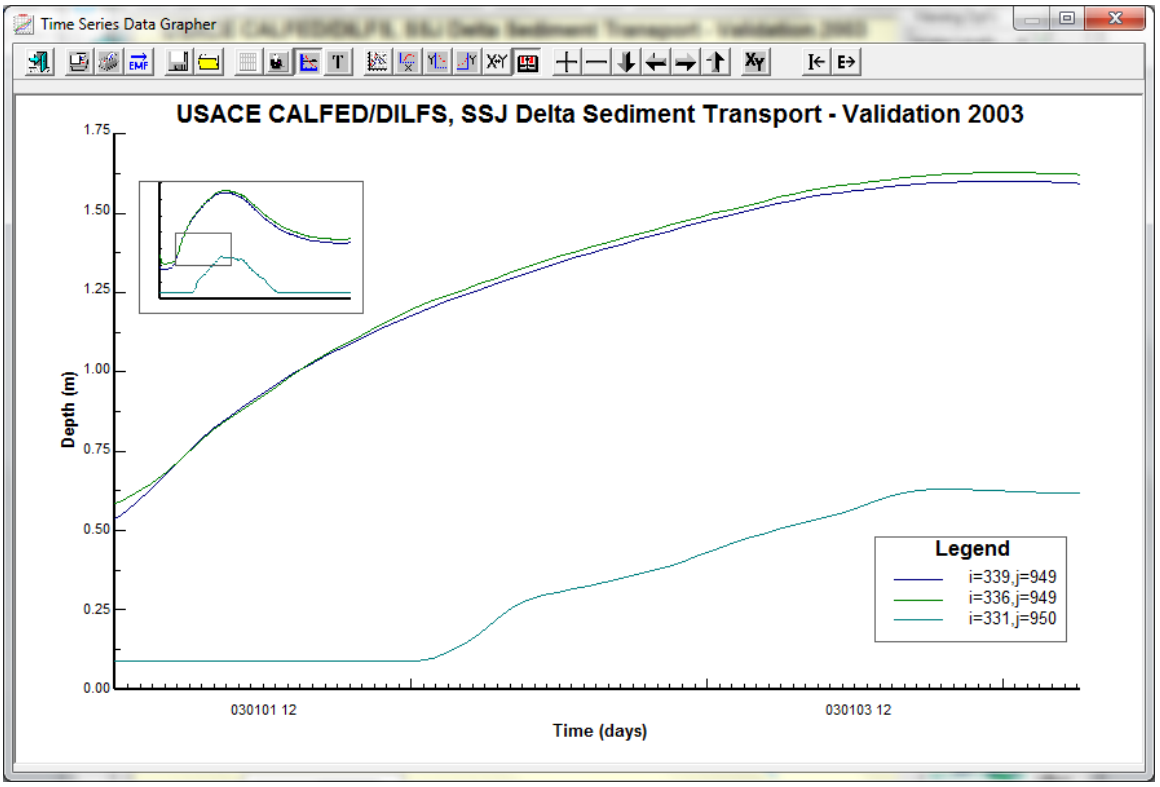


Figure 10-14 TSP Utility: Zoom Inset

The *Zoom Inset Options* are as shown in Figure 10-15. The user must switch on or off this setting from the *Control Line Plotting Options* which is accessed by RMC the legend and selecting “Series Options” then the *Zoom Inset Frame Options* and checking the *Show Zoom Inset Frame* box.

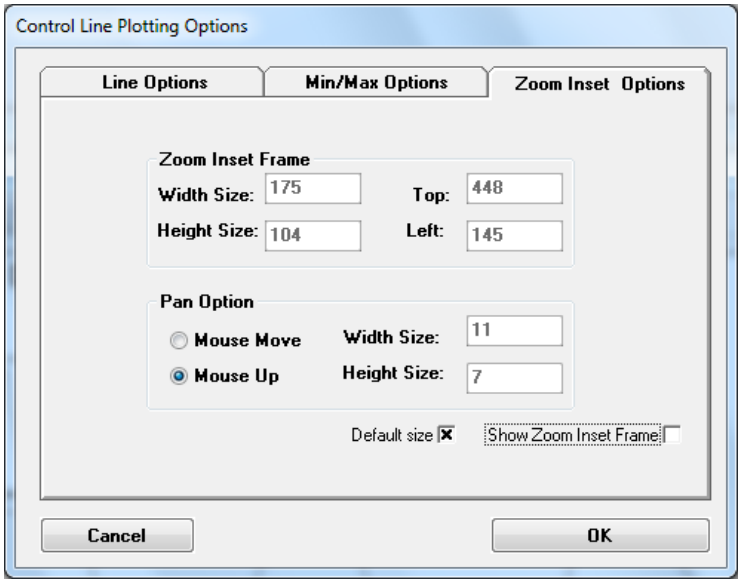


Figure 10-15 TSP Utility: Zoom Inset Options

10.3 Annotated Time Series

A new function in the *Time Series Plotter* is the ability to add a sub-frame with a geo-referenced map and annotations to illustrate the locations of time series stations. This is accessed with the "Show Plan Annotations" button in Figure 10-16. Options from this button include setting the *Plan Annotation Options* which has the same function as the *Annotation* tab in the *2D Display Options* in *ViewPlan*. Here the user may select posting or label files, as well as overlay files to be loaded and displayed in the sub-frame as shown in Figure 10-16 . Other options in the dropdown button include zoom to extents, load or hide a geo-referenced background, and hide or display the Annotation viewer. The *Save picture* option allows the user to save the sub-frame as a .jpg file.

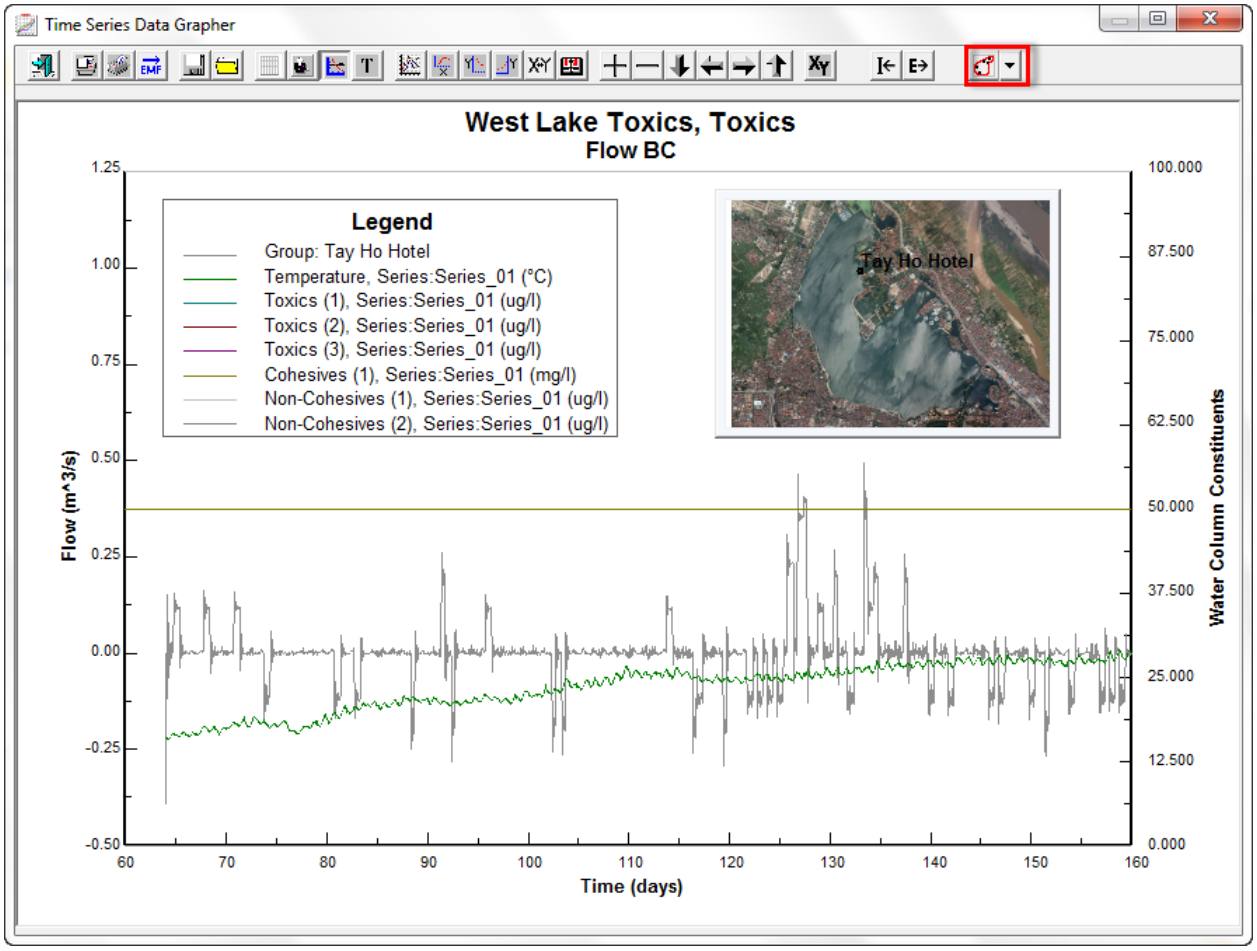
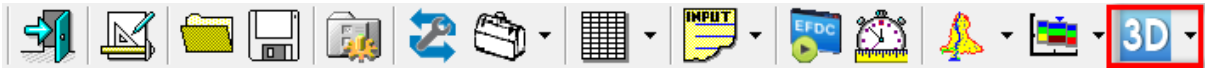


Figure 10-16 TSP: Show Plan Annotations Feature.

11 View3D



Release of EE7.1 provided the capability to view the model domain in three dimensions with the tool *View3D* as shown in Figure 11-1.

Much like the *ViewPlan* mode, the user may select which view option is desired in the *View Options* drop down. Each option provides a different interface in the right-hand column. When viewing water surface the user may toggle water surface and bottom elevation, as well as adjusting the degree of opacity of each for better visualization. The options currently available include: water surface, water column, velocities, bed shear, sediment bed, bed heat, oil spill.

The *Grid* frame is available in all options and provides the user with the ability to switch between views of the grid at the water surface, flat grid, grid at bottom, 3D grid and no grid. The opacity of the grid may also be adjusted by the viewer to suit the visualization effect required. A frame box may also be toggled on and off to help the viewer interpret the viewing angle.

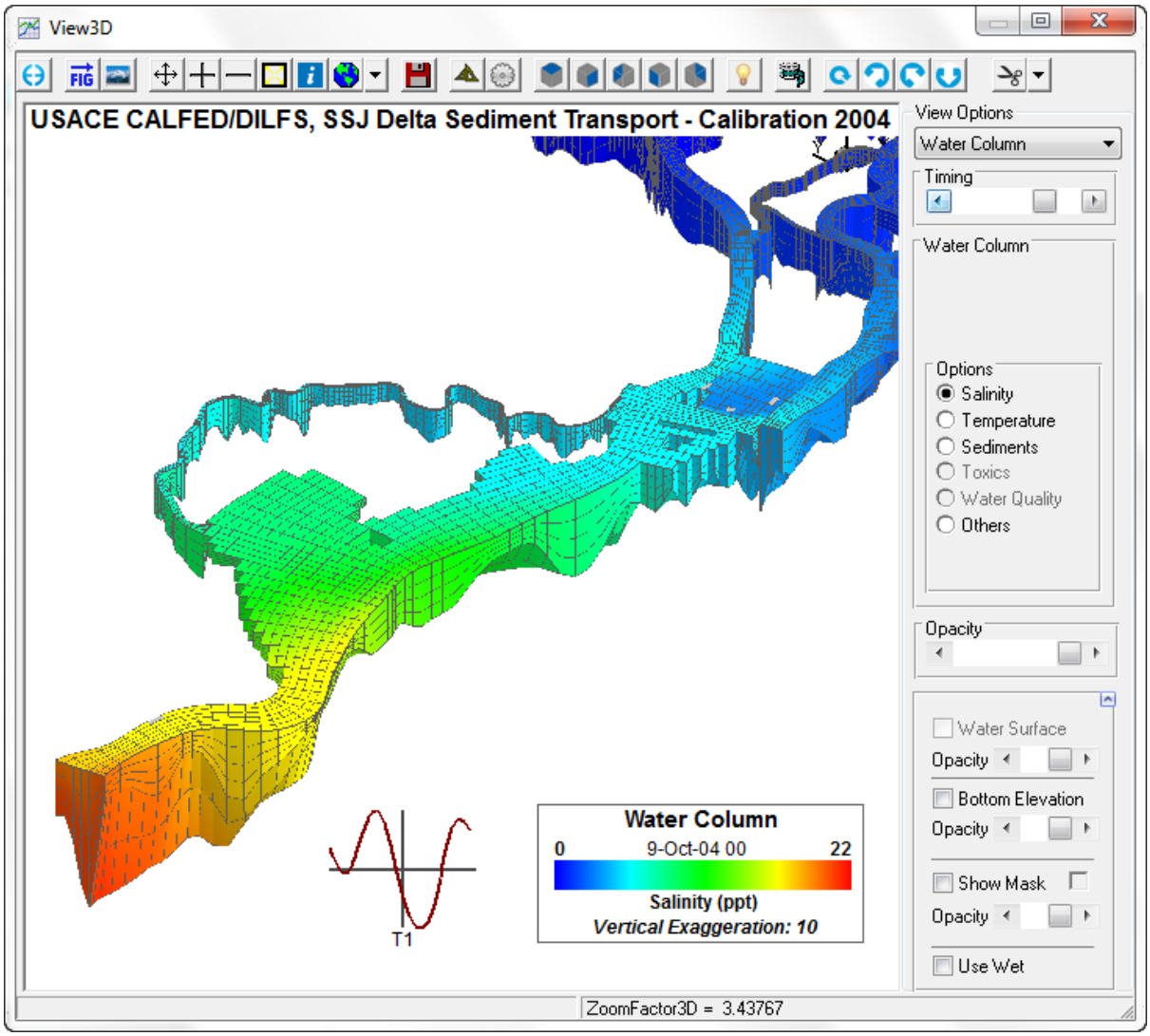


Figure 11-1 View 3D: Viewing Options

A number of further options are available in the toolbar at the top of the form. Those that differ from *ViewPlan* are described in Table 11-1 below.

Table 11-1 Summary of View3D toolbar.

General Functions	
	Exit to main screen.
	Save image to file
	Copy image
	Toggle Pan
	Zoom In
	Zoom Out
	Zoon Extents
	Information Inquire
	Background image display
	Display options for color ramp, vectors, grid lines, overlays, etc.
	Export the current view to a Windows enhanced metafile (EMF).
	Configuration of 3D rotation and camera
	View XY plane from above
	View from East to West
	View from West to East
	View from North to South
	View from South to North
	Adjust light settings and toggle light on and off
	Animation tool. Output animations to the screen and/or AVI files.
	Animate Z axis rotation
	Animate index I blanking
	Animate index J blanking
	Animate index K blanking
	Flight Path Utility
	Blanking Controls

Right clicking the legend also provides the user with a number of configuration options in the *3D View* tab of the *Plan View Display Options* form as shown in Figure 11-2. When using 3D the user has the ability to display two color ramps simultaneously, a primary and secondary color ramp. Within each of these options the user may define the color ramp cut offs or whether to auto-color based on the parameter min and max. The user may also simply display a gray color ramp or used a fixed color. The gradient fill option blends the color from one cell to another providing a more realistic appearance.

The *View Options* frame is used to set the elevation of the text and overlay in the 3D elevation viewing mode. The user should adjust this so that they overlay sits above the model outputs for the view required. The *Overlay 3D Elevation* sets the default elevation to set overlays when they do not have elevations associated with them. A "Z offset for WC bottom" sets the vertical offset when displaying the water column and bottom surface.

Due to horizontal scales typically orders of magnitude larger than the vertical scales, the user may adjust the scale factor with the *Vertical Scale Magnification* option.

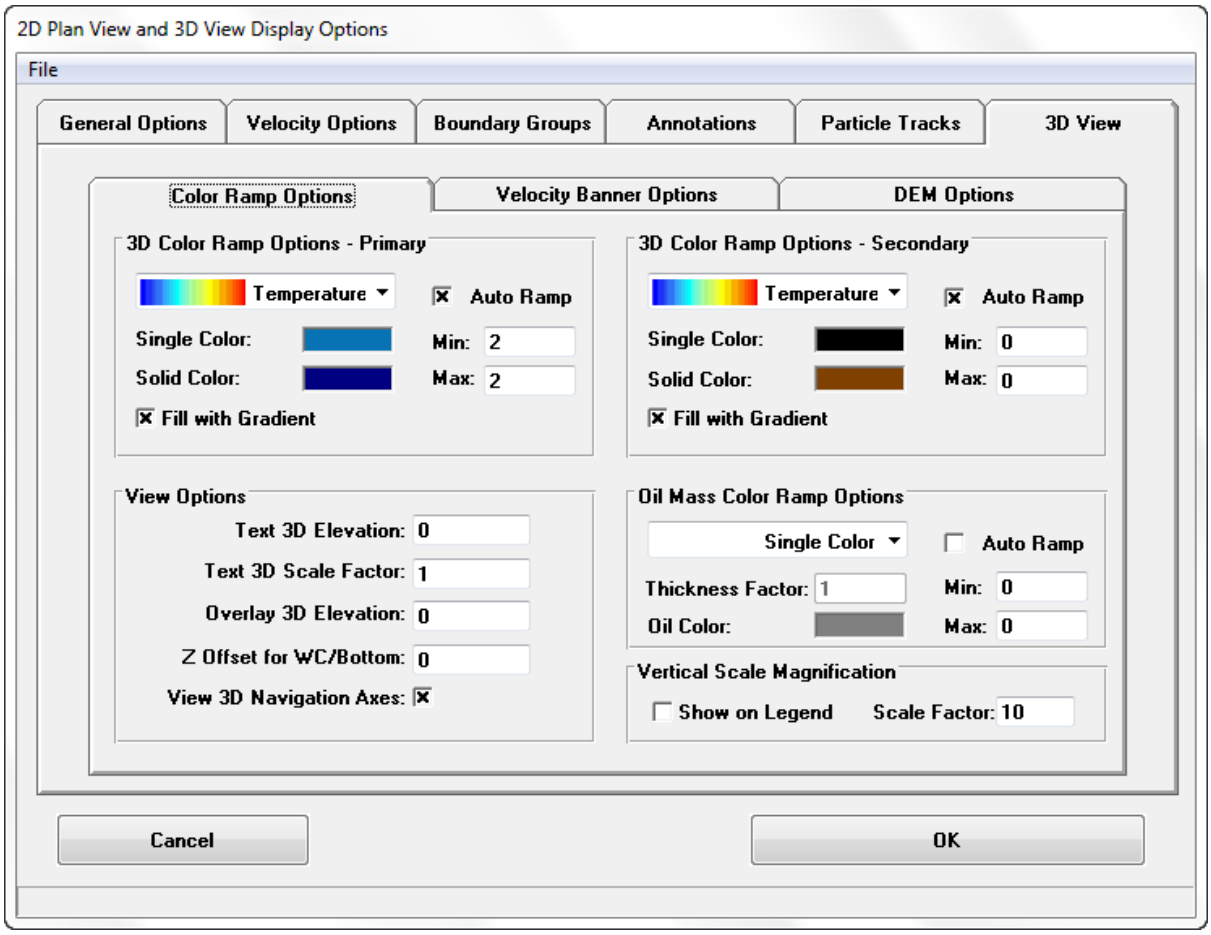


Figure 11-2 View 3D: Display Options

The *Oil Mass Color Ramp Options* also provides the color ramp, single color and auto ramp functions. A *Thickness Factor* may be applied to increase the vertical exaggeration of the oil spill for visualization purposes.

11.1.1. Velocity Banners

The *Velocity Banner* feature is accessed from the *Viewing Options* in *View3D*. Effectively this displays the 3-D velocities in a profile for each cell, something like a “wind sock”, which may be colored by depth or magnitude. A pole runs between the bottom and water surface from which the velocity profiles face in different directions as shown in Figure 11-3.

The user may fix the color of the banner using the *Display Options* | *3D View tab* | *3D Color Ramp Options*.

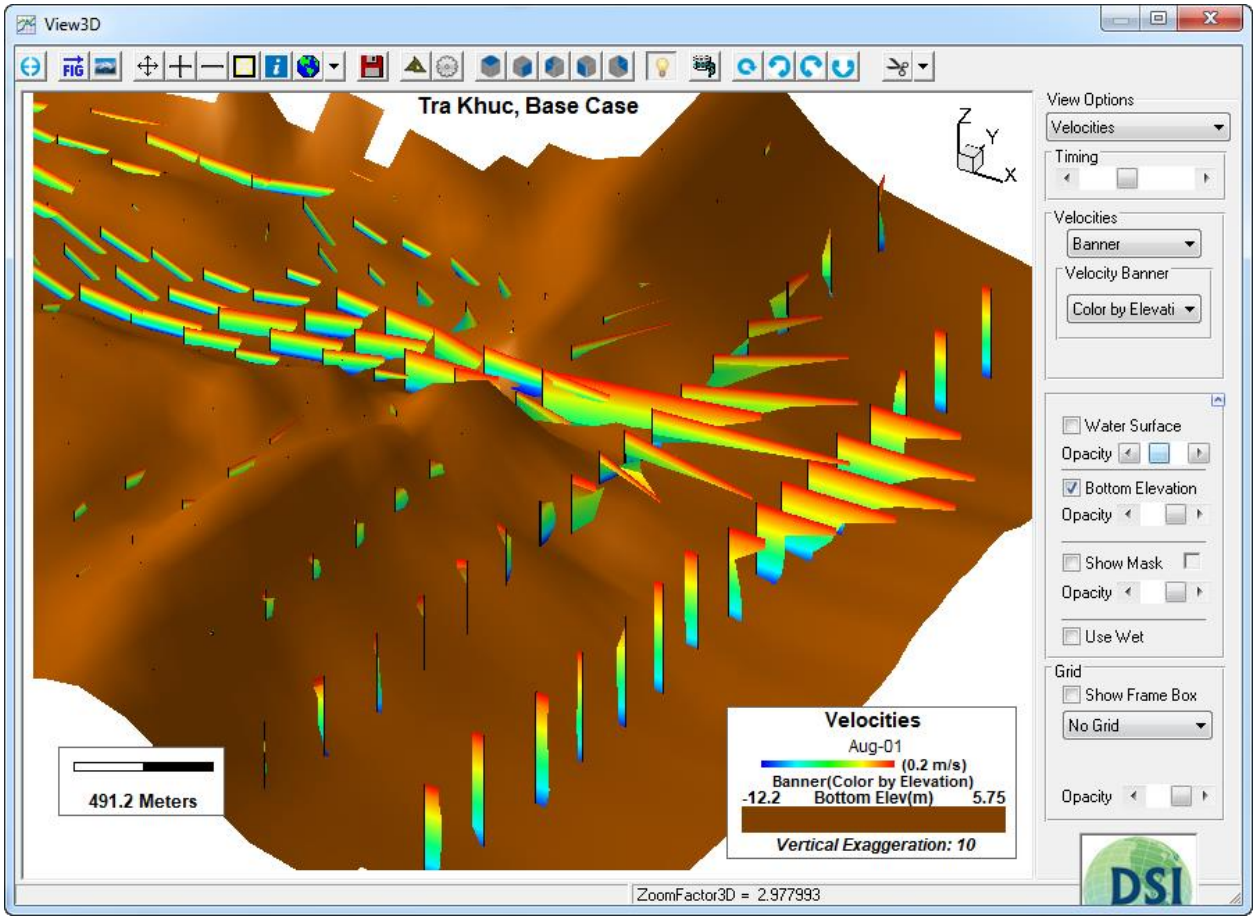


Figure 11-3 View3D: Velocity Profiles.

The user should navigate to *Display Options* | *3D View* | *Velocity Banner Options* tab to set where the profiles begin and set the frequency that they are displayed on the grid. The user may also set the color of the pole and pole thickness as shown in Figure 11-4.

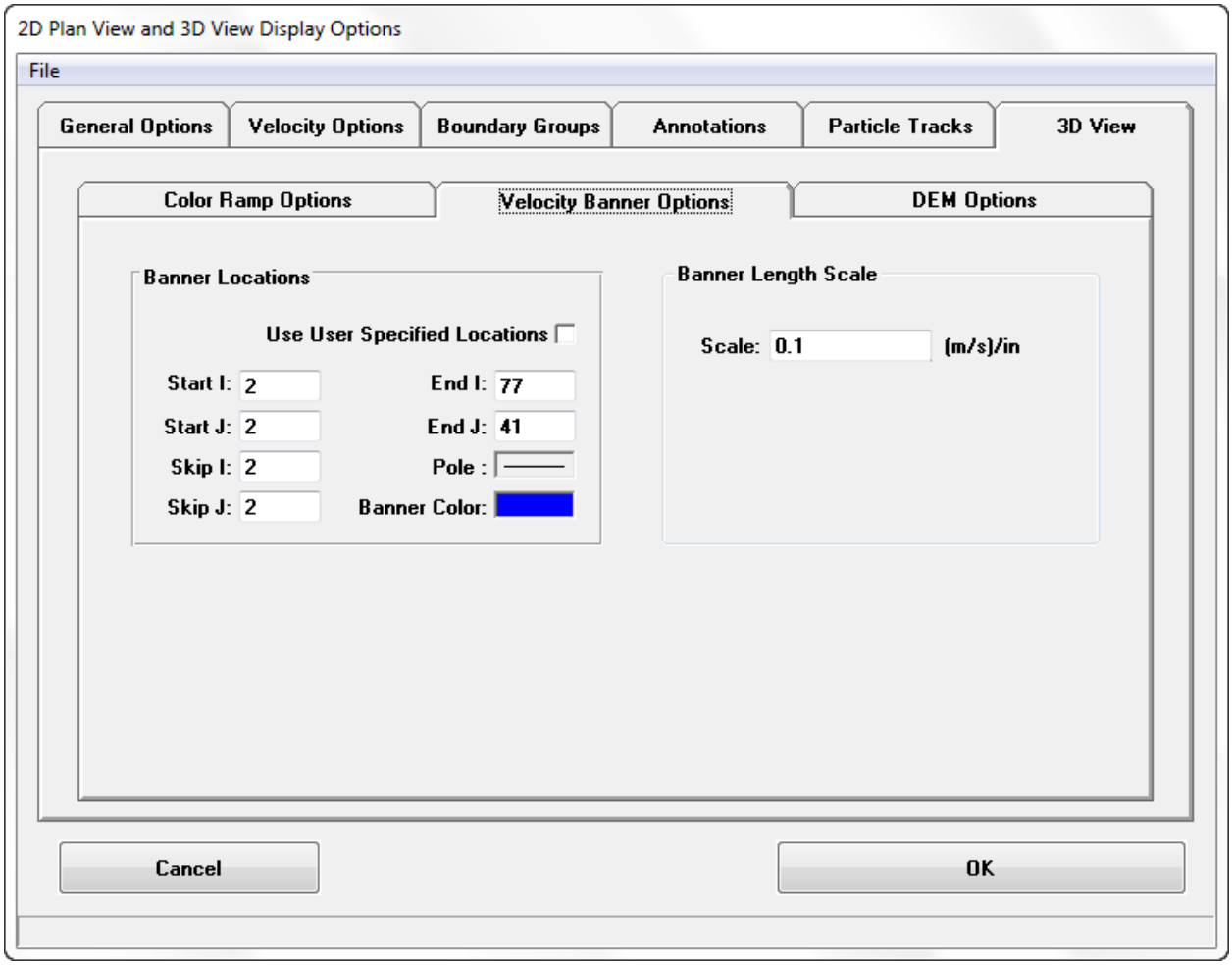


Figure 11-4 View3D: Velocity Profiles display options.

As in the *ViewPlan* for 2D plan view, the *Velocity Profile* setting allows the user to define where to start and end the profiles as well as set the skip function to skip a certain number of cells. The user may also use the *Velocity Labeling Locations File* which will set the points for display. Velocities will only display at the locations set in the file.

11.1.2. Flight Path Animation

The flight path animation tool allows the user to define and edit a flat path and then create an automated animated sequence through the model domain in *View3D*. To create a flight path select *Show/Hide Flight Path Tools* from the dropdown menu as shown in Figure 11-5. Using the controls now displayed the user may define, save, load and edit a flight path.

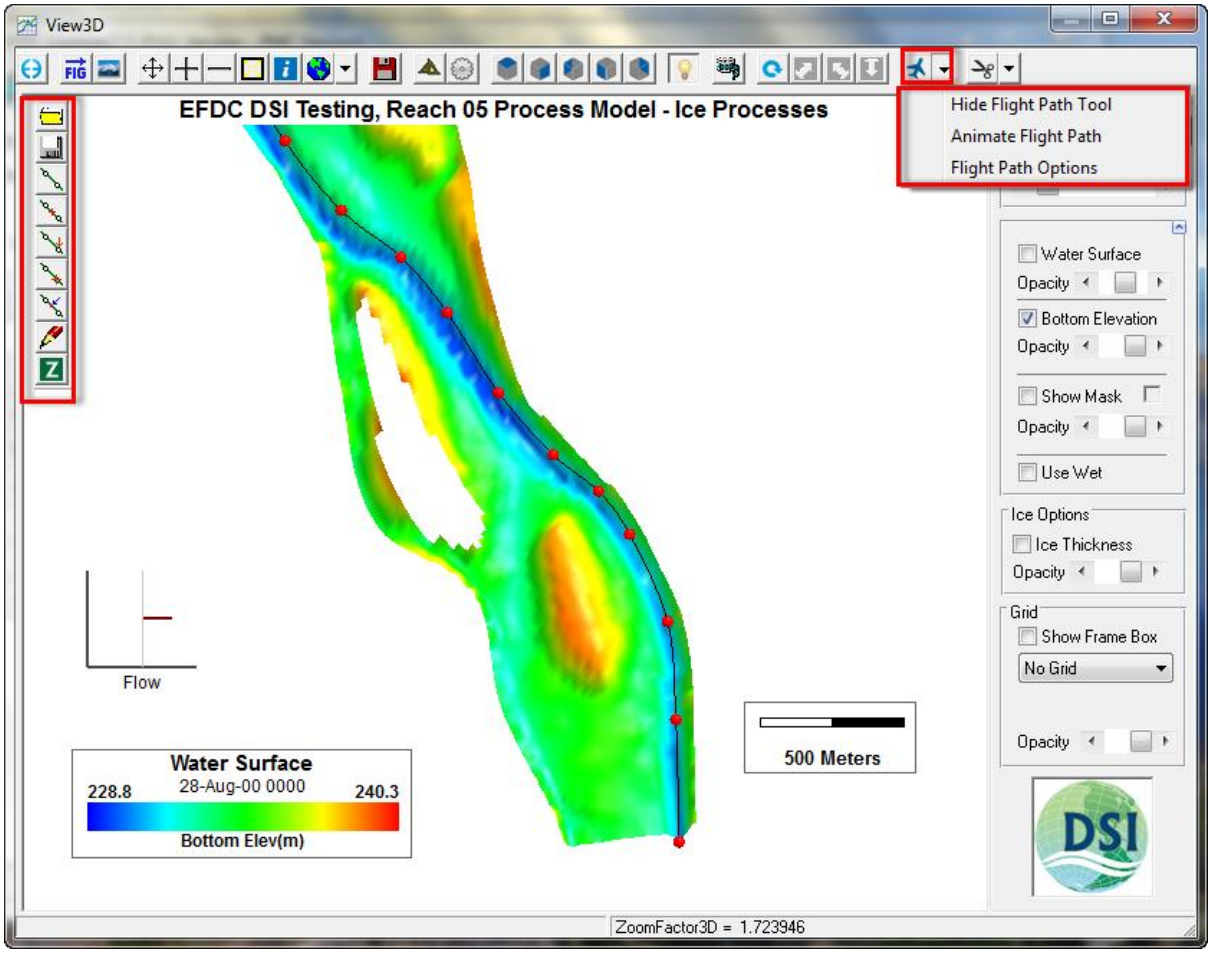


Figure 11-5 View3D: Flight Path Animation.

Controls for the flight path are used to draw a spline that can be either a straight line or a curved line and must contain at least two points. In the main toolbar there are a number of icons that may be used to define the flight path that are explained in Table 11-2.

Table 11-2 Flight Path Polyline Tool Buttons.

	Open an existing flight path polyline.
	Save a flight path polyline. It is necessary to select this icon for the polyline to be saved.
	Define a new flight path. LMC in the workspace creates a point. Moving the mouse and LMC in another location creates another point with a line joining the two points. RMC to end drawing the polyline.
	Delete previously created line. When this icon is selected, clicking on a line will delete it.
	Insert points on an already created line. Once a point is inserted it can be moved to reshape the line.
	Move points on a polyline.
	Delete points on a polyline.
	Edit a point. Edit the x, y, z location of a point as well as the roll.
	Assign a constant elevation to the flight path. Sets all the vertical elevations of the flight path at one time as opposed to editing it point by point.

The process for creating a flight path are as follows:

1. Select the flight path tool drop down *Show Flight Path Tool*
2. EE will reset to plan view to draw the flight path in the horizontal plan view
3. LMC on the first point of the path and LMC for each subsequent point on the path
4. RMC to end the line
5. Move or delete points as required using the *Move* and *Delete* buttons. Note that it is not possible to pan in this mode except by using the arrow keys
6. Select the *Z* button to set the vertical position of the flight path
7. Use the *Edit Point* button to further adjust the vertical position or roll of any point
8. Rotate the model to ensure the flight path is vertically and horizontally correct
9. Save the flight path

A number of options are available to user to define the flight path. These options are available from the *Animate Flight Path* from the dropdown on the toolbar and shown in Figure 11-6. The user can first decide whether or not to display the flight path polygon using the *Show* check

box. It is also possible to adjust the animation steps to provide a smoother visualization. Camera height and angle may also be set. *FovY* is field-of-view in the y dimension i.e. vertical angle, which may be set to maximize the image size on the screen.

The user may select various options for the path color when editing the lines as well as colors displayed when a line is selected or dragged.

The user can select the spline type to be used for the flight path. The default is the *Catmull–Rom spline* in which the original set of points make up the control points for the spline curve. "B spline" may also be selected, in which the curve does not remain on the original control points. The user may also select the spline checkbox to switch on or off the use of splines.

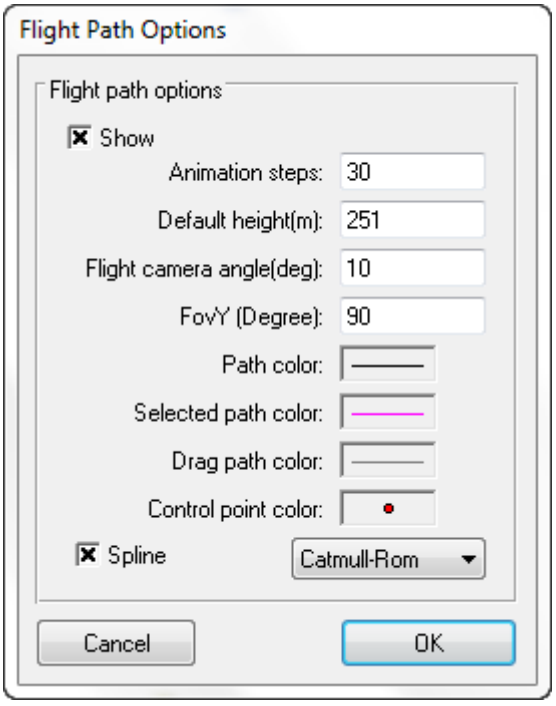


Figure 11-6 View3D: Flight Path Options

To animate a flight path load the existing flight path from the *Show Flight Path* tool. Then select the *Animate Flight Path* from the dropdown on the toolbar. It is possible to animate to an AVI or just animate to screen. For either option the parameters show in Figure 11-7 should be set.

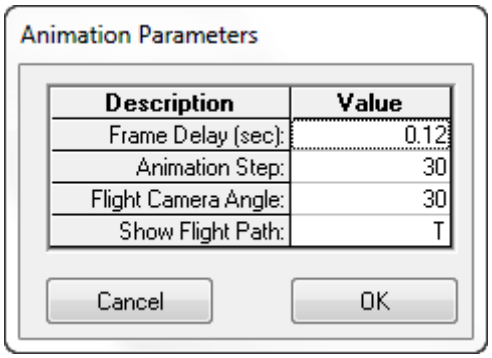


Figure 11-7 View3D: Flight Path Animation Parameters.

Selecting OK will animate the flight path as shown in Figure 11-8

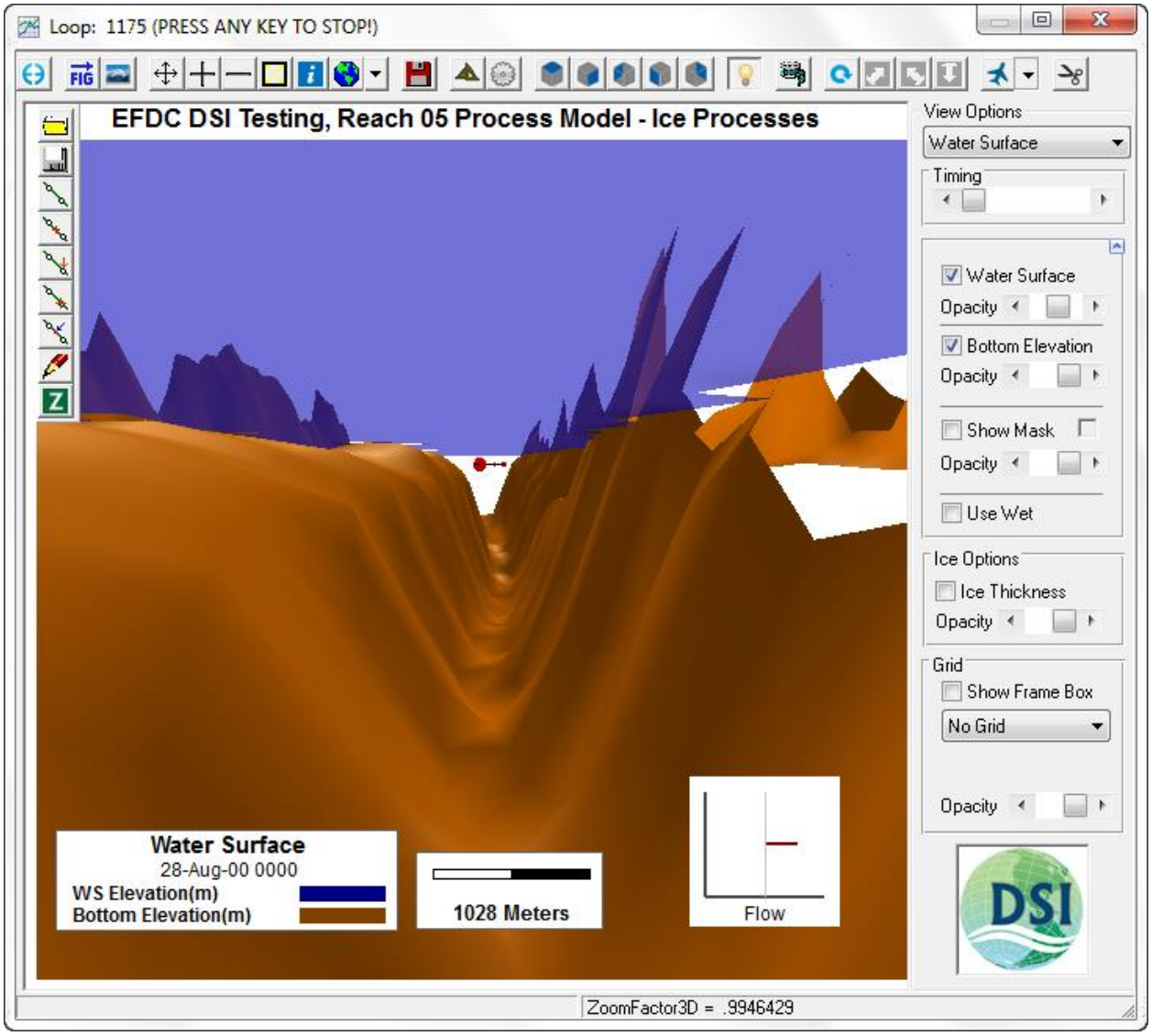


Figure 11-8 View3D: Flight Path Animation.

11.1.3. 3D Blanking and Clipping

The user may view certain cells along the I, J index by using the blanking option in *View3D*. This is accessed by the *Show Blanking* check box on *Water Column* frame. This allows the user to blank between certain I and J indices to see inside the model domain as shown in Figure 11-9. In this case the model is only being viewed between 2 and 73 on the I index, and between 2 and 23 on the J index.

Clipping may also be applied in much the same way with the *By C* option. This value is the same as that set for the color ramp which will only display the parameter selected between certain values. However, if the user selected the *Clipped by Value* checkbox then those cells will be blanked or clipped from the display. This function now clips the cell corners whereas previously it clipped the whole cell based on the cell centroid value.

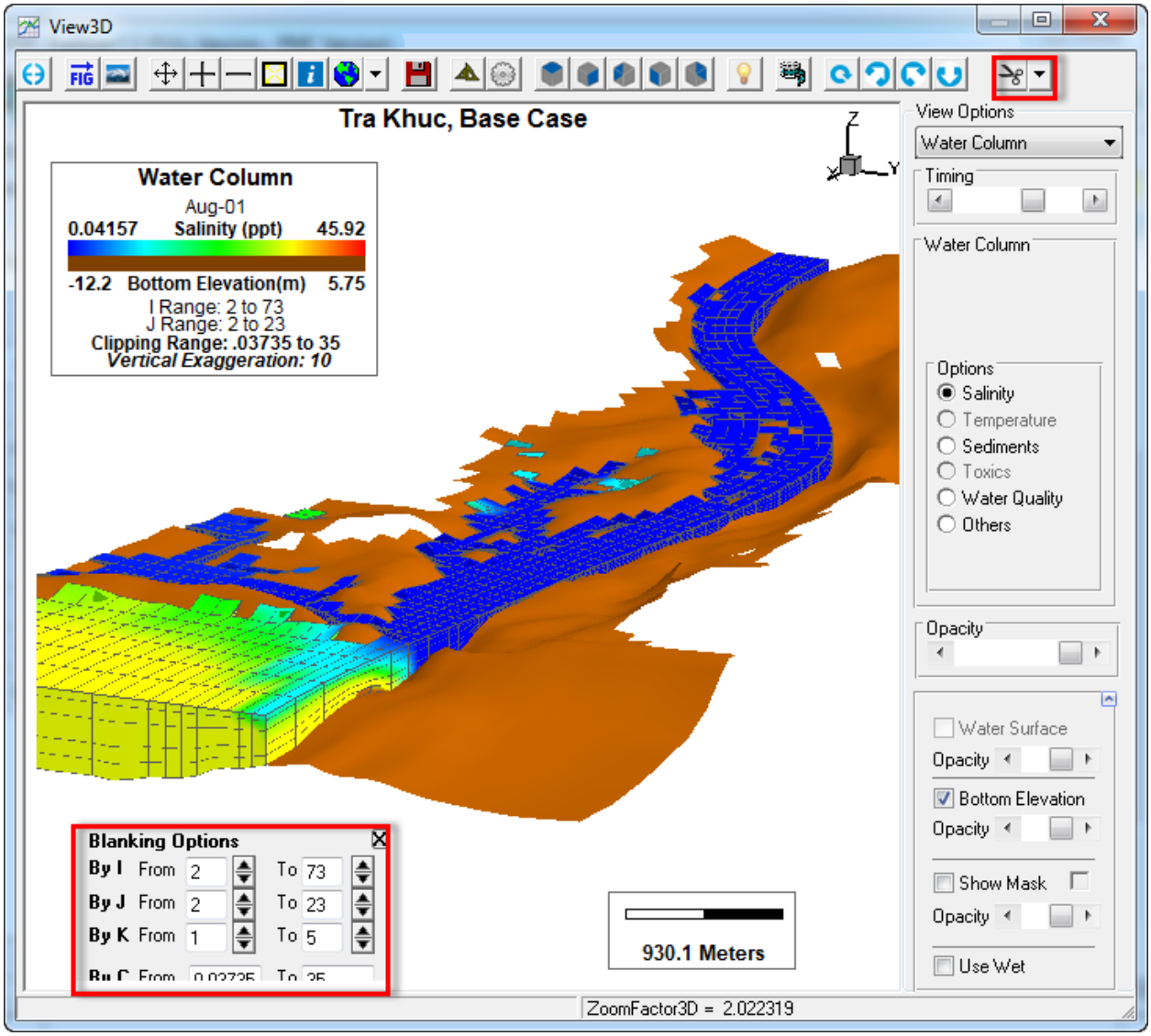


Figure 11-9 View3D: Blanking and Clipping.

11.1.4. Background Images in 3D

Several options are available to provide backgrounds for models in EE. If the user has a 3D surface of the area surrounding the model domain, this may be loaded to provide a land surface background. Alternatively, the user may simply use an automatically generated flat terrain as described below.

11.1.4.1 *Loading Digital Elevation Models*

RMC the *Legend* or select the *Display Options* button to show the *Display Options* and select *3D View* | *DEM Options* as shown in Figure 11-10. If the user has a polyline P2D file or any comma or tab delimited X, Y, Z file then they should first create an overlay file from this with the *Interpolate DEM from Polyline* button. This interpolation process can be quite slow, but once the DEM is generated it loads quickly. The native DEM file is *.TB2 format and should be loaded with *Browse* button. The user must also select *View* from the DEM Option frame. Clicking OK will then display the background file.

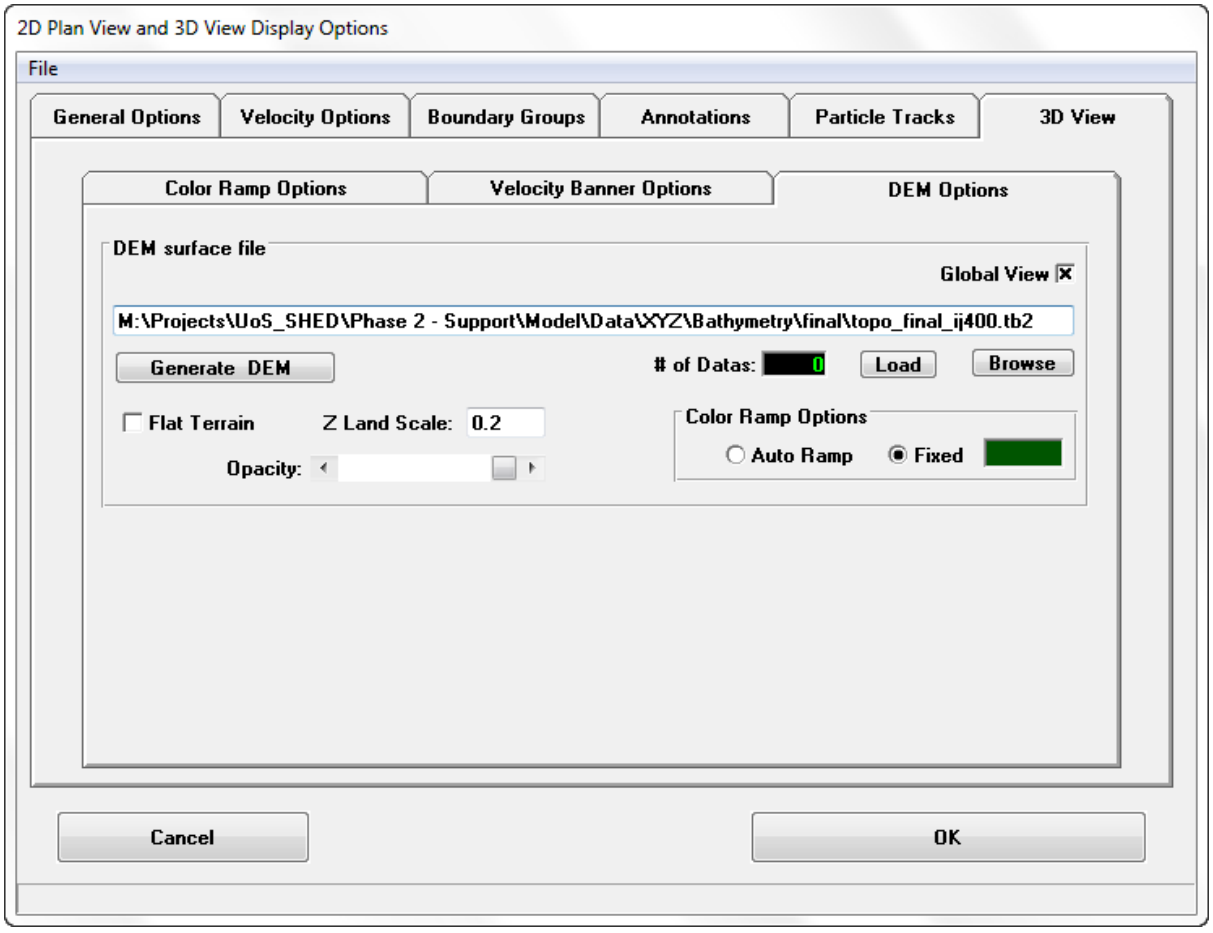


Figure 11-10 View3D: DEM Options.

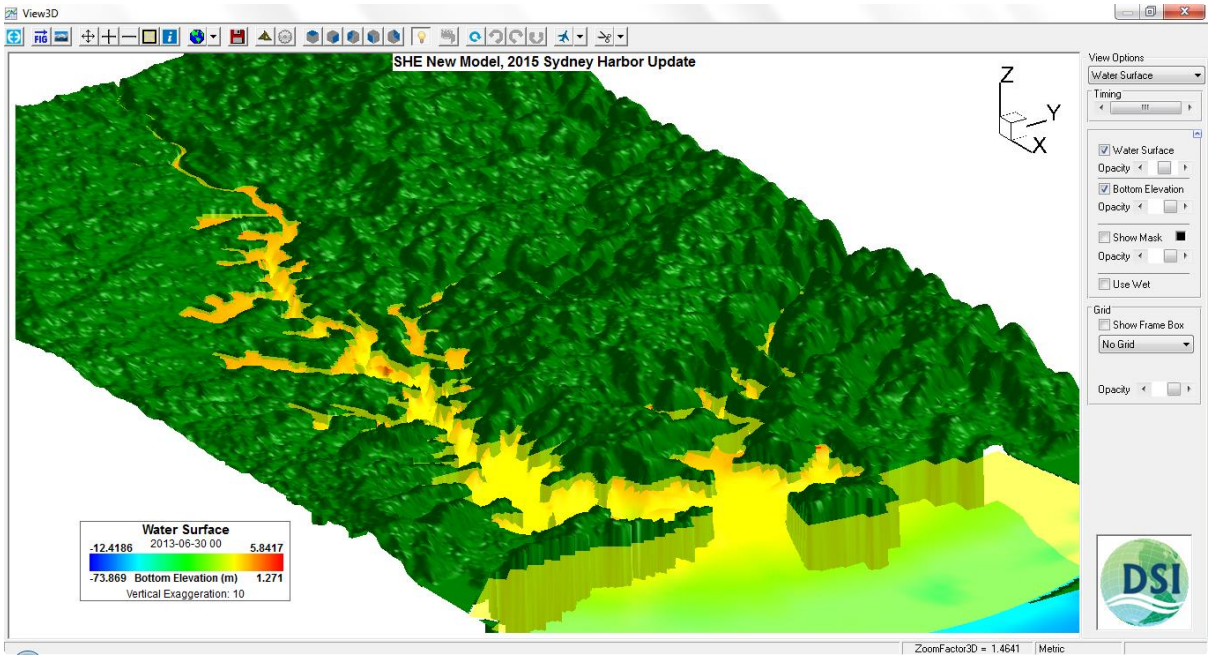


Figure 11-11 View3D: DEM display.

11.1.4.2 *Flat Terrains*

If the user does not have a XYZ file they can use the flat terrain option and change the fixed color.

Viewing options are managed in the *DEM Options* frame shown in Figure 11-10. Select the *View* checkbox to display the DEM. When *OK* pressed the user is prompted for the *Generate Flat Surface Options* as shown in Figure 11-12. Here the resolution and and dimensions of the flat terrain may be set. These may need to be adjusted when working with a coastal domain that has an open boundary.

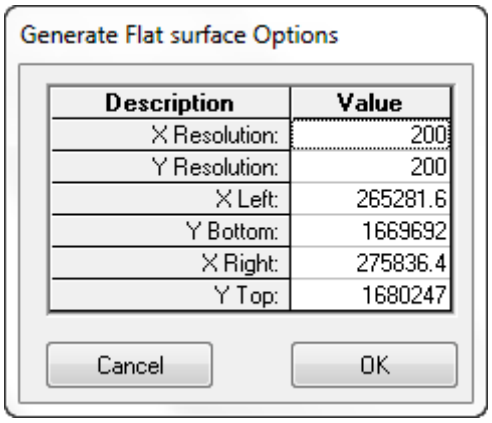


Figure 11-12 View3D: DEM Terrain – Generate Flat Surface Options.

The flat terrain generated is shown in Figure 11-13.

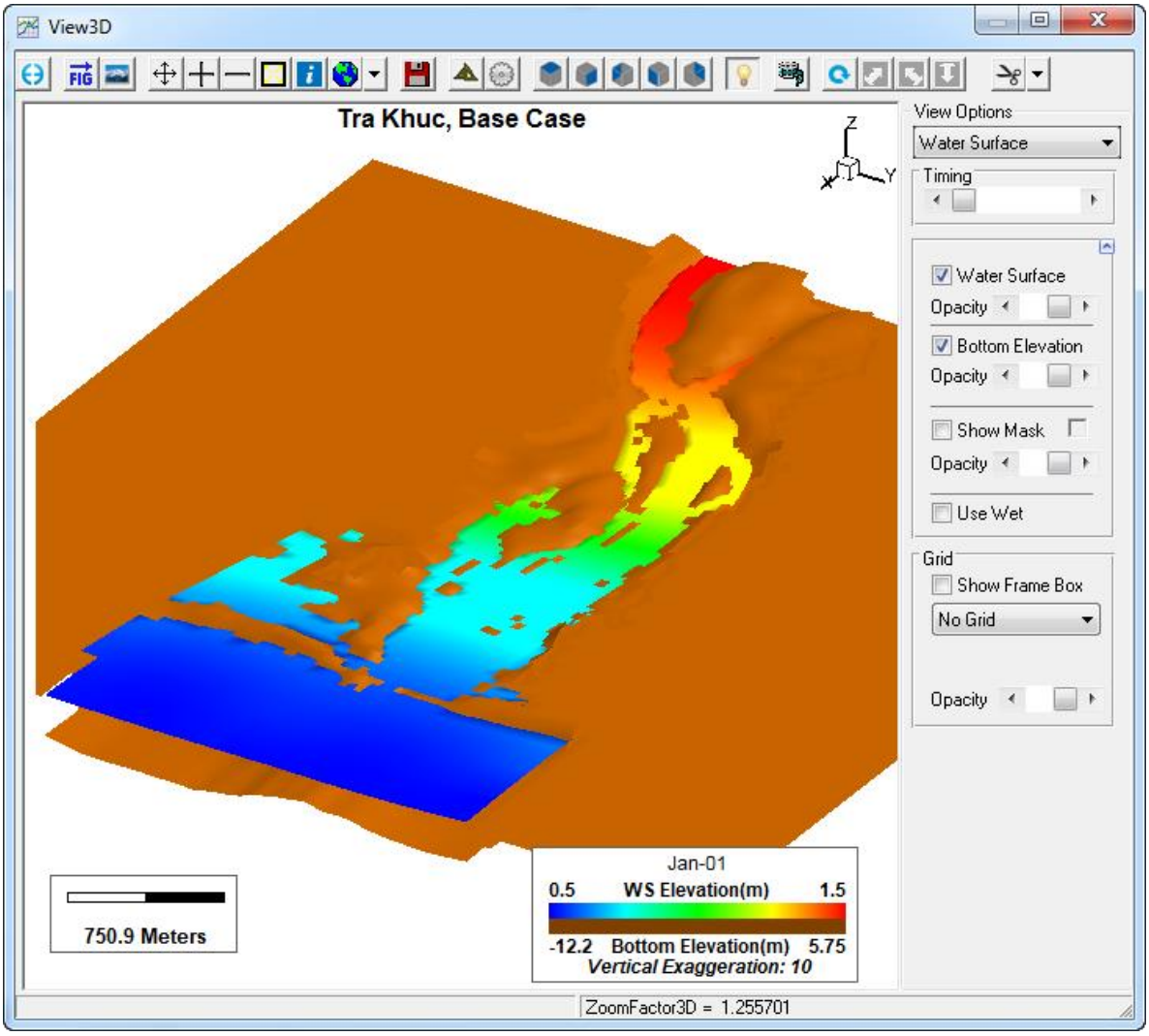


Figure 11-13 View3D: DEM Terrain.

Scale may be adjusted as well as the background color. In order to move the level of the flat terrain up or down to an appropriate level, the user may set the *Land Elevation.* In this case it is set to 1, which will move the land up be one meter.

11.1.5. Sediment Bed Viewing

The sediment bed in the 3D Viewer can be viewed in EE7.2 and later releases. This displays the sediment volume similar to the way the water column is viewed. The user may select from the same display options in terms of top layer, thickness, sediment mass, sediment fraction, porosity, d50 and delta (change in thickness) as in *ViewPlan* as shown in Figure 11-14.

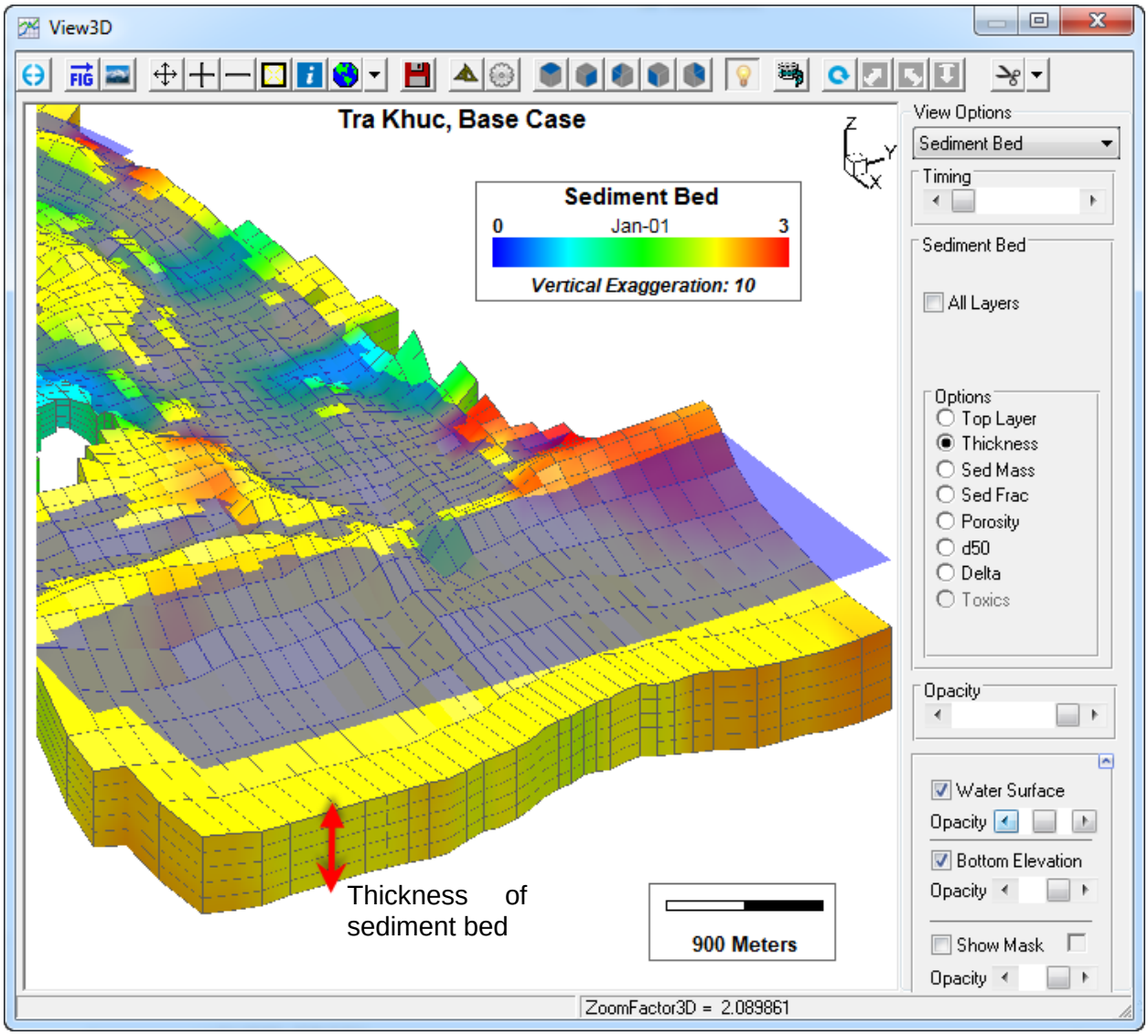


Figure 11-14 View3D: Sediment Bed View.

11.1.6. Mouse Inquire Information

Use mouse to click to see the cell information for each layer is now available much as it is in the *ViewPlan* mode. This function is accessed with the enquire button as shown in Figure 11-15.

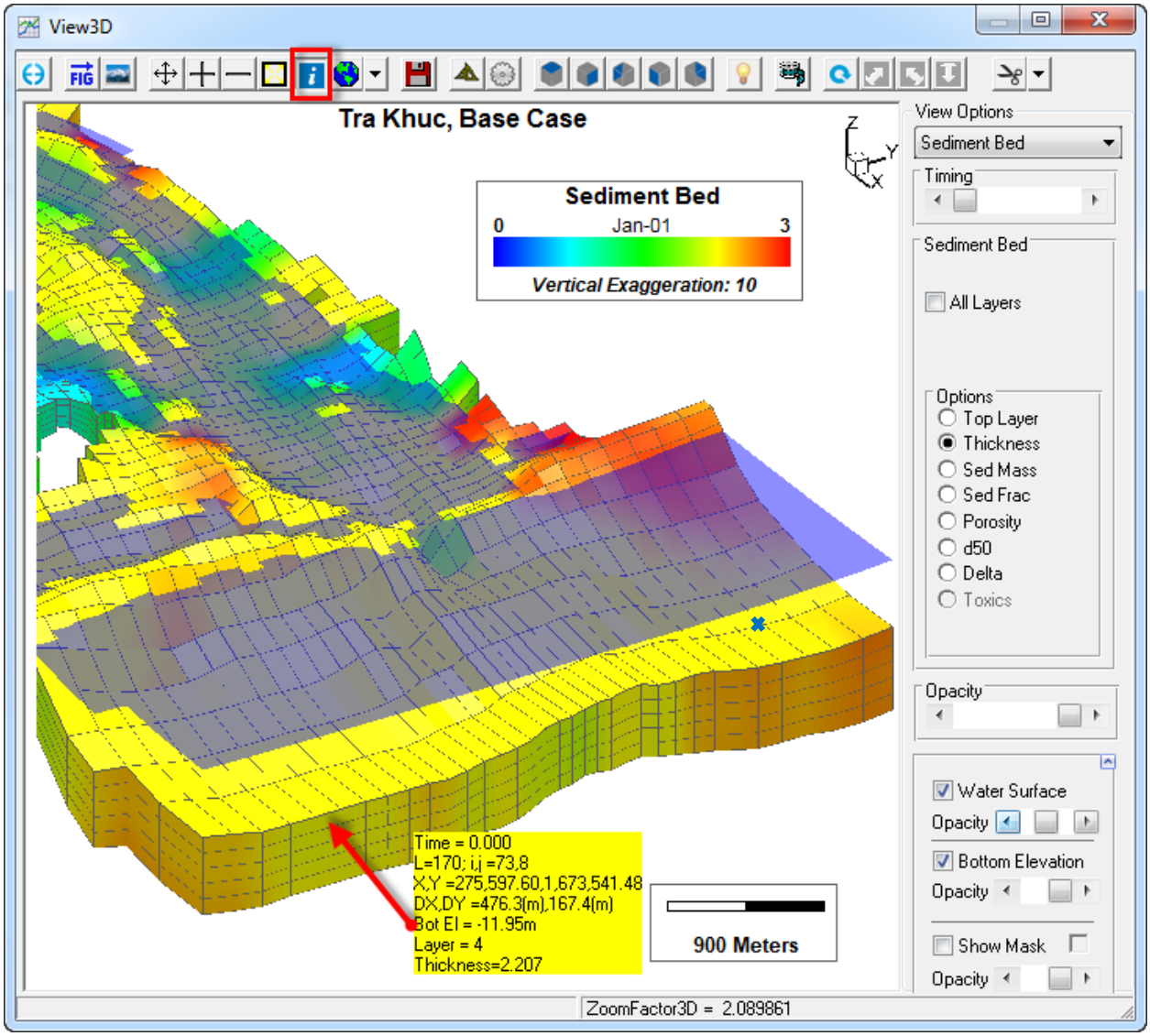


Figure 11-15 View3D: Mouse Enquire Button.

11.1.7. Particle Tracks and Oil Spill Viewing Options

Display options for particles in *View3D* are similar to those in *ViewPlan*, including display by elevation or group. In the particle tracking sub-model of prior to EE7.3, particles that reach the model boundary stick to the edge of the boundary as shown in Figure 11-16. This is to show that the particles have left the model domain and are no longer being calculated. Another method now provides the user the option of allowing the particles to disappear if the the *Hide After Exit* check box is selected as shown in Figure 11-17.

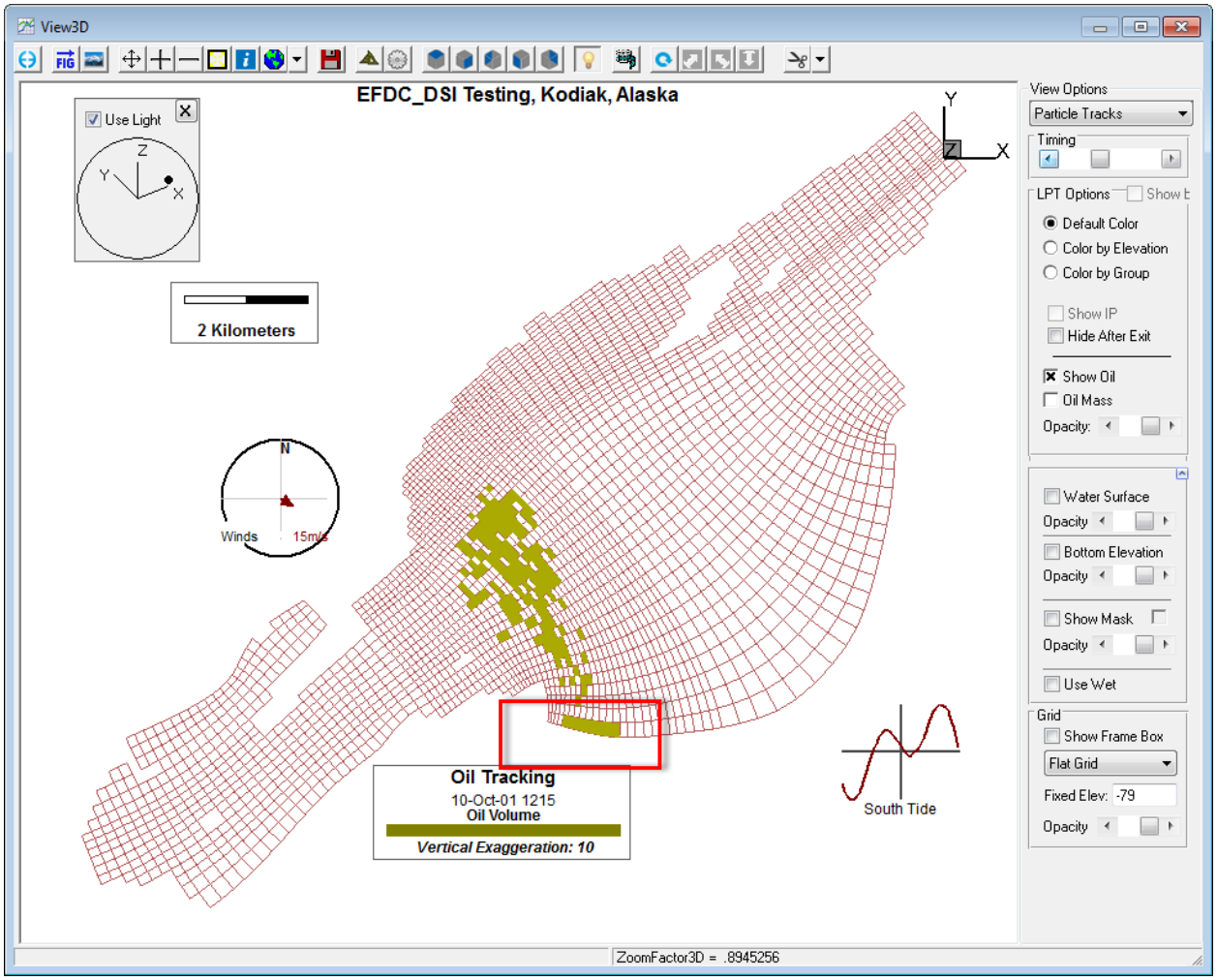


Figure 11-16 View3D: Oil spill still displayed after exiting model domain.

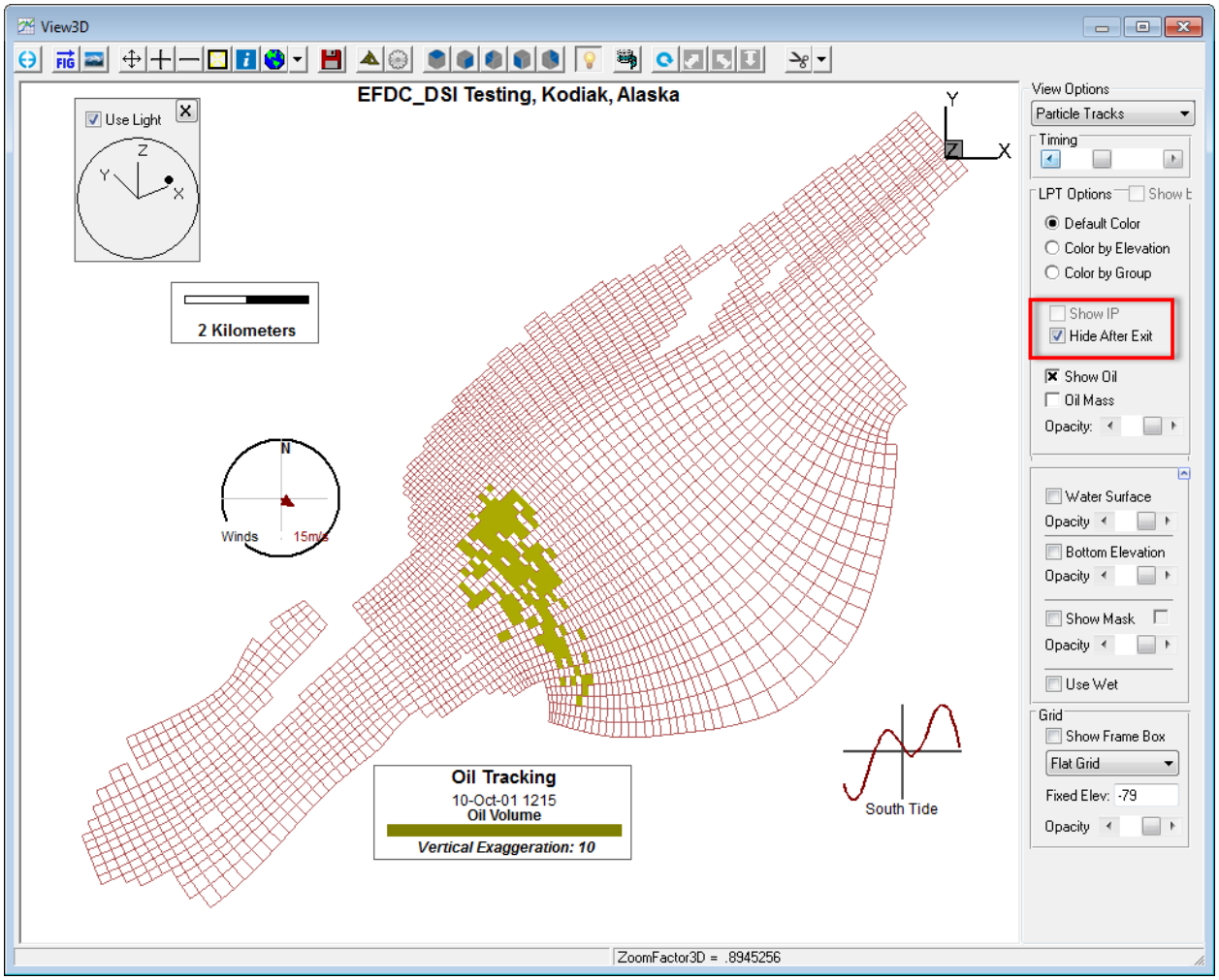


Figure 11-17 View3D: Oil spill hidden after exiting model domain.

If the the checkboxes *Show Oil* and *Oil Mass* are both selected then the mass of the oil is displayed using a color profile as shown in Figure 11-18.

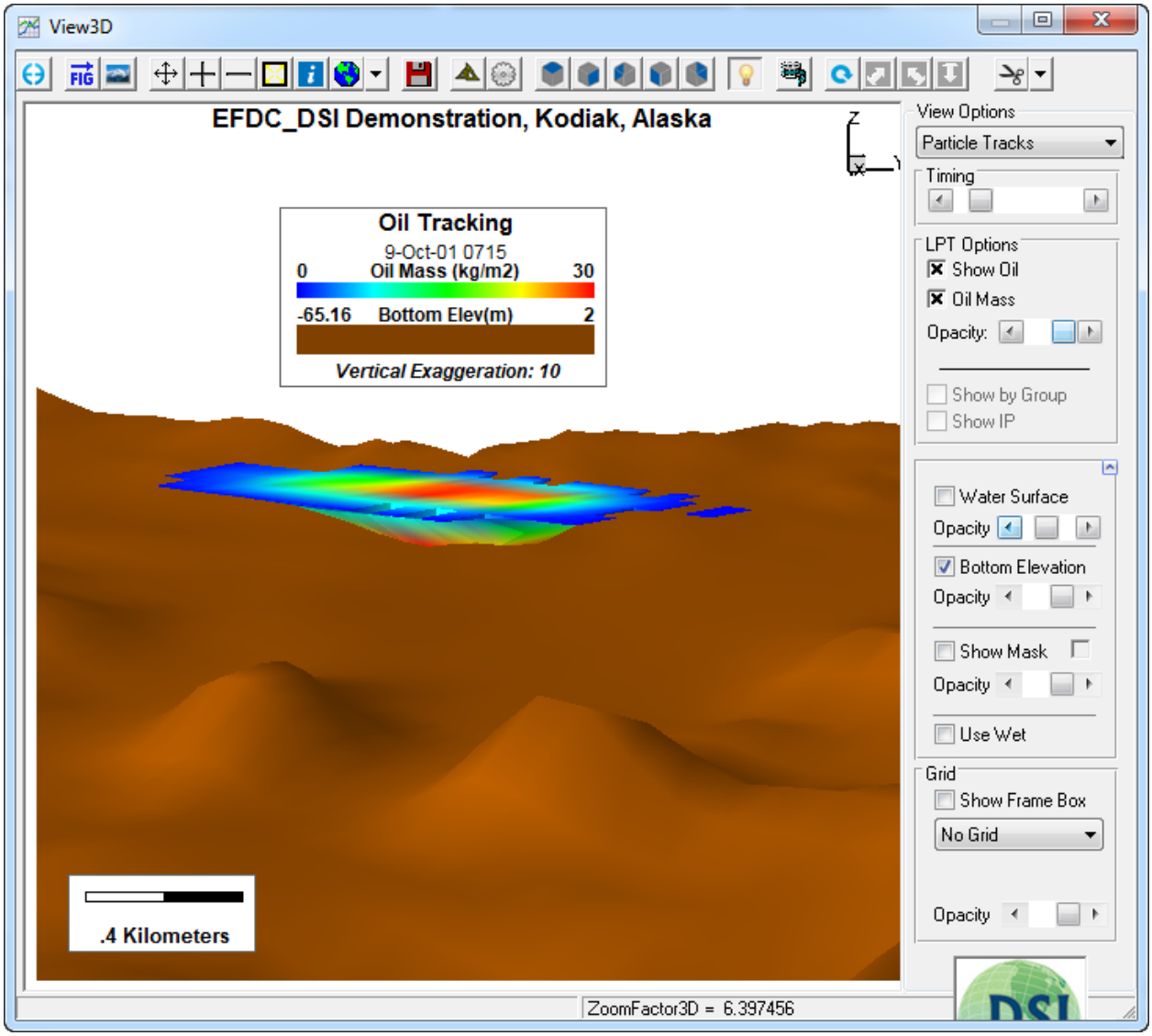


Figure 11-18 Oil Spill Tracking in View3D.

11.1.8. Import Structures in 3D

EE8 allows users to import pre-defined structures for visualization using *View3D* tool. This allows users to get a better feel for the model domain by displaying significant engineering structures in the model domain. Any COLLADA file can now be imported to 3D View as an overlay file. COLLADA stands for “COLLAborative Design Activity” and is an interchange file format for interactive 3D applications. It has a .DAE extension (digital asset exchange) add defines an open standard XML schema for exchanging digital assets among various graphics. COLLADA has been adopted by ISO as a publicly available specification, ISO/PAS 17506.

EE handles the COLLADA file as an overlay file which can be accessed in the *Display Options* menu shown in Figure 11-19. To add the 3D structure to *View3D* the user can either *Browse* to the file, or use more convenient “drag and drop” function to import *.DAE file directly to intended location in *ViewPlan* or *View3D*.

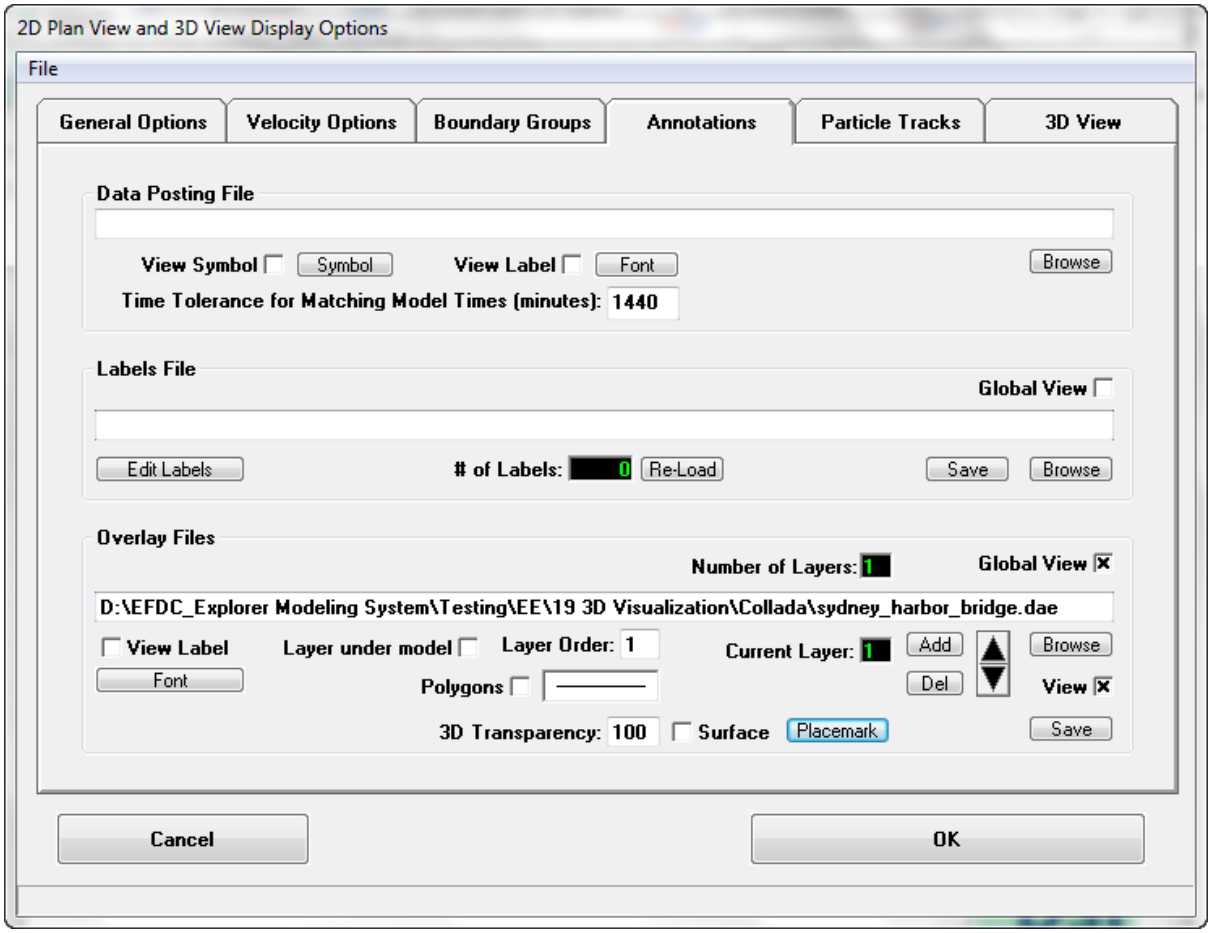


Figure 11-19 3D Structure: Setting overlay file.

Once loaded in, the users have extra flexibility such as adjusting X,Y,Z co-ordinates, re-scaling the object in three dimensions as well as changing the angle of rotation using the *Placemark* button as shown in Figure 11-20. This button may also be accessed with the Alt U key stroke from *ViewPlan* or *View3D*. In *View3D* the user may select the *Move* checkbox to allow the object to be dragged on the screen by the cursor to place it more exactly. An example of a complex structure that can be loaded into View3D is shown in Figure 11-21.

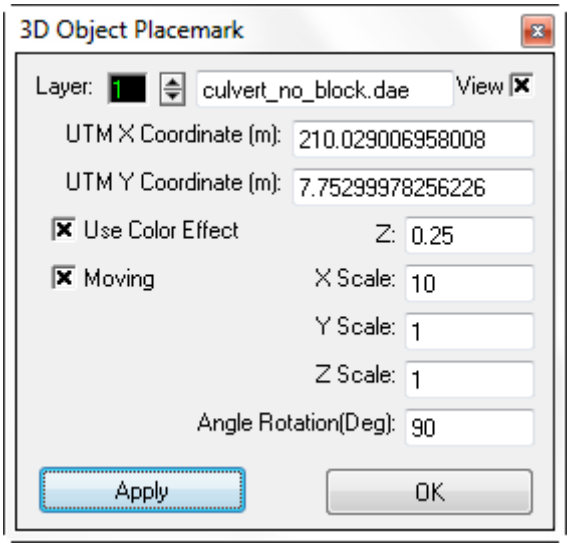


Figure 11-20 3D Structure: Setting place marks.

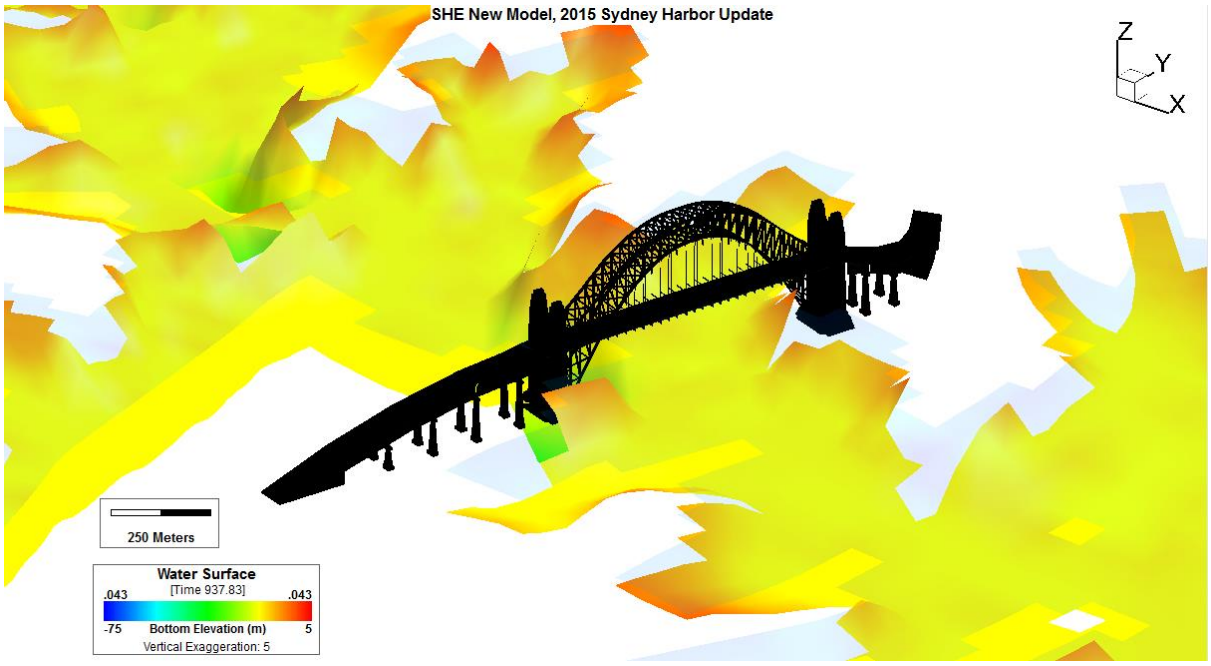


Figure 11-21 COLLADA file in View3D.

11.1.9. 3D Symbols

It is possible to display symbols as 3D images. These are selected in the normal way from *Display Options* | *Annotations* | *Edit Labels* | *Edit (Selected Labels)* as shown in Figure 11-22 and Figure 11-23.

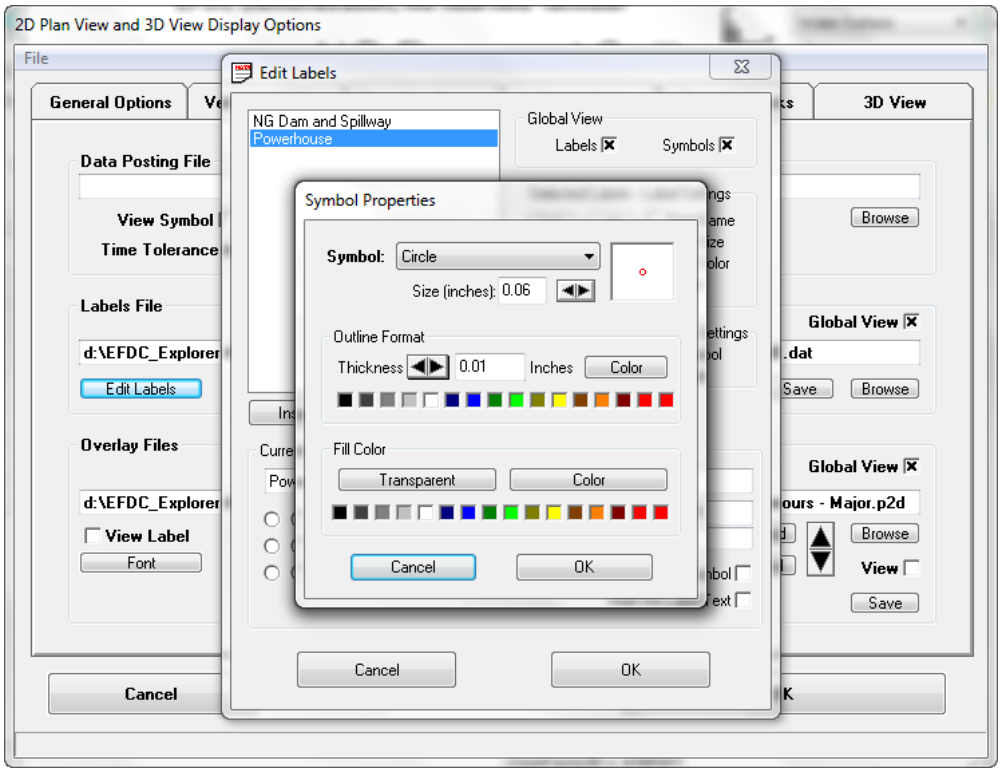


Figure 11-22 View3D Display Options: Configuring 3D symbols.

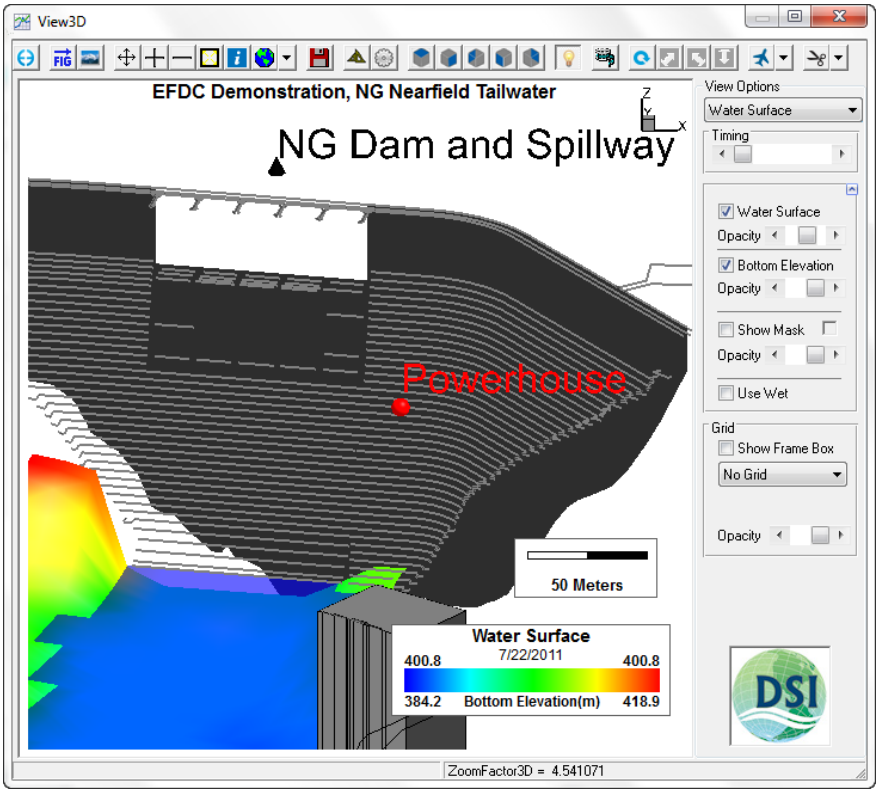


Figure 11-23 View3D Display Options: 3D triangle symbol.

11.1.10. Light Effects

Light effect in the View 3D form may be toggled on and off with the light bulb icon shown in Figure 11-24. This function allows the user to define the direction and intensity of the light source. The user may click the black dot in x, y, z sphere to move the source of light. The light color may also be set for both ambient light and diffuse light by clicking on the color box to the right of these options.

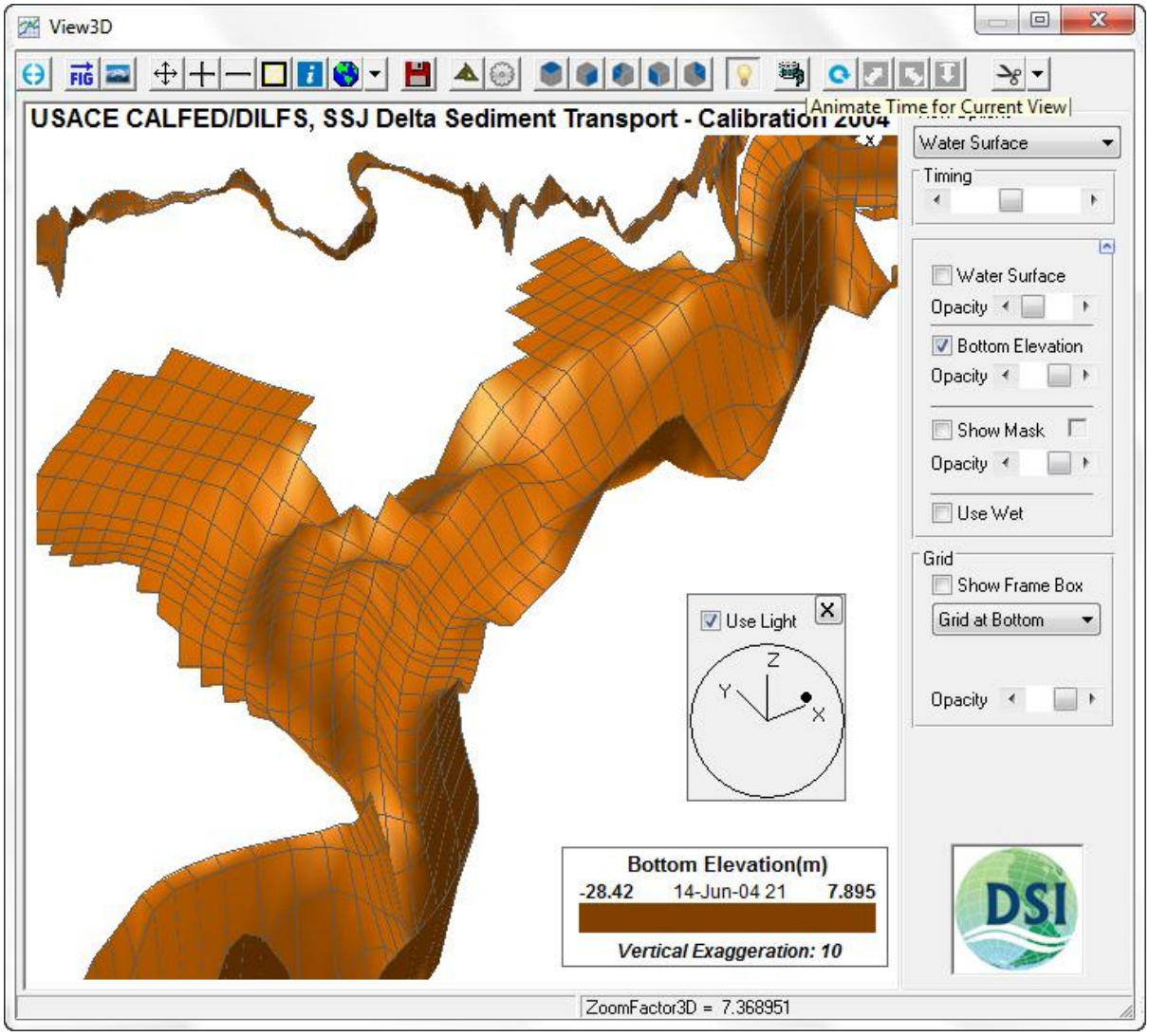


**Figure 11-24 View 3D: Light Effects **

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Appendix A

EFDC Internal Array Visualization Instructions

The only thing the user need to do to use EFDC_Explorer to plot and otherwise visualize arrays from within EFDC is to modify the EEXPORT subroutine in the section of code listed in the following textbox. To turn on the feature, use the following statement:

```
IF(.TRUE..AND.JSEXPLORER.EQ.0)THEN
```

To turn off the feature, use

```
IF(.FALSE..AND.JSEXPLORER.EQ.0)THEN
```

There are basic two types of output, one for time static arrays and one for those that vary as the model progresses (the standard case). Depending on the temporal nature of the array, the user must code the loops inside the IF/THEN block for time static arrays and outside/below the IF/THEN block for time variable arrays. The basic code for outputting the arrays is very simple and examples for both are shown in the text box. The user must make sure the flags are set right in order for EFDC_Explorer to correctly handle the arrays. The EFDC_INT.OUT file is a binary file for efficient reads and disk storage. The following is the basic structure:

DimFlag, TimeFlag	Two Integer*4 flags
ArrayName	One Character*8 name to identify the array
The array	Loop over the appropriate dimensions andoutput the array.

Remember to recompile after any changes to the source code.

A section of code from the EEXPOUT subroutine in EFDC.

```
      INTEGER*4 VER
      CHARACTER*8 ARRAYNAME

C**********************************************************C
C
      ! *** INTERNAL ARRAYS
      IF(.TRUE..AND.JSEXPLORER.EQ.0)THEN

        ! *** TIME STATIC ARRAYS
        IF(N.LT.(2*NTSPTC/NPSPH(8)))THEN
          OPEN(97,FILE='EFDC_INT.OUT',STATUS='UNKNOWN')
         CLOSE(97,STATUS='DELETE')
          OPEN(97,FILE='EFDC_INT.OUT',STATUS='UNKNOWN',
     &         ACCESS='SEQUENTIAL',FORM='BINARY')
          WRITE(97)VER  ! FILE FORMAT VERSION #
          WRITE(97)1    ! # OF TIME VARYING ARRAYS

          ! FLAGS: ARRAY TYPE, TIME VARIABLE
          ! ARRAY TYPE:    0 = L            DIM'D
          !                1 = L,KC         DIM'D
          !                2 = L,0:KC       DIM'D
          !                3 = L,KB         DIM'D
          !                4 = L,KC,NCLASS  DIM'D
          ! TIME VARIABLE: 0 = NOT CHANGING
          !                1 = TIME VARYING

          WRITE(97)0,0
          ARRAYNAME='WVKHV'
          WRITE(97)ARRAYNAME
          DO L=2,LA
            WRITE(97)WVKHV(L)
          ENDDO

        ENDIF

        ! *** TIME VARYING ARRAYS
        WRITE(97)2,1
        ARRAYNAME='QQ'
        WRITE(97)ARRAYNAME
        DO L=2,LA
          DO K=0,KC
            WRITE(97)QQ(L,K)  ! Turbulent Intensity (
          ENDDO
        ENDDO

      ENDIF
C
```

Appendix B

Data Formats

Data Format B-1 P2D Polyline/Polygon File.

Any number of polylines/polygons can reside in the same file. Each one begins with a single line header that is used as the "ID" of the polyline/polygon. Next comes the data in either 2D or 3D format, the importing automatically handles either. Finally the polyline/polygon is finished (not necessarily "closed") by an "*" in column 1. There is no difference between polyline and polygon in a P2D file, only how the data are treated by the application reading the file.

Example

```
------------------------
Polyline                                                    Test
609115.69390674                    3643612.72035394            0
608828.057738002                    3642922.39387814           0
608569.186338242                    3642232.06904821           0
608396.604307826                    3640908.94563456           0
...
...
...
601493.350247946                    3628713.19503985           0
601464.586301899                    3627763.99747289           0
601522.11337106                    3627016.14485373            0
601522.11337106                    3626642.21854415            0
```

Data Format B-2 TB2 XYZ Gridded Data.

This format is a compact way to store regular grid X, Y and Z data. It contains three lists, each with the number of data points as the first value in the list. The file contains a list of the X coordinates, a list Y coordinates and a list of the Z's.

Example

```
------------------------
195                                                         Nx
-97516.6484
-96511.4938
-95506.3391
-94501.1845
...
...
...
113                                                         Ny
656865
657873.929
658882.857
659891.786
...
...
...
22035                                                       Nz
-28.337
-28.346999
-28.1969829
```

Data Format B-3 DX Polyline/Polygon File.

The DX format is described by the following outline. The vertical bar "|" indicates the actual left side of the file.

```
Notes       |Contents of line beginning
            \/Column 1 of the actual file
-----------------------------------------------
Line01      |Description Line                    ] Header lines  (1st line in the file)
Line02      |#Polylines                          ]
Line01p1    |Index,nPts,branch,type,angle,deltax }Header lines for each polyline
Line02p1    |PolyLine ID                         }   branch,angle,deltax are only used
Line03p1    |PolyLine Desc                       }   for special cases but must be input.
Line01p1n1  |X,Y  or X,Y,Z   ] Data Points       }   Type = 0 Undefined (defaults to open)
Line02p1n2  |X,Y  or X,Y,Z   ]   Loop over nPts  }   Type = 1 Polyline (Open)
Line03p1n3  |X,Y  or X,Y,Z   ]                   }   Type = 2 Polygon  (Closed)
...
Linennp2nn  |X,Y  or X,Y,Z   ]                   }End of Polyline Block
Line01p2    |Index,nPts,branch,type,angle,deltax }Header lines for each polyline
Line02p2    |PolyLine ID                         }
Line03p2    |PolyLine Desc                       }
Line01p2n1  |X,Y  or X,Y,Z   ] Data Points       }
Line02p2n2  |X,Y  or X,Y,Z   ]   Loop over nPts  }
Line03p2n3  |X,Y  or X,Y,Z   ]                   }
...
Linennp2nn  |X,Y  or X,Y,Z   ]                   }End of Polyline Block
```

Example

```
HR Test Sections
 2
1,10,0,1,0,0
XS022
Geomorph XS022
55907.8741026827,910678.007703451,956.594
55907.2507413484,910678.789637365,956.594
55906.6273800142,910679.57157128,956.604
55906.0040186799,910680.353505195,956.574
55905.3806573456,910681.135439109,956.404
55904.7572960113,910681.917373024,956.884
55904.1339346771,910682.699306939,957.434
55903.5105733428,910683.481240853,956.954
55902.8872120085,910684.263174768,956.764
55902.2638506742,910685.45108683,957.564
2,22,0,1,0,0
XS023
Geomorph Transect XS023
55867.0383902119,910642.619457468,964.856
55866.7836519141,910643.049699725,961.236
55866.6308089354,910643.307845079,959.236
55866.2741753184,910643.910184238,957.346
55865.7646987227,910644.770668752,957.116
55865.255222127,910645.631153266,957.146
55864.7457455313,910646.49163778,957.146
55864.2362689356,910647.352122294,957.106
55863.7267923399,910648.212606807,956.976
55863.2173157442,910649.073091321,957.126
55862.7078391485,910649.933575835,956.716
55862.1983625528,910650.794060349,956.606
55861.688885957,910651.654544862,956.646
55861.1794093613,910652.515029376,956.696
55860.6699327656,910653.37551389,957.206
55860.1604561699,910654.235998404,957.606
55859.6509795742,910655.096482917,957.836
55859.1415029785,910655.956967431,957.716
```

Data Format B-4 Measured 3D Velocity Point Data (TecPlot).

```
TITLE = "ADCP IJK Data"
VARIABLES = "X", "Y", "Z", "V_E_W", "V_N_S", "V_U_D"
ZONE I= 257, J=1, K=1,F=POINT, T="Avg Vel"
56730.14 906511.90 288.38 -0.0443 -0.1582 -0.0001
56731.96 906530.90 288.42 -0.0570 -0.1482 -0.0001
56730.98 906533.00 288.45 -0.0599 -0.1219 -0.0001
56731.46 906535.70 288.49 -0.0578 -0.1042 -0.0001
56733.52 906512.90 288.39 -0.0548 -0.1524 -0.0001
...
...
...
```

Data Format B-5 I, J, K Formatted Measured 3D Velocity Data (TecPlot).

```
TITLE = "ADCP IJK Data"
VARIABLES = "X", "Y", "Z", "V_E_W", "V_N_S", "V_U_D"
ZONE I=40, J=26, K=1, F=POINT, T="Velocity"
56728.00 906511.00 288.54 -0.0457 -0.1574 0.00000
56732.00 906511.00 288.51 -0.0492 -0.1563 0.00000
56736.00 906511.00 288.52 -0.0549 -0.1551 0.00000
56740.00 906511.00 288.73 -0.0530 -0.1438 0.00000
...
...
...
```

Data Format B-6
Apply Cell Properties via Polygons: Assign Value Using Vertical Profiles

```
This file is used to assign a vertical and horizontally variable initial
condition field for any of the water column parameters.  The data is
basically a series of vertical profiles at any number of locations.  The
following summarizes the input format for one location.  This can be
repeated for as many locations as needed.

Line0:  Data file description

Loop over the number of vertical profile locations
   Line 1: ID                (any user defined ID)
   Line 2: Xc, Yc, nPts      (nPts in vertical profile)
   Loop over nPts
      Input Depth            (Positive depths below water surface, m)
   End Loop
   Loop over nPts
      Input Parameter_Value  (units dependent on parameter)
   End Loop
End Loop
```

Data Format B-7 Time series observation data file structure

Data file containing observation data (*.wq,*.dat)

```
10993  USGS_Speedy, Salinity, PPT
01-Jul-1999 00:00  27.7
01-Jul-1999 01:00  27.6
01-Jul-1999 02:00  27.8
01-Jul-1999 03:00  27.8
01-Jul-1999 04:00  27.7
01-Jul-1999 05:00  27.6
01-Jul-1999 06:00  27.6
01-Jul-1999 07:00  27.5
01-Jul-1999 08:00  27.5
01-Jul-1999 09:00  27.5
01-Jul-1999 10:00  27.6
01-Jul-1999 11:00  27.8
01-Jul-1999 12:00  27.9
01-Jul-1999 13:00  28
01-Jul-1999 14:00  27.9
01-Jul-1999 15:00  28
01-Jul-1999 16:00  28.1
01-Jul-1999 17:00  28.1
.....................
```

Description

First line: 10993: number (**N**) of data points; USGS_Speedy, Name, Units: title for some meaning (here: station name, water temperature in Celsius degree). This text is only used for labeling.

Second line to N lines: Date (Gregorian) (Julian date is also OK, for example the 370th day counting from 01-Jan-1999), time and parameter's value. The Gregorian date format can be in any format that Windows recognizes as a date. EE uses the last parameter in the line as the data value.

The data lines are repeated for all N data points.

Data Format B-8 Vertical Profile for Calibration/Validation Profiles.

```
SE 03, Unknown (Unknown) @  573370.8125,  3009112
 14           18-Aug-1999 11:25
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500   6.000
   6.300   6.300   7.100   7.100   7.400   9.000  10.300  10.300  11.200  11.700  12.300  14.200  16.500  17.500
 12           29-Sep-1999 10:15
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000
   1.600   1.600   1.600   1.700   1.700   1.700   1.700   1.700   1.700   1.700   1.700   1.700
 8            18-Oct-1999 12:05
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000
   0.100   0.100   0.100   0.100   0.100   0.100   0.100   0.100
 13           29-Nov-1999 11:33
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500
   6.300   6.700   7.500   8.400  10.600  12.100  12.700  12.700  16.700  17.000  17.100  17.200  18.400
 9            15-Dec-1999 11:00
   0.000   0.500   1.000   1.500   2.000   2.000   2.500   3.000   3.500
   8.500   8.600   8.600   8.800   9.300   9.700   9.900  10.700  11.480
 13           13-Jan-2000 10:54
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500
   4.100   4.400   8.900  10.300  12.900  14.100  15.100  15.100  15.300  15.100  16.800  17.000  17.200
 13           24-Feb-2000 12:26
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500
  23.800  24.000  24.000  24.200  24.200  24.200  24.200  24.200  24.300  24.300  24.300  24.300  24.400
 14           23-Mar-2000 10:52
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500   6.000
  21.000  21.000  21.100  21.200  21.200  21.300  21.300  21.300  21.300  21.300  21.300  21.700  21.700  21.700
 14           10-Apr-2000 11:26
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500   6.000
  21.600  21.700  21.800  21.800  21.900  22.600  22.700  22.700  22.900  23.000  23.200  23.300  23.500  23.600
 15           01-May-2000 12:00
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500   6.000   6.500
   1.800   1.800   2.200   4.000   5.100   6.300   7.200   7.200   8.700   9.400   9.600   9.700   9.800   9.900   9.900
 13           08-May-2000 11:15
   0.000   0.500   1.000   1.500   2.000   2.500   2.500   3.000   3.500   4.000   4.500   5.000   5.500
   2.300   2.500   2.700   2.700   2.800   2.900   2.900   2.900   2.900   3.200   3.900   3.900   4.200
 11           25-May-2000 12:15
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500
   6.100   6.100   6.100   8.200  10.900  13.700  16.100  16.100  18.700  21.200  21.600
 12           07-Jun-2000 11:30
   0.000   0.500   1.000   1.500   2.000   2.500   2.500   3.000   3.500   4.000   4.500   5.000
  18.400  18.700  19.100  19.300  20.100  20.600  20.600  21.100  21.700  22.500  23.100  23.500
 11           26-Jun-2000 13:20
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500
  17.900  17.890  18.300  21.450  21.640  21.710  21.790  21.790  21.960  22.230  22.340
 14           06-Jul-2000 11:20
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500   6.000
  30.900  20.300  21.500  21.800  22.200  22.600  23.100  23.100  23.600  23.700  24.000  24.000  24.000  24.000
 13           21-Aug-2000 11:55
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500
  15.400  15.300  15.300  15.400  15.400  15.400  15.500  15.500  15.700  15.700  16.100  16.100  16.700
 14           18-Sep-2000 12:28
   0.000   0.500   1.000   1.500   2.000   2.500   3.000   3.000   3.500   4.000   4.500   5.000   5.500   6.000
  21.280  21.210  21.240  21.930  22.060  22.200  22.380  22.380  22.230  22.560  22.470  22.850  23.000  23.040
```

First line: Title for some meaning (here: station name, coordinate) This field is not used by EE.

Second line: number of data points on the vertical profile (i.e., 14); Date (Gregorian) (i.e., 18-Aug-1999) and time (i.e., 11:25)

Third line: the water depths (from the water surface) of the points along the vertical profile (the depth unit must correspond to the model unit: meters)

Fourth line: the observation water quality values of the points. Units – Must correspond to the model units.

Repeat lines 2-4 for each profile.

Data Format B-9 Polygon DSM Format

The "Polygon" Digital Sediment Model (DSM) format is file that contains any number of polygons that define an area followed by a data block that contains the sediment data. The polygon ID and the data block ID's must match. The data block consists of a line for each depth (beginning at the surface or 0.0 depth) for which data exists. On each line the user must include the depth (m), thickness (m), porosity, and then the grain size. The number of grain size classes and the associated size breaks are determined by the space delimited data of the label line (see example). The number of grain size classes and their sizes must be the same for every sediment data block in the file. However, the size classes can vary from file to file or project to project to meet the project needs.

```
POLY ID1
X,Y
X,Y
X,Y
...
...
...
END
DATA ID1
depth thick density porosity Size1mm Size2mm Size3mm ... SizeNmm
Depth 1 data
Depth 2 data
...
...
...
END
```

Example

```
POLY L125.01
56694.6 901917.7
56697.3 901937.2
56678.3 901944.2
56674.4 901924.5
END
DATA L125.01
depth thick density porosity 50mm 19mm 9.5mm 0.85mm 0.25mm 0.075mm 0.065mm 0.023mm 0.013mm 0.004mm 0.003mm
0.000 0.152 1749.3 0.500 1.00000 0.99998 0.99988 0.98257 0.90709 0.70847 0.67613 0.41642 0.28167 0.09082 0.06417
0.152 0.152 1787.6 0.490 1.00000 0.99996 0.99976 0.97702 0.89186 0.68776 0.65569 0.40369 0.27499 0.09192 0.06584
0.305 0.152 1826.7 0.479 0.99999 0.99991 0.99956 0.97004 0.87468 0.66569 0.63396 0.38991 0.26726 0.09230 0.06695
0.457 0.152 1866.7 0.469 0.99998 0.99982 0.99921 0.96138 0.85544 0.64229 0.61100 0.37518 0.25857 0.09199 0.06749
0.610 0.152 1907.6 0.459 0.99996 0.99964 0.99864 0.95077 0.83405 0.61762 0.58686 0.35958 0.24902 0.09101 0.06747
0.762 0.152 1949.4 0.449 0.99990 0.99933 0.99772 0.93794 0.81045 0.59174 0.56162 0.34323 0.23869 0.08941 0.06692
END
```

Data Format B-10 Sediment Cores with Grainsize DSM Format

The "Sediment Cores with Grainsize" Digital Sediment Model (DSM) format is file that contains any number of cores. Each core can have any number of samples at depths below the top of the core. For each core the user must locate the core in horizontal (XY) space. The "Z" requested is the top of the core elevation (i.e. the bottom elevation at the core location).

```
Title Line
"Discrete" Flag: Used to determine data file format
Loop over groupings of cores (Loop Terminated by the "END" statement)
  ID, #Cores
  X Y Z #Depths
  Loop over #Depths
    Thickness, Porosity, SpecGrav, nFractions
    Loop over nFractions
      Max Grainsizes (um)
    EndLoop
    Loop over nFractions
      %Finer
    EndLoop
  EndLoop
END --> End Core Definitions with an "END"

Example  ------------------------------------------------------------------------
Tra Khuc Soil Sampling Results
Discrete
Core01   2
265814.3  1673167.4  5.90  4
0.25 56.00 2.66 9
20000 15000 7500 3500 1250 375 175 75 30
100.00 94.40 89.50 84.10 79.00 36.00 16.50 10.00 2.20
0.65 56.00 2.66 6
1250 375 175 75 30 8
100.00 90.80 70.20 33.00 17.80 9.00
5.00 56.00 2.65 7
7500 3500 1250 375 175 75 30
100.00 96.80 84.80 24.70 11.30 4.10 0.90
8.20 56.00 2.66 7
7500 3500 1250 375 175 75 30
100.00 97.60 87.60 29.30 13.90 5.20 1.00                          ! *** End of 1st Core
265808.1  1673613.9  0.13  2
0.25 56.00 2.66 9
20000 15000 7500 3500 1250 375 175 75 30
100.00 94.40 89.50 84.10 79.00 36.00 16.50 10.00 2.20
0.85 56.00 2.65 7
7500 3500 1250 375 175 75 30
100.00 96.50 83.10 27.70 17.10 4.20 0.00                          ! *** End of 2nd Core
```

Data Format B-11 Water Quality Point Source Loading Concentration Time Series File

```
C ** Caloosahatchee TMDL, WQPSLC Concentration Time Series FILE,  DDD  5/27/2008 10:28:34 AM
C **    This file is only used by EE to generate mass loadings for EFDC
C **    INPUT UNITS (mg/l) EXCEPT:  TAM(mol/l), FCB(MPN/l).
C **
C **   MWQPSR(NS) TCPSER(NS)  TAPSER(NS)  RMULADJ(NS)  ADDADJ(NS)
C **   TWQPSER(M,NS)  WQPSSER(M,NWV= 1: 7,NS)
C **                  WQPSSER(M,NWV= 8:14,NS)
C **                  WQPSSER(M,NWV=15:22,NS)
C **
C **  Time      CHC      CHD      CHG      ROC      LOC      DOC      ROP
C **            LOP      DOP      P4D      RON      LON      DON      NHX
C **            NOX      SUU      SAA      COD      DOX      TAM      FCB      MAC
C **
     732   86400        0        1        0 ! *** S79
  2922.000    0.0000    0.0000    0.1958    1.3686    1.3686   10.9485    0.0096
              0.0113    0.0113    0.0944    0.1130    0.4181    0.5989    0.0585
              0.2984    0.0000    0.0000    0.0000    6.2596    0.0000   49.9881    0.0000
  2923.500    0.0000    0.0000    0.1988    1.3360    1.3360   10.6880    0.0105
              0.0123    0.0123    0.1139    0.1220    0.4514    0.6465    0.0612
              0.3824    0.0000    0.0000    0.0000    6.5626    0.0000   49.9881    0.0000
  2924.500    0.0000    0.0000    0.2007    1.3255    1.3255   10.6037    0.0104
              0.0121    0.0121    0.1073    0.1214    0.4490    0.6432    0.0543
              0.3633    0.0000    0.0000    0.0000    6.4734    0.0000   49.9881    0.0000
  2925.500    0.0000    0.0000    0.2027    1.3934    1.3934   11.1468    0.0109
              0.0127    0.0127    0.0994    0.1212    0.4484    0.6423    0.0554
              0.3291    0.0000    0.0000    0.0000    6.3515    0.0000   49.9881    0.0000
  2926.500    0.0000    0.0000    0.2047    1.4910    1.4910   11.9281    0.0118
              0.0138    0.0138    0.0910    0.1234    0.4565    0.6538    0.0605
              0.2974    0.0000    0.0000    0.0000    6.4037    0.0000   49.9881    0.0000
```

Data Format B-12 External Windwaves

```
C **  Tra Khuc, FILE: WAVE.INP
C **  Version: EFDC_Explorer7.1.1 : Ver 140606
C **  Specify Information For Wave-Current Boundary Layer and Wave Induced
Flow
C **
C **  I   J    WVHEI    WANGLE    WVPER    WVLEN    WVDISP
C **
C **  I,J Cell Indices
C **
C **  WVHEI  = Wave Height (m)
C **  WANGLE    = Wave Angle (Degrees from East)
C **  WVPER    = Wave Period (seconds)
C **  WVLEN    = Wave Length (meters)
C **  WVDISP = Wave Energy Dissipation In (m/s)**3
C **
    52     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    53     3        0.0000      0.0000      0.0000      0.0000      0.0000
    54     3        0.0000      0.0000      0.0000      0.0000      0.0000
    55     3        0.0000      0.0000      0.0000      0.0000      0.0000
    56     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    57     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    58     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    59     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    60     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    61     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    62     3        0.0000      0.0000      0.0000      0.0000      0.0000
    63     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    64     3        0.0000      0.0000      0.0000      0.0000     -0.0090
    65     3        0.0000      0.0000      0.0000      0.0000      0.0000
```

Data Format B-13 External Windwaves

```
C ** JULIAN DAYS CORRESPONDING TO EXTR. WIND FILES
C ** FOR SWAN MODEL RUN IN CASE OF UNSTEADY WAVES
    225.0000
    225.0417
    225.0833
    225.1250
    225.1667
    225.2083
    225.2500
    225.2917
    225.3333
    225.3750
    225.4167
    225.4583
    225.5000
```

Data Format B-14 Harmonic Constants (harcont.ee)

```
C **  NSTA = Number stations
C **  MTIDE  = Number of tidal harmonic constants of this station
C **  xcoord = Longitude / X - coordinate
C **  ycoord = Latitude / Y - coordinate
C **  DATUM = Datum  in meters
C **  TIMEZONE = Time zone
C **  AMPCON = Conversion factor to convert amplitude to meters
C **  PHACON = Conversion factor to convert amplitude to degrees
C **  NAME = Station name
C **  NSTA
C **  MTIDE  XCOORD  YCOORD  DATUM  TIMEZONE  AMPCON  PHACON  NAME
C **  ID   CONSTITUENT AMP(cm)  PHA(deg)
       2     38       0   0   1.77799999713898  8     .01   1   Johor
  92     M2     2232.0000     21.00
 108     S2     2635.0000     48.90
  80     N2     3275.0000    358.30
  43     K1     3481.0000    204.30
 150     M4     3661.0000     82.20
  28     O1     2874.0000    209.20
 184     M6     3884.0000    138.80
 134     MK3    4096.0000     97.70
 160     S4     3603.0000    258.50
 146     MN4    4925.0000    192.70
  83     NU2    2428.0000    351.20
 197     S6     4315.0000    228.90
  76     MU2    2168.0000     10.90
  75     2N2    2165.0000    335.00
```

Data Format B-15 ISER.INP for ISICE = 1

```
C ** PROJECT NAME, ISER.INP TIME SERIES FILE
C **   FILE ISER.INP - EXTERNALLY SPECIFIED ICE COVER
C **   CONTROL AND TIME SERIES DATA REPEATING NISER TIMES.
C        **                        HEADER:                MISER(NISER),
TCISER(NISER),TAISER(NISER),RMULADJC,RMULADJT
C **       MISER    = NUMBER OF DATA, TCISER=TIME CONVERSION TO SEC,
TAISER=ADDITIVE TIME ADJ
C **
          4   86400       0       1       0
   240.001    0.3500
   240.010    0.5500
   240.020    0.7500
   240.030    0.6000
          6   86400       0       1       0
    20.000    0.0000
   238.000    0.4000
   252.000    0.5700
   283.000    0.7200
   290.000    0.5500
   320.000    0.3400
```

Data Format B-16 ICEMAP.INP for ISICE = 1

```
C ** , icemap.inp , Ice time series weightings for each cell and NISER Series
C **
C **
C       I    J    Weighting Fraction by Series
 1
  5113.000  5478.000
        2  107    3  0.000159  0.999672  0.000170
        3  108    3  0.000159  0.999671  0.000170
        4  109    3  0.000159  0.999671  0.000170
        5  110    3  0.000160  0.999670  0.000170
        6  107    4  0.000231  0.999520  0.000249
        7  108    4  0.000232  0.999519  0.000249
        8  109    4  0.000232  0.999518  0.000250
        9  110    4  0.000233  0.999517  0.000250
       10  107    5  0.000313  0.999346  0.000341
       11  108    5  0.000315  0.999344  0.000341
       12  109    5  0.000316  0.999342  0.000342
       13  110    5  0.000317  0.999340  0.000343
       14  107    6  0.000413  0.999135  0.000453
```

Data Format B-17 ISTAT.INP for ISICE = 2

```
C ** ISTAT FILE
C ** TIME SERIES ON THE STATUS OF ICE ON/OFF VIA ICECOVER
C ** FIRST DATA LINE: MISER(N),TCISER(N),TAISER(N),N=1(only one)
C ** NEXT DATA LINES: TISER(M,N),RICECOVS(M,N), M=1:MISER
C ** TISER: TIME, RICECOVS = 1: ON/0:OFF
         5   86400       0
     0.000    0.0000
   240.000    1.0000
   242.000    0.0000
   243.000    1.0000
   300.000    0.0000
```

Data Format B-18 ICE.INP for ISICE = 3 & 4

```
* ICE THICKNESS INITIAL CONDITIONS
*
*    I     J     THICKNESS[M]
     3   207      0.300
     3   208      0.300
     3   209      0.300
     3   210      0.300
     3   211      0.300
     3   212      0.300
     4   207      0.300
     4   208      0.300
     4   209      0.300
     4   210      0.300
     4   211      0.300
     4   212      0.300
     5   207      0.300
     5   208      0.300
```

Data Format B-19 Windmap

```
C ** Ver=80
C **  Caloosahatchee TMDL, FILE: WNDMAP.INP
C ** 002
C ** 003
C ** 004
      6
 2920.037  2921.000    1.000    0.000    0.000
 2921.000  2922.000    1.000    1.000    0.000
 2922.000  2924.037    1.000    1.000    1.000
 2924.037  3276.328    0.900    1.000    1.000
 3276.328  3368.787    0.900    1.000    0.000
 3368.787  3654.996    0.900    0.000    0.000
  2   3   3 0.59541 0.23848 0.16611
  3   4   3 0.59704 0.23938 0.16359
  4   5   3 0.59775 0.24085 0.16140
  5   6   3 0.59741 0.24294 0.15965
  6   7   3 0.59610 0.24537 0.15853
  7   8   3 0.59448 0.24769 0.15783
  8   9   3 0.59233 0.25028 0.15739
  9  10   3 0.58954 0.25331 0.15715
```

Data Format B-20 East West Connectors

```
C  FILE MAPPGEW.INP PROVIDES INFORMATION FOR MAKING MANUAL GRID
CONNECTIONS
C              IN THE EAST-WEST (I+1 AND I-1) DIRECTION
C
C NPEWBP     =  NUMBER OF EAST-WEST CONNECTIONS
C IWPEW,JWPEW =  I,J INDICES OF THE CELL CONNECTED AT THE WEST FACE
C IEPEW,JEPEW =  I,J INDICES OF THE CORRESPONDING CELL CONNECTED AT THE
EAST FACE
C
C   IWPEW   JWPEW   IEPEW   JEPEW  (REPEATED NPEWBP TIMES)
      9
      55    77    11    19
      55    76    11    20
      55    75    11    21
      55    74    11    22
      55    73    11    23
      55    72    11    24
      55    71    11    25
      55    70    11    26
      55    69    11    27
```

Data Format B-21 SEDZLJ Input Files
BED.SDF

```
# VAR_BED Bedload  Nequil LAYMAX  ISEDTIME  IMORPH IFWAVE MAXDEPLIM#
    1              1         0          7          1             0          0          1.0
# ITBM NSICM  Array Parameters #
   8    8
# ZBSKIN (>0 sets Zo in [um]) TAUCONST [dynes/cm^2] (> 0 to set constant)  ISSLOPE #
   10.0  0.0   0
# D50 of Size Class D50(K) [um] #
 10.0  22.0  222.0  375.0  750.0  4000.0  22.0  222.0
 # Critical Shear for Erosion TAUCRS(K) [dynes/cm^2] #
 1.0   2.2   1.6    2.4    3.8    30.8    2.2   1.6
# Critical Shear for Suspension TCRDPS(K) [dynes/cm^2] #
 1.0   2.2   2.4    3.5    12.5   98.8    2.2   2.4
# Sediment Bed Size (um) Tables #
 2.0  222.0  432.0  1020.0  2400.0  2600.0  3360.0  6000.0  8520.0
# Critical Shear for Erosion of Bed Surface [dynes/cm^2]  #
 0.50  2.27   2.96    4.17    5.88    6.07    6.72    8.48    9.75
#  Erosion rates for active and deposited Layers [cm/s] #
 1.00E-09 5.97E-05 5.97E-04 5.96E-03 5.95E-02 5.94E-01 5.94E+00 5.93E+01 !    2 Micron
 1.00E-09 5.97E-05 5.97E-04 5.96E-03 5.95E-02 5.94E-01 5.94E+00 5.93E+01 !  222 Micron
 1.00E-09 3.65E-04 2.16E-03 1.27E-02 7.49E-02 4.42E-01 2.61E+00 1.54E+01 !  432 Micron
 1.00E-09 2.01E-04 1.14E-03 6.51E-03 3.71E-02 2.11E-01 1.20E+00 6.85E+00 ! 1020 Micron
 1.00E-09 1.12E-05 8.40E-05 6.28E-04 4.70E-03 3.51E-02 2.63E-01 1.96E+00 ! 2400 Micron
 1.00E-09 7.94E-06 6.01E-05 4.55E-04 3.45E-03 2.61E-02 1.98E-01 1.50E+00 ! 2600 Micron
 1.00E-09 2.11E-06 1.67E-05 1.31E-04 1.04E-03 8.17E-03 6.44E-02 5.08E-01 ! 3360 Micron
 1.00E-09 2.03E-08 1.67E-07 1.44E-06 1.24E-05 1.07E-04 9.28E-04 8.02E-03 ! 6000 Micron
1.00E-09 1.19E-09 2.75E-09 1.69E-08 1.47E-07 1.33E-06 1.22E-05 1.11E-04 ! 8250 micron
```

Data Format B-22 SEDZLJ Input Files
ERATE.SDF

```
#  Layer Thicknesses                #
  0.00  0.00  15.00  15.00  15.00         15.00  15.00
 # Active layer thickness multiplier
 2.0
#  Core 1 - Riverine Sediments - Critical Shear Stress (dynes/cm^2)  #
 3.0  3.0  3.0  3.5  4.0  5.0  999.0
#  Bulk Density (g/cm^3)            #
 1.35 1.40 1.45 1.50 1.55 1.60 1.60
 # Water and sediment density (g/cm^3)  #
 1.0 2.65
#  Particle Size Distribution        #
  13. 57. 24. 5. 1. 0. 0. 0.
  13. 57. 24. 5. 1. 0. 0. 0.
  13. 57. 24. 5. 1. 0. 0. 0.
  13. 57. 24. 5. 1. 0. 0. 0.
  13. 57. 24. 5. 1. 0. 0. 0.
  13. 57. 24. 5. 1. 0. 0. 0.
  13. 57. 24. 5. 1. 0. 0. 0.
#  Initial Bed Erosion Rates         #
       0.0
 1.000E-09 1.000E-09 1.000E-09 1.000E-09 1.000E-09 1.000E-09 1.000E-09 1.000E-09
1.000E-09 1.000E-09
       2.0
 2.460E-09 2.460E-09 2.460E-09 1.706E-04 8.020E-05 2.260E-05 3.700E-09 3.700E-09
3.700E-09 3.700E-09
       4.0
 1.460E-03 1.460E-03 1.460E-03 5.790E-04 2.120E-04 7.980E-05 7.660E-08 7.660E-08
7.660E-08 7.660E-08
       8.0
 8.680E-03 8.680E-03 8.680E-03 1.910E-03 5.580E-04 2.820E-04 1.580E-06 1.580E-06
1.580E-06 1.580E-06
       16.0
 5.150E-02 5.150E-02 5.150E-02 6.280E-03 1.470E-03 9.950E-04 3.280E-05 3.280E-05
3.280E-05 3.280E-05
      32.0
 3.060E-01 3.060E-01 3.060E-01 2.070E-02 3.890E-03 3.510E-03 6.770E-04 6.770E-04
6.770E-04 6.770E-04
      64.0
 1.820E+00 1.820E+00 1.820E+00 6.820E-02 1.030E-02 1.240E-02 1.400E-02 1.400E-02
1.400E-02 1.400E-02
     128.0
1.080E+01  1.080E+01  1.080E+01  2.250E-01  2.710E-02  4.380E-02  2.900E-01  2.900E-01
2.900E-01 2.900E-01
```

Data Format B-23 SEDZLJ Input Files
ENSIGHT.SDF

```
#
C Input file for Ensight output
C values of 1 turns on the variable and 0 turns off the variable
C         1   U  - X velocity
C         2   V  - Y velocity
C         3  TAU - Shear Stress on active layer
C         4  D50 - Averaged particle size
C         5  CBL - Bedload concentration
C         6  SED - Suspended sediment concentration
C       7  CHC - cyanobacteria
C       8  CHG - diatom algae
C       9  CHD - green algae
C       10  ROC - refractory particulate organic carbon
C       11  LOC - labile particulate organic carbon
C       12  DOC - dissolved organic carbon
C       13  ROP - refractory particulate organic phosphorus
C       14  LOP - labile particulate organic phosphorus
C       15  DOP - dissolved organic phosphorus
C       16  P4D - total phosphate
C       17  RON - refractory particulate organic nitrogen
C       18  LON - labile particulate organic nitrogen
C       19  DON - dissolved organic nitrogen
C       20  NHX - ammonia nitrogen
C       21  NOX - nitrate nitrogen
C       22  SUU - particulate biogenic silica
C       23  SAA - dissolved available silica
C       24  COD - chemical oxygen demand
C       25  DOX - dissolved oxygen
C       26  TAM - total active metal
C       27  FCB - fecal coliform bacteria
C       28  CO2 - Dissolved Carbon Dioxide
C       29  MAC - macroalgae
C       30  HEAT- Heat Content
C         31  TEMP- Temperature (K)
C  U    V   TAU  D50  CBL  SED
C  CHC  CHD  CHG  ROC  LOC  DOC  ROP  LOP  DOP  P4D  RON   LON  DON
C  NHX  NOX  SUU  SAA  COD  DOX  TAM  FCB  CO2  MAC  HEAT  TEMP
   1    1    1    1    1    1
   0    0    0    0    0    0    0    0    0    0    0    0    0
   0    0    0    0    0    0    0    0    0    0    0    0
```

Data Format B-24 SEDZLJ Input Files
CORE_FIELD.SDF

```
   9     !  Number of Sedflume cores
1000000000
```

Appendix C

Derived Constituents for Water Quality

```
' *** Water Quality
' ***  1) CHC - cyanobacteria
' ***  2) CHG - diatom algae
' ***  3) CHD - green algae
' ***  4) ROC - refractory particulate organic carbon
' ***  5) LOC - labile particulate organic carbon
' ***  6) DOC - dissolved organic carbon
' ***  7) ROP - refractory particulate organic phosphorus
' ***  8) LOP - labile particulate organic phosphorus
' ***  9) DOP - dissolved organic phosphorus
' *** 10) P4D - total phosphate
' *** 11) RON - refractory particulate organic nitrogen
' *** 12) LON - labile particulate organic nitrogen
' *** 13) DON - dissolved organic nitrogen
' *** 14) NHX - ammonia nitrogen
' *** 15) NOX - nitrate nitrogen
' *** 16) SUU - particulate biogenic silica
' *** 17) SAA - dissolved available silica
' *** 18) COD - chemical oxygen demand
' *** 19) DOX - dissolved oxygen
' *** 20) TAM - total active metal
' *** 21) FCB - fecal coliform bacteria
' *** NWQV+1) MAC - fecal coliform bacteria

' *** Water Quality Derived Parameters

  ' *** 23 - Total Organic Carbon
  TOC = CHC + CHG + CHD + ROC + LOC + DOC

  ' *** 24 - Total N
  TN = AlgaeRatio(N:C)*(CHC + CHG + CHD) + RON + LON + DON + NHX + NOX

  ' *** 25 - Total P
  TP = AlgaeRatio(P:C)*(CHC + CHG + CHD) + ROP + LOP + DOP + P4D

  ' *** 26 - Chlorophyll a
  Chl-a = AlgaeRatio(Chla:C)*CHC + AlgaeRatio(Chla:C)*CHG + AlgaeRatio(Chla:C)*CHD

  ' *** 27 - Total Organic N
  TON = AlgaeRatio(N:C)*(CHC + CHG + CHD) + RON + LON + DON

  ' *** 28 - Total Inorganic N
  TIN = NHX + NOX

  ' *** 29 - Total Organic P
  TOP = AlgaeRatio(P:C)*(CHC + CHG + CHD) + ROP + LOP + DOP

  ' *** 48 - Carlson's Trophic State Index

  ' *** 49 - Particulate Organic Carbon
  POC = CHC + CHG + CHD + ROC + LOC

  ' *** 50 - Particulate Organic Phosphorus
  POP = AlgaeRatio(P:C)*(CHC + CHG + CHD) + ROP + LOP
```

```
' *** 51 - Particulate Organic Nitrogen
PON = AlgaeRatio(N:C)*(CHC + CHG + CHD) + RON + LON

' *** 52 - Total Suspended Solids (Inorganic + Organic)  TSS(io)
TSSio = TSED + TSND + CHC + CHG + CHD + ROC + LOC

' *** 53 - Total Kjedahl Nitrogen (TKN)
TKN = PON + NHX
TKN = AlgaeRatio(N:C)*(CHC + CHG + CHD) + RON + LON + NHX

' *** 54 - Total Dissolved Phosphorous (DOP+PO4)
TDP = DOP + PO4

' *** 55 - Macroalgae per unit area  ug Chl-a/m2
MAC/m2 = AlgaeRatio(Chla:C)*MAC/Thickness[Layer 1]
```